

Moderated Poster Session 1  
Adrenal, Kidney & Ureter  
Monday, October 13  
1300-1430

**MP-01.01**

**A Novel Laparoscopic Adrenalectomy via Transumbilical Approach: Focus on Technique**

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**Introduction and Objectives:** To evaluate the feasibility, safety and efficacy of a quick technique for novel laparoscopic adrenalectomy via transumbilical approach.

**Materials and Methods:** Between May 2010 and December 2013, 49 patients including 26 males and 23 females, with a mean age of 41.5 (range 21 to 67) years, were subjected to transvaginal NOTES-assisted laparoendoscopic single-site surgery adrenalectomy (TN-LESS-A) or suprapubic-assisted laparoendoscopic single-site surgery adrenalectomy (SA-LESS-A) in our center. Of them, 26 cases had left adrenal tumors, 22 cases had right adrenal tumors, and 1 case had bilateral adrenal tumor combined with right renal tumor. The mean diameter of adrenal tumor was 3.5 (range 0.8 to 6.8) cm. A quick technique was used in all the procedures. The bigger specimens were removed through an extended incision at posterior vaginal fornix or an enlarged pubic hairline.

**Results:** Twenty-one TN-LESS-As and 28 SA-LESS-As were all successfully completed. The median operative time was 45 (range 30 to 60) mins. The median estimated blood loss was 30 (range 20 to 90) ml. The patients resumed ambulation and oral diet within 2 days after surgery. Hospitalization duration was between 5 and 9 days with a mean postoperative stay of 6.5 days. There was no major intraoperative or postoperative complication in each patient. The scars were hidden within the umbilicus or not detectable because of the pubic hairs. During the follow-up, there was no tumor recurrence in any of the patients, and marked clinical improvements appeared in all the patients with a hormone-secreting tumor. However, only one patient with adrenal cortical carcinoma died after 8 months postoperatively because of multiple metastases in his body.

**Conclusion:** This quick technique for novel laparoscopic adrenalectomy via transumbilical approach is feasible and safe with less blood loss, minimal invasion, less operative time, fewer complication, short convalescence and good cosmetic result.

**MP-01.02**

**Conditional Survival Analysis for Patients with Adrenal Cortical Carcinomas: Results from Population-based Data**

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**Introduction and Objectives:** Surgical excision is essential management for adrenal cortical carcinomas (ACC) and offers the best change of cure. This study examine the effect of conditional survival (CS) on future survival probability after surgical excision in patients with ACC.

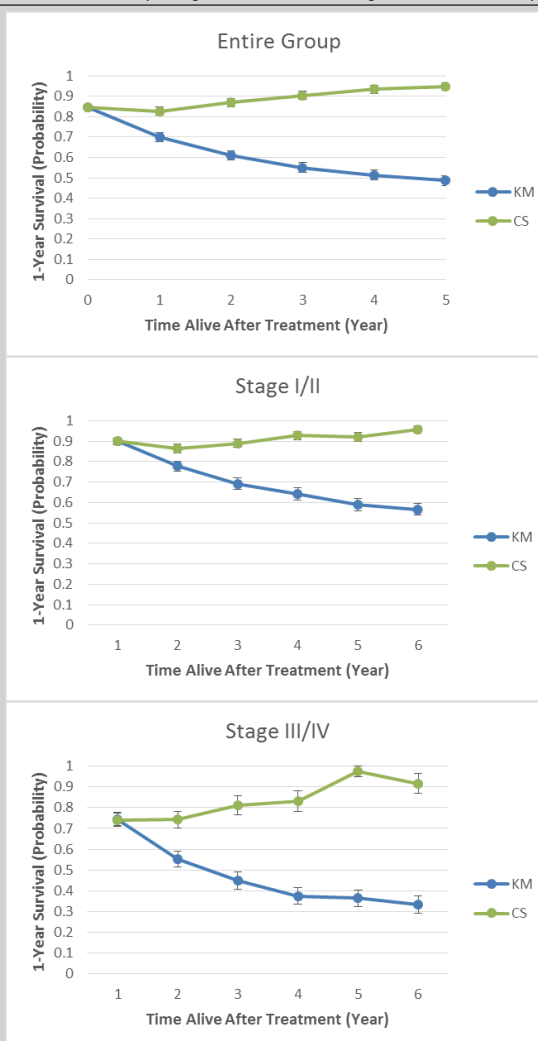
**Materials and Methods:** Using population-based Surveillance, Epidemiology, and End Results database (1988-2010), 531 patients with ACC who underwent surgery were reviewed. Cox proportional hazard models were used to assess the prognostic factors at baseline. CS estimates were calculated in the entire cohort and according to Union for International Cancer

Control (UICC) stage.

**Results:** Median age of entire cohort was 53 years. Most patients had stage II disease (59%), followed by stage III (21%), IV (15%) and I (6%). Median OS was 66 months and 5-year OS rate was 51.4%. Multivariate analysis showed that age and tumor stage had significant effect on OS ( $P < 0.01$ ). One-year OS for the entire cohort at 1, 2, 3, 4, 5 years was 70%, 60.9%, 55%, 51.4% and 48.7%, respectively. CS of an additional year given survival to 1, 2, 3, 4, 5 year was 82.7%, 87%, 90.4%, 93.4% and 94.9%, respectively. Patients who survive >24 months after their initial treatment had improved 1-year CS. The difference between Kaplan-Meier estimates and CS estimates was more pronounced in stage III/IV disease (Figure 1).

**Conclusion:** CS may provide more meaningful life expectancy predictions for survivors of ACC than conventional survival outcomes. Specially, even amongst patients with advanced disease

**MP-01.02, Figure 1. Conditional and Actual 1-Year Survival Rates for Patients Who Have Already Survived 0–5 Years for the Entire Group, Stage I/II Disease and Stage III/IV Disease, Respectively**



after surgery, a more favorable prognosis can be achieved after surviving for 2 years.

#### MP-01.03

##### **Metallic Ureteral Stent in the Management of Benign and Malignant Ureteric Obstructions: Five Years Experience**

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**Introduction and Objectives:** The reconstructive management of ureteric strictures, both of benign and malignant natures may not be desirable or possible due to various risk factors. The emergence of the metallic ureteral stents had enhanced treatment armamentarium with intention of reducing the interval of stent change and increase radial strength. We review our five-year experience of Resonance stent in the management of ureteric obstructions, to assess the patency rates and adversity for the metallic stents in both malignant and benign strictures. **Materials and Methods:** Retrospective analysis of patient with Resonance stent insertions between 2009 and 2014 were carried out. Patients demographic, etiology of occlusion, stent patency and complications were recorded. The necessity for stent change was identified by deterioration of renal function patients with bilateral stents.

**Results:** Forty-four Resonance stents were inserted in 24 patients, of whom 14 had bilateral stents, ten unilateral and six reinsertions (all bilateral). The mean age was 60 years (Range 38-79). Fourteen men and 10 women were treated, with 28 stents in malignant obstructions and 16 benign causes. The aetiology of the malignancy includes gastrointestinal, gynaecological and prostatic cancers; and the benign causes were post-radiation strictures, post-operative ischemia and ketamine bladder contractures. The mean follow-up was 28 months (3 to 58 months). The longest functioning stent was 50 months. The stent patency for both malignant and non-malignant causes was 75%. Half of the patients with malignancy were deceased with functioning stent. The mean interval of change due to obstruction was 11 months (7 to 23 months). The non-malignant group had patency interval of 16 months (3 to 50 months) and mean interval of change is 12 months in 3 patients. Recurrent urinary tract infections were observed in five patients, sepsis recorded in two patients and one stent migration noted.

**Conclusion:** The Resonance metallic ureteral stent is an effective alternative to polymer-based stents for select patients with upper urinary tract obstruction. The potential advantages of metallic ureteral stents include greater tensile strength allowing increased dwell times. Our results demonstrated good stent efficacy in both malignant and benign ureteric obstructions.

Stringent monitoring of renal functions is warranted to prevent stent occlusions.

#### MP-01.04

##### **Twin Ureteric Stents for the Management of Malignant Ureteric Obstruction**

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**Introduction and Objectives:** Malignant ureteric obstruction carries a poor prognosis and represents an endourological challenge due to failure of conventional ureteric stents and the problems associated with long-term percutaneous nephrostomies. The insertion of twin ureteric stents (TUS) represents a minimally-invasive option when single stents have failed. In this study we reviewed our experience and outcomes with TUS.

**Materials and Methods:** Our prospectively-maintained electronic stent register was interrogated to identify TUS procedures performed at our tertiary referral institution. Our technique involves the simultaneous insertion of two 8-French ureteric stents over two superstiff guidewires which are deployed through a 10-French dual-lumen catheter, after coaxial dilatation of the ureters. Nephrostomies, when present, are clamped on the operating table and removed the following day after a nephrostogram confirmed drainage. Pre- and post-operative renal function, technical failure of TUS and patients' survival were the main measured outcomes.

**Results:** Nineteen TUS procedures (12 insertions and 7 changes) were performed. Bilateral TUS was performed in 16 out of 19 cases. Indications were advanced cancers of prostate (37%), rectum (37%), ovaries (21%) and bladder (5%) causing ureteric obstruction with failure of conventional stents. The procedure was technically successful in all patients. Eleven patients had pre-operative nephrostomies for a median time of 17 days. All of them were rendered free of nephrostomies postoperatively. Serum creatinine levels remained stable in the post-operative period and upon discharge (average 235  $\mu\text{mol/L}$  versus 223  $\mu\text{mol/L}$  post-nephrostomy prior to TUS insertion,  $p=0.32$ ). Nine patients died at a median 91 days post-TUS insertion or exchange. Three of them developed signs of stent failure (rising serum creatinine and worsening hydronephrosis on imaging) and had a median life expectancy of 45 days, compared to 156 days for those with stable or improved serum creatinine ( $p<0.05$ ).

**Conclusion:** TUS is an effective and technically successful option for the management of malignant ureteric obstruction, with the majority of patients dying of the underlying condition after a relatively short period of time, however with functioning stents. TUS allows discharge of the majority of patients without nephrostomies in situ. Early stent failure is a

poor prognostic sign, which is associated with even shorter survival.

#### MP-01.05

##### **Endopyelotomy for the Treatment of Pelviureteric Junction (PUJ) Obstruction: Outcomes over a 9-Year Period**

Aldiwani M, Abroaf A, Browning A, Biyani C

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**Introduction and Objectives:** Many urologists consider laparoscopic pyeloplasty as the gold standard treatment for patients with PUJ obstruction due to higher reported success rates. However, endopyelotomy in selected patients still enjoys good outcomes with some advantages. The aim of this audit was to assess outcomes in patients undergoing endopyelotomy at our unit.

**Materials and Methods:** A retrospective analysis of all patients undergoing endopyelotomy for PUJ obstruction at our centre July 2003 – February 2012 was performed. Renograms were analysed pre and post operatively. Success was defined as marked/moderate symptom relief and/or relief of obstruction on renogram. Complications and long-term outcomes were recorded where available.

**Results:** Twenty eight patients underwent endopyelotomy during the prescribed time period, 36% Male, 64% Female; mean age 57 (range 22-88). Three patients were lost to follow-up. Eighty four percent of patients (21/25) had moderate or marked symptom relief. Ninety two percent (23/25) of patients had complete resolution of obstruction. One patient ultimately required nephrectomy due to a poor functioning obstructed kidney. Complications included stent symptoms in 4 patients (16%), urinoma in 1 patient and urosepsis in 1 patient. Of those patients who did not have successful outcome ( $n=4$ ), 2 had previous lap pyeloplasty, 1 patient had crossing vessel on CT and 1 patient failed to attend subsequent follow-up.

**Conclusion:** The majority of patients had a successful outcome after undergoing this minimally invasive, relatively simple (compared to lap pyeloplasty) procedure. Endopyelotomy achieves acceptable results in a carefully selected group of patients and should be considered as a treatment option for patients with PUJ obstruction.

#### MP-01.06

##### **Transmesocolic Pyeloplasty for Left Side Pelvi-Ureteric Junction Stenosis: A Pioneer Initial Experience in Nepal**

Joshi R

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**Introduction and Objectives:** To evaluate the initial outcome of Transmesocolic (TMC) laparoscopic pyeloplasty.

**Materials and Methods:** This is an initial observational study of our cases. We started Laparoscopic Urosurgery from 2009. After gaining experience in retroperitoneal approach we started transperitoneal approach and transmesocolic approach for left PUJ stenosis. Till date we have performed more than 25 transperitoneal (colon reflecting) and 6 TMC pyeloplasty. TMC was started from Jan 2013 to Jan 2014 at Kathmandu Medical College. All patients underwent Fenger's pyeloplasty and three patients had pyelolithotomy as well. We have used conventional laparoscopic instruments for all the procedure. All patients were followed by intravenous urography (IVU) at 3 months. Patients were explained about the procedure and conversion rate. Written and verbal consent were taken. Patients with complicated UTI, sepsis were excluded from the study.

**Results:** Six patients underwent TMC left pyeloplasty. There were 4 female and 2 male patients with age ranging from 21-28 years. Operating time ranged from 120-150 minutes with mean of 132 minutes. Per-operative blood loss was minimal ranging from 10-20 ml. Return of bowel movement was within 1-2 days. All patients were discharged within 2 days and drain removed with 2-3 days. Two patients were discharged on 2<sup>nd</sup> day but drain was removed on 3<sup>rd</sup> day in outdoor. Follow up of 5 patients showed improvement in their pain status and IVU showed no re-stricture or stenosis.

**Conclusion:** Transmesocolic approach for left PUJ obstruction is feasible and seems to have low morbidity with shorter hospital stay. Comparison with Colon-reflecting pyeloplasty and larger number of cases are needed for better statistical evaluation.

#### MP-01.07

##### The 'Anderson-Hynes Pyeloplasty': Are We All Really on the Same Page?

Adam A, Smith G

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**Introduction and Objectives:** The Anderson-Hynes Pyeloplasty (AHP) is commonly performed in the management of pelvi-ureteric junction (PUJ) obstruction. This procedure was initially described using an L-shaped incision on the dilated renal pelvis. This resulted in an inferior pelvic flap which created a funnelled, dependent anastomosis across the reconstructed PUJ. Although the AHP has edged its name in the annals of Urology, we hypothesise that the AHP has become a polymorphic entity within the recent literature, which may not always be referring to the same procedure.

**Materials and Methods:** An Ovid MEDLINE(R) and Ovid OLDMEDLINE(R) database search from 1946 to 20th March 2014, was performed for the terms 'Anderson and Hynes' and 'Anderson-Hynes', with

restriction to the *title* of journal article. Each respective author's descriptive images (figures/photographic) of technique performed was compared to the original procedure. Non-English figure legends were translated using online translational tools.

**Results:** In total, 57 articles (33 English/24 Non-English) were retrieved. Only 29/57 papers had referenced the original procedure. Operative images were present in 17/57 of the articles. Within these papers, only 7/17 articles depicted both the L-shaped cut and pelvic flap. Although all papers were describing dismembered pyeloplasties, diverse variations were observed in the remainder 10/17 articles; including a V, I and Q shaped pelvic cut.

**Conclusions:** The Anderson-Hynes pyeloplasty is often not performed as was originally described. Over the decades, various authors have performed variations to the original incisions. These variations may affect outcomes. To assist with this nomenclature, an 'incision specific' classification system has been proposed for the dismembered pyeloplasty.

#### MP-01.08

##### D Track: A New Software to Track Inserted DJ Stents

Zaidi Z, Khan R

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**Introduction and Objectives:** A Double J Stent (DJS) is a temporary device commonly used in endourological practice. Due to their concealed nature, DJS are often forgotten, and remain within the body beyond their required duration. "Forgotten" DJS is a common problem worldwide and may lead to various complications, ranging from local tissue reaction to sepsis. Given the serious nature of potential complications, keeping track of each inserted DJS is essential. The objective was to design a software which would keep track of DJ stents inserted and send reminders to primary team when to remove them.

**Materials and Methods:** An electronic reminder program for follow up of patients with in situ DJS was piloted at the Indus Hospital, Karachi. The Management Information System (MIS) at our institute records all patient events on an electronic database. A tool was developed whereby the MIS automatically captures all DJS insertions done at our Hospital and calculates an expected date for removal, which was pre-defined to be 3 months from the date of insertion. Additional features integrated into this software include the generation of reports indicating the patients who are overdue, and also provides us information about DJ stents due removal in the coming weeks and months. This is important as it helps prevent missed stents one of the key objectives of the software. Hence is the name of the software D-track or device tracking software. One week prior to the

calculated removal date, an SMS reminder is automatically sent by the MIS to the cell phone of a member of the primary team. Patient medical records are then reviewed and depending on clinical requirements, patients are contacted and scheduled to come back to hospital for removal of the stent.

**Results:** Over a period of 1 year since deployment, no patient with DJ stent has been forgotten. All stents have been removed within time frame set. A timely reminder generated by the software allows physicians to contact patients and call them for DJ removal. When run through the data base of 1753 DJ stents historically placed, 67 missed stents were recognized and retrieved successfully.

**Conclusion:** D track is a low cost software allowing timely removal of DJ stents by keeping track of all stents placed, plan their removal, send reminders for forgotten stents and hence helps in decreasing morbidity associated with forgotten stents. The software design is such that all this is achieved without any human interface. This software can be used to keep track of any device inserted in patients.

#### MP-01.09

##### High-pressure Balloon Dilation for the Treatment of Iatrogenic Ureteral Stricture after Gynecologic Surgery

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**Introduction and Objectives:** We report the safety and efficacy of high-pressure balloon dilation for the treatment of iatrogenic ureteral stricture after gynecologic surgery.

**Materials and Methods:** Between October 2007 and September 2013, a total of 28 ureteral strictures in 26 patients were confirmed to be caused by ureteral injury sustained during gynecologic surgery. We inserted a double-J stent immediately. After 1 month, 30-atm high-pressure balloon dilation was performed with a retrograde approach and a double-J stent was placed again after the treatment. Three months later, the double-J stent was removed.

**Results:** The mean length of the obstructed ureteral segment was 2.4 cm (range 0.5 to 8.0). Balloon dilation was performed once in 22 ureters, and repeated once or twice in 6 ureters after a few months. The mean follow-up duration was 17 months (range 5 to 42). The overall success rate was 75% (21/28). The mean length of the obstructed segment was longer in failed cases (3.3 cm) than in successful cases (2.2 cm). The success rate was 100% (14/14) in patients with benign gynecologic disease. No major complications occurred in any of the cases.

**Conclusion:** High-pressure balloon dilation is a safe and efficient method for the treatment of iatrogenic ureteral stricture after gynecologic surgery.

## Moderated Poster Session 2

### BPO/LUTS

Monday, October 13  
1300-1430

#### MP-02.01

##### The Association between Air Pollution and Benign Prostatic Hyperplasia

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<sup>3</sup>Gachon University, Incheon, South Korea

**Introduction and Objectives:** Potential explanations for large geographic differences in BPH prevalence may be due to genetics or lifestyle and environment. However, little is known about the role of environmental factors in benign tumors associated with BPH. Therefore, we focused on the air pollution as an environment factor affecting BPH and attempted to explain the regional differences in BPH prevalence in relation to air pollution.

**Materials and Methods:** The present study used data from two large population studies: community- and university hospital- based survey which covered nine major areas of South Korea, a total of 56 institutions, and 1,734 participants in those areas. This study attempted to determine the mean emissions of air pollutants with National Air Pollutants Emission 2010 report and TeleMetering System by the National Institute of Environmental Research within the Korean Ministry of Environment.

**Results:** The mean IPSS of total group was 15.25 in the low-level area group and 18.58 in the high-level group and the differences among the area groups were statistically significant ( $p<0.01$ ). The comparison of IPSS among age groups by three area groups adjusted for age revealed that in all the age groups the area groups displayed higher IPSS scores when they had higher air pollutant emissions ( $p<0.05$ ).

**Conclusion:** It can be inferred that if air pollution as an environmental factor affects

malignant neoplasm of the urinary tract, it will have an effect on benign tumor as BPH. Accordingly, the differences in such association could explain variations in BPH prevalence among geographic areas. These findings were the first study to demonstrate the association between air pollution and BPH. Our findings demonstrated increased risks of BPH severity among men exposed to higher concentrations of air pollution.

#### MP-02.02

##### Correlation between Prostate-Urethral Angle and Intravesical Prostate Protrusion on Uroflowmetry and International Prostatic Symptoms Score (IPSS) in BPH Patients

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**Introduction and Objectives:** Lower urinary tract symptoms (LUTS) are one of the most common complaints in aging men, and benign prostate obstruction is one of the most common causes. It is believed that the increase in prostate volume is not related to degree of LUTS. Several studies have reported the importance of anatomical factors in evaluating men with LUTS such as Intravesical prostatic protrusion (IPP) and Prostate urethral angle (PUA). We investigated the correlation between IPP and PUA as anatomical factors that might influence LUTS based on IPSS and Uroflowmetry. The aim of this study is to investigate the correlation between intravesical prostatic protrusion and prostate urethral angle in BPH patient with LUTS based on IPSS and uroflowmetri.

**Materials and Methods:** The study was done on 71 BPH patients with LUTS who attended the urologic clinic of Hasan Sadikin Hospital and matched the inclusion and exclusion criteria. Patient's age, IPSS, PSA, transrectal ultrasound for prostate measurement, uroflowmetry and residual volume were recorded for statistical analysis. Patients were then classified based on prostatic-urethral angle ( $<35^\circ$  dan  $\geq 35^\circ$ ) and

degree of intravesical prostatic protrusion (grade I  $< 5\text{mm}$ ; II  $5\text{--}10\text{mm}$ ; III  $> 10\text{mm}$ ). We used Mann-Whitney test for comparative study, and Spearman correlation test for correlative study. P-values of  $<0.05$  were considered statistically significant.

**Results:** There is an increase in total IPSS ( $22.31 \pm 4.87$ ;  $p<0.001$ ) in the PUA  $\geq 35^\circ$  include in IPSS storage and IPSS voiding. It is followed by decrease in Quality of life (QoL:  $4.02 \pm 0.75$ ;  $p<0.001$ ), decrease in Qmax ( $7.28 \pm 2.11$ ;  $p<0.001$ ), decrease in Qave ( $5.44 \pm 2.06$ ;  $p<0.001$ ), decrease in voided volume ( $184 \pm 79.55$ ;  $p=0.04$ ) also increase in PVR ( $55.30 \pm 29.69$ ;  $p=0.014$ ). Increase in degree of IPP is related to increase in total IPSS, IPSS storage and IPSS voiding: ( $p<0.001$  dan  $0.027$ ) and decrease in Qmax and also increase in PVR ( $p<0.025$ ,  $p<0.033$ ). There is significant correlation between PUA and IPP with LUTS ( $p<0.001$ , with (r) correlation coefficient 0.670 and 0.640).

**Conclusion:** Prostatic-urethral angle and intravesical prostatic protrusion have significant correlation with LUTS in BPH patients based on IPSS and Uroflowmetry. These two kinds of conditions seem to influence LUTS as anatomical factors in BPH patients and Prostate urethral angle has stronger correlation than intravesical prostatic protrusion.

#### MP-02.03

##### The Association between Metabolic Syndrome and Benign Prostatic Enlargement Depends on Prostate Cancer Status

Bhindi B, Kulkarni G, Alibhai S, Hamilton R, Toi A, Trachtenberg J, Zlotta A, Finelli A, Fleshner N

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**Introduction and Objectives:** Metabolic syndrome (MetS) is associated with both benign prostatic hyperplasia and simultaneously an increased risk of prostate cancer (PC) and high grade disease. The interplay between MetS and these two separate pathologies remains to be characterized. Our objective was to evaluate the association between MetS and benign prostatic

MP-02.01, Table 1. IPSS of area groups according to National Air Pollutants Emission 2010

| Area groups | Criteria 1* | Criteria 2** | Province                                 | N (1734) | IPSS of total group†  | IPSS of Age groups        |                           |                           |                   |
|-------------|-------------|--------------|------------------------------------------|----------|-----------------------|---------------------------|---------------------------|---------------------------|-------------------|
|             |             |              |                                          |          |                       | $\geq 50$ & $<60^\dagger$ | $\geq 60$ & $<70^\dagger$ | $\geq 70$ & $<80^\dagger$ | $\geq 80^\dagger$ |
| High        | $> 100$     | $> 10$       | Gangwon, Ulsan                           | 187      | $18.58 \pm 8.50^{\S}$ | $17.2 \pm 8.0$            | $18.7 \pm 8.4$            | $18.4 \pm 8.4$            | $22.2 \pm 10.6$   |
| Medium      | 31 - 100    | 1 - 10       | Incheon, Daegu, Daejeon, Gyeonggi, Busan | 1077     | $17.45 \pm 7.52^{\S}$ | $16.7 \pm 6.5$            | $17.8 \pm 7.3$            | $17.3 \pm 7.7$            | $18.1 \pm 8.7$    |
| Low         | $< 31$      | $< 1$        | Seoul, Gwangju                           | 470      | $15.25 \pm 7.93$      | $14.0 \pm 6.9$            | $15.4 \pm 8.3$            | $15.3 \pm 7.8$            | $15.5 \pm 8.4$    |

\* SUM \*\* SOx; kg/year/person, National Air Pollutants Emission

†  $p < 0.05$ , ‡  $p < 0.01$  for ANOVA. § same letters indicate no statistical significance based on Duncan multiple comparison



enlargement (BPE) separately in men with and without PC diagnosis.

**Materials and Methods:** Men undergoing prostate biopsy were identified using our institutional database. Men undergoing active surveillance or with prior history of TURP were excluded. The most recent biopsy was used in men with multiple biopsies, in order to best reflect underlying cancer status. MetS required any 3 of 5 components (obesity, diabetes/impaird fasting glucose, hypertension, low HDL-cholesterol, and high triglycerides). The primary outcome was TRUS-measured prostate volume (PV). The secondary outcome was International Prostate Symptom Score (IPSS). The Kruskal-Wallis test and multiple linear regression and logistic regression analyses adjusting for age, ethnicity, alpha-blocker use, and 5-alpha reductase inhibitor use, were performed.

**Results:** The cohort included 2428 men, of whom 1402 (57.7%) were found to harbour PC. PV increased in a graded manner is association with the number of MetS components present. Among men without PC, MetS was associated with PV (0 MetS components: 43.0cc (IQR=35.0-57.0) vs. 5 components: 59.5cc (IQR=38.0-83.5);  $p=0.003$ ) but not in men with PC ( $p=0.28$ ). Among men without PC, men with 5 MetS components had a higher median IPSS compared to 0 components (13 (IQR=7-19) vs. 8 (IQR=4-12)), although significance was not attained ( $p=0.18$ ). There was no association between MetS and PV or IPSS among men with PC. In multiple linear regression analyses, MetS was associated with PV among men without PC ( $p=0.027$ ) but not among men with PC ( $p=0.21$ ). Multivariable associations between MetS and IPSS did not reach significance.

**Conclusions:** MetS is associated with larger prostate volume, with the largest effect among men shown not to have underlying PC. It is an interesting observation that MetS promotes PC in some men, and promotes BPE in other men. This study raises interesting questions about the impact of MetS on the biology of the prostate.

#### MP-02.05

##### Time-Response Relationship and Clinically Meaningful Improvement of Lower Urinary Tract Symptoms Secondary to Benign Prostatic Hyperplasia (LUTS/BPH) During Tadalafil Treatment

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<sup>3</sup>Eli Lilly and Company, Indianapolis, USA

**Introduction and Objectives:** Physicians currently lack evidence of the onset of clinically meaningful improvement (CMI) in patients being treated with tadalafil for lower urinary tract symptoms secondary to benign prostatic

hyperplasia (LUTS/BPH). This post-hoc analysis identified the cumulative proportion of men who achieved CMI as a function of time on therapy.

**Materials and Methods:** This is a post-hoc analysis of pooled data from four randomized, double-blind, placebo-controlled, 12-week studies investigating tadalafil 5 mg (N=742) or placebo (N=735) in men aged  $\geq 45$  years of age with International Prostate Symptom Score (IPSS)  $\geq 13$ . CMI was defined as an improvement in total IPSS of  $\geq 2$  for patients with moderate LUTS at baseline,  $\geq 6$  for patients with severe LUTS at baseline and  $\geq 3$  for the overall population. All studies measured IPSS at Weeks 4, 8, and 12; additionally, two studies collected IPSS data at Week 1 and one study collected IPSS data at Week 2.

**Results:** At the 12-week study endpoint, 69% of men on tadalafil vs. 55% on placebo achieved a CMI across all four studies ( $\geq 3$  IPSS improvement;  $p<0.001$ ). Among men who achieved CMI at Week 12 with tadalafil, 55-64% achieved CMI by Week 1, 58% by Week 2, 75-84% by Week 4, and 82-87% by Week 8 ( $\geq 3$  IPSS improvement). Time-to-onset of CMI was generally shorter for men with moderate vs. severe LUTS at baseline.

**Conclusion:** Of men who respond to tadalafil therapy (69% of the study population), approximately 60% respond within 1-2 weeks and approximately 80% respond within 4 weeks.

#### MP-02.07

##### Comparative Analysis of Benign Prostatic Hyperplasia Managements between Urologists and Non-Urologists: Korean Nation-Wide Health Insurance Database Study

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**Introduction and Objectives:** Using Korean nationwide health insurance data, we compared the current management of benign prostatic hyperplasia by urologists and non-urologists.

**Materials and Methods:** We obtained patient data from the national health insurance system on the treatment of BPH. New patients diagnosed with BPH in 2009 were divided into two groups depending on whether they were diagnosed by a urologist (U group) or a non-urologist (NU group).

**Results:** A total of 390,767 individual were

newly diagnosed BPH in 2009. Of these, 240,907 (61.7%) patients were in U group and 149,860 (38.3%) in NU group. Among the non-urologists, internists were most common (57.7%), followed in descending order by general surgery (8.6%), dermatology (7.6%), orthopedics (6.9%), and family medicine (6.8%). The comparative analysis showed that the NU group was older, had more comorbidities, and had a tendency to live in a rural area. They were also more likely to be receiving medical aid and to be treated at a primary hospital or a public health center. The rate of all initial evaluation tests, except serum creatinine, was significantly lower in the NU group. The initial prescription rate was higher in the U group, whereas the prescription period was longer in the NU group. With regards to the initial drugs prescribed, the use of alpha-blockers was common in both groups. However, the U group was prescribed combination therapy of an alpha-blocker and 5 alpha-reductase inhibitor as the second choice, whereas the NU group received monotherapy with a 5 alpha-reductase inhibitor. During the 1-year follow-up, the incidence of surgery was significantly different between the U group and the NU group (3,213 vs. 646, respectively).

**Conclusion:** There is a distinct difference in the diagnosis and treatment of BPH by urologists and non-urologists in Korea. The difference may have adverse consequences for BPH patients. The urological society should take a leadership role in the management of BPH and play an educational role for non-urologists, as well as urologists.

#### MP-02.08

##### Hemi Transurethral Resection for Benign Prostates (H-TURP): Analysis of Results

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**Introduction and Objectives:** TURP is the gold standard surgical technique for treatment of symptomatic benign prostatic hyperplasia. However, large and vascular prostate may represent a surgical challenge both in open as well as in endoscopic resection. We assessed the role of Hemi TURP (H-TURP) as an alternative technique to the standard complete endoscopic resection or open prostatectomy.

**Materials and Methods:** We retrospectively studied patients with enlarged prostate who underwent H-TURP over a period of 5 years. The H-TURP was performed on patients for either high co-morbidity, intra-operative severe bleeding or to reduce the resection time. Prostate resection carried out using monopolar diathermy in all. We compared the results of pre-operative IPSS, QoL, flow test, PSA, and U&Es to those obtained after treatment.

**Results:** Total of 31 patients underwent H-TURP, mean age was 73 years. Thirteen of

**MP-02.08**, Table 1. Comparing our H- TURP results with the EAU quoted results for standard TURP and open prostatectomy.

| Results                         | H-TURP series     | EAU quoted figure for standard TURP | EAU quoted figure for open prostatectomy |
|---------------------------------|-------------------|-------------------------------------|------------------------------------------|
| IPSS improvement                | 51.5 %            | 70.6% 1                             | Up to 86% 4                              |
| QoL (mean) or percent.          | 0.93 at 12 months | 1.2 at 10 years 2                   | Up to 87% 4                              |
| Qmax improvement                | 195 %             | 125% 1                              | 375% 4, 5, 6                             |
| Average Tissue resected (gr)    | 28.86             | Variable                            | Variable                                 |
| PSA reduction                   | 49.54%            | Not reported.                       | Not reported.                            |
| Catheter time and Hospital stay | 3 days            | Variable                            | Variable                                 |
| Operative time                  | 42.8 min          | Not available                       | Not available                            |
| TURP syndrome                   | 0%                | 1.1% 3                              | Not applicable                           |
| Transfusion requirement         | 0%                | 8.4 % 1                             | 7-14% 4, 7                               |
| Re-treatment rate               | 3.2%              | 1-2% per year                       | 0% (at 5 years) 4, 5, 6                  |

these patients had H-TURP for acute urinary retention and 18 for bothersome lower urinary tract symptoms which failed initial pharmacotherapy. No episodes of TURP syndrome was noted (serum sodium mean reduction from 140 to 139), and no patient required blood transfusion. Patients had an average of 12 months post operative follow-up. The mean IPSS reduction is 51.5% and QoL score came down from average of 3.7 to 0.93 post-operatively. The mean Qmax improved from an average of 10.11 to 19.75ml/sec (195% improvement rate). The PSA was reduced 49.5% after H-TURP. Only one patient required further resection (3.2%) at 12 months time.

**Conclusion:** Although this is a small study, however H-TURP appears to be an effective technique in comparison to the gold-standard TURP and open prostatectomy published data. The H-TURP offered clear objective and subjective improvements with a low complication rate and may be suitable for patients with large or vascular prostate. However, the long-term durability of this technique needs to be addressed with longer term follow-up.

#### MP-02.09

##### Hybrid Bipolar Prostatectomy for the Large Prostate

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**Introduction and Objectives:** To evaluate safety and efficacy of a new hybrid technique for bipolar prostatectomy in large prostates without morcellation.

**Materials and Methods:** Twenty patients with BOO due to BPH were included from June 2011 to July 2012. Fifteen patients were catheterized for refractory retention, five had severe

LUTS. We excluded patients with prostate cancer, neurogenic bladder, previous prostatic surgery, urethral stricture, and UTIs. Prostate size was  $95.94 \pm 25$  gm. We used the bipolar high-frequency generator UES-40SurgMaster (Olympus Winter & Ibe GmbH, Hamburg, Germany). The procedure starts by developing a plane between the adenoma and the surgical capsule using the button loop to open the anterior commissure and to make an incision at 5 and 7 o'clock then proceeding to enucleation with the resection loop and tip of resectoscope until each lobe is attached only to the bladder neck, then the resection loop is employed and the lobes are sequentially resected. The button loop is employed again at the end of the procedure for evaporation of residual tissues and hemostasis. Perioperative parameters were recorded and the patients were followed up by IPSS, PVRU and Q-max at 1, 6, and 12 months.

**Results:** Mean operative time was  $146.88 \pm 30.92$  min, mean catheterization time  $3.81 \pm 1.47$  days, and postoperative hospital stay was  $4.25 \pm 1.76$  days. One patient required blood transfusion, no TUR syndrome, no clot retention. Hemoglobin reduction was  $16.04\% \pm 5.85$ . PSA reduction was  $82.22\% \pm 12.2$  ng/mL. All patients had spontaneous voiding postoperatively. The mean Q-max after catheter removal, 1 month, 6 months and 12 months was  $15.95 \pm 4.51$ ,  $16.65 \pm 4.97$ ,  $16.62 \pm 3.84$  and  $18.15 \pm 4.53$  ml/s, respectively. The mean IPSS at 1 month, 6 months and 12 months was  $4.75 \pm 2.35$ ,  $3.87 \pm 1.25$  and  $4.13 \pm 1.36$  respectively. The mean post voiding RU after catheter removal, 1 month, 6 months and 12 months was  $6.56 \pm 17.96$ ,  $12.56 \pm 17.25$ ,  $2.14 \pm 5.79$  and  $3.21 \pm 7.23$  ml, respectively.

**Conclusion:** The hybrid bipolar prostatectomy is safe and effective in relieving bladder outlet

obstruction by the large prostate.

#### MP-02.10

##### Clinical Course of Patients Receiving Anti-Platelets Therapy Who Underwent Thulium Laser Enucleation of the Prostate

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**Introduction and Objectives:** With the progressive ageing of the population, the prevalence of vascular disease as the prevalence of benign prostatic enlargement is increasing. In recent years, laser prostatectomy is emerging as a replacement for the standard transurethral resection of prostate (TURP). The aim of this study was to evaluate the clinical course of patients receiving anti-platelets therapy who underwent Thulium Laser Enucleation of the Prostate.

**Materials and Methods:** From September 2011, we started a prospective study on patients who underwent ThuLEP. All candidates for surgical therapy show lower urinary tract symptoms (LUTS) and obstruction due to a BPH. This work evaluated the surgical outcomes of 42 ThuLEP in patients taking anti-platelet therapy to 50 procedures performed in patients who have never taken anti-platelet agents before surgery.

**Results:** The study group included 39 patients who had been on anti-platelet monotherapy with ASA 100 – 300 mg and 3 in Ticlopidine. In the study group 37 procedures (88%) were performed under spinal anesthesia, while 5 were performed under general anesthesia. There was no significant difference in operative time. A comparison test between groups was not statistically significant with regards to the decrease in haemoglobin concentration. Transfusional support was required in one procedures performed in patients taking anti-platelet therapy, and in no procedures in controls. There were no adverse cardiac events (myocardial infarction, angina, cardiovascular failure, hypovolemic shock). One patient in the treatment group required re-intervention to ensure haemostasis. During this period one case of haematuria happens two week in every of the two group that was treated conservatively. No further bleeding or cardiac events were recorded.

**Conclusion:** In this study we demonstrated for the first time that patients undergoing ThuLEP who continued taking anti-platelet agents had no significantly increased incidence of perioperative bleeding-associated morbidity compared with those who were not taking any anti-platelet medication.

## MP-02.11

**Sexual Outcome of Patients Undergoing Thulium Laser Enucleation (ThuLEP) for BPH: First Study of 110 Patients**

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**Introduction and Objectives:** To assess the effect of ThuLEP on sexual function in a group of patients with LUTS secondary to BPH. To assess whether ThuLEP has any effect on retrograde ejaculation reduction.

**Materials and Methods:** Prospective study that analyzes changes in sexual function and urinary symptoms in a group of 110 consecutive patients that underwent ThuLEP. To assess the changes on erection and ejaculation, urinary symptoms and their interference on quality of life, three validated questionnaires were used: ICIQ-MLUTSsex, AUA questionnaire and QoL index of the ICC. Patients were evaluated before surgery and 3 and 6 months after ThuLEP. Patients with previous abdominal surgery were excluded. Statistical analysis was performed by the Student t and chi-square Test and logistic regression analysis. For all statistical comparisons significance was considered at  $p < 0.05$ .

**Results:** Mean age was 67.83 years. The table shows the changes in scores on the questionnaires employed. No significant differences were observed between the number of patients about erectile function before and after surgery. The percentage of patients with conserved ejaculation increased with ThuLEP to 52.7%.

**Conclusions:** ThuLEP causes an improvement in the scores of questionnaires that assess urinary symptoms and the interference of the symptoms in the quality of life of patients. Although endoscopic management of BPH (TURP and new technologies) causes most of patients have retrograde ejaculation, in patients who underwent ThuLEP there is a conserved ejaculation.

## MP-02.12

**Transurethral Thulium Laser Vapo-Resection versus Transvesical Open Enucleation for Prostate Adenoma Greater Than 80 G: A Study of 78 Patients**

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**Introduction and Objectives:** Prostate adenomas greater than 80 ml have traditionally been treated with open prostatectomy or transurethral resection by skilled resectionists. This procedure may involve considerable blood loss, morbidity, prolonged hospital stay and recovery time. We compare transurethral Thulium laser enucleation (ThuLEP) of the prostate to open prostatectomy for the surgical management of large prostate adenomas.

**Materials and Methods:** A total of 78 obstructed patients with a prostate larger than 80 ml on transrectal ultrasound undergo ThuLEP or open prostatectomy. All patients were assessed preoperatively and postoperatively. Patient baseline characteristics, perioperative data and postoperative outcome were compared. All complications were noted.

**Results:** Forty eight patients underwent ThuLEP, and 30, open prostatectomy. Mean patient age is 78 in the ThuLEP group and 72 in the open adenomectomy group. Mean preoperative transrectal adenoma volume is 126 ml and 115 ml respectively. Mean Hemoglobin loss was significantly less ( $p < 0.05$ ), and catheterization time ( $p < 0.05$ ) and hospital stay ( $p < 0.05$ ) were significantly shorter in the ThuLEP group. Adverse events ( $p < 0.05$ ) were more frequent in the open prostatectomy group such as prolonged haematuria, surgical infection, fever and analgesic drugs use. None of the Thulium group patients needed blood transfusions in contrast to 6 patients (20%) in the prostatectomy group.

**Conclusion:** ThuLEP and open prostatectomy are equally effective procedures for removal of large prostatic adenomas. ThuLEP resulted in

significantly less perioperative morbidity and may become the endourological alternative to open prostatectomy.

## MP-02.13

**Irritative Symptoms after Thulium Laser Enucleation of the Prostate (ThuLEP): Our Personal Experience**

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**Introduction and Objectives:** A recent technological advance for the surgical treatment of BOO has been the Thulium laser. The objective of the present study was to definite the immediate irritative symptoms following the endoscopic surgical treatments for benign prostatic obstruction with the Thulium laser enucleation of the prostate (ThuLEP).

**Materials and Methods:** From January 2012 to December 2013 we start a prospective study on patients who underwent ThuLEP in order to investigate the immediate post-operative irritative symptoms. All candidates for surgical therapy present of lower urinary tract symptoms (LUTS) and obstruction due to a prostate gland enlargement. All these patients have LUTS refractory to medical management. Thulium laser prostatectomy was performed by Quanta Cyber TM Thulium Laser.

**Results:** A total of 200 patients performed the surgical treatment. The patients' mean age was 67.4 years. Mean preoperative prostatic adenoma volume was 78 mL. Forty-six of the patients have an indwelling urethral catheter. Mean total operative time, including cystoscopy, enucleation and morcellation was 84 min. The average Foley catheter-time was respectively of 36 hours. IPSS, QoL and F Max improvement don't vary considering prostate volume, the presence of an indwelling urethral catheter and a significant PVR. In all, 4% of the patients reported urge incontinence 7 days after the procedure that reduce to 0.5 % at 30 days after the procedure and disappeared in the follow-up period. Urgency was present in the 12% of the patients 7 days after the procedure that reduce to 5 % at 30 days after the procedure and disappeared in the follow-up period. Mean follow-up was of 14 months (range 1-24 months). The result in reduction of IPSS, improvement of QoL, F Max were maintained in time.

**Conclusion:** ThuLEP and the reduced depth of coagulation zone in the prostate tissue could explain the reduction of these symptoms in the immediate post-operative period. So, the use of ThuLEP can reduce the frustrating irritative symptoms in the immediate post-operative period, with equivalent results in the relief of LUTS and uroflowmetry parameter in the long period follow-up.

MP-02.11, Table 1.

| Outcomes Data                       | Pre-Op         | At 3 months      | At 6 months      |
|-------------------------------------|----------------|------------------|------------------|
| IPSS                                | 18.1<br>SD 6.7 | 6.8<br>SD 5.3    | 4.3<br>SD 4.9    |
| QoL                                 | 4.2<br>SD 1.4  | 2.1<br>SD 1.7    | 1.4<br>SD 1.6    |
| IIEF 5                              | 16.8<br>SD 4.4 | 16.4<br>SD 3.1   | 18.1<br>SD 3.8   |
| Erection conserved<br>ICIQ-MLUTSsex | 87/110         | 100%             | 100%             |
| Ejaculation conserved<br>MSHQ-EJD   | 74/110         | 58 (52.7%) / 76% | 58 (52.7%) / 76% |



## MP-02.14

**Holmium Laser Enucleation of the Prostate: Validation of a Virtual Reality Simulator**

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**Introduction and Objectives:** Holmium laser enucleation of the prostate (HoLEP), though proven an effective and worthwhile surgery, is a difficult operation to learn associated with a long learning curve. Virtual reality simulation with its novel and varied capabilities may allow the learning curve to be shortened. However, validation is required to assess its potential for use in training. The objective of this study was to assess the Face, Content, and Construct validity of the VirtaMed UroSim™ simulator (VirtaMed AG, Zurich, Switzerland). **Materials and Methods:** This prospective observational study recruited 42 participants, comprising of expert HoLEP surgeons (>100 HoLEPs, n=6), Endourological trainees without HoLEP experience (n=18), and HoLEP novices (n=18). All participants received the same educational package on HoLEP, comprising of didactic lectures on technique, instructional videos, and videos of live surgeries. They then completed a 15-minute familiarisation exercise consisting of a cystoscopy exercise with the VirtaMed UroSim™ simulator (VirtaMed AG, Zurich, Switzerland) before carrying out a full enucleation on a standardised 60cc prostate. Data was collected using in-built simulator metrics and a quantitative questionnaire. The Mann-Whitney U test was used to compare expert, trainee, and novice groups. **Results:** Experts had a significantly increased enucleation efficiency (grams enucleated/hour) compared to Endourological trainees (p<0.001) and to Novices (p<0.001), with trainees significantly outperforming novices (p<0.001). Intra-operative complications were also increased in the novice group, with external sphincter damage being twice as common among novices

compared to experts.

86% of participants agreed that simulator based assessment is essential for patient safety, 61% thought that the overall experience was similar to the real life setting, and 87% agreed that there was a role for a validated VR simulator for use in HoLEP training. Eighty two percent thought it feasible to incorporate simulation into training programmes, and 68% of non-experts reported that the simulation session had improved their skills.

**Conclusion:** This study demonstrated construct, face, and content validity for this novel Virtual reality HoLEP simulator. The majority of participants also agreed that it was a feasible and acceptable training model.

## MP-02.15

**Prospective Randomized Controlled Trial Comparing GreenLight 180-W XPS PVP and Transurethral Resection of Prostate for the Treatment of Benign Prostate Obstruction (The Goliath Study): One-Year Analysis of Sexual Function and General Health Status Stratified by Prostate Size**  
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**Introduction and Objectives:** To prospectively evaluate at one year, changes in sexual function and general health status by prostate size in men undergoing XPS or TURP in a large RCT for benign prostate obstruction.

**Materials and Methods:** A total of 291 patients at 29 sites in 9 European countries were randomized 1:1 to undergo XPS or TURP.

MP-02.15, Table 1.

| Event                                   | Total                    |                          |         | XPS                     |                         |                         | TURP                    |                         |                         | p value |        |       |
|-----------------------------------------|--------------------------|--------------------------|---------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|---------|--------|-------|
|                                         | XPS                      | TURP                     | P value | Small (≤40)             | Medium (40-80)          | Large (≥80)             | Small (≤40)             | Medium (40-80)          | Large (≥80)             | Small   | Medium | Large |
| <b>Retrograde ejaculation - de novo</b> | 91<br>(66.9%)<br>(n=136) | 89<br>(66.9%)<br>(n=133) | 1.000   | 39<br>(73.6%)<br>(n=53) | 47<br>(64.4%)<br>(n=73) | 5<br>(50.0%)<br>(n=10)  | 35<br>(58.3%)<br>(n=60) | 46<br>(73.0%)<br>(n=63) | 8<br>(80.0%)<br>(n=10)  | 0.113   | 0.356  | 0.350 |
| <b>IIEF-5 Total at Baseline</b>         | 13.2<br>± 7.6<br>(n=132) | 13.7<br>± 7.5<br>(n=129) | 0.639   | 13.0<br>± 8.2<br>(n=53) | 13.1<br>± 7.3<br>(n=69) | 15.2<br>± 5.5<br>(n=10) | 13.6<br>± 7.5<br>(n=58) | 14.2<br>± 7.7<br>(n=61) | 11.2<br>± 7.0<br>(n=10) | 0.705   | 0.429  | 0.173 |
| <b>IIEF-5 Total at 12 Months</b>        | 12.8<br>± 7.5<br>(n=128) | 14.1<br>± 8.2<br>(n=119) | 0.189   | 12.5<br>± 8.0<br>(n=50) | 12.5<br>± 7.2<br>(n=68) | 16.3<br>± 6.8<br>(n=10) | 14.9<br>± 7.1<br>(n=55) | 13.5<br>± 9.1<br>(n=57) | 13.7 ± 10.1<br>(n=7)    | 0.118   | 0.501  | 0.537 |



General health status was evaluated by the EQ-5D-3L questionnaire (index score and visual scale) at baseline, 3 weeks, 6 months and 1 year. Erectile function and ejaculatory status were analysed at baseline and at one year. The results were stratified by a baseline TRUS prostate volume ( $\leq 40$ ml, 40-80ml, and  $\geq 80$ ml).

**Results:** Of the 291 patients, 269 received treatment. After one year, 95.5% of the treated patients remained in the trial (130 XPS and 127 TURP). De novo retrograde ejaculation occurred in 66.9% after XPS and 66.9% post TURP ( $p=1.000$ ), with no significant difference for any prostate size (Table 1). Mean IIEF-5 score at baseline was similar in both groups,  $13.2 \pm 7.6$  for XPS, versus  $13.7 \pm 7.5$  for TURP ( $p=0.639$ ). At 1 year, there was a minor decrease in mean IIEF-5 after XPS to  $12.8 \pm 7.5$  and a minor increase with TURP to  $14.1 \pm 8.2$ , with no significant difference between XPS and TURP ( $p=0.189$ ), for any prostate size (Table 1). The mean EQ-5D-3L baseline index score and visual scale were similar in XPS ( $0.8 \pm 0.2$ ;  $74.5 \pm 15.1$ ) and TURP ( $0.8 \pm 0.2$ ;  $72.9 \pm 17.9$ ). Similar improvements in quality of life were seen with both techniques at 3 weeks and were maintained at 1 year.

#### Conclusions:

1. There are no deleterious effects to general health or erectile function in the first year after either XPS or TURP.
2. The incidence of de novo retrograde ejaculation is similar with XPS and TURP and the difference was not statistically significant for any prostate size.

#### MP-02.16

##### The Use of the 180W XPS 'Greenlight' Laser in Men with Refractory Urinary Retention

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**Introduction and Objectives:** Urinary retention is an established marker of disease progression in benign prostatic hyperplasia (BPH) and an indicator for surgical intervention, with monopolar transurethral resection of prostate (TURP) remaining the gold-standard treatment. Photoselective-vaporisation of prostate (PVP) has proven to be effective for men with BPH however few reports exist of use in men with urinary retention. The aim of this study was to determine the safety and efficacy of PVP 180W XPS 'Greenlight' laser prostatectomy (XPS-GLLP) in men with refractory urinary retention in our institution.

**Materials and Methods:** From a prospectively maintained database, patients with refractory urinary retention undergoing XPS-GLLP between October 2010-2013 were identified. Notes were reviewed retrospectively for age, PSA, presenting symptoms, intra-operative

details, complications and outcomes.

**Results:** A total of 155 patients were identified with a mean age of 74 years (52-93 years) and PSA of 7.3 (0.2-89). Pre-operatively 14 patients were undertaking intermittent self-catheterisation and 141 catheterised, for a median duration of 5 months (0.5-28 months). Thirty nine patients had high-pressure systems and 16 had known prostate cancer. Intra-operatively the mean lasing time was 25:42 minutes and mean energy used was 203,397kJ. All patients were catheterised routinely post-procedure. The mean hospitalisation was 23 hours (12 hours-10 days). Within 30-days, 25 (16.1%) patients had documented complications (3 with urinary retention, 12 with urinary-tract infections, 8 with significant haematuria). Of these patients, 18 (11.6%) required re-admission, of whom two required surgical intervention for ongoing haematuria. At a median 18 months (5-39 months) follow-up, the catheter-free rate was 88.5%, with mean IPSS/QOL scores of 4 and 1 respectively at 6 months post-operatively. Ten (6.5%) patients have required secondary treatments (1 bladder neck incision, 2 urethral dilatations and 7 further XPS-GLLP for residual tissue).

**Conclusions:** To our knowledge, this is one of the first reports of the use of XPS-GLLP in the management of refractory urinary retention. We show XPS-GLLP to be an effective treatment option for these patients, with short hospitalisation and catheterisation times, high safety profile and almost 90% success rate. XPS-GLLP therefore offers an excellent alternative to traditional TURP in men with refractory urinary retention.

#### MP-02.17

##### First-in-Human Clinical Experience of a Novel Treatment for BPH: Image Guided Robotically-Assisted Waterjet Ablation of the Prostate

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<sup>2</sup>Procept Biorobotics, Redwood City, USA

**Introduction and Objectives:** To report the initial clinical experience in males with symptomatic BPH undergoing ablation of the prostatic adenoma using a novel robotic water-jet technology (Aquablation™, PROCEPT BioRobotics, Redwood Shores, CA).

**Materials and Methods:** The Aquablation System incorporates a console, a fixed robotic arm and a probe that delivers a high power saline stream designed to selectively excise prostatic adenoma under the direct visual guidance of real-time trans-rectal ultrasound (TRUS). To obtain hemostasis standard electro cautery can be used as necessary.

**Results:** Sixteen males with symptomatic BPH were enrolled and 15 males were treated with Aquablation. Monitored data are available on all 15 males treated. The mean age was 73 years (SD=7.1, 59-86) and baseline prostate size of 54g (SD=17.4, 27-85). A median lobe was present in 6 of the 15 subjects treated. All procedures were technically successful with a mean operative time of 45 minutes (SD=17.9, 25-94) and Aquablation treatment time of 8.2 minutes (6.5, 2.0-22.3). No cases of urinary incontinence or erectile dysfunction were reported. Other peri-operative complications were comparable to those observed with other available BPH therapies. Quality of life and functional outcomes are summarized in the following table.

**Conclusions:** The preliminary results from this initial study show promise for this treatment as a potential ablative treatment option for BPH. Further patient data will be required to validate this early clinical experience.

#### MP-02.18

##### Greenlight XPS-180W Laser Vaporization of the Prostate for Benign Prostatic Hyperplasia: A Global, Multi-Center Study Including 1053 Patients, Analysis of Complication Rates and Outcomes at 2 Years According to Prostate Size

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MP-02.17, Table 1.

| Variable                  | Baseline<br>n = 14*    | 1 month<br>n = 14*       | 3 month<br>n = 10*       |
|---------------------------|------------------------|--------------------------|--------------------------|
| IPSS                      | 23.1                   | 11.8                     | 7.8                      |
| Mean Change from Baseline | (4.9, 16 - 33)<br>-    | (6.8, 3 - 27)<br>-11.3   | (4.6, 3 - 19)<br>-14.7   |
| Quality of Life           | 5.0                    | 2.6                      | 1.5                      |
| Mean Change from Baseline | (0.9, 4 - 6)<br>-      | (1.9, 0 - 6)<br>-2.4     | (0.8, 0 - 3)<br>-3.6     |
| Qmax (ml/s)               | 8.6                    | 13.8                     | 16.3                     |
| Mean Change from Baseline | (2.0, 4.8 - 12.1)<br>- | (6.1, 7.5 - 27.9)<br>5.2 | (5.4, 9.4 - 26.0)<br>7.9 |

\*Matched patients. One patient was not available for follow up at 1 month

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**Introduction and Objectives:** The aim of this study was to evaluate the safety and outcomes at 2 years of the new Greenlight XPS-180W laser system (AMS, Minnetonka, MI, USA) for the treatment of BPH in a large multicenter cohort. In addition analysis of safety and efficacy was assessed according to prostate size.

**Materials and Methods:** A total of 1053 patients underwent Greenlight laser photoselective vaporization in 6 international centers between 2010 and 2012. Perioperative variables IPSS, QoL, Qmax, post-void residual (PVR) and PSA levels were recorded at baseline, 3-6-12-24 months post-operatively when available. Patients were stratified according to prostate volume  $\pm$  80 cc.

**Results:** Patients' mean age was 70.17 $\pm$ 9.23 years, with a PV of 74.54 $\pm$ 45.1 cc and PSA of 4.38 $\pm$ 5.00 ng/ml. In between prostate size groups, age (70.45 vs. 67.36 years), PSA (5.8 vs. 3.9 ng/ml), IPSS (22.5 vs. 20.8), Qmax (6.1 vs. 7.1 ml/s) and PVR (267 vs. 202 ml) were significantly different ( $p < 0.001$ ). Mean operative time (96.4 vs. 55.8 min), laser time (55 vs. 28 min) and energy usages (402.8 vs. 211

kJ) were all significantly increased in prostates >80cc. Conversion to TURP was more prevalent in patients with larger glands (11.2% vs. 1.1%;  $p < 0.001$ ). All other rates of complications including perforation (0.6% vs. 0.6%), bleeding obscuring vision (3.8% vs. 6.7%), bleeding hematuria (0.5 vs. 0%) and blood transfusion (0.3% vs. 0.5%) were relatively low and without significant incidence between groups. Significant improvements compared to baseline were noted in all key clinical outcome parameters postoperatively at 3, 6, 12 and 24 months without significant difference between the two groups. At 24 months compared to baseline, median IPSS decreased from 21.0 to 4.2 while Qmax increased from 6.9 to 19 and QoL improved from 5 to 1. Mean PSA decreased from 4.4 to 2.7 ( $p < 0.05$ ).

**Conclusion:** XPS systems provide safe and effective tissue vaporization with significant clinical relief of BPH obstruction associated with objective outcomes and PSA reduction at 24 months confirming the mid-term durability of this procedure regardless of prostate size. Nevertheless, prostate volume has direct implications on operating parameters and was associated with a higher conversion to TURP.

#### MP-02.19

#### Management of Benign Prostatic Obstruction Using Bipolar Plasma Kinetic Energy in Coagulopathic Patients: Initial Series

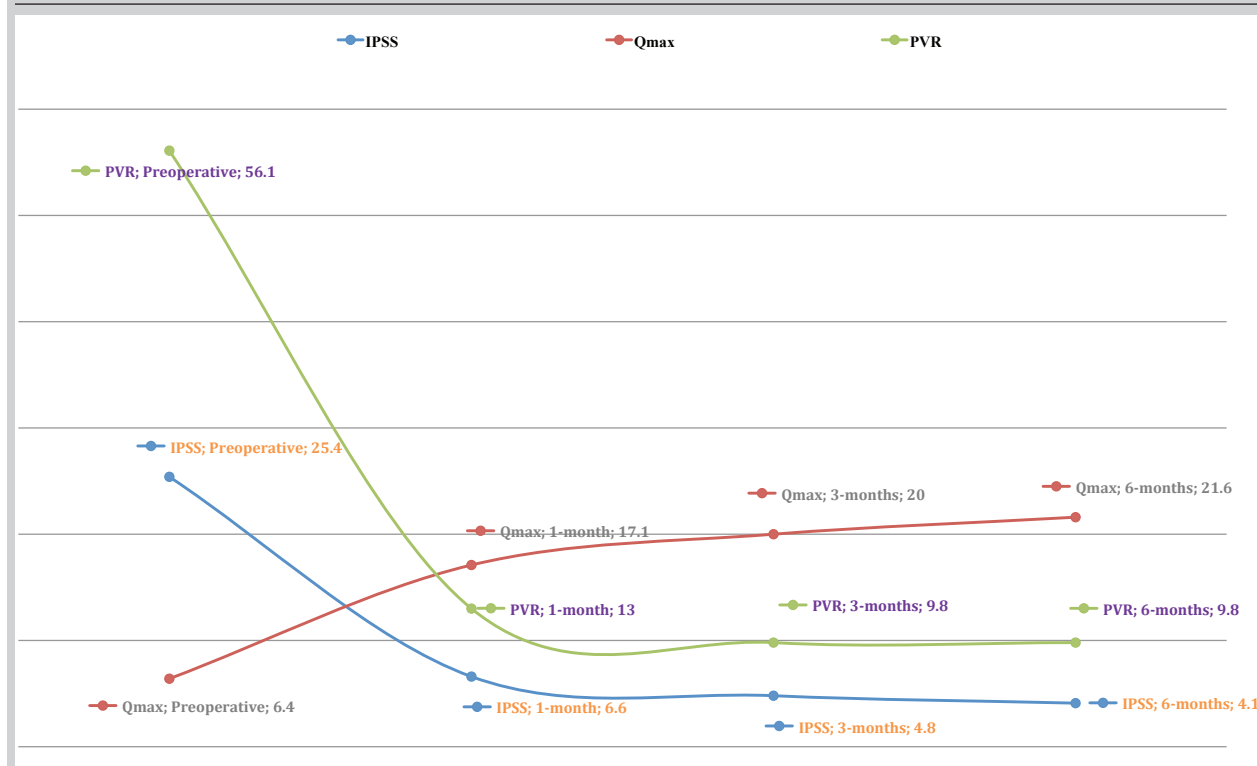
Abou- Taleb A, Khalil M, Sebaey A, Shaher H, Abdelbaky A, Noureldin Y  
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**Introduction and Objectives:** To assess feasibility, safety and efficacy of bipolar plasma kinetic energy for management of benign prostatic obstruction (BPO) in patients on anticoagulant therapy (ACT) or platelet aggregation inhibitors (PAIs).

**Materials and Methods:** After institutional review board approval, transurethral bipolar plasma kinetic enucleation of the prostate was performed in patients presenting with BPO and on concomitant ACT or PAIs. Patient demographics and perioperative data were collected. Moreover, the type of ongoing ACT or PAIs was recorded. Patients' Baseline and follow-up data were compared in terms of the International Prostate Symptoms Score (IPSS), peak flow rate (Qmax) and residual urine volume (PVR) at 1, 3, and 6 months using paired *t*-test.

**Results:** Between December 2012 and February 2014, 30 patients underwent transurethral bipolar plasma kinetic enucleation of the prostate was included. Seven patients were on oral ACT (Warfarin) whereas 23 patients were on PAIs,

MP-02.19, Figure 1. Follow-Up Data of the Improvements in IPSS, Qmax (ml/s), And PVR (ml) from the Baseline at 1, 3, and 6 Months



including 18 on Acetyl Salicylic Acid (ASA) and 5 on Clopidogrel Bisulfate. Patients on Warfarin were shifted to low molecular weight heparin preoperatively; meanwhile Clopidogrel or ASA was continued perioperatively. The median age was 68.5 years (range: 60-80), with median prostate volume of 60cc (30-100), median PSA of 2.4 ng/ml (1-4), median preoperative hemoglobin (Hb) of 12.8 gm/dL (11-15). Five patients (16.7%) presented with preoperative indwelling urethral catheter secondary to refractory urinary retention. The mean preoperative IPSS, Qmax, and PVR were  $25.4 \pm 0.6$ ,  $6.4 \pm 0.3$  mL/sec, and  $56.1 \pm 3.0$  mL, respectively. Only one patient (3.3%) had intraoperative bleeding necessitated blood transfusion. Two patients (6.7%) failed to void after catheter removal. No patients developed either thromboembolic complications or post transurethral resection (TUR) syndrome. Postoperative Hb was comparable to the preoperative level (12 vs. 12.8,  $p > 0.05$ ). Mean hospital stay and catheter time were  $25.7 \pm 10.6$  hours, and  $16 \pm 4.1$  hours, respectively. IPSS, Qmax, and PVR significantly improved at all follow-up times (all  $p < 0.001$ ) (Figure 1).

**Conclusion:** Transurethral Bipolar Plasma kinetic Enucleation of the Prostate seems to be feasible, safe and effective in management of coagulopathic patients with BPO. It is associated with less bleeding, short catheterization time and hospital stay.



Moderated Poster Session 3  
Prostate Cancer:  
Detection & Screening  
Monday, October 13  
1300-1430

MP-03.01

**Early Detection of Clinically Significant Prostate Cancer: Low Free Prostate-Specific Antigen Levels as a Criterion for Prostate Biopsy in Patients with Low Total Prostate-Specific Antigen Levels**

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**Introduction and Objectives:** The present study investigated whether applying the fPSA ratio (%fPSA) as a criterion for prostate biopsy in patients with low PSA ( $\leq 4.0$  ng/mL) could facilitate early diagnosis of clinically significant prostate cancer.

**Materials and Methods:** Subjects comprised 9,522 patients (age range, 40-79 years) who underwent screening for prostate cancer between July 2001 and June 2011. Patients with %fPSA  $\leq 12\%$  and PSA 2.0-10.0 ng/mL with no gaps in prostate biopsy histopathological diagnosis (n=260) were divided into the following two PSA groups: low (PSA: 2.0-4.0 ng/mL) and mildly elevated (PSA: 4.1-10.0 ng/mL). Malignancy was evaluated based on Gleason scores of biopsy specimens. Statistical significance was assessed using Pearson's  $\chi^2$  test.

**Results:** Median age (range), PSA, and %fPSA were 66 years (49-79 years), 4.5 ng/mL (2.1-9.8 ng/mL), and 9.9% (3.5-12.0%), respectively. A total of 129 patients (49.6%) were diagnosed with prostate cancer based on biopsy of all patients in the low PSA (n=114, 43.8%) and mildly elevated PSA (n=146, 56.2%) groups. No significant differences were

observed between the low PSA group (43.9%) and the mildly elevated PSA group (54.1%;  $p=0.1010$ ) in prostate cancer detection rate. Patients were then further classified as low, medium, or high risk (Gleason scores  $\leq 6$ , 7 and  $\geq 8$ , respectively) based on biopsy histopathological diagnosis. However, no significant differences were observed between the low and mildly elevated PSA groups ( $p=0.9974$ ) in malignancy. Overall detection rate in the medium- and high-risk patients was 76%, and highly malignant prostate cancer was efficiently detected.

**Conclusion:** If %fPSA is low, clinically significant prostate cancer is likely to be diagnosed, even in patients with low total PSA (2.0-4.0 ng/mL). A high detection rate of 76% was achieved in medium- and high-risk patients, demonstrating that highly malignant prostate cancer was efficiently detected. In addition to existing criteria such as age and total PSA, including %fPSA as a criterion for prostate biopsy could facilitate early diagnosis of clinically significant prostate cancer, even in patients with low total PSA.

MP-03.02

**Performance of Serum Prostate-specific Antigen Isoform [-2]proPSA (p2PSA) and the Prostate Health Index (PHI) in a Chinese Hospital-based Biopsy Population**

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Medicine, Winston-Salem, USA

**Introduction and Objectives:** The performance of serum p2PSA and its derivative, the prostate health index (PHI), in detecting prostate cancer (PCa) from prostate biopsy has been extensively evaluated in the United States and Europe. However, their performance in Chinese patients has not been reported.

**Materials and Methods:** We recruited consecutive patients who underwent prostate biopsy in three tertiary hospitals in Shanghai, China, during 2012-2013. Serum total PSA (tPSA), free PSA (fPSA), and p2PSA were measured. Discriminative performance of PCa and high-grade PCa were assessed using the area under the receiver operating characteristic curve (AUC) and detection rate. High-grade PCa was defined as having a Gleason Score of 4+3 or worse.

**Results:** Among 636 patients who underwent prostate biopsy (mean age=68.8 years), 274 (43.1%) and 158 (24.8%) were diagnosed with PCa and high-grade PCa, respectively. The AUC for discriminating PCa and non-PCa was 0.81, 0.75, 0.84, and 0.88 for tPSA, fPSA, p2PSA, and PHI, respectively, and the AUC for discriminating high-grade PCa from all others was 0.83, 0.78, 0.84, and 0.86 for tPSA, fPSA, p2PSA, and PHI, respectively. The difference in the AUC between PHI and tPSA was 0.07 for discriminating PCa,  $P=1.94 \times 10^{-6}$ , and 0.03 for discriminating high-grade PCa,  $P=0.06$ . In the patients with tPSA 2-10 ng/mL, the difference in the AUC between PHI and tPSA was larger for discriminating PCa (0.71-0.51=0.20,  $P=8.08 \times 10^{-3}$ ) and for discriminating high-grade PCa (0.69-0.54=0.15,  $P=0.33$ ). The added value of PHI over tPSA was also demonstrated by PCa detection rate (Table 1). For example, in patients with tPSA 2-10 ng/mL where the average detection rate of PCa and high-grade PCa was 17.6% and 6.3%, these rates raised to 60% and 40% if they were in the highest quartile of PHI.

**Conclusion:** PHI provided added value over tPSA in discriminating PCa and high-grade PCa in men who underwent prostate biopsy in China.

MP-03.02, Table 1. Detection rate of prostate cancer and high-grade prostate cancer based on tPSA and PHI

| tPSA level (ng/mL) | No. of Patients | Detection rate (95%CI) of PCa based on PHI |                 |                  |                  | Detection rate (95%CI) of high-grade PCa based on PHI |                 |                  |                  |
|--------------------|-----------------|--------------------------------------------|-----------------|------------------|------------------|-------------------------------------------------------|-----------------|------------------|------------------|
|                    |                 | All                                        | Low(Q1)         | Mid(Q2-Q3)       | High (Q4)        | All                                                   | Low(Q1)         | Mid(Q2-Q3)       | High (Q4)        |
| 2.0-10.0           | 222             | 17.6 (12.8-23.2)                           | 8.4 (3.7-15.9)  | 23.0 (15.8-31.4) | 60.0 (14.7-94.7) | 6.3 (3.5-10.4)                                        | 2.1 (0.3-7.4)   | 8.2 (4.0-14.6)   | 40.0 (5.3-85.3)  |
| 10.1-19.99         | 178             | 36.5 (29.4-44.1)                           | 6.8 (1.4-18.7)  | 44.4 (35.4-53.6) | 70.0 (34.7-93.3) | 14.6 (9.8-20.7)                                       | 2.3 (0.1-12.0)  | 18.6 (12.1-26.5) | 20.0 (2.5-55.6)  |
| $\geq 20.0$        | 215             | 75.8 (69.5-81.4)                           | 10.0 (0.3-44.5) | 46.9 (34.3-59.8) | 93.6 (88.2-97.0) | 53.0 (47.1-59.8)                                      | 10.0 (0.3-44.5) | 28.1 (17.6-40.8) | 67.4 (59.0-75.0) |

tPSA: total PSA; PHI: Prostate Health Index; Q: quartile; PCa: prostate cancer

## MP-03.03

**The Rotterdam Prostate Cancer Risk Calculator: Improved Prediction with More Relevant Pre-Biopsy Information, Now in the Palm of Your Hand**

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**Introduction and Objectives:** The Rotterdam prostate cancer risk calculator (RPCRC, [www.prostatecancer-riskcalculator.org/](http://www.prostatecancer-riskcalculator.org/) [www.uroweb.org](http://www.uroweb.org)) has been developed to risk stratify

potential candidates for a prostate biopsy. To improve the user-friendliness and accessibility the risk calculator has been transformed to a mobile application (app, Figures 1 right). Here we assess the change in predictive capability when using more relevant pre biopsy information in the form of a decision tree incorporated into the app (Figure 1 left).

**Materials and Methods:** Analyses are based on the biopsy outcome of 3,600 men screened for the first time and 2,910 men with a previous PSA test/biopsy within ERSPC, section Rotterdam. Predictive capability of models within the RPCRC app (1: PSA alone, 2: PSA+DRE, 3: PSA+DRE+DRE assessed volume, 4:

PSA+DRE+TRUS+volume) were assessed in terms of discrimination (C-statistic) for both predicting the probability of PC on biopsy and serious PC (defined as > T2B and/or Gleason  $\geq 7$ ).

**Results:** Applying model 1 to model 4 resulted in AUC's of from 0.69 respectively 0.79 for predicting PC and AUC's for predicting serious PC of 0.74 respectively 0.86. Similar data for men with a previous PSA test/ Biopsy were 0.62 respectively 0.69 for predicting PC and 0.69 respectively 0.82 for predicting serious PC (Table 1), confirming that including more relevant information increases predictive capability.

MP-03.03, Table 1 and Figure 1.

| No previous biopsy          | Predicting PC on biopsy (AUC) | 95% CI    | Predicting serious PC on biopsy (AUC) | 95% CI    |
|-----------------------------|-------------------------------|-----------|---------------------------------------|-----------|
| 1. PSA                      | 0.69                          | 0.67-0.71 | 0.74                                  | 0.71-0.77 |
| 2. PSA+DRE                  | 0.73                          | 0.71-0.75 | 0.82                                  | 0.79-0.84 |
| 3. PSA +DRE+DRE ass. volume | 0.77                          | 0.75-0.79 | 0.85                                  | 0.82-0.87 |
| 4. PSA+DRE+TRUS+volume      | 0.79                          | 0.77-0.81 | 0.86                                  | 0.84-0.88 |
| Previous PSA test/biopsy    |                               |           |                                       |           |
| 1. PSA                      | 0.62                          | 0.59-0.64 | 0.69                                  | 0.65-0.74 |
| 2. PSA+DRE                  | 0.64                          | 0.62-0.67 | 0.78                                  | 0.74-0.82 |
| 3. PSA +DRE+DRE ass. volume | 0.68                          | 0.65-0.71 | 0.81                                  | 0.78-0.85 |
| 4. PSA+DRE+TRUS+volume      | 0.69                          | 0.66-0.71 | 0.82                                  | 0.79-0.86 |

**Conclusion:** The Rotterdam risk calculators, based on the robust data from the ERSPC, section Rotterdam, were developed with the prime objective of helping to reduce unnecessary biopsies and the over-diagnosis of indolent prostate cancers. The new mobile app tool takes this one step further, providing doctors and patients with an increasingly powerful tool which is easy to use and always accessible.

#### MP-03.04

##### **Validation of PSA-Age Volume (PSA-AV) Formula for Predicting Prostate Biopsy Outcomes in a UK Cohort**

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**Introduction and Objectives:** Predictive tools for prostate cancer diagnosis have been devised since PSA alone is an inaccurate marker. A novel formula [Patel et al. *Urology* 2013; 81(3): 602-606] incorporates PSA, age, prostate volume and ethnicity to predict positive biopsy rates. PSA-AV score obtained using these variables with a cut-off value of 700 is considered significant. We evaluated the predictive value of this score in a UK cohort of men undergoing transrectal prostate biopsy.

**Materials and Methods:** Men who underwent TRUS biopsies were retrospectively included in the study. PSA-AV score was calculated using PSA level, age and prostate volume for each patient. In African-Caribbean men, a multiplication factor of 0.9 was used to adjust for increased risk, in keeping with the original study. Receiver Operating Characteristic (ROC) curve analysis was performed to assess the diagnostic utility of the PSA-AV formula.

**Results:** Data was analysed for 301 men, 46% Caucasian and 34% African-Caribbean. Median age was 68; PSA 9.7; prostate volume 43cc and cancer was detected in 50.16% (151/301). Positive biopsies and Gleason scores for different PSA-AV values are summarized (Table 1). ROC curve analysis showed good discriminative ability with AUC=0.75 (0.70–0.80; 95% CI),  $p < 0.001$ . For a cut-off value of 700, the PSA-AV score had a sensitivity of 98% and specificity of 17% with positive predictive value (PPV) of 54%. With a cut-off of 500, sensitivity was 91%, specificity 29% and PPV 56%.

**Conclusion:** The PSA-AV formula has good predictability and higher PPV than PSA. A cut-off value of 700 has high sensitivity to exclude

positive biopsy. In our study, the significance of the score improved when a lower cut-off was used. It has a role in cancer detection risk counseling for men with raised PSA.

#### MP-03.05

##### **Poor Glycemic Control of Diabetes Mellitus Is Associated with Higher Risk of Prostate Cancer Detection in a Biopsy Population**

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**Introduction and Objectives:** We evaluated the impact of glycemic control of DM on prostate cancer detection in a biopsy population.

**Materials and Methods:** We retrospectively reviewed the records of 1,368 men who underwent prostate biopsy at our institution. We divided our biopsy population into three groups according to their history of DM, and their Hemoglobin A1c (HbA1c) level: a no-DM (DM-) group; a good glycemic control (DM+GC) group (HbA1c < 6.5%); and a poor glycemic control (DM+PC) group (HbA1c ≥ 6.5%). For subgroup analyses, the DM+PC group was divided into a moderately poor glycemic control (DM+mPC) group (6.5 ≤ HbA1c < 7.5%) and a severely poor glycemic control (DM+sPC) group (HbA1c ≥ 7.5%). Results: Among 1,368 men, 338 (24.7%) had a history of DM, and 393 (28.7%) had a positive biopsy. In men with negative biopsies, the mean prostate-specific antigen (PSA) level was significantly different among the groups ( $P = 0.015$ ), and the figure was 9.10 ng/ml in DM- group, 7.52 ng/ml in DM+GC group, 7.28 ng/ml in DM+PC group. In the multivariate analysis, the DM+PC group was significantly associated with a higher rate of overall prostate cancer detection in biopsy subjects compared to the DM- group (OR = 2.313,  $P = 0.001$ ). In subgroup analysis, DM+sPC group (HbA1c ≥ 7.5%) was significantly related to a higher risk of high-grade diseases compared to the DM- group (OR = 2.446,  $P = 0.048$ ).

**Conclusion:** In men with negative biopsies, the mean PSA level was significantly different according to the history of DM, and the HbA1c level. Poor glycemic control of DM was associated with a higher risk of prostate cancer

detection, including high-grade disease, in the biopsy population.

#### MP-03.06

##### **Impact of the U.S. Preventive Services Task Force Recommendations against PSA-based Screening on Prostate Biopsy and Cancer Detection Rates**

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**Introduction and Objectives:** In May 2012, the U.S. Preventive Services Task Force (USPSTF) released recommendations against routine Prostate-Specific Antigen (PSA) screening for prostate cancer (PC). Our study objective was to examine if our institutional biopsy rates and cancer detection rates have changed following the USPSTF recommendation using a time-series analysis.

**Materials and Methods:** We examined our institutional prostate biopsy database from Oct. 2008 to Jun. 2013. Biopsies for active surveillance or solely targeting MRI-detected lesions were excluded. Low risk PC (LRPC) was defined as no Gleason pattern ≥4, ≤3 cores involved or ≤1/3 of total number of cores involved, and no core with >50% cancer involvement. High grade PC (HGPC) was defined as Gleason 7-10. A time-series analysis using interventional auto-regressive integrated moving average (ARIMA) models with step intervention functions were conducted to examine the effect of the recommendations on number of biopsies performed and cancer detection per month.

**Results:** Within the study period, 3408 biopsies were performed and 1601 (47.0%) PCs were detected (LRPC = 563 (16.5%); HGPC = 914 (26.8%)). The median for biopsies per month decreased from 64 (IQR=58-78) before recommendations to 34 (IQR=27-39) afterward ( $p=0.003$ ), while median number of patients undergoing their first-time biopsies decreased from 45 (IQR=41-57.5) to 24 (IQR=20-32,  $p=0.025$ ). The median number of LRPCs detected per month decreased from 10 (IQR=8-14) to 5 (IQR=4-7,  $p=0.012$ ), while the median number of HGPCs detected per month decreased from 17 (IQR=15.5-21) to 10 (IQR=9-11,  $p<0.001$ ).

**Conclusion:** Since the USPSTF recommendations, the number of biopsies performed (total and first-time biopsies), based on referrals from our catchment area, have decreased. This is likely due to decreased PSA-screening and referral by general practitioners. Although encouraging that less low risk PCs are being diagnosed, the magnitude of sudden decrease in detection rate of Gleason 7-10 PCs is concerning.

**MP-03.04**, Table 1. Positive Biopsy Findings for Different PSA-AV Score Categories

| PSA-AV score | Positive Biopsy (%) | Gleason 6 | Gleason 7 | Gleason >7 |
|--------------|---------------------|-----------|-----------|------------|
| <300         | 105/163 (64.4%)     | 17        | 53        | 35         |
| 300 - 500    | 33/81 (40.7%)       | 7         | 20        | 6          |
| 500 - 700    | 10/29 (34.5%)       | 5         | 5         | 0          |
| >700         | 3/28 (10.7%)        | 0         | 2         | 1          |



**MP-03.07****Demographics, Clinical and Pathological According to the Prostate Biopsy Scheme: A Prospective Multicenter Study**

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**Introduction and Objectives:** Current data support the utility of extended and saturation schemes in TRUS prostate biopsy instead of the traditional sextant scheme proposed by Hodge. The aim of this study is to compare diagnosis features (demographics, clinical and pathological) between four different numerical schemes of TRUS-prostate biopsies used in daily clinical practice.

**Materials and Methods:** We conducted during one year (2010) a prospective multicenter epidemiological study. We analyzed the performance of prostate biopsy in 25 public hospitals in Spain. We grouped the patients depending on the number of cores taken during TRUS-prostate biopsy ( $\leq 6$ , 7-12, 13-19,  $\geq 20$  cores). Demographics, clinical and pathological features in each group were analyzed. Statistical analysis with STATA vs. 10.0 and contingency tables using Chi Square test were performed.

**Results:** We included 4087 patients with prostate cancer. The frequency of use of prostate biopsy schemes: 12.97% ( $\leq 6$  cores), 63.35% (7-12 cores), 16.54% (13-19 cores) y 3.35% ( $\geq 20$  cores). There were differences ( $p < 0.001$ ) between groups in age, comorbidity, PSA value and digital rectal examination (DRE). Older patients with more comorbidity and higher suspicion of prostate cancer (higher levels of PSA and suspicious DRE) were frequently observed in less extensive schemes. Prostate volume didn't show differences between the groups ( $p = 0.057$ ). Gleason  $\geq 7$  were frequency observed in the group in which  $\leq 6$  cores were obtained ( $p < 0.001$ ). Organ confined disease was higher in the saturation group ( $\geq 20$  cores) while advanced disease was predominantly in the group where 13-19 cores were taken and metastatic stages in the  $\leq 6$  cores. In terms of progression of localized prostate cancer (d'Amico criteria) the highest ratio of high risk localized prostate cancer was observed in the saturation group ( $\geq 20$  cores).

**Conclusions:** Sextants TRUS prostate biopsies schemes continue to be used in daily clinical practice for certain patients (fragile and with advanced disease) in order to reduce the morbidity associated with this procedure. This study shows that currently there isn't

a consensus protocol of prostate biopsy in our country. It appears that obtaining more adequate tissue sampling by increasing the number of cores doesn't result in an increase of low risk prostate cancer in comparison with less extensive schemes.

**MP-03.08****Evaluating the Clinical Utility of Template-Guided Transperineal Biopsy on Treatment Recommendations**

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**Introduction and Objectives:** Evaluate the impact on management decision-making of template-guided transperineal biopsy of the prostate when compared to a standard trans-rectal ultrasound biopsy of the prostate.

**Materials and Methods:** Retrospective analysis of 152 men with prostate cancer who were diagnosed on standard TRUS biopsy and then went onto have a Template Prostate Mapping (TPM) biopsy. The management recommendation after each biopsy result was prospectively recorded.

**Results:** Median age at TPM biopsy was 63 years (range 39 to 82). Among all, 79/152 (52%) transitioned to a higher risk cancer, 26/152 (17%) transitioned to lower risk cancer and 47/152 (31%) had no change in risk category following TPM biopsy. After TPM biopsy, 43/65 men had a change in recommendation from Active Surveillance to active treatment, 44/87 men had a change in the type of active treatment they were offered and 41 men had no change in treatment recommendation made. Eight men, who were offered active treatment or Surveillance, were discharged after their TPM biopsy due to a 'no cancer' diagnosis.

**Conclusion:** Template-guided transperineal biopsies, when compared to TRUS biopsies, carry significant clinical utility by having a direct impact on management recommendations made to men with prostate cancer. It has a high rate of transitioning men from surveillance to active treatment recommendations although a significant number were provided reassurance in following a surveillance approach rather than a radical therapeutic approach.

**MP-03.09****Transperineal Template Biopsies as Part of Active Surveillance: Medium-Term Follow-Up**

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**Introduction and Objectives:** Active surveillance (AS) is a valid treatment option for men with low risk prostate cancer. Its effectiveness relies on correct stratification of disease risk

at the outset. Several AS series using TRUS biopsies report a 30% conversion to radical treatment with medium term follow-up. Transperineal template biopsies (TTBs) enable detailed sampling and in some cases targeting of suspicious areas within the prostate. At our institution we offer TTBs to patients prior to embarking on AS and this study aimed to review the progress of our AS cohort.

**Materials and Methods:** Since 2005 we have kept a prospective database of all patients undergoing TTBs from 2005 – 2013. Patients were identified who were/are on AS and had had a TTB, either as a primary diagnostic biopsy or for re-staging following a trans-rectal biopsy (TRUS).

**Results:** A total of 185 men were commenced on AS following a TTB. They had a median age of 66 years (range 42-78) and median PSA of 7 (range 0.40 – 22). The mean number of cores taken was 52 with an average positive core number of 2 per patient. Twenty patients are no longer on AS and have embarked on the following treatment options: brachytherapy (12), hormones and radiotherapy (2), surgery (4), HIFU (2). At a median of 31 months (range 4 – 66 months) follow-up, 165 patients (89.1%) are still on AS.

**Conclusions:** TTBs in patients considered for AS result in more accurate risk stratification. This means men with genuinely significant disease are treated with radical therapy without undue delay and men with insignificant disease can be reassured.

**MP-03.10****Transperineal Prostate (TP) Biopsies: The First Prospective Evaluation of Patient Reported Experience and Effects on Symptoms and Life Style**

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**Introduction and Objectives:** Many urologists are choosing the transperineal biopsy approach (TP) for detection of prostate cancer, with alleged higher detection and negligible infection rates compared to the transrectal approach. There is no published PROM data to assess patient reported experience and effects on symptoms. We aimed to prospectively assess their occurrence using a validated PROM tool.

**Materials and Methods:** Using the PROBE PROM tool, validated for TRUS biopsies as part of the ProtecT study, we collected data prospectively in four centres in 2013. All patients undergoing TRUS or TP biopsies were



**MP-03.10**, Table 1. Demographics and Symptoms scores.

|                                                                               | TRUS biopsy (n= 228) | TP biopsy (n = 201)   | Difference TRUS-TP |
|-------------------------------------------------------------------------------|----------------------|-----------------------|--------------------|
| Age (years)                                                                   | 66.7 ± 8.1 (42-88)   | 63.9 ± 7.9 (36-83)    | p=0.265 (ns)       |
| PSA (ng/ml)                                                                   | 13.5 ± 16.3 (1-116)  | 11.2 ± 8.4 (0.2-53.2) | p=0.000 (s)        |
| Prostate Volume (ml)                                                          | 56.4 ± 32.1 (7-211)  | 56.4 ± 36.1 (6-210)   | p=0.496 (ns)       |
| Symptom scores presented as the mean of the difference (follow-up - baseline) |                      |                       |                    |
| IPSS                                                                          | -0.61 ± 5.35 (ns)    | -0.23 ± 4.05 (ns)     | p=0.50 (ns)        |
| Quality of life                                                               | -0.36 ± 1.21 (ns)    | -0.08 ± 1.22 (ns)     | p=0.06 (ns)        |
| IIEF-5                                                                        | -2.95 ± 6.92 (s)     | -1.96 ± 6.86 (s)      | p≤0.01 (s)         |
| Sexual desire (worse/much worse) since biopsy                                 | 14.5% (n=33)         | 28.3% (n=62)          |                    |
| Pain                                                                          |                      |                       |                    |
| Pain during period following biopsy                                           | 28.1% (n=64)         | 46.8% (n=94)          |                    |
| Patients "little or not affected" by pain                                     | 76.5% (n=49)         | 91.4% (n=86)          |                    |
| Patients that required painkiller prescription by GP                          | 8.2% (n=18)          | 8.8% (n=20)           |                    |
| Patients' experience                                                          |                      |                       |                    |
| Patients describing procedure as 'uncomfortable'                              | 19.2% (n=42)         | 23.2% (n=53)          |                    |
| Patients unhappy to have repeat biopsy                                        | 11% (n=25)           | 10% (22)              |                    |
| Patients describing procedure as a minor intervention                         | 93.9% (n=214)        | 82.6% (n=181)         |                    |
| ns: Not significant, s: Significant                                           |                      |                       |                    |

asked to complete the questionnaires immediately after the procedure and at follow-up.

**Results:** A total of 655 patients were included in the study, of these 429 of patients in total completed both questionnaires (228 for TRUS and 201 for TP biopsy). Outcomes and demographics are shown in Table 1. Twice the numbers of cores were taken for TP biopsies (12.27 vs. 27.1), yet, there was no clinically significant difference in IPPS from before to after biopsy in both groups. However, there was significant change in IIEF score and sexual desire following both procedures, more so for TRUS. Pain was experienced in both groups in days after biopsy with only little impact on patients' life.

**Conclusion:** This study reports the first prospective PROM-based assessment of patients' experience and effects on symptoms of TP biopsies. Despite acquiring more biopsies, TP appears to have similar impact to TRUS. Patients should be warned of the effect of both techniques on sexual desire and erectile function.

#### MP-03.11

##### Side-Effects and Complications of Transperineal Prostate (TP) Biopsies: The First Prospective Evaluation Using a Validated Patient Reported Outcome Measures (PROM) Tool

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**Introduction and Objectives:** Transrectal ultrasound guided biopsies of the prostate (TRUSP) are standard for detection of prostate cancer (CaP). Increasing sepsis rates have turned many urologists to using the TP approach with alleged higher detection rates and negligible infection rates. There is no published

PROM data to assess side-effects and complications of TP biopsies. We aimed to prospectively assess their occurrence using a validated PROM tool.

**Materials and Methods:** Using the Probe PROM tool, validated for TRUSP biopsies as part of the ProtecT study, we collected data prospectively in four centres between February and November 2013. All patients undergoing TRUSP or TP biopsies were asked to complete

**MP-03.11**, Table 1. Demographics and side effects.

|                                                         | TRUS biopsy (n= 228) | TP biopsy (n=201)     | Difference TRUS-TP |
|---------------------------------------------------------|----------------------|-----------------------|--------------------|
| Age (years)                                             | 66.7 ± 8.1 (42-88)   | 63.9 ± 7.9 (36-83)    | p=0.265            |
| PSA (ng/ml)                                             | 13.5 ± 16.3 (1-116)  | 11.2 ± 8.4 (0.2-53.2) | p=0.000            |
| Prostate Volume (ml)                                    | 56.4 ± 32.1 (7-211)  | 56.4 ± 36.1 (6-210)   | p=0.496            |
| Side effect profile at follow-up                        |                      |                       |                    |
| Haematospermia                                          | 64.5% (n=147)*       | 63.2% (n=127)*        |                    |
| Haematuria                                              | 71.5% (n=163)        | 74.6% (n=150)         |                    |
| Haematochezia                                           | 30.7% (n=70)         | 10.0% (n=20)          |                    |
| Acute urinary retention                                 | 7.5% (n=17)          | 5% (n=11)             |                    |
| GP review post procedure                                | 11.8% (n=33)         | 11.9% (n=24)          |                    |
| Fever                                                   | 12.7% (n=29)         | 6.5% (n=13)           |                    |
| Antibiotics by GP for suspected UTI                     | 9.2% (n=21)          | 9.0% (n=18)           |                    |
| *Patients who had not had sexual activity were excluded |                      |                       |                    |

the questionnaires immediately after the procedure and at follow-up.

**Results:** A total of 655 patients were included in the study, of these 65% (429) of patients in total completed both questionnaires (228 for TRUS and 201 for TP biopsy). The side effect profile and demographics can be seen in Table 1. There was one confirmed case of sepsis in the TRUS group, and 4 patients had clot retention in the TP group (1.99%). More than twice the numbers of cores were taken for TP biopsies (12.17 vs. 27.1), yet, subjective infection and urinary retention rates were measured significantly less in the TP group.

**Conclusion:** This study reports the first prospective PROM based assessment of side-effects and complications from TP biopsies. Despite acquiring more biopsies TP appears to have a similar side effect profile to TRUS with fewer septic events and – surprisingly – lower urinary retention rate.

#### MP-03.12

##### **Comparison between Two Different Schemes of Prostate Biopsy: Classic Sextant Biopsy vs. Volume-Dependent Prostate Biopsy**

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**Introduction and Objectives:** Prostate biopsy (PB) schemes vary widely between centers.

There isn't a consensus in the number of cores, nor sampling sites. Currently, the use of a 12 cores scheme, an extended or a saturation prostate biopsy, is controversial. The objective of this study is to compare the detection rate of prostate cancer (PC) obtained by a classic double sextant scheme (DSS) versus the volume-dependent scheme (VDS).

**Materials and Methods:** We performed a prospective study including all patients with an indication of PB in the last 5 months of 2013 (n = 74). Patients were divided in two groups according to the order of inclusion. DSS (n = 44) was performed by a 6 core per lobe needle biopsy, while the VDS (n = 30) consist of 6, 7 or 8 core per lobe biopsy depending in the prostate volume (<25, 25-45 and >45 cc). In both schemes additional cores were obtained from suspect areas in the digital rectal examination (DRE) or transrectal ultrasound (TRUS). We used SPSS 17 software to create the contingency tables and to compare the means.

**Results:** The study groups were comparable in all variables (age, PSA, prostate volume, PSA density, suspect DRE, suspect TRUS, Gleason score and post-biopsy complication rate) except in the rate of PC detection, being higher in the VDS group (70% vs. 43.2%, P 0.033). The subgroup analysis shows that the PC detection rate was similar in both biopsy schemes when

the prostate volume was under 25 cc (near 84%), but it was higher in the VDS when the prostate volume was 25-45 cc (88.9% vs. 42.9%) and over 45 cc (50% vs. 25%).

**Conclusions:** In our study, the volume-dependent scheme of prostate biopsy shows a better prostate cancer detection rate with similar post-biopsy complications especially when the prostate volume was over 25 cc.

#### MP-03.13

##### **Comparative Analysis of MRI-Targeted/Fusion-Guided vs. Transperineal Template-Saturation Prostate Biopsy: First START and PIRADS Conform Prospective Evaluation**

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**Introduction and Objectives:** Multiparametric-MRI (mp-MRI) and MRI-targeted prostate biopsy improve detection of clinically significant prostate cancer (PCa) but standardized prospective evaluation is limited. Here we comparatively analyze MRI-targeted/fusion-guided prostate biopsy versus transperineal saturation biopsy on a per-patient level.

**Materials and Methods:** A total of 294 consecutive men with suspicion of prostate cancer (108 primary and 186 repeat biopsies) enrolled in 2013 underwent 3T mp-MRI according to ESUR guidelines and transperineal prostate biopsy. Systematic transperineal cores (mean 24) were placed independently of MRI suspicion and MRI-targeted cores to allow evaluation according to Standards of Reporting for MRI-targeted Biopsy Studies (START) criteria. Software registration was used to perform fusion-guided biopsies (mean 4.1 cores). The highest Gleason score from each biopsy method was compared. Student-t-tests and McNemar tests were used to compare detection rates of both modalities.

**Results:** Overall, 150 cancers and 86 significant cancers (Gleason score  $\geq 3+4$ ) were diagnosed. Saturation biopsy missed 18 significant tumors, while MRI-guided targeted biopsies did not detect 11. Targeted biopsy alone overlooked 90% of Gleason 3+3 tumors. McNemar tests for detection of significant cancers by one modality against the other modality were not statistically significant both for primary and repeat biopsies (p=0.08 and p=0.5).

**Conclusions:** Compared to transperineal saturation as reference, MRI/US-fusion-guided biopsy alone detected as many significant tumors as saturation biopsy. Simultaneously, the targeted approach mitigated the detection

of lower-grade disease and may consequently prevent overtreatment.

#### MP-03.14

##### **Multiparametric Magnetic Resonance Imaging (mpMRI) of Prostate Cancer**

##### **Lesions: How Much Do We Have to Learn?**

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**Introduction and Objectives:** The introduction of functional mp-MRI imaging has enabled imaging to evolve from having a limited role in local staging of prostate cancer to being able to detect tumours with a relatively high sensitivity and specificity. This study is aimed at determining the accuracy of multiparametric Magnetic Resonance Imaging (mpMRI) during the learning curve of radiologists in a tertiary-referral cancer centre using MRI targeted, transrectal ultrasound guided transperineal fusion biopsy (MTTP) for validation.

**Materials and Methods:** Prospective data on 340 consecutive patients was collated. Patients underwent mpMRI read by two radiologists in line with ESUR standards followed by MTTP biopsy of the lesion (targeted biopsy). A 5-point likert scale of probability was used to determine lesions suspicious for cancer, with scores  $\geq 3$  taken as a positive MR-target. We compared sequential groups to determine the learning curve. Statistical analysis was performed with chi-square correlation test.

**Results:** We detected a positive mpMRI in 64 patients from Group A (91%) and 52 patients from Group B (74%). Prostate cancer (CaP) detection rate on mpMRI increased from 42% (27/64) in Group A and 81% (42/52) in Group B (p value 0.003). CaP detection rate by targeted biopsy increased from 27% (17/64) in Group A and 63% (33/52) in Group B (p value 0.001). The negative predictive value of MRI for significant cancer (> Gleason 3+3) was 88.9% in Group B vs. 66.6% in Group A (see Table 1).

**Conclusions:** We demonstrate an improvement in detection of CaP for MRI reporting over time, suggesting a learning curve for the technique. Despite an improved negative predictive value for significant cancer, this did not reach a level whereby biopsy can be avoided in MR negative cases.

MP-03.14, Table 1.

|                                                                                | First 70 pt 2012 (Group A) | Last 60 pt 2012         | First 70 pt 2013        | Second 70 pt 2013       | Last 70 pt 2013 (Group B) | p value |
|--------------------------------------------------------------------------------|----------------------------|-------------------------|-------------------------|-------------------------|---------------------------|---------|
| <b>N positive (Likert scale <math>\geq 3</math>) MRI (% all patients)</b>      | 64 (91%)                   | 51 (85%)                | 56 (80%)                | 55 (78%)                | 52 (74%)                  | 0,007   |
| N patients Likert scale 5                                                      | 35                         | 12                      | 13                      | 17                      | 22                        |         |
| N patients Likert scale 4                                                      | 19                         | 15                      | 16                      | 16                      | 22                        |         |
| N patients Likert scale 3                                                      | 10                         | 24                      | 27                      | 22                      | 8                         |         |
| <b>CaP detection rate in positive MRI group (N)</b>                            | 42% (27/64)                | 53% (27/51)             | 64% (36/56)             | 73% (40/55)             | 81% (42/52)               | 0,003   |
| <b>CaP detection rate in positive MRI group by sector biopsy (N)</b>           | 39% (25/64)                | 45% (23/51)             | 64% (36/56)             | 71% (39/55)             | 77% (40/52)               | 0,004   |
| CaP significance (GZ / AZ / RZ)* sector biopsy (detection %)                   | 25 / 12 (19%) / 5 (8%)     | 23 / 10 (20%) / 5 (10%) | 36 / 15 (27%) / 8 (14%) | 39 / 16 (29%) / 8 (14%) | 40 / 15 (29%) / 10 (19%)  |         |
| <b>CaP detection rate in positive MRI group by target biopsy (detection %)</b> | 17/64 (27%)                | 15/51 (29%)             | 23/56 (41%)             | 24/55 (44%)             | 33/52 (63%)               | 0,001   |
| out of Likert scale 5 (detection %)                                            | 12/35 (34%)                | 5/12 (42%)              | 9/13 (69%)              | 10/17 (59%)             | 21/22 (95%)               |         |
| out of Likert scale 4 (detection %)                                            | 5/19 (26%)                 | 4/15 (27%)              | 8/16 (50%)              | 8/16 (50%)              | 10/22 (45%)               |         |
| out of Likert scale 3 (detection %)                                            | 0/10 (0%)                  | 6/24 (25%)              | 6/27 (22%)              | 6/22 (27%)              | 2/8 (25%)                 |         |
| CaP significance (GZ / AZ / RZ)* target biopsy (detection %)                   | 17 / 10 (16%) / 5 (8%)     | 15 / 9 (18%) / 5 (10%)  | 23 / 12 (21%) / 7 (12%) | 24 / 17 (31%) / 9 (16%) | 33 / 27 (52%) / 14 (27%)  | 0,003   |
| <b>N negative (Likert scale &lt;3) MRI (% all patients)</b>                    | 6 (9%)                     | 9 (15%)                 | 14 (20%)                | 15 (21%)                | 18 (26%)                  | 0,008   |
| <b>CaP detection rate in negative MRI group (N)</b>                            | 33% (2/6)                  | 33% (3/9)               | 30% (4/14)              | 40% (6/15)              | 39% (7/18)                |         |
| CaP significance (GZ / AZ / RZ)* in negative MRI (detection %)                 | 2 / 2 (33%) / 1 (17%)      | 3 / 2 (22%) / 0         | 4 / 2 (14%) / 1 (7%)    | 6 / 3 (20%) / 2 (13%)   | 7 / 2 (11%) / 0           |         |

CaP = prostate cancer; N = number of patients; \* Classification based on UCL criteria: GZ = green zone, AZ = amber zone, RZ = red zone. p value obtained comparing Group A with Group B

### MP-03.15

#### Correlation of High PIRADS Score on Three-Tesla Magnetic Resonance with in-Gantry Magnetic Resonance Guided Biopsy

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**Introduction and Objectives:** Prostate cancer detection is a difficult process despite the various modalities available. The current standard of practice is based on stratifying risk using Prostate Specific Antigen (PSA), digital rectal examination (DRE) and performing a transrectal ultrasound (TRUS) or transperineal (TP) guided biopsy. Recent advances in three-tesla multiparametric magnetic resonance imaging (MP-MRI) technology and the availability of in-gantry MRI guided biopsies (MRGB) have added another diagnostic tool to help facilitate this. We review MRGB performed on high Prostate Imaging Reporting and Data System (PIRADS) score lesions in a single centre retrospective study.

**Materials and Methods:** There were 77 patients (mean age of 63) with high PIRADS score (4 and 5) that underwent in-gantry MRGB. All the biopsies were performed utilizing dynacad prostate biopsy system on a three-tesla MRI scanner by an urologist with assistance of an experienced radiologist. Two to three samples were obtained from each lesion using an MRI compatible 18-gauge biopsy needle. Three experienced pathologists evaluated the samples and provided the results and Gleason score in each positive sample.

**Results:** Out of the total 77 high PIRADS patients, 54 were PIRADS score 4 (70%) and 23 PIRADS score 5 (30%). There were 22 positive biopsies for adenocarcinoma of prostate with Gleason's score of 3+3=6 or higher. Out of the 54 PIRADS score 4 lesions, 13 were positive (24%) and out of 23 PIRADS 5 lesions, 9 were positive (39%). The remaining 55 biopsies were negative for prostate cancer.

**Conclusion:** We present our series of MRGB in patients with a high PIRADS score for prostate cancer. While this diagnostic paradigm is in its infancy stages, MRGB was positive in 24% of PIRADS 4 and 39% of PIRADS 5 lesions in this series.

### MP-03.16

#### The Role of Magnetic Resonance Imaging as a Patient Selection Tool for Initial Prostate Biopsy in Cases with a PSA Value in the Gray Zone and a Normal Digital Rectal Examination

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**Introduction and Objectives:** With the change of the role of prostate biopsy from pure cancer detection to detection of clinically significant cancer, the timing of initial prostate biopsy could vary (immediate or deferred). If MRI has the potential to exclude clinically significant cancer, this could be a useful tool to offer deferral of biopsy, resulting in avoidance of unnecessary prostate biopsies. Moreover, several reports have suggested that transrectal biopsy may be insufficient to detect anterior cancer, so a transperineal approach should be selected to evaluate the diagnostic accuracy of MRI. In this study, the characteristics of prostate cancers detected by the transperineal approach in patients with normal MRI findings, PSA values in the gray zone, and normal digital rectal examinations (DREs) were analyzed, and whether patients with normal MRI could be offered deferral of biopsy was investigated.

**Materials and Methods:** A total of 302 participants (median age, 66 years) with PSA values of 4–10 ng/mL and normal DREs who underwent transperineal 16–20 cores biopsy were studied. All patients underwent 1.5-T

multiparametric MRI, and MRI findings were assessed by 2 radiologists using a 5-point scale; 163 subjects who were assigned a score 1 or 2 were diagnosed as having normal MRI findings. Significant cancer was defined as 1 or 2 positive cores and Gleason score  $\leq 3+4$ .

**Results:** Prostate cancer was diagnosed in 34 of the 163 patients (20.9%) with normal MRI findings. The number of positive cores was 1 or 2 in 33 cases (97.1%) and 3 in one patient. The Gleason score was 6 in 20, 3+4 in 6, 4+3 in 4, and 4+4 in 4, resulting in 25 of 34 (73.5%) with insignificant cancer. No significant differences were found between the clinically significant cancer group and the clinically insignificant cancer group in PSA, PSAD, or percent free PSA.

**Conclusion:** Most cancers with normal MRI were clinically insignificant, and men with mildly elevated PSA levels and normal DREs could be offered deferral of biopsy. Further investigation to exclude significant cancers before biopsy in cases with normal MRI is needed.

### MP-03.17

#### MRI/3D TRUS Fused Targeted Biopsies for Detecting Anterior Prostate Cancer

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**Introduction and Objectives:** To evaluate the capability of magnetic resonance imaging (MRI) and three-dimensional transrectal ultrasound (3D-TRUS) image fused targeted biopsy (TB)



to diagnose anterior prostate cancers (APCa).

**Materials and Methods:** In period from 1/2010 to 7/2013, we retrospectively analyzed 211 consecutive patients with rising prostate specific antigen (PSA) and MRI suspicious APCa who underwent MRI/3DTRUS elastic fusion guided TB. The MRI pre-biopsy protocol included 3D T2-weighted (T2w) and diffusion weighted imaging (DWI) with apparent diffusion coefficient (ADC) calculated from b50 and b1000. Additional b2000 sequences was used. APCa was defined as >50% of MRI tumor volume (MTV) located anteriorly to the urethra. Suspicious APCa were highlighted on T2w images and segmented on workstation (Urostation® Koelis). MR/TRUS elastic image fusion guided TB were performed. Number of TB, maximum cancer core length (MCCL), cancer core involvement (CCI), primary Gleason grade (GG) and Gleason score (GS) were registered. Cancer detection rate (CDR) according to three degree of cancer suspicions on MRI was calculated. The concordance between GS and GG on TB and prostate specimen was assessed in 70 patients treated by radical prostatectomy (RP).

**Results:** Histopathological prostate cancer was affirmed in 170/211 (81%) patients. The median (range) number of TB per patient was 2 (1-5) and of these, 2 (1-4) were cancer positive. The mean±SD of MCCL and CCI in TB were 9±4.5 mm and 57±26%. CDR according to degree of cancer suspicions on MRI was 96/114 (96%) for high, 46/57 (80%) for medium and 10/35 (29%) for low degree of cancer suspicion. MTV median (range) was 1.2 (0.1-24) ml. ADC for positive and negative TB (mean±SD) were 78.4±15.7 (95% CI: 76-80) and 92±12 (95% CI: 88-96) [ $10^{-5}\text{mm}^2/\text{s}$ ],  $p<0.0001$ . Concordance of primary GG pattern and GS and between TB and RP was and 90%, ( $\kappa=0.7$ ) and (77%), ( $\kappa=0.64$ ).

**Conclusion:** MRI/3D TRUS elastic fusion guided TB technique is accurate method in diagnosing APCa.

#### MP-03.18

**Whole-Gland Magnetic Resonance Imaging (MRI)-Guided Transurethral Ultrasound Ablation of Prostate Cancer: Preliminary Outcomes of a Phase I Clinical Trial**  
Billia M<sup>1</sup>, Burtynyk M<sup>2</sup>, Pahernik S<sup>3,4</sup>, Roethke M<sup>3,4</sup>, Relle J<sup>5</sup>, Hafron J<sup>5</sup>, Schlemmer H<sup>3,4</sup>, Romagnoli C<sup>1</sup>, Chin J<sup>1</sup>

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**Introduction and Objectives:** MRI-guided transurethral ultrasound ablation (TULSA) is a

new minimally-invasive technology to treat localized prostate cancer (PCa), aiming to provide good local disease control with low side-effect profile. This modality consists of a transurethral device emitting planar ultrasound and generating a continuous and precise volume of thermal coagulation shaped to conform to the prostate, using real-time MRI monitoring and active temperature feedback control. The aim of this prospective, multi-center phase I clinical trial is to determine the safety and feasibility of MRI-guided TULSA.

**Materials and Methods:** A total of 30 patients with low-risk PCa (cT1c-T2a, N0, M0; PSA≤10ng/ml; GS≤6) are enrolled. Under general anesthesia, suprapubic catheter (SPC) is inserted and left in for 2 weeks. The TULSA device (PAD-105, Profound Medical Inc.) is inserted manually and positioned precisely in the prostatic urethra with MRI guidance (3-Tesla unit). Treatment planning is performed with therapeutic intent of conservative whole-gland ablation. Treatment is delivered under continuous MR thermometry feedback control, and patients recover with an outpatient protocol. Primary endpoints are safety and feasibility, with follow-up to 12 months. Complete clinical monitoring is 5 years, including serial PSA, TRUS biopsy and QoL questionnaires (IPSS, IIEF).

**Results:** All patients were treated with no intraoperative complications. Median treatment time was 36 (range: 24 - 61) min and prostate volume 44 (21 - 95) cc. Spatial control of the thermal ablation was within ±1.3 mm and contrast-enhanced MRI confirmed the resulting conformal non-perfused volume. Clavien II complications included urinary tract infections (7 patients), and epididymitis (1), all resolved. Clavien I complications included hematuria (11), and acute urinary retention after SPC removal (4) resolving with prolonged- or recatheterization. Median PSA decreased from 5.8 (1.3 - 9.4) ng/ml to 0.7 (0.1 - 3.1) ng/ml at 1 month (n=22) and remaining stable to 0.7 (<0.1 - 1.2) ng/ml at 6 months (n=11). Normal micturition resumed after SPC removal, with median IPSS and IIEF values returning to baseline at 3 months.

**Conclusion:** MRI-guidance enables accurate planning and real-time dosimetry and control of the thermal ablation volume. Preliminary results indicate that MRI-guided TULSA is safe and clinically feasible with a well-tolerated, low side effect profile.

#### MP-03.19

**Histoscanning Accuracy: Single Centre Based Analysis Matching Each Quadrant in Prostate Specimens Slices with this New Imaging Technique**

Busto Martin L<sup>1</sup>, Pertega S<sup>2</sup>, Simmons L<sup>3</sup>, Arumainayagam N<sup>4</sup>, Shamsuddin A<sup>4</sup>, Winkler M<sup>4</sup>

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**Introduction and Objectives:** Prostate Histo-scanning™ ('PHS') is a novel technique using computer-aided analysis of raw data from 3D-transrectal ultrasound (TRUS), to identify and visualize cancer foci within the prostate. The aim of this study was to assess the accuracy of PHS in detecting and ruling-out clinically significant prostate cancer lesions using radical whole-mount prostatectomy specimens as the reference standard.

**Materials and Methods:** Sixty eight men (mean age 62 years, mean PSA 11.63 ug/L), with prostate cancer had preoperative 3D-TRUS and Histoscanning analysis performed and subsequently underwent radical prostatectomy (RP). RP specimens were processed by a single experienced histopathologist using 5mm transverse whole mount step sectioned analysis. These transverse RP slices were digitally captured. PHS images were displayed on a 5x5mm grid and 5mm transverse slices allowing for direct comparison with RP histology digital images. Computer software was used to digitally overlay corresponding PHS and RP digital images for each transverse slice, to allow comparison. Each prostate was divided in quadrants (right and left apex, right and left base) for the purposes of analysis. Two-by-two contingency tables were constructed to calculate sensitivity, specificity, positive (PPV) and negative predictive (NPV) values with 95% confidence intervals.

**Results:** PHS had a sensitivity of 58%, specificity of 71%, PPV of 68% and NPV of 61% for all cancer.

**Conclusions:** Pre-operative imaging with Prostate HistoScanning™ may be a useful adjunct in determining location of cancer in order to plan nerve sparing surgery and also help guide focal therapy using ablative technologies.

#### MP-03.20

**Transrectal Ultrasound (TRUS) Guided Prostate Biopsies in Men Aged 75 Years and Over: Does It Affect Management?**

Lee E, Varnavas M, Banerjee G

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**Introduction and Objectives:** Transrectal ultrasound (TRUS) guided prostate biopsy is a key tool used in prostate cancer diagnosis and risk stratification. Infection is a well recognised complication, with sepsis reported as <1%, but the incidence is rising due to increasing antibiotic resistance. With increasing life expectancy and increased PSA testing, more men aged ≥75 years are being investigated for prostate cancer. We therefore reviewed whether TRUS prostate biopsies in men aged ≥75 years affected management decisions.



**Materials and Methods:** All men aged  $\geq 75$  years undergoing TRUS prostate biopsy under local anaesthetic over a 1-year period (January 2012 – December 2012) in a single institution were identified retrospectively. Patients age, indication for biopsy, digital rectal examination (DRE) findings, PSA at biopsy, histology, staging, D'Amico risk stratification, initial and actual treatment at 1 year were recorded.

**Results:** Eighty two (24%) of 340 TRUS prostate biopsies were performed in men aged  $\geq 75$  years, of which 56 men (68%) were diagnosed with prostate cancer compared to 130 men (50%) aged  $< 75$  years. Mean PSA was 31.3 (range 2.3 to 389.0) and DRE abnormal in 38 men (68%). Fifty one men (91%) were classed D'Amico intermediate or high risk. Nineteen men (63%) aged 75-79 years had treatment with curative intent compared to only six men (23%) aged  $\geq 80$  years. Twenty men (36%) were treated with radical radiotherapy, five (7%) with active surveillance of which one went on to have radiotherapy, twenty (36%) with hormonal therapy alone and eleven (20%) with watchful waiting. Only one of eight men who took part in clinical trials was  $\geq 80$  years. Biopsy upgraded the risk group in 15 men (27%) and all 8 men aged 75-79 years had radical radiotherapy, compared to 2 men (29%) aged  $\geq 80$  years.

**Conclusion:** Prostate biopsies in men aged 75-79 years does affect management decisions compared to men aged  $\geq 80$  years. Therefore, prostate biopsies in this age group are only recommended for men suitable and willing to have curative treatment or participate in clinical trials.

#### MP-03.21

##### High-Intensity Focussed Ultrasound in the Treatment of Localised Prostate Cancer: Focal Salvage Transition Rates

Guillaumier S<sup>1</sup>, Dickinson L<sup>1</sup>, Punwani S<sup>1</sup>, Stone H<sup>1</sup>, McCartan N<sup>1</sup>, Thiruvell M<sup>1</sup>, Maru M<sup>1</sup>, Hindely R<sup>2</sup>, Emberton M<sup>1</sup>, Ahmed H<sup>1</sup>

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<sup>2</sup>Basingstoke and North Hampshire Hospital, Basingstoke, UK

**Introduction and Objectives:** Focal therapy has shown encouraging low rates of genitourinary side-effects in the treatment of localised prostate cancer. As biochemical failure is difficult to define discussion has centred on rates of transition to whole-gland local therapy and need for systemic therapy. We report on our registry experience of over 1,000 men treated with transrectal HIFU.

**Materials and Methods:** Our independent academic HIFU registry incorporates a total of 830 patients with low, intermediate and high

risk, stage T2a-T3aN0M0 prostate adenocarcinoma treated between 2004 and 2012 in a primary setting. Of these, 509 had focal HIFU treatment (Sonablate 500); 313 had whole-gland treatment. We defined our composite failure as need for whole-gland therapy and/or systemic therapy.

**Results:** Of 509 men undergoing focal HIFU therapy, 84 required redo-HIFU (17%). None of the patients in this group went on to have radical prostatectomy. One percent (5/509) had salvage radiotherapy and  $< 0.5\%$  (1/509) had salvage chemotherapy due to lung metastases following a renal primary. One percent (5/509) were subsequently treated with androgen deprivation therapy alone. In the same period, 313 patients had whole-gland HIFU treatment. Of these, 40% (124/313) had redo-HIFU and 5% (15/313) had androgen deprivation for treatment failure. Six percent (18/313) underwent salvage radiotherapy and  $< 1\%$  (2/313) were further treated with salvage radical prostatectomy. Finally,  $< 0.5\%$  (1/313) had salvage chemotherapy and  $< 0.5\%$  (1/313) had cryotherapy. The results are tabulated in Table 1.

**Conclusions:** Focal therapy demonstrates reassuringly low short to medium term rates of transition to whole-gland local therapy or systemic therapy in 11%. Longer follow-up will determine rates of metastases and mortality.

MP-03.21, Table 1.

|             | Total | Redo HIFU | Radical Prostatectomy | EBRT | Hormones | Chemotherapy | Cryotherapy |
|-------------|-------|-----------|-----------------------|------|----------|--------------|-------------|
| Focal       | 509   | 84        | 0                     | 5    | 5        | 1            | 0           |
| Whole Gland | 313   | 124       | 2                     | 18   | 15       | 1            | 1           |

## Moderated Poster Session 4

### Pediatric Urology

Monday, October 13  
1300-1430

#### MP-04.01

##### Overactive Bladder Symptom Score (OABSS) Questionnaire for Children

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<sup>3</sup>Toyokawa City Hospital, Toyokawa, Japan

**Introduction and Objectives:** Most children are not conscious of urgency when urgency incontinence occurs, and their parents are unaware of their conditions. We examined questions about three behaviors of children, seen as urgency, with the overactive bladder symptom score (OABSS) questionnaire.

**Materials and Methods:** The OABSS questionnaire with three additional items were given to 191 children (127 boys and 64 girls), 5 to 15 years of age, with complaints of bedwetting, incontinence, and daytime frequency. Additional items were sudden urge to urinate when leaving home (urge to urinate when going out), urgency to urinate while bathing and urinating in the bathroom (urge to urinate while bathing), and behavior of tolerating urges to urinate (tolerating behavior). Each frequency was classified into 6 stages ranging from 0 to 5. We then examined associations between extents of urgency and urge to urinate when going out, urge to urinate while bathing, tolerating behavior, and urgency incontinence.

**Results:** Correlations between the extent of urgency incontinence and extents of urgency, urge to urinate when going out, tolerating behavior, daytime frequency, and nighttime frequency were examined. Correlations with urgency ( $r=0.572$ ), tolerating behavior ( $r=0.512$ ), urge to urinate when going out ( $r=0.485$ ), and urge to urinate while bathing ( $r=0.282$ ) were observed. Daytime frequency ( $r=0.191$ ) and nighttime frequency ( $r=0.0689$ ) showed no correlations. However, the numbers of cases without urgency, urge to urinate when going out, tolerating behavior, and urge to urinate while bathing despite having urgency incontinence were 37, 28, 22, and 12, respectively.

**Conclusion:** The extent of urgency incontinence correlated with extent of urgency, and with the extents of each of the three added items: tolerating behavior, urge to urinate when going out, and urge to urinate while bathing. However, numbers of children answering that they had no urgency, no urge to urinate when going out, no tolerating behavior, and no urge to urinate while bathing despite having urgency incontinence were 37, 28, 22, and 12, respectively. Therefore, urge to urinate while bathing

and tolerating behavior may be useful items for overactive bladder screening in children not able to recognize urgency.

#### MP-04.02

##### New Symptom Scale for Lower Urinary Tract Dysfunction for Children

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**Introduction and Objectives:** Symptom scales (SS) are questionnaires for evaluating and following up of special illnesses. SS used for the diagnosis of lower urinary tract dysfunction (LUTD) in children is named Lower Urinary Tract Dysfunction Symptom Scale (LUTDSS). LUTDSS has 13 questions. In this research, we wanted to identify the questions which are more important for the diagnosis of LUTD and create a simpler SS.

**Materials and Methods:** From October 2010 to April 2012, 91 children between the age of 5 and 15 were evaluated with the suspicion of LUTD. Eleven children with active urinary tract infections were excluded from the study. Forty-eight children were attending to hospital for routine body weight and height control and had no urinary complaints. Their LUTDSS score was  $<9$  and they were designed as control group. Thirty-two children with LUTDSS score  $\geq 9$  were thought as having LUTD and diagnoses were verified with 3-day bladder diaries and 2-time Uroflow-EMG-PVR tests. The answered questions of LUTDSS in patient and control group were compared.

**Results:** Overall, 1, 3, 7, and 8 questions were detected statistically significant for the diagnosis of LUTD. Children with daylight incontinence (first question of LUTDSS) were 49.8 (4.8-510) times, children with enuresis (third question) were 77.6 (6.2-961) times, children with pause while urinating (eighth question) were 96.3 (4.4-2090) times more likely to have LUTD than the opposites. Also pain while urinating (seventh question) was reported as statistically significant. The area under ROC curve created by using 1, 3, 7, and 8 questions was calculated 96.1%. Although it was not statistically significant, sudden need for urinating (tenth question) was thought to be one of the important aspect of LUTD.

**Conclusions:** This means that using only 1, 3, 7, 8 and 10 questions can make diagnose of LUTD. And we formed a simpler SS by using 1, 3, 7, 8 and 10 questions.

#### MP-04.04

##### Pediatric PCNL in the Split-Leg Modified Lateral Position and Using "Adult" Instruments

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**Introduction and Objectives:** We present our experience of percutaneous nephrolithotomy (PCNL) in children with our standard adults' technique, in the split-leg modified lateral position and using a 20.8 Fr adult nephroscope.

**Materials and Methods:** Between January 2007 and August 2010, 5 boys and 2 girls, aged between 5 to 15 years, undergone PCNL. The indication was failure of SWL in 3 cases and complex stones for the others. Operative technique: The children were placed in the same position throughout the whole procedure, with the trunk in the lateral position, the pelvis in oblique position and the lower limbs are split and bent in the lowest position. Retrograde ureteroscopy in 2 children and all ureteral catheter placements were performed with an 8 Fr adult ureteroscope. Caliceal puncture and tract formation using "one step" dilation with 24 Fr Amplatz dilator and sheath, were performed under fluoroscopic control. A 20.8 Fr nephroscope was used.

**Results:** The pediatric PCNL was possible in the split-leg modified position, using standard "adult" 20.8 Fr nephroscope, and 8 Fr ureteroscope. 9 PCNL were performed (3 left, 2 right, 2 bilateral), 5 under spinal-anesthesia and 4 under general anesthesia. Six approaches were through the upper calyx and 3 via a central calyx. Postoperatively, 8 renal units have been rendered stone free. In a renal unit, with multiple locations, the intended goal has been achieved with persistence of parenchymal calcification. No hemorrhagic, splanchnic, or intra-thoracic complication was noted. No transfusion was required. The Clavien score distribution was: no complication in 7 PCNL (77%) and grade 1 in 2 PCNL (23%).

**Conclusion:** The pediatric PCNL technique is safe and possible with the split-leg modified lateral position. In this position, there is no change of position throughout the procedure, so it is time saving and cost-effective with only one set of draping. Moreover, it allows simultaneous antegrade and retrograde approach with rigid instruments. The use of our standard adult instruments, 20.8 Fr adult nephroscope 8 Fr ureteroscope, was possible. Thus, "adult" urologists can also perform pediatric PCNL, with no need of specialized additional instruments.

**MP-04.05**

**Effect of Site, Size and Number of Upper Urinary Tract Stones on the Outcome of Extracorporeal Shock Wave Lithotripsy (ESWL) in Children: A Single Centre Study**  
Muhammad S, Nawaz G, Rahim W, Khan M, Hussain T, Hussain I, Akhter S  
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**Introduction and Objectives:** The treatment of urinary stones has been revolutionized since the first presentation of extracorporeal shock wave lithotripsy (ESWL) in West Germany, in the early 1980's. It is now accepted worldwide because of its being user friendly, noninvasive, with high efficacy rate and wide availability of lithotripters. Our objective was to determine the effect of site, size and number of upper urinary tract stones on the outcome of extracorporeal shock wave lithotripsy (ESWL) in children.

**Materials and Methods:** A total of 56 children with mean age 8.11 years (range 1-16) underwent ESWL for urinary stones. Double J stent was placed prior to ESWL when indicated. Renal stones of 0.5 to 5cm (mean size 1.10cm) in size were treated. The impulse rate per treatment varied from 1300 to 4,000 (mean 3021). Exclusion criteria were coagulopathies, pyelonephritis, outflow obstruction, non-functioning kidney and hypertension. ESWL was performed under general anesthesia in 38 cases. Follow-up period was 3-6 months.

**Results:** A total of 77 stones were treated: 29 calyceal, 34 pelvic, 9 proximal ureteral stones, 4 combined in renal pelvis and proximal ureter and 1 stone in renal pelvis extending into lower pole. Mean stone size was 1.10cm. A total of 77 sessions were performed. The mean number of shock waves per session was 3021. Overall stone-free rate was 85.71% (48/56 patients). Stone clearance rate was 100% for upper (6/6 stones) and mid calyceal stones (2/2 stones), after placing DJ stents as ancillary procedure in one of the patients in each group respectively, as compared to lower pole stones where clearance rate was (12/18 stones) 66.67%. Rate of stone clearance was 33.34% less in lower calyceal stones as compared to upper or middle calyx. For stones larger than 1 cm, number of lithotripsy sessions per stone were more 1.14 (31 sessions in 27 stones) as compared to 1.08 (46 sessions in 50 stones) for stones smaller than 1cm. In stones more than 1cm the failure rate was 8.92% (5/56) as compared to 5.35% (3/56) in stones less than 1cm. The ratio of ancillary procedures was 1:1.46 in stones larger than 1cm, as compared to 1:3 for stones less than 1cm. Need for ancillary procedure for single stone was (5/36) 13.88%, for two stones (1/2) 50%, for three (2/4) 50% and for four (1/1) 100% respectively. Similarly sessions of ESWL also increased with number of stones in a single collecting system, i.e. single session of ESWL was required in 87% of single stones

and 28.57% of two stones of in a single renal unit. On the other hand 80% of the 3 stones in a single system were cleared by 2 sessions of ESWL and for four stones required at least three sessions. Stone clearance rate was also higher 90.47% (38/42 stones) in single stone as compared to 75% (12/16 stones), 80% (12/15 stones) and 0% (0/4 stones) in two, three and four stones in a single unit respectively. Complications were encountered in 13 patients. Ancillary procedures were required in total 17 (29.82%) patients. DJ stenting was required in 13 (22.8%) patients, 3 (5.26%) patients required ureterorenoscopy with DJ stenting and only 1 (1.75%) patient required percutaneous nephrolithotomy after ESWL.

**Conclusion:** ESWL is a safe and effective method for treatment for stones at least up to 1 cm in pediatric population. Rate of stone clearance is less in lower calyceal stones as compared to upper or middle calyx. The number of lithotripsy sessions, along with rates of failure and ancillary procedures increase in stones more than 1cm in size. There is increase in need for ancillary procedures and number of ESWL sessions along with a decrease in rate of stone clearance in case of multiple stones in a single renal unit.

**MP-04.06**

**Laparoendoscopic Single-Site Surgery (LESS) for Treatment of Different Urologic Pathologies in Pediatrics: Single-Center Single-Surgeon Experience**  
Abdel-Karim A, Moussa A, Aboelfotoh A, Elmissery M, Elsalmy S  
*Alexandria University, Alexandria, Egypt*

**Introduction and Objectives:** Laparoendoscopic single-site surgery (LESS) has been recently reported as a valid treatment option for various urologic pathologies in adults. However, the current applications of LESS in pediatric urology is still limited. In this report we present our initial experience of LESS for treatment of different urologic pathologies in pediatrics.

**Materials and Methods:** Twenty two children with mean age of 6.25±4.23 years had LESS for undescended testes (bilateral=7, unilateral=6), varicocele (left=3, bilateral=1), non-functioning kidney (n=1), multicystic dysplastic kidney (n=1) and ureteropelvic junction obstruction (UPJO) (n=3). Both R-port (n=16) and Covidien port (n=6) were inserted through a periumbilical incision. We used pre-bent (with R-port), articulating (with Covidien port) and straight instruments. Pyeloplasty was done through hand-free intracorporeal suturing using 4/0 vicryl. All procedures were done by a single experienced laparoscopist. Data were collected during and after surgery then analysed retrospectively.

**Results:** Thirty nine LESS procedures were performed including orchiopexy (n=10), first stage Fowler- Stephens orchiopexy (n=9), second stage Fowler- Stephens orchiopexy (n=9),

orchiectomy (n=1), varicocelectomy (n=5), nephrectomy (n=2), dismembered pyeloplasty (n=2) and non-dismembered Y-V plasty (n=1). No conversion to conventional laparoscopy, open surgery or adding an extra-port in any patient. Mean operative time was 89.6±22.8 minutes. Mean blood loss was 32.2±22.1 cc. No intraoperative or post-operative complications were reported. Mean hospital stay was 0.52±0.25 days. Mean visual analogue pain scale at discharge was 0.52±0.33. Mean follow-up was 16.6± 6.4 months.

**Conclusion:** LESS is a feasible and safe option for treatment of different urologic pathologies in pediatrics. However, more patients, indications and prospective comparative studies are needed for further evaluation of the role of LESS in pediatric urology.

**MP-04.07**

**Minimally Invasive Percutaneous Nephrolithotomy in Children: Experience from a Single UK Institution**  
Miller S, Taylor A, Ahmad R, Ratan H, Lloyd J, Lemberger J, Scriven S  
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**Introduction and Objectives:** Children with urolithiasis are at increased risk of forming multiple calculi during their lifetime, particularly if they have underlying metabolic or renal disorders such as cystinuria or Dent's disease. Minimising renal trauma and preserving nephrons is essential in children who face the prospect of multiple procedures. Equally, reduced scarring from smaller access sites can facilitate subsequent percutaneous intervention and preserve renal function. Consequently, minimally invasive techniques were developed to address these issues. We present our experience of using mini-PCNL in our paediatric population over the last 4 years.

**Materials and Methods:** Retrospective analysis of 11 mini-PCNLs performed on 10 patients between 1<sup>st</sup> January 2009 and 31<sup>st</sup> December 2013. Patient demographics, stone size and composition, operative data, outcomes and complications were recorded. A patient was considered stone free if no residual fragments were identified on follow-up imaging. Complications were classified according to the Clavien system and outcomes compared to other series in the literature.

**Results:** Ten patients (7 boys, 3 girls) were included, with 11 PCNLs performed (one had two procedures during the study period). Average age was 10.2 years (range 3-16 years). Average operating time was 206 minutes with average length of stay 3.27 days. Four children had staghorn calculi, 3 had renal pelvis stones and 4 had calyceal stones (2 upper pole, 2 lower pole). Stone composition is as follows: 4 cystine, 3 calcium oxalate, 2 carbapatite, 1 calcium phosphate (other) and 1 struvite. Average Stone

size 25.4mm (range 9mm-63mm). Four (40%) patients had complete clearance, but only 3 (27%) required secondary treatment. Overall 73% patients were stone free or had clinically insignificant fragments that required no further treatment. Average blood loss was 2.6g/dL but no patients required blood transfusion. One child had a post-operative UTI (Clavien grade 2). No significant sepsis occurred.

**Conclusions:** Mini-PCNL is a safe, effective approach to treating large renal stones in children. No patients required blood transfusion and only 1 patient needed treatment for post-operative UTI. Stone clearance rates approach those of other contemporary series, with a 27% ancillary procedure rate.

#### MP-04.08

##### **Resource Utilization and Costs Associated with the Diagnostic Evaluation of Nonrefluxing, Primary Hydronephrosis in Infants**

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<sup>1</sup>Seattle Children's Hospital, Seattle, USA;

<sup>2</sup>University of WA, Seattle, USA

**Introduction and Objectives:** The long-term evaluation of postnatal nonrefluxing, primary hydronephrosis presents a dilemma for urologists, as most cases resolve without surgery. We report longitudinal resource utilization and costs associated with the diagnostic evaluation of infants with isolated, primary nonrefluxing hydronephrosis to determine the costs associated with diagnosing a surgical case, and assess the implication using a cost-consequences analysis.

**Materials and Methods:** A retrospective chart review was used to capture resource utilization for all infants less than 6 months of age with hydronephrosis evaluated at our institution over a 5-year period. Infants with confounding urological diagnoses were excluded. Both payer and societal perspectives were utilized. Costs were estimated from resource utilization, including radiographic imaging and clinical encounter types. Data were collected from first clinic visit until surgery or resolution or 3 years, whichever was shortest.

**Results:** Of 165 included patients, surgical rates for grades 1 - 4 hydronephrosis were: 0, 5%, 21% and 74%, respectively. Median costs of identifying a single surgical case per increasing grade of hydronephrosis were: infinite, \$37,600, \$11,741, \$2,124, respectively ( $p < 0.001$ ).

**Conclusion:** The diagnostic evaluation of higher grades of hydronephrosis is significantly more productive in terms of identifying patients requiring surgery than the evaluation of patients with lower grade disease. A more abbreviated diagnostic strategy may be warranted in patients with grades 1-2 hydronephrosis than

the current standard of care. We project for the population in this analysis that a less intensive approach for lower grades of hydronephrosis could save about 24% of costs.

#### MP-04.09

##### **The Double Barrel Shotgun Technique: Description of a Technique for Implantation of Dextranomer Hyaluronic Acid in Children with Vesicoureteral Reflux**

Mesrobian H

Medical College and Children's Hospital of Wisconsin, Milwaukee, USA

**Introduction and Objectives:** We describe a new technique for implantation of dextranomer/hyaluronic acid (Deflux) for the endoscopic treatment of vesicoureteral reflux (VUR) in children. We present our initial experience and analyze early outcomes.

**Materials and Methods:** Following IRB approval, we reviewed a database collected on 100 evaluable children who underwent this procedure between January of 2005 and December of 2012. Mean patient age at the time of the procedure was 69 months and ranged from 6 to 193. The female to male ratio was 97/3. The main indication for the procedure was the presence of documented febrile urinary tract infection(s) (FUTI) and VUR. A positional instillation cystogram (PIC) at the end of each procedure was performed to document resolution of VUR, in addition to visual evaluation. The presence or absence of a FUTI was the primary outcome. Follow-up at 1 month and at 1 year consisted of a tailored pelvic ultrasound to assess the volume of the Deflux mounds and their relation to the ipsilateral ureteral jet.

**Results:** A total of 185 implantation procedures -accounting for bilateral disease- were performed to treat VUR in 100 children. In 162 ureters with low grade VUR, the mean volume of Deflux utilized was close to 1 ml per ureter (range 0.5 to 3). In 23 ureters with high grade VUR, the mean volume/ureter was 1.5 (range 1 to 3.5). In one patient, asymptomatic hydro ureter was encountered at the initial follow-up. There were no immediate complications. At a mean postoperative follow-up of 14 months (range 1-84), 12 patients in the low-grade group had developed a FUTI, for an infection free success rate of 88%. In the high-grade group, the success rate was 71%. It was 83% for the entire cohort.

**Conclusions:** The double barrel shotgun technique is safe and simple while it maintains a high infection-free success rate. In this era of value driven health care, the reduced volume of Deflux needed to achieve these results represents an advantage over other techniques. We are cautiously optimistic and acknowledge the need for long-term follow-up to see if these favorable outcomes persist.

#### MP-04.10

##### **Classic versus 12 O'clock Incision of Posterior Urethral Valve: A Prospective Randomized Study**

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**Introduction and Objectives:** Endoscopic management of posterior urethral valve (PUV) is the gold standard. Various incision techniques and instruments were studied. Our aim is to evaluate the efficacy and safety of two commonly used incision approaches.

**Materials and Methods:** Between September 2007 and September 2012, 68 consecutive patients with PUV were randomly treated with either classic incision at the 5, 7, and 12 o'clock positions (Group I; 33 cases), or incision at the 12 o'clock position (Group II; 35 cases). Primary ablation was performed using either diathermy hook (in 26 patients) or sickle-shaped cold knife (in 42 patients). Urethral catheter was removed one day postoperatively for all cases. Voiding cystourethrogram was done for all patients at 2 months follow-up. All patients with persistent dilatation of posterior urethra on follow-up underwent revision cystoscopy. Recovery of bladder and renal function as well as resolution of vesicoureteral reflux (VUR) and hydroureteronephrosis (HUN) within the 2 treatment groups were compared.

**Results:** The median age at presentation was 6 months (range, day1- 9 years). Median follow-up was 3 years (range, 1-6). No significant difference was present between the two groups at presentation regarding age, renal function, VUR and HUN. Mean operative time was 20 vs. 10 minutes for groups I and II respectively ( $p < 0.001$ ). Although the use of electro-ablation was comparable, revision valve ablation was indicated for 4 cases in the classic incision arm ( $p = 0.035$ ). At last follow-up, the median estimated glomerular filtration rates were 70 and 78 ml/min/1.73 m<sup>2</sup> for groups I and II respectively ( $p = 0.193$ ). Complete toilet training was achieved in 55% of Group I compared to 69% in Group II ( $p = 0.234$ ).

**Conclusion:** Both PUV incision techniques are fairly comparable. However, single incision at the 12 o'clock position provides a shorter ablation time with relatively less morbidity.

#### MP-04.11

##### **High-Risk Oncogenic Human Papilloma Virus Infection of the Foreskin in Prepubertal Boys**

Balci M, Tuncel A, Baran I, Guzel O, Keten T, Aksu N, Atan A

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**Introduction and Objectives:** The high-risk human papillomavirus (HPV) subtypes that are related to penile cancer. The aim of the present study was to evaluate the prevalence and genotypes of HPV infection of the foreskin in asymptomatic prepubertal boys.

**Materials and Methods:** A total of 100 prepubertal healthy boys who underwent a standard circumcision procedure were included in the study. High-risk HPV status was determined by real-time polymerase chain reaction for the genotypes 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, 66 and 68.

**Results:** The median age at the time of surgery was 5.7 years (range, 2 months-9 years). High-risk HPV was detected in 9 foreskins (9%). Positive samples showed are HPV 16 (n=3), 31 (n=2), 39 (n=3) and 51 (n=1).

**Conclusion:** Our results showed that subclinical high-risk HPV infections are found in the foreskin, which could be a reservoir for HPV-associated diseases.

#### MP-04.12

##### **Severe Complications of Circumcision: An Analysis of 72 Cases at Komfo Anokye Teaching Hospital, Kumasi, Ghana**

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<sup>1</sup>Komfo Anokye Teaching Hospital, Kumasi, Ghana; <sup>2</sup>School of Medical Sciences, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

**Introduction and Objectives:** Circumcision is the commonest operation performed on young boys. Like any other operation, it is not without complications that can range from trivial to the most tragic. We report the findings of 72 cases referred for serious complications after circumcision that needed secondary surgical interventions. The aim of this study is to emphasize the important problem of circumcision complications.

**Materials and Methods:** The 72 cases (mean age 36months, range < 1month-63.5 months) with complications of circumcision were reviewed prospectively. Circumcisions were performed at various medical centers by mainly nurses (77.8%), doctors (8.3%) and in environments other than the hospital by traditional circumcisers (20.8%). All the Patients were examined by urologists and the injury category assigned.

**Results:** The most commonly observed complication was urethrocuteaneous fistula, seen in 56 cases (77.8%). The other complications were: glans amputation in 5 (6.9%), iatrogenic hypospadias in four (5.6%), epidermal inclusion cysts in three (4.2%), skin bridges in 2 (2.8%) and excess foreskin in 2 (2.8%). All patients with urethrocuteaneous fistula had simple closure of their fistulas and patients with iatrogenic

hypospadias were repaired by Mathieu (two patients) and tubularized incised plate urethroplasty techniques. Of the glans amputations, three were complete and two were partial. The complete glans amputations were seen much later after the injuries had completely healed with meatal stenosis and disfigured penis. We used buccal mucosa to reconstruct a neo glans penis in each case ensuring acceptable cosmesis and long term urethral opening. The partial amputations had also healed with constriction bands and urethrocuteaneous fistula and in one patient penile distortion. Formal urethroplasty were done in each case with longitudinal incisions made on the lateral aspect of each corpora which were then closed transversely. Adhesion freeing as well as revision were performed in all cases of skin bridges and excess foreskin.

**Conclusion:** Circumcision may be associated with many serious complications whether done at home or in the hospital. To prevent these complications, the operation should be performed by trained and experienced personnel.

#### MP-04.13

##### **Urethral Mobilization and Advancement versus Meatal Advancement and Glanuloplasty Incorporated for Anterior Hypospadias Repair: A Prospective Comparative Study**

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**Introduction and Objectives:** To evaluate the hypospadias repair created by Urethral mobilization and advancement (UMA) versus Meatal Advancement and Glanuloplasty Incorporated (MAGPI) in terms of cosmesis and preventing adverse events in anterior hypospadias repair.

**Materials and Methods:** Between June 2007 and January 2013, 247 consecutive boys (median age, 4 years) were enrolled. Preoperative position of the hypospadias meatus was glandular in 18, coronal in 101, and in the distal shaft in 128 boys. Ventral penile curvature, hypospadias distal urethra and flat/small glans were noted in 201, 33 and 29 patients respectively. UMA (Group I; n=112) and MAGPI (Group II; n=135) techniques were compared as regards cosmesis and complications.

**Results:** Both groups were comparable regarding patients' age and severity of hypospadias. After a median follow-up period of 2 years (range: 1 - 5), urethrocuteaneous fistula/dehiscence was noted in 3 patients in Group I vs. 7 patients in Group II (p=0.355). No urethral stenosis was encountered. The independent risk factors for secondary surgery were distal meatal location (OR 5.54, CI 95% 0.99-30.82), hypospadias distal urethra (OR 10.32, CI 95% 2.12-50.14) and flat/small glans (OR 14.88, CI 95% 3.32-66.61). A cosmetically normal slit-like meatus was obtained by single procedure

in 93.8% of patients in Group I vs. 81.5% in Group II (p=0.004). Normal circumcised penis was achieved in 92% and 93.3% of patients in groups I and II respectively (p=0.807).

**Conclusions:** There are no significant differences in complication rates between the two techniques, and the UMA technique is usually of better cosmesis.

#### MP-04.14

##### **Objective Long-Term Evaluation after Bladder Autoaugmentation with Rectus Muscle Backing**

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**Introduction and Objectives:** Bladder autoaugmentation with rectus muscle backing has been demonstrated to be an efficient surgical technique for bladder augmentation. Our aim was to objectively evaluate the long-term outcomes and to define value of this procedure.

**Materials and Methods:** Between August 1999 and June 2004, autoaugmentation was performed in 29 patients (18 girls and 11 boys) aged 4 – 13 years (median 8). Indication was neurogenic bladder with small capacity (no less than 60% of that expected for age) and poor compliance (myelomeningocele in 18, posterior urethral valve in 6, tethered cord in 3 and sacral agenesis in 2 patients). Detrusorectomy usually involved the whole upper half of the bladder. Prolapsed bladder urothelium was hitched to both rectus muscles to prevent its retraction and to offer easier bladder emptying with muscle contractions. Of the 29 patients, 22 participated, 3 were unavailable and 4 did not participate in complete evaluation.

**Results:** At the median follow-up of 27 months (ranged 9 to 49), bladder volume was increased significantly in all 22 patients and ranged from 190 to 462 mL (median 322 mL). Now, at the median follow-up of 135 months (ranged from 94 to 164), bladder volume ranged from 296 to 552 mL (median 417 mL). All patients had normal bladder compliance for age and confirmed continence. Voluntary voiding was achieved in 17 patients without residual urine; of 12 patients who used clean intermittent catheterization, only 5 were not able to empty bladder voluntarily.

**Conclusion:** Detrusorectomy with rectus muscle hitch and backing can be used successfully for bladder autoaugmentation. This is minimally invasive, completely extraperitoneal, simple and safe to perform. Voluntary contractions of the rectus muscle enable bladder acting as "pseudodetrusor". Main disadvantage is that this technique is indicated only in selected cases with small bladder capacity and without anomaly of anterior abdominal wall.

**MP-04.15****Disparities in Presentation and Outcomes of Children, Adolescents and Young Adults with Renal Cell Carcinoma: An NCDB Database Study**

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**Introduction and Objectives:** We analyzed age-related differences in presentation and outcomes of children, adolescents and young adult patients with renal cell carcinoma (RCC) using data from the National Cancer Database (NCDB). We hypothesize that children have delayed diagnosis leading to higher stage and decreased survival.

**Materials and Methods:** The NCDB is a nationwide oncology data set that captures 75% of newly diagnosed cancer cases from over 1400 facility based cancer registries in the US. This database was queried for patients under 30 years of age with RCC diagnosed between 1998-2011 with at least a 5 years follow-up. Cases were examined according to age (10-15yrs, 15-21yrs and 21-30yrs) in relation to American Joint Committee on Cancer stage, grade, histology, tumor size, treatment, insurance status, income, race and mortality. Statistical analysis included both univariate and multivariate logistic regression.

**Results:** Of 1,358 patients, 77 (5.7%) were younger than 15, 142 (10.5%) were between 15-21 years old, and 1,139 (83.9%) were between 21-30 years old. Patients under 15 years compared to those over 21 presented with higher stage (26.2% vs. 10% stage 4,  $p<0.0001$ ), higher grade tumors (10% vs. 4.6% grade,  $p=0.003$ ), larger tumor size (67.8mm

vs. 52mm,  $p<0.001$ ), and higher incidence of lymph node positivity (60% vs. 35.1%,  $p=0.002$ ). Papillary RCC was more common in the younger age group (17.8% vs. 7.8%,  $p<0.001$ ). Race, region, insurance status was not significantly different between the groups. We also discovered that a significantly higher proportion of children and adolescents underwent lymph node dissection (51% vs. 14.9%,  $p<0.001$ ), chemotherapy (14.1% vs. 4.8%,  $p=0.001$ ) and immunotherapy (8% vs. 3%,  $p<0.001$ ). In multivariate logistic regression analysis, stage, grade, margin status, and insurance were associated with increased mortality.

**Conclusions:** RCC is rare in patients younger than 15. Due to late diagnosis children present with higher grade and higher stage tumors and therefore have worse prognosis. Screening for microscopic hematuria may help identify these patients earlier and reduce mortality.

**MP-04.16****Ureteroscopy in Children: Is Routine Pre-Operative Ureteral Stent Insertion Necessary?**

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**Introduction and Objectives:** The role of pre-operative stent insertion before flexible and rigid ureteroscopy in children is debatable. We aimed to assess the outcome after rigid and flexible ureteroscopy in children who did not have routine pre-operative ureteral stent insertion, focusing on the ability to advance the ureteroscope up to the kidney and complication rate.

**Materials and Methods:** A total of 165 children (175 renal units) with urolithiasis were

managed in our institution over a 11 year period (2002 to 2013). Forty nine of them underwent flexible or rigid ureteroscopic procedures. Their age range was 2 to 15 years (mean 8.6 years; median 9 years). Twenty seven patients were male and 22 were female. Fifteen children underwent retrograde intrarenal surgery with flexible ureteroscope for renal stones and 34 children underwent lithotripsy with rigid ureteroscope for ureteral stones. Stone size ranged from 3 to 18 mm (mean 8.5 mm). Forty six of the 49 patients did not have pre-operative ureteral stent insertion. The procedures were performed under general anaesthesia in the standard manner. Ureteral dilators were used in 15 patients to improve access into the ureter. Ureteral access sheath was used in 2 patients. Analysis of prospectively collected data was done, focusing on rates of failure to complete the procedure and immediate complications.

**Results:** We were able to complete the procedure by passing the ureteroscope up to the renal pelvis in 41 of the 46 patients (89.1%) who did not have pre-operative ureteral stent insertion. Fifteen (32.6%) of patients required ureteral dilatation. We failed to advance the ureteroscope in 5 patients (10.8% failure rate); mainly due to complex stone burden. One patient urine extravasation requiring post-operative stent insertion (2.2% minor complication rate) and 1 patient developed post-operative sepsis (2.2% major complication rate).

**Conclusion:** Flexible and rigid ureteroscopy in children can be safely performed without routine pre-operative ureteral stent insertion. Our overall success rate in advancing the ureteroscope up to the kidney is 89.1%. The morbidity rates are low.

## Moderated Poster Session 5

### Urinary Incontinence

Monday, October 13  
1300-1430

#### MP-05.01

##### Microstimulation of Hippocampus Inhibits Micturition Reflex in Urethane-Anesthetized Rats

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**Introduction and Objectives:** Hippocampus is known to be involved in emotion and memory functions. Although recent brain imaging studies have indicated the activation of hippocampus during micturition control, there is no report that directly examines the role of hippocampus in micturition reflex. The purpose of this study is to investigate how direct stimulation of hippocampus affects the micturition reflex.

**Materials and Methods:** A total of 8 female Sprague-Dawley rats weighing 300g under urethane anesthesia (1.2 g/kg, subcutaneous administration) were used in this study. The bladder was cannulated through urethra using a PE-50 catheter, and isovolumetric bladder contractions were recorded. After the catheter insertion, rat was fixed on a stereotaxic frame and then the brain was exposed by removing a small piece of skull. A concentric tungsten electrode was inserted stereotaxically into the hippocampus following the coordinates of the rat brain map by Paxinos and Watson. An electrical stimulation (0.2 ms pulse width and 20 V) with frequencies from 1 to 200 Hz was applied to the hippocampus to evaluate its effect on isovolumetric bladder contraction. After the stimulation, the brain was resected and preserved in the formalin for histology.

**Results:** An electrical stimulation of 20 Hz frequency maximally inhibited the isovolumetric bladder contractions with the microelectrodes in the dorsal areas of hippocampus. The inhibition lasted for a much longer time than the stimulation duration. Intraperitoneal administration of naloxone (1 mg/kg) shortened the duration of inhibition, but it did not completely eliminate the inhibition. The stimulation never induced any excitatory response.

**Conclusion:** These results suggest that the electrical activation of neurons in the hippocampus inhibits the micturition reflex. Understanding the brain function in micturition control is important for the treatments of bladder dysfunctions caused by brain damages or disorders.

#### MP-05.02

##### Brindley Device Results the Treatment in Patients with Spinal Cord Injury

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**Introduction and Objectives:** To analyze the clinical and urodynamic results as well as to evaluate the complications arising from the SARS implantation with extradural technique in the treatment of patients with spinal cord injury (SCI) associated with neurogenic detrusor overactivity (NDO).

**Materials and Methods:** We made a descriptive study of the results obtained from 104 patients diagnosed SCI with NDO. These patients had been implanted since 2009. Descriptive variables and clinical variables were taken into account, frequent and descriptive analysis was made using SPSS 15 statistical program.

**Results:** Patients had an average age 38 ± 10 years, 91% were male. Trauma was the main cause of the lesions (99%). The patients had an average time of lesion evolution of 110 ± 86 months. In total, 90% of the patients showed ASIA A classification, 60% located at dorsal level. The urinary infection rate before SARS was 91%, and 15% after the procedure (p < 0.09). Before SARS all patients suffered urinary incontinence, after 86% of the patients has been cured. The patients with autonomic dysreflexia before the procedure was 66% after the SARS only 5% (p < 0.07). After SARS 94% of the patients had a bladder capacity greater than 300 ml. In 6% of the patients it was necessary to implant of a readjustable suburethral mesh due to stress urinary incontinence. There was 91% of patients that achieved an effective voiding, with a residual post-voiding lower than 50ml. In addition, 59% of the patients used the device for erectile function and 88% of the cases use for the bowel function. Two cases of infection in the receiver block have been shown four and five months after the implant. These cases have been solved by withdrawal of the device and subsequent provision of antibiotic therapy. Failure of the device was shown in one of the patients six months later, which was repaired without negative effects by substitution of the receiver block. Due to the increased experience, complications have been meaningfully reduced, as well as the infections, following the routine use of prophylactic antibiotics both in the pre and post-surgery phase.

**Conclusion:** Since 2009, our group has been carrying out the SARS implant using the extradural technique. Our experience in Latin America turned us into the group with the largest series of extradural technique implants

worldwide, we found very good clinical and urodynamic results in SARS implantation with extradural technique and the posterior rhizotomy, leading to improvement of urinary incontinence, urinary infection rate and autonomic dysreflexia in adequately selected patients with NDO due to SCI. Moreover, it proves to be a safe procedure, given the scarce complications it shows, being most of them easily solved.

#### MP-05.03

##### Are We Dosing Our Patients Right in the Management of Neurogenic Detrusor Overactivity (NDO) of Spinal Cord Injury (SCI)?

Suman D

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**Introduction and Objectives:** Anticholinergics are used as the first line of management to control the NDO of SCI. Based on symptoms, dose is often doubled or additional anticholinergics added to achieve clinical control of leaks. Though good symptom control is often achieved with anticholinergics but the same degree of improvement is less frequently documented urodynamically. We undertook this study to ascertain if the optimal dose for symptom relief means the same thing as the dose required for adequate suppression of NDO.

**Materials and Methods:** Fifty patients (age range 20-55 years, 30 males and 20 females) of SCI with documented NDO, undergoing treatment with tolterodine (long acting) 8 mg at bedtime and well controlled in terms of leaks per week (0-4) were included. All patients underwent routine UDS and IWT at 1 month of treatment.

**Results:** All patients presented their 3-day bladder diaries to show good control of leaks (0-4 per week) and were subjectively satisfied with the outcome of oral anticholinergic therapy. IWT was found to be still positive in 30% of patients. UDS showed persistence of NDO in 40% of patients. Maximum pressure of NDO above 40 cm was seen in 30% of patients. Interpretation: Symptom control in the form of prevention of leaks is not akin to objective urodynamic control in patients with NDO after SCI. Persistence of high pressure storage is an important factor in ongoing pressure induced morphological changes in patients with SCI. Thus, the goal of oral anticholinergic therapy should be to achieve adequate suppression of NDO and achieve low pressure storage in such patients.

**Conclusion:** Even oral anticholinergic treatment in the management of NDO should be titrated with IWT and urodynamics to ensure adequate storage pressure control rather than rely only on the symptomatic improvement in terms of reduction of leaks. Failure to achieve such objective optimal control should be a

factor to consider alternate treatment options.

#### MP-05.04

##### **Are Baseline Characteristics Predictive of Sacral Neuromodulation Test Stimulation Response in a Large Multicenter Trial?**

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<sup>1</sup>Metro Urology, Woodbury, USA; <sup>2</sup>Female Pelvic Medicine, Grand Rapids, USA; <sup>3</sup>MetroHealth Medical Center, Cleveland, USA; <sup>4</sup>Stanford University, Stanford, USA; <sup>5</sup>Scott and White Healthcare, Temple, USA; <sup>6</sup>University of Kansas, Kansas City, USA; <sup>7</sup>University of Oklahoma, Oklahoma City, USA; <sup>8</sup>University of Washington, Seattle, USA; <sup>9</sup>University of California, Irvine, USA; <sup>10</sup>Medtronic, Minneapolis, USA

**Introduction and Objectives:** This sub-analysis of the InSite Trial aims to characterize the association of baseline subject characteristics with response to sacral neuromodulation (SNM) test stimulation in a prospective, multicenter post-approval study. Subjects with bothersome symptoms of overactive bladder (OAB) including urinary urge incontinence (UI) and/or urgency-frequency (UF), who had failed at least 1 anticholinergic medication and had at least 1 medication not tried were included.

**Materials and Methods:** Subjects completed test stimulation with an implanted tined lead and external neurostimulator for a mean of 14 ± 3 days. Therapeutic success for UI was defined as a ≥50% improvement in average leaks/day and for UF as a ≥50% reduction in voids/day or a return to normal (<8 voids/day). Those that met criteria went on to receive the implantable neurostimulator. Subjects with both UI and UF required therapeutic success for just one indication to be implanted. Logistic regression was used and test stimulation response was regressed separately for UI and UF subjects on baseline characteristics, including age, race, gender, previous medications tried, years since diagnosis, daily voids or leaks, and medical history at baseline.

**Results:** A total of 340 subjects completed test stimulation; 339 subjects were implanted with the tined lead and 1 received a temporary lead. Mean age was 57 years and 91% were female. After test stimulation, 82% of UI subjects and 65% of UF subjects had therapeutic success. UI subjects were more likely to be test stimulation responders if they had previously tried 1 to 2 OAB medications as compared to 3 or more (OR(Odds Ratio)=2.08, p=0.045), and if they had a constipation history at baseline (OR=2.25, p=0.045). UF subjects were more likely to be test stimulation responders if they had a history of diarrhea at baseline (OR=2.36, p=0.015). No relationship of previous medication use with test stimulation response for UF subjects was observed.

**Conclusions:** This multicenter study shows that fewer prior medication trials (<3 vs. ≥3) is a predictor of a successful test stimulation response in UI patients. Constipation and diarrhea history were observed to be related to positive test stimulation response in UI and UF patients, respectively.

#### MP-05.05

##### **The Effect of Animated Biofeedback Training in Treating Patients with Voiding Dysfunction**

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King Fahad Specialist Hospital, Dammam, Saudi Arabia

**Introduction and Objectives:** To determine the effect of animated biofeedback training in treating patients with voiding dysfunction in the urology department of King Fahd specialist Hospital, Dammam.

**Materials and Methods:** A retrospective single centre study for all patients with voiding dysfunction who underwent animated biofeedback sessions training using urostym machine between March 2010 and Feb 2014. Study included all patients who underwent Biofeedback training age 5 years and more and completed at least 6 sessions of biofeedback. Success was considered as completely cured or 50% or more improvement of the presenting complaint.

**Results:** Among 80 patients who underwent biofeedback training sessions during the mentioned period, 13 patients were excluded due to failure to complete at least 6 sessions of training. There were seventeen males (25.4%) versus 50 females (74.6%). Mean age was 29.94 (range 6-69 years). Diagnosis included 33 dysfunctional voider, 13 with stress urinary incontinence, others 21 (31.3%). The major complaints were analysed as follow: 41 patients with urinary incontinence, 46 patients with frequency /urgency, 22 patients with recurrent UTI, 18 patients with high post void residual, and 17 patients with Nocturnal enuresis. The results were:

1. Forty one patients with urinary incontinence, 2 were dry after treatment (4.9%), 29 improvement (70.7%) with success of overall 75.6% and 10 patients no change (24.4%).
2. Frequency/urgency resolved in 4 out 46 patients (8.7%), improved in 30 patients (65.2%) with overall success of 73.9% and 12 patients with no change (26.1%).
3. Recurrent UTI improved in 18 out of 22 patients (81.8%) with 4 patients didn't improve (18.2%)
4. Nocturnal enuresis resolved in 3 out 17 patients (17.6 %), improved in 8 patients (47.1%), with overall success of 64.7% and 6 patients didn't show any improvement (35.3 %).

All the results were statistically significant with overall success rate of 74.8%.

5. Post void residual improved in 12 out 18 patients (66.7 %) with 6 patients (33.3 %) no change in residual.

**Conclusion:** Treatment of patients with voiding dysfunction using biofeedback urostym training is associated with improved urinary incontinence, urgency frequency symptoms, nocturnal enuresis episodes and decreased urinary tract infection rates.

#### MP-05.06

##### **Conservative Management for Postprostatectomy Urinary Incontinence: Cochrane Systematic Review and Meta-Analysis**

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<sup>4</sup>Cochrane Incontinence Group, Aberdeen, UK

**Introduction and Objectives:** Urinary incontinence (UI) is common after both radical prostatectomy and transurethral resection of the prostate. Conservative management incorporates numerous interventions such as: pelvic floor muscle training (PFMT) with or without biofeedback; electrical stimulation; lifestyle changes; penile clamps; pads; and combinations of these interventions. However, there is uncertainty concerning treatment efficacy and optimal timing of delivery. The aim of this updated Cochrane Systematic Review is to determine if conservative interventions are effective at treating and preventing UI in men after prostate surgery.

**Materials and Methods:** Randomised controlled trials (RCT) comparing conservative interventions for postprostatectomy UI were searched (5<sup>th</sup> February 2014) using the databases: Cochrane Incontinence Group Specialised Register; CENTRAL; EMBASE; CINAHL; ClinicalTrials.gov; and WHO ICTRP. Two independent review authors performed abstract and full-text screening, data extraction and risk of bias assessment. The quality of evidence was assessed using GRADE methodology. Primary outcomes for this review included: number of incontinent men; quality of life; adverse events; and cost-effectiveness.

**Results:** We assessed 764 reports, from which 92 reports of 52 trials were included in the review. The trials included 4545 men, of whom 2736 received an active treatment. For men with UI after surgery, there was no statistically significant difference in the number of incontinent men receiving PFMT with or without biofeedback (57% with UI versus 62% in the control group, Risk Ratio (RR) 0.85, 95% CI 0.60 to 1.22) after 12 months. This was reflected in a lack of difference in quality of



life measures after 12 months assessed using the ICIQ-SF (MD -0.50, 95% CI -1.35 to 0.35). In contrast, for men who received an intervention to prevent incontinence, there was a statistically significant difference in the UI rate at 12 months (10% versus 32%, RR 0.32 95% CI 0.20 to 0.51). However, risk of bias showed methodological limitations: the quality of evidence using GRADE methodology was moderate.

**Conclusion:** The evidence on whether or not conservative interventions are effective at treating or preventing UI post-prostatectomy is conflicting. Rigorous and adequately powered RCTs are still needed to obtain a definitive answer.

#### MP-05.07

##### **Do We Need to Change Our Counseling to Patients Undergoing AUS for Post Prostatectomy Incontinence?**

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**Introduction and Objectives:** Urinary incontinence post radical prostatectomy (PPI) is a common functional complication with significant impact on the patient quality of life. The artificial urinary sphincter (AUS) remains the gold standard surgical treatment for PPI. Historically, literature reported AUS outcome in both neurogenic and non-neurogenic incontinence with high complications and re-operation rates. Aim to present a modern AUS experience in treating PPI only patients from a large volume single centre in the UK.

**Materials and Methods:** A retrospective review of patients underwent AUS implantation (AMS 800) from 2007-2013. Data were collected on patient demographics, infections, erosion rate, mechanical failure, reoperation rate and continence rates. All patients included were strictly post-radical prostatectomy.

**Results:** Eighty four AUS were implanted over a 6-year period. Patient's age ranged between 51-78 (median 69, mean 69.25). Mean follow-up 2.1 years, Median 2 years, and range 3 months to 6 years. 36.9% (31/84) reported that they are completely dry with no pads; 41.6% (35/84) of patients are socially continent (using 1 pad/day), 19% (16/84) are using >2 pads / day. Two lost follow-up. Reoperation rate was 8.3% only (7/84) including 5 mechanical failures (6%). Three abdominal wound infections, one of which required admission and removal of device. Three perineal wound infections and 2 UTIs. One patient had an erosion (1.2%). No urethral atrophies reported. Bladder over activity developed in 6% of patients. Recurrence of bladder neck stricture occurred twice in 1 patient after the AUS implantation.

**Conclusion:** AUS outcome in patients with post radical prostatectomy is very good with low complications and re-operation rates.

Historical re-operation figures to counsel patients should be avoided. However, the limitation of our results is relatively short follow-up.

#### MP-05.08

##### **To What Extent Does Radiotherapy Affect the Outcome of Bulbar Artificial Urinary Sphincter Implantation in the Treatment of Post-Prostatectomy Sphincter Weakness Incontinence?**

Bugeja S, Frost A, Andrich D, Mundy A

University College London Hospitals, London, UK

**Introduction and Objectives:** This study evaluates the impact of radiotherapy on the functional outcome and explantation rate of the bulbar artificial urinary sphincter (AUS) in the management of sphincter weakness incontinence (SWI) following prostate cancer treatment.

**Materials and Methods:** A total of 188 men (mean age 67.9 years) underwent AUS implantation at our institution between 2006 and 2013 for SWI following prostate cancer treatment. These were analysed in 2 groups: Group A, post-radical prostatectomy (RP), n=116; Group B, post-radiotherapy, n=72 (adjuvant to RP in 44). There were 151 primary implantations. Thirty seven were revisions for malfunction, infection or erosion. Mean follow-up was 19.2 months (3.5-92.8 months). Functional outcome was assessed only in patients with follow-up of at least 1 year (67 in Group A; 44 in Group B; mean 27.6 months; range 12.0-92.8 months).

**Results:** Eleven out of 116 (9.1%) unirradiated patients had their device explanted for infection and/or erosion compared to an explantation rate of 13.9% (10 of 72) in the irradiated group. This occurred significantly earlier in Group B (mean 9.2 vs. 20.5 months). Three non-irradiated patients had their device replaced for malfunction compared to 1 in the irradiated group. In terms of continence, at a minimum of 1 year follow-up there was interestingly no difference between irradiated and irradiated patients (79.5% vs. 80.6%) This may be related to the fact that more patients in Group A underwent revision surgery (23.8% vs. 15.9%). In fact, of the 13 incontinent in Group A, 5 had at least one previous AUS (38.5%) compared to only 1 out of 9 in Group B (11.1%). Detrusor overactivity was a commoner cause for recurrent incontinence in unirradiated patients (46% vs. 22.2%) possibly indicating a more careful selection of patients referred for AUS after radiotherapy.

**Conclusion:** Radiotherapy is associated with an increased incidence of AUS infection and erosion which occurs much earlier following implantation. Nonetheless, in carefully selected and appropriately counselled patients, a functional outcome comparable to unirradiated patients is achievable.

#### MP-05.09

##### **Surgical Outcomes of Artificial Urinary Sphincter Revision and Replacement**

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**Introduction and Objectives:** AUS failure can be due to urethral atrophy/erosion, device failure, or infection. The purpose of this study was to characterize the results of AUS revisions and replacements.

**Materials and Methods:** From 1988-2012, 261 men underwent 388 AUS placements (214), revisions (76), or replacements (98). Revision was performed for recurrent/persistent SUI. Replacement was performed for urethral erosion/infection or mechanical failure. End-points included reoperation, worsening SUI, urethral erosion, and infection.

**Results:** Mean age at time of final surgery was 69.1 years and median follow-up 34.6 months with no difference in follow-up among the groups (p=0.16). AUS replacement was associated with higher rates of mechanical failure (p=0.036) and urethral erosion (p<0.001) than virgin AUS placement. AUS replacement after urethral erosion was associated with a higher rate of subsequent urethral erosions (p<0.001) than virgin AUS placement, while AUS replacement after mechanical failure had no higher rate of persistent SUI (p=0.980), mechanical failure (p=0.112), or urethral erosion (p=0.332). There was no difference between the virgin AUS placement group and the AUS revision group with regard to persistent SUI (p=0.244), mechanical failure (p=0.310), urethral erosion (p=0.448), or overall failure (p=0.336). Median times to revision and replacement after virgin AUS placement were 33.1 months and 48.6 months, respectively. After initial AUS placement, the rate of subsequent revision was 21% and replacement was 17%. Among AUS replacements, median time to erosion was 5.7 months and median time to mechanical failure was 33.1 months.

**Conclusions:** AUS replacement is associated with higher rates of mechanical failure and urethral erosion. AUS revision surgery is as safe, effective, and durable as virgin AUS placement.

#### MP-05.10

##### **Comparing the Effectiveness between Mid Urethral Retropubic TVT and Transobturator TOT or TVT-O in Mixed Pattern Urinary Incontinence with Stress Incontinence Predominating**

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<sup>2</sup>Ashford and St Peter's Hospital NHS Foundation Trust, Surrey, UK; <sup>3</sup>Salisbury District Hospital NHS Foundation Trust, Salisbury, UK

**Introduction and Objectives:** We compared

the effectiveness of mid urethral Retropubic TVT and transobturator TVT-O or TOT performed in single centre in female patients with mixed pattern urinary incontinence with stress Incontinence predominating.

**Materials and Methods:** A detailed study of the operating log and the patients follow-up notes was performed to identify the patients treated with midurethral tapes for mixed pattern urinary incontinence with stress incontinence predominating. From March 2006 till March 2011, 74 females patients were identified with symptoms of mixed pattern urinary incontinence with stress incontinence predominating. Thirty eight out of 74 (51.35%) underwent retro pubic TVT insertion, 28/74 (37.83%) underwent transobturator (TVT-O/TOT tapes), 9.45% (7/74) miniarc and 1 TVT Secure. Fifteen out of 74 (20.27%) patients had no prior urodynamic study. Only 3/59 (5.08%) patients who underwent urodynamic study showed mild bladder overactivity and rest had sensory urgency. Of these 3, 2 underwent transobturator tape and 1 miniarc.

**Results:** In the TVT (Retropubic) group 60.52% (23/38) patients were fully continent while 8/38 (21.05%) had partial improvement even with anti cholinergics. Out of these 23 patients 10 (43.47%) needed anti cholinergics while 5/23 (21.73%) needed Botulinum toxin in addition. While in the Transobturator group 24/28 (85.71%) patients were completely continent without complications and only 7/24 (29.16%) needed anti cholinergics and 1 (4.1%) needed Botulinum toxin. One out of 28(3.57%) had partial improvement. 3.57% (1/28) had late failure after initial complete response for 3years. Two out of 3 patients which showed bladder overactivity on urodynamics study and underwent transobturator tape had complete response without any additional treatment. Six out of 38 (15.78%) patients in TVT group presented post op with dysfunctional voiding compared to none in TOT/TVT-O group. Out of these, 2 needed tape division and 1 needed chronic SIC and 1 pt needed early SIC only. In TVT arm 3/38 (7.89%) had AUR all had chronic diff voiding 2 on SIC and 1 Partial response while none in TOT group. In TOT/TVT-O group 3.57% (1/28) Perennial pain needing tape division and 3.57% (1/28) short-term leg pain was seen but not in TVT group.

**Conclusion:** Transobturator tapes are more efficacious and should be considered as first choice in such category.

#### MP-05.11

##### **A Pattern of Detrusor Overactivity in Urodynamic Analysis Is Associated with Lower Urinary Tract Symptoms But Not with Renal Function Deterioration in Neurogenic Bladder Dysfunction**

**Song S**, Lee D, Sohn M, Hong B, Jeong I, Park S, Kim K, Kim C, Kim M, Kim K  
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**Introduction and Objectives:** We aimed to evaluate whether the pattern of DO (phasic or terminal) on urodynamic study is associated with lower urinary tract symptoms or renal function deterioration in pediatric neurogenic bladder patients.

**Materials and Methods:** We retrospectively analyzed the medical records of pediatric patients who underwent urodynamic evaluations and were diagnosed with a neurogenic bladder in our hospital during the last two years. After a thorough review of the urodynamic data in our study cohort, we categorized DO as phasic versus terminal or single versus multiple DO. Renal function deterioration was defined by photon defects on a DMSA renal scan.

**Results:** Forty-eight patients were included in this analysis (Table 1) and comprised cases of spina bifida (29.2%), tumors of the central

nervous system (16.7%), and other neurologic diseases (14.6%). DO was observed in 83.3% of all patients. The pattern of DO was revealed as phasic (77.5%) or terminal (22.5%) and as either single (10%) or multiple (90%). Patients with terminal DO complained of urgency or urge incontinence more frequently than those with phasic DO. No patients with single DO but all patients with multiple DO had urgency or urge incontinence ( $p=0.029$ ). The pattern of DO was not related to the occurrence of photon defects on a DMSA scan. After adjustment for bladder compliance, the detrusor pressure at maximal cystometric capacity was found to be the only significant predictive factor for the presence of a photon defect on a DMSA scan in multivariate regression analysis (OR=1.15, CI 1.01-1.32,  $p=0.027$ ).

**Conclusion:** Urgency or urge incontinence is more prevalent when DO shows a multiple or terminal rather than a single or phasic pattern. A significant predictor of renal function loss was the maximal cystometric capacity but not the DO pattern.

MP-05.11, Table 1. Patient Characteristics

| Patient Data (n=48)                                                                                                                                                           | Mean $\pm$ SD       | Range  | Median |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------|--------|
| Age (years)                                                                                                                                                                   | 10.83 $\pm$ 5.58    | 1-18   | 12.0   |
| Uroflowmetry                                                                                                                                                                  |                     |        |        |
| Qmax (ml/sec)                                                                                                                                                                 | 10.03 $\pm$ 10.24   | 0-34   | 7.0    |
| VV (ml)                                                                                                                                                                       | 113.73 $\pm$ 123.21 | 0-492  | 76.0   |
| PVR (ml)                                                                                                                                                                      | 110.35 $\pm$ 164.65 | 0-520  | 12.5   |
| Urodynamics                                                                                                                                                                   |                     |        |        |
| Detrusor profiles                                                                                                                                                             |                     |        |        |
| PdetQmax (cmH <sub>2</sub> O)                                                                                                                                                 | 33.29 $\pm$ 25.37   | 0-109  | 30.50  |
| MCC (ml)                                                                                                                                                                      | 221.31 $\pm$ 158.16 | 25-520 | 185.0  |
| PdetMCC (cmH <sub>2</sub> O)                                                                                                                                                  | 6.79 $\pm$ 9.19     | 0-38   | 0.00   |
| DO profiles                                                                                                                                                                   |                     |        |        |
| MDOP (cmH <sub>2</sub> O)                                                                                                                                                     | 42.26 $\pm$ 43.35   | 0-164  | 32.0   |
| Volume at DO                                                                                                                                                                  | 86.98 $\pm$ 109.16  | 0-521  | 56.0   |
| Qmax, peak flow rate; VV, voided volume; PVR, post void residual; MCC, maximal cystometric capacity; DO, detrusor overactivity; MDOP, maximum detrusor overactivity pressure. |                     |        |        |

## Moderated Poster Session 6

### Penis/Testis/Urethra, Benign Disease

Monday, October 13  
1435-1600

#### MP-06.01

##### Catheter Associated Urethral Stricture: Not an Uncommon Occurrence

Gyasi-Sarpong K<sup>1</sup>, Appiah K<sup>2</sup>, Azorliade R<sup>2</sup>, Aboah K<sup>1</sup>, Nyamekye B<sup>2</sup>, Otu-Boateng K<sup>2</sup>, Amoah G<sup>2</sup>, Maison P<sup>2</sup>, Arthur D<sup>2</sup>, Opoku-Antwi P<sup>2</sup>, Frimpong-Twumasi B<sup>2</sup>

<sup>1</sup>Kwame Nkrumah University of Science and Technology, Kumasi, Ghana; <sup>2</sup>Komfo Anokye Teaching Hospital, Kumasi, Ghana

**Introduction and Objectives:** Urethral catheters are employed for diagnostic and therapeutic purposes. However its use may be associated with long term morbidity for the patient. Scanty data exist in Ghana on catheter related strictures. The aim of this study is to highlight the serious problem of urethral strictures developing as a result of catheterization either from traumatic catheterization, allergic reaction to catheters or infection and to make recommendations to reduce the incidence of catheter associated urethral strictures.

**Materials and Methods:** Between October 2012 and January 2014, a total of 28 patients had urethroplasty done for urethral strictures which were as a result of urethral catheterizations. The mean age was 45.3 years (range 12 to 80). This represented 22.2% of all strictures treated with urethroplasty at the Komfo Anokye Teaching Hospital, Kumasi. A database was kept prospectively for all patients.

**Results:** Strictures resulting from traumatic catheterizations were all in the bulbar urethra whereas non traumatic catheterizations were more likely to cause strictures in both penile and bulbar urethra. Traumatic catheterization caused singular strictures (100%) whereas non traumatic catheterization injury was more likely to cause multiple strictures (75%). Traumatic catheterization injuries were more likely to cause strictures of less than 3cm (100%) whereas strictures of non traumatic origin were more likely to be more than 3cm (62.5%). Non urologic indications accounted for most catheterizations resulting in these catheter related strictures. Laparotomy for General surgery indications accounted for more than half (57.1%) of these catheter related strictures. Two patients on regular catheter change for refractory urine retention for BPH whiles waiting for surgery also developed long strictures.

**Conclusions:** Catheter associated urethral strictures are common and non traumatic strictures are more complex compared to traumatic strictures. There is need to sensitize health care

providers to avoid passing catheters for short procedures and to use good quality catheter materials under strict asepsis when there is an indication for catheterization.

#### MP-06.02

##### Identifying Patients at Risk of Failure after Direct Vision Internal Urethrotomy Combined with Self-Dilation for Short Segment Strictures Anterior Urethra

Mahmoud O, Fakher I, A. Elbakry A, Harraz A, Tharwat M, El-Assmy A, Mosbah A  
*Urology and Nephrology Center, Mansoura, Egypt*

**Introduction and Objectives:** Patients underwent direct vision internal urethrotomy (DVIU) for short strictures of the anterior urethra might be kept on self-dilation in attempt to decrease recurrence rate. We prospectively evaluated this patient cohort to identify the subgroup requiring further surgery.

**Materials and Methods:** Between October 2013 and March 2014, patients attending at our tertiary referral center for self-dilation after DVIU were interviewed and underwent assessment for their voiding pattern using validated LUTS scores, uroflowmetry, postvoiding residual urine (PVR) and flexible cystoscopy. Patients characteristics including duration and frequency of self-dilation, maximal catheter diameter used for self-dilation and urine cultures were determined. Further patients' demographics at time of surgery were retrospectively collected. We defined patients requiring further intervention when they had moderate to severe LUTS (I-PSS more than 7) and/or maximal flow rate less than 10 ml/min with unsatisfactory quality of life because of urinary symptoms.

**Results:** A total of 70 patients attended the outpatient clinic for self-dilation from which 49 patients completed their evaluation. A total of 21 (42.9%) patients were defined as requiring further intervention. These patients showed significantly higher PVR (median 50 cc vs. 30 cc) ( $p < 0.001$ ). This group of patient were older (median age was 58 vs. 49.5 years) ( $p=0.03$ ) and had previous history of DVIU (62.5% vs. 33.3%) ( $p=0.03$ ). On the other hand, patients' age, BMI, associated comorbidities, stricture length, site or duration of catheterization and preoperative positive cultures did not show significant association with outcome. Similarly, the duration of self-dilation and catheter size did not show significant association with the outcome. On multivariate analysis, only patients with previous history of DVIU are at risk of failure (OR: 3.437; 95%CI: 1.1 – 12.358;  $p = 0.05$ ).

**Conclusions:** A significant portion of patients undergoing self-dilation after DVIU showed unsatisfactory outcome. Patients with previous history of DVIU are at higher risk of treatment failure.

#### MP-06.03

##### Combined Inlay and Onlay Buccal Mucosa Urethroplasty for Long and Narrow Bulbar Urethral Strictures

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**Introduction and Objectives:** Long urethral strictures with a very narrow lumen pose an immense challenge for buccal mucosa augmentation urethroplasty. Larger discrepancy in the size of the graft and the native urethral plate makes the graft vulnerable for contracture and fibrosis. Increasing the width of the urethral plate by a vertical midline mucosal incision and applying an additional inlay buccal graft may lessen the discrepancy and help in improving the adequacy of the urethral lumen.

**Materials and Methods:** Eighteen patients between the age group of 28 to 65 years (mean 47 years) were operated for a combined inlay and onlay buccal mucosa graft urethroplasty from April 2011 to June 2013. Inlay graft was applied either dorsally or ventrally depending upon the urethrotomy incision. The inlay graft length was between 2.5 to 5 cm (mean 3.52 cm), and the onlay graft length varied between 4.7 to 7.6 cm (mean 5.97 cm). Dorsal urethrotomy with ventral inlay was done in 5 (27.8%) and ventral urethrotomy with dorsal inlay was done in 13 (72.2%) patients respectively. Previously operated patients and traumatic strictures were excluded from this retrospective study. Combined grafting was only done in the bulbar urethra. Per-urethral catheter was kept for an average duration of 36 days.

**Results:** Results were analyzed on the basis of pre- and post-operative uroflowmetry and any kind of post-operative instrumentation was considered as failure of the procedure. The mean follow-up duration was 630 days. Stricture length varied from 1.2 to 4.5 cm. (mean 2.74 cm). All the 18 patients managed to have significantly better flow rates after surgery and till date we haven't encountered any failures. The mean maximum flow after surgery was 23.82 ml/sec. Two patients developed a transient fistula through the operative site which resolved after further catheterization for 3 weeks and did not require any instrumentation till date.

**Conclusion:** Combined inlay and onlay graft technique improves the success rates of long and very narrow bulbar strictures. Incising the scarred urethral plate with inlay graft augmentation helps in achieving adequacy of the urethral lumen.

#### MP-06.04

##### Gapometry Urethrometry Index: Can It Predict Need for Inferior Pubectomy during Progressive Perineal Urethroplasty

Mandal A, Bhogesh S, Mavuduru R, Singla K

**MP-06.04, Table 1.** Univariate analysis of predictors of extent of surgery in repair of PFUDD

|                                | Group 1 (n= 29) | Group 2 (n=14) | Odds Ratio (95% CI)   | p-value |
|--------------------------------|-----------------|----------------|-----------------------|---------|
| Length of urethral defect (cm) | 1.990±1.014     | 3.305±0.877    | 1.310 (0.671 - 1.950) | <0.001  |
| Length of bulbar urethra (cm)  | 7.688±1.642     | 6.998 ±0.671   | -.689 (-1.615 - .236) | 0.141   |
| Gapometry Urethrometry index   | 0.304±0.211     | 0.472±0.114    | -.168 (-.290 - -.045) | 0.008   |

**MP-06.04, Table 2.** Multivariate analysis of significant predictors of extent of surgery in repair of PFUDD

|                              | $\beta$ coefficient | Wald Test | OR (95% CI)             |
|------------------------------|---------------------|-----------|-------------------------|
| Length of bulbar urethra     | 1.386               | 1.715     | 3.99 (0.502 - 31.823)   |
| Gapometry/Urethrometry index | 2.465               | 3.711     | 11.76 (0.958 - 144.542) |
| Constant                     | -3.226              | 9.115     | 0.040                   |

Postgraduate Institute of Medical Education and Research, Chandigarh, India

**Introduction and Objectives:** This study was conducted with an aim to identify preoperative predictors of need for inferior pubectomy during progressive perineal urethroplasty (PPU).

**Materials and Methods:** Data on 43 patients with isolated pelvic fracture urethral distraction defect (PFUDD) was analyzed. The gapometry urethrometry index was calculated as the ratio of distraction length and bulbar urethral length on preoperative combined ascending and descending urethrogram. The patients were divided into Group A (men who required inferior pubectomy) and Group B (men who could be managed without inferior pubectomy). Univariate and multivariate analysis were used to identify predictive aspects of the evaluated parameters using ROC analysis.

**Results:** On univariate analysis, longer distraction length and higher gapometry index were significant predictors for need for inferior pubectomy. However, on multivariate analysis, gapometry index remained strong and independent predictor compared to distraction length. ROC curve for gapometry index showed an optimal cutoff of 0.365 with sensitivity of 92.9% and specificity of 31% (Table 1 and Table 2).

**Conclusion:** Higher gapometry index and longer urethral distraction length are predictors for need for inferior pubectomy during perineal urethroplasty. The gapometry index is a stronger predictor with an optimal cut off index of 0.365.

#### MP-06.05

##### **A Novel Cell Based Therapy Using Buccal Epithelium Expanded and Encapsulated in Scaffold (BEES-HAUS): Hybrid Approach to Treat Urethral Stricture**

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**Introduction and Objectives:** We describe the feasibility of a novel cell based technique using buccal epithelium expanded and encapsulated in Thermogelation Polymer (TGP) scaffold for the treatment of urethral stricture. In this procedure autologous cultured buccal epithelial cells embedded in TGP were instilled at the stricture site after a wide endoscopic urethrotomy.

**Materials and Methods:** Four Patients with urethral stricture were included in the study after informed consent. The Thermogelation Polymer (TGP) used in the study is a co-polymer composed of thermo-responsive polymer blocks. TGP is in liquid state at lower than sol-gel transition temperature but turns to gel immediately upon heating and returns to liquid state again when cooled. The lyophilized TGP vial was obtained from Nichi In Biosciences (P) Ltd, Chennai, India. A buccal mucosal biopsy was taken and was transported to the laboratory in phosphate buffer saline (PBS) and TGP at 4°C. Buccal epithelial cells were separated after enzymatic digestion. The cells were cultured along with TGP for 10-12 days. After cell culture, wide urethrotomy was done dorsally from 9 o'clock to 3 o'clock position. The cultured

buccal epithelial cells suspended in TGP were instilled to cover the entire urethrotomy site.

**Results:** Per urethral catheter was removed after 3 weeks and all voided well with good stream. A check urethroscopy was done at 1 month. It revealed that stricture site was healthy, covered by pinkish epithelium and negotiated with 17 Fr cystourethroscope easily. A repeat check scope at 6 months revealed same findings. No patient required auxiliary procedure.

**Conclusions:** Our initial case series showed that novel cell based therapy using and BEES-HAUS is a promising alternative for the open substitution buccal graft urethroplasty. It is possible to get the benefits of open substitution buccal urethroplasty with this endoscopic technique without donor site morbidity.

#### MP-06.06

##### **Tunica Vaginalis as a Vascular Bed for Onlay Buccal Mucosal or Minipatch Skin Grafts for Salvage Reconstruction of Failed Hypospadias Repair: A Preliminary Report**

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**Introduction and Objectives:** To evaluate the reconstruction of a neourethra in crippled hypospadias in one stage; the graft was implanted on the vascular surface of appropriately dissected tunica vaginalis, then the graft is laid in an onlay fashion to complete the neourethra. To our knowledge this is the first report on the use tunica vaginalis as a vascular bed for onlay free grafts.

**MP-06.05, Table 1.**

| sno | Age (yrs) | Prior interventions | Graft size (cms) | Initial cell count | Final cell count (million) | Stricture length (cm) | Follow-up | Auxillary procedure |
|-----|-----------|---------------------|------------------|--------------------|----------------------------|-----------------------|-----------|---------------------|
| 1   | 38        | VIU (twice)         | 0.5x0.5          | 0.3                | 1.4                        | 2.5 cm                | 1 yr      | Nil                 |
| 2   | 28        | Nil                 | 2x1              | 1.56               | 11                         | 2 cm                  | 7 mo      | Nil                 |
| 3   | 70        | BMG urethroplasty   | 2.5x1.5          | 0.76               | 10                         | 2 cm                  | 6 mo      | Nil                 |
| 4   | 37        | VIU                 | 2x1.5            | 2.24               | 9.2                        | 2.5 cm                | 6 mo      | Nil                 |



**Materials and Methods:** Nine patients with cripple hypospadias were operated upon for urethral reconstruction. Patients' age was 5 - 8 (mean  $6.2 \pm 1.09$ ). All patients have 1 - 4 previously failed repairs. A buccal mucosa graft 2 - 4 cm long was used in 6 patients and mini-patch skin graft 1.5-2.5 cm long was used in 3 patients. All grafts were implanted in onlay fashion constituting the ventral wall of the neourethra. A vascularized tunica vaginalis flap was dissected and transposed to act as a vascular bed for the grafts. The grafts were fixed on the vascular surface of the tunica vaginalis flap using 6/0 polygalacten sutures avoiding potential dead space in between. Urethral catheter 8 - 10 French was left for 10 - 14 days. Patients were followed-up for 4 - 6 months.

**Results:** The successful repair was achieved in all cases. No fistulas, meatal stenosis or strictures were encountered along 4 - 6 months follow-up. Cosmetic results were good in 7 cases and acceptable in 2 due to preoperative massive scarring of penile skin and glans deformities.

**Conclusion:** The use of tunica vaginalis as a vascular bed for free grafts either buccal mucosa or small skin patch is feasible and promising provided that tunica has good vasculature. The technique facilitate implantation of the free graft as an onlay flap for the repair of crippled hypospadias in a single stage. It is necessary to evaluate the technique in a large number of patients.

#### MP-06.07

##### Short Term Result of Substitution Urethroplasty Using Free Ileal Mucosa in Patients with Long Stricture of Anterior Urethra

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*Dept. of Urology, Thanjavur Medical College, Thanjavur, India*

**Introduction and Objectives:** We evaluated the results of dorsal onlay ileal mucosal graft urethroplasty in long segment anterior urethral stricture.

**Materials and Methods:** Twenty five patients underwent ileal mucosal urethroplasty for long

segment anterior urethral stricture between December 2007 and June 2013. In all the cases the oral cavity was not suitable for mucosal harvest. A single long segment graft was used in all the patients from the proximal bulbar to the meatal end.

**Results:** Of the 25 patients with a mean follow-up of 41 months (range 3 to 66) 22 patients (88%) were having satisfactory post-operative uroflow with no other urethral manipulation and 3 (12%) patients had short recurrent stricture, which was treated by additional procedures. The mean maximal urine flow rate improved from 7.5 ml/sec to 18.6 ml/sec after the surgery. One patient developed sub acute intestinal obstruction which was managed conservatively.

**Conclusion:** In patients with long stricture of anterior urethra with unsuitable oral cavity for buccal mucosal harvesting, free ileal mucosa graft urethroplasty can be tried and this gives excellent short-term results.

#### MP-06.08

##### Improving Testicular Blood Flow with Electroacupuncture in an Experimental Rat Testicular Torsion Model

**Acar O<sup>1</sup>, Esen T<sup>1,2</sup>, Colakoglu B<sup>3</sup>, Camli M<sup>2</sup>, Cakmak Y<sup>4</sup>**

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**Introduction and Objectives:** In our previous work, we have demonstrated that testicular blood flow can be enhanced by electroacupuncture in healthy volunteers. In the present study, we applied the same technique in an experimental testicular torsion model to observe a potential improvement in testicular perfusion.

**Materials and Methods:** Twelve wistar albino male rats were randomly assigned into two groups (6 rats each in the electroacupuncture and manual acupuncture groups). Three dimensional color doppler ultrasonography was performed with a hockey-stick vascular probe

that has a frequency ranging between 6-15 MHz. Testicular perfusion was quantified by the serial volumetric analyses of 1 mm-thick cross-sectional images. Electroacupuncture was applied for 5 minutes at a frequency of 10 Hz via the needles that were inserted across the acupuncture points on the T12 dermatome and tibial nerve territory. In the manual acupuncture group, acupuncture needles were inserted in the same manner but electrical stimulation was not applied. We initially took baseline ultrasonographic recordings of both testicles. Then, the left testicle was twisted 180 degrees clockwise after which ultrasonography was repeated on both sides. Ipsilateral and contralateral ultrasonography were conducted after electroacupuncture and manual acupuncture, respectively in each group. Baseline, post-torsion and post-intervention images in both groups were compared in terms of the volume occupied by the vessels.

**Results:** In the study group, the difference between baseline and post-torsion testicular perfusion was statistically significant on both sides. After electroacupuncture, we observed a significant improvement in the ipsilateral and contralateral testicular blood flow. In the manual acupuncture group, blood flow diminished significantly on both sides after testicular torsion. However, testicular perfusion, neither ipsilateral nor contralateral, did not show a statistically significant change after manual acupuncture.

**Conclusion:** Electroacupuncture can improve testicular blood flow bilaterally in a rat model of incomplete unilateral testicular torsion.

#### MP-06.09

##### Hematuria in Patients Under 40 Years Old

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**Introduction and Objectives:** Hematuria is common in urological practice. Large studies describe the prevalence of disease, mostly involving cancer and urolithiasis in the elderly patient. Little research has been conducted on

**MP-06.08, Table 1.** Testicular Blood Flow Measurements and the Differences between the Mean Recorded Values

| Mean values                    | Vascularity in the left testis (cm <sup>3</sup> ) |              |                   | Vascularity in the right testis (cm <sup>3</sup> ) |              |                   |
|--------------------------------|---------------------------------------------------|--------------|-------------------|----------------------------------------------------|--------------|-------------------|
|                                | Baseline                                          | Post-torsion | Post-intervention | Baseline                                           | Post-torsion | Post-intervention |
| Electroacupuncture group (n=6) | 0.097±0.034                                       | 0.001±0.002  | 0.075±0.026       | 0.113±0.053                                        | 0.055±0.047  | 0.091±0.041       |
|                                |                                                   | P=0.001      | P=0.001           |                                                    | P=0.008      | P=0.007           |
| Manual acupuncture group (n=6) | 0.088±0.018                                       | 0.010±0.012  | 0.006±0.008       | 0.098±0.009                                        | 0.086±0.011  | 0.076±0.021       |
|                                |                                                   | P=0.0001     | P=0.314           |                                                    | P=0.012      | P=0.091           |

the prevalence of hematuria in patients under 40 years old. This study describes a large cohort of young patients to gain insight in the distribution and prevalence of urological diseases associated with hematuria.

**Materials and Methods:** All records of patients with hematuria in the period January 2003 to May 2013 were collected retrospectively. Patient characteristics, outcome of diagnostic studies and diagnoses were analyzed.

**Results:** A total of 642 patients with hematuria were included of which 94% were male. Median age was 44 years (range 17-65); with 40% (n=257) under 40. Urinary specimens were positive (at least 1-5 RBCs per HPF) in 43% (272/633). A cause for hematuria on ultrasound was found in 11% (43/400), KUB in 5.5% (18/328), CT-IVU in 19% (56/291) and urethroscopy in 17% (71/424). Overall, in 28% of patients a cause for hematuria was found. In 1.9% of patients under 40 years old a malignancy was detected. Urethral stricture disease (4.7%), trauma (4.7%) and Chlamydia trachomatis infection (3.1%) were also common causes for hematuria in patients under 40. In patients with gross hematuria a urological cause was found more often than in patients with microscopic hematuria (50.5% vs. 17.0%,  $p<0.0001$ ). In 9.2% of all patients a nephrological cause was suspected.

**Conclusion:** In almost 30% of patients presenting with hematuria, a treatable cause is found. As in older patients, hematuria is a symptom that should not be ignored and requires a urological work-up. In opposition to elderly patients, only 24% of urological diagnoses consist of urolithiasis and malignancies. Therefore, in patients under 40, attention

should be aimed at urethral strictures, congenital anomalies, trauma and STD's as well as nephrologic pathology.

#### MP-06.10

##### A Smart-Phone App for Actual (Not Relative) Length, Girth and Angle Measurements of the Erect Penis by Patients and Surgeons Garcia M

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**Introduction and Objectives:** It is often desirable to measure the actual length, width, or curvature of an object such as the erect penis. However, owing to the difficulty with achieving natural erection in a clinic setting, this is not feasible. Instead, we depend on photographic images captured by the patient. Photographic images are distorted by both lens angle and distance from the camera. A simple yet accurate means of measurement would facilitate pre-op counseling, and document pre-op length and angle, both for research and medico-legal purposes.

**Materials and Methods:** We describe an App that allows one to measure *actual length*, using either a live smartphone image or from an archived 2-D image. The App eliminates angle distortion by guiding the user to position the camera to yield a true anterior/posterior view. The User selects a reference object (RO) and places this beside the target. A novel RO may be used, or, one can be selected from a menu of objects of known diameter. A virtual caliper device is superimposed on a target and manipulated by the user, to measure the on-screen length of the target. The caliper is calibrated to the RO, and thereafter, the caliper measures the

length of any object in the same plane as the RO in real dimensions. A working prototype of the App was programmed and used to calculate the actual length, width, and curvature of various targets (whose actual length/curvature is known). Measurements were repeated 6X, by 6 non-surgeons. Measurements were also made from archived 2-D images (i.e. post-hoc), with equal precision and accuracy.

**Results:** Actual length and width were compared to measured length/area. Accuracy and precision were calculated. Provided that the target occupies at least 30% of the image field of view, accuracy was >98%, regardless of units or magnification. Precision was 95-99%.

**Conclusion:** Used on a smartphone, the described App serves important mobile-health needs: a means by which patients can record both body appearance and actual-size measurements in a natural setting. This App can be useful to guide pre-op planning and counseling, and to document pre-op measurements.

#### MP-06.11

##### Efficacy and Tolerability of Andro-Penis Extender Device in Patients with Penile Dysmorphism Nowroozi M, Ayati M, Radkhah K, Jamshidian H, Amini E

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**Introduction and Objectives:** A majority of men seeking penile enhancement techniques have a normal sized penis and usually suffer from penile dysmorphism. Although penile dimensions fall within the normal range in these patients they have the perception of inadequate penile size. Such patients should be discouraged from undergoing invasive procedures for penile enhancement which are associated with numerous complications and high dissatisfaction rate. However less invasive techniques including penile extenders are not associated with serious complications and may also be beneficial from a psychological perspective. We conducted this study to assess the efficacy and safety of a widely used penile extender, Andro-Penis (Andromedical, Madrid, Spain).

**Materials and Methods:** Between December 2010 and March 2012, 34 men complaining of small penis were considered for enrolment. Patients younger than 20 years of age, those with a major psychological disorder, penile curvature and reduced manual dexterity were excluded. Eleven men did not meet inclusion criteria and from 23 subjects who entered the study 20 completed the study. Eligible participants were instructed to use the penile extender for at least 4h/day for 3 months. Penile dimensions including flaccid and stretch lengths were measured at baseline and after 1, 2 and 3 months. Patients were also counseled how to measure penile length during erection.

MP-06.09, Table 1. Patients under 40 with hematuria

|                              | Age <40 yr<br>(n= 257) | Age ≥40 yr<br>(n=385) | Significance*<br>(p-value) |
|------------------------------|------------------------|-----------------------|----------------------------|
| <b>Sex</b>                   |                        |                       |                            |
| Male                         | 225 (87.5%)            | 381 (99.0%)           | <0.01                      |
| <b>Diagnosis</b>             | 70 (27.2%)             | 112 (29.1%)           | 0.66                       |
| Urolithiasis                 | 12 (4.7%)              | 40 (10.4%)            | 0.01                       |
| UTI                          | 15 (5.8%)              | 12 (3.1%)             | 0.11                       |
| Urethral sticture            | 12 (4.7%)              | 12 (3.1%)             | 0.40                       |
| Malignancy                   | 5 (1.9%)               | 31 (8.1%)             | <0.01                      |
| Trauma                       | 12 (4.7%)              | 0 (0.0%)              | <0.01                      |
| Congenital                   | 6 (2.3%)               | 4 (1.0%)              | 0.21                       |
| STD                          | 8 (3.1%)               | 0 (0.0%)              | <0.01                      |
| Other                        | 15 (5.8%)              | 12 (3.1%)             | 0.11                       |
| <b>Nephropathy suspected</b> | 28 (73.7%)             | 10 (26.3%)            | 0.16                       |

\*) Calculated with Fisher's exact test

**Results:** Flaccid and stretched penile length increased from  $88.5 \pm 12.3$  and  $119.3 \pm 16.6$  cm at baseline to  $95.8 \pm 12.9$  and  $143.4 \pm 10.6$  cm after 3 months. Moreover participants reported erected penis length of  $12.5 \pm 19.5$  and  $13.7 \pm 21.8$  cm at baseline and after 3 months respectively. Increase in penile length was statistically significant. Nevertheless no significant increase in penile girth was noted in this study. More than 50% of patients were satisfied and reported acceptable improvement.

**Conclusion:** Penile extenders as a minimally invasive treatment with modest efficacy should be considered for men seeking treatment for a short penis. This method is associated with few and minor complications and may also be beneficial from a psychological perspective.

#### MP-06.12

##### Current Trends in the Management of Penile Fractures: Results from a European Multicentric Study

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**Introduction and Objectives:** To review pre-operative diagnostic evaluation, surgical treatment and related outcomes of penile fracture.

**Materials and Methods:** A retrospective analysis of 137 patients from 7 different European Academic centres with penile fracture between 1996 and 2013 was performed. The parameters assessed were age, diagnostic tools including imaging, timing of surgical intervention, length of tunica albuginea defect, surgical technique, and postoperative care.

**Results:** The mean age of the patients was 38.96 (SD 13.55). All patients underwent routine clinical examination. Among them, 82/137 patients (59.85%) underwent penile Doppler ultrasound and 5 patients MRI. Mean time between ER admission and surgical intervention was 780.6 minutes (SD 31). There were 126/137 patients (91.97%) treated surgically

whereas 11/137 (8.03%) were managed conservatively. The mean length of tunica albuginea defect was 15.32 mm (SD 8.30). Mean IIEF-score was 15.09 (SD 7.8) and 16.85 (SD 8.96) after one and three months, respectively. If the surgical intervention was performed more than 8.23 hours after admission, erectile function significantly worsened in the follow-up period ( $p < 0.05$ ). A postoperative penile curvature was observed in 5/137 (3.64%) patients.

**Conclusions:** Penile fracture is a rare event but represents a urological emergency. Clinical examination and penile US Doppler should be performed in all patients. Delaying the surgical intervention for more than 8.23 hours from the presentation in ER resulted in significantly decreased IIEF scores in the follow-up period.

#### MP-06.13

##### Penile Grafting for Benign Causes: An Analysis of Outcome

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**Introduction and Objectives:** Common conditions such as hidradenitis suppurativa, balanitis xerotica obliterans, Peyronies' disease can cause impairment of sexual function.

Lack of penile skin can result in impairment of penile function. Grafting techniques have been developed to maintain penile length and function. We present results of our three year series focusing on patient outcomes for skin grafting in patients undergoing penile grafting for benign causes.

**Materials and Methods:** Over a three year period, 11 cases under a single surgeon undergoing penile reconstruction were reviewed. Patients were followed-up for a maximum of 36 months. Split thickness skin grafts were applied to glans/coronal defects. Full thickness skin grafts were used for shaft defects. We examine underlying pathology, comorbidities, graft take, sexual function, patient satisfaction and complications.

**Results:** One hundred percent were cosmetically satisfied both grafted. One hundred percent of grafts had good take with no requirement for further surgery. Grafting did not affect penile function. There was minimal loss of penile length. One hundred percent were satisfied with the outcome of the procedure. Complications included meatal stenosis in 50% of grafted patients. One patient required further scrotal debulking. There were no contractures.

**Conclusions:** For cases of benign pathology, penile grafting does result in patient satisfaction with good overall cosmesis and erectile function. Good outcomes depend on having an experienced specialist surgeon and team who are able to handle complications when they arise.

#### MP-06.14

##### Evaluation of Extra-Corporeal Shock Wave Therapy for the Patients of Peyronie's Disease: A Preliminary Report

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**Introduction and Objectives:** The non-invasive treatment options available for P.D are oral POTABA, Intra-lesional Steroids, Verapamil ointment, etc., if there is no gross penile deformity. Apart from surgical treatment, there is no specific treatment for the plaques. In this preliminary report, I tried to evaluate the use of ESWT for the painful erections and erectile dysfunction.

**Materials and Methods:** Twenty two men in the age group of 24-37 years (mean 28 years) treated between January 2009 and July 2011 are included in the present study. Seventeen patients had a single plaque while 5 patients had 2 or more plaques. Those patients who have tried conservative treatment, but were non-responders, are included in the present study. Evaluation consisted of routine and specific investigations such as Colour Penile Doppler and soft tissue X-ray of the penis. ESWT consists of 8 weekly treatment of 20 min duration at the intensity of 1. The results are evaluated at baseline, 8 weeks, 12 weeks and 24 weeks after the therapy. All the patients continued Verapamil ointment thereafter. For evaluation, I have employed the International Index of Erectile Function (IIEF5) questionnaire, VAS (Visual Analogue Scale). The treatment was performed on an out-patient basis and was well tolerated.

**Results:** Fourteen patients reported significant improvement of mean VAS score, mean IIEF5 score at 12 weeks and 24 weeks. The mean plaque size reduced by 1.2-1.8 cm but the curvature degree did not improve significantly. Seventeen patients reported that the pain had significantly reduced after the therapy.

**Conclusions:** ESWT can significantly improve the painful erections and to some extent, E.D. in the patients with Peyronie's Disease. The ESWT can also stabilize the plaque size and may reduce the progression of the disease and needs long-term follow-up.

#### MP-06.15

##### Does Early Insertion of a Malleable Penile Prosthesis Allow Later Upsizing of Cylinders in Patients Presenting with Ischaemic Priapism?

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**Introduction and Objectives:** Prolonged ischaemic priapism refractory to conventional medical and surgical intervention, results in

corpus cavernosum smooth muscle necrosis, followed by fibrosis manifesting as erectile dysfunction and penile shortening. Early insertion of a malleable penile prosthesis is easier to perform and also allows the option of an interval exchange to an inflatable penile prosthesis after a few months. The aim of this study was to assess whether upsizing of cylinders is still possible at the time of exchanging to an inflatable prosthesis.

**Materials and Methods:** Over a 24 month period 10 patients with refractory ischaemic priapism underwent an early (within 2 weeks) insertion of a malleable (Coloplast Genesis®) penile prosthesis. The mean age was 42.7 years (range 35-69) and the median duration of priapism was 7 days (range 5-10). The aetiology was sickle cell disease (2 patients), idiopathic (4 patients) and antipsychotic agents (4 patients). Following a median period of 5.7 months (range 3.8-11.5) all of these patients underwent exchange of their penile prosthesis to a 3 piece inflatable (AMS 700® or Titan Coloplast®) prosthesis and the size of the cylinders compared.

**Results:** During the reoperation, it was noticed a median upsize in the length of the cylinders of 1 cm in either one or both corporal bodies (range 0-2cm) as shown in Table 1. Fifty percent of the patients had deliberate undersizing at the initial operation due to a distal shunt. The mean IIEF-5 score preoperatively was 24 (range 23-25). After the initial insertion of the penile prosthesis the satisfaction rate according to the IIEF-5 was 80% compared to 90%, 3 months following the exchange of prosthesis.

**Conclusion:** Although some priapism patients have a shorter implant inserted initially due to previous shunt procedures, following a period of resolution the cylinders can be upsized at the elective exchange.

#### MP-06.16

##### Human Papilloma Virus Infection in Penile and Genitalia Tumors: Atypical Aspects

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**Introduction and Objectives:** One of the most interesting characteristics about the HPV infection is the polymorphism of the clinical appearance.

**Materials and Methods:** We prospectively started a study in 2002 to see the correlations between HPV infection and atypical clinical appearance at Urology and Dermatology Clinics. During 10 years we included in this analysis 128 patients with clinical condiloma acuminata of penile and genital areas.

**Results:** In 25 cases of the 128, we observed atypical clinical appearance photo documented: Atypical age: 4 cases (three children below 10 years old and one elder 84 years old); Atypical localization 4 cases: endo-urethral (one), external urethra meatus (two), sublingual (one); Atypical clinical subtypes found 17 cases: sub clinical types (4), Bowen papillomatosis (one) and together with invasive carcinoma (one), Eritroplasia Queyrat (3) and with simultaneous invasive carcinoma (one) pigmented condyloma acuminata (3), giant genital warts (2), Busche-Lowenstein condyloma acuminata (2). In one case we found a low risk infection that developed invasive carcinoma.

**Conclusion:** Male HPV infection takes a diversity of aspects regarding the localization, size, and evolution and transmission way of the lesion. This involves a careful attention in practice and a multidisciplinary approach of such cases. This requires a team management including urologists, dermatologists, pathologists and oncologists.

#### MP-06.18

##### The Effect of Acupuncture on Relieving Pain after Inguinal Surgeries

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**Introduction and Objectives:** Postoperative pain is one of the most prevalent and bothersome issues found in the surgical department. Nowadays, there are various methods of acupuncture used for relieving pain without the complications found in some routine postoperative analgesics. These methods could be especially useful for high risk patients prone to complications from analgesics, such as transplantation recipients. The aim of this study was to evaluate the efficacy of electro-acupuncture on postoperative pain control after inguinal surgeries.

**Materials and Methods:** Ninety male patients, who were referred to our department with indications of inguinal surgery, were included in the study and randomly divided into two groups, such as acupuncture and control. We used electroacupuncture for the acupuncture group and no actual acupuncture (but placed needle electrodes similar to the acupuncture group) for the control group. Postoperative pain was quantified by a blind observer in both groups using a visual analogue scale (VAS) standard score before being compared.

**Results:** Pain intensity and analgesic use were significantly higher in the control group ( $P < 0.05$ ). In the acupuncture group, the VAS pain scores were significantly lower than the control group at 0.5, 1 and 2 hours post operation. When the opioid related side effects were compared for each group, the results showed that the number of subjects who experienced dizziness in the acupuncture group was significantly lower than the control group ( $P < 0.05$ ).

**Conclusions:** Acupuncture in patients, after inguinal surgery, can reduce the need of analgesics, which also directly reduces the complications that may occur when analgesics are used in relieving pain postoperatively.

#### MP-06.19

##### Penis Rupture: A Topic for Interdisciplinary Consideration

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**Introduction and Objectives:** The purpose of the interdisciplinary cooperation between urological surgery and physics is the development of a physical simulation tool to be used by surgeons in order to give prognosis of possible penis rupture at a certain degree of deviation of the penis and to take prophylactic action.

**Materials and Methods:** For the physicist it

MP-06.15, Table 1.

| Patients | Right corporal cylinder 1 <sup>st</sup> operation | Right corporal cylinder at exchange | Left corporal cylinder 1 <sup>st</sup> operation | Left corporal cylinder at exchange |
|----------|---------------------------------------------------|-------------------------------------|--------------------------------------------------|------------------------------------|
| 1        | 19                                                | 19                                  | 18                                               | 19                                 |
| 2        | 21                                                | 22                                  | 20                                               | 22                                 |
| 3        | 21                                                | 23                                  | 21                                               | 23                                 |
| 4        | 22                                                | 23                                  | 22                                               | 23                                 |
| 5        | 21                                                | 22                                  | 21                                               | 22                                 |
| 6        | 18                                                | 21                                  | 19                                               | 21                                 |
| 7        | 22                                                | 23                                  | 22                                               | 23                                 |
| 8        | 23                                                | 23                                  | 22                                               | 23                                 |
| 9        | 20                                                | 22                                  | 20                                               | 22                                 |
| 10       | 21                                                | 22                                  | 21                                               | 22                                 |
| Median   | 21                                                | 22                                  | 21                                               | 22                                 |

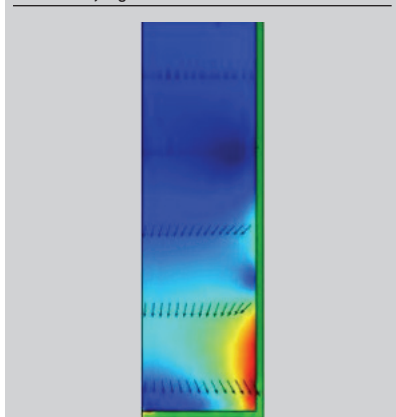


was the first challenge to translate the human organ of the penis into a physical model. Starting and marginal parameters had to be defined, whereby some of them had to be proceeded on the assumption, as physical data of the human living tissue have rarely been measured up to now, such as, f.i. the modulus of elasticity of the tunica, the mass of the penis in erect state and the maximum stress, the tunica can be subdued to. The algorithm and its dependencies had to be developed.

MP-06.19, Figure 1.

$$\varphi = \frac{3}{6b + \frac{Fl}{k}} \left[ a + \sqrt{a^2 + \frac{2}{3} \left( \frac{Fl}{k} - 1 \right) \left( 6b - \frac{Fl}{l} \right)^2} \right]$$

MP-06.19, Figure 2.



**Results:** This paper is a first step of mathematical-physical simulation with the assumption of a 100% filled rigid penis. The calculation (see Figure 1) gives proof of the hypothesis that the fibre-load-angle of the penis is 12 degree (see Figure 2), much less than 30 degrees, which was the assessment of the authorities of urology up to now.

**Conclusion:** Physical simulation is able to provide the surgeon with a simple instrument to calculate and forecast the risk of the individual patient, based upon dependencies of geometry of the differential geometrical body of the penis.

#### MP-06.20

##### Endourologic Management of Urologic Hydatid Cysts

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**Introduction and Objectives:** We present our experience with an endoscopic sterilization, evacuation, and cavity sclerosis of urologic hydatid cysts, via a transurethral transvesical approach in a retrovesical hydatid cyst (RVHC), and a percutaneous endoscopic indirect approach in renal hydatid cysts (RHC).

**Materials and Methods:** Since January 2007, 6 patients (4 men and 2 women) presented with urologic hydatid cyst (5 renal and 1 retrovesical). The mean age was 39 years (13 to 68). The mean cyst diameter was 15 cm. 2 cyst had a fistula with the pelvi-caliceal system. They

received 800 mg daily of albendazole-therapy 3 months preoperatively and postoperatively. Operative technique: for RVHC, cystoscopy is performed using a 20.8 Fr nephroscope. For RHC, the patients are placed in the split-leg lateral modified position. A percutaneous tract, with a 24 Fr Amplatz sheath, is performed through a calyx opposed to the cyst. Endoscopy locates the cyst bulge, which is punctured with a 36-cm-18-gauge needle through the nephroscope. A 20 % saline solution is used as a sclerocidal agent. After balloon dilation, the nephroscope is introduced into the cyst, and the hydatid material is aspirated. For RHC a drain is inserted through another direct percutaneous access. A nephrostomy tube is inserted in the renal pelvis. A 14 Fr and 18 Fr Foley catheters are respectively inserted in RVHC and the bladder. Postoperatively, the cystic cavity was treated by instillation of povidone-iodine.

**Results:** The endoscopic approach and treatment of urologic hydatid cysts was possible with a mean operative time of 135 min. The mean postoperative hospitalization was 6 days. Cystoscopy confirmed a healing of the bladder-wall. Nephrostography documented the sealing of the communication with the cystic cavity. After a mean follow-up of 41 months (5 to 69) with ultrasound and/or CT scan, the patients are free of symptoms with a retracted calcified residual cavity.

**Conclusion:** Endoscopic approach was effective for the treatment of urologic hydatid cysts with lower morbidity than open surgery. We think that there is less risk of dissemination of hydatid material, but only a larger series will tell.

Moderated Poster Session 7  
Prostate Cancer:  
Localized Disease  
Monday, October 13  
1435-1600

**MP-07.01**

**Direct Comparison of the Ability to Predict Final Gleason Score between Magnetic Resonance Imaging Based Targeted Prostate Biopsy and Systematic 14-Core Biopsy**

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**Introduction and Objectives:** The aim of this study is to directly compare the ability to predict radical prostatectomy (RP) Gleason score (GS) using MRI-based targeted prostate biopsy (MRBx) compared to systematic 14-core biopsy (S14Bx).

**Materials and Methods:** A retrospective analysis was performed using data from 53 Japanese patients who were diagnosed using MRBx and S14Bx and then underwent RP from 2011 to 2013. The biopsy protocol was a combination of S14Bx and MRBx. S14Bx includes anterior sampling and MRBx protocol includes 4-core targeted biopsies per one suspicious area. We excluded 5 patients who had neo-adjuvant treatment and 6 patients who had no cancer using S14Bx or MRBx, and the remaining 42 patients were analyzed. We used the highest GS. Biopsy GS was evaluated separately using S14Bx and MRBx. In RP specimens, GS was

evaluated in each cancer focus. GS was subdivided into three groups (6, 7 and 8-10). The ability to predict RP GS was directly compared between S14Bx and MRBx.

**Results:** The median age and PSA were 66 years and 7.5 ng/ml. Table 1 shows the biopsy GS from S14Bx and MRBx. Table 2 shows downgrading, concordance, and upgrading rates between biopsy GS and RP GS. The concordance rate was not significantly different between S14Bx and MRBx (60% vs. 74%). S14Bx tended to underestimate RP GS compared to MRBx with borderline significance ( $p = 0.055$ ). **Conclusion:** MRBx required two-thirds fewer sampling cores, but had the same ability to predict final GS and reduce grade underestimation compared to S14Bx.

**MP-07.02**

**Deferred Active Treatment in Active Surveillance Patients with Prostate Cancer: Triggers for Intervention**

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**Introduction and Objectives:** Active surveillance (AS) has emerged as a strategy to avoid or postpone active treatment (radical prostatectomy, radiation therapies) in early, low/intermediate-risk prostate cancer (PCa) without missing the opportunity for cure. Triggers for considering active treatment vary. We assessed the outcome of men with prostate cancer on AS and identify the triggers for active treatment in the cohort studied.

**Materials and Methods:** We retrospectively reviewed 189 patients on AS between

2006-2012. Data were collected from electronic systems. Patients were followed at 3-monthly intervals with clinical questionnaire, PSA, and were recommended to switch to deferred active treatment based on the triggers below.

**Results:** A total of 189 men with PCa from three different Trusts were followed with a mean of 22.6 months (5-60). Average age was 64.1 years (41-92). In total, 152 (80.4%) were low risk disease, defined by D'Amico criteria, 37 (19.6%) were intermediate risk. PSA was checked at a rate of 3.29 (1-4) times per year, 27 (14.2%) men had their biopsy repeated at a mean interval of 20.7 months (12-60) and 7 (3.7%) men had MRI at 18 month intervals according to their Trusts protocols. Of those who had re-biopsy, 3 (1.5%) had their biopsy upgraded and 5 (2.6%) had their biopsy downgraded. 30 (15.8%) had their management changed to active intervention, because of PSA rise, biopsy upgrade, patients' choice, and MRI upstaging (43.3%, 23.3%, 23.4%, 10%). Cancer specific survival and overall survival at this short follow-up was 100%.

**Conclusion:** At a short follow-up, AS is a safe strategy in low-risk and selected intermediate risk PCa, reducing overtreatment. PSA rise is the most common parameter utilised for commencing active treatment.

**MP-07.03**

**Outcomes in Prostate Cancer Diagnosed by Transurethral Resection of the Prostate: A Database Analysis in England**

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**Introduction and Objectives:** Incidental prostate cancer (IPC) is historically believed to be indolent, but some studies show mortality of >20% after 10 years. We evaluated prostate-cancer and all-cause mortality in IPC, and compared this to localised non-incidental prostate cancer (NIPC).

**Materials and Methods:** Men diagnosed with prostate cancer between 2000 and 2008 were identified from the national cancer registry data in England, and were linked to Hospital Episode Statistics (HES) data, which contains information on procedures undertaken within England. Men with Stage III/IV disease and those who died within 6 months of diagnosis were excluded from the study population. IPC was defined as all men who had a transurethral resection of the prostate (TURP) for benign

**MP-07.01, Table 1. Biopsy Gleason Scores in Systematic 14-Core Prostate Biopsy and Magnetic Resonance Imaging-Based Targeted Prostate Biopsy**

|      |         | S14Bx    |          |         |          |
|------|---------|----------|----------|---------|----------|
|      |         | GS 6     | GS 7     | GS 8-10 | Total    |
| MRBx | GS 6    | 3        | 2        | 0       | 5 (12%)  |
|      | GS 7    | 8        | 18       | 1       | 27 (64%) |
|      | GS 8-10 | 0        | 7        | 3       | 10 (24%) |
|      | Total   | 11 (26%) | 27 (64%) | 4 (10%) | 42       |

GS, Gleason score; S14Bx, systematic 14-core prostate biopsy; MRBx, magnetic resonance imaging based targeted prostate biopsy

**MP-07.01, Table 2. Downgrading, Concordance, and Upgrading Rates between Biopsy Gleason Score and Radical Prostatectomy Gleason Score**

|             | Systematic 14-core Bx | MRI targeted Bx | P     |
|-------------|-----------------------|-----------------|-------|
| Downgrading | 2 (5%)                | 5 (12%)         | 0.055 |
| Concordance | 25 (60%)              | 31 (74%)        |       |
| Upgrading   | 15 (36%)              | 6 (14%)         |       |

MRI, magnetic resonance imaging; Bx, prostate biopsy

**MP-07.03**, Table 1. Results from the multivariable analysis (PC=type of prostate cancer, RR=rate ratio, CI=confidence interval)

| PC   | All-cause mortality RR (95%CI, P-value) | Prostate-cancer mortality RR (95% CI, P-value) |
|------|-----------------------------------------|------------------------------------------------|
| NIPC | 1.0                                     | 1.0                                            |
| IPC  | 1.01 (0.96-1.05, P=0.789)               | 0.70 (95%CI 0.65-0.75, p<0.001)                |

disease, 60 days before or 14 days after the date of diagnosis of prostate cancer. The rest of the men were classified as NIPC. Kaplan-Meier survival probabilities were calculated. Mortality rate ratios (RR) comparing IPC and NIPC were calculated using Poisson multivariable regression, adjusting for age-group, co-morbidities, year-of-diagnosis, and radical treatment received within 1 year of diagnosis.

**Results:** A total of 192,960 men were included: 6767 (3.5%) had IPC, 186 193 (96.5%) NIPC. Median follow-up was 4.7 years (0.5-11.0). Men with IPC tended to be older, with less co-morbidities (p-value<0.001). Prostate-cancer cumulative mortality (at 10 years) in IPC was 17% and 19% for NIPC (logrank p-value<0.001). After multivariable adjustment, prostate-cancer mortality RR was 0.70 (95%CI 0.65-0.75, p-value<0.001) (Table 1).

**Conclusions:** A significant proportion (17%) of men with IPC will die from prostate cancer after 10 years. Nevertheless, IPC has 30% reduced rate of prostate-cancer mortality compared to NIPC. Better risk stratification is required in IPC.

#### MP-07.04

##### Watchfull Waiting (WW) for Incidental Carcinoma of the Prostate (CAP): Long-Term Results

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**Introduction and Objectives:** The treatment of incidental carcinoma of the prostate is still controversial. The aim of this retrospective study was to evaluate the long-term survival in patients with incidental CAP diagnosed after TURP or open prostatectomy and followed with observation only.

**Materials and Methods:** The records of 93 patients with incidental CAP diagnosed from 1976 to 1984 were reviewed. Mean patients' age was 68.2 years. All patients included presented at our clinic because of lower urinary tract symptoms. Fifty six out of 93 patients (60.2%) were diagnosed to have CAP after open prostatectomy while 37/93 (38.8%) after TURP. The specimens were reviewed by a senior pathologist and re-staged according to TNM. Fifty two patients had T1a while 41 had T1b CAP. According to Mostofi grading system, 48 patients had G1 tumours, 34 had G2, 5 G3 and 6 Gx. After surgery patients did not receive any additional treatment and were

followed according to WW protocol. Minimum follow-up was 10 years. Seventy five patients had 15 and 10 patients had 20 years follow-up respectively.

**Results:** Ten and 15 years overall survival (OS) was 76% and 50.1% while the disease specific survival (DSS) was 86% for T1a and 85.7% for T1b. The 10 years DSS for G1-G2 tumours was 86% compared to 37.5% for G3. Fourteen out of 93 (15%) patients progressed: 4 patients had local and 10 systemic progression. Mean time to progression was 7.5 years. Twelve patients died because of the disease and 2 are still alive with metastases. All G3 tumours progressed. In G2 and G1s progression occurred in 5/34 (14.7%) and 4/48 (8.3%) respectively.

**Conclusions:** WW strategy can be proposed to patients with incidental CAP, especially those with low and intermediate grade (G1-G2). Incidental, High grade CAP should be treated more aggressively.

#### MP-07.05

##### What Is the Intensity and Cost of Managing Patients on Active Surveillance for Prostate Cancer?

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**Introduction and Objectives:** Growing evidence supports active surveillance as a mean to reduce over-treatment of prostate cancer. The strategy may per se result in "over-treatment" or "over-management" owing to close and long-standing follow-up. We investigated the burden of active surveillance in patients with localized prostate cancer.

**Materials and Methods:** The study comprises 317 patients with low or intermediate-risk prostate cancer followed on a prospective, single arm active surveillance cohort. The primary outcomes were number of patient contacts, PSA tests, biopsies, in-hospital admission owing to biopsy complications treatment for BOO, and patients eventually undergoing curative treatment. The secondary outcome was cost. The cumulative incidence of discontinued AS was analysed using a competing risk model with both amendment to watchful waiting and death as competing events. The cost of active surveillance was compared to the hypothetical cost of curative RP for all patients using a

Markov model.

**Results:** The 5-year cumulative incidence of discontinued active surveillance in a competing risk model was 39.5%. During the first 5 years of active surveillance patients underwent a median of 2 sets of biopsies, and the number of annual patient contacts including PSA tests was 3-4. Thirty eight of the 406 biopsy sessions led to hospital admission. Eighty seven out of 317 patients required treatment for BOO. The total cost of active surveillance following median 3.7 years follow-up was DKK 51.836 per patient. Assuming all patients had undergone primary radical prostatectomy the cost difference after a median of 3.7 years favours active surveillance with a net benefit of DKK 28.586 per patient (35.5% reduction).

**Conclusion:** The concept active surveillance entails a significant burden on patients, materialized in terms of close clinical follow-up, re-biopsies, and a substantial risk of ending up undergoing delayed definitive therapy. Still, active surveillance seems feasible as an initial strategy for low risk- and selected intermediate risk prostate cancer patients.

#### MP-07.07

##### Clinic Audio-Recording Reduces Decision Regret in Men Undergoing Treatment for Prostate Cancer: A Comparative Cohort Study

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**Introduction and Objectives:** Patients undergoing treatment for localised prostate cancer have a variety of treatment options to choose from, all with varying side effects. The life expectancy of these patients is long and as such the patients will spend many years carrying the burdens and benefits of the decisions they have made. Therefore it is important that decisions on treatments are shared between patient and physician.

**Materials and Methods:** In 2012, we initiated consultation audio-recordings where patients are given a CD of their consultation to keep and replay at home. We conducted a prospective study of patient satisfaction, quality of life (QOL) and decision regret at 12 months follow-up using posted validated questionnaires for the audio-recording (AR) patients and a control cohort. Mann Whitney U tests, Chi squared tests, Odds ratio, Numbers needed to treat and binary logistic regression analysis were all used and performed using SPSS version 21.

**Results:** Thirty-nine of 59 patients in the AR group, and 27 of 45 patients in the control group return the questionnaires. Patient demographics were similar in both groups, age, PSA,

D'Amico risk status, treatment received, employment status, ethnicity were all similar, only educational status (more patients left school after 16 yo in the audio-recording group) was statistically different ( $p = 0.04$ ). Decision regret was lower in the audio-recording group (11/100) vs. control group (19/100) ( $p = 0.05$ ). The odds ratio for having no regret was 0.216 (CI 0.07 – 0.67), with NNT to prevent regret being 4. Multivariate regression analysis showed that receiving audio-recording was strongest predictor for absence of regret over potency and incontinence). Thematic analysis of free text comments revealed overwhelmingly positive replies (77 positive vs. 5 negative). The 3 most common reasons for satisfaction was increased information recall, increased confidence in their decision and an ability to share information with their family and friends.

**Conclusion:** The study has shown that audio-recording clinic consultation reduces long-term decision regret, increases patient information recall, understanding and confidence in their decision. The NNT is low at 4 to prevent one patient having decision regret. There is great potential for further expansion of this low-cost intervention.

#### MP-07.08

##### **Results of Surgical Treatment of Patients with Clinically Insignificant Prostate Cancer Classified according to Epstein Criteria**

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**Introduction and Objectives:** Methods of treatment of patients (pts) with clinically insignificant prostate cancer (CIPC) are on debate. Several studies have demonstrated that established Epstein criteria could be inaccurate in prediction of morphological stage and lead to underestimation of the disease. The aim of the study was to assess morphological results and outcome of surgical treatment of pts with CIPC.

**Materials and Methods:** Retrospective analysis of 1430 pts after radical prostatectomy (RPE) and pelvic lymph node dissection (PLND) since 1998 till 2014 was done. CIPC according to Epstein criteria ( $\leq 2$  positive biopsies;  $< 50\%$  of tumor tissue in biopsy specimen; absence of Gleason patterns 4 and 5; PSA  $\leq 10$  ng/ml and clinical stage  $\leq T2a$ ) was verified in 185 (12.9%) pts. Mean age was  $62.6 \pm 6.1$  years; mean PSA level – 6.85 ng/ml; mean percentage of positive biopsy cores (PPBC) – 11.6%; mean prostate volume –  $38.6 \pm 20.8$  cm<sup>3</sup>; clinical stage T1a-T1c was identified in 124 (67%) patients T2a – in 61 (33%) patients.

**Results:** Pathological stage pT0 was verified in 4 (2.2%) pts; pT2a – in 25 (13.5%); pT2b – in 8 (4.3%); pT2c – in 129 (69.7%); pT3a-T4

– in 19 (10.3%). Lymph node metastases (pN+) were identified in 2 (1.1%) pts. Thus, bilateral disease was found in 80% of pts, stage migration from localized to locally advanced or pN+ was observed in 21 (11.4%) pts. Morphological Gleason score migration with presence of 4 or 5 patterns was found in 26 (14.1%) pts. Median follow-up time was  $33.5 \pm 23.9$  (3-115) months. Biochemical recurrences occurred in 14 (7.6%) pts, death – in 5 (2.7%) cases. 5-year biochemical progression-free survival (BPFS) was  $83.5 \pm 5.7\%$ . In Cox regression model only PPBC significantly correlated with BPFS ( $\beta = -0.06$ ;  $p = 0.04$ ). 5-year overall and cancer-specific survival were  $96.4 \pm 3.5\%$  and  $97.5 \pm 2.5\%$ , respectively.

**Conclusions:** Epstein criteria are not accurate in prediction of unilateral disease in prostate cancer pts. In pts with CIPC morphological examination revealed locally-advanced and poor differentiated tumors were observed in 11.4% and 14.1% respectively. Surgical treatment of CIPC could be associated with satisfactory oncologic outcome.

#### MP-07.09

##### **Comprehensive Evaluation of Learning Curve for Minimally Invasive Radical Prostatectomy**

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**Introduction and Objectives:** To estimate the learning curve of minimally invasive radical prostatectomy (MIRP) in terms of biochemical recurrence (BCR) and positive surgical margin (PSM) from the start of our surgical prostatectomy program in 1998.

**Materials and Methods:** Evaluation of a prospectively collected MIRP database for the estimation of PSM and BCR trends during a 15-year period. Prostatectomies were performed by 9 surgeons. Positive margin was defined as cancer cells at inked margins. Clinical, pathologic, and outcome data were collected for men who underwent pure laparoscopic (LRP) and robotic assisted radical prostatectomy (RARP) between 1998 and 2013. BCR was defined as serum PSA  $> 0.2$  ng/ml and rising or start of secondary therapy. Surgical learning curve was assessed with the application of Kaplan-Meier curves, Cox regression model, CUSUM and logistic model in order to define the “tipping point” of surgical improvement.

**Results:** We evaluated 5547 patients with localized prostate cancer treated with MIRP; 3846 underwent LRP and 1701 RARP. The overall risk of PSM of LRP was 25%, 20% and 17% for the first 50, 50 to 350 and  $> 350$  cases, respectively. For the same population,

the 5-year BCR rate decreased from 21.5% to 16.7% before and after index case 350. After developing the technique of LRP, the overall team experience became more reliable after approximately 800 cases. RALP started 3 years after the LRP program (after approximately 250 LRPs). The PSM rate decreased from 21.8% to 20.4% after case 100. The corresponding 5-year BCR rate decreased from 17.6% to 7.9% after case 100. In multivariable analysis, predictors of BCR included preoperative PSA, gleason score, extraprostatic disease, seminal vesicle invasion and number of operations ( $p < .0001$ ). Patients harboring PSM showed higher BCR risk (23% vs. 8%).

**Conclusion:** Our experience confirms the link between surgical experience and oncologic outcomes in radical prostatectomy. We found a lower BCR risk at two years after 350 cases and 100 cases for LRP and RARP, respectively.

#### MP-07.10

##### **Adverse Oncologic Outcomes after Robotic Assisted Radical Prostatectomy (RARP) in French Patients with Low Risk Prostate Cancer Candidates for Active Surveillance (AS)**

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**Introduction and Objectives:** Encouraging oncologic results of low risk prostate cancer (PCa) series have been reported in US series; however, this has not been thoroughly evaluated in European centers. The aim of this study is to evaluate oncologic outcomes in French patients with low risk prostate cancer who were candidates for AS but elected to undergo RARP. We additionally estimated predictors of pathologic upgrading/upstaging and variable associated with biochemical failure.

**Materials and Methods:** We retrospectively analyzed a prospectively collected database of 1552 consecutive patients treated with RALP at Institut Montsouris since 1998. We studied 405 men who had low risk PCa as defined by MSKCC criteria (PSA  $\leq 10$  ng/ml, clinical stage  $\leq 2a$ , biopsy Gleason sum  $\leq 6$ , number of positive cores  $\leq 2$  and maximum percent core involvement  $\leq 50$ ). Patient characteristics, pathologic data and biochemical recurrence were compared between upgraded/upstaged and non upgraded/upstaged patients. Multivariable modeling was used to estimate predictors of pathologic upgrade/upstage and variables associated with biochemical recurrence.

**Results:** We observed pathologic upgrading at RALP of 48% and pathologic upstaging of



13%. Overall PSM rates were reported in 66 (16%), 16.8% pT2 and 52.6% pT3 ( $p=0.001$ ). Maximum percent core involvement was the only variable significantly predicting pathologic upgrade/upstage in multivariate analyses. The median follow-up was 28 months, and b-DFS was 92% at 5 years.

**Conclusions:** Almost half of French patients in our series with low risk prostate cancer who meet criteria for AS treated with RARP experience upgrading/upstaging. Pathologic upgrade/upstage significantly impacts on BCR free survival. French patients with low risk prostate cancer should be counseled about increased oncologic risk when deciding for AS as a treatment option.

#### MP-07.11

**Perioperative, Pathological and Functional Outcomes of Robotic Radical Prostatectomy: 500 Consecutive Cases with a Minimum 12-Month Follow-Up**  
D'Elia G, Emiliozzi P, Iannello A, Cardi A  
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**Introduction and Objectives:** This study examines perioperative, pathological and functional outcomes as well as complications of robotic radical prostatectomy at a large community-setting center with a quality assurance program.

**Materials and Methods:** Perioperative data and functional and pathological results of 500 consecutive patients who underwent robotic radical prostatectomy were prospectively collected. Perioperative outcome measures included: operative time, estimated blood loss, transfusion rate, complication rate according to modified Clavien system, median hospital stay, mean catheterization time. Pathologic outcome measures encompassed positive surgical margin rate and biochemical recurrence free survival (PSA < 0.2). Return of continence was evaluated at 1, 3, 6 and 12 months (continent 0 pads; incontinent 1 or more pads). Return of potency was evaluated at 1, 3, 6 and 12 months with IIEF-5 scores in 402 patients who underwent a nerve-sparing procedure (mean age 61 years; range 36-70).

**Results:** Mean age was 64.1 years (36-73). Mean body mass index (BMI) was 26.6. Median preoperative PSA level was 6.9 ng/ml. Mean operative time was 146 minutes. Mean estimated blood loss was 160 cc. Blood transfusion was needed in 8 patients. Median hospital stay was 3 days, mean catheterization time was 8.1 days. According to the modified Clavien system, grade III complication rate was 2.6%, whereas minor complication rate was 17%. Positive surgical margin rate was 12.8% for pT2 disease and 29% for pT3 disease. Overall biochemical recurrence free survival is 95%. Complete continence at 1, 3, 6, and 12 months was 57%, 88%, 94% and 98%,

respectively. Mean age of the 402 patients who underwent a nerve-sparing procedure was 61 years (range 36-69). At 1, 3, 6 and 12 months return of potency (IIEF-5 > 21) with or without the use of oral medications was achieved in 6%, 22%, 51% and 68%, respectively.

**Conclusion:** Robotic radical prostatectomy has a low perioperative complication rate and acceptable outcomes in terms of positive surgical margins and maintenance of urinary continence and erectile function.

#### MP-07.12

**Perioperative, Pathological and Functional Outcomes in Robotic Radical Prostatectomy Patients with Prostate Weight More Than 100 Gr**  
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**Introduction and Objectives:** This study examines perioperative, pathological and functional outcomes as well as complications of robotic radical prostatectomy patients with prostate weight more than 100 gr.

**Materials and Methods:** Out of 500 consecutive robotic radical prostatectomy patients, 40 had a prostate specimen weight more than 100 gr. Perioperative data and functional and pathological results were prospectively collected. Perioperative outcome measures included: operative time, estimated blood loss, transfusion rate, complication rate according to modified Clavien system, median hospital stay, mean catheterization time. Pathologic outcome measures encompassed positive surgical margin rate and biochemical recurrence free survival (PSA < 0.2). Return of continence was evaluated at 1, 3, 6 and 12 months (continent 0 pads; incontinent 1 or more pads). Return of potency was evaluated at 1, 3, 6 and 12 months with IIEF-5 scores in 32 out of 40 patients who underwent a nerve-sparing procedure.

**Results:** Mean age was 67.1 years (58-71). Mean body mass index (BMI) was 28.6. Median preoperative PSA level was 9.9 ng/ml. Median prostate weight was 118 gr (100-188). Mean operative time was 160 minutes (110-220 min). Mean estimated blood loss was 200 cc. Blood transfusion was needed in 1 patient. Median hospital stay was 4 days, mean catheterization time was 9.4 days. According to the modified Clavien system, grade III complication rate was 2.5% (rectal injury), whereas minor complication rate was 20%. Positive surgical margin rate was 13.3% for pT2 disease (4 of 30 pts) and 30% for pT3 disease (3 of 10 pts). Overall biochemical recurrence free survival is 87.5% at mean follow-up of 16.9 months. Complete continence at 1, 3, 6, and 12 months was 48%, 80%, 88% and 92%, respectively. Mean age of the 32 patients who underwent a nerve-sparing procedure was 64 years (range 58-70). At 1, 3,

6 and 12 months return of potency (IIEF-5 > 21) with or without the use of oral medications was achieved in 3.1%, 9.3%, 31% and 58.9%, respectively.

**Conclusion:** Robotic radical prostatectomy in prostate specimen weight more than 100 gr has a low perioperative complication rate and acceptable pathological and functional outcomes.

#### MP-07.13

**Dorsal Vein Complex Ligation-Free Technique for Laparoscopic Radical Prostatectomy: A Single-Center Experience of Chinese**

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**Introduction and Objectives:** To describe our experience in control bleeding from the dorsal vein complex (DVC) during laparoscopic radical prostatectomy (LRP).

**Materials and Methods:** This retrospective study utilized prospectively collected data from December 2009 to June 2013 for 138 patients with localized prostate cancer who underwent LRP. Data of 90 patients in whom DVC bleeding was controlled without ligation (Group 1) were retrospectively compared with those of 48 consecutive patients in whom DVC bleeding was controlled with suture ligation (Group 2). Surgical, oncological, and functional outcomes were considered, with special emphasis on estimated blood loss (EBL) and postoperative continence.

**Results:** Operative time was significantly shorter in Group 1 than in Group 2 (153.7 vs. 173.8 min;  $p = 0.005$ ). No significant difference in EBL was noted between two groups (213 vs. 207.6 mL). Ratio of positive apical surgical margin was almost the same (2.2% vs. 2.1%). Postoperative continence rates at 1, 3 and 6 months in Groups 1 and 2 were 63.3% vs. 43.9%, 81.1% vs. 63.4%, and 88.8% vs. 73.1% respectively, these differences were statistically significant. But at one year after operation, the continence rates were not much difference between two groups. During follow-up, tumor recurrence or metastasis were not observed in any patient.

**Conclusions:** Ligation-free is a safe and convenient technique to control DVC bleeding, the prostatic apex can be clearly exposed, while preserving the surrounding anatomy. Operative time was decreased and the level of blood loss was acceptable. Use of this technique may enhance functional outcomes in the early stage, but its long-term therapeutic effects still need large series with longer follow-up studies to be fully evaluated.

**MP-07.14****Effects of Tadalafil Treatment Post Bilateral Nerve-Sparing Radical Prostatectomy: Quality of Life, Psychosocial Outcomes and Treatment Satisfaction**

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**Introduction and Objectives:** We report secondary outcomes on quality of life (QoL) and treatment satisfaction data from a multicenter, randomized, double-blind, double-dummy, placebo-controlled trial (NCT01026818) primarily evaluating the efficacy of tadalafil once-daily (OaD) or on-demand (*pro re nata*, PRN) treatment started early after nerve-sparing radical prostatectomy (nsRP).

**Materials and Methods:** Patients ≤68 years with adenocarcinoma of the prostate (Gleason≤7, normal preoperative erectile function [EF]) were randomized post-nsRP 1:1:1 to 9-month treatment with tadalafil 5mg OaD, tadalafil 20mg PRN, or placebo, followed by 6-week drug-free washout and 3month open-label tadalafil OaD treatment. Secondary outcomes reported include changes in patients' and partners' Expanded Prostate Cancer Index Composite (EPIC26) and Erectile Dysfunction Inventory of Treatment Satisfaction (EDITS), and Self-Esteem and Relationship (SEAR) questionnaires (pairwise comparisons, MMRM, adjusting for treatment, visit, treatment-by-visit interaction, age group, country, baseline score). LS means (95%CI) are reported.

**Results:** A total of 423 patients were randomized to tadalafil OaD (N=139), PRN (N=143), or placebo (N=141); 57 (41.0%), 58 (40.6%), and 50 (35.5%) aged 61-68 years. At the end of double-blind treatment (EDT), patients' EPIC sexual domain scores improved significantly with tadalafil OaD versus placebo (Table 1: Treatment effect [95%CI]: 9.6 [3.1,16.0]; p=0.004); comparisons of PRN versus placebo at EDT, and both comparisons versus placebo after open-label OAD treatment (EOL) were not significant. Only in older patients (61-68 years), EPIC urinary incontinence domain scores also improved significantly with tadalafil OaD versus placebo (overall treatment effect across all visits 8.3 [0.4,16.1]; p=0.040). Treatment satisfaction increased significantly in both tadalafil groups at EDT, EDITS total scores increased significantly with OaD and PRN versus placebo (p=0.005 and 0.041). At EOL, improvement was significant for tadalafil OaD versus placebo only (p=0.035). No significant

differences were observed for SEAR.

**Conclusion:** These QoL data complement previously reported positive effects of tadalafil OaD on EF recovery at EDT.

**MP-07.15****Metabolic Syndrome and Oncologic Outcomes in Men Undergoing Radical Prostatectomy for Prostate Cancer**

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**Introduction and Objectives:** Metabolic syndrome (MetS) is associated with an increased risk of prostate cancer (PC) overall and high grade disease on biopsy pathology. In the present study, our objective was to determine if MetS is associated with adverse pathology and risk of treatment failure in men undergoing radical prostatectomy (RP).

**Materials and Methods:** Patients undergoing RP (2004-2013) were identified using our prospectively maintained institutional database. Salvage RPs and men who received neo-adjuvant therapies were excluded. MetS required any 3 of 5 components (obesity, diabetes or impaired fasting glucose, hypertension, low HDL-cholesterol, and high triglycerides), and was ascertained using electronic chart review. The outcomes were PC stage and grade on final RP pathology, and RP treatment failure, defined by a post-RP serum PSA ≥0.2, or use of adjuvant or salvage therapies such as radiation or androgen deprivation therapy. Multivariable logistic regression models, Kaplan-Meier analyses, and Cox-proportional Hazards models were used.

**Results:** The final cohort consisted of 1939 men, of which 439 (22.6) had MetS. Median follow-up was 36 months. On RP pathology, there were 663 (34.2%) men with extraprostatic disease (≥pT3), 505 (26%) with Gleason

**MP-07.14, Table 1.** LS Mean Changes [95% CI] in EPIC and EDITS Scores with Tadalafil OaD, Tadalafil PRN and Placebo from Baseline to End of Double-Blind and Open-Label Treatment Periods (EDT, EOL)

|                                                                                               | Tadalafil OaD        | Tadalafil PRN      | Placebo            |
|-----------------------------------------------------------------------------------------------|----------------------|--------------------|--------------------|
| <b>EPIC sexual domain score (age-group by treatment interaction: p = 0.083)</b>               |                      |                    |                    |
| EDT                                                                                           | +27.5 [21.6, 33.4]** | +20.7 [15.3, 26.1] | +18.0 [12.1, 23.8] |
| EOL                                                                                           | +36.6 [30.0, 43.1]   | +32.6 [26.6, 38.6] | +33.4 [27.0, 39.8] |
| Men ≤60 years                                                                                 | +30.1 [23.2, 36.9]   | +31.2 [24.8, 37.6] | +24.9 [18.2, 31.6] |
| Men 61-68 years                                                                               | +34.0 [26.0, 42.0]   | +22.1 [14.6, 29.5] | +26.5 [18.5, 34.4] |
| <b>EPIC urinary incontinence domain score (age-group by treatment interaction: p = 0.084)</b> |                      |                    |                    |
| EDT                                                                                           | +34.1 [29.3, 38.9]   | +31.1 [26.7, 35.5] | +30.6 [25.9, 35.3] |
| EOL                                                                                           | +37.4 [32.6, 42.3]   | +35.5 [31.1, 40.0] | +35.4 [30.7, 40.2] |
| Men ≤60 years                                                                                 | +33.0 [27.7, 38.3]   | +34.6 [29.6, 39.7] | +35.8 [30.5, 41.1] |
| Men 61-68 years                                                                               | +38.5 [32.2, 44.8]*  | +32.0 [26.2, 37.9] | +30.2 [24.0, 36.5] |
| <b>EPIC urinary irritative/obstructive domain score</b>                                       |                      |                    |                    |
| EDT                                                                                           | +13.8 [11.5, 16.1]   | +13.3 [11.2, 15.4] | +12.3 [10.0, 14.5] |
| EOL                                                                                           | +13.9 [11.5, 16.2]   | +13.8 [11.7, 15.9] | +12.3 [10.0, 14.6] |
| <b>EPIC bowel domain score</b>                                                                |                      |                    |                    |
| EDT                                                                                           | +5.9 [3.7, 8.2]      | +6.3 [4.2, 8.3]    | +6.5 [4.3, 8.7]    |
| EOL                                                                                           | +6.9 [4.7, 9.1]      | +6.5 [4.5, 8.5]    | +6.8 [4.6, 8.9]    |
| <b>EPIC hormonal domain score</b>                                                             |                      |                    |                    |
| EDT                                                                                           | +1.7 [-0.8, 4.3]     | +2.7 [0.4, 5.1]*   | -0.2 [-2.7, 2.3]   |
| EOL                                                                                           | +2.5 [0.1, 4.9]      | +2.9 [0.8, 5.1]    | +3.0 [0.7, 5.4]    |
| <b>EDITS total score</b>                                                                      |                      |                    |                    |
| EDT                                                                                           | +2.2 [2.0, 2.4]**    | +2.1 [1.9, 2.3]*   | +1.9 [1.7, 2.1]    |
| EOL                                                                                           | +2.5 [2.3, 2.8]*     | +2.4 [2.2, 2.6]    | +2.3 [2.0, 2.5]    |

Data are from mixed model for repeated measures (MMRM), including baseline score, treatment, country, visit (EDT, EOL), age group (men ≤60yrs, men 61-68yrs), and visit-by-treatment interaction. Age-group-by-treatment interaction was included only if significant at the 10% level. For men ≤60 years and 61-68 years, the overall treatment effect presented includes all visits from baseline to EOL.

\*\* p<0.01, \* p<0.05 versus placebo (MMRM)

≤6 disease, 1321 (68.1%) with Gleason 7 disease, and 113 (5.8%) with Gleason 8-10 disease. There were 357 (18.4%) considered as RP treatment failures. MetS was associated with an increased risk of extraprostatic disease (adjusted OR=1.29, 95%CI=1.03-1.62,  $p=0.030$ ) and Gleason 8-10 disease (adjusted OR=1.61, 95%CI=1.06-2.45). Although there was a higher rate of treatment failure among men with MetS (19.1% vs. 18.2%), differences did not reach statistical significance in Kaplan Meier analyses and Cox models.

**Conclusion:** Metabolic syndrome is associated with an increased risk of harbouring extraprostatic and high-grade disease. Longer follow-up will be required to see how this translates into treatment failure outcomes.

#### MP-07.16

##### **Peritoneal Fenestration Effectively Prevents Symptomatic Lymphocele after Laparoscopic Radical Prostatectomy with Extended Pelvic Lymphadenectomy**

Kono Y, Nishihara D, Matsuoka T, Yano T, Utsunomiya N, Okada T, Kawakita M  
*Kobe City Medical Center General Hospital, Kobe, Japan*

**Introduction and Objectives:** To investigate the incidence and the risk factors of symptomatic lymphocele after laparoscopic extraperitoneal radical prostatectomy (LRP) with extended pelvic lymphadenectomy (ePLND).

**Materials and Methods:** Among 197 patients who underwent LRP with ePLND between April 2008 and October 2013, risk factors including age, BMI, PSA, clinical stage, pathological stage, total operative time, lymph node dissection (LND) time, node count, with/without clipping ( $n=185/13$ ), with/without peritoneal fenestration ( $n=74/124$ ), drainage duration, drainage amount, and with/without administration of low-molecular-weight heparin (LMWH) ( $n=104/94$ ) were analyzed.

**Results:** Symptomatic lymphoceles were found in 20 patients (10.1%) on follow-up. DVT and PE were seen in 10% each. Between symptomatic lymphocele (+) group and symptomatic lymphocele (-) group, the mean age (67.2 vs. 67.9 yr,  $p=0.914$ ), BMI (23.8 vs. 24.2 kg/m<sup>2</sup>,  $p=0.474$ ), PSA (11.3 vs. 10.7 ng/ml,  $p=0.666$ ), total operative time (322 vs. 349 min,  $p=0.165$ ), LND time (71.1 vs. 72.0 min,  $p=0.849$ ), node count (17.1 vs. 16.1,  $p=0.433$ ), drainage duration (3.4 vs. 3.3 days,  $p=0.987$ ), and drainage amount (277 vs. 292 ml,  $p=0.556$ ) were not significantly different. Clinical and pathological stages are also similar in two groups ( $p=0.522$  and  $0.807$ , respectively). Significantly higher incidence of symptomatic lymphocele on chi-squared test was detected in the un-fenestration group compared to the fenestration group (62.4 vs. 37.3%,  $p=0.031$ ), whereas no significant differences were observed

in the incidences of symptomatic lymphocele with/without clipping (93.4 vs. 6.5%,  $p=0.110$ ), and with/without LMWH (52.5 vs. 47.4%,  $p=0.246$ ).

**Conclusion:** Peritoneal fenestration after ePLND has proved to effectively prevent symptomatic lymphocele.

#### MP-07.17

##### **Fluorescence-Targeted Laparoscopic Lymph Node Dissection in Prostate Cancer**

Hruby S<sup>1</sup>, Englberger C<sup>1</sup>, Lusuardi L<sup>1</sup>, Schätz T<sup>1</sup>, Kunit T<sup>1</sup>, Hager M<sup>2</sup>, Janetschek G<sup>1</sup>  
<sup>1</sup>Dept. of Urology, Paracelsus Medical University, Salzburg, Austria; <sup>2</sup>Dept. of Pathology, Paracelsus Medical University, Salzburg, Austria

**Introduction and Objectives:** Recently we have introduced fluorescence by means of Indocyanin Green (ICG) in addition to radio-guided dissection using Technetium 99 (Tc99) for laparoscopic pelvic sentinel node dissection (Urology 2012). Since ICG also visualizes the lymphatic vessels, it allows to better understand the lymphatic drainage of the prostate. Hence we could show that ICG alone gave equal results compared to TC99. As evolution we now left the sentinel concept to completely remove this template.

**Materials and Methods:** A total of 38 consecutive men with intermediate and high risk prostate cancer have undergone targeted Lymphnode dissection during laparoscopic radical prostatectomy. The previous transrectal TRUS-guided Injection of 2ml ICG into each lobe is now replaced by a transperineal approach. After removal of the complete ICG visualize Lymphnode template of each side, a standard extended LN-dissection was added as control. The Equipment for ICG –Visualisation was provided by Storz. All Lymphnodes were evaluated by 250  $\mu$ m sections and immunohistochemistry. Data were collected prospectively and analyzed retrospectively.

**Results:** Transperineal Injection allowed for precise deposit of the tracer within the peripheral zone without any periprostic extravasation. Fluorescence stained (F+) nodes were found on both sides in all patients except one. In total 596 nodes ( $17.9 \pm 8.4$ /patient) were removed, of which 473 nodes ( $14.3 \pm 8.51$ /patient) were F+. LN Metastases were found in 15 patients (39.5%), of which 2 patients (5.3%) had solitary micrometastases. In addition 3 patients (7.9%) the LN contained Tumor Cell Cluster. There was not a single patient where non-stained metastases were found in addition to F+ Metastases. In the one patient where there was no staining at all a solitary metastases was found in the template of the extended PLND. Met outside the template of extended PLND occurred in 5 patients (27.8% of N+). Since we always perform extended PLND, when there is no Fluorescence staining at all, no

metastases was overlooked with our concept.

**Conclusion:** Fluorescence-targeted lymphnode dissection allows to identify the lymphatic drainage of the prostate with great reliability. It proved to be more precise than extended lymphnode dissection in patients with intermediate and high grade prostate cancer.

#### MP-07.18

##### **Safety, Objective Response Rate and Long-Term Oncologic Outcomes of Docetaxel Neoadjuvant Chemotherapy followed by Radical Prostatectomy for Patients with Intermediate- and High Risk Prostate Cancer: A Single-Center Study**

Nosov A<sup>1</sup>, Petrov S<sup>1</sup>, Reva S<sup>1</sup>, Mamijev E<sup>1</sup>, Soroka S<sup>1</sup>, Veliev E<sup>2</sup>, Moiseenko V<sup>3</sup>  
<sup>1</sup>N.N. Petrov Research Institute of Oncology, Saint Petersburg, Russia; <sup>2</sup>Russian Medical Academy of Postgraduate Education, Moscow, Russia; <sup>3</sup>Clinical Institute of Oncology, Saint Petersburg, Russia

**Introduction and Objectives:** Effective treatment of prostate cancer (PCa) at high risk of recurrence is a challenging problem. Neoadjuvant treatment provides various benefits for these patients, such as the PSA response, decreased tumor volume, and higher survival rate. However, hormonal treatment followed by surgery has limited value. On the other hand, neoadjuvant chemotherapy has become a healthcare standard in treating certain malignancies. According to recent data, neoadjuvant docetaxel chemotherapy followed by radical prostatectomy may improve clinical outcomes without a significant increase of toxicity.

**Materials and Methods:** Forty-four patients were involved in this study, with a 10-year follow-up period, on average. In the combined (chemotherapy and surgery) treatment group (21 patients), 126 cycles of chemotherapy were administered. We assessed the RPE results in patients with intermediate and high risk of PCa (PSA>10 ng/ml, Gleason score 7 and more, or clinical stage cT2c and more) after weekly treatment with docetaxel (36 mg/m<sup>2</sup> for up to 6 cycles, 21 patients) and compared them with those in the second cohort (23 patients) who met oncologic inclusion criteria but received RP only. The long-term oncologic outcomes in both groups of the patients are reported.

**Results:** Toxicity has been mostly grade 1-2 in intensity and grade 3 and more complication rate does not exceed 10%. A statistically significant of more than 50% reduction in PSA level (pre- vs. post-chemotherapy) was observed in 52.4% cases. During the 11.4-year follow-up period, improvement in cancer-specific survival (CSS) was noted in 90% of patients from the neoadjuvant chemotherapy group, as compared with 60.9% in radical prostatectomy only group. The biochemical recurrence-free survival (BCR) was 68.5% and 37.7%, and the overall



survival (OS) was 75.5% and 54.6% in the combined treatment group and surgery only group, respectively. However, the differences in BCR and OS were not statistically significant.

**Conclusions:** The use of neoadjuvant chemotherapy represents a safe and practicable treatment strategy resulting in reduced prostate volume and PSA level. Neoadjuvant docetaxel chemotherapy followed by radical prostatectomy was associated with higher observed BCR and OS, as compared with a surgical treatment only group. A statistically significant improvement of CSS is found in the combined treatment group. Therefore, the benefits of this treatment modality need to be validated for feasible implementation in the modern standard practices of prostate cancer treatment.

#### MP-07.19

##### **A Tale of Tails: A Novel Approach to Cytokine Immunotherapy**

**Sakellariou C<sup>1</sup>**, Smolarek D<sup>1</sup>, Elhage O<sup>1</sup>, Ukimura O<sup>2</sup>, Gill I<sup>2</sup>, Smith R<sup>1</sup>, Galustian C<sup>1</sup>, Dasgupta P<sup>1</sup>

<sup>1</sup>MRC Centre for Transplantation, King's College, London, UK; <sup>2</sup>Keck School of Medicine, University of Southern California, Los Angeles, USA

**Introduction and Objectives:** Prostate cancer (PCa) progression can arise due to the immunosuppressive tumour microenvironment. We have previously shown that IL-15, unlike other cytokines, such as IL-2, can expand CD8 and NK cells when they are exposed to PCa cells. We hypothesized that if IL-15 could be localized in the cancerous prostate, it will have greater efficacy and less toxicity than drugs administered systemically. Localized immune stimulation can boost systemic immunity to clear distant metastases. Therefore, we created a localizable form of IL-15 to study its effects on NK and CD8 T cells with and without PCa cells present.

**Materials and Methods:** A modified form of IL-15 was produced enabling conjugation to a cytotopic "tail" structure. The "tail" consists of a thiol-reactive region interacting with free cysteines on IL-15, a cationic region binding negatively charged cell membranes, and a hydrophobic region entering cell membranes. Cytotopic "tailed" IL-15 can thus localize to any lesion where it is injected. The tailed IL-15 was compared with non tailed IL-15 in assays measuring T cell proliferation, and NK and CD8 T cell expansion.

**Results:** Tailed IL-15 induced greater proliferation (up to 20 fold) of a murine T cell line, CTLL2, than non-tailed IL-15. Its activity was also greater or equivalent to non-tailed IL-15 in its ability to expand NK cells.

**Conclusion:** A tailed form of IL-15, has greater activity in expanding in immune effector cells than conventional IL-15 and its potential for

regioselective action makes it a promising new agent for PCa immunotherapy.

#### MP-07.20

##### **Salvage Robot-Assisted Laparoscopic Radical Prostatectomy after High Intensity Focused Ultrasound Failure**

**Manea C<sup>1</sup>**, Crisan N<sup>1</sup>, Logigan H<sup>1</sup>, Coroi T<sup>1</sup>, Ivan C<sup>1</sup>, Coman I<sup>2</sup>

<sup>1</sup>Robotic Surgery Center, Municipal Clinical Hospital, Cluj-Napoca, Romania; <sup>2</sup>Iuliu Hatieganu University of Medicine and Pharmacy, Cluj-Napoca, Romania

**Introduction and Objectives:** High Intensity Focused Ultrasound (HIFU) therapy is effective in carefully selected cases. Before applying a therapeutic procedure to patients with prostate cancer, it is necessary to define the best indications for treatment in daily clinical practice as primary therapy. Our presentation describes the surgery steps of the radical excision of the prostate gland after initial HIFU treatment and short time results.

**Materials and Methods:** Out of 192 patients diagnosed with prostate adenocarcinoma and treated with HIFU performed using Sonablate<sup>®</sup> 500 in the last five years, we used robot-assisted laparoscopic radical prostatectomy (RALP) for 5 cases with post-HIFU local recurrence.

**Results:** None of the interventions required a conversion to conventional surgery. The average operative time was 125 minutes, the mean blood loss was 105 ml and no patient was transfused. The histopathological results were stage pT2a in one case, pT2b in two cases, and pT2c in two cases. We recorded a Gleason score of 7 (3+4) for three patients and of 7 with primary pattern 4 for two patients. All cases had adenocarcinoma with ductal features. The resection margins were negative. Two cases didn't have urinary incontinence, one obtained erection 3 weeks postoperatively, and the other three cases had urinary incontinence only on effort, but no erectile function. Mean prospective monitoring time was 5 months.

**Conclusions:** Patients with low pre-HIFU PSA level and a Gleason score greater than 7 with primary pattern 3 and aggressive histological patterns such as ductal carcinoma are not candidates for safety HIFU therapy. These patients' relapse rate is of over ninety percent. Salvage first-line RALP for biopsy-proven local recurrence of prostate cancer after HIFU is feasible and safe. In our experience, the small number of cases and the short monitoring time limit the determination of the rate of subsequent morbidities, compared to the primary indication of radical prostatectomy.

#### MP-07.21

##### **Full Functional Length Urethral Sphincter Preservation during Open Radical Prostatectomy: Will It Improve the Continence Rate? Shimpi R**

*Dept. of Urology, Ruby Hall Clinic, Pune, India*

**Introduction and Objectives:** The full length of the urethral sphincter is the key for urinary continence after Radical Prostatectomy. As demonstrated by various studies, the Intra-Prostatic Urethra between the apex and verumontanum is an important part of the urethral sphincter complex. My modified technique is aimed at preserving the muscular part of the Intra-Prostatic Urethra in low volume disease.

**Materials and Methods:** Ninety eight patients within the age group of 54-67 and treated between 2000 and 2012 were analyzed. Twenty three patients with low volume disease were chosen for Full Functional Length Urethral Preservation (FFLU) while 75 had Non-FFLU. Continence rate was assessed at 1 month and 3 months after the catheter removal. The proximal urethral tissue was sent for frozen section.

**Results:** The continence rate at one month after the catheter removal was defined as no pad at all or one protective pad was used 79.9% with FFLU and that with Non-FFLU was 68.2%. After 3 months, the continence rate rose to 96.7% and 80.4% respectively. The other factors considered in the study are Age, PT stage, prostatectomy Gleason Score. The positive surgical margin rate was 1.6% with FFLU while the one with Non-FFLU was 2.2%.

**Conclusion:** The improved Urinary Continence Rate achieved by doing this technique shows that this simple manoeuvre of Full Functional Length Urethral Preservation (FFLU) in low volume disease should be attempted wherever possible.

#### MP-07.22

##### **Prevalence and Risk Factors of Contralateral Extraprostatic Extension in Men Undergoing Radical Prostatectomy for Localized Unilateral Disease at Biopsy: A Global Multi-Institutional Experience**

**Bienz M<sup>1</sup>**, Camacho A<sup>1</sup>, Hueber P<sup>2</sup>, Liberman D<sup>3</sup>, Al-Hathal N<sup>3</sup>, Al-Enizi A<sup>3</sup>, Mouraviev V<sup>4</sup>, Canda A<sup>5</sup>, Adom M<sup>4</sup>, Balbay M<sup>5</sup>, Albala D<sup>4</sup>, Latour M<sup>3</sup>, El-Hakim A<sup>3</sup>, Saad F<sup>2</sup>, Zorn K<sup>2</sup>

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**Introduction and Objectives:** Interfascial nerve-sparing technique during RARP may be performed on the contralateral side of unilaterally diagnosed prostate cancer. Unsuspected bilateral disease could be associated with



extraprostatic extension. We aim to assess the incidence and risk factors of contralateral EPE (cEPE) and contralateral positive surgical margins (cPSM) in patients diagnosed preoperatively with unilateral disease.

**Materials and Methods:** This multicenter cohort consisted of 331 men diagnosed with unilateral PCa who underwent RARP. Localization and occurrence of positive cores from biopsy, cEPE, cPSM and SVI was noted. cEPE<sup>+</sup> and cEPE<sup>-</sup> groups were compared for preoperative predictive parameters.

**Results:** Pathology reported cPCa in 50.2% and cEPE in 4% of the cohort. PSA levels of cEPE<sup>+</sup> and cEPE<sup>-</sup> patients was 6.4 µg/L (5.1-14.6) and 5.2 µg/L (4.0-7.1) respectively (p=0.026). Also proportion of positive cores

(p=0.189), maximum cancer involvement in a core (p=0.168), clinical stage (p=0.327), Gleason score (p=0.178) and TRUS size (p=0.411) was assessed. Lastly, in the pT3 subgroup, the frequency of positive biopsies at the apex increased with contralateral cancer invasion (p=0.007).

**Conclusion:** Despite the 50% chance of bilateral disease, the risk of cPSM associated with cEPE is only 1% in the cohort. Contralateral nerve-sparing procedures may be considered safe in patients with unilateral disease on preoperative biopsies.

MP-07.22, Table 1.

| (%)  | cPCa <sup>-</sup> , cEPE <sup>-</sup><br>n=165 [50%] | cPCa <sup>+</sup> , cEPE <sup>-</sup><br>n=153 [46%] | cPCa <sup>+</sup> , cEPE <sup>+</sup><br>n=13 [4%] | P-value |
|------|------------------------------------------------------|------------------------------------------------------|----------------------------------------------------|---------|
| PSM  | 14.5                                                 | 26.8                                                 | 38.5                                               | 0.008   |
| cPSM | 0.0                                                  | 10.5                                                 | 23.1                                               | <0.001  |
| SVI  | 3.0                                                  | 5.2                                                  | 38.5                                               | <0.001  |

## Moderated Poster Session 8

### Bladder Cancer: Invasive and Superficial Diseases

Monday, October 13  
1435-1600

#### MP-08.01

##### **Hematuria as Screening Test for Bladder Invasion by Carcinoma of the Cervix Can Decrease the Use of Staging Cystoscopy**

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*Dept. of Urology, Stellenbosch University and Tygerberg Hospital, Tygerberg, South Africa*

**Introduction and Objectives:** A recent study suggested that hematuria can be used as a screening test to detect urinary bladder invasion in women with cervical cancer. The study of 130 patients (42% had hematuria) showed sensitivity 100%, specificity 60.3%, positive predictive value (PPV) 7.4%, negative predictive value (NPV) 100% and accuracy 61.5% (Asian Pac J Cancer Prev 2012;13(10):4931-3). The objective of this study was to evaluate hematuria as screening test to predict the presence of bladder invasion in women with cervical carcinoma.

**Materials and Methods:** In total, 241 women with carcinoma of the uterine cervix were evaluated January 2012 through April 2013 using midstream urinalysis, cystoscopy and bladder biopsy to assess malignant bladder invasion. Urine erythrocyte and leukocyte counts were quantified using the Fast Read method during microscopy (counting the cells under a grid cover slide).

**Results:** The mean patient age was 49.7 (range 20.5 to 87) years. The clinical stage of cervical cancer was T1 in 29.7%, T2 in 42.2% and T3 in 23.4%. Human immunodeficiency virus (HIV) testing was positive in 27.2%. Hematuria (erythrocytes >1000/ml) was present in 41.4% and leukocyturia (>1000/ml) in 49.5%. Bladder invasion (confirmed on histology) was found in 8.3% of cases. The sensitivity, specificity, PPV, NPV and accuracy of hematuria and leukocyturia in detecting bladder invasion is shown in Table 1.

**Conclusion:** Using significant hematuria (erythrocyte count >100 000/ml) as a screening test for bladder invasion in cervical carcinoma can avoid 78% of staging cystoscopies while missing 7% of cases with bladder invasion. This may be an acceptable option for the management of women with T3 cervical cancer in resource constrained regions, because the treatment does not differ significantly if T3 is cystoscopically upstaged to T4.

#### MP-08.02

##### **Optical Biopsy with Confocal Laser Endomicroscopy (CLE): A New Technology for Diagnosis of Bladder Cancer – Preliminary Results of a Phase II Pilot Study**

Brausi M<sup>1</sup>, Peracchia G<sup>1</sup>, De Luca G<sup>1</sup>, Viola M<sup>1</sup>, Swartz F<sup>2</sup>

<sup>1</sup>Dept. of Urology, AUSL Modena, Carpi, Italy;

<sup>2</sup>Mauna Kea Technologies, Paris, France

**Introduction and Objectives:** Confocal laser endomicroscopy is a new endoscopic imaging technology that could complement white light cystoscopy by providing *in vivo* bladder histopathology. We evaluated confocal laser endomicroscopy by imaging normal, malignant appearing inflammation and suspicious bladder mucosa areas in a pilot study.

**Materials and Methods:** Patients scheduled to undergo transurethral resection of bladder tumors were recruited during a 2-month period. After standard cystoscopy fluorescein was administer intravesically and/or intravenously as a contrast dye, a 2.6 mm probe based confocal laser endomicroscope was passed through a 17.5 F flexible scope and/or a 24 F rigid cystoscope to image normal or abnormal appearing areas before and after TUR. The images were collected with 488 nm excitation at 8 to 12 frames per second. The endomicroscopic images were compared with standard hematoxylin and eosin analysis of TUR and biopsies of bladder tumor specimen and suspicious areas.

**Results:** Nine patients were recruited at our center in the last 2 months and treated in 3 OR sessions. Six patients had low grade tumors, 1 high grade, 2 patients had dysplasia (mild and severe) and inflammation. Endomicroscopic images demonstrated clear differences between

normal mucosa and low and high grade tumors. In normal urothelium larger umbrella cells are seen most superficially followed by smaller intermediate cells and the less cellular lamina propria. Low grade papillary tumors demonstrate densely arranged but normal-shaped small cells in multiple layers (>6). High grade tumors show markedly irregular architecture and cellular pleomorphism. Some artifacts were often present which were delete during slide preparation.

**Conclusions:** We report one of the first experience *in vivo* done in europe of confocal laser endomicroscopy in the urinary tract. Differences among normal urothelium, low grade and high grade tumors was observed. We defined specific CLE image interpretation criteria for *in vivo* characterization of inflammation, dysplasia, cis and tumor.

#### MP-08.03

##### **Results of a Randomized Controlled Trial Comparing Intravesical Combined Chemohyperthermia with Mitomycin-C versus BCG for Adjuvant Treatment of Patients with Intermediate and High Risk Non-Muscle Invasive Bladder Cancer**

Arends T, van der Heijden A, Witjes A  
*Radboudumc, Nijmegen, The Netherlands*

**Introduction and Objectives:** Despite regular adjuvant intravesical treatments, the incidence of recurrences in non-muscle invasive bladder cancer (NMIBC) is still high. Therefore, new treatment options are most welcome. This randomized, controlled, multicentre trial compares combined chemohyperthermia using Mitomycin-C (C-HT) with BCG as adjuvant treatment for intermediate and high risk NMIBC.

**Materials and Methods:** Between 2002 and 2012, 190 NMIBC patients were eligible for inclusion and randomized between 1-year C-HT (induction: six weekly treatments and six maintenance treatments) and 1-year BCG immunotherapy (induction: six weekly and nine maintenance treatments). Median follow-up was 22 months in the C-HT group and 18 months in the BCG group. Recurrence-free survival (RFS) in all papillary NMIBC was the primary objective. The secondary objective was

MP-08.01, Table 1.

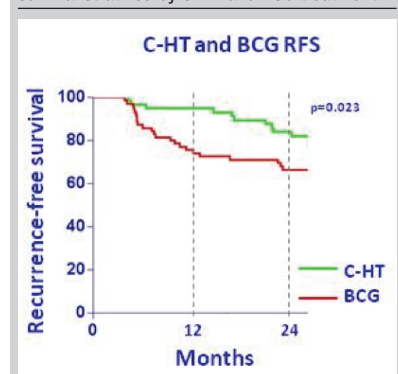
|                             | Erythrocytes<br>>1000/ml | Erythrocytes<br>>100 000/ml | Leucocytes<br>>0/ml (any) | Leukocytes<br>>10 000/ml | Erythrocytes<br>>1000/ml + any<br>leukocytes |
|-----------------------------|--------------------------|-----------------------------|---------------------------|--------------------------|----------------------------------------------|
| <b>Sensitivity</b>          | 55%                      | 35%                         | 90%                       | 60%                      | 58.8%                                        |
| <b>Specificity</b>          | 65.6%                    | 79.6%                       | 13.1%                     | 58.8%                    | 67.8%                                        |
| <b>PPV</b>                  | 12.6%                    | 13.5%                       | 8.6%                      | 11.7%                    | 17.2%                                        |
| <b>NPV</b>                  | 94.2%                    | 93.1%                       | 93.5%                     | 94.2%                    | 93.5%                                        |
| <b>Accuracy</b>             | 64.7%                    | 75.9%                       | 19.5%                     | 58.9%                    | 66.9%                                        |
| <b>Cystoscopies avoided</b> | 63.9%                    | 78.4%                       | 12.9%                     | 57.3%                    | 44.8%                                        |

the safety-analysis of both treatments.

**Results:** A total of 190 patients were included; 132 patients were eligible for RFS-analysis and 184 subjects were included in the safety-analysis. Baseline characteristics were equally divided. Two-years RFS was 78.3% en 66.7% for the C-HT and BCG group, respectively (N=132, p=0.023, Figure 1). Although most side-effects didn't differ significantly between the two groups, systemic side-effects (fever, fatigue, arthralgia) were more reported in the BCG group. Bladder pain and -spasms, however, were more frequent in the C-HT group. Muscle-invasive progression was low in C-HT and BCG treatment: 1.7% and 2.8%, respectively.

**Conclusion:** C-HT is a safe and effective treatment option in patients with intermediate and high risk papillary NMIBC. The 2-years RFS is significantly higher in the C-HT arm in comparison to the BCG-arm. Furthermore, the systemic side-effects are higher in the BCG group. Accordingly, one should consider the option of C-HT treatment in intermediate and high risk NMIBC patients.

MP-08.03, Figure 1. The recurrence-free survival stratified by C-HT and BCG treatment



#### MP-08.04

##### A Randomized Control Study to Evaluate the Efficacy, Safety and QOL in Low-Dose Bacillus Calmette-Guerin Instillation Therapy for Non-Muscle Invasive Bladder Cancer

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**Introduction and Objectives:** The optimal dose of intravesical bacillus Calmette-Guerin (BCG) in the treatment of non-muscle-invasive bladder cancer (NMIBC) is controversial. To determine if a low dose (40mg) is not inferior to the standard dose (80mg) in the BCG induction intravesical instillation are associated with less toxicity and higher QOL.

**Materials and Methods:** After transurethral resection, intermediate- to high-risk NMIBC patients were randomized to low dose (40mg) or standard dose (80mg) of BCG instillation induction therapy (weekly, 8 times). The primary endpoint was the non-inferiority in the low dose group with the null hypothesis of a 15% decrease in the complete response (CR) rate. The secondary endpoints were recurrence-free survival, overall survival, compliance of protocol treatment, adverse events and quality of life (QOL) using EORTC QLQ C30.

**Results:** In an intention-to-treat analysis of 136 patients, CR rate of the low dose (40mg) and standard dose (80mg) of BCG were 79% (95% C.I.:0.70-0.88) and 85% (95% C.I.: 0.77-0.92), respectively. A Dunnett-Gent analysis revealed that null hypotheses of inferiority of the CR rate in the low dose group could not be rejected (P=0.119). And there was no significant difference in recurrence free survival in those two group (Log rank test, p=0.916) as well as overall survival and compliance of protocol treatment. In the comparison of the frequency of adverse events, the low dose group had a significantly lower frequency in fever (p=0.001) and micturition pain (p=0.047). Furthermore, the low dose group showed significantly higher QOL score in general QOL, role functioning and functional impairment than those in the standard dose group.

**Conclusions:** The equivalent efficacy of low dose BCG therapy could be acceptable even if the null hypotheses of inferiority of the CR rate in the low dose group could not be rejected. The benefit of the low dose BCG therapy was associated with lower toxicity and higher QOL.

#### MP-08.05

##### Is GA Cystoscopy and Bladder Biopsies Necessary after Intravesical BCG Treatment Transitional Cell Bladder Cancer and if the Bladder Is Visually Clear Can GA Cystoscopy Be Substituted by Flexible Cystoscopy?

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**Introduction and Objectives:** Bacillus Calmette-Guerin (BCG) is routinely introduced to the patients after endoscopic resection of high grade superficial urothelial tumours. Rigid cystoscopies and biopsies under general anaesthetic often performed to assess effectiveness of BCG therapy. Our aim is to assess the necessity of GA cystoscopies and biopsies in accordance to their compatible histopathological findings.

**Materials and Methods:** The study included 73 consecutive patients with high grade superficial bladder cancer who underwent full induction and then maintenance course of BCG. The data was collected retrospectively. Cystoscopic and histopathological findings were evaluated and compared for consistency. Cystoscopic findings were categorized as clear, scar, red patch or recurrence. The histopathological findings were categorized as chronic inflammation or positive histology.

**Results:** Different patients had different numbers of cystoscopies and biopsies done. A total of 510 cystoscopies were done. Numbers of follow-up cystoscopies done ranged from 4 to 15. On cystoscopic finding; 269 (52.74%) where clear, 45 (8.82%) where scar, 152 (29.80%) where red patch, 43 (8.64%) where recurrence.

**Conclusion:** A total of 510 cystoscopies were done. Only one histology was positive in clear category. The histopathological results were compatible with the cystoscopic visual findings. This is enough evidence to show the association of negative finding from histopathological results in accordance with clear cystoscopic findings. Patient could therefore preferably avoid this procedure. We recommend that GA cystoscopy should be substituted with flexible cystoscopy. Subsequent GA cystoscopy and biopsies can be instituted in accordance with the flexible cystoscopic findings. This would improve patient safety and care, improve staff efficiency and reduce staff time and costs.

MP-08.05, Table 1.

| Histology  | Total (%) | Biopsy not done (%) | Chronic inflammation (%) | + Histology (%) |
|------------|-----------|---------------------|--------------------------|-----------------|
| Cystoscopy |           |                     |                          |                 |
| Clear      | 269       | 231 (85.87)         | 37 (13.75)               | 1 (0.37)        |
| Scar       | 45        | 10 (22.2)           | 27 (60)                  | 8 (17.7)        |
| Red Patch  | 152       | 10 (6.6)            | 121 (79.6)               | 21 (13.8)       |
| Recurrence | 44        | 0                   | 12 (27.3)                | 32 (72.7)       |
| Total      | 510       | 251 (49.2)          | 197 (38.6)               | 62 (12.2)       |

**MP-08.06****Managing High-Risk Non-Muscle Invasive Bladder Cancer with Intravesical Mitomycin-C Hyperthermia: A Seven-Year Experience**

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**Introduction and Objectives:** Intra-vesical Bacillus Calmette–Guérin (BCG) therapy is the current standard of care following transurethral resection of high-risk non-muscle invasive bladder cancer (HR-NMIBC). However, up to 50% of patients who receive BCG treatment will experience non-response, recurrence or disease progression. The alternative, radical cystectomy, may not be suitable for a large proportion of patients due to its inherent morbidity. We report our seven-year experience of intra-vesical Mitomycin-C hyperthermia (MMC-HT), and establish its role as a viable alternative in this patient cohort.

**Materials and Methods:** Prospective data was collected for 107 patients with HR-NMIBC treated with MMC-HT between June 2006 and January 2014 from a single institution. A total of 103 patients completed induction treatment and underwent urine cytology, cystoscopy and bladder-biopsies at three-months. Responders were continued on maintenance MMC-HT instillations. Response rates at three-months, tumour progression, survival and side effects were noted. Progression was defined as development of muscle invasion, distant metastases, requirement for cystectomy or radiotherapy and death from bladder cancer.

**Results:** A median follow-up of 40 months (3-92 months) revealed 72% of patients had a complete response at three-months. A further 10% achieved partial response and 18% developed disease recurrence. No patients suffered a Clavien-Dindo complication above two. Five-year overall survival was 61.9%, and disease specific survival was 85.2%. Progression free survival five-year survival was 46.5%. Twenty-one patients underwent radical cystectomy. Nineteen patients had T0 or organ-confined disease and two patients had pathological T3 disease. Only one patient developed disease recurrence following cystectomy.

**Conclusion:** MMC-HT has comparable five-year survival to radical-cystectomy in the management of HR-NMIBC following BCG-failure. It is well tolerated and can be delivered effectively in a regional centre. For those suitable patients who fail MMC-HT, radical-cystectomy remains a potentially curative option.

**MP-08.07****Activity and Safety Analysis of TMX-101 0.2% and 0.4% for Non-Muscle Invasive Bladder Cancer in a Multicentre Phase I Study**Arends T<sup>1</sup>, Lammers R<sup>1</sup>, Falke J<sup>1</sup>, Cornel E<sup>2</sup>, Vergunst H<sup>3</sup>, de Reijke T<sup>4</sup>, Witjes A<sup>1</sup><sup>1</sup>*Radboudumc, Nijmegen, The Netherlands;*<sup>2</sup>*Ziekenhuis Groep Twente, Hengelo, The**Netherlands;* <sup>3</sup>*Canisius Wilhelmina Ziekenhuis,**Nijmegen, The Netherlands;* <sup>4</sup>*Academisch Medisch Centrum, Amsterdam, The Netherlands*

**Introduction and Objectives:** TMX-101 is the liquid form of imiquimod, a toll like receptor 7 agonist, for intravesical instillation and is effective *in vitro* against urothelial non-muscle invasive bladder cancer (NMIBC). In Part I of this Phase I study (N=16), the safety of TMX-101 was demonstrated. In this Part II, the activity of TMX-101 was analyzed in low-grade NMIBC patients. Furthermore, the safety of TMX-101 was evaluated.

**Materials and Methods:** Seven subjects underwent a marker-lesion TURBT and six weekly instillations with TMX-101 0.2% or 0.4% to determine the effective biological dose (EBD, = complete response in >2 patients). Subjects were evaluated weekly on adverse event (AE's), pharmacokinetics (PK) and pharmacodynamics (PD). Cystoscopy two weeks after the last instillation confirmed the efficacy of TMX-101 on the marker lesion.

**Results:** Eighty seven percent reported at least one AE. All events were of ≤ grade 2 intensity (CTCAE 4.2). No clinical significant changes in laboratory parameters and/or vital signs were observed, during or after treatment. An EBD could not be defined as none of the subjects achieved a complete response. Maximum plasma concentration was 75.1 ng/ml in the 0.4% dose group. No drug accumulation was observed. In the PD-analysis, urinary IL-1ra

represents the most sensitive and uniform response after TMX-101 instillation (Figure 1). No dose-dependent differences in cytokine concentrations were measured.

**Conclusion:** TLR-7 agonists are effective in urothelial carcinoma in preclinical research. In Part II of this Phase I study, the EBD could not be determined since no subject achieved complete response. The safety of TMX-101 has been demonstrated in 23 patients (Part I: 16 + Part II: 7). IL-1ra could be valuable as urinary biomarker in future developments. New doses, and other NMIBC-subgroups, should be tested to define the EBD. A pilot study in CIS patients is currently ongoing and results are expected shortly.

**MP-08.08****Is Regular Follow-Up Necessary in Patients with Low Grade Bladder Tumors after 5-Years of Disease Free Status?**

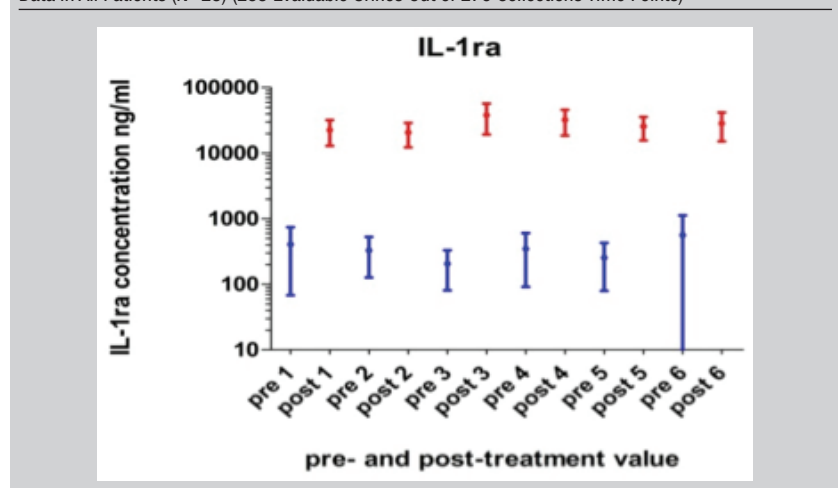
Palou J, Emiliani E, Gausa L, Gaya J, Rodríguez O, Gavrillo P, Kanashiro A, Parada R, Esquena S, Villavicencio H

*Fundació Puigvert, Barcelona, Spain*

**Introduction and Objectives:** There is no evidence enough to assess the evolution of low grade bladder tumors after five years without recurrences. The highest rate of recurrences has been reported during the first two years after the initial treatment, moreover one of the main reasons for a strict follow-up is the risk of progression. Current European guidelines recommend the follow-up for low grade tumors using minimally invasive tests. The aim of this study was to assess the incidence and evolution of the recurrences of NMIBC and 5-years of disease free status (low and intermediate risk tumours included).

**Materials and Methods:** We performed a retrospective study including 704 patients with grade I-II NMIBC. The mean age was 65.8 years and

**MP-08.07, Figure 1.** Mean and 95%-CI of Pre- and Post-Instillation Concentration of Pooled IL-1ra: Data in All Patients (N=23) (253 Evaluable Urines Out of 276 Collections Time Points)





MP-08.08, Table 1.

| Year                        | 1    | 2    | 3    | 4    | 5    | >5   |
|-----------------------------|------|------|------|------|------|------|
| Overall recurrence (%)      | 11.6 | 24.9 | 32.1 | 35.5 | 37.4 | 40.4 |
| Recurrence distribution (%) | 28.6 | 61.5 | 79.3 | 87.6 | 92.2 | 100  |

the median follow-up was  $65.7 \pm 34$  months. Patients with CIS associated were excluded. The evolution, current disease status and the recurrences after 1, 2, 3, 4, 5 and >5 years were analyzed in patients with  $\geq 5$  years follow-up.

**Results:** A total of 449 patients had  $\geq 5$ -year follow-up recorded, 231 (79.3%) of them, free of tumour at first 3-year period. Only 22 patients (3.1% of all the series and 7.7% of all patients that presented recurrence) had their recurrences after a 5-year disease free period. Moreover all of them were low grade.

**Conclusions:** After a 5-year period of disease free status, the incidence of recurrence of NMIBC grade I-II (low and intermediate risk) is low. In this group, it is reasonable to perform minimally invasive test or focus the follow upon clinical symptoms.

#### MP-08.10

##### Psoas Muscle Mass as a Predictor of Survival in Patients Who Undergo Radical Cystectomy

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**Introduction and Objectives:** Morphomic measures of abdominal muscle mass are reportedly accurate predictors of post-operative outcome in patients who undergo major abdominal surgeries. We evaluated the association between CT scan-based total psoas muscle mass area (TPA) and operative outcome of radical cystectomy (RC) and urinary diversion reconstruction.

**Materials and Methods:** A total of 442 bladder cancer patients who underwent RC and urinary diversion reconstruction at Department of Urology, University of Michigan between 2007 and 2012, were enrolled. Cross-sectional areas of the left and right psoas muscles at the level of L4 were determined and summed to generate TPA. Outcome measures were short-term outcome including length of hospital stay (LOS), 30-day readmission rate (RR) and complication rate (CR); Intermediate-term outcome including 90-day RR and CR as well as 6-month mortality; and long-term outcome including 3-year overall survival (OS). Multivariate logistic and Cox proportional-hazards regression models were used to determine predictors of operative outcome and OS, respectively.

Predictive accuracy of different survival models were also compared using Receiver Operating Characteristic (ROC) curves.

**Results:** Mean age of participants and follow-up time was 66 y/o (range, 31–91) and 2.3 yrs (range, 0.04–6.3), respectively. Mean TPA was 2363 mm<sup>2</sup> (range, 552–4322). Mean LOS was 10 days. 30-day RR and CR were 63/466 (13.5%) and 196/440 (44.5%), respectively. TPA was not associated with any early operative outcome measures. ( $P > 0.05$ ) 90-day RR and CR were 95/466 (20%) and 238/466 (51%), respectively. Thirty patients (6.4%) died within first 6-months from surgery of whom, 21 (70%) died of UCB. TPA was not associated with any intermediate operative outcome measures. ( $P > 0.05$ ) 3-year OS rate was 0.62 (range, 0.56–0.67). TPA (HR= 0.96, 95%CI: 0.93–0.98;  $P = 0.005$ ) was significantly associated with 3-year OS. Patients with TPA at highest quartile had 0.18 survival advantage over those with TPA at lowest quartile 3 years from RC. ( $P = 0.02$ ) Predictive accuracy of 3-year survival model was increased from 0.71 to 0.72 when TPA was included in the model.

**Conclusion:** Muscle mass does not seem to predict early morbidity and mortality following RC but it is an independent predictor of long-term survival; regardless of age, pathologic stage, and comorbidities.

#### MP-08.11

##### Bladder Neck Involvement Predicts both Recurrence and Progression in Non-Muscle Invasive Bladder Cancer

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**Introduction and Objectives:** About a half of non-muscle invasive bladder cancers (NMIBCs) recur and approximately 15% of these cases progress. We previously reported that the bladder neck involvement (BNI) was an independent risk factor for progression in primary NMIBC patients in retrospective and prospective cohorts (Fujii, et al., Eur Urol 1998; Kobayashi and Fujii, et al., Urol Oncol 2013). In recent years, some Cox proportional hazards models (PHM) have been proposed that handle clustered and multiple event data. The Prentice-Williams-Peterson counting process (PWP-CP) model is a conditional model that analyzes multiple ordered events by stratification. Here, we examine whether BNI is a predictive factor for recurrence and progression in NMIBC including primary and recurrent tumors.

**Materials and Methods:** Between 2000 and 2010, 410 patients who underwent transurethral resection (TUR) and were pathologically diagnosed with Ta or T1 urothelial carcinoma of the bladder were enrolled in this prospective study. We examined the significance of clinicopathological factors including BNI to predict recurrence and progression by multivariate analysis, including the PWP-CP model.

**Results:** During the study period, the 410 patients underwent TUR 761 times (368 for primary and 393 for recurrent tumors). The multivariate Cox PHM revealed that BNI was associated with both recurrence and progression. The multivariate PWP-CP model revealed that tumor number, pathologic stage, histologic grade, and BNI were associated with disease recurrence, and histologic grade 3 and presence of BNI were independent predictors for progression. Recurrence probability at 5 years was 74.7% in patients with BNI, compared to 60.2% in patients without BNI ( $p = 0.0043$ ). Progression probabilities at 5 years in patients with or without BNI were 18.7 and 5.3%, respectively ( $p < 0.0001$ ).

**Conclusion:** This study is the first to show that BNI is an independent risk factor for both recurrence and progression in NMIBC including primary and recurrent tumors.

MP-08.11, Figure 1.

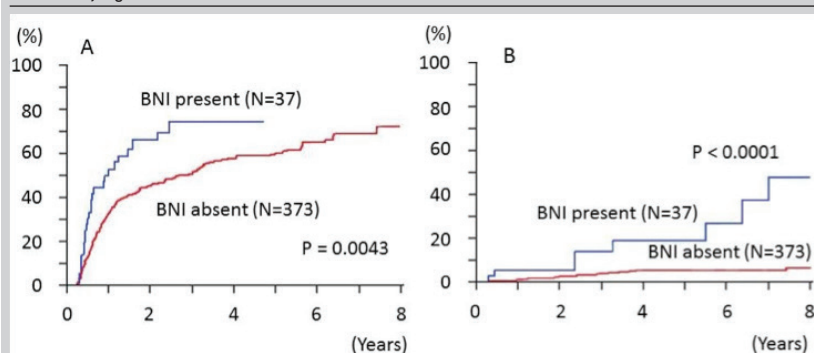


Figure. Time to recurrence (A) and progression (B)

**MP-08.12****Radical Cystectomy for Bladder Cancer: Comparison of Early Surgical Complications after Laparoscopic, Open and Combined Surgery in a Single-Site Cohort**

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**Introduction and Objectives:** Radical cystectomy (RC) is the standard method for treatment of muscle-invasive and locally advanced bladder cancer (BCa). Several less invasive approaches have been suggested recently, including totally laparoscopic radical cystectomy (LRC), hand-assisted LRC (HALRC), and robotic cystectomy. However, despite significant improvements in surgical techniques, the overall occurrence of perioperative complications is still high. Our purpose in this study was to evaluate peri- and postoperative morbidity and functional results of LRC in a single-site cohort of patients, comparing it with the standard open approach (ORC) and laparoscopic cystectomy with open urinary diversion (HALRC).

**Materials and Methods:** A prospective analysis was performed in 51 patients with muscle-invasive and locally advanced BCa, who underwent RC between February 2012 and March 2014, in N.N. Petrov Research Institute of Oncology, Saint-Petersburg. The final cohort included 21 ORC, 21 LRC and 9 HALRC patients. The mean age of patients was 64 (38-81) years, and it did not differ in all groups. The pathological stages were similar in all groups. Multivariable logistic and median regression was performed to evaluate the surgery duration, perioperative and postoperative (30-d and 90-d) complications according to Clavien classification, readmission rates, and length of stay (LOS) – both overall and in ICU.

**Results:** The duration of surgery for LRC and HALRC was longer than that for ORC (398 min vs. 468 min vs. 243 min, respectively). Despite that, there was no statistically significant influence of surgery type on intraoperative complications - 14.3% in ORC group, 11.1% in HALRC and 4.7% in LRC patients. Major complication rates (Clavien grade  $\geq 3$ ; 23.8% vs. 33.3% vs. 19.4%) were similar in all groups. However, LRC had 4.0-fold lower rate of minor complications (Clavien grade 1-2) as compared to ORC (4.7% vs. 19.0%). LRC had a significantly shorter LOS (27.8 d vs. 32.6 d vs. 22.6 d in ORC, HALRC and LRC groups, respectively), but there were no significant differences in ICU stay (5.1 d vs. 3.1 d vs. 2.1 d). The morbidity rate was equal to one patient per group (medium rate - 5.8%). The common transfusion rate during and after surgical intervention was 19.6% and was higher in ORC group (33.3% vs. 4.7% in LRC); moreover, intraoperative bleeding was lower in minimally invasive techniques - the average volume of

blood loss was 285 ml in LRC and did not vary between HALRC and ORC groups - 468 and 577 ml, respectively. The greatest diversities were observed in the occurrence of gastrointestinal complications (firstly - ileus) with significantly better outcomes in LRC patients - 14.2%, as compared to 47.6% and 55% in ORC and HALRC, respectively.

**Conclusions:** We found that LRC is a safe method associated with lower blood loss, decreased rate of postoperative ileus and reduced LOS as compared with ORC. Using a population-based cohort, we found that laparoscopic surgery for bladder cancer reduces the occurrence of minor complications (mainly due to decreased bleeding and gastrointestinal complication rate) with no influence on major complications.

**MP-08.13****Radical Cystectomy (RC) in Octogenarians: Long-Term Experience of Two High Volume Institutions**Brausi M<sup>1</sup>, Selli C<sup>2</sup>, De Luca G<sup>1</sup>, Rossi A<sup>2</sup>, Peracchia G<sup>1</sup>, Viola M<sup>1</sup><sup>1</sup>Dept. of Urology, AUSL Modena, Carpi, Italy;<sup>2</sup>Dept. of Urology, University of Pisa, Pisa, Italy

**Introduction and Objectives:** The objectives of the study were to evaluate morbidity, overall survival (OS) and disease specific survival (DSS) of radical cystectomy (RC) in octogenarians in two urological centers.

**Materials and Methods:** From 2000 to 2012, 1010 patients with infiltrative or recurrent high grade T1 TCC of the bladder received RC and urinary diversion in 2 Italian institutions. A total of 170 out of 1010 patients (16.8%) were 80 years old or older. The mean age was 83.2 years: M/F: 128/42. ASA score was used for classifying preop. risk. ASA 2: 56/170 (33%), ASA 3: 75/170 (44.1%), ASA 4: 39/170 (23%). There were 113 patients out of 170 (66.5%) who received uretero-cutaneostomy (UCS) as a diversion while 42/170 (25%) had Bricker, 14/170 patients (8.3%) had an orthotopic neobladder, 1/170 patient had an ureterosigmoidostomy (0.5%). P stage was: T0: 1 patient (0.5%). Tis+Ta-T1: 25/170 patients (14.7%); T2b: 35/170 (20.6%); T3a: 32/170 (19%); T3b: 45/170 (26.5%); T4: 32/170 (19%). Grade G3: 153/170 pts (90%), G2: 17/170 (10%). Thirty three patients did not receive pelvic lymph adenectomy (salvage RC). Twenty nine out of 137 patients (21%) were N+(pT3-T4); 125/170 patients (73.5%) were in intensive care Unit (ICU) for 1-6 days. Eighty one out of 170 patients (47.6%) were transfused. The average blood unit received was 3.5 U.

**Results:** The mean follow-up was 44.5 months (21-118 months). Peri-operative mortality was 7.6% (13/170). Mean hospital stay was 14.5 days (7-35 days). The complication rate (medical and surgical) was 43%. 8.3% of patients

required a second operation. Medical and surgical complications by ASA were: ASA2 = 11.8%, ASA 3 = (50%), ASA4 = 38% respectively. The medical complication rate by surgical approach: extraperitoneal = 40.4%, peritoneal = 27%; surgical complication rate: extraperitoneal route = 12.8%, trans-peritoneal approach = 30% (p < 0.001). Complication rate by diversion: UCS = 26% Bricker = 49.2%, Orthotopic = 45% (p < 0.001). OS: After 1 year = 60%, 2 years = 43.6%, 3 year = 40%. DSS was 63.3% at 1 year, 51.2% at 2 years and 50% after 3 years.

**Conclusions:** The results of our study support the use of RC in octogenarians. Mortality and complications were acceptable. Major complications were correlated with high ASA score (3-4), type of urinary diversion (Bricker) and surgical approach (intra-peritoneal).

**MP-08.14****Robotic Radical Cystectomy with Intracorporeal Urinary Diversion: Impact on an Established Enhanced Recovery Protocol**

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**Introduction and Objectives:** Enhanced recovery protocols (ERP's) incorporating evidence based interventions have been demonstrated to accelerate patient recovery after radical cystectomy. The ERP was introduced at our institution in 2004 and focuses on reduced bowel preparation and standardised feeding and analgesic regimens. The present study evaluates the impact of the introduction of robotic radical cystectomy with intracorporeal reconstruction on an already established ERP.

**Methods and Materials:** We examined the impact of the introduction robotic radical cystectomy with intracorporeal reconstruction to our ERP. Data was obtained from our prospectively updated database. A total of 112 patients were examined, 56 open radical cystectomies and 56 robotic assisted cystectomies. The primary outcomes measured were inpatient stay and complication rate.

**Results:** The demographics of the two groups showed no significant difference in age, gender distribution, American Society of Anesthesiologists grade. A significant reduction in total complication rate was observed in the robotic cystectomy group versus the open group (21% vs. 31%, p < 0.001). In addition the median inpatient stay for the robotic group was 7 days versus 13 days (p < 0.001).

**Conclusions:** Introduction of robotic radical cystectomy and intracorporeal reconstruction to an established enhanced recovery protocol improves peri-operative morbidity and significantly reduces in-patient stay.

**MP-08.15****Enhancing Outcomes: The Role of an Enhanced Recovery Programme in Robotic Assisted Radical Cystectomy**Nair R<sup>1</sup>, Pai A<sup>1</sup>, Ayres B<sup>1</sup>, Sooriakumaran P<sup>2</sup>, Perry M<sup>1</sup>, Issa R<sup>1</sup><sup>1</sup>St. Georges Hospital, London, UK; <sup>2</sup>University of Oxford, Oxford, UK

**Introduction and Objectives:** We have previously reported on ERP in open radical cystectomy (ORC) and shown that it is safe and not associated with an increase in complications or readmissions. Further, it is associated with a reduction in ICU and hospital stay and duration of postoperative ileus. The recent introduction of Robotic Assisted Radical Cystectomy (RARC), with its perceived benefit of minimal invasiveness, has led us to question whether patients who have their radical cystectomy in the environment of ERP would also experience this added benefit. In this study we compare the peri-operative results of the last 50 RARC patients with the last 50 ORC. The same ERP protocol was implemented in all patients.

**Materials and Methods:** From 2010, RARC was offered as the first line treatment for all bladder cancer patients with an indication for bladder extirpation. A prospective review of the last 50 ORC and 50 RARC patients were compared. Pre-operative (age, sex, comorbidity), intraoperative (diversion type, fluid loss, blood transfusion, conversion and number of lymph node dissections-LND) and post-operative (length of stay LOS, nodal yield and pathological T stage) variables were recorded. Thirty-day complication rates were documented according to the Clavien-Dindo classification.

**Results:** The 2 groups did not differ significantly in their preoperative variables or pathological stage. Patients in the RARC cohort were more likely to have continent diversion, less intraoperative fluid loss (RARC: 533mls, ORC: 1700mls,  $p=0.0064$ ), shorter LOS (RARC: 9 days, ORC: 10 days,  $p=0.03$ ), and higher lymph node yield (RARC 13.5, ORC 12,  $p=0.01$ ). The RARC patients had significantly lower transfusion and overall 30-day complications rates ( $p=0.0013$ ).

**Conclusions:** We have shown that robotic surgery offers an added value to patients undergoing radical cystectomy for bladder cancer in addition to the benefits gained from enrolling in an Enhanced Recovery Programme. This is likely due to the minimally invasive nature of robotic surgery, and thus an attenuation of its physiological insult, which is the cornerstone of ERP theory.

**MP-08.16****Function of the Orthotopic Ileal versus Sigmoid Neobladders: Continence and Urodynamic Evaluation**Mahmoud M<sup>1</sup>, Mousa E<sup>1</sup>, Farid M<sup>1</sup>, El-helalyH<sup>1</sup>, Metwally M<sup>1</sup>, Khalaf I<sup>1</sup>, Abdel-Khalik M<sup>1</sup>, Kamal M<sup>2</sup><sup>1</sup>Al-Azhar University, Cairo, Egypt; <sup>2</sup>El-Fayoum Faculty of Medicine, El-Fayoum, Egypt

**Introduction and Objectives:** To comparatively evaluate the voiding functional results of sigmoid versus W-ileal neobladder reconstruction after radical cystoprostatectomy.

**Materials and Methods:** Since Jan., 2004, to Feb. 2014 we have performed 162 radical cystectomy; of them, 38 patients with orthotopic diversion were available for evaluation with a mean follow-up of 34 months (range 9-113). We have performed clinical, urodynamic, radiographic and metabolic evaluation twice during the first year and yearly thereafter. Flowmetry, cystometry and pressure-flow test were performed. The upper urinary tract was evaluated by renal ultrasound. Entero-ureteral reflux, entero-urethral stricture and pouch capacity were evaluated by cysto-urethrography. Urethral pressure profile was done in diurnal incontinence.

**Results:** Clinically, frequency of micturition by day was less in the IN group (5 times/d), while in the SN group was (7 times/d). Day time continence was achieved in 85% of IN patients and in 77% of SN patients, while 82% IN patients and 58% of SN patients were continent by night respectively. Urodynamically the average reservoir capacity of the SN is (375 ml) was lower than the IN (520 ml) with filling pressure of 14 cm H<sub>2</sub>O and 20 cm H<sub>2</sub>O for the IN and SN group respectively. The majority of patients void by the Valsalva maneuver yet achieved good peak flow rates (SN group 15.5 ml/sec and IN group 11 ml/sec mean flow) with patients in both groups empty their reservoirs adequately with post void residual (PVR) less than 10% of the void volume. Maximum urethral closure pressure on urethral pressure profile was  $44.2 \pm 1.6$  (range, 32 to 71) cm water in patients with daytime continence, while lower in patients who were incontinent during the day and night; ( $16 \pm 2.1$  cm water).

**Conclusion:** Completely detubularized sigmoid bladder is less compliant and more contractile, yet; it's functionally and urodynamically comparable to detubularized Ileal neobladder. The total high incidence of incontinence in our study may be due to iatrogenic sphincter injury or non-nerve sparing procedures.

**MP-08.17****Implications of Circulating Tumor Cells on Adjuvant Chemotherapy Decision-Making in Patients with Urothelial Carcinoma of the Bladder Treated with Radical Cystectomy**Soave A<sup>1</sup>, Riethdorf S<sup>2</sup>, Weisbach L<sup>1</sup>, Minner S<sup>3</sup>, Engel O<sup>1</sup>, Hansen J<sup>1</sup>, Chun F<sup>1</sup>, Dahlem R<sup>1</sup>, Pantel K<sup>2</sup>, Fisch M<sup>1</sup>, Rink M<sup>1</sup><sup>1</sup>Dept. of Urology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany;<sup>2</sup>Institute of Tumorbiology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany; <sup>3</sup>Dept. of Pathology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany

**Introduction and Objectives:** Circulating tumor cells (CTC) in urothelial carcinoma of the bladder (UCB) patients treated with radical cystectomy (RC) are predictors for unfavorable outcomes. Administration of adjuvant chemotherapy is associated with improved survival in some UCB patients. The aim of this study is to evaluate if determination of CTC may help in patient counseling regarding administration of adjuvant chemotherapy.

**Materials and Methods:** We prospectively collected data of 181 UCB patients treated with RC without neoadjuvant chemotherapy at an academic center between 2007 and 2012. Preoperatively collected blood samples (7.5 ml) were analyzed for CTC using the CellSearch<sup>®</sup> system (Veridex, USA). We correlated clinicopathologic parameters and outcomes with administration of adjuvant chemotherapy and according to the CTC status. Cox regression models evaluated associations with disease recurrence and cancer-specific survival, respectively.

**Results:** In total, 40 patients (22.1%) had presence of CTC (mean CTC number:  $11 \pm 34$ ; median: 1) prior to RC. Adjuvant chemotherapy (86.9% Cisplatin-based) was administered to 46 patients (25.4%) with a median number of 4 cycles (range: 1-6). Administration of adjuvant chemotherapy was significantly associated with younger age, advanced pT-stage, higher grade, lymph node metastasis, presence of LVI and positive margin status (all  $p$ -values  $\leq 0.039$ ). Patients with presence of CTC received more often adjuvant chemotherapy (CTC pos vs. neg: 35% v. 18%;  $p=0.023$ ). At a mean follow-up of 30 months, CTC presence was associated with disease recurrence and cancer-specific mortality (both  $p<0.001$ ), respectively, in patients without adjuvant chemotherapy. In contrast, in patients who received adjuvant chemotherapy CTC status was not associated with outcomes. In multivariable analysis that adjusted for age, pT-stage, tumor grade, lymph node metastasis, LVI and margin status, presence of CTC was an independent predictor for disease recurrence (HR 3.94; 95%CI 1.77-8.77;  $p=0.001$ ) and cancer specific mortality (HR 3.83; 95%CI 1.56-9.38;  $p=0.003$ ), respectively, in patients without adjuvant chemotherapy administration.

**Conclusions:** Presence of CTC is a strong predictor for outcomes in UCB patients treated with RC without adjuvant chemotherapy. Thus, CTC status may be used for patient counseling and decision-making regarding further therapies in patients who otherwise would not be counseled regarding adjuvant therapies. Moreover, adjuvant chemotherapy may abrogate the impact of CTC on survival.



## MP-08.18

**Adjuvant Cisplatin-Based Combination Chemotherapy for Lymph Node-Positive Urothelial Carcinoma of the Bladder following Radical Cystectomy: A Retrospective International Study of More than 1500 Patients**

Lucca I<sup>1,2</sup>, Rouprêt M<sup>3</sup>, Kluth L<sup>4</sup>, Rink M<sup>4</sup>, Tilki D<sup>5</sup>, Fajkovic H<sup>1</sup>, Kassouf W<sup>6</sup>, Hofbauer S<sup>1</sup>, De Martino M<sup>1</sup>, Karakiewicz P<sup>7</sup>, Briganti A<sup>8</sup>, Trinh Q<sup>9</sup>, Seitz C<sup>1</sup>, Fritsche H<sup>10</sup>, Burger M<sup>10</sup>, Lotan Y<sup>11</sup>, Kramer G<sup>1</sup>, Shariat S<sup>1,11,12</sup>, Klatte T<sup>1</sup>

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**Introduction and Objectives:** To compare outcomes of patients with lymph node-positive urothelial carcinoma of the bladder (UCB) treated with or without adjuvant cisplatin-based combination chemotherapy (AC) after radical cystectomy (RC).

**Materials and Methods:** We retrospectively analyzed 1,523 patients with lymph node-positive UCB, who underwent RC with bilateral pelvic lymph node dissection. All patients had no evidence of disease after RC. AC was administered within 3 months. Competing-risks models were applied to compare UCB-related mortality.

**Results:** Of the 1523 patients, 874 (57.4%) received AC. The cumulative 1-, 2- and 5-year UCB-related mortality rates for all patients were 16%, 36% and 56%, respectively. Administration of AC was associated with an 18% relative reduction in the risk of UCB-related death (SHR 0.82,  $p=0.005$ ). The absolute reduction in mortality was 3.5% at 5 years. The positive effect of AC was detectable in patients  $\leq 70$  years, in women, in pT3-4 disease, and in those with a higher lymph node density and lymphovascular invasion. This study is limited by its retrospective and non-randomized design, selection bias, the absence of central pathologic review and lack in standardization of lymph node dissection and cisplatin-based protocols.

**Conclusion:** AC seems to reduce UCB-related mortality in patients with lymph node-positive UCB after RC. Younger patients, women and those with high risk features such as pT3-4 disease, a higher lymph node density and lymphovascular invasion appear to benefit most. Appropriately powered prospective randomized trials are necessary to confirm these findings.

## MP-08.19

**Impact of Neoadjuvant Chemotherapy on Complications of Minimally Invasive Radical Cystectomy**

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**Introduction and Objectives:** Neoadjuvant chemotherapy before minimally invasive radical cystectomy (MIRC) is considered a standard of

care in muscle invasive or recurrent, high-risk, non-muscle invasive bladder cancer. We evaluated the impact of neoadjuvant chemotherapy on morbidity and mortality after MIRC.

**Materials and Methods:** Using a prospectively maintained, single institution database, we evaluated 135 patients who underwent MIRC (laparoscopic,  $n=100$ ; robotic,  $n=35$ ) between 2007 and 2013 with at least 90 days of follow-up. Complications were analysed and graded according Clavien's classification system. Logistic regression models were used to evaluate the impact of neoadjuvant chemotherapy on post-operative complications. Kaplan-Meier methods with log rank test were used for cancer-specific survival probabilities and differences between each group (radical cystectomy with or without neoadjuvant chemotherapy).

**Results:** The median age of patients was 66 years of age and 86% had a Charlson index  $\geq 2$ . Of them, 62 received neoadjuvant chemotherapy (54.5% MVAC). Overall, 119 patients (88%) developed 179 complications, mainly of infectious (63.7%) or gastro-intestinal (28.1%) origin within 90 days of surgery. The most frequent grade of complications was grade 2 ( $n=55$ ), followed by grade 1 ( $n=38$ ) and grade 3-4 ( $n=22$ ); three patients (all without neoadjuvant chemotherapy) died before day 90 after cystectomy. Neoadjuvant chemotherapy had no impact on the incidence of post-operative complications but was associated with less positive nodes ( $p=0.004$ ) compared to patients without neoadjuvant chemotherapy. In univariate and multivariate analysis, higher T stage was correlated with an increased incidence of complications (Clavien 2 or more). Median duration of follow-up was 17.2 months. Overall survival rates were 83% and 80% at 2 years in patients with and without neoadjuvant chemotherapy, respectively.

**Conclusion:** Neoadjuvant chemotherapy does not have an impact on postoperative morbidity nor mortality. Longer follow-up is needed to evaluate the impact of neoadjuvant chemotherapy on oncologic outcomes.



## Moderated Poster Session 9

### Stones: Surgical Management and New Technologies

Monday, October 13  
1435-1600

#### MP-09.01

##### External Validation of S.T.O.N.E Nephrolithometry Scoring System

Noureddin Y<sup>1,2</sup>, Elkoushy M<sup>1,3</sup>, Andonian S<sup>1</sup>  
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**Introduction and Objectives:** To perform external validation of S.T.O.N.E Nephrolithometry as an emerging surgical scoring system of patients undergoing percutaneous nephrolithotomy (PCNL).

**Materials and Methods:** All PCNL cases performed by a single endourologist (SA) from 2009 till 2013 at McGill University Health Center were reviewed. Out of 185 cases, 155 were chosen in this study after exclusion of 2<sup>nd</sup> look PCNL cases and cases where the percutaneous access was established by interventional radiologist. CT scans of these patients were reviewed for the five parameters of the S.T.O.N.E Nephrolithometry scoring system (stone size, tract length, obstruction, number of involved calyces and the stone essence). The score was then calculated and correlated with the stone free status (as the main goal of PCNL), in addition to the estimated blood loss (EBL), operative time and the length of hospital stay (LOS). **Results:** A total of 155 cases were included. The mean age was 54.9±1.2 (17-85), with 100 (64.5%) males and 55 (35.5%) females. The mean stone score was 7.67±0.1 (5-13), with mean stone volume of 609.8±48.4 (25-4030) mm<sup>3</sup>, mean Hounsfield unit of 887.7±25.3 (222-1766), mean tract length of 97.3±1.9

(53-175), mean BMI of 26.9±0.5 (17.2-51), mean operative time of 100.1±2.8 (60-240) min and mean LOS of 4.2±0.3 (1-18). The overall stone free rate after the primary procedure was 71.6%. The S.T.O.N.E score significantly affected the stone-free status ( $p=0.008$ ) and the EBL ( $p=0.003$ ). There was good correlation between the S.T.O.N.E Nephrolithometry score and operative time ( $r=0.4$ ;  $p=0.00$ ) and LOS ( $r=0.3$ ;  $p=0.001$ ). Therefore, the higher the S.T.O.N.E Nephrolithometry score, the longer the operative time, the higher the estimated blood loss, the longer the LOS, and the lower the stone-free status.

**Conclusion:** This study confirms external validation of the S.T.O.N.E Nephrolithometry scoring system as a preoperative predictor of PCNL outcome.

#### MP-09.02

##### Utility of the Guy's Stone Score and Nephrolithometric Nomogram Based on Computed Tomographic Scan Findings for Predicting Percutaneous Nephrolithotomy Outcomes

Valente P, Castro H, Vila F, Borges R, Lindoro J  
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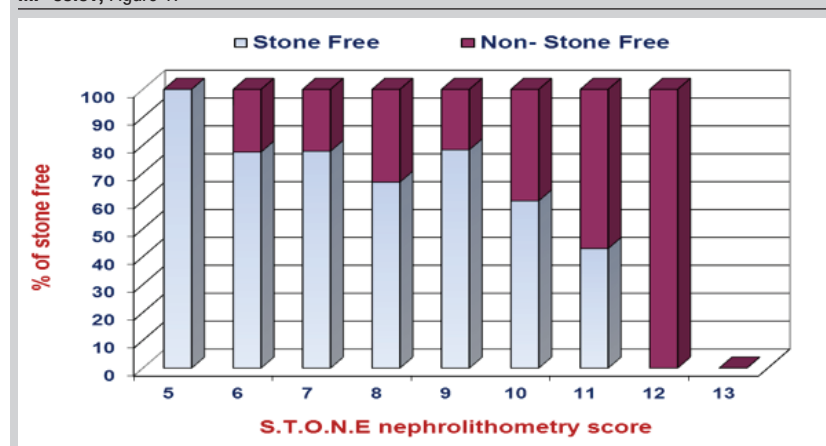
**Introduction and Objectives:** The aim of this work is to evaluate the utility of Guy's Stone Score and Nephrolithometric Nomogram of CROES based on CT scan findings for predicting percutaneous nephrolithotomy outcomes.

**Materials and Methods:** The authors reviewed retrospectively all cases of PCNL performed in our center. Using CT scan results, all cases were classified according Guy's Stone Score (GS) and it was calculated the chance of treatment success using the Nephrolithometry Nomogram of CROES (NNC). Outcomes and complications (Clavien Classification System—CS) were analysed. Success was defined as no residual fragments or residual stone fragments < 4mm at follow-up imaging. The data analyses were made using IBM SPSS Statistics 20.0.

**Results:** Since September 2008 to June 2013, were performed 81 PCNL for primary treatment of renal stones. Of them 24.7% were classified as GS1, 4.9% as GS2, 44.4% as GS3 and 25.9% as GS4. Among the characteristics of the groups, the authors found significative differences in mean length of calculus (GS1=22.8mm; GS2=29.4mm; GS3=36.7; GS4=47.1mm,  $p<0.0001$ ). The mean of operative time was different among groups (GS1=159.4min.; GS2=126min.; GS3=139.5min.; GS4=174.7min.), with statistically significance ( $p=0.013$ ). The success rate after first PCNL was 50% in GS1, 100% in GS2, 52.8% in GS3 and 47.6% in GS4, ( $p=0.169$ ). The rate of repeat PCNL as adjuvant treatment was 15% in GS1, 11% in GS3 and 19% in GS4 ( $p=0.709$ ). The global success rate after adjuvant therapies was 85% in GS1, 100% in GS2, 91.7% in GS3 and 76.2% in GS4 ( $p=0.359$ ). The rate of relevant complications was 5% in GS1 (blood transfusion-CSII), 0% in GS2, 8.3% in GS3 (renal abscess-CSIIIb, Pyelonephritis-CSII, blood transfusion-CSII) and 19% in GS4 (Pyelonephritis-CSII, Urosepsis-CSIVb and 2 blood transfusion-CSII). Using the NNC the cases were divided into two groups (A:<60% vs. B:≥60% chance of treatment success). Group A included 45.7% of cases. Only calculus length had statistically significant difference (A:48.8mm vs. B:26.9mm,  $p=0.0001$ ). The complications rate was A=16.5% vs. B=4.5%,  $p=0.084$ . The success rate after first PCNL was A=48.6% vs. B=56.8%,  $p=0.508$ . The global success rate after adjuvant therapies was 75.7% in Group A vs. 95.5% in Group B with statistically significance ( $p=0.011$ ).

**Conclusions:** The Guy's Stone Score and Nephrolithometric Nomogram of CROES based on CT scan findings are good instruments for predicting PCNL outcomes. The GS4 and the Group of <60% of chance of treatment success using the referred Nomogram are associate with higher rate of complications and a lower global success rate. These prognostic tools may be useful in patient counseling, optimize treatment selection and plan surgery.

MP-09.01, Figure 1.



#### MP-09.03

##### Diagnostic Performance of Low-Dose Non Enhance CT with Iterative Reconstruction in Diagnosis of Urolithiasis

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**Introduction and Objectives:** Low-dose CT (LDCT) is a promising option for diagnosing urinary stones, but it can substantially increase image noise. We evaluated the efficacy of iterative reconstruction (IR) technique for reducing image noises on LDCT and the diagnostic

performances especially in urologic perspective.

**Materials and Methods:** A total of 197 stones from 119 patients who performed non-enhanced CT with both standard dose and LDCT with IR (LDCT-IR) were enrolled. Interpretations were performed in the two scans for stone characteristics (size, volume, location, housefield unit(HU), skin-to-stone distance (SSD)), radiation (dose-length product (DLP), effective dose (ED)), image noise (objective, subjective). Inter-observer agreements were assessed between urologist and radiologist with kappa analysis. All comparisons were performed after dividing all stones into 3 groups; all size, larger than 3mm and less than 3mm.

**Results:** No statistical differences were found in stone characteristics between the two scans except HU in the all size group. The average DLP and ED were 394.94mGy and 5.92mSv in CCT, and 92.88mGy and 1.39mSv in LDCT-IR, respectively. The average dose reduction was 76.6% ( $p < 0.001$ ). In the LDCT-IR, the sensitivity, specificity, positive predictive value, negative predictive value, accuracy for the urologist and radiologist reviewer have no significant differences for diagnosing stones of larger than 3 mm ( $p > 0.05$ ). Inter-observer agreement of LDCT-IR between the two reviewers was high with kappa values ranging from 0.901 to 1.000 in all groups. Objective image noise was higher on LDCT-IR ( $p < 0.01$ ), but no significant differences of subjective noise was found.

**Conclusions:** LDCT-IR showed significant radiation reduction while maintaining image quality as an attractive option in urologic perspective to diagnosis of urinary stones.

#### MP-09.04

##### A Prospective Randomized Trial about Music Listening to Decrease Discomfort during SWL

**Bosio A, Destefanis P, Alessandria E, Buffardi A, Bisconti A, Gontero P, Fontana D, Frea B**  
*Dept. of Urology, Città della Salute e della Scienza, Molinette Hospital, Turin, Italy*

**Introduction and Objectives:** SWL can cause discomfort and pain. Reducing pain during SWL is also important to limit pain-induced movements and excessive respiratory excursions. The objective of our study was to evaluate if music listening can have a role in reducing patient discomfort during SWL.

**Materials and Methods:** We proposed to all patients treated by SWL from February 2011 to October 2012 to participate in a prospective randomized trial. A total of 3500–4000 SW were delivered at 20–24 kV according to individual tolerance. Patients randomized in Group A listened to music during the first part of the treatment (first 1800 SW), patients in Group B during the second. Patients were asked to fill in a questionnaire and a visual analogue scale (VAS) for pain concerning each part of the treatment.

**Results:** All the 70 patients (34 in Group A, 36 in Group B) appreciated the chance to listen to music during SWL and would choose music listening again in case of further treatments. Ninety four percent considered music useful to relieve pain or discomfort. Mean VAS score decreased during music listening in Group A ( $4 \pm 2.17$  vs.  $5.24 \pm 2.36$ ;  $p = 0.03$ ), but not in Group B. The significant increase in mean VAS score during the second part of the treatment observed in Group A ( $p = 0.01$ ), was not observed in Group B ( $p = 0.15$ ). The need for analgesics was reduced during the listening part of the treatment in both groups (from 29% to 26% in Group A and from 22% to 19% in Group B), although not significantly ( $p = 0.41$  and  $p = 0.38$ ). SW number in the second part of the treatment was higher in Group B (mean 2097 SW vs. 2056 SW), but not significantly ( $p = 0.42$ ).

**Conclusion:** The great majority of patients appreciated music listening during SWL and considered music helpful in relieving pain. During music listening, mean VAS score decreased in the first part of the treatment and did not increase in the second, the need for analgesics tended to decrease and there was a trend to tolerate a higher number of SW.

#### MP-09.05

##### The Effect of Kidney-Protective Treatment Protocols on Clinical Renal Injury during Extracorporeal Shockwave Lithotripsy

**Ng C, Luke S, Lee W, Teoh J, Hou S**  
*The Chinese University of Hong Kong, Hong Kong, China*

**Introduction and Objectives:** We would like to assess the effects of different treatment protocols on acute renal injury in human subjects during extracorporeal shockwave lithotripsy (ESWL).

**Materials and Methods:** Adult patients with renal stone  $\leq 15$ mm and planned for ESWL were recruited and randomized to follow one of the protocols, (1) 80% power (19.2kV) from the beginning till the end of treatment (Control); (2) the first 100 shocks at 40% power (9.6kV), followed by SWs at 80% power till the end; (3) the first 100 shocks at 40% power, followed by a 3-minute pause and then further SWs at 80% power till the end; (4) the first 100 shocks at 80% power, followed by a 3-minute pause and then further SWs at 80% power till the end. MRI or NCCT was performed on Day 2 after ESWL to assess renal haematoma. Spot urine samples were collected before (baseline), immediately (post-treatment), then at Day-2, Week-6 and Week-12. Urinary markers N-acetyl-  $\beta$ -D-glucosaminidase (NAG), albumin, Neutrophil gelatinase-associated lipocalin (NGAL) and Interleukin-18 (IL-18) for comparison.

**Results:** A total of 320 patients were recruited.

The baseline and treatment parameters of the 4 groups were comparable. Results showed no significant difference between groups in the incidence of haematoma. However, there was a trend that Group 3 and 4 patients tend to have lower rate of haematoma. Urinary NAG, albumin and IL-18 levels increased significantly after ESWL compared to baseline levels in all groups ( $p < 0.005$ ). No significant difference was observed in the level of markers between groups, however, albumin levels dropped back to normal more rapidly in Group 3 and 4 than Group 1 and 2.

**Conclusions:** No significant decrease of renal injury was observed in patients treated with low energy shockwave pretreatment protocol and pause-protection protocol modified ESWL compared to those received standard protocol. However, pause-protection protocol (Group 3 & 4) treated patients have shown a trend of decreased incidence of haematoma and accelerated normalization of acute kidney injury markers.

#### MP-09.06

##### Impact of Shock Wave Lithotripsy on Renal Function

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**Introduction and Objectives:** To evaluate if shock wave lithotripsy (SWL) impacts renal function, and to assess any cumulative effect of repeated treatments.

**Materials and Methods:** A total of 324 patients who had at least one SWL with serum creatinine measurements before and after, from 2002-2013, were included in this study. The patients were divided into 3 groups based on number of SWL treatments (1, 2-5, >5). The estimated glomerular filtration rate (eGFR) was used as a marker of overall renal function. For each patient, the baseline eGFR was compared to values after at least 1 SWL using a paired t-test. ANOVA was used to compare any change in renal function before and after SWL. Multivariate linear regression was used to identify other variables that might impact renal function.

**Results:** There was a mean of 3.5 years between serum creatinine measurements, and patients had a mean age of 49 years at the time of their first treatment; 60% of patients were male. Sixteen percent of patients underwent 1 SWL treatment, 69% underwent 2-5 treatments, and 15% underwent more than 5 treatments. Overall, there was a decrease in eGFR seen after SWL, but the absolute change was small (84.1 to 81.4,  $p = 0.02$ ). There was no significant difference in change in renal function in patients treated with a single SWL compared to those with multiple treatments ( $p = 0.26$ ). Younger

females were more likely to have a reduction in renal function after SWL. Stone location, length of time between creatinine measurements, number of SWL treatments, and BMI did not impact change in renal function.

**Conclusions:** While some, especially younger females, demonstrated a small deterioration in renal function over time, the number of SWL treatments did not correlate with this change. This suggests that even multiple SWL treatments have negligible impact on global renal function.

#### MP-09.07

##### New Is Silver and Old Is Gold: Not True for Siemens Third Generation Electromagnetic Lithotripters

Choo Z, Liu Z, Tan Y, Lee Y

Tan Tock Seng Hospital, Singapore

**Introduction and Objectives:** Extracorporeal shock wave lithotripsy (ESWL) is an important modality in the treatment of urinary lithiasis. Siemens Modularis (SM) and Siemens Lithoskop (SL) are both third generation electromagnetic lithotripters from the same manufacturer with SL being a newer model with a wider focal zone and inline targeting capability which have a potential for higher stone free rates after a single session. This study aims to compare stone free rates after first session of ESWL between SM and SL.

**Materials and Methods:** Between August 2012 and September 2013, we retrospectively reviewed 100 ESWL cases performed with SM and SL each. All stones were less than 2cm and were first session ESWL. The stones

were localized, and treated with the appropriate number of shocks and energy level as per guidelines. All patients were followed-up within 30 days with the same radiological imaging used before and after the procedure. The operators were all doctors trained in operating the lithotripters.

**Results:** The mean stone size for SL in the kidney and ureter were 9.8 mm (SD 3.7) and 9.9 mm (SD 4.2) respectively with 54% located in the kidney and 46% in the ureter. The mean stone size for SM was 10.4 mm (SD 3.8) in the kidney and 10.1 mm (SD 3.5) in the ureter with 58% located in the kidney, and 42% in the ureter. Overall stone free rate for both kidney and ureter combined was 45% for SL and 20% for SM and this was significant with a p-value of 0.001.

**Conclusion:** The initial clinical experience with SL shows higher stone free rate after one session of ESWL compared to an older electromagnetic lithotripter, SM.

#### MP-09.08

##### Lessons Learned After 558 Procedures with Flex-Xc Digital Flexible Ureteroscope Multescu R, Geavlete B, Georgescu D, Geavlete P

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**Introduction and Objectives:** New digital flexible ureteroscopes offer certain advantages by comparison to their predecessors. We aimed to retrospectively analyze the ureterorenoscopic procedures performed with the Storz Flex-Xc model in order to evaluate its particularities.

**Materials and Methods:** Between May 2012 and January 2014, all the flexible ureteroscopic procedures performed with Storz Flex-Xc were analyzed. A total number of 5 ureteroscopes were used: (the first and last ones previously used in another center and 3 new ones).

**Results:** A total of 558 procedures were performed on 510 patients: first endoscope used on 62 procedures (55 patients), second one on 96 procedures (90 patients), third one on 151 procedures (139 patients), the fourth on 159 procedures (143 patients) and the last one, still operational on 90 procedures (83 patients). Ureteral access sheath was used in 71% of the cases. The endoscopes were used for 51, 67.1, 107.7, 107.2 and 69 hours, respectively. Difficulties to effectively access the stone were encountered in 0.4% of the cases. Overall stone free rate was 92.8% after one, 96.9% after two and 97.8% after three procedures. Major repairs were needed after optical system chip failure (first endoscope), significant damages of the outer coating (second one) and severe deterioration of the deflecting mechanism (third and fourth endoscopes).

**Conclusions:** The digital Storz Flex-Xc seems to be a durable model of flexible ureteroscope. It offers excellent maneuverability and visibility, translating in great effectiveness.

#### MP-09.09

##### Retrograde Flexible Ureteroscopy: Experience on 1000 Cases

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**Introduction and Objectives:** Nowadays flexible ureteroscopy is, in many centers, a routine procedure. The aim of this study was to evaluate the indications, limits and efficacy of flexible ureteroscopy on a significant number of cases.

**Materials and Methods:** Between January 2002 and January 2014, 1000 diagnosis and treatment retrograde flexible ureteroscopic procedures were performed at "Saint John" Emergency Clinical Hospital. We retrospectively reviewed the indications, endoscopes' types, procedural efficacy and complications rates.

**Results:** A fiberoptic first generation Storz flexible ureteroscope was used in 194 cases, a digital Flex-Xc in 588 cases, a fiberoptic Wolf Cobra in 68 cases and a digital Olympus URF-V in 150 cases. A total of 9.8% of the procedures were diagnostic, 2.4% therapeutic for upper urinary tract tumors and 87.8% for pyelocaliceal lithiasis (associated or not with other pathologies such as pyelocaliceal diverticulum or infundibulum stenosis). During the diagnostic procedures inspection of the entire upper urinary tract was possible in 91% of the cases (89 patients). Stone free rate in lithiasis

MP-09.07, Table 1. Stone free rates according to lithotripter type for different stone locations

| Stone Free (ie: absence of radiologically reported evidence of targeted stones within 30 days after 1st session ESWL) | Lithoskop     | Modularis    | p-value |
|-----------------------------------------------------------------------------------------------------------------------|---------------|--------------|---------|
| Gender:                                                                                                               |               |              |         |
| Male                                                                                                                  | 26 (62%)      | 15 (75%)     | 0.4005  |
| Female                                                                                                                | 17 (38%)      | 5 (25%)      |         |
| Renal Upper pole                                                                                                      | 3/13 (23.0%)  | 7/13 (53.8%) | NA      |
| Renal Middle pole                                                                                                     | 2/8 (25.0%)   | 4/11 (36%)   |         |
| Renal Lower pole                                                                                                      | 15/33 (45.0%) | 2/34 (5.9%)  |         |
| Overall Stone Free Percentage for Renal Stones                                                                        | 37%           | 23%          | 0.091   |
| Upper ureter                                                                                                          | 19/40 (47.5%) | 7/40 (17.5%) | NA      |
| Middle ureter                                                                                                         | -             | 0/1 (0.0%)   |         |
| Lower ureter                                                                                                          | 6/6 (100.0%)  | 0/1 (0.0%)   |         |
| Overall Stone free Percentage for Ureter Stones                                                                       | 54%           | 17%          | 0.001   |
| Overall Stone Free Percentage for Both Kidney and Ureter                                                              | 45%           | 20%          | 0.001   |



cases was 94.8% (832 cases) after one procedure, 97.9% (860 cases) after two procedures and 98.8% (867 cases) after three procedures. Complication rate was 19.2%, 16.2% Clavien I and II, 3% Clavien III, 0% Clavien IV and V.

**Conclusions:** Retrograde flexible ureteroscopic approach is an efficient diagnostic and treatment method for upper urinary tract pathology. Technological progress during the last years modified method's indications. The safety of this procedure is very good, most of the complications being minor.

#### MP-09.10

**Predictive Factors for Stone Clearance following Ureteroscopy and Laser Lithotripsy**  
Marri R<sup>1</sup>, Housami F<sup>1</sup>, Malki M<sup>2</sup>, McIlhenny C<sup>1</sup>

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<sup>2</sup>University Hospitals of Leicester, UK

**Introduction and Objectives:** Semi-rigid ureteroscopy and flexible ureterorenoscopy have a high stone clearance rate and low morbidity and mortality rate. The aim of this study is to identify patient, stone and procedure factors predictive of stone clearance.

**Materials and Methods:** Data was collected prospectively for all emergency and elective therapeutic semi-rigid ureteroscopy or flexible ureterorenoscopy for stone disease between November 2010 and October 2013. Patient, stone and procedure parameters were recorded at the time of ureteroscopy. Stone clearance was evaluated with x ray or CT-KUB depending on the opacity of presenting stone.

**Results:** During the study period 305 patients underwent therapeutic ureteroscopy of which 243 (85.6%) were stone free at 3 months follow-up. Patient related parameters of age, gender, BMI and co-morbidities had no affect the stone clearance rate. Clearance rates were highest in lower or mid ureter 96.6% as compared to upper ureter 81.2%, renal pelvis 76.7%, upper/mid pole 78.4% or lower pole 77.8% stones ( $p=0.001$ ). There were no significant differences in number of stones, stone composition or mean density but clearance correlated with lower total stone volume ( $p<0.001$ ). Procedure related parameters showed a correlation between the stone clearance rate and lower laser energy ( $p=0.001$ ), shorter operative time ( $p=0.013$ ) and retrieval of stone fragments ( $p=0.002$ ).

**Conclusion:** Stone location and total volume impact on the rate of stone clearance, whereas stone composition and density seem less relevant in the era of laser lithotripsy. Retrieval of fragments improved clearance rates mainly for renal stones.

#### MP-09.11

**Predicting Surgical Time in Ureteroscopic Stone Surgery**

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**Introduction and Objectives:** Semirigid ureteroscopy and flexible ureterorenoscopy have a high stone clearance rate and low morbidity and mortality rate. However variation in surgical time exists which has an impact on planning surgical lists. The aim of this study is to identify patient, stone and surgeon factors predictive of the surgical time.

**Materials and Methods:** Data was collected prospectively for all emergency and elective therapeutic semi-rigid or flexible ureteroscopy for stone disease between November 2010 and October 2013. Patient, stone and procedure parameters were recorded at the time of ureteroscopy.

**Results:** During the study period, 320 patients underwent therapeutic ureteroscopy. We excluded 12 bilateral ureteroscopies and 3 ureteroscopies for encrusted stents. Data was analysed for the remaining 305 procedures. The patient's age correlated with longer surgical time ( $R=0.198$ ,  $p<0.001$ ). We found no significant effect of patients' gender, BMI or ASA grade on the surgical time. Procedures performed by trainees were slightly longer,  $50\pm18$  minutes, than those performed by consultants,  $45\pm24$  minutes. There were no significant differences in the surgical time by stone location, chemical composition or the presence of ureteric stent. However the surgical time correlated to the number of stones ( $R=0.232$ ,  $p<0.001$ ), total stone volume ( $R=0.424$ ,  $p<0.001$ ) and mean stone density ( $R=0.209$ ,  $p<0.001$ ). The surgical time to treat a single stone was  $43\pm22$  minutes compared to  $56\pm22$  minutes for multiple stones ( $p<0.001$ ). The surgical time to treat a stone volume less than 500cc (which corresponds to 10mm diameter calculus) was  $42\pm19$  minute compared to  $68\pm26$  minutes for larger stone volume ( $p<0.001$ ). The surgical time to treat stones of density less than 800HU was  $43\pm21$  minutes compared to  $54\pm24$  minutes for stones of higher density ( $p<0.001$ ).

**Conclusion:** The number of stones, total stone volume and mean stone density correlated with the surgical time for therapeutic ureteroscopy. The data is useful when planning surgical lists to estimate the time required for each procedure.

#### MP-09.12

**Routine Use of Ureteral Access Sheath during Flexible Ureteroscopy for Urolithiasis Treatment: Is It Feasible?**

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**Introduction and Objectives:** Use of ureteral access sheath during retrograde flexible ureteroscopy is still an issue of debate. The aim of our study was to evaluate the particularities of routine use of such a device.

**Materials and Methods:** We prospectively evaluated 200 patients treated by retrograde flexible ureteroscopic approach for single pyelocaliceal stones between 1 and 2 cm in size: 100 in which a Cook Flexor 10/12F ureteral access sheath was used and 100 in which the procedure was performed without it. Pre-ureteroscopy stenting necessities, intraprocedural characteristics, stone-free rates and complications were evaluated and compared.

**Results:** Among all the patients, 7% of the first group and 10% of the second one were already JJ stented. Due to difficulties to ascend the access sheath a supplementary 8% of the cases from the study group were also stented, while in the second group impossible ureteral passage of the flexible ureteroscope imposed this maneuver in 2% of the cases. Intraprocedural visibility was better when the UAS was used (mean score 4.5 vs. 3.9). Perioperative complications rate and stone free rate were similar among the two groups (13% vs. 9% and 96% vs. 97% respectively). Septic complications were significantly reduced in the study group (30% vs. 55.5%). No late ureteral stenosis was encountered.

**Conclusions:** Routine but careful use of ureteral access sheath does not increase the complication rates specific for the flexible ureteroscopic approach. It offers some clear advantages regarding access, visibility and seems to be associated with less septic complications, probably by maintaining a low pressure.

#### MP-09.13

**Preventing Retrograde Stone Displacement during Pneumatic Lithotripsy for Ureteral Calculi Using Lidocaine Jelly**  
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**Introduction and Objectives:** To assess the efficacy of lubricating jelly instillation proximal to the ureteral calculi during lithotripsy on the prevention of retrograde stone displacement and the stone-free rate.

**Materials and Methods:** A total of 110 patients with ureteral calculi of less than 2 cm were randomized into 2 groups: jelly instillation ( $n=55$ ) and controls ( $n=55$ ). Ureteroscopy was performed using a 9.8F semirigid ureteroscope. A 5F ureteral stent was advanced beyond the stone. Lidocaine jelly (2 mL) was instilled, and lithotripsy was done with a Swiss Lithoclast. A 5F ureteral catheter was left in place for 24



hours, and patients were followed up at 24 hours with radiography of the kidneys, ureters, and bladder and at 2 weeks with sonography. **Results:** Both groups were comparable in terms of mean age and stone size. Stone or stone fragment migration occurred in 18.8% of the treatment group and 44.2 of the controls, statistically significantly different ( $P=0.009$ ). The stone-free rate was 81.3 and 55.8 in the treatment and control groups, respectively. The rates did not improve after 2 weeks, and the difference was not statistically significant between the two groups ( $P=0.384$ ). The mean operative time was also comparable between the two groups.

**Conclusion:** Lidocaine jelly instillation proximal to the ureteral calculi during lithotripsy is an effective method to prevent retrograde stone displacement.

#### MP-09.14

##### **Predicting Risk of Complications following Ureteroscopy and Laser Stone Fragmentation**

Housami F<sup>1</sup>, Malki M<sup>2</sup>, Marri R<sup>1</sup>, McIlhenny C<sup>1</sup>

<sup>1</sup>Forth Valley Royal Hospital, Larbert, UK;

<sup>2</sup>University Hospitals of Leicester, UK

**Introduction and Objectives:** Semi-rigid ureteroscopy and flexible ureterorenoscopy have a high stone clearance rate and low morbidity and mortality rate. The aim of this study is to identify patient, stone and procedure factors predictive of complications or readmission.

**Materials and Methods:** Data was collected prospectively for all emergency and elective therapeutic semi-rigid ureteroscopy or flexible

ureterorenoscopy for stone disease between November 2010 and October 2013. Patient, stone and procedure parameters were recorded at the time of ureteroscopy. Complications and readmission were identified and recorded at outpatient review in 3 months.

**Results:** During the study period, 305 patients underwent therapeutic ureteroscopy. Thirty five patients (12%) had recorded complications or unplanned readmission within 3 months. These included urosepsis in 6.9%, pain in 3.0% and urinary retention in 1.3%. In terms of patient related parameters, there were no significant differences in patients' age, gender, BMI or comorbidities. Patients with positive pre-operative urine culture were more likely to develop post-operative sepsis ( $p=0.001$ ) despite appropriate antibiotic cover. As for stone related parameters, there were no significant differences in stone location, number of stones, total stone volume or mean stone density. Finally procedure related parameters showed no differences in ureteric access, ureteroscope type, operative time or presence of stent. Readmission was more likely if there were residual fragments ( $p=0.001$ ) or if stone was fragmented to dust with no retrieval ( $p=0.009$ ) usually due to difficult access.

**Conclusion:** Although risk is relatively low, urosepsis remains a significant complication post ureteroscopy. Pre-operative urine culture is essential to guide choice of antibiotic.

#### MP-09.15

##### **Intracorporeal Lithotripsy in Renal Stones Treatment: Dust or Extractable Fragments?**

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R, Geavlete P

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**Introduction and Objectives:** Pyelocaliceal calculi flexible ureteroscopic approach raises problems related with operative time, associated morbidity and costs, especially by potential endoscope damage.

**Materials and Methods:** Five series, each of 20 patients with single pyelocaliceal lithiasis were analyzed: Group I with calculi < 1 cm fragmented to dust, Group II with calculi < 1 cm with lithotripsy in fragments, Group III with calculi of 1-2 cm fragmented to dust, Group IV with calculi of 1-2 cm with lithotripsy in fragments, Group V with calculi of 1-2 cm fragmented to dust until they reached 1 cm, and lithotripsy in fragments afterwards. In all cases a flexible Story Flex-Xc ureteroscope and Ho:YAG lithotripsy were used.

**Results:** Ureteral access sheath was used in 70% of the cases. Mean stone volume in groups I and II, and groups II, IV and V were similar. Success rate in all groups was statistically similar. Mean operating time was 39 min in Group I, 21 min in Group II, 112 min in Group III, 72 min in Group IV and 51 min in Group V. Minor complications occurred in 7 cases, while a single major complication occurred in Group IV.

**Conclusions:** The optimal lithotripsy method of calculi < 1 cm seems to be in extractable fragments. Larger calculi should be fragmented to dust until they reach 1 cm and then the lithotripsy should be continued into extractable fragments.

## Moderated Poster Session 10

### Kidney & Ureteral Cancer: Treatment

Monday, October 13  
1435-1600

#### MP-10.01

##### Outcomes after Percutaneous Renal Ablation for Small Renal Masses

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**Introduction and Objectives:** Partial nephrectomy (PN) currently is considered the gold standard for treatment of T1 renal tumors. However, complications associated with PN and the high incidence of benign lesions after surgery have led to more interest in minimally invasive strategies, such as radiofrequency ablation (RFA) and cryoablation (CA). We assessed disease characteristics and oncological outcomes after RFA and CA therapies for small renal masses.

**Materials and Methods:** We retrospectively studied of 127 patients who underwent 129 percutaneous ablation procedures (86 CA and 43 RFA) for presumed renal cancers at UCSF in the period between 2005 and 2012. Sociodemographics, tumor characteristics, histology, and oncological outcomes were compared by between RFA and CA. Follow-up was monitored with serial imaging and recurrence was defined as enhancement on imaging or receipt of salvage therapy. Outcomes were evaluated with life tables and Cox proportional hazards regression adjusted for age, BMI, type of ablation procedure, maximum tumor size, and number of comorbidities.

**Results:** At first ablation, median age was 69 years for 72 (57%) men and 55 (43%) women. Seventeen percent had history of RCC and 19% underwent pre-ablation biopsy. Intra-ablation histology was benign for 29% of patients. Patient and tumor characteristics did not differ by type of ablation. Complications were reported in 7 procedures (5.4%): burnt skin, lower extremity DVT, intraoperative afib, abortion of a procedure and 3 perinephric hematomas. One-third had high serum creatinine after ablation. During a median follow-up of 13 months (IQR 3-33), 13 patients underwent salvage therapies. Salvage-free survival rate was 89% at 12 months and 82% at 36 months. Salvage treatment rates and types did not differ between CA and RFA ( $p=0.71$ ). Larger maximum tumor dimension was associated with a higher risk of undergoing salvage treatment ( $p<0.01$ ). Prior

history of RCC demonstrated a non-significant trend toward higher risk of salvage.

**Conclusion:** CA and RFA showed comparable outcomes in a small retrospective study. Maximum tumor size was associated with risk of salvage treatment. Ablation procedures remain viable options for patients with multiple or recurrent small tumors as well as for patients with tumors thought to be less fit for conventional treatment.

#### MP-10.02

##### Nephron Sparing Fast Track Surgery in Kidney Cancer

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National Cancer Institute, Kiev, Ukraine

**Introduction and Objectives:** Nephron sparing surgery (NSS) is widely used in renal cancer and is an alternative to nephrectomy in T1 stage tumors. Surgery with fast recovery, called "fast track surgery" (FTS) approach is widely accepted approach in surgery nowadays, although in NSS there is not much research done. Our objective was to analyze FTS approach in nephron sparing surgery for renal cell carcinoma (RCC).

**Materials and Methods:** Randomized control trial was initiated to compare the "Fast track surgery" approach versus standard care in patients who had partial nephrectomy for RCC from 01/01/2013. Inclusion criteria's were: single side kidney tumor (T1N0M0) that is not involving kidney pelvic system; absence of significant comorbidities. Seventy patients were randomized in the study: on 35 in the study and control groups. Both groups were equal in mean patients age, male/female distribution, mean size of the tumor, overall GFR and ECOG status. All the surgeries were done through transabdominal approach without central ischemia.

**Results:** There were no intraoperative complications recorded. Blood loss was  $264 \pm 126$  ml in study and  $302 \pm 147$  in control group ( $t$ -test;  $p > 0.3$ ). Postoperative complications were recorded in 4 patients: 2 (5.7%) in study group and 2 (5.7%) in control group. Complications were not higher than grade 3 by Clavien classification. For pain evaluation 10 points scale was used and pain levels in the study group were insignificantly lower to control:  $2.9 \pm 1.1$  points versus  $3.5 \pm 1.6$  points ( $t$ -test;  $p = 0.21$ ). Postoperative hospital stay was significantly shorter for study group  $3.8 \pm 1.7$  days versus  $6.9 \pm 1.8$  days ( $t$ -test;  $p < 0.001$ ). No early complications that needed admittance to the hospital were recorder in early postoperative period.

**Conclusions:** Our preliminary data showed promising results of FTS approach comparing to standard care in patients who underwent PR for RCC. This multimodal approach gives significant advantage in postoperative hospital

stay with the same levels of postoperative complications and pain intensity.

#### MP-10.03

##### Does Anatomical Tumor Complexity Affect Renal Function after Clamless Partial Nephrectomy? A Functional Analysis Using Renal Scintigraphy

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**Introduction and Objectives:** Renal function as assessed by glomerular filtration rate (GFR) would decrease to a greater extent after partial nephrectomy (PN) using hilar clamping in patients with tumors of higher anatomical complexity which is reportedly associated with longer ischemic time. To preserve renal function as much as possible even in complicated cases, we perform clamless PN for small renal mass using a gasless single-port laparoendoscopic surgical technique (Kihara K. Eur Urol Suppl. 2010). Here, we investigated associations of anatomical tumor complexity with renal function after clamless PN.

**Materials and Methods:** Between 2000 and 2011, 119 Japanese patients received gasless single-port PN and were followed up 1 year or longer. Excluding 5 patients in whom hilar clamping was used, 114 (84 male and 30 female) undergoing clamless PN were enrolled in this study. Anatomical tumor complexity was assessed using preoperative aspects and dimensions used for an anatomical (PADUA) classification, and estimated GFR (eGFR) was calculated by the Modification of Diet in Renal Disease equation. Multivariate linear regression analyses were carried out to determine clinical parameters that influence %change of eGFR at 1 year after PN (% $\Delta$ eGFR). Moreover, pre-operative and 1 year postoperative split renal function was assessed by (<sup>99m</sup>Tc<sup>m</sup>)-diethylene-triamine-penta-acetic acid renal scintigraphy in 45 patients. Correlations of clinical parameters with %change of split renal function in an operated renal unit after PN (% $\Delta$ SRF) were assessed by Spearman's rank correlation test.

**Results:** Median age, body mass index, preoperative eGFR, operation time, blood loss, tumor size and PADUA score were 58 year, 23.3 kg/m<sup>2</sup>, 98.8 ml/min/1.73 m<sup>2</sup>, 206 minute and 213 ml, 2.3 cm, and 7, respectively. Median % $\Delta$ eGFR was 12.2%. Among clinical parameters, tumor size was significantly and independently associated with % $\Delta$ eGFR; PADUA score and elements other than tumor size were not associated with % $\Delta$ eGFR. Median % $\Delta$ SRF was

21.7%. Tumor size was significantly correlated with %ΔSRF ( $p = 0.364$ ,  $p = 0.014$ ).

**Conclusion:** Tumor complexity except tumor size is unlikely to affect renal function after clampless PN for small renal mass. Only volume loss of renal parenchyma may influence renal function after clampless PN.

#### MP-10.04

##### **Times of Transition: Results of Open Partial Nephrectomy Procedures from Four Centres in the UK**

**Shahzad S**<sup>1</sup>, **Shafik A**<sup>2</sup>, **Hanna L**<sup>3</sup>, **Parkin J**<sup>4</sup>, **Waymont B**<sup>3</sup>, **Ojha H**<sup>1</sup>, **Chen T**<sup>2</sup>, **Makar A**<sup>2</sup>  
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**Introduction and Objectives:** Latest advancements in technology have revolutionized the management of renal tumours. Nephron sparing surgery (NSS) has become more prevalent with early detection. In recent trials, overall survival and reduction in renal function is scrutinized comparing the NSS with Radical nephrectomy. Although more recently minimally invasive techniques have become more available, in many centres Open Partial Nephrectomy (OPN) is still being performed. We report our novel experience with OPN in 4 centres over the last decade.

**Materials and Methods:** This was a retrospective multicentre study analysing data from 187 patients who underwent OPN for suspected renal cell carcinoma between 2003 and 2013. Data was obtained using appropriate OPCS codes. Indication, ischaemia time, change in the renal function, hospital stay and complications were recorded. Clavien-Dindo classification was used to record the complications.

**Results:** Median age was 61 years, predominantly males (59.3%). Median follow-up was 6 years. Multifocal tumours in 2.1%. All procedures were unilateral. Cold ischaemia was used in 81.2% ( $n=152$ ) and warm ischaemia in 18.7% ( $n=35$ ). Median ischaemia time was 19 minutes. Median blood loss was 150mls and Median length of hospital stay was 4 days. Positive margins were reported in 4.2% ( $n=8$ ). Mean 3-month estimated GFR declined by 13.2%.

**Conclusions:** Nephron sparing surgery is indeed evolving. There is increased drive towards minimally invasive techniques using laparoscopic and robotic technology with segmental vascular dissection and zero ischaemia. Whilst minimally invasive techniques are constantly perfected and expertise is being developed, OPN remains an important and feasible surgical option in treatment of renal tumours.

#### MP-10.05

##### **Transitioning from Open to Robot Assisted Partial Nephrectomy: Do Benefits Outweigh the Challenges?**

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**Introduction and Objectives:** The transition from open (OPN) to robotic assisted partial nephrectomy (RA-PN) is challenging and may deter those with limited laparoscopic experience. This study analyses whether patient safety can be assured, equivalent oncological outcomes can be achieved and quantifies the learning curve required.

**Materials and Methods:** A prospective single centre study of 292 patients who underwent partial nephrectomy was performed between January 2002 and January 2014. One hundred RA-PN patients (mean age: 58, range 32-85) were matched against 192 OPN patients (mean age: 58, range 19-82). Peri-operative and oncological outcomes were compared.

**Results:** Median follow-up for RA-PN and OPN was 18 months (1-62) and 77 months (1-138) respectively. Twenty consecutive robotic cases were required to achieve a warm ischaemic time of less than 20 minutes and 10 cases to achieve a console time of less than 120 minutes. RA-PN has favourable outcomes with respect to median length of stay (RA-PN: 4 days (2-11), OPN: 6days (3-35);  $p<0.0001$ ), positive surgical margins (RA-PN: 6%, OPN: 13%;  $p=0.09$ ) and estimated blood loss (RA-PN: 308mls OPN: 998mls;  $p<0.0001$ ). Clavien-Dindo scores (I + II) were lower in the RA-PN group (4%) versus the OPN group (11%;  $p=0.05$ ). None of the robotic cases were converted to an open procedure. There was no difference in major complication rate or pre-and post-operative estimated glomerular-filtration rates. Five-year recurrence free survival was 100% in the RA-PN group and 94.2% in OPN group.

**Conclusion:** The robotic approach to partial nephrectomy compares favourably with open surgery. It is therefore feasible to transition from OPN to RA-PN whilst maintaining safe peri- and post-operative outcomes, and comparable oncological results during and beyond the initial learning curve.

#### MP-10.06

##### **100 Cases of Robotic Assisted Partial Nephrectomy: Which Surgical Approach**

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**Introduction and Objectives:** The importance of the role of robotic assisted partial nephrectomy (RAPN) is now becoming increasingly accepted. Despite many specialist upper tract laparoscopic surgeons employing the

extraperitoneal approach and some described benefits in terms of reduced surgical morbidity in comparative series, thanks to perceived technical difficulties, this approach is rarely described and practiced for RAPN. We report our first 100 cases of RAPN, selecting surgical approach on tumour anatomy alone.

**Materials and Methods:** We prospectively collected data regarding the first 100 partial nephrectomies carried out in our centre over a 3-year period. Data included lesion location (as per Nephrometry Score), surgical technique and approach, operative and post-operative complications.

**Results:** Average tumour size was 30.3 mm. Eighty seven percent of the cases were carried out via a retroperitoneal approach despite a wide variety of tumour locations, only those most anterior and hilar necessitating a trans-peritoneal procedure. Mean operating time was 144 mins with an average blood loss of 77 ml and warm ischaemia time of 23 mins. Transfusion was infrequently required (4%); there were three conversions to a radical procedure and 3 instances of delayed haemorrhage requiring radiological embolisation. The median hospital stay was 1 night.

**Conclusions:** Our data suggests that in a high volume tertiary centre, RAPN is a safe, reproducible and truly minimally invasive surgical option for appropriate tumours whatever their anatomy and with the necessary experience can be predominantly carried out using a retroperitoneal approach which may benefit the patient even further.

#### MP-10.07

##### **Repeat Robotic Partial Nephrectomy for Complex Renal Tumors: Characteristics and Renal Functional Outcomes**

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**Introduction and Objectives:** Multifocal and hereditary kidney cancer conditions often require multiple ipsilateral partial nephrectomies. Repeat and salvage open renal surgeries have been shown to have higher blood loss and complication rate compared to first time renal surgery. Consequently, many surgeons avoid minimally invasive techniques in the setting of repeat renal surgery. We present the characteristics and short-term renal functional outcomes of patients who underwent a repeat robotic partial nephrectomy (RRPNx) at the National Institutes of Health.

**Materials and Methods:** A prospectively maintained database was reviewed to identify

patients who underwent complex multifocal partial nephrectomies between January 2007 and December 2013. Patients who had undergone  $\geq 2$  ipsilateral renal or adrenal surgery, the second one being an RRPNx were selected. Clinical characteristics, surgical parameters and renal functional outcomes both preoperative (preop) and at 3-month follow-up were collected and compared to patients undergoing initial robotic partial nephrectomy (iRPNx).

**Results:** A total of 125 patients underwent robotic partial nephrectomy between January 2007 and December 2013. Of these, 20.8% (26/125) underwent a repeat robotic partial nephrectomy (22 were 2<sup>nd</sup> time cases and 4 were 3<sup>rd</sup> time cases). Sixteen patients underwent previous open ipsilateral surgery, 8 had previous

minimally invasive ipsilateral procedures, 4 underwent prior ipsilateral thermal ablation and 4 had combined ipsilateral therapies. Four cases were converted from robotic to open and, of those, 1 was also converted from partial to radical nephrectomy. Mean age for this group was  $48.8 \pm 12.8$  (28-75) years. Mean BMI was  $30.2 \pm 5.2$  (22.9-40.4). Mean number of tumors resected was  $4.3 \pm 5.8$  (1-29), mean surgery time was  $366.6 \pm 122.1$  (180-652) minutes, mean EBL was  $1426 \pm 1769$  cc (200-8500). Mean preop creatinine in the RRPNx cohort was  $1.02 \pm 0.3$  mg/dl (0.58-1.63) compared to  $0.95 \pm 0.3$  for those undergoing iRPNx. At 3-month follow-up, creatinine was  $1.1 \pm 0.3$  mg/dl (0.33-2.2) in the RRPNx group compared to  $1.0 \pm 0.3$  mg/dl (0.47-2.36) for the iRPNx

population. Mean change in creatinine from preop level to three-month follow-up was 0.09 mg/dl for RRPNx vs. 0.05 mg/dl for iRPNx ( $p=0.3$ ).

**Conclusions:** RRPNx is safe and feasible in highly selected patients. At three-month follow-up renal function preservation is excellent with respect to preoperative levels and to iRPNx patients.

#### MP-10.08

##### **Robotic Multiplex Partial Nephrectomy: The National Cancer Institute Experience with Robotic Partial Nephrectomy for 3 or More Tumors in a Single Kidney**

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**MP-10.09**, Table 1. Clinicodemographics and Perioperative Outcomes

|                                                  | Robotic Nephroureterectomy | Laparoscopic Nephroureterectomy | p-value |
|--------------------------------------------------|----------------------------|---------------------------------|---------|
| Cohort size                                      | 43                         | 104                             |         |
| Follow-up (range, months) <sup>†</sup>           | 9.7 [9.1—14.2]             | 31.3 [13.9—49.4]                |         |
| Age (range, years) <sup>†</sup>                  | 69.9 [65.9—75.8]           | 72.6 [65.8—81.8]                | 0.12    |
| TUMOR CHARACTERISTICS                            |                            |                                 |         |
| Affected Collecting System                       |                            |                                 |         |
| Right                                            | 23 (54%)                   | 61 (59%)                        | 0.59    |
| Left                                             | 20 (47%)                   | 43 (41%)                        |         |
| Location of Tumor <sup>††</sup>                  |                            |                                 |         |
| Renal Pelvis                                     | 30 (70%)                   | 64 (62%)                        | 0.74    |
| Ureter                                           | 22 (51%)                   | 55 (53%)                        |         |
| Renal pelvis and ureter                          | 9 (21%)                    | 15 (35%)                        |         |
| Neo-Adjuvant Chemotherapy (NAC)(yes)             | 16 (37%)                   | 36 (35%)                        | 0.85    |
| Overall Pathologic (pT) Stage                    |                            |                                 |         |
| pT0 (Downstaged, endoscopy)                      | 2 (5%)                     | 2 (5%)                          | 0.46    |
| pT0 (Downstaged, NAC)                            | 3 (7%)                     | 5 (5%)                          |         |
| pTis                                             | 1 (2%)                     | 3 (3%)                          |         |
| pTa                                              | 12 (28%)                   | 33 (32%)                        |         |
| pT1                                              | 15 (35%)                   | 23 (22%)                        |         |
| pT2                                              | 3 (7%)                     | 11 (11%)                        |         |
| pT3/pT4                                          | 7 (16%)                    | 27 (26%)                        |         |
| Positive Pathologic Lymph Nodes (cN+)            | 1 (2%)                     | 14 (13%)                        | 0.07    |
| PERIOPERATIVE OUTCOMES                           |                            |                                 |         |
| Operative Time (hours) <sup>†</sup>              | 4.8 [3.6—5.8]              | 3.8 [3.3—4.7]                   | 0.007   |
| EBL (mL) <sup>†</sup>                            | 200 [50—325]               | 200 [100—400]                   | 0.16    |
| Post-Operative Hospital Stay (days) <sup>†</sup> | 4.0 [3.0—6.3]              | 2.0 [2.0—4.0]                   | 0.0001  |
| Positive Surgical Margins (yes)                  | 0 (0%)                     | 5 (5%)                          | 0.32    |
| Overall Complication Rate (Clavien II—V)         | 8 (19%)                    | 19 (19%)                        | -       |
| (Clavien III—V)                                  | 1 (2%)                     | 5 (5%)                          |         |

<sup>†</sup>Continuous variables are expressed as median values with interquartile ranges

<sup>††</sup>Several patients had tumor in multiple locations.



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**Introduction and Objectives:** Robotic partial nephrectomy is a well-established technique in the management of renal cell carcinoma and now used more frequently for complex multifocal disease. The Urologic Oncology Branch at the National Institutes of Health has many patients with hereditary multifocal renal cell carcinoma conditions that require partial nephrectomy. Herein we present renal functional outcomes after robotic surgery on a single kidney with three or more lesions which we have termed a "Robotic Multiplex Partial Nephrectomy" (RMxPNx).

**Materials and Methods:** A prospectively maintained database from the Urologic Oncology Branch of the National Cancer Institute, NIH was reviewed to evaluate patients who had undergone robotic assisted multifocal partial nephrectomies between January 2007 and November 2013. We chose resection of 3 or more masses in a single kidney as the definition for MxPNx. Data compiled included preoperative (preop) serum creatinine (SCr), postoperative (postop) SCr daily and at 3 month postop follow-up. Additionally renal function was assessed using eGFR (CKD-EPI- Creatinine 2009 formula), and differences were reported as percent change in eGFR from the preop value. Differences in SCr, eGFR and clinical variables were compared.

**Results:** From 407 partial nephrectomies performed from 2007 to 2013, 54 patients were identified who underwent RMxPNx (mean age 46, range 20 to 84). Mean BMI was 30.5 (range 22.5 to 41.6). Mean number of tumors removed was 8.63 (range 3 to 52). Hilar occlusion was very rarely employed with only 18% of cases demonstrating a mean warm ischemic

time of 23.3 minutes (range 15 to 37 minutes). Mean blood loss was 1434mL (range 250 to 8500). The robotic to open conversion rate was 11% with all 6 conversions occurring in the first 20 patients of the series. Mean preoperative creatinine and eGFR were 1.02±0.25mg/dL and 68.9±20.8mL/min respectively. At 3 month follow-up the mean creatinine increase from baseline was 0.038mg/dL (p=0.148) and mean decrease in eGFR was 2.04mL/min (p=0.331).

**Conclusions:** Robotic multiplex partial nephrectomy is a safe and feasible approach in highly selected patients with multifocal renal masses. This minimally invasive approach for nephron sparing surgery results in excellent preservation of preoperative renal function.

#### MP-10.09

##### Single-Docking Technique as a Simplified Approach for Robotic-Assisted Nephroureterectomy, Bladder Cuff Excision, and Retroperitoneal Lymph Node Dissection in Upper-Tract Urothelial Carcinoma

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**Introduction and Objectives:** To describe a novel, simple single-docking approach for robotic nephroureterectomy with bladder cuff excision and retroperitoneal lymph node dissection (RPLND) and compare those results to traditional laparoscopic approach.

**Materials and Methods:** After obtaining Institutional Review Board approval, 104 and 43 consecutive patients who underwent laparoscopic and robotic nephroureterectomy, respectively, for upper-tract urothelial carcinoma (UTUC) from 2003 to 2013 were evaluated.

**Results:** The single-docking approach was performed in 43 patients. Robotic RPLND was

associated with greater lymph node procurement than laparoscopic dissection (median 23.0 versus 13.0 nodes, p<0.0001). The robotic approach had somewhat longer operative times (median 4.8 versus 3.8 hours, p=0.007) and length of stays (median 4.0 versus 2.0 days, p=0.0001). Other comparators (Clavien complication rate, positive surgical margins) favored the robotic approach but were not statistically significant in this study population.

**Conclusion:** We describe a single-docking robotic technique that is feasible for nephroureterectomy with bladder excision and RPLND with improved lymph node counts in comparison to laparoscopy.

#### MP-10.10

##### Validation of a Predictive Nomogram for Urine Leak in Complex Partial Nephrectomy

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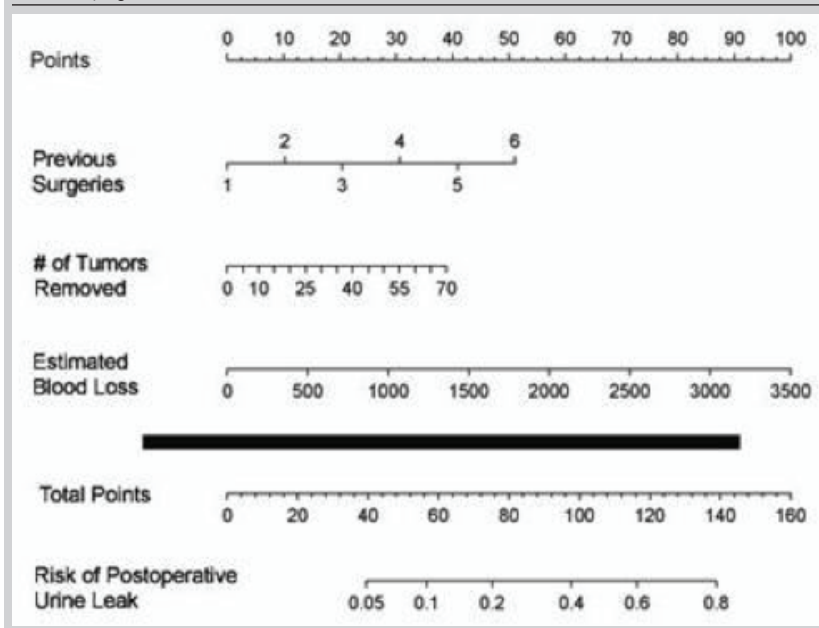
**Introduction and Objectives:** Urine leak after partial nephrectomy is a complication more frequently encountered with increasing tumor complexity. We developed a nomogram to predict postoperative urine leaks after complex partial nephrectomy and validated it with an independent set of patients.

**Materials and Methods:** A cohort of 130 multifocal complex partial nephrectomies performed between January 2008 and September 2012 was divided into a training (50 most recent patients) and validation set (80 prior patients). Urine leak was defined as elevation of JP drain creatinine significantly above serum after post op day 2. For the predictive model used in the nomogram, parameters that were

MP-10.09, Table 2. Lymph Node Dissection

|                                                               |           | Robotic Nephroureterectomy | Laparoscopic Nephroureterectomy | p-value |
|---------------------------------------------------------------|-----------|----------------------------|---------------------------------|---------|
| Lymphadenectomy performed                                     |           | 30 (70%)                   | 51 (49%)                        | 0.03    |
| Lymph Nodes Removed, Overall                                  | (average) | 23.0 [19.0—31.0]           | 13.0 [7.0—22.5]                 | <0.0001 |
|                                                               | (range)   | {9—61}                     | {5—39}                          |         |
| Lymph Nodes Removed, Pelvic                                   | (average) | 4.0 [2.0—8.8]              | 5.0 [3.0—8.3]                   | 0.89    |
|                                                               | (range)   | {2—39}                     | {3—13}                          |         |
| Categorical Breakdown (total LNs procured)<br>(# of patients) |           |                            |                                 |         |
| 1-5 Nodes                                                     |           | 0 (0%)                     | 9 (18%)                         | 0.003   |
| 6-10 Nodes                                                    |           | 1 (3%)                     | 10 (20%)                        |         |
| 11-20 Nodes                                                   |           | 9 (30%)                    | 16 (31%)                        |         |
| ≥ 21 Nodes                                                    |           | 20 (67%)                   | 16 (31%)                        |         |

MP-10.10, Figure 1.



significantly associated on logistic regression with postop urine leak ( $p < 0.05$ ) or with a trend towards significance ( $p < 0.1$ ) were included.

**Results:** The mean age of the training cohort was 49.5 years old with a mean of one prior surgery and 13.1 tumors removed per kidney per operation. The validation cohort had a mean age of 48.6 years, 0.9 previous surgeries, and 6.7 tumors removed per operation. Postoperative urine leak was seen in 28 patients. A nomogram with good predictive characteristics ( $c$ -index=0.85) was generated from the training set based on blood loss, number of tumors resected, and number of prior kidney procedures (Figure 1). An optimal probability cutoff for assignment of patients into the “likely to leak” category was determined to be 25% on the nomogram based on the training cohort. The nomogram correctly predicted postop urine leak 88% of cases in the training cohort. This nomogram was then applied on the 80 patient validation cohort and correctly predicted leak in 74% of cases.

**Conclusions:** Risk of urine leak increases with increased blood loss, number of resected tumors and number of prior renal surgeries. A validated nomogram strongly predicts post-operative urine leak for complex partial nephrectomy with good accuracy in both the training and validation cohorts.

#### MP-10.11

##### Functional Results of Nephron-Sparing Approach for Tumors Larger Than 7cm in Patients with RCC

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**Introduction and Objectives:** Nephron-sparing surgery (NSS) for kidney tumors more than 7cm in size are indicated for selected patients in high-volume centers. Radical nephrectomy remains the mainstream approach for tumors larger than 7cm. Organ sparing approach and evolution of surgical technique pushes the limits of nephron-sparing surgery (NSS) over the range of 7cm nowadays. Objective of the study was evaluate functional and oncological outcomes of NSS in tumors more than 7cm.

**Materials and Methods:** Database search identified 56 patients with RCC lesions > 7cm that underwent PN from 2008 to 2013. Mean age of the group was  $52.9 \pm 13.0$  years. Major indication for nephron-sparing approach was 55% of calculated functional parenchyma in kidney with tumor. Mean size of the tumor was  $93.6 \pm 25.6$ mm. Mean R.E.N.A.L. nephrometry score was  $9.5 \pm 1.8$  with 55.4% of patients with scores 10-12. Mean functional renal parenchyma volume (FRPV) calculated on CT imaging was  $68.0 \pm 9.7\%$ . Baseline mean GFR was evaluated with nephroscintigraphy and was  $86.9 \pm 18.8$ ml/min ( $40.0 \pm 11.7$  on tumorside).

**Results:** All surgeries were done in open fashion through subcostal transabdominal approach. No ischemia during resection was performed in 41 (73.2%) cases. Fifteen (26.8%) patients were operated with central ischemia (mean duration  $15.7 \pm 5.3$ ). Mean intraoperative blood loss was  $483 \pm 365$ cc and in 3 (5.4%) cases blood transfusions were needed. There were 2 (3.6%) intraoperative spleen injuries, followed by splenectomy. Five (8.9%) postoperative complications were recorded: 3 (5.4%) cases of urinary fistula and 2 (3.6%) cases of pyelonephritis. Mean follow-up was  $21.8 \pm 4.5$ .

Local recurrence was diagnosed in 3 (5.4%) patients at 12, 13 and 24 months postop respectively. Overall kidney function decreased to  $77.5 \pm 19.0$  ml/min (10.8%) at 3 months and to mean  $72.7 \pm 30.2$  ml/min (16.3%) at 12 months ( $p = 0.06$ ). There was a 37% decline in function of operated kidney ( $25.2 \pm 8.1$ ml/min) at 3 months and 36% ( $25.6 \pm 13.4$ ml/min) at 12 months respectively ( $p < 0.05$ ), showing the contralateral kidney effect on overall function.

**Conclusions:** NSS in tumors more than 7cm allows GFR preservation with preserving overall kidney function and slight decrease (19.7%) in GFR of operated kidney with acceptable intra-operative (8.6%), postoperative (11.4%) complication rate and low recurrence rate (5.7%).

#### MP-10.12

##### Oncological and Functional Outcomes of Nephron-Sparing Surgery versus Radical Nephrectomy in cT1b-4 RCC: A Single-Institution Matched Analysis

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**Introduction and Objectives:** The present retrospective cohort-study investigates the role of NSS in RCC stage cT1b or larger by means of a case-matched analysis of nephron-sparing surgery (NSS) vs. radical nephrectomy (RN). Oncological and functional outcomes are analysed.

**Materials and Methods:** Our single-institution RCC database was queried to select patients with clinical stage T1b-4 N0 M0. All imaging of patients who underwent RN was reviewed and only patients deemed eligible for NSS were included. After case-matching, 152 patients (76 per group) were included in the final analysis. Cancer specific survival (CSS), clinical progression free survival (CPFS) and overall survival (OS) were calculated using Kaplan-Meier analysis and differences between groups assessed by Log-rank test. Cox proportional hazard regression models were constructed. Proportions were compared by Fisher exact tests and Wilcoxon rank-sum tests used to compare continuous variables.

**Results:** Groups were case-matched (Table 1). ECOG performance status and median follow-up (RN: 79 months (24.1 – 100.5) vs. NSS: 38.5 months (20.5 – 72.1)) differed significantly between groups. Five-yr CPFS, CSS and OS were 77.2%, 90.5% and 86.4% in the RN-group and 83.6%, 91.1% and 82.0% in the NSS-group ( $p = 0.55$ ;  $p = 0.33$  and  $p = 0.33$ , respectively). In the multivariate Cox model, surgical method was confirmed not to be a significant predictor of CPFS, CSS and OS. We found a significantly higher loss of eGFR in the RN group ( $14.1$  ml/min per  $1.73\text{m}^2$ ) vs. the NSS group ( $5.4$  ml/min per  $1.73\text{m}^2$ ) ( $p < 0.03$ ).

MP-10.12, Table 1.

|             | Total group (n = 152)  | RN group (n = 76)      | PN group (n = 76)      |         |
|-------------|------------------------|------------------------|------------------------|---------|
|             | Median [IQR range] (%) | Median [IQR range] (%) | Median [IQR range] (%) | p-value |
| Age         | 63,1 [54,5 - 71,1]     | 62,1 [52,1 - 71,7]     | 63,8 [55,8 - 70,6]     | 0,64    |
| Gender      | Female 46 (30,3%)      | 25 (32,9%)             | 21 (27,6%)             | 0,60    |
|             | Male 106 (69,7%)       | 51 (67,1%)             | 55 (72,4%)             |         |
| BMI         | 26,2 [24,1 - 29,3]     | 26,2 [24,1 - 28,4]     | 26,3 [24,1 - 29,75]    | 0,74    |
| ECOG        | 0 108 (71,1%)          | 63 (82,9%)             | 45 (59,2%)             | 0,002   |
|             | 1 39 (25,7%)           | 13 (17,1%)             | 26 (34,2%)             |         |
|             | 2 5 (3,3%)             | 0 (0%)                 | 5 (6,6%)               |         |
| CCI         | 0 69 (45,4%)           | 43 (56,6%)             | 26 (34,2%)             | 0,05    |
|             | 1 17 (11,2%)           | 7 (9,2%)               | 10 (13,2%)             |         |
|             | 2 34 (22,4%)           | 14 (18,4%)             | 20 (26,3%)             |         |
|             | 3 32 (21,1%)           | 12 (15,8%)             | 20 (26,3%)             |         |
| eGFR pre-op | 66,1 [56,8 - 82,5]     | 64,58 [56,79 - 74,89]  | 69,43 [26,6 - 87,6]    | 0,21    |
| cT          | cT1b 113 (74,3%)       | 52 (68,4%)             | 61 (80,3%)             | 0,40    |
|             | cT2 27 (17,8%)         | 16 (21,1%)             | 11 (14,5%)             |         |
|             | cT3a 9 (5,9%)          | 6 (7,9%)               | 3 (3,9%)               |         |
|             | cT4 3 (2,0%)           | 2 (2,6%)               | 1 (1,3%)               |         |

**Conclusion:** In patients with cT1b-4 RCC, oncological outcomes did not differ significantly between RN and NSS. However, renal function preservation is better in patients undergoing NSS. Our data suggest that NSS can be considered an oncologically safe procedure in larger RCC. The decision to perform NSS in RCC cT1b or larger should be based on tumor localization and technical feasibility rather than on tumor stage.

#### MP-10.13

##### Functional Results of Radical Nephrectomy in Patients with Clinically Localized Renal Cell Carcinoma and Normal Contralateral Kidney Function

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**Introduction and Objectives:** To assess functional results of radical nephrectomy in patients with clinically localized renal cell carcinoma (RCC) and normal contralateral kidney function.

**Materials and Methods:** We analyzed medical data of 426 consecutive patients (1991-2011) with clinically localized RCC and normal contralateral kidney function who underwent radical nephrectomy (open – 211 (49.5%), laparoscopic – 215 (50.5%)). Median age was 57 years. A male to female ratio was 1.1:1. Median tumor diameter was 5.0±2.2 cm. No regional and distant metastases were diagnosed before surgery in any of the cases. Median initial glomerular filtration rate (GFR) was 83.8 ml/min/1.73 m<sup>2</sup> (<60 ml/min/1.73 m<sup>2</sup> – 82 (19.2%)). Concomitant diseases adversely affecting renal function were in 139 (32.6%), obesity – in 168 (39.4%) cases. Median follow-up was 50.0±12.3 months.

**Results:** Acute kidney injury (AKI) was registered in 110 (25.8%) cases (acute dialysis was required in 9 (2.1%) patients). Independent risk factors of AKI included diseases adversely affecting renal function (OR=0.226 (95%CI:0.073-0.699); p<0.0001) and initial GFR<45 ml/min/1.73 m<sup>2</sup> (OR=0.066 (95%CI:0.037-0.117); p<0.010). CKD stage ≥3 was observed in 174 (40.8%) cases. Continuing decline of GFR≥4 ml/min/1.73 m<sup>2</sup>/year in patients with CKD stage ≥3 occurred in 12 (2.8%) cases. Independent risk factors of CKD stage ≥3 included age ≥60 years (OR=0.261; 95%CI:0.149-0.459; p<0.0001), presence of diseases adversely affecting kidney function (OR=0.288, 95%CI:0.154-0.539, p<0.0001), tumor size <4 cm (OR=0.411, 95%CI:0.330-0.492, p<0.0001), and AKI following nephrectomy (OR=0.119, 95%CI:0.057-0.248, p<0.0001).

**Conclusion:** Functional results of radical nephrectomy in patients with clinically localized RCC and normal contralateral kidney function are not satisfactory and require careful selection of candidates for non organ sparing surgery.

#### MP-10.14

##### Renal Cell Carcinoma with Inferior Vena Caval Tumour Thrombus: Is Surgery Better Than Chemotherapy? An Update

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**Introduction and Objectives:** Renal cell carcinoma extends into the inferior vena cava in 10% of patients. Caval thrombectomy is rarely performed; peri-operative mortality is 10%, with complication rates of 30%. There are no papers examining long-term survival.

We present a 13-year series and review survival outcomes compared to chemotherapy alone.

**Materials and Methods:** We analysed 23 patients undergoing nephrectomy and caval thrombectomy under a single surgeon and 11 patients receiving sunitinib alone. Data were collated on extent of tumour thrombus, cardiac bypass (CPB) and hypothermic circulatory arrest, pathological stage and tumour grade, complications and survival.

**Results:** A total of 23 patients underwent nephrectomy and excision of caval thrombus. Forty seven percent required bypass. At the end of 21 months, 57.1% with Fuhrman grade 2 tumours were alive, 83.3% with Fuhrman grade 3, 66% Fuhrman grade 4. Complications included: AF 4%, renal failure 14.2%, PE 4%. No complications were experienced in 71%. There was one perioperative death within 30 days of surgery (4.7%). The mean survival for patients undergoing chemotherapy was 5 months. Seventy two percent had metastatic disease at the time of treatment. Twenty eight percent had no metastatic disease at initiation of chemotherapy, mean survival was 5.5 months. The mean survival with surgery is at least 32 months, 66% are still alive.

**Conclusions:** We conclude an aggressive surgical approach improves patient outcomes and prolongs survival. We demonstrate this surgery is not associated with as high a complication rate. We compare our results with other published data.

#### MP-10.15

##### GM-CSF May Predict the Response of Metastatic Renal Cell Carcinoma to Tyrosine Kinase Inhibitor Therapy

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**Introduction and Objectives:** To identify predictive markers for the response of metastatic renal cell carcinoma (RCC) to tyrosine kinase inhibitors (TKIs), we conducted a prospective study.

**Materials and Methods:** Patients with histologically proven RCC with at least one measurable metastatic lesion were enrolled in this study. Blood samples were collected before treatment and the plasma levels of 27 cytokines were measured. Tumor response was assessed eight to 12 weeks after the start of TKI treatment.

**Results:** Thirteen patients (11 males / two females) with a median age of 63 years received sunitinib (eight cases), sorafenib (one case), or axitinib (four cases). Partial response (PR) was achieved in five patients (38%), stable disease (SD) in four (30%), and progressive disease (PD) in four (30%). Plasma granulocyte macrophage colony-stimulating factor (GM-CSF)



in PR cases was significantly higher than that in SD or PD cases ( $p=0.012$ ).

**Conclusion:** GM-CSF may be a predictive biomarker of the response of RCC to TKI treatment, suggesting that TKIs could exert clinical effects not only through suppression of VEGF but also through the immune system.

#### MP-10.16

##### **The Treatment of Patients with Metastatic Renal Cell Carcinoma with Angiogenesis Inhibitors: Will They Continue with the Same Treatment after One Year?**

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**Introduction and Objectives:** Anti-angiogenic therapy is currently the standard treatment for patients with metastatic renal cell carcinoma (mRCC). According to the published clinical studies, approximately 20-25% of the patients usually abandon the treatment due to side-effects. With this study, we intended to describe the one-year incidence of treatment dropout and to analyze the determinants of interruption of patients treated in the real clinical practice.

**Materials and Methods:** We evaluated 59 patients with mRCC that had been receiving treatment with angiogenesis inhibitors (Sunitinib, Pazopanib or Temsirolimus) in the Department of Urology of a University Hospital, between January 2008 and November 2013. Data regarding socio-demographic characteristics, performance status (Karnofsky), comorbidities (ACE-27), tumor histology, TNM stage, treatment regimen, side-effects and treatment dropout were collected from patients' clinical records. We computed Kaplan-Meier curves and estimated age and gender-adjusted hazard ratios with a cox regression analysis in order to identify the dropout determinants.

**Results:** Median patient age was 66 years (58-74) and median patient Karnofsky performance status at diagnosis was 90 (80-90). The one-year incidence of treatment dropout was 37.3%

( $n=22$ ) and the main causes for this interruption were: drug toxicity [ $n=7$  (11.9%)], disease progression [ $n=13$  (22.0%)], patient refusal [ $n=1$  (1.7%)] and complete response [ $n=1$  (1.7%)]. Although 86.5% ( $n=51$ ) reported side effects, only 11.9% abandoned the treatment for that reason. Patients with Karnofsky performance status lower than 80 [HR=3.9 (1.5–10.0)], ECOG higher than 1 [HR= 2.9 (1.0–8.4)] and grades 2 or 3 of ACE-27 comorbidity scale [HR= 3.0 (1.2–7.8)] revealed a statistically significant higher risk of treatment interruption. Paradoxically, those that reported more than 1 side effect demonstrated a lower risk of dropout [HR=0.3 (0.1–0.8)] and in this group the proportion of cases with disease progression was lower.

**Conclusion:** Approximately 62% of the patients with mRCC remained in treatment with angiogenesis inhibitors after one year, despite the high rate of side-effects. The main reason for the interruption was RCC progression and was closely related to the patients' previous health-status.

#### MP-10.17

##### **Is There a Need for Cytorreductive Nephrectomy in Metastatic Renal Cell Carcinoma?**

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**Introduction and Objectives:** To assess the role of cytorreductive nephrectomy (CN) in metastatic renal cell carcinoma (mRCC).

**Materials and Methods:** Medical records of 1333 patients with mRCC were analyzed. Patients' median age was 54 (16-89) years. A male-to-female ratio was 1:2.4. Tumor venous thrombosis was diagnosed in 212 (15.9%), N+ – in 678 (50.9%), multiple metastatic sites – in 431 (32.3%) cases. CN was performed in 883 (66.2%) patients. Systemic therapy was administered in 706 (52.9%) cases (antiangiogenic targeted therapy – 79 (6.0%), cytokines – 627 (47.0%)). The remaining 627 (47.0%) patients received supportive care only. Patients' characteristics were imbalanced between CN group and conservative management group in terms

of rate of N+ and number of metastatic sites. Median follow-up was 18 months.

**Results:** CN was associated with significant increase of median overall survival (10.4 vs. 24.7 months (HR 2.026 (95%CI:1.733-2.369),  $p<0.0001$ )). The survival advantage following surgery was significant in all subgroups of patients receiving antiangiogenic, cytokine and symptomatic therapy. Independent risk factors of overall survival were pN+ (HR 0.648 (95%CI:0.510-0.823),  $P<0.0001$ ), tumor venous thrombosis (HR 1.47 (95%CI:1.012-2.012),  $p=0.043$ ) and incomplete removal of the primary tumor (HR 2.053 (95%CI:1.534-2.747),  $p<0.001$ ). Survival advantage following surgery was significant in patients with 0-1 risk factors ( $p<0.0001$ ). Cytorreductive nephrectomy did not improve prognosis in patients with  $\geq 2$  risk factors ( $p=0.786$ ).

**Conclusion:** CN in carefully selected mRCC patients with 0-1 risk factors significantly increases median survival and should be considered regardless of the type of planned systemic therapy.

#### MP-10.18

##### **Bladder Tumour Recurrences after Curative Surgery for Upper Tract Urothelial Carcinoma in Patients without Prior Bladder Urothelial Carcinoma: A Look at Clinicopathological Predictive Factors**

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**Introduction and Objectives:** Current literature has reported concomitant carcinoma-in-situ (CIS) and higher tumour grade to have a higher bladder tumour recurrence post-curative nephroureterectomy. We aimed to assess whether histologic multifocality and smoking status, as well as other clinicopathological variables affected the bladder tumour recurrence rate.

**Materials and Methods:** All patients who underwent curative nephroureterectomy for non-metastatic upper urinary tract cancer (UTUC) at our centre over a period of 5 years were retrospectively identified. Patients with prior bladder tumours were excluded. Univariate log rank tests and multivariate Cox regression

**MP-10.18, Table 1.** Clinicopathological variables and their effects on bladder tumour recurrences

|                                                                 | Univariate         |         | Multivariate       |         |
|-----------------------------------------------------------------|--------------------|---------|--------------------|---------|
|                                                                 | HR (95% CI)        | P value | HR (95% CI)        | P value |
| Histologic multifocality<br>- No vs. Yes                        | 0.28 (0.09 - 0.94) | 0.04    | 0.30 (0.08 - 1.11) | 0.071   |
| Tumour grade<br>- High grade vs. low grade                      | 1.50 (0.40 - 5.66) | 0.55    | 1.2 (0.25 - 6.18)  | 0.786   |
| CIS<br>- Yes vs. No                                             | 1.79 (0.47 - 6.81) | 0.393   | 0.95 (0.18 - 4.60) | 0.92    |
| Smoking status<br>- Ex-smoker and current smoker vs. non-smoker | 1.32 (0.40 - 4.36) | 0.645   | 1.35 (0.35 - 5.15) | 0.665   |



models were used to examine the significance of the variables on the incidence of bladder tumour recurrence.

**Results:** Histologic multifocality was found to be a predictor for bladder tumour recurrence ( $p=0.04$ , HR 0.28, 95% CI 0.09-0.94). However on multivariate analyses, histologic multifocality did not reach statistical significance. Interestingly, CIS, tumour grade and smoking status were not found to be significant predictors for bladder tumour recurrences on both univariate and multivariate analyses. The median time to tumour recurrence was 13 months; with most recurrences within the first 12 months postoperatively (72.7%), and majority within the first 24 months (90.9%).

**Conclusion:** Histologic multifocality was predictive for bladder tumour recurrence post-nephroureterectomy, with majority of tumour recurrences within the first 24 months. Close bladder surveillance with flexible cystoscopy is needed for timely detection and treatment of such recurrences.

#### MP-10.19

##### Adjuvant Chemotherapy for Lymph Node-Positive Upper Tract Urothelial Carcinoma following Radical Nephroureterectomy: A Retrospective International Multicenter Study

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A<sup>14</sup>, Novara G<sup>15</sup>, Shariat S<sup>1,12,16</sup>

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**Introduction and Objectives:** To evaluate the effect of adjuvant chemotherapy (AC) on

mortality after RNU for upper tract urothelial carcinoma (UTUC) with positive lymph nodes and to identify patient subgroups that are most likely to benefit from AC.

**Materials and Methods:** We retrospectively analyzed data of 263 patients with lymph node-positive UTUC, who underwent full surgical resection. One hundred seven patients (41%) received 3 to 6 cycles of AC, while 156 (59.3%) were treated with surgery alone. UTUC-related mortality was evaluated using competing-risks regression models.

**Results:** In all patients (Tall N+), administration of AC had no significant impact on UTUC-related mortality on univariable ( $P=0.49$ ) and multivariable ( $P=0.11$ ) analysis. Further stratified analyses showed that only N+ patients with pT3-4 disease benefited from AC. In this subgroup, AC reduced UTUC-related mortality by 34% ( $P=0.019$ ). The absolute difference in mortality was 10% after the first year and increased to 23% after 5 years. On multivariable analysis, administration of AC was associated with significantly reduced UTUC-related mortality (SHR 0.67,  $P=0.022$ ). Limitations of this study are the retrospective non-randomized design, selection bias, absence of a central pathologic review and different AC protocols.

**Conclusion:** AC seems to reduce mortality in patients with pT3-4 lymph node-positive UTUC after RNU. This subgroup of lymph node-positive patients could serve as target population for an adjuvant prospective randomized trial.

## Moderated Poster Session 11

### Men's Health

Tuesday, October 14  
1300-1430

#### MP-11.01

##### Impact of Chronic HCV

##### Infection on Male Fertility

Mohamed Abdel Hamid A<sup>1</sup>, Khalaf Fath Elbab T<sup>1</sup>, Mohamed Galal E<sup>1</sup>, Khairy Salem H<sup>2</sup>  
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**Introduction and Objectives:** Hepatitis C virus (HCV) infection is gaining attention as a global health problem especially in Egypt which has the highest prevalence. Treatment with peg interferon alfa and ribavirin (PEGIFN/RBV) is the standard of care for chronic hepatitis C. There is some evidence that sperm quality may be impaired in HCV patients. Furthermore limited data is available on the effect of PEGIFN/RBV treatment on human sperm quality and male fertility as some reports have linked HCV infection to altered gametogenesis found in HCV patients. Aim: To study the impact of chronic HCV infection and standard antiviral treatment on male fertility via estimation of seminal parameters, DNA integrity and reproductive hormonal serum levels.

**Materials and Methods:** Thirty male patients proved pathologically as chronic hepatitis C complaining of infertility (with mean age 35±3.6 years). In addition to 15 healthy volunteers were studied. In all subjects semen samples were analyzed for motility, morphology, and concentration. Furthermore we analyzed sperm DNA integrity via chromatin structure assay which is expressed as the DNA fragmentation index (DFI). A DFI larger than 30% is considered abnormal. In addition to estimation of hormonal levels of (follicle-stimulating hormone, luteinizing hormone, free testosterone, prolactin and DAHES) before treatment and after 12, 24 and 48 weeks of combination therapy of pegyinterferon + ribavirin.

**Results:** The sperm concentration, the mean sperm motility, were significantly decreased in HCV patients compared to that of controls (P 0.001) (P 0.03) respectively, also the normal morphology was significantly reduced in HCV patients (P 0.04) with respect to that of controls. An impairment of spermic morphology occurred, while other seminal parameters did not change significantly during antiviral treatment. Data on sperm DNA integrity were available in 20 patients. At baseline, 5 of 20 patients (25%) had a DFI >30%. The median DFA increased very markedly during treatment (from 20.5% before to 65.2% at 24 weeks of treatment) and remained elevated at 48 weeks. Hormonal pattern of patients did not

significantly change after treatment except free testosterone, prolactin and DAHES in INF responders. Advanced liver fibrosis and high viral load were risk factors for male infertility.

**Conclusion:** HCV infection has a negative impact on male fertility through impaired reproductive hormones, quantitative and qualitative alterations of spermatogenesis with DNA abnormalities. Advanced liver fibrosis and high viremia were risk factors for male infertility. Successful treatment will improve hormonal pattern.

#### MP-11.02

##### Nomogram to Predict the Outcomes of Intracytoplasmic Sperm Injection (ICSI)

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**Introduction and Objectives:** Although many studies have shown that the success of ICSI depends on a number of male factors, female factors and laboratory procedures, no one has analyzed these variables collectively.

**Materials and Methods:** We prospectively analyzed 1112 ICSI cycles performed for male or female infertility. We used multiple factors to build the first nomogram to predict the probability of clinical pregnancy of ICSI. All ICSI procedures were performed in the same institute using the same technique and the same operators. Eleven variables (male factors, female factor, and laboratory factors) were included in a multiple logistic regression analysis. The final nomogram was internally validated with bootstrap analysis.

**Results:** A young maternal age and an increased number of good quality embryos increased the likelihood of clinical pregnancy. The nomogram was found to have a concordance index of 0.652 and was well calibrated.

**Conclusion:** We constructed the first nomogram that is accurate to predict the probability of clinical pregnancy of ICSI. The nomogram still needs to be validated externally.

#### MP-11.03

##### Assessment of Testicular Perfusion Prior to Sperm Extraction Predicts Success Rate and Decreases the Number of Required Biopsies in Patients with Non-Obstructive Azoospermia

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**Introduction and Objectives:** There have been several attempts to decrease the morbidity of testicular sperm extraction and increase the

likelihood of sperm retrieval (SR). In this study we assessed the role of power and color Doppler ultrasonography (US) in patients with azoospermia prior to testicular sperm extraction.

**Materials and Methods:** One hundred and thirty consecutive patients with azoospermia were assessed in this prospective study. Testicular artery resistive index (RI) was measured using color Doppler US. Based on a semi-quantitative method, the results of power Doppler US were graded into 3 categories: grade 1, no visible vessels; grade 2, between one and three detectable vessels; grade 3: more than three detectable vessels. The location of each visible vessel was also recorded as upper, middle or lower third of the testis.

**Results:** Seventy four patients with non-obstructive azoospermia (NOA) and 27 with obstructive azoospermia (OA) fulfilled the study criteria. OA patients revealed a significantly lower RI and higher intratesticular perfusion compared with NOA patients. Among NOA patients with grade 3 testicular perfusion, 60% required less than 3 biopsies for successful sperm retrieval (SR) whereas in less than 18% of patients with grade 1 perfusion, sperm was retrieved prior to the third sampling. We also noted a correlation between the presence of visible vessels in each segment and the probability of successful SR during biopsy from the corresponding segment.

**Conclusions:** Our data indicates that a semi-quantitative, simplified Power Doppler US assessment is capable of localizing areas containing viable sperm with the potential to direct biopsies to specific sites and subsequent decrease in the number of required biopsies.

#### MP-11.04

##### A Modified Single-Armed Technique for Microsurgical Vasoepididymostomy

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**Introduction and Objectives:** The aim of this study is to evaluate the effectiveness and safety of a modified single-armed suture technique for microsurgical vasoepididymostomy (MVE) in patients with epididymal obstructive azoospermia.

**Materials and Methods:** From September 2011 to December 2012, microsurgical two-suture longitudinal intussusception vasoepididymostomy were performed using our modified single-armed suture technique in 52 men with epididymal obstructive azoospermia at our hospital. Postoperative patency rate, postoperative impregnation rate and complications were followed.

**Results:** Among all the cases followed-up more than 10-25 months after operation, 41 patients (79%) had sperm (>10<sup>4</sup>/ml<sup>-1</sup>) in the semen. The

median sperm density was  $18.2 \times 10^6/\text{ml}^{-1}$ . Thirteen patients' wives (25%) achieved the natural pregnancy. Postoperative complications, such as scrotal hematoma in 2 cases, were observed in the study.

**Conclusions:** Our modified single-armed suture technique for MVE is an effective and somewhat simplified method for treatment of epididymal obstructive azoospermia.

#### MP-11.05

##### Phosphorus as Predictive Factor for Erectile Dysfunction in Middle Aged Men: A Cross Sectional Study in Korea

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**Introduction and Objectives:** Higher serum phosphate levels are associated with atherosclerotic disease and an increased risk of cardiovascular events. However, the association of phosphate with erectile dysfunction (ED) is not well known. We evaluate the influence of serum phosphorus level on ED and the association with other clinical factors.

**Materials and Methods:** Between March and September 2013, 1899 police officers aged 40-59 years who had participated in a health examination were included. No one had a definite history of neurogenic or vasculogenic ED. All men underwent a detailed clinical evaluation using the Korean version of International Index of Erectile Function-5 (IIEF-5) questionnaires. Serum prostate-specific antigen (PSA), testosterone, phosphorus, body mass index (BMI), evaluation of metabolic syndrome (MetS), and transrectal ultrasonography were also performed.

**Results:** The median age was 53.0 years, and median value of phosphorus and testosterone were  $3.50\text{mg/dL}$ ,  $4.58\text{ng/mL}$ , respectively. The median total IIEF-5 score was 18. Serum phosphorus level ( $r=0.108$ ,  $P<0.001$ ) showed the highest correlation coefficient with IIEF except age, followed by total prostate volume (TPV) ( $r=-0.065$ ,  $P<0.001$ ). Serum testosterone level had no significant correlation with the IIEF. Phosphorus level showed a weak but significant correlation with IIEF score after adjusting for age, testosterone, and MetS ( $r=0.044$ ,  $p=0.048$ ). Using logistic regression analysis, age, phosphorus, and MetS were

predictive factors for moderate to severe ED in the univariate analysis ( $p<0.001$ , OR 1.125, 95% CI 1.093-1.159,  $p=0.007$ , OR 0.702, 95% CI 0.542-0.908,  $p=0.010$ , OR 1.372, 95% CI 1.078-1.745, respectively). The PSA, testosterone, BMI, and TPV had no predictive value for ED. Age and MetS were independent predictors for moderate to severe ED ( $p<0.001$ , OR 1.122, 95% CI 1.089-1.156,  $p=0.048$ , OR 1.283, 95% CI 1.003-1.641) in the multivariate analysis.

**Conclusion:** Our data showed that phosphorus had a correlation with ED. Phosphorus is a significant predictor for ED and the powerful factor that can be easily modified in middle-aged men. Handling of phosphorus level in men may be considered one of the useful modalities against the development of ED.

#### MP-11.06

##### Use of Erectile Dysfunction as a Tool for Early Diagnosis of Cardiovascular Diseases

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**Introduction and Objectives:** Myocardial infarction and ischemic stroke are the main causes of death in adult males. In addition, the identification of erectile dysfunction (ED) may indicate the presence of these disorders in a patient in a latent form, according to some authors. We determined the prevalence of cardiovascular diseases and risk factors in patients seeking medical help with erectile dysfunction at an outpatient clinic.

**Materials and Methods:** The 108 men with ED were enrolled in the study. All patients filled the International Index of Erectile Function questionnaire (IIEF). Waist circumference, blood pressure, blood chemistry and penile duplex Doppler ultrasound were assessed. Also medical history was studied for the presence of cardiovascular diseases that were previously diagnosed.

**Results:** The average age of the patients was  $52.7 \pm 8.6$  years. Duration of erectile dysfunction at the time of enrollment was 44.5 (range: 1-180) months. According to medical history the hypertension has been previously diagnosed in 19 (17.5%) patients, whereas elevated blood pressure was detected in 46 (42.5%) patients

during examination. Coronary artery disease was previously diagnosed in 5 (4.6%) males. Type 2 diabetes was previously diagnosed in 16 (14.8%) patients. However, fasting glucose level was higher  $6.1\text{ mmol/l}$  in 29 (26.8%) patients. History of acute cerebrovascular accidents and intermittent claudication were in 2 patients (1.8%) and 1 (0.9%) patients, respectively. Decreased reactive hyperemia index and the development of systemic endothelial dysfunction were found in 45 (41.6%) patients. Increasing the level of total cholesterol, LDL and triglyceride levels were observed in 68, 79 and 46 cases, respectively (62.9%, 73.1% and 42%, respectively).

**Conclusions:** Metabolic changes indicating atherosclerosis and other life-threatening diseases in men with erectile dysfunction marked much more frequently than in the general population. Therefore, we have shown that ED is an early marker of systemic diseases in men. These data should be used as a tool for prevention of premature mortality from avoidable causes.

#### MP-11.07

##### Erectile Dysfunction, Testosterone Level and Abdominal Obesity

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**Introduction and Objectives:** Testosterone (TST) plays an important role in erectile dysfunction (ED) and may have influence on the development of metabolic syndrome. The number of men with abdominal obesity (AO) which constitutes a serious health risk is continuously growing. This study examined the association among ED, TST level and metabolic syndrome in men with AO at different waist circumference.

**Materials and Methods:** We examined 216 men, 42 years to 78 years old. AO (waist circumference over 94cm) had 198 men and 18 men were as control group without AO. Complete urological evaluation and internal evaluation were carried out in every patient.

**Results** We found some degree of ED in 74.7% of men with AO. With growing AO there were

MP-11.07, Table 1. AO and TST level.

| Testosterone level | Grade 0, < 94 cm |        | G 1, 94-101 cm |        | G 2, 102-109 cm |        | G 3, 110-119 cm |        | G 4, 120 + cm |        | Total |        |
|--------------------|------------------|--------|----------------|--------|-----------------|--------|-----------------|--------|---------------|--------|-------|--------|
|                    | no               | %      | no             | %      | no              | %      | no              | %      | no            | %      |       |        |
| <7.0nmol/l         | 0                | 0.00   | 2              | 4.35   | 3               | 4.76   | 9               | 15.52  | 7             | 22.58  | 21    | 9.72   |
| 7.0-10.5           | 1                | 5.56   | 9              | 19.57  | 24              | 38.10  | 22              | 37.93  | 16            | 51.61  | 72    | 33.33  |
| 10.5-14.0          | 1                | 5.56   | 17             | 36.96  | 14              | 22.22  | 11              | 18.97  | 4             | 12.90  | 47    | 21.76  |
| >14.0nmol/l        | 16               | 8.89   | 18             | 39.13  | 22              | 34.92  | 16              | 27.59  | 4             | 12.90  | 76    | 35.19  |
| Total              | 18               | 100.00 | 46             | 100.00 | 63              | 100.00 | 58              | 100.00 | 31            | 100.00 | 216   | 100.00 |

more and severe forms of ED. In the control group without AO 50% of men had ED (33% had a mild form, 17% medium and nobody had a severe form ED). We found a strong correlation between TST level and AO (Table 1). There were more patients with low level of TST with growing AO. We found that nearly 89% of men without AO had a TST level over 14nmol/l and none of these men had a low level of TST (below 7.0 nmol/l). With growing AO there were more men with a low level of TST. Ninety eight out of 198 (49.5%) men with AO and 1/18 (5.5%) men without AO had testosterone deficiency syndrome (TDS). Metabolic syndrome was diagnosed in 105/198 (53.0%) men with AO. No men without AO had MetS.

**Conclusion:** Men over 40 with ED and AO have a higher incidence of TDS and MetS and should also be examined for TDS and MetS. It seems to be reasonable to divide men into four groups according waist circumference. There are significant differences among groups in ED, TDS and MetS.

#### MP-11.09

##### **Proteomics in Diabetic Patients with Erectil Dysfunction and Treatment with 5-Phosphodiesterase Inhibitors**

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**Introduction and Objectives:** Endothelial dysfunction is one of the first symptoms of erectile dysfunction (ED) and is closely related to atherosclerosis and risk factors such as diabetes mellitus (DM), characterized by inflammatory and oxidative advanced state. Vardenafil is one of the more effective 5-phosphodiesterase inhibitors known in patients with DE and DM. However, at present, it is speculated on the potential pleiotropic effects on inflammation, oxidative stress or endothelial homeostasis. The purpose of this study is to determine the plasma proteome of patients with DM and the effect of vardenafil administration in the expression of proteins related to inflammatory, oxidative stress and cellular homeostasis.

**Materials and Methods:** Seventeen patients with ED (IIEF-EF  $8.8 \pm 0.7$ ) and DM were recruited for the study. They received 20 mg of vardenafil one or two times per week for 12 weeks. Each patient's plasma was obtained before and after the 12 weeks of treatment and was analyzed by a bi-dimensional electrophoresis. The plasma proteome obtained before and

after the treatment were compared and analyzed with mass spectrometry.

**Results:** After the treatment with vardenafil, erectile function of patients increased significantly (IIEF-EF  $18.3 \pm 2.0$   $p = 0.05$ ). The expression levels of ceruloplasmin isoform 1 (antioxidant protein) and A1 apolipoprotein isoform 5 (endothelial protective protein) increased significantly after the treatment with vardenafil. Furthermore two isoforms of alpha-antitrypsin and betatropomyosin were significantly reduced after treatment. We also observed a significant negative correlation between plasma levels of beta-tropomyosin and IIEF-EF score. Elevated levels of beta-tropomyosin in plasma indicate cell damage and loss of cellular regenerative capacity.

**Conclusions:** Vardenafil treatment may exert a protective endothelial role through the reduction of systemic inflammation, oxidative stress and the increase of the cellular regenerative capacity.

#### MP-11.10

##### **Venous Leakage Treatment Revisited: Pelvic Venoblation Using Aethoxysclerol under Air Block Technique and Valsalva Maneuver Herwig R<sup>1</sup>, Sansalone S<sup>2</sup>**

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**Introduction and Objectives:** We evaluated the effectiveness of pelvic vein embolisation with aethoxysclerol in airo-block technique for the treatment of impotence due to venous leakage in men using sildenafil for intercourse. The aim of the procedure was to reduce or eliminate the use of sildenafil.

**Materials and Methods:** A total of 49 patients with veno-occlusive dysfunction, severe enough for the need of PDE5 inhibitors for vaginal penetration, underwent pelvic venoblation with aethoxysclerol. The mean patient age was 53.5 years. Venous leaks were identified by Color Doppler Ultrasound after intracavernous alprostadil injection. Under local anesthesia a 5F-Angiopot was inserted antegrade into the deep dorsal penile vein. The pelvic venogram obtained with deep dorsal venography was included. Aethoxysclerol 3% as sclerosing agent was injected after air-block under valsalva maneuver in three consecutive steps. A 5F-angiography catheter was placed in the vein of major outflow from the penis. Success was defined as the ability to achieve vaginal insertion without the aid of any drugs, vasoactive injections, penile prosthesis, or vacuum device. Additionally, a pre- and posttherapeutical IIEF-5 score was performed.

**Results:** At a 12 month follow-up 40 out of 49 patients (81.63%) reported to have erections sufficient for vaginal insertion without the use of any drug or additional device. Four

(8.16%) patients did not report any betterment. Follow-up Color Doppler Ultrasound revealed a new or persistent venous leakage in 8 (16.33%) of the patients. No serious complications occurred.

**Conclusions:** Our new pelvic venoblation technique using aethoxysclerol in air-block technique was effective, minimally invasive, and cost-effective. All patients were able to perform sexual intercourse without the previously used dosage of their PDE5 inhibitor. This new method may help in patients with contraindications against PDE5 inhibitors, in patients who cannot afford the frequent usage of expensive oral medication or those who do not fully respond to PDE5-inhibitors.

#### MP-11.11

##### **Early Experiences with Single Session Low-Intensity Shock Wave Therapy for Erectile Dysfunction: Multi-Center Study Jung G<sup>1</sup>, Ha S<sup>1</sup>, Seo J<sup>1</sup>, Park S<sup>1</sup>, Seo K<sup>2</sup>, Eom M<sup>3</sup>, Rhee H<sup>3</sup>**

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**Introduction and Objectives:** Shock wave therapy is a novel treatment option for erectile dysfunction (ED), and has been proved to be effective in some studies. But the treatment methods were empirical and there is no established shock wave treatment method for ED yet. We analyzed our experiences with short-course single session Low-intensity shock wave therapy (LSWT) for ED.

**Materials and Methods:** Forty eight consecutive ED patients without any history of pelvic surgery, trauma and/or irradiation who underwent single session LSWT from December 2012 to July 2013 at the 3 centers were included in this study. We applied LSWT to the patients using ED-1000TM (Medispec Ltd.). Single session treatment consisted with 8-time treatment: twice a week, for 4 weeks. Erectile function was assessed with self-administrated International Index of Erectile Function-Erectile Function Domain (IIEF-EFD) score and Erection Hardness Score (EHS) before and 1 month after treatment. The patients' satisfaction to the treatment was assessed by Global Assessment Questionnaire (GAQ) 1 month after the treatment.

**Results:** Mean IIEF-EFD score increased from 12.6 points to 17.6 points after treatment ( $p < 0.001$ ). There were 22 patients (45.8%) who had a 5-point or greater increase in IIEF-EFD score and 34 patients (70.8%) who answered positively to GAQ after treatment. Among 28 patients who couldn't achieve erection hard enough to penetrate initially (EHS  $\leq 2$ ), 39.3% (11/28) could achieve penetration after the treatment (EHS  $\geq 3$ ). No adverse events were



reported from the treatment.

**Conclusion:** Single session LSWT improved erectile function of ED patients without any adverse events.

#### MP-11.12

##### **Penile Prosthesis Placement in Patients with a History of Total Phallic Construction**

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**Introduction and Objectives:** Outcomes following penile prosthesis placement in patients with a history of total phallic construction are not well described.

**Materials and Methods:** Retrospective review penile prosthesis placement in patients with prior total phallic construction. Gortex sleeve corporal construction was utilized in all patients.

**Results:** Twenty-five patients underwent neophallus prosthesis placement at a mean 34.4 years of age. Prosthesis placement occurred an average 42 months following phallic construction and follow-up was a mean of 60 months. Malleable prostheses were placed in 17 patients and inflatable in 8; implants were bilateral in 92%. Eight percent experienced operative complications including a bladder injury (1) and phallic flap arterial injury (1). Post-operative complications occurred in 24% at a median 5.9 months following placement. Four prostheses (16%) were explanted secondary to infection or erosion and two additional required revision. Of the explanted prosthesis one was later replaced without further complication. Seventy-six percent of patients were sexually active following prosthesis placement.

**Conclusions:** Penile prosthesis placement is possible in patients with prior phallic construction. Although complications rates appear to be increased in this population compared to historic controls of normal anatomic males, the majority in this series were sexually active following prosthesis placement. This demonstrates the utility of prosthesis implantation in these difficult patients.

#### MP-11.13

##### **Penile Prosthesis Implantation in Surgical Treatment of Severe Peyronie's Disease with Buccal Mucosa Graft**

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**Introduction and Objectives:** To present our experience with management of patients with Peyronie's disease. Younger patients with severe

Peyronie's disease associated with erectile dysfunction (ED) require the reconstruction of the cavernosal body (CB), preservation of the vascular elements of the CB and implantation of the penile prosthesis.

**Materials and Methods:** Nineteen younger patients with impaired axial rigidity due to severe Peyronie's disease and partial, or complete ED, underwent surgical reconstruction from September 2011 to November 2013. Color Doppler Ultrasound (CDU), revealed that majority of patients had normal arterial flow through CBs, while three patients had moderately impaired flow. Circumcision incision line was used to approach the defect on the cavernosal bodies. Multiple buccal mucosa grafts (BMG) were used as patches in order to straighten the CB curvature and to prevent the narrowing phenomenon. The mucosal side of the BMG was positioned to face the penile prosthesis, inside the CB, in that way, buccal epithelium replaces the endothelium. Semirigid, AMS-Spectra<sup>®</sup> prosthesis, 9mm in diameter, was used in all patients, in order to enable the straightening of the penis and to prevent the postoperative recurrent curvature. The use of thinner prostheses enables genuine erections in younger patients.

**Results:** The average patients' age was 43 years (range 26 -57 years). Average follow-up was 14 months (range 6-24 months). All patients were satisfied by cosmetic and functional results. None of the patients had voiding problem. Fifteen patients (78.9%) reported improved erection after the surgery, while four patients had to induce the erection with sildenafil. There were no cases with recurrent curvature, or penile shortening. In addition, there were no cases with the rejection of the prosthesis.

**Conclusion:** In the management of Peyronie's disease in the younger men, BMG provides satisfactory functional and cosmetic results. The orientation of the buccal epithelium towards the prosthesis is important. Dartos tunica provides better vascular bed for the BMG than the erectile tissue of the CBs. The use of thinner prosthesis preserve additional erection in younger patients and also provides good axial rigidity.

#### MP-11.14

##### **Different Ways to Support Recovery of Erectile Function after Nerve-Sparing Radical Prostatectomy: Concepts for Rehabilitation in Germany**

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**Introduction and Objectives:** Despite objective data regarding rehabilitation of erectile function (EF) after nerve-sparing radical prostatectomy (nsRP) the "gold-standard"-treatment is still under debate. The aim of this study was

to evaluate the distribution of the different treatment options in Germany.

**Materials and Methods:** Between 10/2010 and 11/2013 we performed a questionnaire sent to all urologist (outpatient/general and university hospitals/rehabilitation hospitals) in Germany. The survey consists of different questions, e.g. if and what kind of therapy the urologist choose to support rehabilitation of EF after nsRP. Further questions dealt with the frequency, duration and "optimal" start of the chosen therapy.

**Results:** Until today 262 urologists completed and returned the questionnaire. The distribution was: urologists in hospitals n=110, outpatient/ambulatory n=148, with 24% performing surgical treatment, and urologist in rehabilitation hospitals n=4. Overall 50% of the urologists are performing radical prostatectomy on a regular basis. The question about the "rehabilitation concept" showed 39 different treatments within this group. To increase EF after nsRP PDE5-inhibitors were mostly administered (88%): 45% "on demand" vs. 55% on a daily or regular basis  $\geq 3$  times/week. The use of penile injection therapy, MUSE or VCD was 32%, 6% and 30% respectively. In 56% the treatment started within the first weeks after surgery and was performed until the patient regained potency in 46%. Only 14% of the urologists didn't choose any "active" kind of rehabilitation treatment for EF recovery after nsRP.

**Conclusion:** Lots of different therapeutic concepts are currently performed in Germany to increase EF recovery after nsRP. The use of PDE5-inhibitors is the most chosen treatment option. Despite the published data regarding effectiveness the optimal treatment seems to be still unknown.

#### MP-11.15

##### **"Kiel Concept": Low-Dose Sildenafil for Rehabilitation of Erectile Function after Nerve-Sparing Radical Prostatectomy: A Two-Year Follow-Up**

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**Introduction and Objectives:** Several treatment regimens for rehabilitation of erectile function after nerve-sparing radical prostatectomy (nsRP) are currently discussed. In previous prospective studies we showed nocturnal penile tumescence and rigidity (NPTR) in 95% of the patients in the early phase after nsRP.

**Materials and Methods:** Forty three sexual active patients were operated by nerve-sparing retropubic radical prostatectomy. All patients completed an IIEF-5 questionnaire concerning erectile function preoperatively. A measurement of NPTR (Rigi-Scan<sup>®</sup>) was carried out in the following night after catheter removal.

Twenty three patients with preserved nocturnal erections detected during NPTR-recordings received sildenafil 25mg/d at night. A control of 18 patients underwent follow-up without PDE-5-inhibitors. IIEF-5 questionnaire was performed 6, 12, 24, 36, 52, 78 and 104 weeks after operation.

**Results:** Forty one of 43 patients (95%) showed 1-5 erections during the first night after catheter removal. In the group of daily sildenafil the IIEF-5 score decreased from preoperative 21.2 mean score to 3.6 at 6 weeks, 3.9 at 12 weeks, 6.1 at 24 weeks, 10.4 at 36 weeks, 14.5 at 52 weeks, 19.4 at 78 weeks and 21.0 mean score at 104 weeks after prostatectomy. In the control group preoperative IIEF-5 mean score 21.2 decreased to 3.8 at 6 weeks, 3.9 at 12 weeks, 5.3 at 24 weeks, 6.1 at 36 weeks, 9.3 at 52 weeks, 13.4 at 78 weeks and 14.8 at 104 weeks. Statistical evaluation showed a significant difference in IIEF-5 score and time to recovery of erectile function ( $p < 0.05$ ).

**Conclusion:** The measurement of NPTR after nsRP showed erectile function even at the "first" night after catheter removal. In case of early penile erections the cavernous nerve was well preserved during surgery providing good neuronal integrity. Daily low dose PDE-5 inhibitors lead to a significant improvement of recovery of erectile function and is supportive to the organic rehabilitation.

#### MP-11.16

##### Premature Ejaculation in University Students: Prevalence and Relation with Mood Disorders

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**Introduction and Objectives:** Actual prevalence of premature ejaculation (PE) in young males is not well described. Some recent studies (Shindel et al) showed that healthy sexuality can be protective against depressive symptoms in medical students. The aim of this study was to assess self-reported PE and the relation with depressive or anxious symptoms in a young male population.

**Materials and Methods:** Students from a public university were invited to participate in an online questionnaire. Participants completed a demographic survey and sexual history and sexual bother was evaluated with the question: "How do you characterize your sexual function at this time?". A validated scale (Hospital Anxiety and Depression Scale – HADS) evaluated depression – score of  $\geq 9$  in the HADS depression dimension – and anxiety – score of  $\geq 9$  in the HADS anxiety dimension. Subjects completed Premature Ejaculation Diagnostic Tool (PEDT) and a score of 9 or 10 indicated

probable PE and a score of  $\geq 11$  indicated PE. Descriptive statistics, Mann-Whitney U and Chi square were used to analyze responses.

**Results:** A total of 231 male subjects (mean age 24.2y) completed the questionnaire. Demographic data is available in Table 1. Depressive symptoms were present in 10 (5.5%) and anxiety symptoms in 62 (34.1%) subjects. The presence of PE was self reported in 40 (22.0%) and probable PE in 17 (9.3%) subjects. The subjects that had depression score in HADS showed more sexual bother ( $p=0.028$ ), but didn't present more PE ( $p=0.515$ ). Subjects with anxiety symptoms had more sexual bother ( $p=0.042$ ) and more PE ( $p=0.030$ ).

**Conclusion:** Our study showed a high prevalence of PE in young male population. There is a relation with mood disorders to disturbances in normal sexual function. Anxiety disorders, more than depression, play an important role in

sexual dysfunction, namely in PE.

#### MP-11.17

##### Are Estimated and Stop-Watch Intravaginal Ejaculatory Latency Time Surely Interchangeable in Assigning Lifelong Premature Ejaculation?

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**Introduction and Objectives:** To assess the clinical utility of estimated intravaginal ejaculatory latency time (eIELT) compared to stop-watch intravaginal ejaculatory latency time (sIELT), in assigning lifelong premature ejaculation (PE) based on International Society for Sexual Medicine (ISSM) Guidelines.

MP-11.16, Table 1.

|                                                                                             |            |
|---------------------------------------------------------------------------------------------|------------|
| Age, in years (min-max)                                                                     | 24 (17-65) |
| <b>Sexual Orientation</b>                                                                   | n (%)      |
| Heterosexual                                                                                | 201 (87.0) |
| Homosexual                                                                                  | 15 (6.5)   |
| Bisexual                                                                                    | 15 (6.5)   |
| <b>Relationship status</b>                                                                  | n (%)      |
| Single, no actual relation                                                                  | 87 (37.7)  |
| Single, with relation                                                                       | 119 (51.5) |
| Living together                                                                             | 12 (5.2)   |
| Married                                                                                     | 12 (5.2)   |
| Divorced                                                                                    | 1 (0.4)    |
| Age of first sexual intercourse, in years (min-max)                                         | 18 (13-30) |
| Never had sexual relations, n (%)                                                           | 49 (21.2)  |
| <b>Number of sexual partners in the last six months</b>                                     | n (%)      |
| 0                                                                                           | 67 (29.0)  |
| 1                                                                                           | 135 (58.4) |
| $\geq 2$                                                                                    | 29 (12.6)  |
| <b>Subjective global assessment of sexual function</b><br>(* - classified as sexual bother) | n (%)      |
| Satisfied, no desire for change                                                             | 97 (42.0)  |
| Most satisfied, desire change                                                               | 53 (22.9)  |
| Unsatisfied, no desire for change                                                           | 5 (2.2)    |
| Unsatisfied, desire change *                                                                | 52 (22.5)  |
| Sexual dysfunction *                                                                        | 7 (3.0)    |
| Sex is not an issue for me                                                                  | 17 (7.4)   |
| <b>PEDT (total n: 182)</b>                                                                  | n (%)      |
| No PE                                                                                       | 125 (68.7) |
| Probable PE                                                                                 | 17 (7.4)   |
| PE                                                                                          | 40 (17.3)  |
| Legend: PEDT – Premature Ejaculation Diagnostic Tool                                        |            |

**Materials and Methods:** A total of 118 healthy men between the ages of 30 and 70 years with a history of lifelong PE, not erectile dysfunction, were recruited between July 2010 and August 2011 at five institutions in Korea. At the initial visit, all subjects underwent preliminary assessments including a medical and sexual history, vital signs, physical examination, and self-administration of the eIELT, International Index of Erectile Function erectile function domain (IIEF-EF), Premature Ejaculation Profile (PEP) and Premature Ejaculation Diagnostic Tool (PEDT) questionnaire. During the 1-week study period, subjects were requested to experience sexual intercourse at least two times, and to record the time from foreplay to beginning intercourse (FTIT) using a stop-watch and sIELT. PE was diagnosed using the ISSM guidelines.

**Results:** Geometric mean eIELT and sIELT were 3.3 and 2.8 minutes, respectively. eIELT and sIELT correlated reasonably well (Pearson correlation coefficient=0.512;  $p<0.001$ ). However, eIELT was overestimated by 1.2 min, compared to the sIELT ( $p=0.046$ ). Also, eIELT was less correlated with each PEP questionnaire, PEP index score and overall sexual act time (OSAT), when compared to the sIELT (each  $p$  value  $<0.05$ ). Accuracy of eIELT was lower (67.9%,  $p=0.001$ ), and eIELT of 2 minutes was the most acceptable cut-off value (sensitivity 64.0%, specificity 65.6%), compared to diagnose PE by sIELT.

**Conclusion:** Our results showed that eIELT was overestimated, and lower clinical utility

compared to sIELT. Our study suggests that eIELT and sIELT cannot be directly interchangeable, and the flexibility may be needed to use eIELT instead of sIELT, in men who are likely to seek treatment for lifelong PE. More research is needed.

#### MP-11.18

##### **Diagnostics of Penile Microcirculation Disorders in Patients with Vasculogenic Erectile Dysfunction**

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**Introduction and Objectives:** Duplex Doppler ultrasonography (DDU) is the conventional test for penile haemodynamic evaluation for erectile dysfunction (ED) screening. We used since 2007 a new method for vasculogenic ED diagnosis - Laser Doppler Microcirculation Floumetry (LDMF).

**Materials and Methods:** Ninety six patients were examined – 69 men (mean age  $56\pm5.2$ ) suffered arteriovenous form of ED and 27 healthy men (mean age  $28\pm4.6$ ) as control group. Patients with diabetes mellitus, severe arterial hypertension were excluded. Initially all men undergone DDU with pharmacological test, then tested by LDMF with laser analyzer LAKK-02 (“LAZMA”, Russia) and helium-neon laser of LGN-207 B-type with 0.63 microns wave length.

**Results:** Slow rhythms with frequency of 1-10

oscillations per minute prevailed in the control group. These rhythms were accompanied with the maximum intravascular resistance and active microcirculations mechanism. We considered these blood flow parameters as a physiological norm of the peripheral microcirculation and it correlated with the dynamics fluctuations of the blood flow. The patients with ED had intravascular circulatory injuries. It characterized by abrupt decrease of the blood flow and statistically significant fluctuations of the erythrocytes velocity, decreasing of the myogenic tone and increasing of the neurogenic tone in the precapillary net. The microcirculation's effectiveness index was also decreased. There was the decreasing of the blood perfusion intensity in cavernous tissue, blood volume reduction in the arterioles, and the congestion in venules with increasing of the erythrocytes amount. These data indicated that the patients with vasculogenic ED had the spastic form of microcirculation injure - decreasing of blood inflow due to cavernous tissue vessels spasm. In these cases the microcirculation index and sphygmic fluctuations amplitude decreased, but the slow oscillations amplitude and vasomotory activity increased.

**Conclusion:** LDMF is a new method of vasculogenic ED diagnosis. In comparison with DDU this method allows to check the microcirculation mechanism and evaluate blood flow disturbances directly in the cavernous tissue. It helps to study the pathogenic changes caused by vasculogenic ED and thus improves the quality of diagnosis.

## Moderated Poster Session 12

### Prostate Cancer: Markers, Prevention and Staging

Tuesday, October 14  
1300-1430

#### MP-12.01

#### **Incidence of Prostate Cancer (PCa) in 340 Hypogonadal Men Treated with Testosterone Undecanoate Injections (TU) for Up To 7 Years: Observational Data from a Registry Study**

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**Introduction and Objectives:** Concerns regarding the safety of testosterone treatment, particularly regarding PCa in middle-aged and elderly men, still hamper the use of testosterone in hypogonadal men. In this study, we investigated prostate parameters incl. incidence of PCa in hypogonadal patients on long-term treatment with TU.

**Materials and Methods:** In a prospective, cumulative registry study, 340 men (age: 57.37 ± 7.03 years) with testosterone ≤12.1 nmol/L received TU 1000 mg every 12 weeks following an initial interval of 6 weeks for up to 7 years. Prostate volume (PV) and PSA were measured and digital rectal examination (DRE)/transrectal ultrasound (TRUS) performed before treatment initiation and then regularly every 3-6 months. In case of suspected PCa, biopsies were performed.

**Results:** PV increased from 28.96 ± 10.41 to 29.88 ± 13.85 ml by model-adjusted 2.59 ± 0.2 ml ( $p < 0.0001$ ). This increase was statistically significant compared to the previous year for the first four years. PSA increased from 1.74 ± 0.94 to 1.96 ± 1.03 ng/ml by model-adjusted 0.23 ± 0.52 ng/ml ( $p < 0.0001$ ). Fifty three biopsies were performed in testosterone-treated patients. Of these, 5 (9.4%) were positive and 48 (90.6%) negative. The proportion of PCa in testosterone-treated patients in our registry study was 1.5% with an incidence of 30.7 per 10,000 patient years. In hypogonadal patients without testosterone treatment, 314 biopsies were performed. Of these, 111 (35.4%) were positive and 203 (64.6%) negative. In eugonadal patients, 584 biopsies were performed. Of these, 263 (40.4%) were positive and 321 (55%) negative. In total, 898 prostate biopsies were performed in our practice from 2004 through 2013, of which 379 (42.2%) were positive and 519 (57.8%) negative.

**Conclusions:** Long-term treatment with TU

in hypogonadal men undergoing regular monitoring according to EAU guidelines does not increase the incidence of PCa.

#### MP-12.02

#### **Genetic Variations of the ADIPOQ Gene, Adiponectin and Risk of Prostate Cancer**

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**Introduction and Objectives:** Adiponectin secreted by adipose tissue has been implicated in prostate carcinogenesis. Genetic variations in *ADIPOQ* are thought to influence the activity of adiponectin, thus relating to cancer occurrence.

**Materials and Methods:** In this hospital-based case-control study of 917 prostate cancer (PCa) cases and 1,036 cancer-free controls, we evaluated the association of single nucleotide polymorphisms (SNPs) in *ADIPOQ* gene with risk of PCa and adiponectin levels. Variants of *ADIPOQ* gene were genotyped by Taqman polymerase chain reaction method. The plasma adiponectin concentrations were measured by enzyme-linked immunosorbent assay (ELISA).

**Results:** We found that *ADIPOQ* rs3774262 variant AA genotype was associated with both decreased PCa risk [adjusted odds ratio (OR): 0.66, 95% confidence interval (CI) = 0.48-0.92] and increased plasma adiponectin levels ( $p = 0.036$  and  $0.043$ ), with significant difference by tumor grade, clinical stage and aggressiveness. A significant interaction between *ADIPOQ* rs3774262 and body mass index (BMI) was observed in modifying the risk of PCa ( $p = 6.7 \times 10^{-3}$ ). *ADIPOQ* rs266729 and rs182052 were not related to PCa risk or plasma adiponectin levels.

**Conclusion:** Our data support *ADIPOQ* rs3774262 may be prognostic factors for PCa development in combination with plasma adiponectin levels. It may contribute to the molecular basis for the association between obesity and PCa, but underlying complex mechanisms require further studies.

#### MP-12.03

#### **Virus-Encoded microRNAs hsv1-miR-H18 and hsv2-miR-H9-5p: Valuable Diagnostic Biomarkers for Prostate Cancer**

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**Introduction and Objectives:** miRNAs may be potential cancer biomarkers as stable miRNAs have been detected in biological fluids such as serum, plasma, and urine. In terms of prostate cancer, although several studies have reported on the use of putative miRNAs in body fluids

as diagnostic and prognostic markers, the significance has been limited. This study is to examine whether the expression of urinary cell-free miRNAs is different in patients with prostate cancer and controls with benign prostatic hyperplasia.

**Materials and Methods:** A total of 750 urine samples from patients with prostate cancer, benign prostatic hyperplasia, and prostate biopsy were used. After miRNA array analysis from urine to select candidate urinary miRNAs, ten candidates were validated in independent cohort. Finally matched samples from cases and control were used to evaluate correlation between candidate miRNA in prostate tissue, urine and serum. Urinary miRNAs were detected using miRNA array and real time PCR, and the diagnostic value was evaluated by Receiver operating characteristics analysis.

**Results:** There were significant differences in the expression of cell-free hsa-miR-615-3p, ebv-miR-BART4, hsv1-miR-H18, hsv2-miR-H9-5p, and hsa-miR-4316 between benign prostatic hyperplasia controls and prostate cancer patients ( $P < 0.05$ ). In particular, hsv1-miR-H18 and hsv2-miR-H9-5p levels were higher in urine from prostate cancer patients ( $P < 0.001$ ). The levels of hsv1-miR-H18 and hsv2-miR-H9-5p in urine showed an area under the curve of 0.790, with a sensitivity of 78.2% and a specificity of 68.5%, and an area under the curve of 0.826, with a sensitivity of 69.2% and a specificity of 80.8%, respectively. For patients in the PSA gray zone and transrectal biopsy, hsv2-miR-H9-5p showed a better diagnostic performance than serum PSA. There was a significant correlation between hsv1-miR-H18 and hsv2-miR-H9-5p in tissue and urine samples ( $r = 0.369$ ,  $P < 0.001$ , and 0.552,  $P < 0.001$ , respectively).

**Conclusions:** Virus-encoded miRNAs were associated with prostate cancer, and hsv1-miR-H18 and hsv2-miR-H9-5p may be important urinary diagnostic markers for prostate cancer even in patients with PSA gray zone.

#### MP-12.04

#### **New Perspectives in Urinary Markers of Prostate Cancer: Role of MMP9, hK3 and PSMA Gene Expressions**

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**Introduction and Objectives:** The evaluation of urinary markers for prostate cancer (PCa) diagnosis has become an important field of research. Our goal is to evaluate the role of urinary matrix metalloproteinase-9 (MMP9); kallikrein-3 (hK3) and prostate-specific membrane antigen (PSMA) gene expressions in the diagnosis and prognosis of PCa.

**Materials and Methods:** We conducted a case-control study with 26 patients with PCa and 28



patients with low-risk for PCa diagnosis after the ethics committee approval. Urine samples were collected after prostatic massage of patients during the outpatient visit to the urology department at the University of Sao Paulo and analyzed by qRT-PCR. Expression levels were compared between patients with and without PCa and according to the prognostic factors, as Gleason score, PSA, clinical stage.

**Results:** Expression levels of urinary markers in PCa cases relative to patients without PCa are shown in Figure 1. The MMP9 showed an overexpression in 95.4% of with an average overexpression of 8.3 times. When we analyzed genes simultaneously, we found that overexpression of MMP9 and PSMA was related to a Gleason >6 in 100% of patients (p: 0.036). The PSMA presented underexpression pattern in cases of PCa, however, in patients with poorer prognostic factors, it showed an overexpression pattern: 3.19 times (Gleason >6) and 3.45 times (PSA>10). hK3 presented an underexpression pattern in cases of PCa and lack of association with prognostic factors. None of the genes correlate with prostate size.

**Conclusions:** We have shown that there is an upregulation of MMP-9 and a downregulation of hK3 and PSMA in PCa cases. In addition, we demonstrated a prognostic relevance of PSMA expression as demonstrated by the significant correlation with poorer prognostic factors.

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**Introduction and Objectives:** ERG overexpression, a result of ERG genomic alterations, is the most common validated prostate cancer (pc) gene alteration. ERG typing will impact its utility in biological classification. Evaluations of ERG oncogenic activation are providing new insights into the ERG based stratification of pc in diverse ethnic populations around the world. This study provides a comprehensive evaluation of ERG oncoprotein in 909 whole-mount prostatectomy specimens in the context of multifocal disease, age and ethnicity.

**Materials and Methods:** For comparative evaluation of ERG frequencies, ERG oncoprotein expression was analyzed in whole-mounted sections of 491 Caucasian American (CA) and 418 African American (AA) patients who underwent radical prostatectomy at Walter Reed National Military Medical Center. ERG oncoprotein was evaluated by immunohistochemistry, an excellent surrogate of ERG gene fusions in pc.

**Results:** Consistent with our previous observa-

expression. Significantly increased ERG oncoprotein frequency was noted in pc of both young CA and AA patients.

**Conclusions:** ERG-based stratification defines molecular subtypes of pc. Association of ERG negativity with high grade tumors and poor prognostic features in AA warrants ERG inclusion in the development of informative marker panels to aid in diagnostic, prognostic and targeted therapy settings. ERG typing significantly contributes to the understanding of causal alterations resulting in higher incidence and associated mortality among AA patients.

#### MP-12.06

##### Systematic Review of Prostate Cancer Risks and Association with Consumption of Fish and Fish-Oils: Analysis of 495,321 Participants

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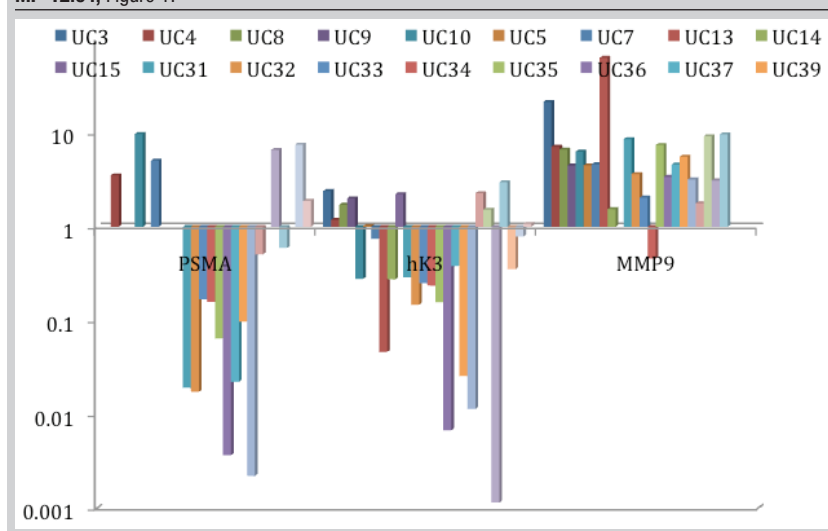
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**Introduction and Objectives:** The role of fish-oils in inflammation entails potential role in inhibiting carcinogenesis and favourable outcomes for cancer symptoms. This systematic-analysis of the literature aims to review current evidence for the role of dietary-fish and fish-oil intake in prostate-cancer (PC) risk, aggression and mortality.

**Materials and Methods:** A systematic-review of studies exploring evidence for PC-risk, aggression and mortality associated with dietary-fish and fish-oil intake was conducted through PubMed, MEDLINE and Embase. Outcome-measures were PC-risk, aggression and mortality were extracted for analysis.

**Results:** A total of 495,321 (37-studies) participants were investigated. These revealed various relationships regarding PC-risk (n=31), aggression (n=8) and mortality (n=3). Ten studies considering PC-risk found some significant inverse trend related to fish or fish-oil intake. One found a dose-response relationship where-as greater intake of long-chain-polyunsaturated-fatty-acids increased risk of PC when considering crude odds-ratios (p=0.014). Three studies addressing cancer aggression identified a significantly relationship with reduced risk of aggressive disease. This applied when considering the greatest intake of total fish (OR 0.56 (95% CI 0.37–0.86)), dark fish and shellfish-meat (p<0.0001), EPA (p=0.03) and DHA (p=0.04). Three studies investigating fish-consumption and PC-mortality identified a significantly reduced risk. Multivariate-OR (95% CI) were 0.9 (0.6–1.7), 0.12 (0.05–0.32) and 0.52 (0.30–0.91) at the highest measured fish-intake.

MP-12.04, Figure 1.



#### MP-12.05

##### ERG Typing of Prostate Cancers in the Context of Multi-Focal Tumors, Age and Ethnicity in 909 Whole-Mount Prostatectomy Specimens

Dobi A<sup>1,2,3</sup>, Young D<sup>1,2,3</sup>, Rosen P<sup>1,2,3</sup>, Farrell J<sup>1,2,3</sup>, Degon M<sup>1,2,3</sup>, Chen Y<sup>1,2,3</sup>, Cullen J<sup>1,2,3</sup>, Petrovics G<sup>1,2,3</sup>, Srivastava S<sup>4</sup>, Kagan J<sup>4</sup>, McLeod

tions, this comprehensive evaluation of ERG in a larger cohort further confirmed significantly lower frequencies of ERG oncoprotein expression (p<0.0001) in the index tumors of AA (28.6%) as compared to CA (51.1%) patients. The difference was more pronounced in higher grade tumors of AA patients with majority of Gleason 8-10 index tumors lacking ERG

**Conclusion:** Fish and fish-oil do not show a consistent role in reducing PC incidence, aggression and mortality. Results suggest that type of fish consumed and fish-oil ratio are significant considerations. Findings demonstrate potential for incorporating awareness of fish and fish oil consumption into public-health campaigns for primary and secondary prevention.

#### MP-12.07

##### The Preliminary Study of Genistein in the Chinese Prostate Biopsy Population

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**Introduction and Objectives:** Genistein is one of the main soy isoflavone in our daily diet. There were some studies both *in vivo* and *in vitro* had proved that high concentration of serum

genistein may relate to the low morbidity and mortality of prostate cancer in Asian population. Since there are few studies of genistein in Chinese population, we performed this study to preliminary evaluate the associations among serum genistein, epidemiologic factors and prostate cancer in Chinese population.

**Materials and Methods:** Between 2012 and 2013, 98 men over the age of 40, underwent prostate biopsy for prostate cancer (PCa) at Huashan Hospital, Shanghai, China. Clinical information, epidemiologic information and blood samples were collected prior to biopsy for each patient. All patients underwent 10-core ultrasound guided transperineal prostate biopsy, the pathology results were collected after biopsy. Measure the serum genistein concentration of the blood samples and analyze the results along with the clinical and epidemiologic information.

**Results:** Among the 98 patients, 46 (46.9%)

were diagnosed with prostate cancer. The serum genistein concentration of non-PCa patients (913.6 ng/ml) were significantly higher than PCa patients (639.6 ng/ml) ( $P < 0.05$ ). As for other epidemiologic factors, we only found significant difference in age between PCa and non-PCa patients (72.5 vs. 68.1) ( $P < 0.05$ ). In the univariate analysis, we found age and serum genistein were related with prostate cancer ( $P < 0.05$ ). In multivariate analysis, we found age, smoking history and serum genistein were related with prostate cancer ( $P < 0.05$ ).

**Conclusion:** High concentration of serum genistein may have a protective effect for prostate cancer.

#### MP-12.08

##### Relationship between Volume-Adjusted Prostate-Specific Antigen and Pathologic Gleason Score

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Queen Mary Hospital, University of Hong Kong, Hong Kong

**Introduction and Objectives:** We investigate the relationship between pathologic Gleason score of prostate cancer and volume-adjusted serum prostate-specific antigen (PSA) concentration estimates based on pre-operative magnetic resonance imaging (MRI).

**Materials and Methods:** All prostate cancer patients who underwent MRI prior to Robot-assisted Laparoscopic Radical Prostatectomy (RaLRP) in our hospital from October 2011 to February 2014 were included. Data was retrieved from a prospectively collected database of all patients treated with RaLRP for prostate cancer. The height (H), width (W) and length (L) of the prostate on T-2 weighed MRI were measured by the radiologist and the MRI-estimated volume was calculated according to the ellipsoid formula ( $H \times W \times L \times \pi/6$ ). The volume-adjusted serum PSA was derived from PSA divided by MRI-estimated prostate volume. The volume-adjusted serum PSA was then compared with the pathologic total Gleason score by using Spearman correlation coefficient.

**Results:** A total of 84 patients (average age 65.9 years old, range 46 – 77 years old) were included. The mean MRI-estimated prostate volume was  $41.6 \pm 37.5 \text{ cm}^3$ . Of the 84 patients, 29 (34.5%) patient had Gleason score of 6, 40 (47.6%) had Gleason score of 7; 8 (9.5%) had Gleason score of 8 and 7 (8.3%) had Gleason score of 9. The mean volume-adjusted serum PSA is  $0.2737 \pm 0.18 \text{ ng/mL/cm}^3$ . The Gleason score was significantly correlated with volume-adjusted serum PSA ( $\rho = 0.48$ ;  $p = 0.000004$ ) (Figure 1).

**Conclusion:** Volume-adjusted serum PSA has shown a significant correlation with pathologic Gleason score of prostate cancer and can potentially contribute to risk stratification of significant prostate cancer.

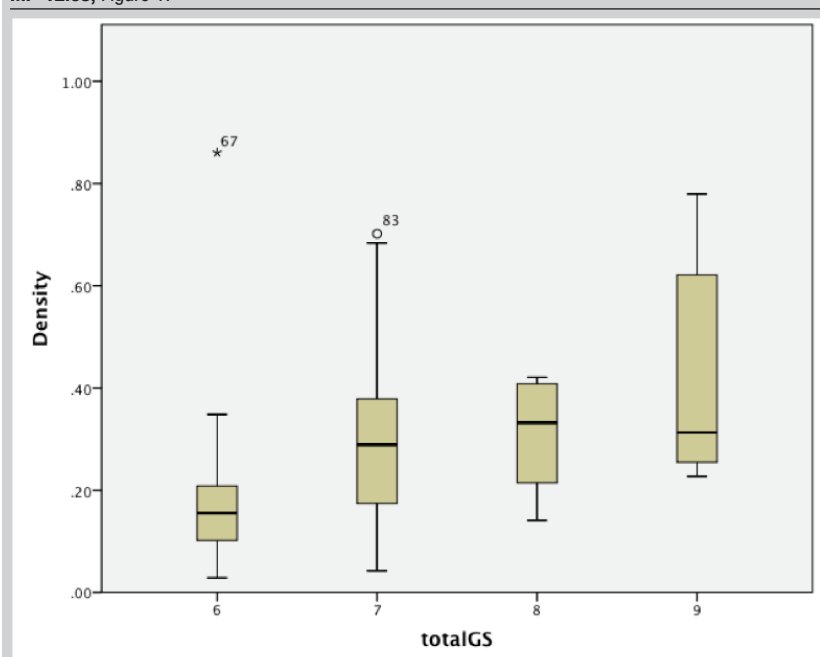
**MP-12.07, Table 1.** The Epidemiological Features and Serum Genistein Concentration of Chinese Prostate Biopsy Population

| Variables                                             | Overall (n=98) | PCa (n=46)    | non-PCa (n=52) | P value |
|-------------------------------------------------------|----------------|---------------|----------------|---------|
| <b>Age (year)</b>                                     |                |               |                |         |
| Mean (SD)                                             | 70.1 (8.9)     | 72.5 (8.4)    | 68.1 (8.9)     | 0.014a  |
| <b>Family history of cancer</b>                       |                |               |                |         |
| Positive n (%)                                        | 52 (52)        | 27 (56.3)     | 25 (48.1)      | 0.414b  |
| <b>Smoking history</b>                                |                |               |                |         |
| Positive n (%)                                        | 38 (38.0)      | 22 (45.8)     | 15 (28.8)      | 0.053b  |
| <b>Total genistein at the time of biopsy (nmol/l)</b> |                |               |                |         |
| Mean (SD)                                             | 773.1 (592.3)  | 639.6 (442.0) | 913.6 (677.6)  | 0.022a  |
| <b>Intake of soy products</b>                         |                |               |                |         |
| High intake (%)                                       | 27 (27.0)      | 11 (23.9)     | 16 (30.8)      | 0.448b  |

**MP-12.07, Table 2.** Univariate and Multivariate Analysis of the Epidemiological Variables at the Time of Biopsy

| Variables                       | Univariate analysis   | P Value |
|---------------------------------|-----------------------|---------|
|                                 | Odds ratio (95% CI)   |         |
| <b>Age</b>                      | 1.062 (1.011-1.116)   | 0.017   |
| <b>Family history of cancer</b> |                       |         |
| Positive vs. Negative           | 1.286 (0.580-2.848)   | 0.536   |
| <b>Smoking history</b>          |                       |         |
| Negative vs. Positive           | 0.442 (0.192-1.018)   | 0.055   |
| <b>Total genistein</b>          | 0.999 (0.998-1.000)   | 0.026   |
| <b>Intake of soy products</b>   |                       |         |
| High vs. Low                    | 0.707 (0.288-1.735)   | 0.449   |
| Variables                       | Multivariate analysis | P Value |
|                                 | Odds ratio (95% CI)   |         |
| <b>Age</b>                      | 1.073 (1.019-1.131)   | 0.01    |
| <b>Smoking history</b>          |                       |         |
| Negative vs. Positive           | 0.386 (0.155-0.959)   | 0.013   |
| <b>Total genistein</b>          | 0.999 (0.998-1.000)   | 0.018   |

MP-12.08, Figure 1.

**MP-12.09****Role of Transition Zone PSA Density in the Detection of Prostate Cancer in Patients with Chronic Prostatitis**

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**Introduction and Objectives:** Prostatic inflammation may cause increased PSA level in some men; we conducted this study to assess the predictive accuracy of transition zone PSA density in the detection of prostate cancer among patients with abnormal PSA and chronic prostatitis.

**Materials and Methods:** Analysis was performed on data derived from TRUS-Biopsy registry database containing 2014 men. Patients with serum PSA level higher than 20 ng/dL were excluded and 1352 patients were assessed. Patients were divided into two groups based on the presence of chronic inflammation in prostate histopathology specimens. Group 1 consisted of 783 patients with no evidence of inflammatory process in histopathology specimens and Group 2 consisted of patients with the presence of chronic inflammatory process. All patients underwent standard TRUS-guided 12 core prostatic biopsy with calculation of total and transition zone prostatic volume. Receiver operating characteristic (ROC) curve analysis was performed and the area under the roc curve (AUC) was calculated to evaluate the diagnostic accuracy of PSA and PSA-related parameters in each group.

**Results:** The AUC for PSA density and Transition zone PSA density in Group 1 patients was

0.748 and 0.769 respectively. In this group of patients at PSA density cut-off value of 0.1, sensitivity and specificity estimates were 92% and 28% respectively and at transition zone PSA density cut-off value of 0.22 the sensitivity and specificity of prostate cancer detection was 90% and 30%. In the second group, The AUC for PSA density and Transition zone PSA density was 0.708 and 0.760 respectively. In this subgroup of patients we noted that at PSA density cut-off value of 0.1, sensitivity and specificity estimates were 90% and 23% respectively whereas at transition zone PSA density cut-off value of 0.22 the sensitivity and specificity of prostate cancer detection was 90% and 40%.

**Conclusion:** transition zone PSA density acts no better than PSA density in detecting prostate cancer in patients without prostatic inflammation, however transition zone PSA density may be considered a valuable tool in detecting prostate cancer among patients with chronic prostatitis and may reduce number of unnecessary biopsies.

**MP-12.10****Stratification of Patients with Intermediate-risk Prostate Cancer**Hong S<sup>1</sup>, Jung J<sup>2</sup>, Lee J<sup>1</sup>, Byun S<sup>1</sup>, Lee S<sup>1</sup><sup>1</sup>Seoul National University Bundang Hospital, Seongnam, South Korea; <sup>2</sup>Catholic St. Mary's International Hospital, Incheon, South Korea

**Introduction and Objectives:** Published reports on the outcomes after definitive primary treatments for intermediate-risk prostate cancer (PCa) suggest that significant heterogeneity may exist among intermediate-risk group. Thus,

we sought to identify appropriate risk stratification system for intermediate-risk PCa.

**Materials and Methods:** Classifying patients according to National Comprehensive Cancer Network (NCCN) risk groups, we reviewed data of 1559 radical prostatectomy (RP) patients who were treated at our institution from 2006 to 2013. For our analyses, intermediate-risk PCa meeting at least one of two following factors were designated as unfavorable intermediate-risk disease: biopsy Gleason score  $\geq 4 + 3$  and/or multiple ( $\geq 2$ ) intermediate-risk criteria present. All other intermediate-risk PCas were designated as favorable intermediate-risk disease. Postoperative outcomes including biochemical recurrence (BCR)-free survivals were calculated and compared via log-rank test and Cox proportional hazards model.

**Results:** In multivariable analysis, biopsy Gleason score  $\geq 4 + 3$  ( $p = 0.003$ ) and multiple ( $\geq 2$ ) intermediate-risk criteria ( $p = 0.005$ ) were observed to be independent predictors of the risk of BCR amongst intermediate-risk group. Favorable intermediate-risk group showed a significantly higher 5-year BCR-free survival than unfavorable group (87.5% vs. 66.5%) ( $p < 0.001$ ). Unfavorable intermediate-risk group showed significantly higher 5-year BCR-free survival than high-risk group (66.5% vs. 47.9%) (log rank,  $p < 0.001$ ) while favorable intermediate-risk group demonstrated significantly lower 5-year BCR-free survival than low-risk group (87.5% vs. 93.9%) (log rank,  $p = 0.002$ ).

**Conclusions:** A significant heterogeneity exists in biochemical outcomes of contemporary patients with intermediate-risk PCa who underwent definitive RP. According to biopsy Gleason score and number of intermediate-risk criteria present, intermediate-risk group should be stratified into favorable and unfavorable disease.

**MP-12.11****Tertiary Gleason Grade as Prognostic Factor of Prostate Cancer Recurrence**

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**Introduction and Objectives:** Tertiary Gleason grade (TGG) pattern in radical prostatectomy (RP) specimens is known to be associated with adverse pathology characteristics and a higher risk of biochemical recurrence (BCR). The aim of this study was to determine whether the TGG pattern in RP specimens can be used as a prognostic factor of BCR-free survival of patients undergoing RP.

**Materials and Methods:** We retrospectively reviewed medical records and histological specimens of 159 consecutive patients treated with RP at our institution. All patients did not receive any hormonal therapy. Each surgical

specimen was reviewed and assessed for the Gleason score (according to 2005 ISUPCC criteria) and the presence of TGG pattern. TGG was defined as the third most prevalent Gleason pattern if it was higher than the two predominant Gleason grades. Biochemical relapse was defined as an increase of PSA level higher than 0.2 ng/ml confirmed with at least two measurements.

**Results:** A presence of TGG in histological specimens was reported in 51 of 159 (32.1%) patients. In 34 cases (66.7%) TGG constituted  $\leq 5\%$  of the whole tumour volume, in 17 cases (33.3%) the volume was  $> 5\%$  of tumour volume. The presence of TGG was statistically significantly associated with extracapsular extension, seminal vesicle invasion, lymph node invasion and positive surgical margins ( $p < 0.05$ ). The 5-year BCR-free survival rates for patients with TGG were  $46.4 \pm 10.2\%$  versus  $76.6 \pm 5.2\%$  when no tertiary Gleason grade was present in the specimens (log-rank test  $p < 0.001$ ).

**Conclusion:** Our findings confirm the statistically significant association of the presence of a TGG with adverse pathologic features and biochemical recurrence.

#### MP-12.12

##### Implication of Ultrasensitive Prostate-Specific Antigen Assay for Prediction of Biochemical Recurrence after Radical Prostatectomy

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**Introduction and Objectives:** The role of post-operative ultrasensitive PSA level (USPSA) in identifying patients at high risk for biochemical recurrence (BCR) after radical prostatectomy (RP) is controversial but might have important implication for selection of candidates for adjuvant therapy. The aim of this study was to determine whether the prostate-specific antigen (PSA) nadir measured with an USPSA assay can be used as a prognostic factor of BCR following RP.

**Materials and Methods:** We retrospectively analyzed 136 patients who underwent RP at our institution and were followed with regular measurements of USPSA levels at 1, 3, 6, 9 and 12 months during the first year, once in 6 months during the second year and annually afterwards. Biochemical relapse was defined as PSA  $> 0.2$  ng/ml confirmed with at least two measurements.

**Results:** Forty eight (35%) of 136 patients recurred during a median follow-up of 57 months. Median PSA nadir of patients without BCR was 0.005 ng/ml, while that of BCR group 0.08 ng/ml ( $p < 0.001$ ). Median time

to achieve a USPSA nadir in BCR group was significantly shorter (median 7.5 weeks) than in the group without BCR (median 14 weeks) ( $p < 0.001$ ). The receiver operator characteristics (ROC) curve defined an optimal cut-off value for USPSA nadir of 0.02 ng/ml, showing a statistically significant difference in BCR after RP. All patients were stratified into two groups based on their USPSA nadir: Group 1, which consisted of patients whose PSA was  $\leq 0.02$  ng/ml ( $n = 87$ ), and Group 2, which consisted of patients whose PSA was  $> 0.02$  ng/ml ( $n = 49$ ). In the first group only 8 of 87 patients (9.2%) experienced BCR, whereas in the second - 40 of 49 patients (81.6%) recurred biochemically ( $p < 0.001$ ). Patients who achieved a USPSA nadir level  $\leq 0.02$  ng/ml had significantly greater BCR free survival (log-rank test  $p < 0.001$ ). Within the first 36 months after RP among patients whose PSA was  $\leq 0.02$  ng/ml, no case of BCR was registered and the 5-year BCR-free survival rates achieved  $96.7 \pm 2.3\%$ . The 3-year and 5-year BCR-free survival rates for patients with USPSA nadir  $> 0.02$  ng/ml were  $58.2 \pm 7.3\%$  and  $32.8 \pm 7.1\%$ . A multivariable Cox proportional hazards model showed that the USPSA nadir  $> 0.02$  ng/ml was a significant predictor of BCR (HR 11.77; 95% CI 5.49 – 25.23;  $p < 0.001$ ).

**Conclusion:** The PSA nadir measured with USPSA assay is a significant predictor of BCR of patients undergoing RP.

#### MP-12.13

##### Is Outcome following Radical Prostatectomy Determined by Race?

Nair R, Pai A, Kulkarni M, Campbell E, Perry M, Anderson C, Issa R

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**Introduction and Objectives:** Racial variations in outcomes following radical prostatectomy (RP) have been attributed to socio-economic deprivation and genetic factors in Afro-Caribbean men from North America. Recurrence is more common, and oncological outcomes are poorer in this group. The last ten years however, has seen both improved imaging and biopsy techniques in the diagnosis and surveillance of prostate cancer. Despite these changes, it is unclear whether the racial variation in prostate cancer outcome still remains. We aim to evaluate disease outcomes in specific racial groups within a metropolitan UK population treated following RP within the National Health Service.

**Materials and Methods:** A prospective single-centre study of 599 patients undergoing RP for localised prostate cancer was performed over between July 2002 and February 2014. Men were divided into three ethnic groups: Asian (AS), Caucasian (CA) and Afro-Caribbean (AC), and clinical and pathological variables compared. Biochemical-recurrence free survival

(BCR) (PSA  $> 0.1$  ng/mL) rates and predictors of BCR were calculated.

**Results:** A total of 599 men underwent RP (CA:  $n = 393$ , AC:  $n = 148$  and AS:  $n = 58$ ). There was no significant difference between patient age (mean, 62.4 years), follow-up (median, 49 months) or preoperative PSA (mean, 8.5 ng/ml). AC-patients had a longer operative time ( $p = 0.0038$ ) and narrow measurable pelvic-inlets ( $P < 0.0001$ ). There were no significant differences in blood loss, tumour stage, histology, volume or margin status between groups. AC-men had significantly lower 5-year BCR-free survival (56.7%) than CA-men (73.1%) or AS-men (80.8%,  $p = 0.005$ ). On multivariate analysis, AC-race was an independent predictor of BCR.

**Conclusion:** In a national healthcare system, despite advances in prostate cancer detection, surveillance and treatment, being Afro-Caribbean is a significant predictor of BCR. Asian men have favourable outcomes, and this should be taken into consideration prior to treatment. These factors do not appear to be explained by histopathological factors. There remains an imperative need to map genetic and environmental mechanisms accounting for different outcomes in this population.

#### MP-12.14

##### Predicting Biochemical Recurrence following Radical Prostatectomy through Tissue Expression of the Insulin Receptor, IGF-1 Receptor and PTEN

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**Introduction and Objectives:** The insulin receptor (INSR) and IGF-1 receptor (IGF-1R) are key components of the Insulin-like Growth Factor (IGF) axis which has been implicated in prostate cancer development and progression through activation of downstream signaling mechanisms including PI3K and AKT. The tumour suppressor gene, PTEN, is a negative regulator of the PI3K pathway. Reduced/absent PTEN expression has been linked to lethal prostate cancer. We examined the expression of the INSR, IGF-1R and PTEN in radical prostatectomy tissue of patients who developed biochemical recurrence (BCR) post-surgery.

**Materials and Methods:** Tissue microarray (TMA) of 32 patients post-radical prostatectomy (16 = BCR, 16 = controls) was stained by immunohistochemistry for the INSR, IGF-1R and PTEN. The TMA was then scored by independent pathologists. The parametric independent samples *t*-test and non-parametric Mann Whitney U test were used to examine the statistical significance of differences in INSR, IGF1-R and PTEN expression between benign and malignant tissues, and Gleason



grade. Correlations between INSR, IGF1-R and PTEN were measured using Spearman's correlation coefficient. Odd ratio's (OR) were also estimated.

**Results:** INSR and IGF-1R expression was significantly increased in malignant compared to benign prostate tissue ( $p < 0.001$ ,  $p = 0.002$  respectively). There was no difference between either receptors expression across Gleason grades ( $p = 0.482$ ,  $p = 0.391$ ) or between patients with BCR versus non-BCR ( $p = 0.929$ ,  $p = 0.289$ ). There was a significant correlation between INSR and IGF-1R expression in both benign (correlation coefficient = 0.836,  $p = 0.0001$ ) and malignant tissue (correlation coefficient = 0.436,  $p = 0.002$ ). There was a trend towards reduced PTEN expression in malignant tissue of BCR patients versus non-BCR patients ( $p = 0.0942$ ). The trend was strongest between Gleason grade 4 disease in BCR vs. non-BCR patients ( $p = 0.0651$ ). When we examined the interaction of PTEN with the IGF-1R, we found patients with high IGF-1R + low PTEN were 3.5 times more likely to experience BCR versus patients with low IGF-1R and high PTEN (OR 3.5).

**Conclusions:** Low PTEN is associated with BCR and this association is strongly modified by high IGF-1R expression. Measurement of these proteins could help to inform appropriate patient selection for post-operative adjuvant therapy and prevent BCR.

#### MP-12.15

##### External Validation of the 2007 and 2013 Updated Partin Tables in a Cohort of Irish Men

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<sup>1</sup>UCD School of Medicine and Medical Science, University College Dublin, Dublin, Ireland;  
<sup>2</sup>UCD School of Mathematical Sciences, University College Dublin, Dublin, Ireland

**Introduction and Objectives:** Our aims were to perform the first external validation of the 2013 Partin tables and to compare the 2007 and 2013 Partin tables in a cohort of Irish men.

**Materials and Methods:** Prospective data collection from 795 men was performed through the Irish Prostate Cancer Research Consortium; those for whom full data was not available were excluded resulting in 414 consecutive cases available for statistical analysis. The predictive accuracy of the 2013 and 2007 Partin tables were assessed by means of receiver operating characteristic (ROC) curves and area under the curve (AUC) values to measure discriminate ability, calibration curves to measure calibration and decision curve analysis to measure clinical benefit.

**Results:** In our cohort, the rates of organ confined (OC) disease, extracapsular extension (ECE), seminal vesicle invasion (SVI) and

lymph node involvement (LNI) respectively were 73%, 17%, 7% and 3% respectively. AUC values for OC, ECE, SVI and LNI were 0.669, 0.592, 0.695 and 0.795 respectively for the 2007 Partin tables and 0.668, 0.600, 0.688 and 0.811 respectively for the 2013 Partin tables. Both the SVI and LNI predictions from the 2007 and 2013 Partin tables are well calibrated.

**Conclusion:** The 2013 and 2007 Partin tables cannot accurately predict OC, ECE and SVI in Irish men and should not be used for this purpose. However, the 2013 Partin tables can accurately predict LNI (AUC = 0.811 - 0.978), indicating that the 2013 Tables represent a useful clinical tool to inform decision-making regarding pelvic lymphadenectomy at radical prostatectomy. The 2013 Partin tables are superior to the 2007 tables for the prediction of LNI in an Irish population, and therefore the 2013 Tables should be preferred by clinicians.

#### MP-12.16

##### The Relationship between Prostate Cancer, Presence of Metabolic Syndrome and Late Onset Hypogonadism

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**Introduction and Objectives:** We aimed to investigate the relationship between Prostate cancer (PCa), presence of Metabolic Syndrome (MetS) and Late Onset Hypogonadism (LOH).

**Materials and Methods:** One hundred and seventy patients who underwent transrectal ultrasonography guided prostate needle biopsy were included in this study. For the diagnosis of MetS, AHA/NHLBI criteria were used. For the diagnosis of LOH, Androgen Deficiency in Aging Males questionnaire and serum total and free testosterone levels were used. Patients were divided into four groups according to the presence of MetS and LOH; Group 1 MetS and LOH, Group 2 with MetS but without LOH,

Group 3 with LOH but without MetS and Group 4 with neither MetS nor LOH.

**Results:** The mean age of the patients was  $63.7 \pm 7.2$  years. In Group 1, 12 patients (37.5%); in Group 2, 5 patients (25%); in Group 3, 11 patients (26.8%) and in Group 4, 14 patients (18.2%) were diagnosed as PCa. Aggressive PCa was determined in 7 patients in Group 1 (21.9%), 2 patients in Group 2 (10%), 5 patients in Group 3 (12.2%) and 5 patients in Group 4 (6.5%). The PCa and aggressive PCa were detected at a higher rate in patients with MetS alone, independent from the presence of LOH (32.7% vs. 21.2%,  $p = 0.109$  and 17.3% vs. 8.5%,  $p = 0.092$ ). Similarly, the PCa and aggressive PCa were detected at a higher rate in patients with LOH alone, independent from the presence of MetS (31.5% vs. 19.6%  $p = 0.074$  and 16.4% vs. 7.2%,  $p = 0.059$ ). There was a statistically significant difference only in between Groups 1 and 4 in terms of detection of PCa (37.5% vs. 18.2%,  $p = 0.031$ ) and aggressive PCa (21.9% vs. 6.5%,  $p = 0.019$ ) (Figure 1).

**Conclusions:** These results highlight the fact that co-existence of MetS and LOH increases the risk of PCa and aggressive PCa.

#### MP-12.17

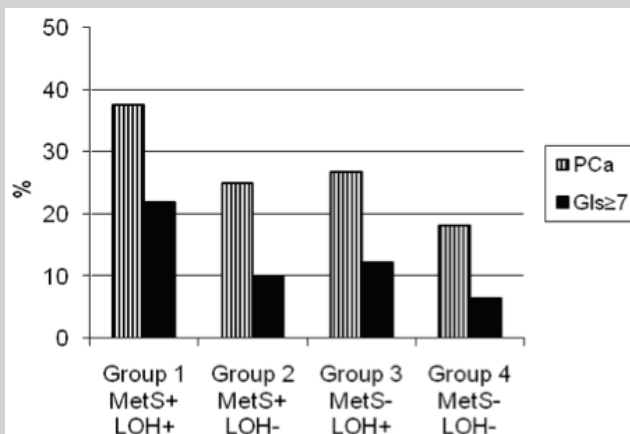
##### Variations in Prostate Biopsy Practice between West and East: Results of Two National Surveys

Wang W<sup>1</sup>, Philpott-Howard J<sup>2</sup>, Nemade H<sup>2</sup>, Sheehan S<sup>2</sup>, Thompson P<sup>3</sup>  
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**Introduction and Objectives:** To our knowledge, this is the first international survey to discover the variation in biopsy practice and policy of antimicrobial prophylaxis for prostate biopsies between west and east.

**Materials and Methods:** A questionnaire of

MP-12.16, Figure 1. PCa and Aggressive PCa Ratio of the Groups



urological practice in relation to prostate biopsy was constructed, using the website *Survey-Monkey*<sup>®</sup>. A link to the survey was e-mailed to members of the British Association of Urological Surgeons (BAUS) and the Chinese Urological Association (CUA), with the permission and co-operation of the Societies. Members were invited to complete the survey on-line.

**Results:** The proportion of urologists who responded was 189 out of 1199 (16%) in the UK and 2168/3200 (68%) in China. For the UK and China surveys respectively: 150/186

(81%) and 1963/2168 (91%) performed transrectal prostate biopsies (TRPB); 64 out of 186 (34%) and 205 out of 2168 (10%) performed transperineal prostate biopsies (TPPB). The commonest antimicrobial prophylaxis regimens were ciprofloxacin & metronidazole or co-amoxiclav & gentamicin (UK) and ciprofloxacin or levofloxacin, and cefotiam (China). The most common antibiotic duration is 1-3 days in both UK and China. A total of 12% urologists in UK recommend that antibiotic policy should depend on pre-biopsy rectal swab culture results

while only 4% urologists in China use pre-biopsy rectal culture. The results of the preference to the biopsy route are summarized as Table 1.

**Conclusion:** This survey indicates that there is a wide variety of opinions and practice in prostate biopsies among urologists who practice in the UK and China. Despite the fact that most urologists agreed that the sepsis risk after transperineal route was low, most urologist still preferred transrectal route. Further consensus, research and debate on the antibiotic policy and biopsy route are necessary.

MP-12.17, Table 1.

| Respondents -                                                                                                                                  |                                                                            | UK                  | China                |
|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------|----------------------|
| Strongly agreed or agreed that:                                                                                                                | in my experience sepsis is a significant problem post-transrectal biopsy   | 107/186 (60%)       | 889/2168 (41%)       |
|                                                                                                                                                | in my experience sepsis is a significant problem post-transperineal biopsy | 15/183 (8%)         | 65/2168 (3%)         |
|                                                                                                                                                | transrectal biopsy should be routinely replaced by transperineal biopsy    | 28/189 (15%)        | 195/2168 (9%)        |
| When consenting for transrectal biopsy, do you believe there is a medico-legal responsibility to offer transperineal biopsy as an alternative? |                                                                            | No<br>126/189 (67%) | No<br>715/2168 (33%) |
| If advised to have a prostate biopsy, would you personally prefer to have a:                                                                   | transperineal biopsy                                                       | 50/181 (28%)        | 932/2168 (43%)       |
|                                                                                                                                                | transrectal biopsy                                                         | 72/181 (40%)        | 911/2168 (42%)       |
|                                                                                                                                                | no preference                                                              | 59/181 (32%)        | 325/2168 (15%)       |

## Moderated Poster Session 13

### Minimally Invasive Surgery

Tuesday, October 14  
1300-1430

#### MP-13.01

##### Arterial Renal Embolization:

##### A Single Center Study

Thorlund M<sup>1</sup>, Wennevik G<sup>1</sup>, Andersen M<sup>1</sup>, Andersen P<sup>2,3</sup>, Lund L<sup>1,3</sup>

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**Introduction and Objectives:** Transcatheter arterial embolization (TAE) is an effective minimal invasive treatment modality, which is associated with low morbidity and mortality. TAE can be performed in patients with low performance status. Furthermore, TAE is well established in endovascular treatment of urological vascular emergencies such as iatrogenic, traumatic or spontaneously with or without underlying pathology. The typical patient undergoing TAE is not well described. This descriptive study aim to present the demographics of patients who underwent renal embolization performed elective or acute in a single center study with 24-hour available embolization.

**Materials and Methods:** All patients who underwent TAE at Odense University Hospital from October 2010 to July 2013 were identified retrospectively. Patients' medical records were examined to determine indication for treatment, procedural details and complications. Patients were divided in to 4 groups; renal cancer, trauma, angiomyolipoma (AML) and others. When elective TEA was performed, selective kidney function was determined by renography. Computed tomography with intravenous contrast confirmed the embolization diagnosis in all cases. When there was indication for embolization renal angiography was performed and followed by embolization if possible. The procedure was performed in local analgesia via the common femoral artery and as super-selective as possible to save as many nephrons as possible. Most common embolization material was coils and vascular plugs but in some cases of coagulopathy or patients on anticoagulation treatment, liquid embolization materials were used.

**Results:** A total of 35 patients were included, mean age 64 years (range 17-95), 8 females and 27 males. Fifteen patients underwent embolization because of renal cancer; 9 elective and 6 acute. Seven traumas were embolized. Five patients with AML, 3 acute and 2 elective, underwent embolization. Finally, 8 patients were treated because of spontaneous bleeding, AV-malformation and aneurisms (3 elective,

5 acute). The post-embolization syndrome occurred in 22 patients (63%) and 6 patients (17%) were re-embolized. One patient had persistent infection and one patient died (3%). Post-embolization nephrectomy was performed in four patients (11%) because of ongoing bleeding.

**Conclusion:** The most common reason for TAE was renal cancer. TAE is a safe modality with few complications both as an acute or elective procedure.

#### MP-13.02

##### Risk Factors and Management of

##### Unruptured Renal Artery Pseudoaneurysms Immediately after Minimally Invasive Partial Nephrectomy

Omae K, Kondo T, Fukuda H, Yoshida K, Takagi T, Iizuka J, Kobayashi H, Hashimoto Y, Tanabe K

Tokyo Women's Medical University, Tokyo, Japan

**Introduction and Objectives:** Renal artery pseudoaneurysm (RAP) after partial nephrectomy (PN) is a rare complication and its presentation is often delayed, which could lead to life-threatening bleeding. Its incidence has been reported to be 1 to 5%, however there are few reports of the risk factors associated with the development of RAP after PN. We have been performing 3-dimensional computed tomography arteriography (CTA) in the early postoperative period to screen for complications after PN since January 2012. Our objectives are to report the risk factors and natural history of unruptured RAP detected by the CTA screening after minimally invasive partial nephrectomy (MIPN), including laparoscopic and robotic PN.

**Materials and Methods:** We retrospectively reviewed the medical records and surgical videos of 101 consecutive patients who underwent MIPN followed by the CTA screening on the 3<sup>rd</sup> or 4<sup>th</sup> postoperative day in our hospital between February 2012 and November 2013. Two radiologists reviewed the CTA images to diagnose RAP. Patient characteristics, tumor specifics and surgical data were analyzed.

**Results:** Screening CTA revealed asymptomatic unruptured RAPs in 22 patients (21.7%) after MIPN. Patients who developed RAPs had larger resected tumors ( $p=0.02$ ) and more frequent intraoperative renal sinus exposure ( $p<0.01$ ) than patients who did not develop RAPs. Multivariate analyses identified intraoperative renal sinus exposure as the only independent risk factor of the development of RAP (renal sinus exposure: OR=4.99, 95%CI 1.11-24.0,  $p=0.03$ ).

**Conclusions:** The rate of development of asymptomatic unruptured RAP detected by the CTA screening in the early postoperative period after MIPN was higher than that of development of postoperative ruptured RAP identified

in our hospital before 2012 and reported in previous studies. Most of the RAPs immediately after MIPN might resolve spontaneously with no intervention. Avoidance of intraoperative deep excision into the renal sinus could be the most important factor in minimizing the development of RAP after MIPN.

#### MP-13.03

##### Accurate Multi-Parametric MRI

##### Monitoring of Focal Therapy with

##### Compensation for Local Deformation

Orczyk C<sup>1,2,3</sup>, Rusinek H<sup>4</sup>, Rosenkrantz A<sup>4</sup>, Valable S<sup>2</sup>, Mikheev A<sup>4</sup>, Villers A<sup>5,6</sup>, Taneja S<sup>7</sup>

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**Introduction and Objectives:** Focal therapy (FT) is an emerging approach for treatment of localized prostate cancer. Multi-parametric (mp) MRI demonstrated capability to monitor the effect of FT procedures. At time of follow-up, the ablated zone (AZ) clearly undergoes local shrinkage. Accurate definition of AZ at follow-up will improve FT evaluation and oncologic safety. We analyzed the volume and shape changes of the gland between pre and post FT MRI by developing a 3D coregistration method to compensate for deformation of the gland in response to FT.

**Materials and Methods:** We studied 10 patients who underwent FT (interstitial laser ablation and photodynamic therapy) within IRB approved trials. All patients underwent preoperative, early control and late postoperative 3T MRI which included T2, T1, diffusion and perfusion weighted sequences. We have developed image registration software to analyze, transfer and model shape changes using a deformable and a rigid body transformation. Alignment between pre- and post-op images of AZ was assessed using the overlap index or Dice index (Di) and the maximum boundary distance, or Hausdorff distance (HD). Correction for deformation was measured using the HD normalized by the volume to transform in mm/cc and automated feature of the software.

**Results:** There was a significant volume decrease D of the gland that averaged 6.49 cc ( $p=0.017$ ) between preoperative and postoperative gland. D was directly correlated ( $\rho=0.738$ ,  $p=0.014$ ) with the ablated volume (7.88,  $p=0.04$ ). We successfully co registered pre operative to post operative MRI in each cases. There was a significant increase D computed with deformable versus rigid transform. Deformable

model achieved a significantly more accurate match of pre- vs. post-FT AZ with deformable ( $Di=0.88$ ,  $HD=1.98$  mm) vs. rigid transformation ( $Di=0.95$ ,  $HD=3.83$  mm). The deformable approach also yielded a higher ( $p=0.019$ ) correction of deformation ( $0.72$  mm/cc) compared to the rigid model ( $0.15$  mm/cc).

**Conclusion:** We described a novel method that compensate in 3D for the local deformation induced by FT. The method achieves better than 2mm of accuracy of aligning pre and post FT MRI. These findings could be implemented in clinical trials for follow-up evaluation and oncologic safety while incorporated in biopsy platforms.

#### MP-13.04

##### **Point of Technique: Reducing Wrong Side Errors for Endourology Procedures**

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**Introduction and Objectives:** Upper tract endourological procedures form a key part of the workload in Urology. The potential for wrong side surgery exists despite the adoption of universal precautions such as the WHO checklist. To reduce such risks even further, we describe a novel simple technique used when inserting a double J stent or performing upper tract endourological procedures.

**Materials and Methods:** We use an adhesive Skintact<sup>®</sup> ECG electrode (Leonhard Lang GmbH, Austria) placed in the corresponding groin on the side of the intended procedure in full view of the whole surgical team at the time of the surgical checklist. This provides a visual aid on the fluoroscopy image to ensure the correct side is being performed, especially when the marked side is often obscured by the surgical drape.

**Results:** Use of this technique is an invaluable tool in the prevention of wrong side surgery during upper tract endoscopic procedures.

**Conclusion:** With an estimated 1 in 10 patients experiencing an adverse event during their hospital stay, patient safety and implementation of measures to prevent wrong side surgery should be paramount. The negative consequences upon the patient and the surgeon should not be underestimated and any intervention or point of technique which will help reduce medical errors should be welcomed.

#### MP-13.05

##### **Laparoscopic Nephrectomy for Adult Polycystic Kidney Disease: Techniques, Complications and a Novel Minimally Invasive Approach**

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<sup>2</sup>NHS Tayside, Dundee, Scotland, UK

**Introduction and Objectives:** Indications for laparoscopic renal surgery are increasing, however benefits in APKD remain uncertain. In cases where kidneys are massively enlarged, laparoscopic approach can be challenging. The data from studies reporting on resection in APKD has not been synthesised. In Tayside a novel technique is used, involving the use of intra-operative ultrasound to assist with cyst puncture for kidney debulking. We aimed to synthesise results from observational studies including a Tayside cohort and assess safety, feasibility and complications.

**Materials and Methods:** We conducted a review of the published literature reporting on laparoscopic nephrectomy in APKD between 1991 and 2013. The criteria from the 'Strengthening the Reporting of Observational Studies in Epidemiology' (STROBE) study were used to assess the quality of reported literature. Unpublished local data including description of a novel retroperitoneoscopic technique was added to the published evidence.

**Results:** Three prospective and twenty-one retrospective studies of low to modest quality (according to the STROBE checklist) were identified, reporting on 327 patients. Transperitoneal approach was the most commonly used. Mean mass of affected kidneys was 1,186g (range 132g – 7,200g). The duration of hospital stay ranged from 1.5 to 12 days; mean 5 days. Operative time ranged from 90-568 minutes, with 13% of patients requiring blood transfusion. There were 26 intra-operative complications and 80 post-operative complications, a rate of 8% and 24% respectively. A total of 17 cases (5%) were converted to open technique. None of the included studies reported any mortality.

**Conclusion:** Laparoscopic approach to excision of APKD is safe and feasible, but complication rate is high. Quality of included studies was poor. Further research is required to clarify the generalisability and validity of different approaches, including greater reporting of patient outcomes with laparoscopic techniques. A modified retroperitoneoscopic technique described in this study could be offered to patients. Higher quality research is required to establish the generalisability of the approaches including standardisation of approach.

#### MP-13.06

##### **Keep Your Head: Three-Port Laparoscopic Radical Prostatectomy Should Be a More Cost-Effective Therapy in Developing Countries like China**

Xu B, Luo C, Zhang Q, Jin J

Peking University First Hospital, Beijing, China

**Introduction and Objectives:** To introduce a more cost-effective therapy of three-port laparoscopic radical prostatectomy (LRP) for prostate cancer (PCa) in developing countries

like China, and compare its results with the conventional four-five port LRP of ours and other minimally invasive techniques performed by others.

**Materials and Methods:** We retrospectively reviewed the data of 112 patients with PCa receiving LRP between January 2011 and December 2013 at our institution. They were divided into Group A (three-port LRP, 24 patients) and Group B (conventional four-five port LRP, 88 patients). Both groups were compared by the perioperative parameters. A learning curve was also analyzed by dividing these 24 patients into the early and late stage.

**Results:** The two groups were comparable with regard to all of the preoperative characteristics. All three-port LRP were performed successfully. There was no conversion to open radical prostatectomy (ORP) or four-five port LRP. In Group A, the mean operative time (OT) was 89.3min, the mean estimated blood loss (EBL) was 132.5ml, the mean hospitalization was 4.2d, and 29.2% of the prostate specimen margins (PSM) were positive. In Group B, the figures were 100.6min, 216.7ml, 5.2d respectively and 33.0% of PSM were positive. The differences of OT and EBL were statistically significant between both groups ( $p<0.05$ ). After undergoing the early stage of a learning curve in three-port LRP, the OT and EBL could be decreased evidently.

**Conclusions:** Three-port LRP is a modified technique, which exhibits superior intraoperative advantages to the conventional LRP. Simultaneously, by making a comparison between three-port LRP with the other minimally invasive techniques, due to its lower cost and less EBL with a shorter learning curve and OT, it should be recommended that three-port LRP is a more cost-effective therapy in developing countries like China. Please keep your head when faced to the hot topic!

#### MP-13.07

##### **Laparoendoscopic Single-Site (LESS) for Treatment of Different Urologic Pathologies: Single-Centre Single-Surgeon Experience**

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**Introduction and Objectives:** LESS is an evolving minimally invasive treatment option in urology. However, LESS needs a steep learning curve and advanced laparoscopic background. We present our single-centre single-surgeon experience of LESS for treatment of different urologic pathologies.

**Materials and Methods:** Between January 2010 and June 2013, 114 consecutive patients had LESS by a single surgeon with advanced laparoscopic skills. LESS procedures included simple nephrectomy ( $n=18$ ), radical nephrectomy ( $n=12$ ), nephroureterectomy (6), pyeloplasty



(n=13), excision of renal cyst (13), ureterolithotomy (8), orchiopexy (n=18), varicocelectomy (n=8), repair of vesicovaginal fistula (VVF) (n=8), repair of vesicouterine fistula (n=5), distal ureterectomy (n=2), adrenalectomy (n=1), sacrocolpopexy (n=1) and bladder diverticulectomy (n=1). Data was collected in a standard data sheet and all procedures were approved by the Ethical Care Committee of our institute.

**Results:** Mean age of patients was 28.6 years while mean BMI was 28.5. Of all the patients, 25.4% had previous abdominal surgery. Upper urinary tract LESS procedures were done in 60% of the patients. Both transperitoneal and retroperitoneal LESS was done in 93% and 7% of the patients respectively. Intraoperative and postoperative complications were reported in 2.7% (Clavien Grade II) and 5.4% (Calvein Grade II 3.6% and IIIa 1.8%) of the patients respectively. All intraoperative complications were reported during the first 30 LESS procedures. Mean blood loss was 69 c.c., while mean operative time was 120 minutes. Adding of one 5-mm extraport, conversion to conventional laparoscopy and conversion to open surgery was reported in 14, 0.9, 0.9% of the cases, respectively. Mean postoperative hospital stay was 1.4 days, while mean Visual Analogue Pain Score at discharge was 1.2. On multivariate analysis, factors significantly associated with longer operative time were history of prior abdominal surgery and high BMI, while reconstructive and pelvic LESS procedures were significantly associated with a higher risk of adding an extraport. At a mean follow-up of 21.7 months, all patients with reconstructive LESS procedures but one (with complex VVF) were successful.

**Conclusions:** In experienced hands, LESS is technically feasible for both ablative and reconstructive urologic applications. Although LESS is still technically difficult, both conversion and complications rates are low.

#### MP-13.08

##### Laparoendoscopic Single-Site Surgery (LESS) and Conventional Laparoscopy for Treatment of Different Upper Urinary Tract Pathologies: Outcome of Prospective Comparative Study

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<sup>1</sup>Alexandria University, Alexandria, Egypt; <sup>2</sup>Tanta University, Tanta, Egypt

**Introduction and Objectives:** There are increasing reports in the literature about role of LESS as an alternative to conventional laparoscopy (CL) for treatment of different urologic pathologies. However, most of these reports are retrospective case series. We present a prospective comparative study between LESS and CL for treatment of different upper urinary tract pathologies.

**Materials and Methods:** This is a prospective

study that included consecutive patients with different upper urinary tract pathologies who had LESS and CL at both Alexandria and Tanta universities from December 2011 to June 2013. Procedures were done by two experienced laparoscopists. Exclusion criteria included absolute contraindications of laparoscopy and children less than 3 years. Data was independently collected by two persons in a standard data sheet. All procedures were approved by the Ethical Care Committee at both institutes.

**Results:** Included in this study were 58 patients (30 patients had 31 LESS procedures and 28 patients had CL). Indications included pyeloplasty (LESS=10, CL=4), simple nephrectomy (LESS=7, CL=6), excision of renal cyst (LESS=7, CL=7; one patient had bilateral LESS renal cysts excision), nephroureterectomy (LESS=2, CL=0), ureterolithotomy (LESS=2, CL=5), radical nephrectomy (LESS=1, CL=1), adrenalectomy (LESS=1, CL=1), ureterectomy (LESS=1, CL=0), pyelolithotomy (LESS=0, CL=3), partial nephrectomy (LESS=0, CL=1). Mean age was significantly less and BMI was significantly higher in LESS group. Mean operative time and blood loss was significantly less in LESS group. Intraoperative complications were reported in 12.6% and 3.3% of CL and LESS groups, respectively. In 54.2, 37.5 and 8.3% of patients in CL group 3, 4 and 5 ports (5-15mm), were used respectively, while 5-mm extraport was added in 3.6% of LESS group. Conversion to open was reported in 8.3% and 3.3% of CL and LESS groups respectively, while none was converted to CL in LESS group. Postoperative hospital stay, analgesic requirement and visual analogue pain scale was significantly less in LESS group. Patient satisfaction regarding postoperative scar was higher in LESS group.

**Conclusions:** LESS offers a safe and efficient alternative to CL for treatment of different upper urinary tract pathologies, with reduced pain, faster recovery, less complications and better cosmetic outcome.

#### MP-13.09

##### Major Vascular Injury in Laparoscopic Urology: Single Center Experience in 5347 Cases

Simforoosh N, Basiri A, Ziaee S, Tabibi A, Nouralizadeh A, Radfar M, Sarhangnejad R, Mirsadeghi A

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**Introduction and Objectives:** Major vascular injury is the most devastating complication of laparoscopy, occurring most commonly during the laparoscopic entry phase. Our goal was to report our experience with major vascular injury during laparoscopic entry with closed- and open-access techniques in urologic procedures.

**Materials and Methods:** All 5347 patients who underwent laparoscopic urologic procedures from 1996 to 2011 at our hospital were included in the study. Entry was carried out using either the closed Veress needle technique or the modified open Hasson technique. Patients' charts were reviewed retrospectively investigating access-related major vascular injury.

**Results:** The closed technique was used in the first 474 operations and the open technique in the remaining 4873 cases. We had three cases with major vascular injury in our patients. They were three men scheduled for nephrectomy without any history of previous surgery. All injuries occurred in the closed-access group during the setup phase with insertion of the first trocar. Injury location was the abdominal aorta in two patients and the external iliac vein in one patient. Management was performed after conversion to open surgery, control of bleeding, and repair of the injured vessel.

**Conclusions:** To our knowledge, this is the first report of laparoscopic entry in urologic procedures. Given the high morbidity and mortality of major vascular injury, its clinically higher incidence in laparoscopic urologic procedures with the closed-access technique leads us to suggest using the open technique for the entry phase of laparoscopy. Using the open-access technique may decrease laparophobia and encourage a higher number of urologists to enter the laparoscopy field.

#### MP-13.10

##### Transvaginal Natural Orifice Transluminal Endoscopic Surgery (NOTES) in Urology: One Single Centre Experience

Zou X, Zhang G, Yuan Y, Xiao R, Wu G, Wang X, Xue Y

First Affiliated Hospital of Gannan Medical University, Ganzhou, China

**Introduction and Objectives:** The feasibility of transvaginal natural orifice transluminal endoscopic surgery (NOTES) has already been demonstrated in urology. However, the experience is limited. We aimed to describe our experience with the transvaginal NOTES in female patients, and to evaluate its feasibility, safety and efficacy.

**Materials and Methods:** Between May 2010 and December 2013, 172 female patients with a mean age of 36.3 (range 28 to 65) years and a median body mass index of 26.2 (range 18.2 to 29.8) kg/m<sup>2</sup>, were subjected to transvaginal NOTES in our center. After induction of general anesthesia, the patients were positioned in lithotomy with ipsilateral lumbar at 60° angle to the floor. In transvaginal NOTES-assisted laparoscopic procedures, a 5-mm trocar and a 10-mm trocar were inserted in the umbilical edge for conventional operating apparatus. A 5-mm or 10-mm trocar was inserted in the posterior vaginal fornix for a 5-mm flexible-tip

0° or 10-mm 30° laparoscope. In pure transvaginal NOTES procedures, a 30-mm incision was made at the posterior vaginal fornix, and a 5mm trocar was introduced into the pelvic cavity guided by a 5-mm forceps. A 5-mm flexible-tip 0° laparoscope was inserted into the pelvic cavity confirming no rectal injury. Then a Zou-Port was introduced at the posterior vaginal fornix. Dissection was performed according to the method of the standard laparoscopy. The intact specimen was extracted transvaginally.

**Results:** Transvaginal NOTES was successfully completed in 172 patients. One patients required conversion to suprapubic-assisted laparoendoscopic single-site surgery because of the rectal injury in pure transvaginal NOTES nephrectomy. Two patients underwent open conversion because of the injury of inferior vena cava (N=1) and the injury of spleen (N=1) during the dissection. The various transvaginal NOTES-assisted laparoscopic procedures performed included adrenalectomy (N=21), nephrectomy (N=124; simple 101, radical 23), nephroureterectomy (N=4), nephron sparing surgery (N=1), and heminephroureterectomy for duplex kidney (N=1). The various pure transvaginal NOTES procedures performed included renal cyst excision (N=5), nephrectomy (N=16; simple 15, radical 1). The mean operative time was 123, 116, 183, 188, and 87 minutes, and estimated blood loss was 162, 94, 137, 175, and 26 ml for transvaginal NOTES-assisted nephrectomy, transvaginal NOTES-assisted adrenalectomy, transvaginal NOTES-assisted nephroureterectomy, pure transvaginal NOTES nephrectomy, and pure NOTES transvaginal renal cyst excision respectively. The mean follow-up of 23.8 (range 2 to 45) months showed hidden umbilicus scar. The umbilical scar of each patient was hidden and difficult to find. The incision in the vagina healed well. No infection in the abdominal or pelvic cavity or celiocoele occurred.

**Conclusion:** Transvaginal NOTES is feasible, safe and effective. It provides a good cosmetic outcome. However, existing instruments need improving for the development of transvaginal NOTES.

#### MP-13.11

##### **Pure Transurethral Natural Orifice Transluminal Endoscopic Surgery (NOTES) for Fenestration and Drainage Treatment of Renal Cyst: Report of Eight Cases**

Zhang G, Xu H, Yang J, Liu M, Xu R, Zou X  
First Affiliated Hospital of Gannan Medical University, Ganzhou, China

**Introduction and Objectives:** To describe the initial clinical experience of pure transurethral NOTES for fenestration and drainage treatment of renal cyst, and evaluate its feasibility and safety.

**Materials and Methods:** From May 2012 to

August 2013, five female patients and three male patients with renal cyst underwent pure transurethral NOTES for fenestration and drainage treatment. The median age was 29.2 (range 22 to 42) years. In this group, 3 renal cysts were on the right side and 5 on the left. All cases were confirmed by B ultrasound, CT scan and IVU. The median diameter of renal cyst was 6.3 (range 5.2 to 7.2) cm. After induction of general anesthesia, flexible ureter scope was transurethrally introduced into renal collect system. Renal cyst was found by B ultrasound guide when necessary. A crisscross incision was cut by Holmium laser to 1.5-2.0cm, and a 7F double J stent was positioned with the proximal end coiled in the cyst cavity, which was removed after 4 weeks.

**Results:** All procedures were successfully completed. Three cysts were found by flexible ureter scope and the others were found by B ultrasound guide. The median operation time was 55 (range 30 to 125) min. There were no intraoperative or postoperative complications. The diameter of cyst decreased at least 1/2 during follow-up (3 to 15 months).

**Conclusion:** Pure transurethral NOTES for fenestration and drainage treatment of renal cyst is safe, feasible, minimal invasive and cosmetic. It is worth selecting the method to treat endogenous renal cyst.

#### MP-13.12

##### **Defining a Learning Curve from Simple to Complex: Adopting Tumour Nephrometry Scores during the First One Hundred Robot Assisted Partial Nephrectomies**

Nair R, Pai A, Macarthur R, Tsavalas P, Issa R, Patel H, Anderson C  
St. George's Hospital, London, UK

**Introduction and Objectives:** Present learning curve data for robot-assisted partial-nephrectomy (RA-PN) has reflected on achieving acceptable warm ischaemic (WIT) and console times (CT). Development of modular training however, must incorporate tumour complexity. We critically evaluate Pre-operative Aspects and Dimensions Used for an Anatomical (PADUA) nephrometry scores against learning curve parameters.

**Materials and Methods:** A prospective study of 100 consecutive patients undergoing RA-PN was performed from August 2008 to February 2014. Individual PADUA nephrometry scores (range 6-12) were evaluated against learning curve parameters and Clavien-Dindo complications (CDC).

**Results:** Twenty consecutive cases were required to achieve a WIT of less than twenty minutes and ten cases to achieve a CT of less than 120 minutes. These cases demonstrated PADUA scores of less than or equal to eight (PADUA ≤8). The first attempted PADUA score-9 was case 23. When comparing individual PADUA

scores, against WIT, CT and CDC, there were statistical differences beyond PADUA score-8. A further five PADUA ≥8 cases (case-number thirty-six) were required to achieve a WIT of less than twenty minutes. An extra ten PADUA ≥8 cases (case number 45) were required to achieve a CT of less than 120 minutes with no significant change in CDC. Between individual PADUA scores, there were no significant differences in blood loss, CDC, positive surgical margins or pre-and post-operative estimated glomerular filtration rate.

**Conclusion:** Nephrometry scoring creates opportunity for RA-PN modular training by providing an objective parameter of difficulty. Only tumours with PADUA scores of less than or equal to eight should be attempted during the first twenty case learning curve.

#### MP-13.13

##### **Is Robotic Partial Nephrectomy Safe for T1b Tumours? A Comparison of the Functional and Oncological Outcomes for T1a and T1b Tumours at a Single Centre**

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Guy's and St Thomas' NHS Trust, London, UK

**Introduction and Objectives:** Partial nephrectomy remains the gold standard in the definitive treatment of T1a renal tumours, with robotic partial nephrectomy (RPN) gaining popularity. To date, only a handful of retrospective studies explore the feasibility of RPN for T1b tumours. We assessed whether outcomes achieved using RPN in T1a tumours could be successfully reproduced in T1b tumours.

**Materials and Methods:** Using a prospective database of 116 elective cases, the peri-operative, oncological, and functional outcomes of 95 T1a tumours were compared with 21 T1b tumours including a single T2a tumour.

**Results:** Mean age was 55.9 years (T1a) and 55.3 years (T1b), mean tumour size 2.59 cm vs. 5.10 cm and mean PADUA scores 7.39 (T1a) vs. 9.08 (T1b) (p<0.01). There were no significant differences in operative times, 179 mins (T1a) vs. 192 mins (T1b), warm ischaemic times (18.0 vs. 19.8 mins) or hospital stay (3.5 days and 3.4 days). Estimated blood loss was significantly decreased in the T1a group (147 mls vs. 256 mls), with no difference in subsequent haemoglobin drop (1.44 vs. 1.45 g/dl). There were 2 positive margins early in the T1a group but no radiological recurrences at mean 16 months. There was 1 conversion to radical nephrectomy in the T1a group but no transfusions or open conversions. Both groups had 1 Clavien grade IIIa complication (angiobolism) with 1 Clavien IIIb (ureteric stent) in the T1a group. Serum creatinine rose by 5.18 mol/l (T1a) and 12.5 mmol/l (T1b) (NS). 65/95 T1a's and 16/21 T1b's were performed for malignancy.

**Conclusion:** In the elective setting RPN can be performed safely on carefully selected T1b tumours to achieve equivalent oncological and functional results to those seen with T1a tumours, therefore potentially extending indications for RPN. It is possible the advantages offered by the Da Vinci Surgical system may overcome the limitations previously posed by the larger T1b tumours with both OPN and LPN.

#### MP-13.14

##### Comparison of Peri-Operative and Pathological Outcomes for Open and Robotic Assisted Laparoscopic Prostatectomy during Early Stages of Surgeon's Learning Curve for Both the Procedures

Nandwani G, Chahal R, Hanchanale V, Singh R, Addla S

Bradford Teaching Hospital NHS Trust, Bradford, UK

**Introduction and Objectives:** Robotic assisted laparoscopic prostatectomy (RALP) has become the current standard of care for prostatectomy. The evidence has accrued from high volume centres performed by experienced surgeons who have plateaued their surgical learning curve. We assessed the outcomes of RALP and open prostatectomy performed by surgeons relatively early in their surgical learning curve.

**Materials and Methods:** We retrospectively compared the first 150 consecutive cases of RALP with similar number of open prostatectomies carried out by surgeons at our institute who all were relatively new to independently performing both these procedures.

**Results:** The preoperative patient characteristics were evenly matched between both the cohorts (Table 1). However, patients in the open cohort had a higher incidence of Gleason 7 or more. There was a higher proportion of T3 disease in the open cohort compared to RALP cohort but the margin positive rate was higher for both T2 and T3 in the open prostatectomy cohort. Clavien-Dindo grade 4 complications were notably higher in the open cohort compared to the RALP cohort. The open cohort also had a strikingly higher proportion of bladder neck strictures (Table 2). There were no immediate returns to theatre or post-operative deaths in either of the study arms.

**Conclusion:** This study reaffirms the superiority of RALP in terms of perioperative complications and pathological outcomes even when performed by relatively inexperienced surgeons. RALP can be safely introduced to a new program and demonstrate early patient benefits.

#### MP-13.15

##### Retroperitoneal Robotic Assisted Dismembered Pyeloplasty: Initial Experience

Drinnan N, Hindley R, Emara A, Ni Raghallaigh H, Barber N

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**Introduction and Objectives:** Robotic assisted pyeloplasty (RAP) has gained wide acceptance around the world as surgical treatment for pelvi-ureteric junction obstruction (PUJO); the advantages of the robotic system thanks to binocular vision and wristed instruments allowing more reproducible and rapid suturing than the more technically challenging minimally invasive alternative of standard laparoscopy. The approach employed in descriptions of RAP is almost exclusively transperitoneal; we report our series of RAP employing only a retroperitoneal approach.

**Materials and Methods:** We prospectively collected data from 2010 to 2013, all patients undergoing a retroperitoneal 4 port RAP employing a standard dismembered technique, parameters recorded prospectively included

operative time, length of stay and peri-operative complications. Outcome data has been collected yearly.

**Results:** In total 46 patients underwent the procedure. The average age was 46 years with an exact 1:1 Right:Left ratio. Forty percent were tertiary referrals and over that time a further 18 laparoscopic procedures were performed. The average operative time was 151.4 minutes (skin to skin) and blood loss 10.5ml. One patient did require conversion to open because of anatomical difficulty. The median length of in hospital stay was 1 day. We have encountered two failures to date, both as a result of re-stricture.

**Conclusions:** The retroperitoneal approach in performing robotic assisted pyeloplasty achieves comparable outcomes with alternative

MP-13.14, Table 1. Preoperative Characteristics of Patients in the 2 Cohorts

|                          | Open           | RALP         |
|--------------------------|----------------|--------------|
| Total number of patients | 150            | 150          |
| Age                      | 65 (43 – 75)   | 63 (44 – 71) |
| PSA                      | 8.3 (0.5 – 32) | 8 (0.9 – 26) |
| Gleason score 6          | 39%            | 61%          |
| Gleason score 7          | 51%            | 36%          |
| Gleason score >7         | 10%            | 3%           |
| Clinical T1              | 19%            | 29%          |
| Clinical T2              | 64%            | 60%          |
| Clinical T3              | 17%            | 11%          |
| Radiological T2          | 61%            | 60%          |
| Radiological T3          | 39%            | 40%          |

MP-13.14, Table 2. Post-Operative Complications and Pathological Outcomes for the 2 Cohorts

|                                            | Open      | RALP     |
|--------------------------------------------|-----------|----------|
| Pathological staging                       |           |          |
| pT2                                        | 39%       | 56%      |
| pT3                                        | 61%       | 44%      |
| Post op complications                      |           |          |
| Bladder neck contracture needing operation | 11 (7.3%) | 0        |
| Incisional hernia repair                   | 2 (1.3%)  | 3 (2%)   |
| Peri-op complications                      |           |          |
| Clavien-Dindo grade 1                      | 4 (2.6%)  | 5 (3.3%) |
| Clavien-Dindo grade 2                      | 3 (2%)    | 0        |
| Clavien-Dindo grade 3a                     | 3 (2%)    | 2 (1.3%) |
| Clavien-Dindo grade 3b                     | 2 (1.3%)  | 2 (1.3%) |
| Clavien-Dindo grade 4                      | 4 (2.6%)  | 0        |
| Margin status                              |           |          |
| Overall margin positive                    | 37%       | 12%      |
| pT2 margin positive                        | 11%       | 4%       |
| pT3 margin positive                        | 54%       | 23%      |



approaches and is associated with a low complication rate and associated surgical morbidity reflected in a very rapid discharge from hospital, most patients being discharged the following day.

#### MP-13.16

##### **Patient's Experience of the Surgeon's Robotic Learning Curve**

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**Introduction and Objectives:** Robotic assisted laparoscopic radical prostatectomy (RALP) is known to be associated with a difficult surgical learning curve. The objective of this study was to understand the impact of a robotic surgical team's learning curve on the patient's apprehensions, expectations and also patients' experience of undergoing an innovative surgical procedure. **Materials and Methods:** Patient questionnaires were posted to the first 60 patients who underwent RALP at our hospital. The questions were divided into 4 domains: 1) Anxiety and expectations about the operation; 2) quality of pre-operative information; 3) patients experience of RALP and 4) views on expensive innovative surgical techniques.

**Results:** To date 50 patients (83.3%) returned the completed questionnaires. All patients who were offered RALP agreed to undergo the procedure, including those initial patients who were clearly informed about the surgeon's lack of robotic surgical experience. The majority (88%) of the patients felt they were well informed pre-operatively. Twenty percent and 58% said they were apprehensive about developing a major or minor complication with RALP respectively. The actual patient perceived major and minor complication rates were 6% and 26%, respectively. With respect to return of urinary continence following RALP, 63% attained this function sooner than or at their expected post-operative phase. The remaining 37% attained urinary continence later than their expectations. Return to normal physical activity was concordant with patients' expectations in 90%. All patients rated their overall robotic surgical experience as excellent or good with 100% of the patients highly recommending robotic surgery to their friends and relatives. All patients felt the extra cost involved in delivering robotic surgery was justified.

**Conclusion:** Our study shows a very good patient satisfaction associated with RALP. Although our belief was that patients undergoing RALP in the initial phase of the surgeon's learning curve would be more anxious and apprehensive, this was not the case. There was no relation between patients' anxiety levels and surgeon's surgical experience. This study further clarifies the role of fully informing patients

pre-operatively and shows that patient can be equally anxious irrespective of the surgeon's position on the ladder of surgical experience.

#### MP-13.17

##### **Transvaginal Natural Orifice Transluminal Endoscopic Surgery (NOTES)-Assisted versus Conventional Laparoscopic Nephrectomy: A Prospective, Non-Randomized Trial at a Single Center** Xue Y, Zou X, Zhang G, Yuan Y, Xiao R *First Affiliated Hospital of Gannan Medical University, Ganzhou, China*

**Introduction and Objectives:** To compare the surgical outcomes of two contemporary series of female patients with benign or malignant kidney disease treated by transvaginal natural orifice transluminal endoscopic surgery-assisted nephrectomy (NOTES-N) or conventional laparoscopic nephrectomy (LN).

**Materials and Methods:** This was a non-randomized prospective comparative study of all female patients undergoing NOTES-N or LN at our institution between September 2011 and October 2013.

**Results:** We enrolled 94 patients in the NOTES-N and 98 in the LN group; the two groups were comparable for all preoperative parameters except for median age. Procedural time, length of postoperative hospital stay, and the rate of intra- and postoperative complications were similar in the 2 groups. Both the visual analogue scale and the postoperative use of analgesics were significantly lower during postoperative days 1, 2, and 3 in patients who underwent NOTES-N, compared with patients who underwent LN. Time to return to normal activities was shorter in the NOTES-N group compared with the LN group. In the NOTES-N group, patients were significantly more satisfied with the cosmetic result ( $P < 0.05$ ) and were reported unaltered sexual function after surgery.

**Conclusion:** This series adds to the existing evidence that NOTES-N is a technically feasible and safe procedure with significantly less pain and faster recovery compared to conventional LN; however, multi-institutional randomized trials are required to confirm benefits.

#### MP-13.18

##### **A Prospective Investigation of the Impact of Transvaginal Natural Orifice Transluminal Endoscopic Surgery (NOTES)-Assisted Laparoscopic Nephrectomy on Female Sexual Function and Quality of Life** Jiang B, Zou X, Wu Y, Zhang G, Yuan Y *First Affiliated Hospital of Gannan Medical University, Ganzhou, China*

**Introduction and Objectives:** To investigate the impact of transvaginal natural orifice transluminal endoscopic surgery (NOTES)-assisted laparoscopic nephrectomy on female sexual

function and quality of life.

**Materials and Methods:** From May 2010 to November 2012, 81 females underwent transvaginal NOTES-assisted laparoscopic nephrectomy. According to the inclusion and exclusion criteria, 42 patients with a mean age of  $36.90 \pm 5.31$  years and a mean body mass index of  $21.67 \pm 2.60$  kg/m<sup>2</sup> were included in the investigation. The female sexual function and quality of life were assessed before and 4 months, 7 months and 1 year after surgery using the Female Sexual Function Index (FSFI) questionnaire and the MOS 36-item Short-Form Health Survey (SF-36), respectively.

**Results:** The mean FSFI of 42 cases preoperatively and 4 months, 7 months and 1 year postoperatively were  $27.74 \pm 4.34$ ,  $27.19 \pm 4.49$ ,  $28.54 \pm 4.23$ , and  $28.68 \pm 4.19$ , respectively. There was no statistically significant difference among them ( $F=1.111$ ,  $P=0.346$ ). Compared with that of preoperation, the physical function, vitality, mental health, body pain, and general health of the patients were improved, but the role-physical, role-emotion and social function were not improved at postoperative month 4 and month 7 ( $P < 0.05$ ). Each item of SF-36 was improved after postoperative year 1 ( $P < 0.05$ ).

**Conclusion:** Transvaginal NOTES-assisted laparoscopic nephrectomy could not cause negative effects on the female sexual function. The quality of life could be improved after operation. The physical function is improved at early stage, and then the psychological function.

#### MP-13.19

##### **Robotic Pyeloplasty: Initial Experience of a Single UK Centre**

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**Introduction and Objectives:** In order to decrease the morbidity and improve outcomes associated with pyeloplasty for the treatment of pelvoureteric junction obstruction, a robotic assisted laparoscopic approach is increasingly being used. We present our experience at a single UK robotic centre with robotic assisted laparoscopic dismembered pyeloplasty.

**Materials and Methods:** Between July 2009 and July 2013, 20 robotic-assisted transperitoneal laparoscopic pyeloplasties were performed by 4 surgeons. Data were obtained from patient case notes, patient charts, and radiographic reports.

**Results:** A total of 20 robot assisted laparoscopic dismembered pyeloplasties were performed in 11 men and 9 women. Mean age at treatment was 37 years (range 16–75) with an average follow-up of 14 months (5–24). The average console time was 2hr:08min (1:30–3:30), mean drop in haemoglobin  $0.67$ g/dL ( $-3.5$ – $+1.3$ ),



mean change in serum creatinine -0.5 (-22-+18). The average time to post-operative catheter removal was 1.64 days (1-2), to post-operative drain removal 2.43 days (2-3), and to discharge 2.8 days (2-4). The average time to stent removal was 6.1 weeks (6-8). At the mean follow-up of 10 no major perioperative complications had occurred and no patients were re-admitted within 30 days of operation. The success rate was 95%, one patient required a second procedure in the form of an open redo pyeloplasty because of continued pain and radiographic evidence of continued obstruction.

**Conclusions:** These results suggest that robotic assisted laparoscopic pyeloplasty is a feasible treatment for PUJ obstruction in a UK centre. Low rates of morbidity, short post-operative stay and high success rates at our centre are comparable with other series. Longer follow-up of this cohort is needed to demonstrate durable and effective outcomes.

#### MP-13.20

##### **Suprapubic-Assisted Laparoendoscopic Single-Site Surgery (SAE-LESS) for Nephroureterectomy**

Liu Q, Zou X, Zhang G, Liao Y, Xue Y, Yuan Y, Xiao R, Wu G, Wang X, Liu F  
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**Introduction and Objectives:** During the past years, suprapubic-assisted laparoendoscopic single-site surgery (SA-LESS) established by ourselves has been used in urology in our center. We report 17 cases of SA-LESS for nephroureterectomy in a patient with right upper urinary tract cell carcinoma.

**Materials and Methods:** Seventeen consecutive patients including 10 males and 7 females, with a mean age of 67.6 years (range 58 to 76), were subjected to SA-LESS nephroureterectomy in our center. There were 8 renal pelvic carcinomas, 4 ureteral carcinomas, one renal pelvic carcinoma combined with ureteral carcinoma, two renal tuberculosis, and one megaureter with empyema included in this study. The patients were administrated general endotracheal anesthesia and secured on the operating table

in lateral decubitus with right side elevated 70°. One 5- and 10-mm trocars were inserted at the medial margin of umbilicus. A 5-mm trocar was inserted into the abdominal cavity below the pubic hairline under the direct vision from a 5-mm flexible-tip 0° laparoscope through the umbilical trocar. The operation was performed using conventional operating apparatus placed in the abdominal trocars, under direct vision achieved by a 5-mm flexible-tip 0° laparoscope placed through the trocar below the pubic hairline. Firstly, the distal ureter was dissected and blocked by a Hem-O-lok. Then, the bladder cuff excision was performed and the incision was sutured. Finally, the ureter was isolated completely and radical nephrectomy was performed according to the method of the standard laparoscopy. The specimen was removed after the incisions below the pubic hairline was enlarged transversely and the rectus abdominis muscle sheath was incised vertically.

**Results:** All the procedures were successfully performed. The median operative time was 160 (range 115 to 220) minutes, and the median estimated blood loss was 150 (range 50 to 250) ml. No major perioperative complication occurred. All the patients resumed ambulation on postoperative day 1. Pelvic drainage tube was removed on postoperative day 2-3. Urethral catheter was removed on postoperative day 6-7 (the tumor patients underwent irrigation of bladder with Pirarubicin). The patients were discharged on postoperative day 8.

**Conclusions:** SA-LESS nephroureterectomy appears to be feasible, safe and effective. The placement of trocar at umbilicus and below the pubic hairline not only decreases the difficulty of operation, but can also leads to good cosmetic results.

#### MP-13.21

##### **Pure Transvaginal Natural Orifice Transluminal Endoscopic Surgery (NOTES) for Nephrectomy: Report of 16 Cases**

Zou X, Zhang G, Xiao R, Yuan Y, Xue Y  
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**Introduction and Objectives:** Pure

natural orifice transluminal endoscopic surgery (NOTES) within urology has largely been limited to experimental animal studies. We describe the initial clinical experience of pure transvaginal NOTES for nephrectomy, and evaluate its feasibility.

**Materials and Methods:** Fifteen female patients with non-function kidney (right 12, left 3), with a median age of 42.6 years (range 32 to 68), underwent pure transvaginal NOTES nephrectomy, and one female patient with right renal carcinoma underwent pure transvaginal NOTES radical nephrectomy. After induction of general anesthesia, the patients were positioned in lithotomy with ipsilateral lumbar at 60° angle to the floor. A 5-mm incision was made at the posterior vaginal fornix, and a 5-mm trocar was introduced into the pelvic cavity guided by a 5-mm forceps. A 5-mm flexible-tip 0° laparoscope was inserted into the pelvic cavity confirming no rectum injury. Then a ZOU-port was introduced at the posterior vaginal fornix. Dissection was performed according to the method of the standard laparoscopic simple and radical nephrectomy. The intact specimen was extracted transvaginally.

**Results:** The procedures were successfully performed in all patients without additional trocars except for 1, who immediately underwent suprapubic-assisted laparoendoscopic single-site surgery conversion for rectal injury during vaginal entry. There was no other major perioperative complication occurred. The median operative time was 180 mins (range 170 to 330). The median estimated blood loss was 165 ml (range 100 to 250). The patients resumed ambulation on postoperative day 1 (3 for the patient with rectal injury). The pelvic drainage was removed on postoperative day 2 to 3. The patients resumed nutrition on postoperative day 2 to 3 (8 for the patient with rectal injury). The patients were discharged on postoperative day 6 (12 for the patient with rectal injury). The median follow-up of 14.5 months (range 2 to 35) showed that the incision in the vagina healed well. No patient experienced sexual dysfunction.

**Conclusion:** Pure transvaginal NOTES for nephrectomy is feasible and effective.

Moderated Poster Session 14  
Reconstruction and Trauma  
Tuesday, October 14  
1300-1430

**MP-14.01**

**High BMI, Previous Urethral Surgery and Early Experience Are Risk Factors for Failure in Open Urethroplasty Due to Penile Strictures**

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**Introduction and Objectives:** To evaluate outcome and possible risk factors for failure of open urethroplasty due to penile urethral strictures.

**Materials and Methods:** Retrospective chart review of 114 patients with penile stricture undergoing open urethroplasty between 2000 and 2011. In 80 patients, a one-stage onlay procedure was used. A majority of these patients had a pediculated penile skin flap while a few had a free buccal mucosal graft. A two-stage procedure using buccal mucosa was performed in 29 patients. Another five patients with panurethral and complex strictures were managed with a perineal urethrostomy. Failure was defined as when further intervention was needed such as clean intermittent dilatation, internal urethrotomy, a new urethroplasty or a fistula reconstruction. Fisher's exact test and chi-square test was used to identify univariate and multivariate predictors of failure and failure over time.

**Results:** Mean ages in the one-stage, the two-stage and the perineal urethrostomy groups were 50, 54, and 74 years, respectively. The success rates in the corresponding groups were 65%, 72% and 80% with follow-up times 44, 29 and 32 months, respectively. In the one-stage group, 28 failures (23 strictures/5 fistulas) were recorded and in the two-stage group, 8 (4 strictures/4 fistulas). Among the five perineal urethrostomies one failure was found. Multivariable analyses disclosed previous internal urethrotomy/urethroplasty and BMI >30 to be significant risk factors for failure in the one-stage group. Failure over time significantly decreased during the study period ( $p=0.002$ ).

**Conclusions:** Both one- and two-stage penile urethroplasty demonstrate a good success rate in line with previous reports. Obesity, previous urethral surgery and early experience appear to be associated with less favorable outcome after open urethroplasty due to penile strictures. Two-stage procedures and perineal urethrostomies could be preferred options in carefully selected patients.

**MP-14.02**

**Urethral Stricture vs. Benign Prostatic Obstruction: A Radiographic Analysis**

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**Introduction and Objectives:** Urodynamic parameters such as peak flow, pressure/flow, and PVR are often used during the evaluation of patients with urinary retention. Patients with urethral stricture disease often demonstrate the same urodynamic picture as patients with benign prostatic obstruction (BPO), potentially leading to incorrect management. In this study, we examine the radiographic parameters on voiding cystourethrogram (VCUG) and retrograde urethrogram (RUG) that identify stricture patients that have concurrent BPO.

**Materials and Methods:** This is a retrospective review of 30 consecutive patients who underwent anterior urethroplasty; 15 who had a diagnosis of BPO and 15 who did not. All surgeries were performed by a single surgeon (SBB) from 2000 to 2012. Patients with BPO were diagnosed by urodynamics or symptomatic relief with an alpha-blocker. Among patients with BPO, 9/15 required TURP/PVP after urethroplasty. Radiographic characteristics on RUG and VCUG were evaluated by 2 independent reviewers (JRE, SBB). Univariate and multivariable analyses were performed.

**Results:** Median age was 50 years in the BPO group and 49 years in the non-BPO group and ( $p=0.88$ ). Features associated with concurrent BPO included narrow bladder neck (BN) width on VCUG (mean 1.0cm vs. 1.8cm,  $p=0.0002$ ), lower BN/prostatic length ratio (0.40 vs. 0.57,  $p=0.01$ ), BN shape (concave vs. convex,  $p=0.003$ ), narrowed prostatic urethra on RUG ( $p=0.016$ ), bladder trabeculation/diverticula ( $p=0.0000003$ ), PVR>180cc after VCUG ( $p=0.00006$ ), and inability to complete VCUG ( $p=0.03$ ). Factors not associated with concurrent BPO and urethral stricture included non-visualization of proximal urethra on RUG ( $p=0.063$ ), enlarged bladder ( $p=0.12$ ), and closed BN on VCUG ( $p=1.0$ ). The multivariable model including BN width, BN shape, trabeculation/diverticula, and PVR showed that each additional variable increased the odds of having BPO 13.4-fold ( $p=0.009$ ).

**Conclusions:** Urethral stricture patients with urethrographic findings of narrow BN, low BN/prostatic length ratio, concave BN on VCUG, narrowed prostatic urethra, bladder trabeculation/diverticula, PVR>180cc, and inability to complete VCUG, should be counseled that they may need subsequent therapy to treat prostatic obstruction in addition to their anterior urethroplasty.

**MP-14.03**

**Substitution Bulbar Urethroplasty Using Buccal Mucosal Graft in Dorsal Onlay Fashion (Barbagli Technique): Results for Patients Treated over a 10-Year Period in a Scottish Regional Referral Centre**

**Khan R**, Kelly L, Palmer M

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**Introduction and Objectives:** Substitution bulbar urethroplasty using buccal mucosal grafting in dorsal onlay fashion (Barbagli technique) was introduced in 1995-96. Over the years, it has become an established technique used worldwide for treating bulbar urethral strictures. We analysed the outcomes for patients undergoing this procedure in our Scottish regional referral centre under one surgeon over a ten-year period (2003-2012).

**Materials and Methods:** This is a retrospective review of all patients, identified from the theatre procedure book, who had bulbar urethroplasty using above technique between years 2003 to 2012. We excluded anastomotic, re-do and peno-bulbar urethroplasty patients. A total of 68 patients were identified but 61 were included in this analysis. Seven were excluded from analysis due to non-availability of complete data or graft placement technique other than Barbagli's.

**Results:** Average age of patients at the time of the procedure was 43.9 years (Range 18 to 79). Average follow-up was 40.7 months (Range 0 to 113). Single buccal mucosal graft was used in all patients. Fifty eight patients (93%) had previous documented intervention for stricture disease with Optical urethrotomy being the most common procedure. Pre- and post-operative urethrogram results were accessible at time of analysis in 95% of patients. Four patients (6.5%) had evidence of leak on first post-procedure urethrogram. These patients had repeat urethrogram subsequently which showed complete healing. Erectile dysfunction was reported by 6 patients (9.8%). Fourteen patients (23%) had recurrence of stricture with average time to recurrence being 29 months (Range 2-106). For the patients with recurrence, 6 out of 14 had recurred by 7 months of follow-up (early recurrence). Two patients needed intervention to allow check cystoscopies for transitional cell cancers and may not be true recurrences.

**Conclusion:** Barbagli technique for bulbar urethroplasty is a reliable and reproducible technique. In our series, re-stricture rate is 23% for a cohort of patients treated over a long, ten-year period. Overall our results are comparable to other contemporary series in literature. This technique using buccal mucosal graft continues to be our technique of choice for first intervention for bulbar urethral stricture.

**MP-14.04****Urethroplasty for Long Segments and Panurethral Strictures: A Multi-Institutional Study**

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**Introduction and Objectives:** Panurethral strictures are the most difficult problems in reconstructive urology. We performed a multi-institutional study involving large numbers of patients to determine practice patterns and success rates after urethroplasty for this difficult population.

**Materials and Methods:** After IRB approval, an online form was used to collect urethroplasty data from 7 urethroplasty centers of excellence. Patients all had stricture length  $\geq 8$  cm and at least 1 year follow-up. Data was collected: age, previous procedures, etiology, length, urethroplasty type, follow-up time, complications and success rates. Success was defined as patients not needing further instrumentation after urethroplasty.

**Results:** Total sample size was 466. Mean age was 51.5 years (16-82). Mean stricture length 12.7 cm (8-24). Overall recurrence rate was 22.7%. Previous urethrotomy, dilation, or failed urethroplasty did not decrease success rates and success rates were similar for all stricture lengths. Patients had 2 or more strictures 16% of the time. Buccal urethroplasties had the highest success rate: 85% (p value <0.001 and an Odds Ratio of 0.43 for recurrence compared to all other surgeries). Second stage Johanson (65% success) and fasciocutaneous (63% success) were less successful. Planned first-stage-only Johanson urethroplasty and salvage perineal urethrostomy were successful 76% of the time and required revision 24% of the time. Combination surgery such as fasciocutaneous and buccal graft surgery was required in 4%. Neither fossa navicularis nor penile urethral involvement predicted higher failure rates. Immediate complication rate was 8%. Late complication rate was 7%, of which chordee and fistula were most common (38% + 22% respectively).

**Conclusion:** Long segment urethroplasty  $\geq 8$  cm has a potentially high success rate of 78% in expert hands, with Kulkarni (dorsal buccal)

urethroplasties being most successful. Success rates are unaffected by previous urethrotomy/urethroplasty, stricture length, or even penile/fossa navicularis involvement in this long-segment and panurethral stricture population.

**MP-14.05****The Everyman Urethroplasty**

Sharma D

Woodlands Hospital, Georgetown, Guyana

**Introduction and Objectives:** Posterior Urethroplasty is a challenging, formidable operation. It will be required for patients who have suffered severe post traumatic urethral injury, oftentimes with fractured pelvis, seen after road traffic, construction site and forestry accidents, or after post infective stricture disease with perineal sepsis and the watering can perineum. Objectives: To develop a simplified Urethroplasty that can be performed by a competent surgeon working in the Third World.

**Materials and Methods:** The simplified technique has been developed and refined during a 30 year experience working with approximately 80 patients. It is a no suture substitution Urethroplasty done in two stages using perineo-scrotal skin to create a proximal neo-urethra. In the first stage the perineo-scrotal skin is passed through the bladder neck and anchored onto the anterior abdominal wall. No sutures are required. Very little dissection is encouraged. The bladder neck and strictured area are stretched to accept an index finger. The second stage proceeds after three months. The skin neo-urethra is transected as high as conveniently possible. A tubularised buccal mucosal graft is attached to the skin tube proximally and the patient's native urethra distally.

**Results:** A total of 77 men and 3 boys have had the operation. They all control the passage of urine and are free of catheters. Two men were happy with the perineal urethrostomy and refused the second stage.

**Conclusion:** The simplified technique can be used to repair the difficult post-infective urethral stricture or the severe post-traumatic urethral damage/disruption associated with pelvic fracture.

**Disadvantages:** It is a staged procedure with the theoretical bother of hair and skin changes rather like BXO in time – 30 years time. By inserting a buccal mucosal tube we limit the amount of skin in the neo-urethra.

**Advantages:** This is a simple straightforward procedure, technically possible even in very young boys. Minimum dissection required hence less risk of damage to the sphincteric mechanism and is do-able by a competent surgeon in the Third World under 3<sup>rd</sup> World conditions.

**MP-14.06****Analysis of an Autologous Tissue-Engineered Graft for Urethral Reconstruction: A Safety Assessment**

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**Introduction and Objectives:** Urethral reconstruction using autologous engineered constructs could provide a superior alternative to native oral mucosa graft, while avoiding excision of larger tissue segments from patient's mouth. The present report sums up safety data of an autologous tissue-engineered oral mucosa graft from preclinical studies. Additional reported clinical safety data from an ongoing observational were also analysed.

**Materials and Methods:** For graft production, patient's oral mucosa cells were generated from a small oral mucosa biopsy and cultured on the surface of a biocompatible scaffold. Karyotype evaluation was performed on cultured cells from 7 patients to assess their chromosomal stability. Cell number used for karyotyping was  $1.9\text{--}2.6 \times 10^6/\text{patient}$ . To examine the tumorigenic risk of the graft *in vivo*, cells from 6 different human donors were injected by intraperitoneal and subcutaneous route into each of ten immunodeficient athymic nude mice. Each injection consisted of  $10^7$  cells  $\pm 2 \times 10^6$  cells, which were injected into each animal on day 1, 18, 25 and 46 of the study. An additional group consisting of ten animals each received cell culture medium as vehicle control. A bio-distribution study was performed to provide data about a possible "spreading" of implanted cells into distinct organs. For this purpose, engineered homologous murine Transplants (size  $1.5 \times 0.5 \text{ cm}$ ) generated from eGFP-transgenic mice were transferred into the peritoneal cavity of histocompatible non-transgenic mice and *vice versa*. Groups of three transplanted animals each were sacrificed after one week, two weeks and four weeks for histological analysis. Additionally, reported safety data from 70 tissue-engineered graft-treated patients suffering from urethral stricture were collected and analysed to assess safety of the graft. Patients were recruited in an ongoing observational study with a 2-year follow-up period for which ethical committee votum was available and a pharmacovigilance system according to international standard was in place.

**Results:** Karyotype analysis did not reveal any genetic instability or chromosomal alterations. The GLP-Tumorigenicity study in nude mice

did not show any sign of macroscopic tumor formation during the life span of 140 days that was 3 months following last cell injection. No malignancy attributable to graft was detected following thorough microscopic tissue evaluation. Biodistribution studies of the murine homologue of the graft revealed no migration of transplanted cells into distant tissues and organs. Reported clinical data of 70 patients demonstrated no peri- or post-operative events related to the engineered graft.

**Conclusions:** Tissue-engineered oral mucosa graft appears to be a safe, viable and superior alternative to native oral mucosa transplants for the treatment of urethral stricture.

#### MP-14.07

##### **Urethroplasty with Autologous Tissue-Engineered Oral Mucosa Graft: Efficacy Results in Two Centers**

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**Introduction and Objectives:** To avoid potential complications associated with oral mucosa excision for urethroplasty, engineered grafts may become valuable alternatives. Here we present our preliminary results in the use of tissue-engineered oral mucosa graft, manufactured from a tiny biopsy in the treatment of urethral strictures in two centers.

**Materials and Methods:** We present reported data of 29 patients with bulbar or bulbopenile urethral stricture recurrence, who underwent one-stage urethroplasty with tissue-engineered graft. Ethical committee votum was available. Excluded in this report are patients with allergy against MukoCell<sup>®</sup> components, patients who had previous radiation or laser therapy at urethral site as well as patients with severe autoimmune disease. The age of patients was between 24 to 86 years. Stricture length in each patient was between 20mm and 78mm.

**Results:** The size of the oral biopsy was about 0.4cmx0.8cm. No complications occurred at the donor site. Three weeks later, the operative procedure was performed. The mean follow-up time was 12.7 months. All patients voided spontaneously after catheter removal, 3 weeks after urethroplasty, and radiographic examinations showed a patent urethra. The mean reported postoperative peak flow was 27.1 ml/sec. Four patients had a recurrence, all of them in the first 12 months after operation. One patient had a postoperative meatus stenosis, while the grafted urethral site with MukoCell<sup>®</sup> remained free of stricture.

**Conclusion:** Our preliminary results regarding urethroplasty with autologous tissue-engineered graft are promising. A higher number of patients and longer term results are necessary to determine if tissue engineered grafts can replace native oral mucosa for urethroplasty in the future.

#### MP-14.08

##### **Tissue Engineered Buccal Mucosal Urethroplasty: Long-Term Clinical Outcomes**

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**Introduction and Objectives:** Buccal mucosa is commonly used as a graft in urethral substitution. In patients with recurrent or complex urethral stricture disease, obtaining sufficient quantity of tissue may not be possible. We previously developed a methodology for culturing tissue engineered buccal mucosa (TEBM) and 3-year clinical outcomes. We now present the findings from long-term (9 years) follow-up.

**Materials and Methods:** Five men with recurrent urethral strictures secondary to lichen sclerosis requiring extensive substitution underwent TEBM urethroplasty with local ethics committee approval. A 0.5cm<sup>2</sup> full thickness buccal mucosal biopsy was taken from each patient. Oral fibroblasts and keratinocytes were isolated, expanded in culture before being seeded on sterilized de-epidermised dermis from donors. The cells were then cultured for 7 to 10 days at air-fluid interface to TEBM grafts. Follow-up was initially performed at 2 weeks, 6 weeks then 3 months to 1 year, 6 monthly to 3 years and then yearly thereafter, by direct visualisation with flexible urethroscopy.

**Results:** Mean follow-up was 111 months (range 110 – 115 months). The grafts initially took in all patients. Subsequently one patient required partial excision of the graft for a hypproliferative tissue reaction. Another required full excision for fibrosis. Of the three remaining patients with intact TEBM, two patients have since required an internal urethrotomy and one has required a urethral dilatation. Two of these three patients continue to perform intermittent self-dilatation. All had patent urethras on cystoscopic examinations within the past year with no adverse mucosal features.

**Conclusion:** As reported previously there was a significant initial risk of an adverse fibrotic reaction several months after implantation of TEBM. This occurred in 2 patients. It is not clear whether this was related to patient factors (i.e. lichen sclerosis) or whether graft related. In the other 3 patients results have been durable for 9 years albeit with the requirement for 1 further minimally invasive procedure. This

study emphasizes the need for clinical evaluation before new technologies are introduced into clinical practice but that TEBM can provide a durable result.

#### MP-14.09

##### **Functional Outcomes after Urethroplasty: Prospective Analysis from a Single Center**

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**Introduction and Objectives:** Urethroplasty is the 'gold standard' treatment of urethral strictures. Different techniques are used and success is generally described as freedom of stricture recurrence. However, patients are not only interested in stricture-free survival, but also want to experience a normalized urinary and erectile function and an improved quality-of-life. Therefore, patient-reported outcome measures (PROMS) are of major interest in this field. The study objective is to prospectively analyze functional outcomes and patient satisfaction after anterior urethral reconstruction.

**Materials and Methods:** We prospectively evaluated 125 patients who underwent anterior urethral stricture repair between August 2009 and February 2014. Preoperatively and at each follow-up, patients received following questionnaires: the International Prostate Symptom Score (IPSS) with the International Prostate Symptom Score Quality Of Life (IPSS-QOL), the Urogenital Distress Inventory Short Form score (UDI-6), the International Index of Erectile Function-5 score (IIEF-5) score and the ICIQ-Lower Urinary Tract Symptoms Quality of Life score (ICIQ-LUTS-QOL). Voiding symptoms, incontinence, erectile dysfunction (ED) and quality-of-life were analyzed using paired sample t-tests.

**Results:** Patients with mild or no baseline ED (IIEF 17-25) had a significant decline in erectile function at first follow-up (p<0.0001). There was a recovery of erectile function between first and second follow-up (p=0.0199) but erectile function did not come back at baseline-levels. Patients with mild or moderate to severe ED (IIEF 5-16) at baseline experienced no significant changes in erectile function. Significant differences were noted between preoperative IPSS-score (p<0.0001), IPSS-QOL (p<0.0001), UDI-6-score (p<0.0001) and ICIQ-LUTS-QOL-score (p=0.0001) and the scores at first follow-up. These differences remain significant at second and third follow-up.

**Conclusion:** Following urethroplasty for anterior urethral strictures, we noticed a significant decline in erectile function in patients with mild or no baseline ED at first follow-up. A recovery was seen after 8.6 months, but there was no full restoration of erectile function. Voiding symptoms improved significantly, and we noticed a major positive impact on quality-of-life,



which remained up to 17.1 months after surgery. This large prospective study shows that success should not merely be defined by the absence of stricture recurrence. Other factors such as erectile function and voiding symptoms contribute to quality-of-life and to success.

#### MP-14.10

##### Patient Reported Quality of Life and Timing of Discharge after Outpatient or Short-Stay Urethroplasty

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**Introduction and Objectives:** In this study we sought to assess quality of life (QOL) of patients after anterior or posterior urethroplasty in the outpatient or short-stay setting.

**Materials and Methods:** We retrospectively reviewed records of 50 consecutive adult patients who underwent anterior or posterior urethroplasty between 9/2012 and 3/ 2014 at our institution. A EuroQol-5 (EQ-5) validated questionnaire was administered 24 hours

after the patient's discharge assessing "Mobility", "Self-Care", "Usual Activity", "Pain" and "Anxiety" on a scale from 1 (severe) to 3 (no problem). An additional question assessing the timing of discharge, as either "too late", "too soon", or "on time" was added to the survey.

**Results:** Mean age of the group was 47.5 years (18-78). Mean length of urethral stricture was 47.2 mm (4-160 mm). The etiology of strictures were idiopathic, 19 (38%), traumatic, 12 (24%), hypospadias failures, 7 (14%), and iatrogenic, 5 (10%). Repairs included excision primary anastomosis (EPA) (14), Kulkarni dorsal onlay (13), augmented anastomotic urethroplasty (AAU) (9), Staged urethroplasty with buccal mucosa graft (BMG) (5), and posterior urethroplasty (4). Nine patients (18%) were discharged postoperatively on the day of surgery, while 41 (82%) stayed overnight. At 24 hours there was an 82% response rate. In the short-stay and the outpatient cohort, 77.7% and 86% respectively felt they were discharged on time. Severe problems with self care, pain, or

anxiety/depression were each reported by only 4.5% of patients, no patients reported severe problems with mobility.

**Conclusion:** In a heterogeneous population of patients including those requiring complex reconstruction for anterior strictures or posterior urethral disruption, performing urethroplasty in the outpatient or short hospital stay setting does not compromise QOL in the early post operative period. Despite early discharge, 86% of patients perceived that discharge timing was appropriate.

#### MP-14.11

##### Treatment of Urethral Strictures in Patients Having Undergone Prior Total Phallic Construction

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**Introduction and Objectives:** The treatment of urethral strictures in patients having undergone a prior total phallic construction is not well defined.

**Materials and Methods:** We retrospectively reviewed patients treated for a urethral stricture at our institution who had undergone a prior total phallic construction. Stricture treatments were tabulated and outcomes assessed.

**Results:** We identified 32 patients who met inclusion criteria. They underwent a phallic construction at a mean 30.2 years of age. The urethra was constructed with a tubularized musculocutaneous forearm free flap (24) or a tubularized full-thickness skin graft (FTSG) (8). One patient underwent excision and primary anastomosis and another a second stage buccal graft urethroplasty at the time of phallic construction. Urethral strictures developed at a median 5.7 months following phallic construction. Stricture location was at the urethral anastomosis in the majority of patients (21); the remaining strictures were either proximal (1) or distal (7) to the anastomosis or pan-neourethral (2). One patient's stricture location was not determined by retrospective review. The patients underwent an average 2.6 stricture treatments. Forty-one percent underwent an initial dilation or internal urethrotomy (DVIU). This was the definitive procedure in only 29% with the remainder undergoing formal reconstruction after the DVIU failed or performing intermittent self-dilations. Patients not undergoing initial DVIU attempt underwent variety of procedures depending on stricture length and location, including a proximal cutaneous urethrostomy (3), meatoplasty (3), perineal urethrostomy (4), single-stage onlay (3), tubed FTSG (1), or a 2-stage reconstruction (1). Following urethral repair 84% of patients were voiding via the neourethra, 12.5% performed either intermittent catheterization or had an indwelling catheter, and one patient voided through a fistula

MP-14.10, Table 1. Number (Percentage) Reporting on EQ-5 and Discharge Time

| EQ-5; n = 44               | 1=Severe          | 2=Moderate        | 3=No Problem      |
|----------------------------|-------------------|-------------------|-------------------|
| <b>Mobility</b>            |                   |                   |                   |
| Outpatient n = 9           | 0                 | 4                 | 5                 |
| Short-Stay n = 35          | 0                 | 18                | 17                |
| <b>All Responders</b>      | <b>0</b>          | <b>22 (50%)</b>   | <b>22 (50%)</b>   |
| <b>Self-Care</b>           |                   |                   |                   |
| Outpatient                 | 0                 | 1                 | 8                 |
| Short-Stay                 | 2                 | 10                | 23                |
| <b>All Responders</b>      | <b>2 (4.5%)</b>   | <b>11 (25%)</b>   | <b>31 (70%)</b>   |
| <b>Usual Activities</b>    |                   |                   |                   |
| Outpatient                 | 1                 | 5                 | 3                 |
| Short-Stay                 | 11                | 14                | 10                |
| <b>All Responders</b>      | <b>12 (27.3%)</b> | <b>19 (43.2%)</b> | <b>13 (29.5%)</b> |
| <b>Pain/Discomfort</b>     |                   |                   |                   |
| Outpatient                 | 0                 | 8                 | 1                 |
| Short-Stay                 | 2                 | 26                | 7                 |
| <b>All Responders</b>      | <b>2 (4.5%)</b>   | <b>34 (77.3%)</b> | <b>8 (18.2%)</b>  |
| <b>Anxiety/Depression</b>  |                   |                   |                   |
| Outpatient                 | 0                 | 0                 | 9                 |
| Short-Stay                 | 2                 | 6                 | 27                |
| <b>All Responders</b>      | <b>2 (4.5%)</b>   | <b>6 (13.6%)</b>  | <b>36 (81.8%)</b> |
| <b>Time</b>                |                   |                   |                   |
| <b>Discharge; n = 42</b>   | <b>1=Too late</b> | <b>2=Too Soon</b> | <b>3=On Time</b>  |
| <b>Timing of Discharge</b> |                   |                   |                   |
| Short-stay n = 33          | 0                 | 4                 | 29                |
| Outpatient n = 9           | 0                 | 2                 | 7                 |
| <b>All Responders</b>      | <b>0</b>          | <b>6 (14.3%)</b>  | <b>36 (85.7%)</b> |

proximal to a more distal stricture.

**Conclusions:** Urethral reconstruction for stricture disease is possible in patients with prior total phallic construction. Initial DVIU may be attempted, but is successful in less than 1/3 of patients not performing self-dilations. Whatever approach is chosen, multiple procedures to attain urethral patency is the rule rather than the exception in this difficult patient population.

#### MP-14.12

##### **Functional Outcome following Open Surgical Correction of Bladder Neck Contractures with Subsequent Implantation of an Artificial Urinary Sphincter**

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**Introduction and Objectives:** We review the functional outcome after open surgical reconstruction of bladder neck contractures and prostatic urethral stenoses with subsequent artificial urinary sphincter (AUS) implantation following treatment of prostate cancer.

**Materials and Methods:** Between 2006 and 2012, 32 patients (mean age 65.5 years) were treated in a single unit. Men with anastomotic contractures following radical prostatectomy only (n=16) or radical prostatectomy and adjuvant radiotherapy (n=6) underwent transperineal revision of the vesico-urethral anastomosis (VUA). Those with prostatic stenoses following external beam radiotherapy, HIFU, cryotherapy, brachytherapy or a combination of these (n=10) were treated by salvage radical prostatectomy. All had failed multiple previous attempts at endoscopic management. A bulbar AUS was subsequently implanted on average 9.1 months after the reconstruction to treat consequent incontinence.

**Results:** The overall stricture-free rate after reconstruction was 87.5% (28 of 32). Only one patient each following radical prostatectomy only and after adjuvant radiotherapy recurred and are carrying out self-dilatation. All the rest, except one who declined, went on to have a bulbar AUS implanted (n=19). All are dry at a mean follow-up of 31.2 months except one unirradiated patient with recurring incontinence due to device malfunction. Eight out of 10 patients (80%) having salvage prostatectomy were also successful. Seven underwent AUS implantation. Four remain dry after their primary implant, 3 are dry however had their devices revised for erosion (1) and malfunction (2). Therefore functional normality (unobstructed and continent) was achieved in 18 of 22 (81.8%) transperineal reconstructions. In irradiated patients with prostatic stenoses, 7 of 10 (70%) were rendered functionally normal after salvage radical prostatectomy albeit with a higher rate of AUS revision as a consequence of the primary irradiation treatment.

**Conclusion:** Surgical correction of recalcitrant BNC after treatment of prostate cancer is feasible and very effective however requires implantation of an AUS to restore functional normality. Salvage radical prostatectomy, though technically challenging, with longer hospital stays and time to catheter-free status, is the treatment of choice in radiotherapy-related prostatic urethral stenoses.

#### MP-14.13

##### **Reconstructive Surgery for Uro-Rectal and Perineal Fistulae**

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**Introduction and Objectives:** This study evaluates the significance of perineal wound infection and post-operative radiological leak, determining their role in fistula recurrence after transperineal repair of uro-rectal and/or uro-cutaneous fistulae, with additional abdominal exposure when necessary.

**Materials and Methods:** Between 2006-2013, 43 transperineal (17 with gracilis flap) and 23 abdomino-perineal repairs (omentoplasty) of uro-rectal and/or uro-perineal fistulae were performed in 58 men. Thirteen were redo procedures. Thirty one patients had undergone radiotherapy, HIFU, cryotherapy, brachytherapy or a combination of these. All underwent fluoroscopic assessment on average 4 weeks post-op prior to urethral catheter removal. In 54 cases the perineal wound was closed over a corrugated drain.

**Results:** Overall success rate was 78% (52 of 66 procedures), all recurrences occurring in irradiated patients, on a background of Crohn's disease or in non-primary procedures. Fourteen perineal wound infections were documented, 13 in patients having a perineal drain. In 11, a fistulous track developed onto the perineum at the drain site. Nine arose from the urinary aspect, only two originating from the bowel side. Three eventually healed spontaneously after 3 months. The rest required salvage surgery. Of the 9 failed urorectal fistula repairs, only one had a persistent urinary leak through the rectum. The rest presented with these fistulae to the perineal drain site. In 20 of 52 cases with a successful outcome (38.5%), urethrogram at 4 weeks demonstrated a leak from the urinary side. This was either into a contained cavity or blind-ending track and was managed conservatively. The catheter was eventually removed on average 84.8 days after surgery. Eighteen patients remained asymptomatic. In 2, this cavity subsequently became infected and ruptured, requiring further salvage surgery.

**Conclusions:** Recurrence after fistula repair is usually from the urinary tract. This most often presents as a recurrent track, not into the rectum, but onto the perineum to the drain site. Consequently we now avoid draining the

perineal wound routinely. Radiological leaks from the urinary system into a blind ending track or cavity are common after this type of procedure and very often can be managed conservatively without compromising the final outcome.

#### MP-14.14

##### **Bulbar Urethral Necrosis**

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**Introduction and Objectives:** Multiple failed anastomotic urethroplasties for Pelvic Fracture Urethral Defects (PFUD) may result in ischemic bulbar necrosis leading to long defects in the urethra. After transection of the bulbar urethra, distal bulbar urethra is dependent on retrograde blood supply from glans and penile urethra in anastomotic urethroplasty. This supply is suboptimal in vasculogenic impotence. During inferior pubectomy the dorsal penile arteries may be traumatized adding further to ischemia. We present our retrospective data analysis of bulbar urethra necrosis.

**Materials and Methods:** Between, 2010 and 2012, 46 patients were referred with bulbar necrosis. This was partial (narrow urethral plate) or complete. All 46 patients had >2 attempts at urethroplasty prior to referral. Choice of surgery was dependant on length of urethra to be created, presence of urethral plate, prepuce and status of scrotum. P1 patients with no urethral plate required circumferential substitution urethroplasty and P2 patients had narrow plate and required augmentation with a vascularised flap.

**Results:** In P1, 25 underwent preputial tube urethroplasty, 4 underwent oral mucosa flap urethroplasty\* and 3 had scrotal dropback. Two patients had dorsal BMG and ventral pedicled penile skin flap. In P2 requiring augmentation, 8 patients had pedicled penile flap as ventral onlay, and 4 had oral mucosa flap.\*In oral mucosa flap midline scrotal incision is made and widened. A 3cm by 7cm buccal graft is applied on dartos. After 6 weeks graft is mobilized on midline scrotal septum, transposed to perineum and then either used as an onlay or tubularised. Follow-up ranges from 5 to 120 months. Nineteen out of 25 patients of preputial tube developed symptomatic diverticulum. Four had proximal anastomotic narrowing. One had incontinence due to bladder neck issues. One developed fistula. Four out of 8 patients of oral mucosa flap were successful. Two developed proximal narrowing.

**Conclusions:** A majority of preputial tube patients do well. Diverticulum formation, post micturition dribble and anastomotic narrowing are common problems. Oral mucosa flap is the new technique where vascularised random

buccal graft flap can be used for substitution. Inferior pubectomy should be performed below the periosium to avoid damage to the dorsal penile arteries.

#### MP-14.15

##### The Role of Bedside Bladder Sonography for Detection of Bladder Trauma

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**Introduction and Objectives:** Bladder trauma is one of the most frequent urologic injury associated with concomitant pelvic trauma. The current gold-standard for diagnosis of traumatic bladder rupture include CT or cystogram, but these imaging modalities may only be performed in stable patients. On the other hand, sonograms are readily available in emergency bed-side settings, since Focused Assessment Sonography for Trauma (FAST) has become a routine procedure. This study aims to evaluate the role of bedside bladder sonography along with retrograde instillation of saline as a novel diagnostic procedure for suspected bladder trauma.

**Materials and Methods:** Prospective evaluation of all patients with suspected bladder injuries admitted to the emergency department of Hasan Sadikin General Hospital, Bandung, Indonesia, from 2010 to 2013. Suspected urethral injury patients were excluded. Along with routine FAST procedure, bedside bladder sonography was performed concurrently with retrograde instillation of normal saline 150-200cc through a Foley catheter. The objective of *real-time* bladder sonogram was to examine the presence of peri-vesical free fluid turbulence and accumulation during saline instillation, which subsequently indicated a suspected bladder perforation. The accuracy of sonographic results were compared with CT, cystogram or intraoperative findings. Time to diagnosis was recorded. Statistical analysis was performed to evaluate the sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV) and diagnostic accuracy.

**Results:** Twenty three patients met the inclusion criteria. The mean age was 27.21 years old, 87% were males. Based on cystogram or intraoperative finding there are 21 patients have bladder rupture. Among these patients, 14 patients had positive result on bladder sonogram, and all confirmed positive on cystogram and operative finding. Nine patients had negative result on bladder sonogram. While 7 among them have positive result on cystogram or intraoperative finding. Analysis revealed 67% sensitivity, 100% specificity, 100% PPV, and 22.2% NPV. Overall diagnostic accuracy of bladder sonogram was 83.5%. Time to diagnosis was significantly shorter in bedside bladder sonogram compared to cystogram or intraoperative finding (11.82 ± 2.99 min vs. 181.30 ±

88.89 min;  $p < 0.05$ ).

**Conclusions:** Bedside bladder sonogram is a useful adjunct procedure for diagnosis of bladder trauma. It is time and cost effective, and can be performed in bed-side emergency setting with acceptable accuracy.

#### MP-14.16

##### Postoperative Complications following Damage Control Nephrectomy Are Associated with Concomitant Abdominal Organ Resection

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**Introduction and Objectives:** Renal injuries account for 3% of all trauma admissions in the US. Severely injured patients who are unstable due to hemorrhage or metabolic derangement require damage control nephrectomy. The purpose of this study is to describe the complications associated with damage control nephrectomy performed for trauma.

**Materials and Methods:** From 2000–2013, 49 patients underwent emergent damage control nephrectomy as a result of blunt or penetrating traumatic injury at a single institution. End-points were re-hospitalization, reoperation or additional procedures due to complications.

**Results:** Median age at time of surgery was 24 (15-82) years and median follow-up was 11 months. Median revised trauma score and injury severity score were 7.84 and 26, respectively. Thirty-two patients underwent additional abdominal visceral surgeries (10 splenectomies, 7 liver resections or repairs, 19 small or large bowel resections or repairs, and 6 distal pancreatectomies). Median length of initial hospitalization was 13 days (range 1-49). Of the 49 patients studied, 7 (14%) died during initial hospitalization. Of the 42 who survived, 9 (21.4%) were re-hospitalized and 7 (16.7%) required reoperation or additional procedures due to complications. The median time until postoperative complication was 40 days. The most common complication was infection or abscess formation (14.2%). Of note, patients who underwent damage control nephrectomy and removal of one or more additional abdominal visceral organs had a higher rate of major complications (Clavien grade ≥ 3) within 30 days than patients who underwent emergent nephrectomy alone (30% vs. 0%,  $p = 0.04$ ).

**Conclusion:** Complications are common among trauma patients who undergo damage control nephrectomies. Among patients who undergo damage control nephrectomy, repair/resection of other abdominal organs is associated with higher incidences of post-operative complications. Patients who underwent damage control nephrectomy alone, however, had very low rates of readmission, reoperation or reintervention.

#### MP-14.17

##### Management of Female Urethral Trauma

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**Introduction and Objectives:** Data on females treated for urethral trauma is from case reports and small series. The management of such patients is still evolving. We present our experience in treating urethral trauma in females and suggest an algorithm for management.

**Materials and Methods:** Ours is a tertiary referral center for Reconstructive Urology for a large population of 1.2 billion. This is an observational, retrospective analysis from January 1995 to July 2013. Out of 1264 urethral surgeries, 344 surgeries were for male posterior urethral trauma and 10 were female urethral trauma. One patient had acute distal urethral transection due to penetrating injury and was repaired primarily. Nine patients had delayed treatment after 3-6 months. Etiology was road traffic accident in 8, and 2 were injured by a collapsed wall. Six patients were prepubertal (4-11 years) and 4 patients post pubertal (23-45 years). Assessment included RGU/MCU, urethroscopy and antegrade cystoscopy. Antegrade cystoscopy is paramount in assessing bladder neck status and presence of stones. Urethral transection was proximal in 2, mid in 7 and distal in 1 patient. Anastomotic urethroplasty was performed in 9 patients. One patient with distal obstruction underwent meatoplasty. No patient had rectal injury. Catheters were kept for 6 weeks. Follow-up was 40.4 to 80 months. Follow-up was 3 monthly and included clinical examination, uroflow and USG.

**Results:** Transpubic approach with posterior pubectomy was performed in all prepubertal girls. Approach in all post pubertal females was transvaginal. In transpubic approach, cystotomy was avoided by incising posterior urethra over a sound through the suprapubic tract or endoscope. Three girls had urethra-vaginal fistula which was closed transabdominally. One girl had anastomotic leak which healed by keeping catheter for additional 2 weeks. One girl with meatoplasty required revision due to stenosis. One girl had borderline flow and was investigated and diagnosed with hypocontractile bladder. All patients were continent and had optimal flow.

**Conclusions:** Female urethral trauma is uncommon. Transection can occur in proximal, mid or distal urethra. Delayed repair should be preferred. Post pubertal patients should be treated with transvaginal repair. Prepubertal girls merit a transpubic approach. Cystotomy can be avoided in transpubic approach.

**MP-14.18****Epidemiology of Adrenal Injuries Requiring Adrenal Surgery**

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**Introduction and Objectives:** Adrenal trauma is extremely rare and current literature is lacking in data from large case series. In this study, we analyze adrenal injuries using the National Trauma Data Bank (NTDB).

**Materials and Methods:** We performed a retrospective analysis of the NTDB from the years 2007-2011. Patient demographics, Injury Severity Score (ISS), mechanism of injury, and blunt versus penetrating trauma, associated injuries and hypovolemic shock were assessed. Multivariable models were used to determine associations with outcomes such as need for surgery, type of surgery mean length of stay, need for ICU, and death.

**Results:** Of the 1,766,606 trauma cases in the data set, 8683 were identified involving one or both adrenal glands. There were 7835 blunt and 663 penetrating injuries, and 184 of these injuries were isolated to the adrenal glands. Of the 8683 adrenal injuries, 80 (0.9%) required surgery. However, none of the 184 isolated adrenal injuries required surgery ( $p=0.42$ ). Factors associated with isolated adrenal injury include lower ISS ( $p<0.001$ ), younger age ( $p<0.001$ ), and penetrating injury ( $p<0.001$ ). No isolated adrenal injuries were associated with death (12% vs. 0%,  $p<0.0001$ ). The most common associated organ injuries were ribs (50.9%), thoracic (50.0%), liver (41.6%), vertebrae (30.9%), kidney (27.8%), and spleen (22.0%). Logistic regression showed that injuries to the thorax ( $p=0.0014$ ) and multiple abdominal injuries ( $p<0.001$ ) were associated with a lower rate of undergoing adrenal surgery. Higher ISS score ( $p=0.007$ ), penetrating injury ( $p<0.001$ ), race (Black) ( $p=0.029$ ) and concurrent injuries to the spleen ( $p<0.001$ ) and intestines ( $p=0.016$ ) were associated with a higher likelihood of requiring adrenal surgery. Older age ( $p<0.001$ ), higher ISS score ( $p<0.001$ ), race (Black, Other) ( $p=0.03$ ,  $p=0.02$ ), penetrating injuries ( $p<0.001$ ) and injuries to the aorta/vena cava ( $p=0.008$ ), vessels ( $p=0.001$ ), thorax ( $p=0.03$ ), ribs ( $p=0.005$ ), stomach ( $p=0.02$ ), liver ( $p=0.03$ ), multiple abdominal injuries ( $p=0.002$ ), and brain/spinal cord ( $p<0.001$ ) were associated with a higher mortality rate.

**Conclusions:** Adrenal injuries are rare,

comprising 0.49% of all traumatic injuries.

In our database, isolated adrenal injuries were not fatal and did not require surgery. Younger age, race (Black), higher ISS score, penetrating injury, and concurrent injuries to the spleen/intestines were associated with a higher likelihood of requiring an adrenalectomy.

**MP-14.19****Outcomes of Iatrogenic Genitourinary Injuries during Colorectal Surgery**

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**Introduction and Objectives:** The purpose of this study is to quantify and categorize genitourinary injuries during colorectal surgery.

**Materials and Methods:** We retrospectively reviewed patients who underwent colorectal surgery at Barnes Jewish Hospital and developed iatrogenic genitourinary complications requiring surgical repair between 2003 and 2013. Endpoints included GU repair failures.

**Results:** There were 75 patients in this series, with a mean age of 57.5 years (22-91) at time of surgery and median follow-up of 16.7 months (0-127). Sixty-four patients had single GU repairs and 11 patients had multiple GU repairs, with 18 patients having a failure of their initial repair. Colorectal procedures included colectomies (16), lower abdominal resections (15), lower abdominal and abdominoperineal resections with total abdominal hysterectomies and bilateral salpingo-oophorectomies (13), and abdominoperineal resections (9). The most common initial GU repairs were cystorrhaphy (24), ureteroureterostomy (22), ureteroneocystostomy with psoas hitch (12), and ureteroneocystostomy (11). Secondary GU repairs included stent placement or PCN for persistent ureteral leak (5), fistula repair (3) stent placement or cystorrhaphy for persistent bladder leak (2), and repair of recurrent ureteral strictures (1). Twenty-seven patients (36%) had prior radiation and 35 patients (47%) had prior chemotherapy. Fifty patients (67%) are alive, with a 30-day mortality rate of 4%. Pre-operative radiation was associated with failure of the GU repair (11/28 vs. 7/47,  $p=0.025$ ). Pre-operative chemotherapy was also associated with GU repair failure (13/35 vs. 5/40,  $p=0.016$ ).

**Conclusion:** Pre-operative radiation and chemotherapy are associated with increased repair failure rates of GU injuries during colorectal surgery.

**MP-14.20****Cyclic Stretch-Induced the Synthesis and Degradation of Extracellular Matrix of Human Bladder Smooth Muscle Cells: Role of MMPs and TIMPs in This Dynamic Balance**

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**Introduction and Objectives:** Mechanical stimuli is an essential factor for growth and development of bladder. The purpose of this study was to emphasize the effects of the cyclic stretch up on HBSMCs proliferation and the metabolism of ECM, and the role of MMPs/TIMPs in this process.

**Materials and Methods:** HBSMCs were seeded onto silicone membrane and subjected to cyclic stretch at ranges of (5%, 10%, 15%) for 16 hours, and then cultured for 8h without any stretch. RT-PCR was used to assay gene expression of ECM (Col1, Col3, Col4, En, Ln, Fn), MMPs (MMP-1, 2, 3, 7, 9) and TIMPs (TIMP-1, 2). Cell proliferation determined by EdU incorporation assay.

**Results:** The expression of Col1 increased in group 10%, and there was a downward trend without statistical significance when the stretch changed from 10% to 15%. Col3 increased gradually as the stretch enhanced, but there is no difference among the three experiment groups. Fn was improved significantly, and the group 10% reached the highest level. The expression of Col4, En, Ln unchanged under stretch. In group 5%, MMP-1, 3 and 7 significantly increased. When the stretch changed from 5% to 10%, MMP-1, 2, 3 and 7 increased, but only MMP-1 and 2 were significant. In group 15%, MMP-1, 2, 3 and 7 changed without statistical significance compared with group 10%. MMP-9 was scarcely affected by stretch. TIMP-1 and 2 did not increase significantly until the stretch was increased to 15%. The proliferation rate of control group and each parameter group (5%, 10%, 15%) was  $7.0\pm1.01\%$ ,  $12.9\pm1.75\%$ ,  $20.7\pm2.03\%$ ,  $19.7\pm1.81\%$ , respectively.

**Conclusion:** This study showed that cyclic stretch can promote the proliferation of HBSMCs in a reasonable range. ECM and MMPs synthesis increased as the stretch rose to an optimal parameter. But in the exceeding range, the increase of TIMPs synthesis would interrupt the balance of MMPs and ECM, led to decrease of cell proliferation and accumulation of ECM. This interesting dynamic balance under cyclic stretch should bring enlightenment to tissue engineering bladder and pathogenesis of bladder diseases.



Moderated Poster Session 15  
Female Urology  
Tuesday, October 14  
1300-1430

**MP-15.01**

**Female Ventral Buccal Mucosa Urethroplasty: Initial Operative Experience and Outcomes**  
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**Introduction and Objectives:** Female urethral strictures are uncommon and present diagnostic and treatment dilemmas. Stricture dilation is often inadequate and definitive surgical repair techniques vary. We present a simple ventral-onlay approach using buccal mucosa and short term outcomes.

**Materials and Methods:** A retrospective chart review of consecutive patients. Stricture etiology, prior treatments, duration of symptoms, urodynamic and imaging results were reviewed. Patient outcomes, complications, and symptom improvements were categorized.

**Results:** Four women aged 53-76yo, underwent ventral buccal urethroplasty between 5/2013-2/2014. Etiologies were idiopathic (1), Foley trauma (2), and radiation (1). At diagnosis, two had urinary retention, two with lower urinary symptoms. One patient demonstrated hydronephrosis, two with bladder diverticula. Mean average urinary flow was 4.9ml/sec, maximum flow 11.3 ml/sec. Mean detrusor pressure was 82 cm/H<sub>2</sub>O (25-126) and residual urine 213ml. All patients underwent trans-vaginal ventral buccal graft urethroplasty for strictures in the mid-urethra. This was covered in 3 layers with peri-urethral and vaginal tissue, with the radiated patient receiving Martius labial fat graft. Average time from diagnosis to urethroplasty was 7.25 years and included an average of 2 internal urethrotomies. No patient suffered renal deterioration. Voiding cystogram was performed 2 weeks post-operatively without evidence of leak. The radiation patient had breakdown of vaginal mucosa, with visible Martius graft, no other post-operative complications were noted. At mean follow-up of 3 months, average residual urine was 21 ml, and mean improvement in American Urological Association Symptom Score was 17.8. No de-novo stress incontinence has been demonstrated. One patient was noted to have narrowing at the proximal anastomosis 4 months later. Rigid cystoscopy confirmed widely patent urethra.

**Conclusion:** Ventral-onlay buccal mucosal graft urethroplasty is simple and effective for treatment of female urethral strictures. Patients have substantial improvement in voiding symptoms and residual urine measurements. Longer follow-up is needed to assess durability, though success in male strictures has been proven.

**MP-15.02**

**A Single Centre Experience in the Management of Symptomatic Urethral Diverticula, with Emphasis on Different Surgical Approaches**

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**Introduction and Objectives:** Clinically significant urethral diverticula are rare, albeit they have been reported to occur asymptotically in up to 6% of adult females. Patients may present with a lump/mass, pain, dyspareunia, voiding difficulties or recurrent infection. We present a single centre experience in the surgical management of such patients.

**Materials and Methods:** We retrospectively reviewed 89 patients referred with symptomatic urethral diverticula. All patients were operated by two surgeons over an eight-year period (October 2004 to November 2012). Follow-up period was ranged from 3-20 months. We find the prone position for the transvaginal approach to be effective for the majority of patients as it allows for excellent access with enhanced exposure. In addition, we have found with recurrent or complex diverticula which extend anteriorly, a dorsal supra-urethral approach with the patient in a supine position is helpful. The surgical approaches are described in detail.

**Results:** Early complications included one urinary tract infection and one Martius graft infection both requiring intravenous antibiotics and settled without further morbidity. A total of 72% of patients were dry and cured post-operatively. Thirteen patients (15%) had de-novo SUI following surgery. Those with bothersome SUI went on to have an autologous sling at six months. All were dry, three (23%) required clean intermittent self-catheterisation. Three patients have had a recurrent residual diverticulum (3.4%) following surgery. Two patients had a repeat diverticulectomy performed via a dorsal approach with good symptomatic relief and remained continent.

**Conclusions:** We have found that the prone position allows excellent exposure and is recommended over a traditional lithotomy positioning, particularly in cases where access may be an issue. The dorsal supra-urethral approach is useful for diverticula that extend anteriorly, e.g. horseshoe-shaped/circumferential, sometimes towards the bladder neck or re-do cases where a previous ventral incision has been made. Both approaches should be part of the reconstructive surgeon's armamentarium.

**MP-15.03**

**The Predictors of Recurrence and Successful Treatment following Vesico-Vaginal Fistula Surgery: A Single Centre Experience at Saint Luc Hospital, Kisantu, Democratic Republic of Congo**

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**Introduction and Objectives:** In this study, we determined the factors contributing to recurrence and successful treatment of VVF in a single Centre at Saint Luc Hospital, Democratic Republic of Congo.

**Materials and Methods:** One hundred-sixty data from vesico-vaginal fistula patients in the saint Luc hospital, Kisantu (DRC) were retrospectively collected between 2007 and 2013. Patients underwent surgery as treatment and followed-up for 3 months when they were evaluated. Successful treatment was defined as dry pads following surgery at 3 months follow-up, while recurrence was defined as wet pads persistently or after a period of dryness following surgery during the same period. The Waldijk-classification system was used prior to treatment to categorized patients as category I, IIAa, IIAb, IIBa, IIBb and III. Location of fistula was differentiated between urethral or other locations. Univariate and multivariate (stepwise method) analysis for recurrence and successful treatment were performed using logistic regression, corrected for pre-operative characteristics (POC), location of fistula (LF), size of fistula (SF) and event of fibrosis (FB). Similar method was used to identify the predictors for recurrence and successful treatment following surgery, corrected for intra-operative characteristics: flap procedure (FP), ureteral reimplantation (UR), urethral repair (URR).

**Results:** The median age was 29.11±9.6 years. Patients were categorized as category I (57.2%), IIAa (10.2%), IIAb (3.6%), IIBa (1.2%), IIBb (7.2%) and III (20.5%). Based on the pre and post-operative Waldijk-classification system, patients were considered completely cured, partially cured (down-staged), persistent and up-staged as 71.7%, 15.67%, 12% and 0.6% respectively. FB and LF were found to be independent predictors for recurrence. Vesico-vaginal patients with fibrosis were 2.95-fold (95% CI 1.31-6.62, p=0.0088) at higher risk for recurrence following surgery in comparison to those without fibrosis, meanwhile fistula at urethral location had 3.57-fold risk (95% CI 1.53 – 8.35, p=0.0033) of recurrence compared to other locations. Similarly, FB and LF

were also proven as independent predictors for successful treatment. Vesico-vaginal fistula patients with fibrosis were 68%-less likely (OR 0.32, 95% CI 0.14-0.73;  $p=0.0065$ ) to reach a successful treatment following surgery in comparison to those without fibrosis. On the other hand, fistula at urethral location were 73%-less likely (OR 0.27, 95% CI 0.11 – 0.63,  $p=0.0024$ ) to end up with successful treatment compared to other locations. None of the intra-operative characteristics (FP, UR and URR) were proven to be predictors for recurrence or successful treatment.

**Conclusion:** This study showed the impact of fibrosis and urethral location on the possibility of recurrence following surgery. A more extensive-particularly technique needs to be considered to treat VVF patients with fibrosis and/or urethral location.

#### MP-15.04

##### Vaginal Flaps for Repair of Vesico-Vaginal Fistulae

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**Introduction and Objectives:** Trans-vaginal repair of vesico-vaginal fistula (VVF) usually involves separation of bladder wall from vagina, freshening of edges, closure in layers and possible interposition of labial fat. We describe the use of vaginal flaps in situations where this is not possible.

**Materials and Methods:** From January 2009 to December 2013, 23 patients underwent trans-vaginal repair of VVF in our unit. Seventeen of these underwent the standard technique and 6 patients needed vaginal flaps. Four of these 6 were obstetric fistulae due to obstructed labour and two were after hysterectomy. One of the hysterectomies was trans-abdominal and the other was laparoscopic. Three of the obstetric and the laparoscopic hysterectomy patient had undergone previous attempts at VVF repair before presenting to our unit. Two of these were trans-abdominal and two were trans-vaginal. The size of the fistula varied from 1cm to 3cm. Reasons for using vaginal flaps were proximity to ureteric orifice in two cases, reduced bladder capacity in two cases, high VVF with previously failed trans-abdominal repair in one and fistula adherent to pubic bone in one case. Vagina was pre-treated with twice daily application of estrogen cream for a period of six weeks to make it supple and vascularised. One sided vaginal flap was used in the two patients with proximity to the ureteric orifice and in the patient with fistula adherent to the pubic bone. Two vaginal flaps (each flap based on each lateral edge of the fistula) were used in the two patients with reduced bladder capacity) and four flaps were used in the patient with high VVF with previously failed trans-abdominal repair.

The technique involved mobilizing the vaginal flap and flipping it in to bridge the defect on the bladder side. Subsequently the vaginal walls would be mobilized to make the second layer of repair.

**Results:** Five out of six repairs worked. The sixth patient with the fistula adherent to the pubic bone needed another trans-vaginal repair after 4 months which was successful.

**Conclusion:** Use of vaginal flaps adds a versatile technique to trans-vaginal repair of VVF in difficult situations.

#### MP-15.05

##### Surgical Repair of Iatrogenic Vesicovaginal Fistula

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**Introduction and Objectives:** Due to the high risk of recurrence, the surgical repair of iatrogenic vesicovaginal fistula is a real challenge, even for experienced urologists. The aim of this study was to evaluate the factors, contributing for the success of the procedure, and to present our current surgical technique, with the new modifications made in the last few years.

**Materials and Methods:** Between March 2004 and March 2014, a total of 22 women underwent surgical repair for iatrogenic vesicovaginal fistula at our institution. Mean patient age was 46.2 years (range 16 to 67 years). The fistula occurred after urological (1) or gynecological (21) operations, performed for benign (10), or malignant (12) diseases. In 9 of these cases the fistula was recurrent, and a few (1 to 4) unsuccessful attempts to close it had been previously done at other institutions. Beside total hysterectomy, due to gynecological malignancies, adjuvant pelvic radiotherapy had been also applied in 8 of the cases. Our standard surgical technique included: low midline laparotomy; sectio alta; insertion of ureteral prostheses; separation of fistula edges from the bladder wall; complete excision of all devitalized tissues; closure of the defect by suturing the vagina; interpositioning of a vascularized omentum flap, and closure of the bladder by a running 3-0 polydioxanone suture. In the last 8 cases the mobilized omentum flap was replaced (in 2 cases), or supplemented (in 6 cases) by the natural collagen implant "Pelvicol™" (Bard).

**Results:** All surgical procedures were done without complications. The postoperative period ran smoothly in all cases. The ureteral prostheses were usually removed on the 7<sup>th</sup> day, and the urethral catheter – on the 10<sup>th</sup> day after surgery. None of the patients experienced fistula recurrence during follow-up.

**Conclusion:** The interpositioning of a vascularized omentum flap and/or a biological collagen implant guarantees definitive closure

of iatrogenic vesicovaginal fistula and prevents fistula recurrences.

#### MP-15.06

##### Formation of a Vesicovaginal Fistula in a Pig Model

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**Introduction and Objectives:** To establish an animal model of a vesicovaginal fistula (VVF), which later can be used in the development of new treatment modalities.

**Materials and Methods:** Six female pigs of Landrace/Yorkshire race were used. The pigs were fully anaesthetised and through open abdominal surgery a vesicotomy was performed. An incision between the bladder and the vagina was made and the mucosa of the two organs were sutured together with soluble sutures. A Durometer Ureteral stent was introduced in to the fistula, secured in the bladder wall, allowing for the formation of a persistent fistula tract. Two pigs were used to evaluate the plausibility of the surgical technique and euthanized directly after the surgery. Six weeks post operatively cystoscopy, examining the fistula *in vivo*, was performed. Furthermore methylene blue was administered in to the bladder and leakage to the vagina was documented by video recording. Thereafter the pigs were euthanized with IV pentobarbital.

**Results:** Two out of four (50%) pigs developed persistent fistulas.

**Conclusion:** This study indicates that the use of pigs in a model of VVF can be an effective and cheap way to create a fistula between the bladder and the vagina.

#### MP-15.07

##### Pelvic Radiation is Associated with Urinary Fistulae Repair Failure and Permanent Urinary Diversion

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**Introduction and Objectives:** Enterourinary fistulae (EUF) and urinary-cutaneous fistulae (UCF) can be treated either with surgical repair by either excision with primary closure of the bladder using an omental, sliding, or muscle flap, or urinary diversion to allow the wound to heal by secondary intent. Permanent urinary diversion is often required to achieve adequate urinary management. Here, we review our experience with non-muscle flap repairs of EUF and UCF.

**Materials and Methods:** We reviewed 86 patients who underwent treatment of EUF or UCF at Barnes-Jewish Hospital between the

years 1998-2013. Of these patients, 39 underwent surgical repair of the communication (22 primary closures, 8 omental flaps, and 9 sliding flaps), while 47 patients underwent either surgical or non-surgical urinary diversion (16 catheters, 7 bilateral percutaneous nephrostomy tubes, 5 SP tubes, 16 ileal conduits, 2 coloconduits, and 1 neobladder). Patient outcomes were assessed including post-operative fistula closure, need for permanent urinary diversion, and urinary incontinence. Multiple variables were examined for association with poor outcomes, such as age, ASA score, diabetes, coronary artery disease, hypertension, obesity, smoking status, excision vs. repair of fistulae, intraoperative urinary/fecal diversion, and prior radiation.

**Results:** The mean age in our series was 59 years (21-87) at time of surgery with median follow-up of 20 months (1-137). Among patients who underwent surgical repair, radiation was associated higher rates of repair failure ( $p=0.001$ ), post-surgical incontinence ( $p<0.0001$ ), and need for permanent urinary diversion ( $p=0.024$ ). Overall, 31 of the 44 radiated patients required permanent diversion (70%), compared to 3 of the 42 non-radiated patients (7%) ( $p<0.0001$ ).

**Conclusions:** Patients who undergo pelvic radiation prior to EUF and UCF repairs are at higher risk for developing repair failure and post-surgical incontinence. Many of these patients eventually require permanent urinary diversion, often after a long and difficult series of surgical repairs. Therefore, EUF and UCF repairs in radiated patients should be undertaken with caution, and patients should be counseled about the possibility of performing permanent urinary diversion as the primary therapy.

#### MP-15.08

##### The Management of a Neobladder Vaginal Fistula following Radical Cystectomy and Orthotopic Ileal Neobladder Formation

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**Introduction and Objectives:** Neobladder-vaginal fistulation (NBVF) is a relatively rare complication following radical cystectomy and orthotopic ileal neobladder (ONB) formation (3-5% of patients). There are few series describing the management and subsequent outcomes of this complex condition that is more likely to occur with vaginal breach at initial surgery. We present a series of three patients over a one-year period undergoing repair of NBVF and the patient outcomes at our institution.

**Materials and Methods:** All patients undergoing repair of NBVF were reviewed prospectively (N=3). All fistulae were located at the neobladder urethral anastomosis and repair was achieved via a transvaginal approach in all cases. After first circumscribing the fistula tract,

a submucosal tissue plane was created to separate the vaginal mucosa from the neobladder prior to closure of the neobladder defect with polyglactin suture material. Unilateral Martius flap interpositioning was performed over the repair followed by vaginal wall closure. A 16Ch two-way urethral catheter was left *in situ* until a post-operative cysto-urethrogram demonstrated the absence of contrast leak.

**Results:** Three patients with a mean age of 62 (range 53 – 74), underwent a primary transvaginal repair of NBVF. Fistulae were diagnosed at initial post-operative cystogram and all patients had an initial trial of conservative management. Mean time from cystectomy to NBVF repair was 395 days (range 222 – 720). Two patients had a watertight repair on post-operative cystogram and the remaining patient demonstrated a small leak, which resolved following 2 weeks of further urethral catheterization. At the 3-month clinic review following NBVF repair, all patients described stress urinary incontinence (SUI) symptoms proven on video-urodynamic assessment without associated neobladder pressure rises. Two patients underwent an autologous fascia mid-urethral sling procedure with symptom resolution. Of these, one performs CISC. The remaining patient is currently awaiting a sling procedure.

**Conclusions:** NBVF repair challenges even the experienced surgeon. Transvaginal repair is preferred over open surgery at our institution if feasible and is worth attempting initially over urinary diversion. SUI is common following repair so patients are counseled that this could occur. Sling procedures can achieve continence in this situation with or without CISC.

#### MP-15.09

##### Objective Findings in Patients Presenting with Pelvic Mesh Complications

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**Introduction and Objectives:** We evaluated the true incidence of mesh related complications among patients presenting with “mesh anxiety”, described objective findings and treatment outcomes in these patients.

**Materials and Methods:** Between August 2012 and March 2014, we identified all patients presented with complaints related to perceived mesh complications. Past operative reports were obtained. Patients were offered work-up including a complete history, pelvic exam, cystoscopy, and urodynamic study, if indicated. If objective findings of mesh extrusion, erosion, exposure, obstruction or tenderness were found, the patients were offered appropriate corrective surgery.

**Results:** A total of 56 patients presented with mesh related complaints. Median age was 57 years (33-85). Median time elapsed between

mesh placement and presentation was 55 months (11-264). Presenting complaints included urinary incontinence in 29/56 patients (51.8%), pelvic pain in 9/56 (16%), combined incontinence and pelvic pain in 8/56 (14.2%), and others (e.g.: recurrent UTI, dysuria, stranguria, hematuria) in 10/56 (18%). Initial procedures were performed due to pelvic organ prolapse (POP) in 12/56 (21.4%), incontinence in 29/56 (51.7%), and combination of POP and incontinence in 15/56 (26.9%). Past operative reports indicated that 55/56 patients had a mesh and one patient had no mesh. Out of 56 patients, 49 had completed full work up in our institution. Of those, objective findings demonstrated mesh-related complications in only 23/49 patients (47%). Complications included extrusion into vaginal epithelium in 8/23 (35%), erosion into urinary tract in 3/23 (13%), point tenderness over the arms of the mesh in 11/23 (48%), and obstruction in 1/23 (4%). A total of 20/49 (40.8%) patients were offered mesh excision: 13/20 (65%) patients underwent an operation, 5/20 (25%) patients are awaiting their procedures and 2/20 (10%) deferred the operation. All other patients were non-surgically managed. All patients treated with mesh excision reported resolution of presenting symptoms.

**Conclusions:** In this study, more than half of the patients presenting with perceived mesh-related complaints did not have objective findings of mesh complications, but rather may have mesh failure or new problem. In this sub-population of patients who have objective findings, mesh removal often alleviates their symptoms.

#### MP-15.10

##### Evaluation Study between Tension-Free Vaginal Tape (TVT) and Transobturator Tape (TOT) in Treatment Stress Incontinence in Female

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**Introduction and Objectives:** To evaluate short and midterm results of tension-free vaginal tape (TVT) and transobturator tape (TOT) for treatment of female stress urinary incontinence (SUI) in Imam Reza Hospital.

**Materials and Methods:** In this prospective randomized clinical trial study at Imam Reza Hospital between March 2008 and October 2010, a total of 100 women with stress urinary incontinence treated with tension-free vaginal tape (TVT) (n=50) or transobturator tape (TOT) (n=50) were included. Preoperative workup including case history, clinical examination, marshal and boney test, urodynamic study with abdominal leak point pressure and post void residue measurement, pad test, TV scan,



ICIQ UI and ICIQ QOL questionnaires were performed. During hospitalization type of anesthesia, operative time, hospital stay and catheter indwelling time were recorded. Postoperatively, continence status and subjective patient satisfaction were evaluated using marshal and pad test, TV scan and ICIQ UI and ICIQ QOL questionnaires in 1, 3, 6, 12 and 18 months follow-up was performed for all the patients.

**Results:** Mean age was  $52.02 \pm 7.37$  yrs for TVT group and  $52.27 \pm 7.34$  yrs for TOT group. The rate of the complications was similar in two groups. Operative time was  $23.50 \pm 9.04$  min in TVT and  $25.00 \pm 9.48$  in TOT ( $P=0.86$ ), hospital stay time was  $1.56 \pm 0.51$  day in TVT and  $1.52 \pm 0.47$  in TOT ( $p=0.76$ ), catheter indwelling time (day) was  $1.58 \pm 0.41$  in TVT and  $1.55 \pm 0.47$  in TOT ( $P=0.651$ ), respectively. 1, 3, 6, 12 and 18 months after intervention, ICIQ UI and ICIQ QOL was filled for all the patients and there was no significant difference between two groups, but the rate of ICIQ (specially ICIQ QOL) after surgery was better in TVT group than TOT group.

**Conclusion:** Our results showed a similar efficacy and safety of TVT and TOT for women with SUI. In 18 months follow-up, TVT was not significantly effective than TOT.

#### MP-15.11

##### A Clinical Evaluation of MiniArc Precise Performed for the Treatment of Stress Urinary Incontinence: Why Are They So Effective?

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**Introduction and Objectives:** This study was designed to evaluate the efficacy of MiniArc Precise for treatment of the SUI and to review of the various self fixing mechanisms of the tapes.

**Materials and Methods:** A total of 32 women patients performed MiniArc Precise for treatment of SUI were enrolled in this study, who actually complained unacceptable SUI and inconvenience in daily life. All patients were  $55.2 \pm 6.3$  years of age. The pad test, urodynamic study, operation time, estimated blood loss (EBL), pain scale (NRS, numeric rating scale), the duration of urethral Foley catheter and hospital stay, BMI (body mass index), postoperative outcomes and life of quality by IIQ-7 questionnaire, etc. were investigated. MiniArc

procedures performed under the general (29 cases), regional (2 cases), or local anesthesia (one case) by one surgeon. The follow-up durations were  $13.8 \pm 4.9$  months. The IIO-7 Questionnaires were obtained by the telephone or a personal interview at the clinic.

**Results:** The weight of 1-hour pad test was 50gm (median). Operation times were  $20.71 \pm 5.06$  minutes. EBL were  $27.34 \pm 21.02$  ml. The postoperative pain scores (NRS) were  $2.75 \pm 0.84$ . They kept urethral Foley catheters for 2 to 24 hours and were discharged on postoperative 1<sup>st</sup> to 4<sup>th</sup> day. The IIQ-7 scores decreased from  $15.25 \pm 3.2$  to  $1.25 \pm 1.9$  ( $p=0.000$ ) at mean 14 months after operation. SUI cured in 28 patients of 32 patients (87.5%). Among these 4 failed cases 3 cases showed improved in IIQ-7 score and degree of incontinence, and two patients were combined with urgency incontinence. So, the cure and the significant improvement below the moderate degree of stress incontinence achieved in 31 patients (96.9%). There were no significant correlations between the BMI and other factors ( $p>0.05$ ).

**Conclusion:** MiniArcPrecise as a single-incision, minimally invasive, mini- and mid-urethral sling was an efficient choice for treatment of SUI and improving the quality of life with less complications and inconveniences. And it is thought that the bleeding while operation might be resulted from making a wrong way getting to the internal obturator muscle, and the shape of the self fixating extremity of the tape may be the most important for successful surgery in terms of normal voiding without incontinence nor retention, and which is need to be discussed in details.

#### MP-15.12

##### Genitourinary Disorders in Women in Postmenopausal Period

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**Introduction and Objectives:** Urogenital disorders (UGD) are found in 30% of women over 55 years old and in 75% of women over 70 years old. The development of UGD during climacterical period is caused by sex hormones deficiency, first of all – oestrogens. One of the widespread methods of treatment for dysuric phenomena is local application of estradiol. There are a number of contraindications to oestrogens

application – oestrogen dependent unknown site neoplasms and thromboembolism. The necessity of an alternative treatment for urogenital disorders in women is evident. The objective of the present research is to estimate the efficiency of a complex therapy including Depantol (dexpantenol 0.0525g and chlorhexidine bigluconate 0.0802) for women of climacteric age suffering from urogenital disorders.

**Materials and Methods:** A total of 50 women at the age of 50-65 were subjected to observation. The objects underwent vaginal secretion pH test, colposcopy, vaginal status index was estimated, urinary inoculation and vaginal smear for flora were analyzed, urination diary was filled in, CUDI, cystoscopy, and vaginal mucous membrane and bladder LDF, as well as pelvis minor organs ultrasound, were examined. Two groups of objects were sorted out. The first control group (C group) was formed by 20 patients who underwent etiotropic therapy for chronic cystitis (Furamag, diuretic herbs, intravesical instillations with Synthomycine emulsion, and B vitamins). In the second group the patients were prescribed additional Depantol – 2 vaginal suppositories per day for 20 days – for vaginal mucous membrane recovery.

**Results:** The control study was carried out 3 and 6 months later; when the results of both – etiotropic and complex therapy – were estimated, gradual relief of the main urogenital disorders symptoms was noted. In the 1<sup>st</sup> group the disease relapsed in 50% of the objects; clinical presentation of atrophic colpitis and cystourethritis continued or even worsened. Much better results were detected in the 2<sup>nd</sup> group. No relapse of urogenital disorders were found in 70% of the patients: no urination disorders, dyspareunia, vaginal xerosis or burning, or abundant unpleasant vaginal discharges. Normal acidity of vaginal medium was kept  $<5$ , mucous membrane atrophy decreased significantly.

**Conclusion:** Age-related oestrogen deficiency plays the leading part in pathogenesis of urogenital disorders, it leads to vaginal and urethral epithelium proliferation abnormality, vaginal secretion changes (lactobacilli extinction, pH increase), and reinfection. Good tolerance, easy usage, and hormone-free composition enable to recommend Depantol to a wide range of patients with mild and medium urogenital disorders during climacterical period.



## Moderated Poster Session 16

### Prostate Cancer: Advanced Disease Tuesday, October 14 1435-1600

#### MP-16.01

##### Androgen Receptor Variant-7 Plays an Essential Role in Development and Progression of Castration-Resistant Prostate Cancer

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**Introduction and Objectives:** The constitutively active, regardless of androgen level, ligand independent androgen receptor variants (AR-Vs) were proposed to be partly responsible for the development and growth of castration-resistant prostate cancer (CRPC). Among the numerous AR-Vs identified so far, androgen receptor variant-7 (AR-V7) is one of the most abundant and the best characterized variants. In the present study, we investigated the role of AR-V7 expression in the development of CRPC and determined whether the expression of AR-V7 in CRPC tissues could predict cancer-specific survival.

**Materials and Methods:** We enrolled 104 newly diagnosed metastatic prostate cancer (PCa) patients, 46 CRPC patients, 20 clinically localized PCa patients and 10 benign prostatic hyperplasia (BPH) patients, who underwent prostate biopsy, transurethral resection of the prostate or radical prostatectomy at our institution. The expression of AR-V7 was assessed by immunohistochemistry in primary tumor for all patients. Cox proportional hazards models were used to evaluate the predictive role of all covariates in the development of CRPC. Time to CRPC and cancer-specific survival curves were estimated using the Kaplan-Meier method and compared by the log-rank test.

**Results:** The AR-V7 detection rate in CRPC was significantly elevated compared with newly diagnosed metastatic PCa ( $P < 0.001$ ), with localized PCa ( $P < 0.001$ ), and with BPH ( $P = 0.001$ ). Multivariate analysis revealed that AR-V7 expression ( $HR = 2.627$ ,  $P = 0.001$ ) and prostate-specific antigen (PSA) nadir ( $HR = 1.012$ ,  $P = 0.019$ ) were independent predictive factors for the development of CRPC after adjusting for age, PSA, Gleason score, stage, time to PSA nadir and PSA half-life (PSAHL). Furthermore, the expression of AR-V7, which inversely correlates with serum PSA level, is associated with a shorter survival in CRPC patients.

**Conclusion:** The protein AR-V7 level in primary tumor can be used as a predictive marker for the development of CRPC and as a

prognostic factor in CRPC patients. Moreover, therapy targeting AR-V7 may help prevent PCa progression and improve prognosis of CRPC patients.

#### MP-16.02

##### Up-Regulation of COL6A1 Results in Cell Migration and Invasion and Predicts the Metastasis of Prostate Cancer

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**Introduction and Objectives:** The effective prognostic biomarkers, specific to the advanced stage, are urgently needed for better prediction and management of prostate cancer (PCa). COL6A1 is one of the three major subunit of collagen VI which is an important Extracellular matrix (ECM) protein, and involves in multiple signaling pathways that regulate apoptosis, proliferation, angiogenesis, fibrosis, and inflammation. Recent studies demonstrated COL6A1 was overexpressed in cancers and associated with tumor progression. In this study, we investigated the biological function and prognostic value of COL6A1 in PCa, found by our group among a panel of differentially methylated genes by DNA methylomic analysis.

**Materials and Methods:** The DNA methylation status of COL6A1 gene was confirmed by bisulfite-sequencing PCR, and expression pattern was determined by qRT-PCR and western blot. Cell invasion and migration was analyzed using transwell and wound healing assay. The expression level of COL6A1 protein in 223 prostate adenocarcinoma and 40 non-cancer tissues were determined by immunohistochemistry.

**Results:** The expression level of COL6A1 gene was negatively correlated with the methylation status of its 2<sup>nd</sup> exon in PCa cell lines. COL6A1 was a positive regulator of both cell invasion and migration and correlated with the expression of MMP-9 and CXCR4. The expression level of COL6A1 protein was significantly higher in patients with advanced stage, as bone and/or lymph node metastasis, than that in patients with early stage of PCa.

**Conclusion:** The up-regulation of COL6A1 results in cell migration and invasion and may serve as a promising biomarker to predict the metastasis of PCa.

#### MP-16.03

##### Microstructural Analysis for Novel Bone Metastatic Model of Prostate Cancer by Micro CT

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**Introduction and Objectives:** We have already established the novel metastatic prostate cancer model in mice. In that model, we used bone metastatic cancer cells directly deprived from the prostate cancer patients, and we can create the similar condition as clinical patients with bone metastasis by using this model. The object of this study is to evaluate microstructural changes in this humanized bone metastatic model of prostate cancer by micro CT analysis.

**Materials and Methods:** Tumor cells were directly isolated from the bone metastatic region of the prostate cancer patients. Isolated cells were named as PCSD. PCSD and Mitrigel were transplanted into the right femurs of Rag2(-/-);γc(-/-) mice by the needle injection via condyle of the femur. Thirteen mice with needle holes and tumor formation were picked up for PCSD group, and 8 mice with needle holes for Control group respectively. The microstructural analysis was carried out by micro-CT using the commercial equipment SkyScan.

**Results:** Significant changes were observed in PCSD group on both 2D and 3D images. The right femur length was shorter than the left femur in both groups, however, the diameter of the bone shaft of the right femur was broadened only in PCSD group ( $p < 0.01$ ). Bone volume and total volume ratio (BV/TV) and bone volume (BV) were significantly decreased in the proximal and distal area in PCSD group ( $p < 0.01$  respectively), however, BV was significantly increased in the cortex bone in the middle of bone shaft in PCSD group ( $p < 0.05$ ). Trabecular bone didn't show much difference between PCSD group and Control group, however, the changes in trabecular bone could be divided into either osteolytic or osteoblastic nature.

**Conclusion:** Osteolytic and osteoblastic or mixed changes were confirmed in all samples. Abnormal bone extension to the lateral side was observed in PCSD group. As a whole, Distal and proximal of the femur have an osteolytic nature, and cortical bone has an osteoblastic nature. The changes of trabecular bone could be divided into two ways of osteoblastic or osteolytic character.

#### MP-16.05

##### Radical Prostatectomy for High-Risk Prostate Cancer: Long-Term Oncological Outcomes

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**Introduction and Objectives:** We analyzed the long-term biochemical recurrence-free survival (BCRFS) and prostate cancer-specific survival (PCSS) in patients with high-risk (HR) prostate cancer (PC) treated with open retropubic radical prostatectomy (RP) according to the number of HR-factors and results of pathological evaluation (specimen-confined (SC) vs. non-specimen-confined (NSC) PC).

**Materials and Methods:** Between 1997 and 2012, 776 patients with HR-PC (clinical stage  $\geq$  T2c and/or biopsy Gleason score 8-10 and/or prostate-specific antigen  $> 20$  ng/ml) underwent RP at our hospital. The study comprised 446 patients who were followed up for  $> 12$  months, had all preoperative and postoperative information available and received no neoadjuvant androgen deprivation therapy. SC disease was defined as pT2-3R0N0 PC. The Kaplan-Meier method with long-rank test was used to evaluate BCRFS and PCSS rates.

**Results:** Median follow-up time was 50 months (IQR: 31-74); 320 patients (71.8%) had one HR-factor, 109 (24.4%) – two HR-factors and 17 (3.8%) – three HR-factors. SC PC was found in 308 (69%) patients. At 5 and 10 years BCRFS was 65% and 62%, the 10- and 15-year PCSS was 92.6% and 82.6%. In SC and NSC groups 5-year BCRFS was 79.6% and 32.7% ( $p < 0.001$ ); 10-year PCSS was 100% and 78.6% respectively ( $p < 0.001$ ). Presence of  $> 1$  HR-factor was a significant predictor of biochemical recurrence (HR 2.916;  $p < 0.001$ ). For patients with 1, 2 and 3 HR-factors 5-year BCRFS was 76.7%, 39% and 35.3%; 10-year PCSS – 97.8%, 85.4% and 64.2% respectively.

**Conclusion:** RP can result in high long-term BCRFS in selected patients with HR-PC. SC disease (radical operation) provides highest BCRFS and PCSS after RP. Presence of  $> 1$  HR-factors results in less favorable oncological outcomes. Substratification of HR-patients and individual risk-assessment are essential.

#### MP-16.06

##### Chronic Kidney Disease in Patients with Prostate Cancer: A New Epidemic?

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**Introduction and Objectives:** The incidence of prostate cancer in Kenya has increased by 75% from the last decade with over 2000 new diagnoses annually, making it only second to esophageal cancer. Chronic kidney disease (CKD) is an important sequela in advanced prostate cancer although data from Africans is scarce and absent from Kenya. We investigated

the incidence of CKD associated with prostate cancer at Kenya's largest Referral Hospital.

**Materials and Methods:** Forty prostate cancer patients were reviewed from January 2013 to December 2013. Clinical presentation, creatinine, PSA, Gleason's score and comorbidities were analyzed.

**Results:** A total of 75% had metastatic disease at diagnosis. Most of the remaining patients had locally advanced disease. Mean age was 72.4 years ( $\pm 10$ ). Six patients were diabetic and 8 were hypertensive. The clinical presentation of these patients is summarized below (Figure 1). A total of 57.5% of the patients were in renal failure at presentation (serum creatinine  $\geq 140$   $\mu$ mol/l) and were subsequently dialyzed; 25% of the patients eventually developed CKD (eGFR  $\leq 60$  ml/min three months post-admission). Factors that were associated with CKD included metastatic disease at admission, hydronephrosis (O.R.=10.5{95% CI 3.1-12.9}) and hemoglobin  $\leq 10$  g/dl (O.R.=24.5{95% CI 5.2-27.4}).

**Conclusion:** A total of 75% of patients presented with advanced disease and highly aggressive tumors, probably one of the highest in African reports. Half of the patients had significant obstructive uropathy leading to CKD. Hydronephrosis and hemoglobin  $\leq 10$  g/dl carried the highest odds for developing CKD. Future studies should look at the outcome of these patients to mitigate on this silent killer.

#### MP-16.07

##### Satisfaction and Tolerability among Men with Advanced Hormone-Dependent Prostate Carcinoma Treated with One-Monthly Subcutaneous Administration of Degarelix in The Netherlands

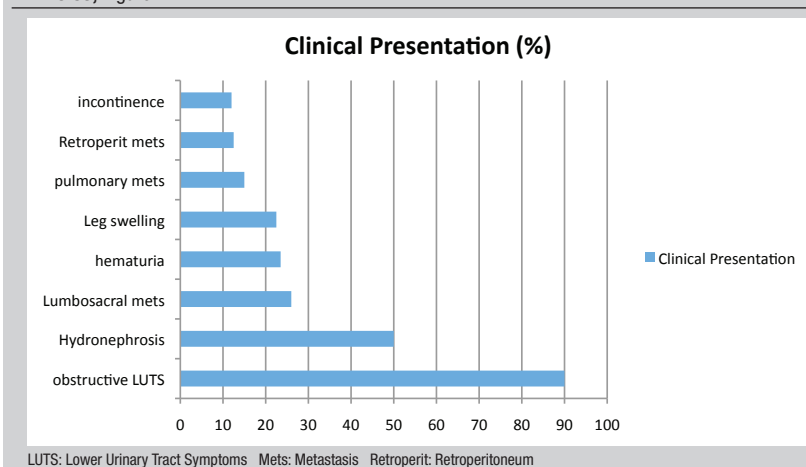
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**Introduction and Objectives:** We assessed the satisfaction and tolerability of patients who received degarelix as primary treatment or after previous hormone treatment.

**Materials and Methods:** Since September 2010, 273 patients with advanced hormone-dependent prostate cancer who were treatment with degarelix, were included in this multi-center phase IV non-interventional trial in the Netherlands. Degarelix was administered monthly subcutaneously for a maximum of three years after written informed consent was obtained. Previous hormone therapy was not a contraindication. Satisfaction was assessed six months after starting treatment and at the end of the study in the subgroup meeting the primary endpoint of PSA progression or dropping out prematurely for other reasons. For this analysis patients were divided into two groups: hormone-naïve versus previous hormone treatment.

MP-16.06, Figure 1.



MP-16.06, Table 1. Baseline Laboratory and Histological Parameters at Diagnosis

| PARAMETER                | AVERAGE           | RANGE              |
|--------------------------|-------------------|--------------------|
| Serum creatinine         | 310.9 $\mu$ mol/L | 72-976 $\mu$ mol/L |
| PSA                      | 410.8 ng/ml       | 27-4,079 ng/ml     |
| Gleason's score          | 8.6               | 6-10               |
| Hemoglobin concentration | 10.2 g/dl         | 3.8-15.2 g/dl      |

**Results:** Mean age was 71.6 years (range, 47–92 years). There were 138 out of 273 (51%) patients who were asymptomatic. PSA level at start of treatment was <10 mg/ml in 33 (12%), 10–20 mg/ml in 37 (14%), 20–50 mg/ml in 63 (23%) and >50mg/ml in 139 (51%) patients, respectively. Two hundred (73%) patients were hormone-naïve. Treatment satisfaction could be evaluated in 196 men. Monthly administration was considered a satisfactory frequency by 162 (82%) men, while 34 (18%) patients perceived it as being too often. No differences were observed between those who were hormone-naïve or previously treated with other hormone therapy (82% vs. 84% satisfaction, respectively). The one monthly nurse visits was perceived to be too often by only 9 patients (5%), without differences between the two patients groups (hormone-naïve vs. hormone-pretreated). End of study satisfaction could be evaluated in 121 patients. Among these, degarelix was poorly tolerated by 13 (11%) patients. The remaining 109 men tolerated treatment very well in 7 (6%), well in 67 (55%) and fairly in 34 (28%) patients, respectively. No differences were observed when patients were stratified according to administration of injections by (specialized) nurse-service, general practitioner or urologist. **Conclusion:** In this study, one monthly administration of degarelix was well accepted among men with advanced hormone-dependent prostate cancer. No differences in patients' satisfaction were observed between those being hormone-naïve or with a history of previous hormone treatment.

#### MP-16.08

##### Delayed Hormonal Therapy Could Be an Option in Selected Patients with Lymph Node Metastases after Surgical Treatment

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**Introduction and Objectives:** Lymph node invasion (LNI) is a poor prognostic factor in prostate cancer (PC) patients (pts) undergone radical prostatectomy (RPE) and pelvic lymph node dissection (PLND) and often require immediate adjuvant hormonal therapy (ADT). The question if ADT could be delayed is controversial. The aim of the study was to assess biochemical progression-free survival (BPFS) in different prognostic subgroups of LN positive PC pts.

**Materials and Methods:** Retrospective analysis of 1430 PC pts undergone RPE and PLND since 1998 till 2014 was done. LN metastases were verified in 229 (16%). Mean PSA level was  $23.8 \pm 23.3$  ng/ml; mean percentage of positive biopsy cores (PPBC) was  $76.4 \pm 27.7\%$ . Clinical stage was T1b-T2c in 111 (48.5%), T3a-T3b – in 118 (51.5%). Biopsy Gleason score  $\leq 6$  was verified in 57 (24.9%) pts; 7 (3+4) – in 58 (25.3%); 7 (4+3) – in 68 (29.7%) and 8–10 – in 37 (16.2%); not assessed in 9 (3.9%) pts. Low risk PC was verified in 5 (2.2%) pts, intermediate risk – in 44 (19.2%) pts and high risk – in 180 (78.6%) pts. Biochemical recurrence (BR) was assessed as elevation of PSA>0.2 ng/ml on three consecutive measurements. Pts with immediate ADT after the operation were excluded.

**Results:** Mean number of LN removed was  $23 \pm 9$  (2–53). Mean follow-up time was  $29.6 \pm 24.9$  months (3–125 months). BR was observed in 91 (39.7%) pts. 3-year biochemical progression-free survival (BPFS) was  $21.1 \pm 4.6\%$ . 3-year BPFS in subgroup of pts undergone extended vs. standard PLND was  $27.9 \pm 6.8\%$  and  $15.3 \pm 5.6\%$ , respectively ( $p=0.02$ ); in pts with  $\leq 2$  positive LN vs.  $>2$  positive LN it was  $33.1 \pm 6.9\%$  and  $6.8 \pm 4.3\%$ , respectively ( $p=0.0002$ ); in pts with positive LN density <15% vs.  $\geq 15\%$  was  $30.3 \pm 6.4\%$  and  $7.3 \pm 4.7\%$  ( $p=0.001$ ) and in pts with presence of metastatic LN extra capsular extension (LN ECE) vs. absence of LN ECE it was  $9.2 \pm 5.2\%$  and  $31.8 \pm 8.2\%$ , respectively ( $p=0.003$ ). In pts with LN metastases only in 1 anatomical region vs. in  $\geq 2$  regions 3-year BPFS was  $33.2 \pm 7.1\%$

and  $9.4 \pm 4.8\%$ , respectively ( $p=0.008$ ).

**Conclusions:** In LN positive pts underwent RPE and extended PLND with  $\leq 2$  positive LN, LN density <15%, absence of LN ECE and only 1 anatomical region involved delayed ADT could be an option.

#### MP-16.09

##### The PREVAIL Study: Primary and Non-Visceral/Visceral Disease Subgroup Results for Enzalutamide-Treated Men with Metastatic Castration-Resistant Prostate Cancer

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**Introduction and Objectives:** Enzalutamide (ENZA), an oral androgen receptor inhibitor, improved overall survival (OS) in metastatic castration-resistant prostate cancer (mCRPC) patients (pts) who had received prior docetaxel.

MP-16.09, Table 1.

|                                                        | With visceral disease         |                 | Without visceral disease      |                 | All patients                            |                 |
|--------------------------------------------------------|-------------------------------|-----------------|-------------------------------|-----------------|-----------------------------------------|-----------------|
|                                                        | ENZA (n=98)                   | Placebo (n=106) | ENZA (n=774)                  | Placebo (n=739) | ENZA (n=872)                            | Placebo (n=845) |
| <b>Median* OS (months)</b>                             | 27.8                          | 22.8            | NYR                           | 30.2            | 32.4                                    | 30.2            |
|                                                        | HR (95% CI)=0.82 (0.55, 1.23) |                 | HR (95% CI)=0.69 (0.57, 0.83) |                 | HR (95% CI)=0.71 (0.60, 0.84); P<0.0001 |                 |
| <b>Median* rPFS (months)</b>                           | NYR                           | 3.6             | 14.1                          | 4.0             | NYR                                     | 3.9             |
|                                                        | HR (95% CI)=0.28 (0.16, 0.49) |                 | HR (95% CI)=0.17 (0.14, 0.22) |                 | HR (95% CI)=0.19 (0.15, 0.23); P<0.0001 |                 |
| <b>Median* Time to cytotoxic chemotherapy (months)</b> | 22.6                          | 6.6             | 28.4                          | 11.6            | 28.0                                    | 10.8            |
|                                                        | HR (95% CI)=0.30 (0.20, 0.45) |                 | HR (95% CI)=0.36 (0.31, 0.42) |                 | HR (95% CI)=0.35 (0.30, 0.40); P<0.0001 |                 |

\*Based on Kaplan Meier estimate; NYR: not yet reached



The PREVAIL study examined the impact of ENZA on OS and radiographic progression-free survival (rPFS) in asymptomatic or mildly symptomatic chemotherapy-naïve men with mCRPC that had progressed on androgen deprivation therapy. Based on statistically significant benefits for ENZA in OS and rPFS, the Data Monitoring Committee recommended stopping PREVAIL at the interim OS analysis. Here we report primary endpoints and subgroup analyses for pts with and without visceral disease (liver and/or lung) at screening.

**Materials and Methods:** This double-blind, multinational study randomized pts 1:1 to ENZA 160 mg/day or placebo. OS and rPFS were coprimary endpoints. Planned sample size was 1680 with 765 deaths to achieve 80% power to detect a target OS hazard ratio (HR) of 0.815 with a 2-sided type I error rate of 0.049 and a single interim analysis at 516 (67%) deaths. Prespecified subgroup analyses evaluated OS and rPFS in pts with and without visceral disease using the same statistical methodologies.

**Results:** A total of 1717 men were randomized (1715 treated), including 204 pts (11.9%) with visceral disease at screening (63.7% lung and 36.3% liver and/or lung). Pts without visceral disease (n=1513) had lower baseline median PSA than pts with visceral disease (46.8 and 72.5 ng/mL, respectively), better performance status (68.9% and 61.8% ECOG PS=0), less lymph node disease (49.8% and 57.8%), but similar rates of bone disease (83.7% and 80.4%). Endpoint results by subgroup are shown in Table 1.

**Conclusions:** Treatment with enzalutamide significantly improved OS and rPFS in men with chemotherapy-naïve mCRPC. Although pts with visceral disease progressed more rapidly regardless of treatment allocation, the benefit of enzalutamide was consistent among pts with or without visceral disease.

#### MP-16.10

##### Enzalutamide Monotherapy: 1-Year Extended Follow-Up of a Phase 2 Study in Hormone-Naïve Prostate Cancer Patients

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**Introduction and Objectives:** The efficacy and safety of enzalutamide monotherapy was assessed in men with any-stage hormone-naïve prostate cancer eligible for androgen-deprivation therapy (ADT). The primary endpoint of PSA response rate ( $\geq 80\%$  PSA decline between baseline and week 25) was 92.5% (Tombal B et al, EAU 2013). The median (range) maximum PSA decline from baseline to week 25 was  $-99.6\%$  ( $-100$ ,  $-86.5$ ). 1-year extended follow-up data are presented.

**Materials and Methods:** In an open-label, single-arm Phase 2 study (NCT01302041), men  $\geq 18$  years with histologically confirmed prostate cancer requiring ADT, non-castrate testosterone ( $\geq 8$  nmol/L), PSA  $\geq 2$  ng/mL at screening, and a life expectancy of  $\geq 12$  months, received 160 mg enzalutamide once daily until disease progression or unacceptable toxicity. Other endpoints included changes in hormone levels, metabolic parameters, bone mineral density (BMD), safety, and quality of life (QoL).

**Results:** Sixty seven men were enrolled. Median (range) age was 73 years (48–86); 38.8% had metastases; 35.8% and 23.9% had undergone prior prostatectomy and radiotherapy, respectively. Fifty four men (80.6%) completed 1 year of treatment with a PSA response rate of 100% and 53 (98.1%) had  $\geq 90\%$  PSA decline from baseline. The median (range) maximum decline in PSA was  $-100\%$  ( $-100$ ,  $-86.5$ ) from baseline to 1 year. Luteinising hormone and testosterone were increased from baseline by 215.2% and 101.7%, respectively. Mean changes from baseline for fasting metabolic variables were:  $+5.0\%$  total cholesterol,  $+8.9\%$  triglycerides,  $-3.5\%$  HbA1c, and  $+19.7\%$  insulin resistance (HOMA-IR). Total body BMD was maintained ( $-0.3\%$  from baseline). The most frequently reported treatment-emergent AEs were gynecomastia (47.8%) and fatigue (38.8%). Seven non-drug-related serious AEs were reported. QoL scores at 1 year demonstrate maintenance of global health status and a decrease in sexual activity and sexual functioning.

**Conclusion:** Extended follow-up of hormone-naïve patients demonstrated sustained PSA reductions up to 1 year of enzalutamide monotherapy. Endocrine and metabolic changes, and AEs were consistent with potent AR inhibition and similar to results reported at 25 weeks.

#### MP-16.11

##### Chemotherapy-Driven Transcriptome Profiles from Pre- and Post-Treatment Prostatic Biopsies of Patients with Advanced Hormone-Naïve Prostate Cancer

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**Introduction and Objectives:** Although chemotherapy for prostate cancer (PCa) can improve patient survival, some tumours are chemo-resistant. As tumour molecular profiles may be predictive of treatment response, we studied the feasibility of transcriptome profiling of pre- and post-treatment biopsies from patients with advanced hormone-naïve prostate cancer treated with docetaxel chemotherapy and androgen deprivation therapy (ADT).

**Materials and Methods:** RNA sequencing (RNA-seq) was performed on prostatic biopsies from four patients with locally-advanced/metastatic PCa before and  $\sim 22$  weeks after docetaxel and ADT initiation. Differentially-regulated genes in treatment pairs were identified, and pathway enrichment analyses undertaken. Flow cytometry was performed on PCa cells to determine the functional effect of docetaxel.

**Results:** RNA-Seq identified 298 and 277 genes up- and down-regulated genes at least 2-fold, respectively (False Discovery Rate [FDR]  $< 0.05$ ), in response to docetaxel plus ADT. Pathway analyses revealed enrichment for cell cycle (but not apoptosis/cell death) expression changes. We demonstrate docetaxel-induced G2/M arrest and sub-G0/G1 accumulation of androgen-responsive LNCaP cells, which is enhanced and diminished by androgen deprivation, respectively. Finally, we demonstrate an inverse correlation between docetaxel-induced G2/M arrest and apoptosis.

**Conclusions:** Identification of chemotherapy-driven PCa transcriptome profiles is feasible and informative. ADT may diminish the cytotoxic effects of docetaxel chemotherapy in the context of hormone-naïve PCa.

#### MP-16.12

##### Cohort Compassionate Use Program with Cabazitaxel Plus Prednisolone in Patients with Metastatic Castration-Resistant Prostate Cancer (mCRPC) Previously Treated with Docetaxel: Data in Thai Patients

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**Introduction and Objectives:** Cabazitaxel is a next generation taxane specifically developed to overcome resistance to older taxanes. Compassionate use program (CUP) was established to provide access to cabazitaxel before the approval and to evaluate safety and tolerability of this agent in a real-world population. The objectives are to allow access to cabazitaxel before it is commercially available in patients with mCRPC whose disease has progressed during or after docetaxel treatment, and who have similar disease and baseline characteristics to those evaluated in TROPIC trial and to document the overall safety of cabazitaxel in these patients.

**Materials and Methods:** Between July 2012 and May 2013, a total of 40 mCRPC patients received cabazitaxel 25 mg/m<sup>2</sup> intravenously every 3 weeks in combination with prednisone/prednisolone 10 mg oral, daily until disease progression, death, unacceptable toxicity, physician's decision or patient's refusal. Granulocyte colony-stimulating factor (G-CSF) was administered as primary and secondary prophylaxis as per physician's discretion.

**Results:** Mean/Median age of the patients was 70.3/72 years (range 50-83) 14 patients (35%) were ≥75 years old; 95% were ECOG PS 0-1. Mean/Median cumulative dose of prior D was 422.7/450.0 (range 180.0-825.0) mg/m<sup>2</sup>. The mean/median time from last dose of D to inclusion was 6.59/3.22 (range 0.7-45.0) months; 61.5% of patients experienced disease progression <3 months after D. Sixty five percent of patients had ≥2 metastatic sites; the most common were bone (92.5%). Mean/Median cycle of 7/7 (range 1-13) cycles of Cabazitaxel+Prednisolone was administered and 12 patients (30%) received ≥10 cycles. Mean/Median relative dose intensity was 100.0%/99.8% (range 96.6-106.5). G-CSF was administered to 77.5% of patients in cycle 1 (2.5% therapeutic, 25% prophylactic). Of all the patients, 97.5% experienced adverse events (all grades). The most common Grade 3-4 AEs included neutropenia 45%, anemia 15%, febrile neutropenia 12.5%, diarrhea 10%, fatigue 7.5%, vomiting 5%, and sepsis 2.5%. Five cases (12.5%) of treatment-related deaths were reported.

**Conclusion:** The CUP provides tolerability and safety data in Thai patients with mCRPC. Treatment with Cabazitaxel+Prednisolone was tolerable, with a manageable toxicity profile consistent with the reported data in TROPIC and CUP data in other part of the world, although the age of the Thai patients were slightly older.

#### MP-16.13

**Effect of Neoadjuvant Docetaxel Chemotherapy Combined with Adjuvant Chemotherapy Compared with Radical Prostatectomy Alone in Patients with High Risk Prostate Cancer**

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**Introduction and Objectives:** Preoperative chemotherapy with androgen deprivation therapy is under investigation, some with promising results. However, the definitive value of preoperative chemotherapy-only therapy remains unevaluated for high risk prostate cancer. We investigated the efficacy of docetaxel treatment before and after radical prostatectomy (RP) in patients with high risk prostate cancer.

**Materials and Methods:** Patients with high risk prostate cancer according to D'Amico criteria were prospectively randomized to one of two treatment groups: neoadjuvant chemotherapy (three cycles) followed by RP (arm A, n=27); or RP alone (arm B, n=21). Patients in arm A were treated with 75 mg/m<sup>2</sup> IV docetaxel every three weeks following premedication with 8 mg of dexamethasone; they then underwent RP two weeks later. After RP, 17 patients in Group A were treated with docetaxel with same schedule. The median observation time was 50.74 months in both arms.

**Results:** There were no differences in the age (66.6 and 64.2 years, p=0.080), pretreatment PSA levels (23.9 and 16.2 ng/mL, p=0.052), clinical stage and Gleason score between the patients in arm A and B, respectively. Grade 1-2 non-hematologic toxicities occurred in 13 patients (81.3%), none with grade 3-4 toxicities. The pathological stages were: T2 in 4 (23.5%), T3a in 8 (47.1%), and T3b in 5 (29.4%) in arm A; T2 in 5 (23.8%), T3a in 8 (38.1%), and T3b in 8 (38.1%) in arm B (p=0.914). Positive margin rate was similar between the arms in pT2 patients while in pT3 patients it was lower in arm A (37.5% vs. 90.9%, p=0.008). PSA response (> 25% reduction) after chemotherapy was observed in 8 patients (50.0%), but pathological downstaging was noted in 2 (12.5%). After RP, 6 and 7 from each group failed to reach undetectable PSA. PSA failure rate was 27.3%, 50.0% in arm A and B respectively (p=0.414).

**Conclusions:** Docetaxel neoadjuvant chemotherapy lowers positive resection margin in patients with high risk prostate cancer. However, pathological downstaging or immediate oncological result was observed only in a limited number of patients who demonstrated substantial PSA response. But there was no difference in biochemical recurrence free survival.

#### MP-16.14

**Sustained Remission after Docetaxel Chemotherapy for Metastatic Castration-Resistant Prostate Cancer**

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**Introduction and Objectives:** Docetaxel (DTX) chemotherapy is a standard

chemotherapy for patients with metastatic castration-resistant prostate cancer (CRPC). Although substantial responses to DTX are frequently reported, sustained remission may not be expected. Here, we present patients who showed sustained remission during DTX discontinuation and analyzed their clinical parameters.

**Materials and Methods:** The criteria of sustained remission include PSA normalization (< 4 ng/ml), no PSA progression (> 2ng/ml PSA increase from nadir), no symptomatic or radiographic progression, DTX discontinuation, and the remission lasting 6months or longer. Patients who received DTX for metastatic CRPC in our institution were reviewed retrospectively and those patients who achieved the sustained remission were retrieved.

**Results:** From April 2005 to September 2013, 57 patients with metastatic CRPC were treated with triweekly DTX (75mg/m<sup>2</sup> every 3 weeks) or weekly low-dose DTX (25mg/m<sup>2</sup> days 2 and 9) with estramustine (560 mg on days 1-3 and 8-10). All patients received oral prednisolone or dexamethasone daily. Among them, 7 patients (12%) achieved the sustained remission lasting from 6 to 50 months after the last DTX (median number of cycles, 10 cycles; range 8 to 11). Median age was 76 years old (range: 59 to 82 years). Median PSA was 19 ng/ml (range: 4.7 to 139 ng/ml). Two patients could discontinue oral glucocorticoids. Six patients had bone metastasis only and a patient had lymph node metastasis only. Extent of bone disease (EOD) was graded as 0 in a patient, 1 in three patients, 2 in a patient, and 3 in two patients. Lower EOD grade was significantly associated with longer duration of the remission (p<0.05). All (7) patients were alive at the time of analysis with the survival time after DTX chemotherapy ranging from 16 to 57 months.

**Conclusion:** Sustained remission is achievable by DTX chemotherapy in a minority of patients with metastatic CRPC. Patients with low EOD grade may have a significant benefit from chemotherapy holiday after completion of a full course of DTX.

#### MP-16.15

**Clinical Effect of Abiraterone Acetate in Korean Patients with Metastatic Castration-Resistant Prostate Cancer (mCRPC) according to ECOG Performance Status**

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**Introduction and Objectives:** The therapeutic novel agents for metastatic castration-resistant prostate cancer (mCRPC) have rapidly expanded in recent years. Abiraterone acetate is a representative novel agent, but there was not enough data for mainly ECOG PS (Eastern

**MP-16.15**, Table 1. Baseline characteristics at initiation of abiraterone therapy

| Chararacteristics                                                                                       | ECOG 1-0 (n=3) | ECOG ≥ 2 (n=17) | p-value |
|---------------------------------------------------------------------------------------------------------|----------------|-----------------|---------|
| <b>Age:</b>                                                                                             |                |                 |         |
| Mean (SD)                                                                                               | 80.33 (5.51)   | 75.18 (7.36)    | 0.203   |
| <b>BMI:</b>                                                                                             |                |                 |         |
| Mean (SD)                                                                                               | 22.69 (1.56)   | 24.41 (3.32)    | 0.56    |
| <b>PSA (ng/ml) :</b>                                                                                    |                |                 |         |
| Mean (SD)                                                                                               | 151.95 (84.16) | 160.04 (257.52) | 0.266   |
| <b>Metastasis site number*</b>                                                                          |                |                 |         |
| 1                                                                                                       | 3 (100)        | 13 (76.5)       | 0.643   |
| 2                                                                                                       | 0              | 3 (17.6)        |         |
| 3                                                                                                       | 0              | 1 (5.9)         |         |
| Previous RT*                                                                                            | 1 (33.3)       | 7 (41.2)        | 0.798   |
| Previous RP*                                                                                            | 0              | 4 (23.5)        | 0.348   |
| <b>Gleason score*</b>                                                                                   |                |                 |         |
| 7                                                                                                       | 0              | 2 (11.8)        | 0.312   |
| 8                                                                                                       | 2 (66.7)       | 5 (29.4)        |         |
| 9                                                                                                       | 0              | 8 (47.1)        |         |
| 10                                                                                                      | 1 (33.3)       | 2 (11.8)        |         |
| <b>Prior therapy:</b>                                                                                   |                |                 |         |
| Duration of ADT                                                                                         | 29 (18.03)     | 35.47 (24.65)   | 0.921   |
| <b>Prior chemotherapy</b>                                                                               |                |                 |         |
| Docetaxel                                                                                               | 3 (100)        | 10 (58.82)      |         |
| Docetaxel+Carbazitaxel                                                                                  | 0              | 4 (23.53)       |         |
| Mitoxantrone                                                                                            | 0              | 1 (5.88)        |         |
| Duration of chemotherapy, cycle                                                                         | 8.33 (9.24)    | 11.82 (8.78)    | 0.479   |
| Mann-Whitney U- test; * Fisher's exact-test                                                             |                |                 |         |
| RT=radiation therapy; RP=Radical prostatectomy; BMI= body mass index;<br>PSA= prostate-specific antigen |                |                 |         |
| PSAD= PSA density; ADT=androgen deprivation therapy                                                     |                |                 |         |

**MP-16.15**, Table 2. Outcomes with abiraterone stratified belong to ECOG PS

| Characteristics                                                                                | ECOG 0-1 (n=3) | ECOG ≥ 2 (n=17) | p-value |
|------------------------------------------------------------------------------------------------|----------------|-----------------|---------|
| Nadir PSA (ng/ml)                                                                              | 94.19 (53)     | 57.45 (78.82)   | 0.452   |
| PSA decline ≥ 50%                                                                              | 1 (33.3)       | 6 (35.3)        | 0.948   |
| Duration of PSA decline ≥ 50% (month)                                                          | 0.68 (1.15)    | 0.74 (1.15)     | 0.925   |
| Side effect*                                                                                   | 0              | 8 (47.1)        | 0.125   |
| PSA response*                                                                                  | 2 (66.7)       | 8 (47.1)        | 0.531   |
| Duration of treatment (month)                                                                  | 10.67 (6.03)   | 6.53 (4.32)     | 0.163   |
| PSA doubling time (month)                                                                      | 6              | 2.31 (1.32)     | 0.019   |
| PSA progression*                                                                               | 3 (100)        | 14 (82.4)       | 0.43    |
| Radiologic progression*                                                                        | 0              | 6 (35.3)        | 0.219   |
| Clinical progression*                                                                          | 0              | 4 (23.5)        | 0.348   |
| * Fisher's exact- test                                                                         |                |                 |         |
| ECOG= Eastern Cooperative Oncology Group; PSA= prostate-specific antigen; OS= overall survival |                |                 |         |
| PSA progression, Radiologic progression, Clinical progression were assessed by PCWG 2 criteria |                |                 |         |

Cooperative Oncology Group Performance Status) ≥ 2.

**Materials and Methods:** We analysed prostate-specific antigen (PSA) nadir level, time to PSA nadir, PSA doubling time, PSA response and modes of progression on abiraterone (PSA, radiologic, clinical) in data of each patients. PSA and radiologic progression were classified as Prostate Cancer Working Group 2 criteria.

**Results:** We reviewed data from 20 mCRPC patients who received abiraterone acetate after docetaxel chemotherapy failure from May 2012 to March 2014 at our institution. There is no significant difference of baseline characteristics who received abiraterone acetate between patients with ECOG PS 1-0 and ≥ 2. It's shown no statistically significant difference of outcomes with abiraterone acetate stratified between patients with ECOG PS 1-0 and ≥ 2 during this period. However, Kaplan-Meier analysis demonstrated that radiologic progression free survival rate and clinical progression free survival rate might be associated with ECOG PS.

**Conclusion:** Although it is the retrospective small size study, favorable response of abiraterone acetate may be observed in patients with ECOG PS 0 or 1 in terms of radiologic progression free survival rate and clinical progression free survival rate.

#### MP-16.16

**Safety Results of the Enzalutamide Expanded Access Program (EAP) in the U.S. and Canada for Patients with Metastatic Castration-Resistant Prostate Cancer (mCRPC) Previously Treated with Docetaxel**  
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**Introduction and Objectives:** In the Phase 3 AFFIRM trial, the androgen receptor inhibitor enzalutamide improved overall survival versus placebo in patients with mCRPC who had received prior docetaxel. This open-label single-arm study monitored safety and provided access to enzalutamide for mCRPC patients pending marketing approval in the U.S. and Canada.

**Materials and Methods:** A total of 507 patients with mCRPC were treated at 54 sites. Patients received enzalutamide 160 mg/d until disease progression, intolerable adverse event (AE), or commercial availability. AEs and other safety measures were assessed on day 1, week 4 and 12, and every 12 weeks thereafter. No efficacy data were collected.

**Results:** Median age was 71 years (range 43-97). Baseline ECOG PS was 0, 1 and 2 in 27.8%, 56.1%, and 15.9% of patients, respectively. Prior prostate cancer treatments included abiraterone (76.0%), and cabazitaxel (28.6%); 24.8% of patients received both prior abiraterone and cabazitaxel. Median enzalutamide treatment duration was 2.6 months (range 0.03-9.07); data following transition to

commercial drug was not collected. Reasons for discontinuation ( $\geq 5\%$ ) were transition to commercial enzalutamide (44.7%), progressive disease (33.7%), AEs (8.1%), and withdrawal by patient (5.9%). Common AEs ( $\geq 10\%$ ) included fatigue (39.1%), nausea (22.7%), anorexia (14.8%), anemia (11.8%), peripheral edema (11.4%), back pain (10.3%), vomiting (10.3%), and arthralgia (10.1%). Serious AEs in  $\geq 1\%$  of patients were disease progression (7.9%), pneumonia (2.0%), asthenia (1.8%), anemia (1.6%), and back pain (1.4%). Drug-related AEs leading to discontinuation occurred in 3.7% of patients. Grade  $\geq 3$  drug-related AEs were reported in 14.2% of patients, and drug-related AEs leading to death occurred in 4 patients: 1 cerebrovascular accident, 2 myocardial infarctions and 1 death not otherwise specified. Seizure was reported in 4 (0.8%) patients, of whom 3 had brain metastases.

**Conclusion:** In this EAP population with progressive mCRPC previously treated with docetaxel, enzalutamide was well tolerated. The safety profile was consistent with AFFIRM. Median treatment duration was shorter than

in AFFIRM (2.6 vs. 8.3 months); duration on subsequent commercial drug was not collected.

**MP-16.17**

**Radical Cystectomy following Radiation Therapy for Prostate Cancer: The Likelihood of Residual Prostate Cancer**

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**Introduction and Objectives:** To evaluate the likelihood of residual prostate cancer following radiation therapy in prostate cancer (PCa) patients who later underwent radical cysto-prostatectomy for muscle invasive bladder cancer.

**Materials and Methods:** We retrospectively reviewed data from patients who received radiation therapy for prostate cancer prior to undergoing radical cystectomy for muscle invasive bladder cancer. Biochemical recurrence following radiation therapy was defined as prostate specific antigen (PSA) of 2.0 ng/ml above nadir.

**Results:** Fifty one patients were included in the study. The prostate was not identified in the pathology specimen of one patient. Five (10%) had remaining prostate cancer in the final pathology (Table 1). Three out of the five patients were considered biochemical recurrence free based on the PSA 2.0 ng/ml above nadir definition. Following radical cystectomy, 4 patients had a PSA of zero. In the remaining 45 patients, prostate cancer was not identified in the prostate. However, two had prostate cancer in the lymph nodes.

**Conclusion:** PSA of 2.0 ng/ml definition of biochemical recurrence might not detect some patients with Prostate Cancer remaining (recurrence or residual) following radiation therapy.

**MP-16.17, Table 1. Patients with PCa remaining after Radiation therapy.**

| Patient | Age | BCR after RT | PCa location          | GS  | PSA following RC |
|---------|-----|--------------|-----------------------|-----|------------------|
| 1       | 72  | Yes          | Right Seminal vesicle | 3+3 | 0                |
| 2       | 75  | No           | Prostate              | 3+3 | 0                |
| 3       | 66  | No           | Prostate              | 3+4 | 0                |
| 4       | 84  | Yes          | Prostate              | 4+4 | 2.1              |
| 5       | 83  | No           | Prostate              | 5+5 | 0                |

BCR; biochemical recurrence. RT; radiation therapy. PCa; prostate cancer. GS; Gleason sum. PSA; prostate specific antigen. RC; radical cystectomy

# Moderated Poster Session 17 Bladder Cancer: Various Topics Tuesday, October 14 1435-1600

## MP-17.01

### Overexpression of the Long Noncoding RNA LOC572558 Impairs *in vitro* Bladder Cancer Cell Proliferation by Regulating p53

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**Introduction and Objectives:** In previous study, we identified a collection of aberrantly expressed long noncoding RNAs (lncRNAs) in bladder cancer by microarray. In the present study, we aimed to explore the expression level and the functional role of LOC572558, one of the most deregulated lncRNAs, in bladder cancer.

**Materials and Methods:** We determined the expression of LOC572558 in 50 pairs of bladder cancer tissue samples and control, as well as in bladder cancer cell line T24 and 5637 by real-time polymerase chain reaction assay. We then defined the biological functions of LOC572558 by CCK-8 assay, flow cytometry, wound healing assay and transwell assay. Using a high-throughput phospho-proteome array, we identified proteins that were phosphorylated and dephosphorylated in bladder cancer cells where LOC572558 expression was upmodulated by shRNA. We confirmed the findings from the array by Western blotting.

**Results:** We demonstrated that LOC572558 expression was markedly decreased in bladder cancer tissues and bladder cancer cell lines. Moreover, ectopic expression of LOC572558

inhibited cell proliferation, induced a S phase arrest in cell cycle and promoted cell apoptosis in T24 and 5637 bladder cancer cell lines. We further verified that over-expression of LOC572558 was associated with dephosphorylation of MDM2 and phosphorylation of p53 protein.

**Conclusion:** These data suggest an important role of LOC572558 in the molecular etiology of bladder cancer and implicate the potential application of LOC572558 in bladder cancer therapy.

## MP-17.02

### High-Throughput Screening of Chemical Compounds in Two Bladder Cancer Cell Lines at Opposite Ends of the Epithelial to Mesenchymal Spectrum

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**Introduction and Objectives:** Epithelial to mesenchymal transition (EMT) may explain bladder cancer progression. We applied high-throughput screening (HTS) of oncology compounds to bladder cancer cell lines at opposite ends of the EMT spectrum to identify novel therapies for these disparate phenotypes.

**Materials and Methods:** We screened RT4 (epithelial) and UMUC-3 (mesenchymal) bladder cancer cell lines against 1,912 oncology-focused drugs using a 48 hr cell proliferation assay with an ATP-based readout (CellTiterGlo), and determined the activity and potency of the

compounds in a dose response manner.

**Results:** A total of 250 compounds inhibited cell proliferation by >70% for both cell lines (Figure 1A) and included many proteasome and topoisomerase inhibitors. There were approximately 90 compounds that produced full dose responses and were more potent for each RT4 or UMUC3 by at least 3-fold change in IC<sub>50</sub>s. Compounds that were potent and more cytotoxic for RT4 included many tubulin modulators and HDAC and HSP90 inhibitors (Figure 1B). Compounds that were potent and more cytotoxic for UMUC3 included many PI3K and MTOR inhibitors (Figure 1C).

**Conclusion:** HTS of a library of oncology compounds is an efficient way to produce a list of novel targets and therapies for cancer. We have identified compounds selective to epithelial or mesenchymal phenotypes in bladder cancer with this approach.

## MP-17.03

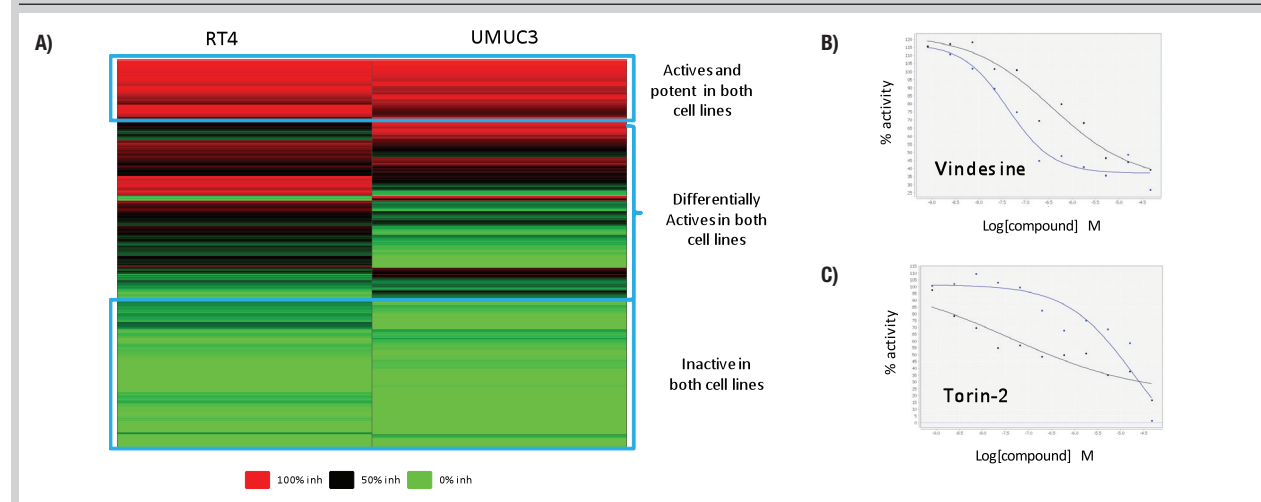
### The Expressions of CK20, p53, c-erb B2, and Ki-67 Were Correlated with the Prognostic Factors of Bladder Urothelial Carcinoma

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South Korea

**Introduction and Objectives:** Among various tumor markers, CK20, p53, c-erb B2, and Ki-67 are most commonly expressed markers and have known as related with tumor prognosis. In this study, we try to evaluate the association between tumor prognostic factors and expression of CK20, p53, c-erb B2, and Ki-67 in bladder urothelial carcinoma (Bca).

**Materials and Methods:** We analyzed 497 patients who was diagnosed as Bca after transurethral resection of bladder tumor for 5 years. The expression of tumor markers of Bca was

**MP-17.02, Figure 1. A) Hierarchical clustering of % inhibition at the maximum concentration tested (40  $\mu$ M) for each bladder cancer cell screened. B) Example of dose responses for a compound differentially cytotoxic for epithelial RT4 cell. C) Example of dose responses for a compound differentially cytotoxic for mesenchymal UMUC3 cell.**





detected by immunohistochemical staining. The association between expression of tumor markers and gender, age, number of tumor, tumor grade, tumor staging, histologic type, distant metastasis and recurrence was analyzed by Chi-square method. And the Spearman correlation was calculated between the expression of CK20, p53, c-erb B2, and Ki-67 in BCa tissues.

**Results:** Men were 389 and women were 108, and median age was 67.2. Recurrent patients were 211. Pathologic findings demonstrated that the expression of c-erb B2 and Ki-67 was associated significantly with tumor invasiveness ( $p=0.003$ ,  $p<0.001$ ) and the expression of p-53 was associated significantly with tumor grade ( $p=0.016$ ). The expression of CK20 and c-erb B2 was associated significantly with pathologic staging ( $p<0.001$ ,  $p=0.001$ ), and the expression of CK20 was associated significantly with distant metastasis ( $p=0.007$ ). The Spearman analysis indicated that the expression of CK20 has a significant positive correlation with that of c-erb B2 ( $p<0.001$ ) and the expression of c-erb B2 has a significant positive correlation with that of Ki-67 ( $p=0.010$ ).

**Conclusions:** The results suggest that the expression of CK20, p53, c-erb B2 and Ki-67 may have benefits the prediction of BCa prognosis.

#### MP-17.04

##### Deterministic Study of the Origin of Bladder Tumors Using Whole Genome Sequencing

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**Introduction and Objectives:** Two competing

theories have been proposed to explain the emergence of bladder tumors. The clonal hypothesis suggests that multifocality arises from a single founder tumor that undergoes intraluminal spread. Contrasting this, the "field effect" hypothesis suggests that these tumors arise from exposure to carcinogens that cause multiple cells to be partially transformed. To better understand this clinically important question we aimed to characterize the molecular evolution of bladder tumors.

**Materials and Methods:** Several samples from a single patient with multifocal bladder tumor were taken during the initial transurethral resection. We sampled the base and apex of one tumor focus, the base of a second tumour focus and the normal mucosa. Samples were sequenced by Illumina next-generation sequencing following exomic enrichment (pairwise, 100x coverage). The resulting data (2.14x108 total bp) was aligned to a reference genome to identify any potential mutations. Approximately 90% of the reads mapped to the reference sequence with PCR duplication rate of ~2%. With strict filtering a total 2,094 single nucleotide variants (SNV) were identified.

**Results:** Approximately half of the SNV's ( $n=934$ ) were shared among all 4 samples, including the normal sample, strongly suggesting that these represent the individual's unique genotype. Of the remaining mutations the vast majority ( $n=926$ ) were shared among all 3 tumor samples, but not the normal mucosa. Only ~60 SNVs were found to be private mutations in the tumor samples. Bootstrap analysis indicated that the tumor samples are significantly closer to each other than either are to the normal sample ( $p<0.001$ ) strongly supporting a single origin for the separate tumors. In addition, our data demonstrated a strong bias towards TpC\* mutations (where a C, preceded by a T, is mutated). Such a mutational pattern is

believed to occur due to over activity of APO-BEC RNA/DNA editing enzymes.

**Conclusions:** From these results we hypothesize that a period of early APOBEC overactivity may have lead to an increase in TpC mutations within the ancestral tumor cell. Once tumor formation commenced, mutation rates rose overall, thereby allowing secondary functional mutations to accumulate.

#### MP-17.05

##### A Prospective Study to Assess the Patterns of Recurrences in Radical Cystectomy Patients to Guide Radiological Surveillance

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**Introduction and Objectives:** Follow-up radiological imaging following radical cystectomy assesses local, systemic and upper tract recurrence and functional outcomes. We prospectively assessed the patterns of recurrences to direct better use of radiological imaging in post cystectomy surveillance.

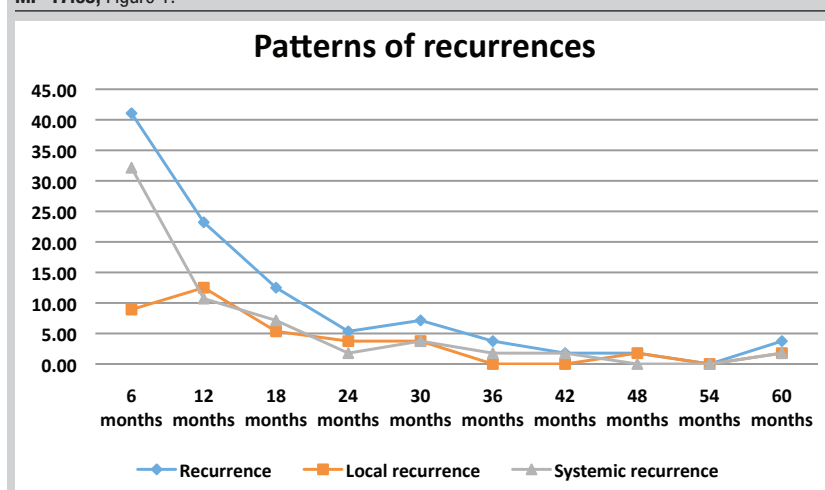
**Materials and Methods:** The data was prospectively collected from January 2006 to December 2013 on 184 patients from our database. Median follow-up was 48 months (1-96). CT chest abdomen and pelvis was done at 6 monthly intervals till 5 years.

**Results:** Recurrences developed in 56 (30.43%) patients on follow-up; of these, local pelvic and nodal recurrence occurred in 21 (11.4%) and 35 (19.02%) had systemic recurrences. Median time to local recurrence was 12 months (1 – 58) and 7 months (1 – 56) for systemic recurrence. The cancer specific survival was 78.8% at 8 years. Majority of recurrences (82.14%) occurred within 2 years and were associated with T3/T4/N+ disease. Upper tract recurrence (0.54%) was rare; obstructive uropathy developed in 16 patients (8.70%) at a median time

MP-17.05, Table 1. Pathological Stage and Recurrence Patterns

| Pathological stage and recurrence patterns |                     |                |                      |                                          |                                  |                                             |
|--------------------------------------------|---------------------|----------------|----------------------|------------------------------------------|----------------------------------|---------------------------------------------|
| Stage                                      | No. of patients (%) | Recurrence (%) | Local recurrence (%) | Median time to local recurrence (months) | Systemic recurrence (%) (months) | Median time to systemic recurrence (months) |
| pT0                                        | 35 (19.02%)         | 2 (5.71%)      | 1 (2.85%)            | 8                                        | 1 (2.85%)                        | 56                                          |
| pT1                                        | 25 (13.58%)         | 3 (12%)        | 1 (4%)               | 24                                       | 2 (8%)                           | 24 (17 – 36)                                |
| pTis                                       | 21 (11.41%)         | 2 (9.52%)      | 1 (4.76%)            | 27                                       | 1 (4.76%)                        | 25                                          |
| pT2                                        | 19 (10.32%)         | 5 (26.31%)     | 3 (15.78%)           | 18 (18 – 22)                             | 2 (10.52%)                       | 22.5 (6 – 39)                               |
| pT3                                        | 29 (15.76%)         | 12 (41.37%)    | 6 (20.68%)           | 8 (5 – 12)                               | 6 (20.68%)                       | 8 (1 – 34)                                  |
| pT4                                        | 12 (6.52%)          | 6 (50%)        | 2 (16.66%)           | 5.5 (1 – 10)                             | 4 (33.33%)                       | 4 (3 – 5)                                   |
| pN+                                        | 43 (23.36%)         | 25 (58.13%)    | 6 (13.95%)           | 11 (4 – 48)                              | 19 (44.18%)                      | 5.5 (1 – 30)                                |
| pN1                                        | 16 (8.69%)          | 9 (56.25%)     | 4 (25%)              | 20.5 (8 – 48)                            | 5 (31.25%)                       | 12 (5 – 22)                                 |
| pN2                                        | 25 (13.58%)         | 16 (64%)       | 2 (8%)               | 4 (4 – 4)                                | 14 (56%)                         | 4 (1 – 30)                                  |
| Mar+                                       | 12 (6.52%)          | 8 (66.66%)     | 1 (8.33%)            | 1                                        | 7 (58.33%)                       | 3 (1 – 9)                                   |

MP-17.05, Figure 1.



of 18 months (6-60) mainly on the left side of which 11 (5.97%) required intervention.

**Conclusions:** Two risk categories were apparent from our data which can be used to tailor radiological follow-up. Patients with low risk (pT1, pTis and pT2) should have a CT scan at 18, 24 and 36 months. Patients at high risk (pT3, pT4 and N+) should have 6 monthly CT during the first 2 years and subsequently annually till 5 years.

#### MP-17.07

##### **Risk-Adapted Screening for Bladder Cancer: Results of Daily Routine Preventive Care of the Urological Health-Service Research Association IQUO, Germany**

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**Introduction and Objectives:** The open-access risk assessment tool RiskCheck Bladder Cancer<sup>®</sup> was used in daily work to check asymptomatic people for the possible presence of bladder cancer over a time interval from 2010 to 2013. This was performed as a preventive care initiative to clarify the risk situation of a urological population in respect to risk exposure and the rate of detectable bladder cancer cases.

**Materials and Methods:** For the scientific evaluation the tool RiskCheck Bladder Cancer<sup>®</sup> was implemented in a secure web-portal of the initiative for quality management in uro-oncology (IQUO) in Germany. Over a time interval of 3 years, 598 people were interviewed, 361 with screening intention and the rest as a historical bladder cancer cohort. The preventive power for risk-adapted screening was proven in the screening cohort. Descriptive analysis and classification tree analysis were performed, significance  $p < 0.05$ .

**Results:** For the screening population (361

cases) the assessment tool has an accuracy of 87.3% with a specificity of 95.2%, a sensitivity of 40.8%, a negative predictive value of 91.1%, a positive predictive value of 57.1%, a false positive rate of 4.8% and a false negative rate of 59.2%. In a classification tree analysis we found as main risk factors age, gender, type of smoker and more or less than 30 pack years of consumed smoking goods. Out of the screened population we have a detection rate of 5.5% and out of the risk population 57.1% for BC whereas 42.9% live under risk without tumor manifestation.

**Conclusions:** We can clearly demonstrate that preventive care in bladder cancer is effective and aim achieving using the Internet tool RiskCheck Bladder Cancer<sup>®</sup> in daily urological work. We have a strong association of detectable risk and presence of bladder cancer in an asymptomatic screening population with such a risk-adapted screening focus. In respect to the developed risk-adapted screening for lung cancer bladder cancer behave equally in relation to smoking. In consequence the tool has been adapted in screening for bladder cancer and lung cancer. It is available for everyone via Internet in 10 languages as an open-access tool and newly on iPad.

#### MP-17.08

##### **Low Grade Diagnosis in T1 Non-Muscle Invasive Bladder Cancer (NMIBC) Patients Reduces Number of Early Repeat Resections (reTUR)**

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**Introduction and Objectives:** Invasion of sub-epithelial connective tissue, that is, the lamina propria (T1 stage) is well recognized recurrence and progression risk factor. According to International Society of Uro-Pathologist diagnosis of

submucosa invasion (T1 stage) characterize aggressive cancer and should be followed by high grade diagnosis. Thus, the number of T1G1 cases should not exceed 5% of all NMIBC. In our series T1G1 is recognized in about 22% of patients. The aim of the study is to assess the recurrence rate and the influence of reTUR on recurrence rate in T1G1 NMIBC patients.

**Materials and Methods:** Data of 501 patients (pts) submitted to TURBT in 2005-2012 in our department due to primary bladder cancer were retrospectively analyzed. One hundred twenty five (25%) pts diagnosed with muscle invasive BC and 178 (35.5%) pts with follow-up period shorter than one year were excluded from further analysis. Median follow-up time was 24.5 (12-87) months.

**Results:** In T1G1 group 25 of 43 (56.8%) pts. had cancer recurrence. Mean (range) time to recurrence was 17.5 (3-68) months. Recurrence rate in reTUR T1G1 vs. T1G1 without reTUR were 42.8% vs. 61% respectively. In comparison the number of recurrence in TaG1 and T1G2 NMIBC were: 12 (33.3%) and 34 (44.7%). Mean (range) time to recurrence in TaG1 and T1G2 groups were: 17.8 (3-53) and 12.9 (3-62) months. Irrespective grade, recurrence rate was higher in T1 stage patients. Low grade diagnosis in T1 patients was one of the factors that influent lower number of reTUR procedures (15.9% in T1G1 vs. 44.7% in T1G2) performed in that group of patients. This may explain higher recurrence number in T1G1 group of patients.

**Conclusion:** Submucosa invasion is an important risk factor of recurrences. ReTUR should be considered in all T1 cases irrespective of pathological grade.

#### MP-17.09

##### **Endoscopic Gold Fiducial Marker Placement into the Bladder Wall to Optimize Radiotherapy Targeting for Bladder-Preserving Management of Muscle-Invasive Bladder Cancer: Clinical Outcomes in Our Series of 27 Patients**

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<sup>1</sup>University of California, San Francisco, USA;

<sup>2</sup>University of Minnesota, Minneapolis, USA

**Introduction and Objectives:** Bladder-preserving chemo-radiotherapy is a management option for carefully selected patients with muscle-invasive bladder cancer. However, the inability to visualize the tumor site during treatment and normal bladder movement limits targeting accuracy and increases collateral radiation. We describe our early experience with a bladder fiducial marker device and protocol we developed.

**Materials and Methods:** Twenty-seven consecutive patients with muscle-invasive bladder cancer (T1-T4) elected bladder-preserving treatment

with maximal TUR, radiation and concurrent chemotherapy. All underwent endoscopic placement of 24-K gold fiducial markers modified with micro-tines (126 [2.9x0.9mm.]; 19 [2.1x0.7mm.]) into healthy submucosa 5-10mm. from the resection margin, using custom-made coaxial needles. Marker migration was assessed for with intra-op bladder-filling cystogram and measurement of distance between markers using a smartphone App (MedMeasure!) we developed. Set-up error and marker retention through completion of radiotherapy was confirmed by on-table portal imaging.

**Results:** Between 1/2007 and 8/2013, a total of 145 markers (3-5 per tumor-site) were placed into 27 patients of mean age 74-years. Two patients elected cystectomy before starting treatment; 19/25 completed chemo-radiotherapy. All (100%) markers were visible with all on-table (portal, cone-beam CT), fluoroscopy, plain film, and CT-scan imaging. In each of two patients, 1-of-4 markers placed at the tumor site fell-out (voided) during the second half of radiotherapy. All other markers (100/102, 98%) were present through the end of radiation therapy. No intra or post-op complications occurred. Use of micro-tined fiducial tumor-site markers afforded a 2 to 6-fold reduction in bladder area targeted with high-dose radiation. Clinical outcomes: (See Table 1.).

**Conclusions:** Placement of the micro-tined fiducial markers into the bladder is feasible and associated with excellent marker retention and

no evidence of marker migration or complications. Our markers improve targeting accuracy and help reduce collateral radiation. Fiducial markers may also afford tumor-site boost dose-escalation, to possibly further increase treatment efficacy.

#### MP-17.10

##### Is Embedding the Entire Prostate in Radical Cystectomy Specimens Essential?

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**Introduction and Objectives:** Radical cystectomy is the standard of care in the management of muscle invasive bladder cancer. There is a lack of standardisation of practise in the way the prostate is processed. We assessed the significant pathological findings in the whole mounts of prostate in radical cystectomy specimens at our institution.

**Materials and Methods:** The data was collected from Jan 2008 to December 2013 on 216 patients from our database. All cystectomies were processed with standard whole mount preparation of the prostate.

**Results:** Of the 216 cystectomies performed, there were 52 females and 2 males who underwent prostate-sparing procedures. Of the remaining 162 cystectomies, prostate cancer was reported in 83 (51.2%) patients; of these, 66 (79.5%) had organ confined disease and 17 (20.4%) had extraprostatic or margin positive disease. Twenty five (30.1%) patients had

Gleason 3+4 or higher grade disease. CIS was noted in 80 (49.3%) of the 162 patients. Of these, 19 (23.5%) had prostatic urethral CIS. Of these, additional stromal invasion of prostate was noted in 8 (10%) cases. The margin was reported as positive in 5 and as denuded in 5 further cases, implying margin positivity as these had associated urethral CIS. In 6 cases the prostate was directly infiltrated by TCC from the bladder (pT4 disease). Overall 108 significant pathologies were noted in the whole mount prostates including prostate cancer and prostatic urothelial neoplasia. Significant pathology needing further management included 17 patients with non-organ confined prostate cancer disease (PSA monitoring) and an additional 16 patients who needed to be considered for urethrectomy.

**Conclusions:** A total of 51.2% of patients had incidental prostate cancer on whole mount studies of the prostate, overall 10.55% of patients had non-organ confined cancer. Prostatic CIS is associated with bladder CIS in 23.5% of cases with margin positivity in 12%. As the entire prostatic urethra and the peripheral zone of the prostate need thorough histological examination, whole mount processing of prostate is a quicker and simpler method than random examination of small blocks. Whole mounts provide superior staging and prognostic information that is relevant for further management and follow-up of patients undergoing radical cystectomy.

MP-17.09, Table 1.

| Sex:                                                                                                           | Age (yrs):                                | Stage:                                          | # markers placed:                                                                 | Started Chemo/Rad:                                                                     | Completed Tx. & one or more f/u cystoscopies                                                                                    |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Men: 19<br>Women: 6<br><br>Total: 27                                                                           | 74.05<br>74                               | T1 (3)<br>T2 (20)<br>T3 (2)<br>T4 (2)           | 145                                                                               | 25 / 27<br><br>(2 opted for cystectomy before starting Tx)                             | 19 / 25<br><br>Mean f/u = 21.33 mo.<br><br>2 Delayed due to: ,<br>hip fx (1); pre-tx. mets (1)<br><br>4 currently receiving tx. |
| No local recurrence:                                                                                           | Local recurrence, managed conservatively: | Local recurrence, Salvage Cystectomy:           | Distant recurrence <i>only</i>                                                    |                                                                                        |                                                                                                                                 |
| 15 / 19 (79%)<br><br>Mean f/u = 20.17 mo.                                                                      | 3 / 19 (15.7%)<br><br>Mean f/u = 32.9 mo. | 1 / 19 (5%)<br><br>Diagnosis/cystectomy @ 4 mo. | 4 / 19 (21%)<br><br>Mean time to Met. Dz: 7.25 mo.                                | (2)<br><b>NED; Age 96:</b><br><br>1/19 (5%)<br>F/u = 17 mo.                            |                                                                                                                                 |
| <b>NED / Bladder Intact after All Tx:</b><br>17 / 19 (89.5%)<br><br>Mean f/u = 22.89 mo. (range: 7.3 - 57 mo.) |                                           |                                                 | Currently in chemo (1)<br>Cystectomy (1)<br>Lost to follow-up (1)<br>Deceased (1) | <b>Metastatic Dz:</b><br>1/19 (5%)<br>Time to recurrence: 5 mo.<br>Local disease: none |                                                                                                                                 |

**MP-17.11****Immunohistochemistry (IHC) to Enhance the Prognostic Allocation of Locally-Advanced and Metastatic Urothelial Cancer (UC) Undergoing First-Line Chemotherapy (CT)**

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<sup>1</sup>Fondazione IRCCS Istituto Nazionale dei Tumori, Milano, Italy; <sup>2</sup>UAB Comprehensive Cancer Center, Birmingham, USA

**Introduction and Objectives:** Knowledge of the expression of molecular drivers may enhance prognostic classification of metastatic UC. We aimed at assessing the expression of multiple molecular biomarkers (BMK) by IHC and their potential to enhance prognostic allocation of patients (pts) with UC.

**Materials and Methods:** We analyzed FFPE tumors from pts undergoing 1<sup>st</sup>-line CT with MVAC for locally-advanced unresectable (LA, T3-4±N+) and metastatic (M) disease between the years 2000 and 2013. The expression of the following panel of BMKs by IHC was evaluated using conventional protocols: ERCC1, EGFR, HER2/neu, VEGFR, PDGFR, p53, p63. Expression levels were dichotomized as positive (2+, 3+) or negative (≤1+). Fisher exact test was used to evaluate the association with response and setting (LA vs. M). Cox regression multivariate (MVA) models evaluated the association with PFS and OS.

**Results:** Since 06/2009, tissues of 88 pts (27 LA, 61 M) underwent IHC. Samples were from primary tumor (N=67) or metastases (N=21). Rates of positive IHC/number evaluable were as follows: ERCC1: 30/66 (45%); HER2: 24/52 (46%); EGFR: 31/54 (57%); VEGFR: 50/66 (76%); PDGFR: 10/63 (16%); p53: 25/56 (45%); p63: 46/53 (87%). HER2 trended for a significant association with higher stage (p=0.079). Median follow-up was 41 months (IQR, 15-64). On MVA, significant results were obtained for VEGFR and PDGFR, in addition to Bajorin score (Table 1). The c-index was 0.68 for both PFS and OS.

**Conclusion:** VEGFR and PDGFR expression appears to confer a divergent prognostic impact in pts receiving 1<sup>st</sup>-line cisplatin-based CT. These data underline the difficulties in defining angiogenesis as a molecular driver and therapeutic target.

**MP-17.12****MicroRNA-140-5p Inhibits Cell Migration and Invasion in Bladder Cancer Cells by Targeting SOX4**

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**Introduction and Objectives:** In recent years, the pathological relevance and significance of microRNAs (miRNAs) in bladder cancer have attracted more and more attention. Increasing evidence has suggested that dysregulation of certain miRNAs may contribute to carcinogenesis and metastasis in bladder cancer. In our previous study, we used miRNA microarray and found miR-140-5p was down-regulated in bladder cancer. Thus, this study was conducted to investigate the biological functions and its molecular mechanisms of miR-140-5p in human bladder cancer cell lines, discussing whether it has a potential to be a promising biomarker and therapeutic target of bladder cancer.

**Materials and Methods:** Four human bladder cancer cell lines and samples from sixty-five patients with bladder cancer were analyzed for the expression of miR-140-5p by quantitative RT-PCR. Exogenous overexpression of miR-140-5p was established by transfecting mimics into T24, EJ, 5637 and J82 cells, after that cell proliferation and cell cycle were assessed by CCK-8 assay, flow cytometry and Colony-forming assay. Cell motility and invasion ability were evaluated by wound healing assay and transwell assay. Tissue immunohistochemistry with antibodies against SOX4 was performed. The target gene of miR-140-5p was determined by luciferase assays, quantitative RT-PCR and western blot. The regulation of epithelial-to-mesenchymal transition by miR-140-5p was analyzed by western blot.

**Results:** miR-140-5p was down-regulated in bladder cancer both in four bladder cancer cell lines and clinical samples compared to normal bladder urothelium. The expression level of miR-140-5p was correlated with tumor stage, tumor grade and lymph node metastasis in bladder cancer. Overexpression of miR-140-5p induced G2/M-phase arrest and suppressed cell growth in CCK-8 and colony-forming assay. miR-140-5p significantly repressed the capability of migration and invasion of bladder cancer cells. In addition, SOX4 was identified as a new target of miR-140-5p and up-regulated in bladder cancer tissues. Furthermore, we demonstrated overexpression of miR-140-5p could inhibit bladder cancer cell epithelial-mesenchymal transition (EMT) phenotype *in vitro*.

**Conclusion:** miR-140-5p can repress the migration and invasion of bladder cancer cells via regulating SOX4. Our data indicate that miR-140-5p could be a tumor suppressor and may have a potential to be a promising biomarker in bladder cancer.

**MP-17.13****'FROG Bladder' – A New Type of Neobladder: The Functional and Metabolic Characteristics**

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**Introduction and Objectives:** There have been various modifications of surgical techniques of Neobladder. I report a new type of bladder, 'The FROG Bladder', which satisfies the principles of continent Urinary Diversion, such as reservoir of satisfactory capacity, low pressure, minimal PVR and minimal surface exposure to avoid metabolic disorders with sufficient length of the ileum for tension free anastomosis. The aim of this study was to develop a Neobladder, with minimal intra-operative and post-operative complications after Radical Cystectomy.

**Materials and Methods:** Thirty three patients (30 males and 3 females) in the age group 45-62 years (Mean 57.5 years.) who had undergone Radical Cystectomy between 1999 and 2010 are included in the present study and were offered 'FROG Bladder' - a type of Neobladder. The salient features of these bladders are- two segments of the ileum of 30 cm each are used and 5 cm of distal most part of the ileum is not detubularized and both the segments are anastomized in such a way that these two non-tubularized segments can be used for tension-free uretero-ileal anastomosis which is stented with 8 Fr RT. The patients were followed-up at 3, 6, 9, 12 months intervals in the form of USG abdomen, Urine Cytology, Serum B12, Uroflowmetry and UDS.

**Results:** The average age of the patients was 62 years intra-operatively, there was no problem in placing the Neobladder in the pelvis as well

**MP-17.11, Table 1.**

| Variable                   | PFS  |           |        | OS   |            |        |
|----------------------------|------|-----------|--------|------|------------|--------|
|                            | HR   | 95% CI    | p      | HR   | 95% CI     | p      |
| VEGFR<br>Pos vs. Neg       | 0.45 | 0.21-1.01 | 0.054  | 0.36 | 0.15-0.85  | 0.019  |
| PDGFR<br>Pos vs. Neg       | 3.32 | 1.28-8.58 | 0.013  | 2.66 | 0.96-7.42  | 0.060  |
| Bajorin score<br>1-2 vs. 0 | 4.24 | 1.94-9.26 | <0.001 | 4.49 | 1.91-10.56 | <0.001 |



as there was no fear of tension on suture line of Uretro-Intestinal anastomosis. Bladder capacity; at six month it was  $245 \pm 150$  ml while at 1 year it was  $290 \pm 220$  ml, PVR at six month was  $10 \pm 30$  (17) at 1 year,  $10 \pm 90$  (36), 24 hr Voiding frequency at 6 month was 11 (8-14) while at 1 year it was 7 (5-10). Day time continence was 76% while at 1 year it was 92%. Night time continence at 6 months was 56% while at 1 year it was 82%. Max flow rate was 16 ml/sec at six months while at 1 year it was 21 ml/sec.

**Conclusion:** The FROG bladder at 1-year follow-up achieved an excellent result with good capacity, minimal PVR and minimal back pressure changes in the upper tracts.

#### MP-17.14

##### **Belgrade Pouch-Orthotopic Neobladder Created from Shorter Ileal Segment**

Aleksic P, Bancevic V, Stamenkovic D, Jovanovic M, Tosevski P, Campara Z, Nikolic I, Milosevic R, Jovanovic D, Milovic N  
*Medical Faculty, University of Defense,*

*Military Medical Academy, Urology Clinic, Belgrade, Serbia*

**Introduction and Objectives:** Neobladder constructed of 40 to 60 cm of ileum provide excellent capacity and low pressure with satisfactory continence and frequency. During the time, increasing neobladder capacity and post voided residual volume lead to metabolic disorders, voiding problems, excessive mucus production, urinary infection, calculosis and other complications. In order to reduce these complications and to preserve satisfactory pouch capacity and continence we created Belgrade pouch- neobladder constructed of shorter ileal segment (average 29 cm).

**Materials and Methods:** In prospective clinical study we included ninety eight patients scheduled for radical cystectomy, and followed up for four years. Inclusion criteria for orthotopic derivatin were according to EAU guidelines for muscle invasive bladder carcinoma. Exclusion criteria were chemotherapy, disease progression, obstructive pulmonal disease and diabetes mellitus. "U" shaped pouch (BP) was

constructed using 24- 35 cm of detubularised terminal ileum. Direct ureteroileal anastomosis was performed. Urethro-pouch anastomosis was created with 6 (5-7) sutures. Pouch was flushed twice per day until hospital discharge. Patients were instructed to void every two to three hours during the daytime, and in the nighttime, to void twice for the first two or three months following the surgery. We recorded delayed complications following the surgery.

**Results:** Fifteen female and eighty three male patients 64 (42-77) years old underwent surgery. Neobladder was constructed using 29 (24-35) cm of terminal ileum. Delayed complications and pouch characteristics during the time are presented in Table 1.

**Conclusion:** Few months after surgery BP can obtain adequate pouch capacity and frequency, small post voided residual urine, satisfactory daytime and nighttime continence, lower incidence of acidosis, calculosis and vitamin B12 deficiency, and small percentage of bicarbonate substitution during four-year period of follow-up.

**MP-17.14, Table 1. Outcome Data Presented as Median (Range) and Number (Percentage)**

|                                | 6 months      | 1 year        | 2 years       | 3 years       | 4 years       |
|--------------------------------|---------------|---------------|---------------|---------------|---------------|
| Day continence (%)             | 72            | 89            | 92            | 89            | 89            |
| Night continence (%)           | 68            | 88            | 89            | 88            | 89            |
| Neobladder capacity (ml)       | 315 (220-378) | 429 (289-515) | 449 (300-585) | 472 (323-659) | 491 (335-680) |
| 24 hours voiding frequency     | 9             | 7             | 6             | 6             | 5             |
| Post-void residual volume (ml) | 15 (0-20)     | 22 (0-41)     | 28 (0-40)     | 50 (0-95)     | 55 (0-190)    |
| Acidosis (%)                   | 0             | 2             | 5             | 7             | 7             |
| Bicarbonate supstitution (%)   | 0             | 0             | 2             | 3             | 3             |
| Vitamin B12 deficiency (%)     | 0             | 0             | 0             | 3             | 4             |
| Pouch calculosis (%)           | 0             | 0             | 3             | 6             | 6             |
| Hydronephrosis (%)             |               |               |               |               |               |
| Bilateral                      | 8 /GrI/       | 3             | 3             | 3             | 4             |
| Onesided                       | 0             | 0             | 3 /GrIII/     | 6 /Gr IV/     | 7 /Gr IV/     |

## Moderated Poster Session 18

### Stones: Surgical Management and New Technologies

Tuesday, October 14  
1435-1600

#### MP-18.01

**Ureteroscopy with Holmium: YAG Laser Lithotripsy for Impacted Upper Ureteral Stones: A Comparative Study between Impacted Stones and Non-Impacted Stones**  
Omori Y, Shinbori M, Oiwa Y, Umeda K, Yokoyama H, Iwabuchi T, Kawano Y, Hariu K, Yamamoto R, Shiga Y  
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**Introduction and Objectives:** Ureteroscopy (URS) for impacted upper ureteral stones is difficult to perform retrogradely because of the severe tortuosity of ureter or ureteral edema and fibrosis. In some cases, it may be difficult to obtain stone clearance intraoperatively. The aim of this study is to evaluate safety and efficacy of URS for impacted upper ureteral stones (single procedure) compared to improving impaction with double-J stent insertion followed by URS.

**Materials and Methods:** Between June 2011 and February 2013, a total of 47 patients with upper ureteral stones (with all patients showing symptoms of ureteral obstruction at least temporarily) were divided into 2 groups. Group 1 included 28 patients who underwent URS for impacted upper ureteral stones (U1) and Group 2 included 19 patients who showed symptoms of pyelonephritis underwent double-J stent insertion for impacted upper ureteral stones followed by URS. Impacted stone is defined as the stone that has remained in the same position for at least two months with hydronephrosis. Postoperative KUB and ultrasonography were used to assess the stone clearance rate, which had residual fragments <2mm.

**Results:** Regarding patient age and stone size there was no significant difference between the two groups. The mean operative time was 64.1±17.3 and 56.5±17.5 min in Group 1 and 2, respectively (P=0.138). Two patients underwent a second session of URS (one in each group). The stone clearance rate at post-operative day one and at post-operative two months were 46.4% vs. 73.7% and 92.8% vs. 94.7% in Group 1 and 2, respectively (P=0.064, P=0.796). There were no significant intraoperative complications between two groups. Transient postoperative fever was encountered in two (7.1%) and three (15.8%) patients in Group 1 and 2, respectively (P=0.345). Three patients (10.7%) of Group 1 experienced steinstrasse.

**Conclusion:** Although it may be difficult to extract all stone fragments intraoperatively,

ureteroscopy for impacted upper ureteral stones is an optimal treatment which is recommended not only for fragmentation using Holmium:YAG laser but also in regard to its fewer complications. In our present study, we conclude that ureteroscopy for impacted upper ureteral stones is a safe and effective procedure.

#### MP-18.02

**Retrograde Intrarenal Surgery Is Superior to SWL in the Treatment of Renal Stones Up to 20mm**

Bosio A, Destefanis P, Alessandria E, Preto M, Santià S, Buffardi A, Palazzetti A, Dalmasso E, Bisconti A, Gontero P, Fontana D, Frea B  
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**Introduction and Objectives:** Retrograde intrarenal surgery (RIRS) has become a common procedure in the treatment of renal stones but SWL still remains the treatment of choice in most cases. The objective of our study was to compare success rates of RIRS and SWL on renal stones.

**Materials and Methods:** We organized a retrospective trial on all RIRS performed in our Stone Center from May 2009 to August 2013 and all SWL performed from January 2011 to August 2013. Single procedure outcomes were determined on ultrasound at 2 weeks. Procedures were considered successful in case of stone free or clinically insignificant residual fragments (CIRF) – defined up to 5 mm – and

unsuccessful in case of larger residual fragments or unmodified stone. Stones size and site and need for further procedures (RIRS or ESWL) were also considered. Student t and Chi-square tests were used for statistical analysis.

**Results:** Results are summarized in Table 1.

**Conclusion:** In our retrospective trial RIRS seemed to be superior to SWL in the treatment of renal stones. Success and stone free rates after RIRS were significantly higher compared to SWL and the need for further treatments significantly lower. Stones located in lower calices and in renal pelvis specially seemed to benefit from RIRS.

#### MP-18.03

**Effect of Obesity on the Success of Retrograde Intrarenal Surgery**

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**Introduction and Objectives:** While retrograde intrarenal surgery (RIRS) has been improving and getting a common procedure in urology practice, many articles evaluate the factors that can affect the success rates of RIRS. But there are few articles evaluating the association of obesity and RIRS. We hypothesized that there could be restriction of mobility in patients with high body mass index (BMI) so it could

MP-18.02, Table 1.

|                                        | RIRS          | SWL           |           |
|----------------------------------------|---------------|---------------|-----------|
| Analysed medical records               | 184           | 372           |           |
| Available post-treatment ultrasound    | 147           | 293           |           |
| Mean stone size                        | 11.9 ± 4.0 mm | 10.1 ± 3.2 mm | p < 0.001 |
| Single procedure:                      |               |               |           |
| Success rate                           | 65%           | 37%           | p < 0.001 |
| Stone size:                            |               |               |           |
| ≤ 10 mm                                | 71%           | 44%           | p < 0.001 |
| 11 - 15 mm                             | 60%           | 18%           | p < 0.001 |
| 16 - 20 mm                             | 45%           | 7%            | p = 0.02  |
| Stone site:                            |               |               |           |
| Upper calices                          | 64%           | 37%           | p = 0.09  |
| Middle calices                         | 56%           | 43%           | p = 0.44  |
| Lower calices                          | 67%           | 48%           | p = 0.03  |
| Renal pelvis                           | 52%           | 23%           | p = 0.02  |
| After a previous SWL                   | 62%           | 39%           | p < 0.001 |
| Stone free rate                        | 34%           | 14%           | p = 0.03  |
| CIRF rate                              | 42%           | 23%           | p < 0.001 |
| Unmodified stone rate                  | 4%            | 35%           | p < 0.001 |
| Further procedure (RIRS or SWL) needed | 23%           | 59%           | p < 0.001 |

be associated with the spontaneous passage of stone fragments. The aim of this study was to evaluate whether obesity has an effect on the final success of RIRS due to association of the restriction on mobilisation.

**Materials and Methods:** We retrospectively reviewed the medical records of patients who underwent RIRS due to kidney stones. Two hundred and twenty-two patients whose BMI values were obtained, were included to this study. According to BMI values, patients were divided into three groups. Group I (n:67): patients whose BMI was <25; Group II (n:91): patients whose BMI was between 25 and 29; Group III (n:64): patients whose BMI was >29. We determined the success rates, age, gender, stone size and location, operation time and postoperative renal colic. All patients were evaluated for final success 1 month after the operation. Success was defined as stone-free or fragment smaller than 3 mm.

**Results:** Mean ages were 34.6, 43.1 and 48.5 years, respectively. ( $p < 0.05$ ). Male/female ratio was 0.7, 0.4, 1.4, respectively. ( $p = 0.004$ ). Mean stone size was 175.8, 180.3 and 195.9 mm<sup>2</sup> respectively ( $p > 0.05$ ). Stone location, operation and scopy times, postoperative hospitalisation times and the renal colic rates were similar in all groups ( $p > 0.05$ ). The final success rates were 70.1%, 79.1% and 73.7% in Group I, II and III respectively, the difference was not statistically significant ( $p > 0.05$ ).

**Conclusions:** With this study we determined that patients in Group III were older than others. Although increased age has accompanied in Group III, BMI did not affect the spontaneous passage of stone fragments and the success of the operation.

#### MP-18.04

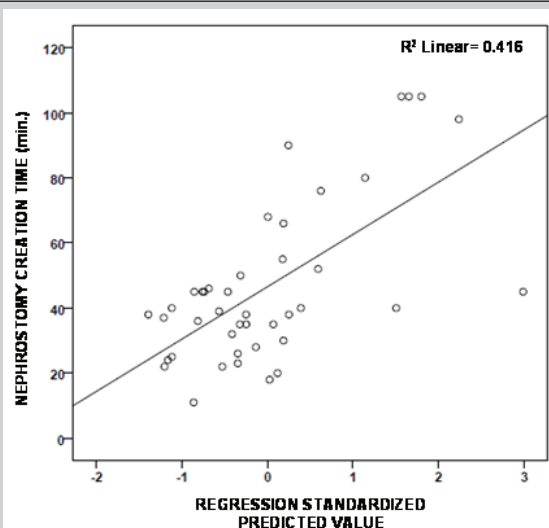
##### Body Mass Index (BMI) Predicts Outcome of Ureteroscopy-Assisted Retrograde Nephrostomy (UARN) for Percutaneous Nephrolithotomy

Atalla C, Wynberg J  
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**Introduction and Objectives:** Several clinical series of retrograde nephrostomy for PCNL have been published over the past 30 years demonstrating good outcomes and safety. We previously reported our adaptation of the Lawson technique, wherein we deploy the puncture wire through a flexible ureteroscope. We herein aim to clarify the performance characteristics of this nephrostomy creation technique.

**Materials and Methods:** IRB approval and informed consent were obtained. UARN procedure was performed as described previously. Data was collected prospectively. Multiple patient and operative factors were evaluated for association with UARN success and nephrostomy creation time: BMI, skin-to-stone distance (SSD), Guy's score, CROES nephrolithometric

**MP-18.04, Figure 1.** Linear regression shows correlation between BMI ( $r^2 = 0.219$ ), stone burden ( $r^2 = 0.094$ ) and use of holmium laser to access calyx ( $r^2 = 0.104$ ; total  $r^2$  linear = 0.416) and nephrostomy creation time using the UARN technique.



score, hydronephrosis, stone burden, location of nephrostomy, exit from a stone-bearing calyx, and use of holmium laser to access calyx.

**Results:** Nephrostomy was successful in 49/52 UARN procedures (94%). Only single access was placed: upper - 18, mid - 27, and lower - 7. Median BMI was 29 kg/m<sup>2</sup> and median time for nephrostomy creation was 39 minutes. Fluoroscopy time for entire PCNL was 84 and 16 seconds for cases # 1-25 and 26-52, respectively. By stepwise linear regression, variables correlating with nephrostomy creation time were BMI ( $r^2 = 0.219$ ), stone burden ( $r^2 = 0.094$ ), and use of holmium laser to access calyx ( $r^2 = 0.104$ ), total  $r^2$  linear = 0.416.

**Conclusions:** UARN is an intuitive, safe procedure that offers dramatic reductions in fluoroscopy times. UARN is best suited to cases requiring only one nephrostomy tract. Upper pole access is commonly performed with a sub-costal technique to navigate the puncture wire below the ribs. Increasing BMI best predicts longer nephrostomy creation times; procedure failure was associated with BMI exceeding 40 kg/m<sup>2</sup>. UARN is a robust technique for nephrostomy creation in appropriately selected patients.

#### MP-18.05

##### Assessment of Ureteric Injury Produced by Access Sheath Use in Flexible Ureteroscopy

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**Introduction and Objectives:** The use of access sheath in flexible ureteroscopy improves flow, endoscopic views and allows easier extraction of fragments. However it carries risks of ureteric injury and stricture. The aim of this study is to identify the prevalence and predictive factors of

ureteric injury with access sheath use.

**Materials and Methods:** Flexible ureteroscopy data was collected prospectively between August 2012 and December 2013. The ureter was inspected at the end of procedure and ureteric injury was graded 0-3 depending on the depth of injury as described by Traxer et al. All patients had ultrasound scan at 3 months for assessment of evidence of hydronephrosis.

**Results:** During the study period 69 patients had flexible ureteroscopy with use of access sheath. Stone locations were in the upper/mid pole in 9 (13%); lower pole in 26 (38%); renal pelvis in 21 (30%); and upper ureter in 13 (19%) patients. The ureter was pre-stented in 30 (44%) cases. Tight ureter was noted in 7 (10%) patients on insertion of access sheath. Mean operative time was 46±22 minutes and mean access sheath time 30±19 minutes. Stones were fragmented in 64 (93%) and extracted in 5 (7%) cases. On inspection of the ureter at the end of procedure the ureter was intact in 29 (42%), grade 1 injury was noted in 36 (52%), grade 2 in 4 (6%) cases and no grade 3 injury was observed. Access sheath time was 35±25 minutes in cases with intact ureter and 27±15 minutes in those with any degree of ureteric injury ( $p = 0.151$ ). No significant hydronephrosis was noted on follow-up ultrasound scans.

**Conclusion:** A minor degree of ureteric injury is noted with access sheath use in more than half the patients. This does not appear to be related to length of access sheath time. The risk of significant stricture causing hydronephrosis was very low in this study.

**MP-18.06****A 20-30mm Single Renal Stone: Percutaneous Nephrolithotomy or Extracorporeal Shockwaves Lithotripsy?**

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**Introduction and Objectives:** Optimal management of a 20-30mm single renal stone is controversial. We compared SWL and PNL efficacy and safety in treatment of such patients.

**Materials and Methods:** Between January 2006 and December 2012, Files of patients treated with SWL or PNL for a 20-30mm single renal stone were reviewed. Exclusion criteria were; history of intervention for this stone within the last 6 months, solitary kidney, anatomical renal abnormality, severe Hydronephrosis or branched stones. We compared patients' criteria (age, sex and BMI), stone criteria (side, location, opacity, largest diameter, and attenuation value), need for retreatment, auxiliary, complications and costs. Success was defined as no residual stones or insignificant residual (4mm or less) 3 months after last SWL session or PNL. Statistical analysis was conducted using t-test for continuous variables and chi-square test for categorical variables.

**Results:** Study included 420 patients, 215 were treated by SWL and 205 were treated by PNL. Table 1 summarizes patients and stones characters. Success rates, need for re-treatment or auxiliary procedures, complication rates and costs are demonstrated in Table 2. PNL showed significantly higher success rate and lower retreatment rate and auxiliary procedures than SWL. The incidence of complications was comparable, however costs were significantly lower in SWL. The efficiency quotient for SWL was 0.39.

**Conclusion:** PNL was more effective than SWL in treatment of 20-30mm single renal stone. However, SWL was less costly but associated with higher re-treatment rate.

**MP-18.07****Comparative Study of Balloon Dilator versus Telescopic Metal Dilators for Tract Dilatation in PCNL for Staghorn Stones and Calyceal Stones**

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**Introduction and Objectives:** To compare the results of balloon versus telescopic metal dilators in cases with calyceal stones or staghorn stones without hydronephrosis.

**Materials and Methods:** This study was prospectively designed over 3 years from 2011 to 2014. All cases were performed in Farwaniya

Hospital, Kuwait. Patients with anticipated difficulty in dilation (Complex staghorn stones, non dilated targeted calyx, stone fully casting the targeted calyx and anterior targeted calyx suggested from CT and identified by failure of guide wire to reach to renal pelvis after puncture) were included in the study and randomized between 2 groups. Intervention: First group

(A) used balloon as a primary dilation technique and metal telescopic dilators as a salvage. In the second group (B) we used telescopic metal dilators as a primary dilatation technique.

**Results:** We identified 97 patients out of 235 cases of PCNL: Group (A), 55 patients underwent balloon dilation as a primary dilatation technique, and Group (B), 42 patients

**MP-18.06, Table 1. Patients and Stone Characters in Both Groups**

| Patients and stone characters in both groups |                             |                    |                    |         |
|----------------------------------------------|-----------------------------|--------------------|--------------------|---------|
|                                              | N. (%)                      | SWL<br>215 (51.2%) | PNL<br>205 (48.8%) | P value |
| <b>Patients characters</b>                   |                             |                    |                    |         |
|                                              | Age (years) Mean ± SD       | 48 ± 11.9          | 50.7 ± 12.2        | 0.02    |
|                                              | Sex                         |                    |                    |         |
|                                              | Male                        | 142 (66%)          | 106 (51.7%)        | 0.003   |
|                                              | Female                      | 73 (34%)           | 99 (48.3%)         |         |
|                                              | BMI ± SD                    | 31.3 ± 5           | 32 ± 7             | 0.367   |
| <b>Stone character</b>                       |                             |                    |                    |         |
|                                              | Side                        |                    |                    | 0.449   |
|                                              | Right                       | 98 (45.6%)         | 101 (49.3%)        |         |
|                                              | Left                        | 117 (54.4%)        | 104 (50.7%)        |         |
|                                              | Location                    |                    |                    | 0.216   |
|                                              | Renal pelvis                | 168 (78.1%)        | 170 (82.9%)        |         |
|                                              | Calyceal                    | 47 (21.9%)         | 35 (17.1%)         |         |
|                                              | Largest diameter (mm ± SD)  | 23.4 ± 2.7         | 24.9 ± 3           | < 0.001 |
|                                              | Opacity                     |                    |                    | 0.4     |
|                                              | Radiopaque                  | 176 (81.9%)        | 174 (84.9%)        |         |
|                                              | Radiolucent                 | 39 (18.1%)         | 31 (15.1%)         |         |
|                                              | Attenuation value (HU ± SD) | 816.3 ± 350.2      | 748.4 ± 352.9      | 0.165   |
| <b>Hydronephrosis</b>                        |                             |                    |                    |         |
|                                              | Yes                         | 136 (63.3%)        | 145 (70.7%)        | 0.104   |
|                                              | No                          | 79 (36.7%)         | 60 (29.3%)         |         |

**MP-18.06, Table 2. Efficacy, Safety and Costs in Both Groups**

| Efficacy, safety and costs in both groups |                     |                    |                    |         |
|-------------------------------------------|---------------------|--------------------|--------------------|---------|
|                                           | N. (%)              | SWL<br>215 (51.2%) | PNL<br>205 (48.8%) | P value |
| Efficacy                                  |                     |                    |                    |         |
|                                           | Success rate        | 74%                | 95.6%              | < 0.001 |
|                                           | Re-treatment rate   | 163 (75.8%)        | 11 (5.4%)          | < 0.001 |
|                                           | Auxiliary procedure | 30 (14%)           | 11 (5.4%)          | < 0.001 |
| Safety                                    |                     |                    |                    |         |
|                                           | Complications       |                    |                    | 0.815   |
|                                           | G 0                 | 188 (87.4%)        | 174 (85%)          |         |
|                                           | G I                 | 12 (5.6%)          | 13 (6.2%)          |         |
|                                           | G II                | 6 (2.8%)           | 9 (4.4%)           |         |
|                                           | G IIIa              | 9 (4.2%)           | 9 (4.4%)           |         |
| Cost                                      |                     |                    |                    |         |
|                                           | \$U.S ± SD          | 835 ± 513          | 1187 ± 137         | < 0.001 |



underwent telescopic metal dilatation. In Group A (55 cases) dilation was successful in 28 cases, overzealous dilation with parenchymal injury in 6 patients, failure and need for metal dilators in 21 cases and failure of procedure was in 1 case after failed telescopic dilatation and extravasation. In Group B (42 cases) dilatation was successful in 33 cases, overzealous dilation was in 5 cases, under dilation with re-dilatation was in 4 cases and no cases had failed dilatation. Differences in mean duration of dilatation, estimated mean of blood loss and blood transfusion rate were statistically insignificant. **Conclusion:** Balloon dilatation has a higher failure rate than metal telescopic dilatation during PCNL for calyceal casting stones or stag-horn stones with little space around the stones.

#### MP-18.08

##### **Supracostal Calyceal Access by Caudal Renal Displacement during Percutaneous Nephrolithotomy (PCNL): A Novel Ureteric Balloon Catheter (UBC) Technique**

**Khan R, Ramsay A, Vint R, Nalagatla S**  
*Monklands District General Hospital, Airdrie, Scotland, UK*

**Introduction and Objectives:** Percutaneous access through the supracostal collecting system during percutaneous nephrolithotomy (PCNL) is often avoided due to the risk of pleural injury. We have developed a novel technique using a ureteric balloon catheter (UBC) which allows us to bring a supracostal collecting system to an infracostal position for percutaneous access. We describe this technique which, to our knowledge, is the only such description using ureteric balloon catheter (UBC) in literature. We also present the up-to-date results for our growing cohort of patients treated using this technique in our centre.

**Materials and Methods:** The technique involves placing patient in a prone or supine position to perform PCNL. Ureteric balloon catheter (UBC) is then placed into the renal pelvis or desired calyx, and a retrograde pyelogram is performed prior to balloon inflation. Gentle traction is then applied to the UBC to displace the kidney caudally. This allows a desired calyx to be accessed in the infracostal position under fluoroscopy guidance. We prospectively collected data for all the patients treated using this technique in our centre. The data was reviewed to assess the success of the technique.

**Results:** To date, UBC technique has been used in a total of 10 cases in our centre. We successfully achieved access in 9 cases (2 supine PCNL and 7 prone PCNL). Supracostal calyces (4 upper pole, 2 interpolar and 3 lower pole) were accessed using UBC technique. This technique was unsuccessful in 1 case. No significant complications were encountered and complete clearance achieved in all cases where access was achieved. Additionally, this technique stabilises

the kidney and reduces movement due to respiration which is advantageous during PCNL.

**Conclusion:** UBC technique is a novel and simple technique to displace supracostal calyces below the 12<sup>th</sup> rib to gain access safely and avoid intra-thoracic complications.

#### MP-18.09

##### **Pulsed Fluoroscopy: Bringing Safety into PCNL Efficacy**

**Durutovic O, Nikic P, Milenkovic-Petronic D, Santric V, Milojevic B, Ladjevic N, Mimic A, Vuksanovic A, Acimovic M, Dzamic Z**  
*Clinic of Urology, Belgrade, Serbia*

**Introduction and Objectives:** Percutaneous nephrolithotripsy (PCNL) presents a gold standard in treatment of large kidney stones. The technique of access to the renal collecting system and appropriate calyx differs between centers. In the beginning access was mainly performed by radiologist. Today access is mostly performed by urologist, by fluoroscopic or/and ultrasound guidance. One of the goals of modern urology in stone disease is how to decrease the level of radiation exposure of patients (because of repeated exposures), but also personnel involved in fluoroscopic procedures. In aim to decrease radiation exposure, techniques of access are improved toward the increased precision and efficacy with decreased radiation exposure.

**Materials and Methods:** In this study we have compared our data in two stages/groups, but also with data from the literature. We have compared the results when access and procedure control was performed with ultrasound and continuous fluoroscopy (Group 1) to the second stage, when ultrasound and pulsed fluoroscopy was used (Group 2). We have compared fluoroscopic time (FT), number of attempts necessary to approach desired calyx and stone free rate between two groups.

**Results:** We found significant difference when compared FT in Group 1 to Group 2 (2.6 min. vs. 1.7 min.). Compared to the available data from the literature, investigating FT achieved at the level of excellence, FT in our study was significantly lower (2.6/1.7 min vs. 8.9 min). The number of attempts necessary to achieved desired calyx did not differ between Group 1 and Group 2. Also, in terms of efficacy, the stone free rate did not differ significantly between Group 1 and Group 2.

**Conclusion:** Fluoroscopic time can be safely reduced with use of ultrasound for precise access to the appropriate calyx. Further improvement can be achieved by pulsed fluoroscopy, as it will not decrease the control and efficacy of the procedure, but will significantly decrease fluoroscopic time. In this manner PCNL will remain its efficacy with increased safety for not only patient but also operating room personnel.

#### MP-18.10

##### **Implementation of Ultra-Mini Percutaneous Nephrolithotomy for Treatment of 2 - 3cm Kidney Stone: A Preliminary Report**

**Shah A, Xu K, Huang J, Bi L, Liu H, Huang H, Shrestha R**

*Dept. of Urology, Sun Yat-sen Memorial Hospital, Sun Yat-sen University, Guangzhou, China*

**Introduction and Objectives:** Percutaneous nephrolithotomy is still a preferred technique for treatment of large (>2cm) renal stone. Recently, retrograde intrarenal surgery (RIRS) and miniature percutaneous nephrolithotomy techniques are increasingly used in the treatment of renal stones. However, they are indicated for the treatment of renal stones (<2cm). We present our preliminary report of treating 2-3 cm renal stone using ultra-mini nephrolithotomy associated with intermittent suction through the modified nephroscopy sheath and retrograde ureteral access sheath.

**Materials and Methods:** We implemented 3.5-F ultra-thin telescope with 6F inner sheath through a modified 10-F closed nephroscopy sheath with adjustable drainage outlet. The stone was disintegrated with a 200-μ holmium laser fibre. Intermittent suction was applied through the nephroscopy sheath outlet for active removal of stone fragments. Peristaltic pump was used for saline infusion. A retrograde 9.5 or 12-F ureteral access sheath was positioned for maintaining low intraoperative renal pelvic pressure and debris drainage. During April 2013 to August 2013, the technique was applied in treatment of 15 renal stones cases (2-3cm). All cases were completed under epidural anesthesia with patient in 45° semi-supine combined with lithotomy position. Ultrasonographic guidance was used for establishment of access tract and confirmation of residual stone at the completion of the procedure. If required, RIRS was associated for removal of any stone fragments (>4mm) inaccessible through the primary nephroscopy tract. A 6-F Double-J stent and an 8-F nephrostomy tube was placed at the end of the procedure.

**Results:** All 15 cases were successfully completed. The mean [SD] stone size was 3.06 (0.31) cm, operative time 123.5 (42.4) mins, intraoperative estimated blood 30.3 (48.0) ml. Nephrostomy tube was removed within 48 hours postoperatively while double-J stent was removed after 4 weeks. The mean [SD] hospital stay was 3.6 (0.5) days. Complete stone clearance was 73.3% with UMP alone while it was 93.3% when associated with RIRS. No major intra- or postoperative complications occurred.

**Conclusion:** Implementation of Ultra-mini percutaneous nephrolithotomy for the treatment of renal stones 2 - 3cm is feasible and safe. The procedure is less invasive and has faster recovery period. Intraoperative retrograde ureteral sheath facilitates simultaneous RIRS,

decrease intrapelvic pressure and facilitates removal of stone fragments.

#### MP-18.11

##### **Safety, Efficacy and Outcomes of Supine Mini PCNL**

**Balakrishnan A, Gowda R**

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**Introduction and Objectives:** The treatment of urolithiasis has undergone a paradigm shift in the past decade. Although a low-risk procedure, ESWL often leads to persistent residual stones. Non ideal pelvicalyceal anatomy and poor durability of the flexible ureteroscope may impact success of RIRS. PCNL though safe and effective, can be associated with serious complications. Decreasing the tract size for PCNL could decrease bleeding and morbidity. Supine mini-percutaneous nephrolithotomy (SmPCNL) can also offer a potential advantage of performing simultaneous ureteroscopy and better airway access. This study was done to evaluate the safety and efficacy of SmPCNL in our patients.

**Materials and Methods:** Between August 2013 and March 2014, 60 consecutive patients underwent SmPCNL by a single surgeon at Apollo hospital, Chennai. Data were collected prospectively including stone size and location, operative time, use of stent, ureteric catheter and nephrostomy tube, hospital stay, stone clearance and complications. All cases were performed using Karl Storz mini PCNL set, in a modified lithotomy position using fluoroscopy. Success was defined as patient free of stones or with residual stone fragments <4mm.

**Results:** There were 32 single, 22 multiple and 6 partial staghorn stones. The mean stone size was 21.1mm. 42 cases were done under general anaesthesia and 18 were done under spinal anaesthesia. Simultaneous URS and lithotripsy was done in 10 cases successfully prior to PCNL. All cases were punctured successfully with easy access to the pelvicalyceal system in the supine position. Stones were successfully cleared in 95% patients including ureteric stones. Mean operative time was 48 minutes. Three patients required additional ESWL for clearing residual stone. DJ stent was placed in 10% patients and ureteric catheter was left *in situ* for 24 to 48 hrs in rest. Nephrostomy was deployed in 1 case. Median hospital stay was 2 days. Overall complication rate was 5%. Two patients had transient fever (grade 1 claviens) and 1 patient had peri nephrostomy urine leakage (grade 2 claviens), managed conservatively.

**Conclusions:** SmPCNL is safe and effective in suitable patients and offers good stone clearance with minimal morbidity.

#### MP-18.12

##### **Comparison the Results and Complications between Standard and Tubeless PCNL**

**Darabi-Mahboub M, Taghavi R, Sadidi J**

*Mashhad University of Medical Sciences, Mashhad, Iran*

**Introduction and Objectives:** Urolithiasis is the most common urological disease. Percutaneous nephrolithotomy (PCNL) is the treatment of choice with lowest side effects for large and complex stones. We evaluated and compared the results and complication of

“conventional PCNL” with placement of temporary nephrostomy tube at the end of procedure and “tubeless PCNL”.

**Materials and Methods:** A total of 160 patients candidate for PCNL were randomized in two groups. Ultrasonography and intravenous pyelography were performed for all patients before PCNL. In all patients after placement of 5 Fr ureteral catheter, PCNL has been performed in prone position with fluoroscopic guidance. In Group I (80 patients) nephrostomy tube placed post operatively and in Group II (80 patients) the site of incision sutured without any tube insertion. In Group II, 6 patients had residual stone and 2 patients had abnormal hemorrhage and nephrostomy inserted. Therefore 72 patients evaluated as tubeless and 88 patients with nephrostomy.

**Results:** Mean stay in hospital in Group I was 3.5 days and in Group II 2.1 days ( $P < 0.001$ ). Mean analgesia requirement in Group I was 190 mg (50 – 350 mg) pethedin and in Group II 105 mg (25 – 275 mg) ( $P < 0.001$ ). Perirenal liquid collection was seen in 3 patients in Group I and in 8 patients in Group II ( $P = 0.005$ ). Febrile infection has been occurred in 4 patients in Group I and in 7 patients in Group II ( $P = 0.005$ ). Urinary leakage found in 2 patients with nephrostomy that need intervention.

**Conclusion:** Tubeless PCNL is an effective and safe method and reduced hospital stay and post-operative pain without serious complications. Nephrostomy tube is indicated only in residual stones, multiple tracts, abnormal hemorrhage and injury of pyelocalical system.

## Moderated Poster Session 19

### Infections & Inflammatory Diseases

Tuesday, October 14  
1435-1600

#### MP-19.01

##### A Single Center Study of Acute Prostatitis after Prostate Biopsy and Quinolone Resistance

Ryu J<sup>1</sup>, Ahn S<sup>2</sup>, Choi J<sup>2</sup>, Kim J<sup>2</sup>, Moon Y<sup>1</sup>, Kim T<sup>1</sup>

<sup>1</sup>Chung-Ang University Hospital, Seoul, South Korea; <sup>2</sup>Kepeco Medical Center, Seoul, South Korea

**Introduction and Objectives:** Complications after prostate biopsy have increased and various causes have been reported. Growing body of evidence on increasing quinolone resistance is of particular concern. We evaluate retrospectively the incidence of infective complication after prostate biopsy and identify the risk factor. **Materials and Methods:** The study population included 1189 patients who underwent a prostate biopsy between January 2007 and December 2012 at our center. Cases in which occurred acute prostatitis within 7 days were investigated. Clinical information included age, prostate specific antigen, prostate volume, hypertension, diabetes, and body mass index. Patients receive 250mg of IV quinolone before the procedure, and 250mg quinolone was orally administered two times daily for 7 days. We use univariate and multivariate analysis to investigate predictive factor for acute prostatitis

**Results:** Acute prostatitis developed in 23 (1.93%) cases. Core number was increasing from 2007 (8 core) to 2012 (12 core) and quinolone resistant bacteria began to appear since 2010 (quinolone resistance era). In univariate analysis, core number  $\geq 12$  ( $p=0.018$ ), BMI  $> 25$  ( $p=0.032$ ), quinolone resistance era ( $p=0.01$ ) were significant factors. However, multivariate analysis adjusted for core number results to be not significant, only BMI  $> 25$  ( $p=0.014$ ) and quinolone resistance era ( $p=0.02$ ) is significantly associated. Ten patients' bacteria were cultured in blood or urine, of which 70% has quinolone resistance. Quinolone resistance was the most important predictive factor.

**Conclusions:** Quinolone resistance is the main cause of post-biopsy infections in our center. We suggest that further evaluation to validate similar trends and novel strategies to find alternative prophylactic agent are necessary.

#### MP-19.02

##### Targeted Antimicrobial Prophylaxis for Trans-Rectal Prostate Biopsy: Does It Reduce Septic Complications?

Tukmachty H<sup>1</sup>, Nemede H<sup>1</sup>, Vohra A<sup>1</sup>, Thompson P<sup>2</sup>

<sup>1</sup>Basildon and Thurrock University Hospital, Basildon, UK; <sup>2</sup>King's College Hospital, London, UK

**Introduction and Objectives:** The rise in infective risk of Transrectal Ultrasound guided Prostate Biopsy (TRPB) is being well recognised in recent years. There remains an unacceptable complication rate of septicemia with no universally accepted antibiotic policy. Targeted antibiotic prophylaxis policy based on rectal flora has been suggested in the recent past. We report the first prospective UK study to our knowledge to assess the resistance of rectal flora to fluoroquinolones and complication rate after TRPB.

**Materials and Methods:** A total of 98 patients underwent TRPB at a secondary referral centre in the UK over a 3 month period, were assessed for pre biopsy MSU, risk factors (international travel and antibiotic use 6 months prior to TRPB) and prospectively followed for complications till 4 weeks. Of these, 60 patients had rectal swab at time of biopsy for culture and antibiotic sensitivities. Oral Ciprofloxacin 500mg bd / 3 days+ Metronidazole 1gm were administered as prophylaxis.

**Results:** Analysis of the rectal swab showed the bacteria to which the patients were exposed. On culture of rectal swab, 56/60 patients (93%) had mixed flora, 3/60 (5%) Gram-negative bacteria (*Escherichia coli*) and 1/60 patient (1.67%) Gram-positive bacteria (*Beta-haemolytic Streptococcus group C*). No Fluoroquinolone-resistant microbial strains were isolated. Mean age 66, Mean PSA 71, Adenocarcinoma Prostate 36%. International travel 23% and previous antibiotic use 0%.

Infective complications: 3/98(3%) with urinary tract infection (*E Coli* – sensitive to Fluoroquinolone). One out of 98 (1.67%) was hospitalized for acute urinary retention and another 1/98 (1.67%) admitted for sepsis (Blood culture: *E Coli* – sensitive to fluoroquinolone). None of the patients with infective complications had prior antibiotic / international travel 6 months previously.

Non-infective complications: 16/98 (16.3%) reported persistent haematuria for > 48 hours

post-biopsy without hospitalization, while 8/16 (8.1%) reported persistent haematospermia at 4 weeks.

**Conclusion:** The alarming rate of fluoroquinolone-resistance in both Gram-negative and Gram-positive organisms identified in recently published studies is inconsistent with our study. Our data also raises questions about the routine use of targeted prophylaxis based on rectal flora. In high risk patients (multiple UTI, indwelling catheters), targeted prophylaxis or trans-perineal prostate biopsies could be considered as a safer alternative.

#### MP-19.03

##### Flare Up the Symptoms in Patients with Chronic Prostatitis/Chronic Pelvic Pain Syndrome: A 27-Month Follow-Up Study

Lee G<sup>1</sup>, Cho I<sup>2</sup>

<sup>1</sup>Dankook University College of Medicine, Cheonan, South Korea; <sup>2</sup>IlsanPaik Hospital, Inje University, Ilisan, South Korea

**Introduction and Objectives:** As far as we know, no studies have reported on seasonal changes of symptoms among patients with chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS) for an extended period of time. We evaluated whether seasonal changes flare up the symptoms of CP/CPPS by serial administration of the National Institutes of Health chronic prostatitis symptom index (NIH-CPSI) questionnaires for a 27-month follow-up.

**Materials and Methods:** A total of 77 men who had evaluated the CP/CPPS symptoms were enrolled. The subjects were requested to complete a NIH-CPSI questionnaire every 2 or 3 months during the regular follow-up. We calculated the mean value from the available scores over that season for each individual patient. If a patient experienced a break-through symptom between the regular follow-up visits, he was encouraged to visit the clinic for the evaluation. The break-through symptoms were defined as a walk-in clinic visit due to flare up of symptoms, or NIH-CPSI total scores higher than 3 times when compared to the previous regular visit and patients wanted new medications for

MP-19.03, Table 1. Clinical characteristics of study subjects

|                          |               |
|--------------------------|---------------|
| No. of patients          | 77            |
| Age (yr)                 | 55.12 ± 11.88 |
| EPS+                     | 9.18 ± 11.19  |
| Initial NIH-CPSI** score |               |
| Pain item 1-3 (sum)      | 3.08 ± 2.04   |
| Pain item 4              | 3.51 ± 2.44   |
| Voiding domain           | 5.52 ± 2.54   |
| Quality of life domain   | 5.41 ± 2.37   |
| Follow up (mo)           | 27.11 ± 10.00 |
| No. of visiting          | 14.68 ± 5.33  |

The data were expressed as mean ± SD.

\*Expressed prostatic massage.

\*\*NIH-CPSI; Korean version of National Institutes of Health chronic prostatitis symptoms index.

MP-19.03, Figure 1.

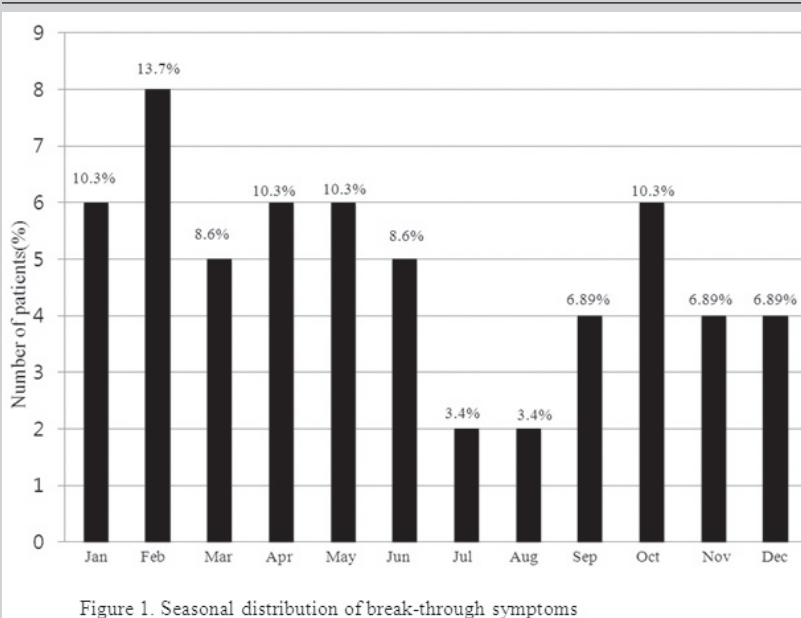


Figure 1. Seasonal distribution of break-through symptoms

ameliorating the flare up symptoms.

**Results:** The clinical characteristics of subjects are shown in Table 1. Fifty-eight patients complained of break-through symptoms during the follow-up. The incidences of break-through symptoms of CP/CPPS in our study were lowest in July and August and highest in February (Figure 1).

**Conclusion:** We observed the symptom changes among CP/CPPS patients for an extended duration. We found that severe pain attack was in winter and lesser during the summer season.

#### MP-19.04

**Isolation of Enterococcus without Associated Pyuria in Midstream Urine Specimens is Common: Caution against Overtreatment**

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<sup>1</sup>North West London Hospitals NHS Trust, London, UK; <sup>2</sup>Ealing Hospital NHS Trust, London, UK

**Introduction and Objectives:** Recent research comparing urine collected from in-out catheters with midstream urine (MSU) from patients with suspected uncomplicated UTI has shown that enterococci present in MSU are commonly absent in urine collected by in-out catheters. This suggests that enterococci are frequent contaminants in MSU and patients may be needlessly treated with antibiotics. As absence of significant pyuria ( $>10$  WCC/mm<sup>3</sup>) may indicate contamination, we investigated the frequency of association of significant enterococcal bacteriuria ( $>10^5$  cfu/ml) with significant pyuria.

**Materials and Methods:** This is a retrospective cross-sectional review of microscopy and

culture results of urines processed in a laboratory of a large district general hospital in a single year. We compared significant pyuria in MSUs with significant growth of *Enterococcus* to those where *E. coli* was isolated, using standard laboratory methods with microscopy and semi-quantitative culture on CLED and chromogenic agar media.

**Results:** Of the 1933 MSU where *Enterococcus* was grown, 64.7% of urine specimens did not have evidence of significant pyuria. By contrast, in 7836 MSUs where *E. coli* was isolated, only 21% of patients did not exhibit significant pyuria. This difference is statistically significant ( $p < 0.01$ , Z test for two population proportions) (Figure 1).

**Conclusion:** Our study shows that significant

pyuria is often absent in MSU where enterococci are isolated, unlike *E. coli* bacteriuria. This suggests that enterococci may be contaminants and not causative of infection. Clinicians should be cautious about initiating antibiotic treatment based solely on the presence of enterococcal bacteriuria if significant pyuria is not present.

#### MP-19.05

**Risk Factor for Chlamydia Infection among Men Attending a Urology Outpatient Clinic**  
Lee G<sup>1</sup>, Cho I<sup>2</sup>

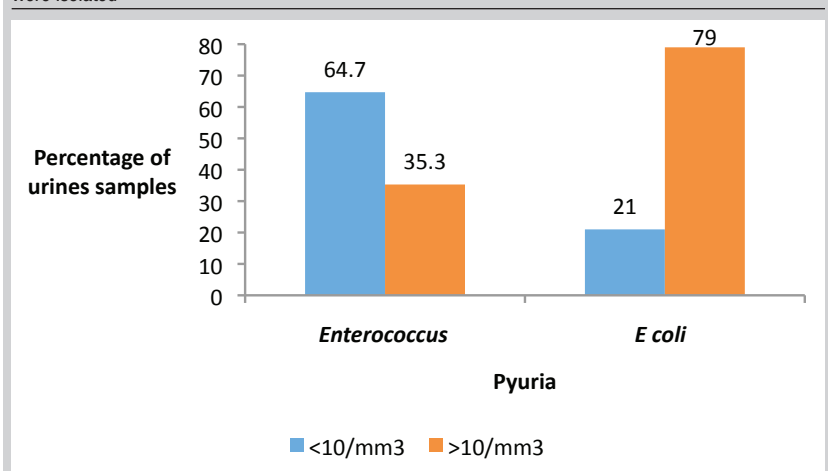
<sup>1</sup>Dankook University College of Medicine, Cheonan, South Korea; <sup>2</sup>IlsanPaik Hospital, Inje University, Ilsan, South Korea

**Introduction and Objectives:** We examined the risk factors of chlamydia infection among men attending a urology outpatient clinic.

**Materials and Methods:** Men, aged 18 to 55 years, were recruited over 4 years. The participants were evaluated by the suggestive symptoms or signs of chlamydia infection such as urethral discharge, genitourinary pain or discomfort, micturition symptoms, and WBC (white blood cell) counts in EPS (expressed prostatic secretion). We evaluated the genitourinary pain or discomfort by using a NIH-CPSI (Chronic Prostatitis Symptom Index) questionnaire. The urination symptoms were evaluated with the NIH-CPSI and IPSS (International Prostate Symptom Score) questionnaires. In house NAAT (Nucleic-Acid Amplification Test) for *Chlamydia trachomatis* (CT) and *Neisseria gonorrhoeae* detection, and WBC counts were performed from the patient's first-void urine. Variable that were statistically significantly different ( $P < 0.05$ ) in bivariate analyses were then put into a multivariate logistic regression model.

**Results:** A total of 804 consecutive participants were recruited. The overall CT prevalence was

MP-19.04, Figure 1. Percentage of MSU with Significant Pyuria where Enterococcus or E coli were Isolated





**MP-19.05**, Table 1. Multivariate analysis of risk factors for *Chlamydia trachomatis* infection

| Risk factors                                                      | OR <sup>1</sup> | (95% CI) <sup>2</sup> | P-values |
|-------------------------------------------------------------------|-----------------|-----------------------|----------|
| Age                                                               | 0.979           | (0.951-1.008)         | 0.154    |
| WBC <sup>3</sup> count in EPS <sup>4</sup> (No/HPF <sup>5</sup> ) |                 |                       |          |
| 0-4                                                               | 1               |                       |          |
| 5-15                                                              | 1.677           | (0.751-3.744)         | 0.207    |
| >15                                                               | 1.982           | (1.006-3.903)         | 0.048    |
| WBC <sup>3</sup> count in urine (No)                              |                 |                       |          |
| 0-1                                                               | 1               |                       |          |
| >1-4                                                              | 4.290           | (2.124-8.664)         | 0.000    |
| >4                                                                | 1.228           | (0.400-3.776)         | 0.720    |
| Urethral discharge                                                |                 |                       |          |
| No                                                                | 1               |                       |          |
| Yes                                                               | 4.048           | (1.565-10.475)        | 0.004    |
| NIH-CPSI <sup>6</sup> pain or discomfort score                    |                 |                       |          |
| 2a <sup>7</sup>                                                   |                 |                       |          |
| No                                                                | 1               |                       |          |
| Yes                                                               | 1.561           | (0.830-2.939)         | 0.167    |
| 2b <sup>8</sup>                                                   |                 |                       |          |
| No                                                                | 1               |                       |          |
| Yes                                                               | 1.956           | (1.035-3.696)         | 0.039    |

<sup>1</sup>OR; adjusted odds ratio, <sup>2</sup>CI; confidence interval, <sup>3</sup>WBC; white blood cell, <sup>4</sup>EPS; expressed prostate secretion,

<sup>5</sup>HPF; high power field, <sup>6</sup>NIH-CPSI; Korean version of NIH-chronic prostatitis symptom index,

<sup>7</sup>2a; pain or burning during urination, <sup>8</sup>2b; pain or discomfort during or after sexual climax (ejaculation).

7.9% (64/804). The WBC numbers in EPS, the WBC counts in the first-voided urine, the presence of urethral discharge, pain or burning during urination, and pain or discomfort during or after sexual climax (ejaculation) were statistically associated with the chlamydial infection in bivariate analysis. However, the urination symptoms from NIH-CPSI or IPSS questionnaires were not definitively correlated with the chlamydial infection. Furthermore, pain or discomfort on perineum, testicle, tip of the penis, and below waist, were not related to the infection. Multivariate analysis found that >15 WBCs in EPS, 1-4 WBCs in, urethral discharge, and pain or discomfort during or after sexual climax (ejaculation) were statistically significant risk factors for chlamydia infection (Table 1).

**Conclusion:** The results may be useful in selective screening processes for chlamydia infection in men with these risk factors who attend urology outpatient clinics.

#### MP-19.06

##### The Use of Intravesical Gentamicin to Treat Recurrent Urinary Tract Infections

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<sup>1</sup>Southmead Hospital, Bristol, UK; <sup>2</sup>Royal Devon and Exeter Hospital, Devon, UK

**Introduction and Objectives:** Recurrent urinary tract infections (UTIs) are a common problem, which negatively impact on quality of life. In a small number of individuals including complex urological patients, UTIs become refractory to standard treatment so hospital admission and intravenous antibiotics are often required. The aim of this study was to assess the use and tolerability of intravesical gentamicin for treating patients with refractory UTIs.

**Materials and Methods:** A two-centre retrospective cohort study of patients treated with intravesical gentamicin was performed over a two-year period. A treatment protocol was developed, reviewed and accepted by the clinical effectiveness committee of both hospitals. Patients were taught to instill the gentamicin (80mg in 50ml of sterile water) into the bladder on a nightly basis by a urology specialist nurse. Inclusion criteria included failure to respond to standard therapy, having 6 or more cultured confirmed UTIs over a 12 month period or at least 1 hospital admission with sepsis. Serum gentamicin levels were taken after 7 days and the treatment was discontinued if the level was >1mg/L. Patients were counseled about the limited evidence base for this treatment.

**Results:** Fourteen patients have been treated with intravesical gentamicin for an average of 16 months. Ten were performing ISC and 2 had suprapubic catheters at the time of instituting treatment. All fourteen patients started on daily 80mg gentamicin, but once control had been achieved one changed to 160mg every other day and another to every 5 days. The serum gentamicin level was <0.3ng/ml on day 7 for all patients and none reported side effects. Of the 12 patients who continued with the instillations, 7 patients have had no further proven UTIs. One patient had a single enterococcus infection, which was resistant to gentamicin. Three patients have had more than 1 coliform or enterococcus infection amenable to single courses of oral antibiotics.

**Conclusion:** This case series has shown that in the small number of adult patients who have multiple symptomatic UTIs that are refractory to conventional treatment, intravesical gentamicin is effective in reducing the frequency of infections. The treatment is well tolerated with no evidence of systemic absorption.

#### MP-19.07

##### Factors Influencing Length of Stay and Suitability for Early Community Discharge in the Management of Acute Pyelonephritis (APN)

Hsu R, Shergill P, Gnanapragasam V

Cambridge University Hospital NHS Foundation Trust, Cambridge, UK

**Introduction and Objectives:** APN is a common urological condition necessitating thousands of admissions a year and often with prolonged hospitalisation. Here we assessed factors that influenced the length of stay and suitability for safe early discharge of APN patients to community care.

**Materials and Methods:** Patients diagnosed with APN (ICD N10) in 2011-2012 to a tertiary hospital were included in a retrospective study. Data collected included patient demographics, Charlson Co-morbidity Index (CCI), admitting speciality, imaging, hospital stay and any intervention.

**Results:** A total of 266 patients were analysed with a median age of 35 years (range 15-91 years). 83.1% were managed by urologists and the rest by different specialities. Mean stay was 5.0 days (range 1-73 days). The mean time to imaging was 0.49 days and 1.43 days for urology and other specialties respectively (p<0.001). 18.8% (50/266) of patients had urinary tract abnormalities on imaging but only 1.9% (5/266) required intervention. 9.4% (25/266) had subsequent imaging following an initial scan but none resulted in an intervention. Patients in non-urological specialties were older (p=0.001) and had higher CCI (p<0.001). A high CCI was also a significant predictor of longer stays regardless of admitting speciality (p<0.001). Admission to non-urological specialties however remained a significant predictor of both delayed imaging (p=0.001) and longer stays (p<0.001) even after correction for CCI.

**Conclusion:** Admitting speciality, co-morbidity and time to imaging are key factors that influence APN management. Rapid urological review and imaging could identify patients suitable for safe early discharge to community management. Urinary tract anomalies or significant co-morbidity however will continue to need specialty specific hospital care.

#### MP-19.08

##### Antibiotic Resistance Patterns in an Emergency Room of a Tertiary Care Center in Mexico

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INNSZ, Mexico City, Mexico

**Introduction and Objectives:** Urinary tract infection (UTI) is one of the most common reasons for emergency room (ER) visits. Antibiotic resistance patterns have changed in recent

years narrowing treatment options, particularly in patients with relevant medical history. We made a retrospective analysis of UTIs diagnosed in the ER of a tertiary care center in Mexico to look for patterns of resistance and predictive factors for extended-spectrum beta-lactamase (ESBL) producing microorganisms.

**Materials and Methods:** We analyzed data from patients with urinary tract symptoms and positive urine culture after ER evaluation from January 2013 to June 2013. Clinical, demographic and microbiological characteristics were analyzed. Patterns of resistance were studied according to the intrinsic spectrum of each antimicrobial agent. Mann-Whitney test was used to compare differences between ESBL isolations. Statistical analysis was performed with SPSS v.20.

**Results:** There were 177 patients included with a total of 198 isolated microorganisms. Mean age was 57.1 years and 74% were women. Eighty three percent were classified as complicated UTI and most were managed as outpatients (80.2%). The most common microorganisms were *E. coli* (71.7%, n=142), *Enterococcus sp* (n=15), *Klebsiella pneumoniae* (n=13) and *Proteus mirabilis* (n=10). Other different bacteria were found in 18 cases. ESBL microorganisms were found in 39.3% of all urine cultures and were more common among those with urological comorbidities ( $p<0.0001$ ), recent hospitalization ( $p=0.008$ ), prior urological procedures ( $p=0.001$ ), recent antibiotic use ( $p=0.036$ ) and those with recurrent UTI ( $p=0.049$ ). Patients requiring hospitalization tended to be more commonly affected by ESBL organisms (56.8% vs. 28%,  $p=0.002$ ). Specific isolated microorganisms showed a 55% resistance to trimethoprim/sulfamethoxazole, 51.5% to quinolones, 31.3% to gentamicin, 18.8% to nitrofurantoin and 9.1% to fosfomycin. Microorganisms resistant to fosfomycin were more frequently isolated in patients with recurrent UTI (16.4% vs. 5.3,  $p=0.017$ ).

**Conclusions:** We found a high prevalence of ESBL producing bacteria in UTI diagnosed in ED. Patients with recent hospitalization, antibiotic use, urological manipulation and recurrent UTI were more frequently infected by this type of microorganisms. Lowest rates of resistance were for nitrofurantoin and fosfomycin. Therefore we suggest these agents should be used for uncomplicated UTI as outlined by recent practice guidelines.

#### MP-19.09

##### The Battle against Catheter-Associated Biofilms

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**Introduction and Objectives:** The long-term use of indwelling catheters can result in persistent infections and blockages with up to 50%

of patients experiencing encrustations and an almost permanent bacterial colonisation of urine. Previous work has shown the presence of biofilms to be an important factor in both infection risk and in the formation of encrustations which lead to catheter blockage. Here we report on an advanced microscopy technique which can be used to examine biofilms directly on catheters, revealing new information on their development and persistence.

**Materials and Methods:** Using a 6-well plate assay and defined artificial urine medium, the attachment of uropathogenic *Escherichia coli*, *Pseudomonas aeruginosa* and *Proteus mirabilis*, and subsequent biofilm formation on silicone and hydrogel latex catheter sections were followed. Resuspended bacterial cells were quantified using culture techniques and epifluorescence microscopy following labelling with the LIVE/DEAD BacLight™ bacterial staining system. Episcopic differential interference contrast (EDIC) microscopy was then used to examine biofilm structures directly on catheter sections, to give pseudo-3D images of biofilms and the surface topography of the catheter material.

**Results:** All analysis methods showed rapid attachment of all bacteria on all catheter materials, with culture analysis indicating  $10^6 \text{ cm}^{-2}$  after 2 h exposure. EDIC microscopy could be successfully used to show the surface topography of the catheter material and interactions with the developing biofilm. Initially individual bacterial cells could be seen to attach before the entire surface was covered within 6 h. With the exception of *P. mirabilis*, after 24 h exposure detachment events could be observed where areas of biofilm had sloughed away, leaving exposed catheter surfaces ready for re-colonisation. On exposure to *P. mirabilis*, an extensive crystalline biofilm was observed with copious amounts of diffuse material and well-defined struvite crystals. Using EDIC microscopy, we were able to track the various stages of biofilm development.

**Conclusions:** Using a range of techniques it was clear that bacteria attach to all catheter material types, resulting in rapid biofilm formation. Increasing our understanding of biofilm development and persistence will aid in the preparation of protocols for control and prevention, including the development of new catheter materials.

#### MP-19.10

##### Endoscopy, Bacteriology or Histology: How to Diagnose Bladder Tuberculosis?

Kulchavenya E, Kholobin D  
Novosibirsk Research TB Institute, Medical University, Novosibirsk, Russia

**Introduction and Objectives:** Bladder tuberculosis (BTB) complicates kidney TB in 45.8 – 84.8% and is one of the most severe complications, and leads to the shrinking of bladder and development of terminal renal failure.

According to the classification, BTB is divided in 4 grades, as well as iatrogenic bladder TB as a complication of the BCG therapy for bladder cancer.

**Materials and Methods:** In order to estimate the value of endoscopy, bacteriology and histology for diagnosis BTB, we analyzed 190 patients suspicious of BTB. All underwent X-ray examination, including intravenous urography and multispiral computer tomography, cystoscopy (excluding patients with cystostoma due to extremely low bladder volume), bladder biopsy. Patients with BTB 4 grade underwent cystectomy with ilioecystoplastic, bladder tissue was also investigated. Bacteriology included luminescent microscopy of the sediment of urine, microscopy of smear colored by Zhiel-Nelsen technique, culture on 3 mediums, PCR-diagnostic, Bactec and GenExpert.

**Results:** Among all 190 patients, bladder tuberculosis (BTB) was confirmed in 18 cases, and non-specific cystitis (NSC) in 172 cases. Decreased volume was revealed in all BTB patients and in 15.7% of NSC patients ( $p<0.001$ ). Trabecules (66.7%), ulcers (11.1%), contact haemorrhages (83.8%), bullous edema (44.4%), deformity of mouth of ureter (94.4%) were found significantly more often in BTB patients. Hyperemia was met in BTB as often as in NSC (accordingly 38.9% and 33.1%). Histology revealed specific TB inflammation in 11.8% only, in other BTB patients lymphoid and eosinophilic infiltration as well as fibrosis prevailed. M.tuberculosis was found by culture in 11.1%, by PCR – in 16.7%, none – by microscopy.

**Conclusion:** There are no endoscopic findings to confirm BTB, but there are some signs to suspect this disease. Histological and bacteriological confirmation is possible in 11.8% only- mostly due to late examination, after long antibacterial therapy.

#### MP-19.11

**Optimized Therapy for Prostate Tuberculosis**  
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Novosibirsk Research TB Institute, Medical University, Novosibirsk, Russia

**Introduction and Objectives:** The therapy of prostate tuberculosis (PTB) is very difficult as a few antibacterial agents are able to distribute to the prostatic tissue and achieve sufficient concentrations at the site of infection.

**Materials and Methods:** Efficiency of optimized scheme for the therapy of prostate tuberculosis (PTB) was estimated in 53 patients in the age of 24 - 69 (on average  $32.1 \pm 7.2$ ). Infiltrative PTB was diagnosed in 24, cavernous - in 29. M. tuberculosis (Mtb) was found in 21 patients (39.6%): in 6 by culture and in 15 – by PCR; histological verification was in 8 (15.1%), in 29 patients (54.7%) diagnosis was established by X-ray. The first group (25

patients) was treated with a standard scheme of chemotherapy: in intensive phase 4 anti-TB drugs daily (isoniazid + pyrazinamid + rifampicin per os and streptomycin i.m.). In phase continuation the therapy includes only isoniazid and rifampicin – in intermittent regime (3 times a week). The second group (28 patients) alongside with standard chemotherapy received ofloxacin during intensive phase, and isoniazid was given as intravenous infusion. The phase continuation was identical in both groups. The efficiency was estimated in 2 and 8 months. The criteria of the efficiency were: resolution of pain, dysuria, pyospermia, negative results of bacteriology.

**Results:** Patients in the 1<sup>st</sup> group had faster positive dynamic concerning pain, dysuria, pyospermia and negative growths on Mtb as well as PCR in ejaculate; in 8 months best results of optimized therapy were as well. In the 1<sup>st</sup> group 77.8% of patients had good results; 6 patients (22.2%) had moderate results, while in the control group, who received standard therapy, only 11 patients (44.0%) showed good results, moderate results – 13 patients (52.0%), one patient (4.0%) had poor result after completion full course of standard therapy.

**Conclusion:** Thus, optimization of the standard therapy by additional administration of ofloxacin improved results of the treatment on 33.8%.

#### MP-19.12

##### Urogenital Tuberculosis as Multivariant Disease

Kulchavenya E

*Novosibirsk Research TB Institute, Medical University, Novosibirsk, Russia*

**Introduction and Objectives:** Urogenital tuberculosis (UGTB) is one of the most common forms of tuberculosis (TB) after pulmonary TB. **Materials and Methods:** With the purpose to estimate clinical features of UGTB, we analyzed historical cases of 131 patients.

**Results:** Among 131 patients with UGTB, 88 (67.2%) had isolated kidney TB (KTB): 10 patients (10.2%) – TB of parenchyma, 35 patients (39.8%) – papillitis, 22 patients (22.4%) – cavernous KTB, 21 patients (21.4%) – polycavernous KTB; in 10 patients alongside with polycavernous KTB male genital TB (MGTB) was diagnosed. In 33 patients (25.2) MGTB only was revealed: in 14 – orchepididymitis, and in 19 – prostate TB. Main clinical features were pain (flank or perineal), dysuria, hematuria, hemospermia, toxicity, but their frequency varied from 0 to 60.0% in different groups. Among all cohort of UGTB asymptomatic course was in 12.2%, among kidney TB – in 15.9%. Every third patient complained of flank pain and dysuria (accordingly 35.2% and 39.8%), 17% presented toxicity symptoms, 9.1% – renal colic, 7.9% – gross-hematuria.

MBT was found in 31.8% in isolated kidney TB as whole. Sterile pyuria was in 25%. The onset of TB orchepididymitis was in 35.7%, hemospermia – in 7.1%, dysuria – in 35.7%. Most common complaints for prostate TB were perineal pain (31.6%), dysuria (also 31.6%), hemospermia (26.3%). MBT in prostate secretion / ejaculate was revealed in this group in 10.5%.

**Conclusion:** UGTB is a heterogeneous disease. Term UGTB is incorrect and incomplete for characteristic of clinical features, as it combines some different forms of the disease with their own clinical features and outcomes. UGTB is a collective term, and its usage may lead to discrepancy, misunderstanding and misdiagnosing.

#### MP-19.13

##### Prostate Inflammation in Patients with Pulmonary Tuberculosis

Kulchavenya E

*Novosibirsk Research TB Institute, Medical University, Novosibirsk, Russia*

**Introduction and Objectives:** Tuberculosis (TB) is a global medical and social problem. The majority of pulmonary TB (PTB) patients are young men, and prostate diseases are common for them.

**Materials and Methods:** Seventy two male PTB patients were enrolled in the study: 26 with fibrocavernous PTB (FCPTB), and 46 with infiltrative PTB (IPTB), as well as 37 healthy voluntaries for control. Expressed prostatic secretion (EPS) and ejaculate were investigated by microscopy and culture. All PTB patients who were diagnosed as prostatitis received alongside with anti-TB chemotherapy prostatotropic treatment with afala (realized active anti-bodies to prostatic specific antigen) and prostanorm (complex herbal drug).

**Results:** In IPTB patients, prostate inflammation was revealed in 32.6%, in FCPTB – in 57.7%. Only 54% of healthy voluntaries had leukocytosis in EPS. Leukocytosis in ejaculate was found in 71.7% of IPTB patients, in 84.6% of FCPTB patients, and in 10.8% in control group. Growth of microflora in EPS/ ejaculate was found in 17.4% of IPTB patients and in 38.5% – in FCPTB patients. M. tuberculosis was found in 4.4% – 6.9% accordingly. Complex afala + prostanorm in PTB patients with prostatitis (both TB and non-specific) decreased inflammation on 30.0% in patients with IPTB and on 26.9% – in patients with FCPTB and improved reproductive function on 15.3% – 35.7 accordingly.

**Conclusion:** Prostate inflammation is a common condition in PTB patients, prostate TB was revealed in 6.9%. Complex afala + prostanorm significantly decrease prostate inflammation as well as improve reproductive function in PTB patients.

#### MP-19.15

##### Differential Diagnosis of Bladder Tuberculosis

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**Introduction and Objectives:** Bladder tuberculosis (BT) is difficult to be diagnosed in time; for a long time it has been misdiagnosed as recurrent cystitis.

**Materials and Methods:** In order to estimate the diagnostic value of spectrum of low urinary tract symptoms (LUTS), the comparison of dysuria was conducted between BT and recurrent cystitis (RC) in 70 female patients: 35 with BT and 35 with RC.

**Results:** Among BT patients acute debut was in 2 (5.7%), 33 patients (94.3%) had chronic suprapubic pain, which increased when bladder filled. Nineteen BT patients (54.3%) had transient hematuria, 11 (31.4%) – gross-hematuria. Antibacterial therapy with nitrofurantoin and fosfomycin was performed on 7 (20.0%) BT patients but was not effective at all, which allowed to suspect urogenital tuberculosis soon. Another 28 BT patients (80.0%) were treated with fluoroquinolone that resulted in rather fast resolution of symptoms, but recurrence in the following few months. Correct diagnosis BT was established in this group in 6.4 years on average. In BT patients, LUTS progressed over time, the bladder volume increased. Nobody had any symptoms due to sexual intercourse. Among 35 women with RC the onset of the disease was acute without any precursor. Eleven patients (31.4%) had correlation between relapse and sexual intercourse. All 35 RC patients had frequency, urgency and burning pain when urinating, 18 patients (51.4%) had terminal hematuria. The efficacy of standard antibacterial therapy or phytotherapy was fast and complete. The significance difference ( $p=0.001$ ) was found in the debut of the disease, correlation with sexual intercourse, character of pain and hematuria.

**Conclusion:** There are significant differences in the spectrum of LUTS in recurrent cystitis and bladder tuberculosis which allow to suspect bladder tuberculosis in time in patients with dysuria.

#### MP-19.16

##### Detection of Pathogens of Sexually Transmitted Diseases in the Removed Prostatic Tissue from Patients with Benign Prostatic Hyperplasia and Prostate Cancer

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**Introduction and Objectives:** There are also known the high distribution of the sexually transmitted diseases (STD), and the possibility of their latent course. We studied the infectivity of

prostatic tissue (PT) by STD pathogens in patients who had undergone transvesical and radical prostatectomy for benign prostatic hyperplasia (BPH) and prostate cancer (PC), respectively.

**Materials and Methods:** There were examined the 195 BPH patients who had undergone transvesical prostatectomy and the 52 PC patients who had had radical prostatectomy. The age of patients varied between 50 and 86 years. As a result of a routine clinical examination, not a single patient demonstrated changes in general analysis of blood or urine, biochemical analysis of blood and also the STD in anamnesis. The infectivity by pathogens of urogenital chlamydia, mycoplasmosis, ureaplasmosis and trichomoniasis was studied using the polymerase chain reaction in the PT removed during the surgical operation.

**Results:** In the PT DNA pathogens of urogenital infections were identified in 46 (23.6±3.0%) patients with BPH and in 11 (21.2±5.7%) patients with PC ( $p>0.05$ ). The Table 1 represents the spectrum of the STD pathogens identified in the PT of the men being operated on for BPH and PC. As a whole, the frequency of DNA detection of STD pathogens in patients with BPH and PC in the removed PT significantly did not differ and was 30.8±3.3% and 23.1±5.8%, respectively. With this, the frequency of DNA detection of mollicutes (*Mycoplasma hominis*, *Ureaplasma urealyticum*) as well as

the associations of microorganisms statistically prevail 2.8 times and 3.8 times, respectively, in the BPH patients than in the PC patients.

**Conclusions:** In 23.6% of patients with BPH and in 21.2% with PC ( $p>0.05$ ) DNA of the STD pathogens are found in the intraoperatively removed PT. In the spectrum of the detected pathogens in the patients with BPH and PC dominates *Trichomonas vaginalis*. The mollicutes and associations of pathogens are significantly more often are found in the patients with BPH than in those with PC. The STD pathogens, present in the PT, which may cause the complications after surgical treatment for BPH and PC demand their identification.

#### MP-19.17

##### Detection of Microsporidiosis in Patients with Chronic Renal Failure

Omar Fadl H<sup>1</sup>, Khairy H<sup>2</sup>, Salem Ahmed Hanafy N<sup>1</sup>, Ali EL Dib N<sup>1</sup>

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**Introduction and Objectives:** Microsporidiosis is an emerging and opportunistic infection causing diarrhea affecting mainly immunocompromised patients. The aim of the work was to detect microsporidial infection in stool

and urine samples from chronic renal failure patients using conventional microscopy with Modified Trichrome stain (MTS) and molecular methods using PCR.

**Materials and Methods:** A cross sectional study was performed on 80 subjects including study group (n=50, chronic renal failure patients) and control group (n=30, apparently healthy volunteers). Fresh stool and urine samples from both groups were subjected to parasitological examination and molecular diagnosis. Each stool sample was examined by microscopy as direct smear for other parasites and Permanent stained smears using (MTS) (Weber Green) for detection of microsporidia spores. The second part will be stored at  $-20^{\circ}$  in eppendorf tubes for molecular diagnosis by PCR. All urine samples was centrifuged at 5000 rpm for 10 minutes and the sediment was divided into two parts; the first part was examined by microscopy as Permanent stained smears using (MTS) (Weber Green) for detection of microsporidia spores. The second part will be stored at  $-20^{\circ}$  in eppendorf tubes for molecular diagnosis by PCR. PCR was used as a confirmatory test after MTS for the detection of DNA of microsporidia in stool and urine specimens.

**Results:** In the study group (n=50), microsporidiosis was revealed in 2 stool samples (4%) using both PCR and Modified Trichrome stain. One stool sample was positive only by MTS staining. No microsporidia spores were detected in the urine samples or the control group. Associated Risk factors in the study group included; anemia in 70%, diabetes in 20%, cortisone treatment in 20%, and malignancy in 4%. Other parasites were detected by MTS in the study group; *Cryptosporidium* oocyst (4%), *Blastocystis* (38%), and *cyclospora* (2%). Microsporidiosis was positive in 2 stool samples of the study group.

**Conclusion:** Microsporidiosis is an important cause of diarrhea affecting mainly immunocompromised patients including those with CRF. Modified Trichrome Stain is a valuable and low cost method for detection of microsporidial spores.

**MP-19.16, Table 1.** Specific Spectrum of STD Pathogens in the Investigation of PT of Men Operated on for BPH and PC

| Pathogen types                                | PT                    |          |                         |          | p-value |
|-----------------------------------------------|-----------------------|----------|-------------------------|----------|---------|
|                                               | PC patients<br>(n=52) |          | BPH patients<br>(n=195) |          |         |
|                                               | abs.n                 | %±SE     | abs.n                   | %±SE     |         |
| Trichomonas vaginalis                         | 8                     | 15.4±5.0 | 27                      | 13.8±2.5 | p>0.05  |
| Mycoplasma hominis,<br>Ureaplasma urealyticum | 2                     | 3.8±2.7  | 21                      | 10.8±2.2 | p<0.05  |
| Chlamydia trachomatis                         | 2                     | 3.8±2.7  | 12                      | 6.2±1.7  | p>0.05  |
| Total                                         | 12                    | 23.1±5.8 | 60                      | 30.8±3.3 | p>0.05  |
| Association of STD pathogens                  | 1                     | 1.9±1.9  | 14                      | 7.2±1.9  | p<0.05  |



## Moderated Poster Session 20

### Prostate Cancer: Basic Science

Wednesday, October 15  
1300-1430

#### MP-20.01

##### Can ERG Protein Expression at Diagnosis Predict Prostate Cancer Progression for Patients Managed on Active Surveillance?

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**Introduction and Objectives:** New biomarkers identifying prostate cancer (PCa) patients with high risk of disease progression during active surveillance (AS) are needed. We aimed to analyse the association between ERG protein expression at diagnosis and risk of progression during AS.

**Materials and Methods:** In total, 265 PCa patients were followed prospectively on AS with prostate-specific antigen (PSA) measurements, clinical examinations including digital rectal examination, and 10-12 core re-biopsies. Immunohistochemical analysis for ERG protein expression was performed on diagnostic sections using a monoclonal antibody (anti-ERG, EPR3864; Roche/Ventana). Patients were labelled as 'ERG-positive' if minimum one tumour focus demonstrated ERG-expression and otherwise as 'ERG-negative'.

The primary endpoint was 'Overall AS progression' during AS defined as minimum one of:

- *Clinical-progression:* increased clinical tumour category  $\geq$  T2b
- *Histopathological-progression:* Increase in Gleason score;  $>3$  positive cores and/or bilateral positive cores on re-biopsies
- *PSA-progression:* PSA doubling time  $<3$  years.

We used multiple cause-specific Cox regression and stratified cumulative incidences for analyzing the risk of progression; competing events were curatively intended treatment, watchful waiting and death without progression. Finally, re-classification diagrams and calibration plots were applied for assessing changes in the predicted risk of progression when adding ERG-status to the multiple regression models.

**Results:** A total of 217 patients had complete diagnostic information; 121 were ERG-negative and 96 were ERG-positive. ERG-positive patients had significantly higher incidences of overall AS progression ( $p<0.0001$ ), PSA-progression ( $p<0.0001$ ) and histopathological progression ( $p<0.0001$ ) when using competing

risk models. The 2-yr cumulative incidence of overall AS progression was 58.6% (95%CI: 48.7%-68.5%) for ERG-positive patients and 21.7% (95%CI: 14.3%-29.1%) for ERG-negative patients. ERG-positivity was a predictor of overall AS progression (HR: 2.45, [95%CI: 1.62-3.72],  $p<0.0001$ ) and histopathological progression (HR: 3.06, [95%CI: 1.78-5.26],  $p<0.0001$ ) in multiple cause-specific Cox regression models. ERG-status significantly altered the predicted risk of progression during AS beyond the other diagnostic parameters.

**Conclusion:** ERG-positive PCa runs a more aggressive course if left untreated. ERG-status assessed at diagnosis can be used as a prognostic marker for progression and may be used to individualise future AS-programmes.

#### MP-20.02

##### IGF-1 Related Pathways and High-Fat Diet Promotion of TRAMP Mouse Prostate Cancer Progression

Xu H, Jiang H, Ding Q  
Huashan Hospital, Fudan University, Shanghai, China

**Introduction and Objectives:** We aimed to examine the effect of high-fat diet on PCa development and progression by comparing tumor incidence and mortality among TRAMP mice fed with high-fat diet or not. Then we investigate the role of IGF-1 related pathway in high-fat diet promotion of TRAMP mouse PCa progression.

**Materials and Methods:** TRAMP (Transgenic Adenocarcinoma of the Mouse Prostate) mice were randomly divided into two groups: the experimental group - high-fat diet group ( $n=42$ ), and the control group - ordinary diet group ( $n=42$ ). High-fat diet group TRAMP mice were fed with high-fat forage to induce obesity and insulin resistance. TRAMP mice of both groups were sacrificed and sampled on the 20<sup>th</sup>, 24<sup>th</sup> and 28<sup>th</sup> week respectively. TRAMP mice were required to fast overnight before sacrifice. Fasting blood glucose was measured when blood was taken. Serum levels of insulin, IGF-1 and IGF-2 were tested by ELISA. Prostate tissue of TRAMP mice was used for both HE staining and immunohistochemical staining of IGF-1 related pathway proteins, including IGF-1R $\alpha$ , IGF-1R $\beta$ , IGFBP1, IGFBP2, IGFBP3, IGFBP5, IGFBP6 and Akt.

**Results:** A total of 13 cases of death were observed in the study, including 3 TRAMP mice (7.14%) from the normal diet group, and 10 TRAMP mice (23.81%) from the high-fat diet group. The mortality of TRAMP mice from high-fat diet group was significantly higher than normal diet group ( $P=0.035$ ). When TRAMP mice were sacrificed at 20<sup>th</sup> week, there were 11 TRAMP mice (78.57%) from high-fat diet group had typical PCa, while there were only 5 TRAMP mice (35.71%)

from ordinary diet group had typical PCa. The tumor incidence at 20<sup>th</sup> week of TRAMP mice from high-fat diet group was significantly higher than normal diet group ( $P=0.022$ ). Blood glucose, serum insulin and IGF-2 levels were not significantly different among normal diet group and high-fat diet group at 20<sup>th</sup>, 24<sup>th</sup> and 28<sup>th</sup> week. But serum IGF-1 level of high-fat diet TRAMP mice was significantly higher than that of normal diet TRAMP mice. The difference was even significant at 28<sup>th</sup> week with  $P$  value of 0.011. In addition, serum IGF-1 level tended to increase with high-fat diet TRAMP mice's age. Immunohistochemical staining of TRAMP mice prostate showed that high-fat diet TRAMP mice had higher positive staining rate of IGF-1R $\alpha$ , IGF-1R $\beta$ , IGFBP3 and Akt than normal diet TRAMP mice. Though other IGF binding proteins (IGFBP1, IGFBP2, IGFBP5 and IGFBP6) were also expressed in TRAMP mice prostate, there was no significant difference of positive staining rate between the two groups.

**Conclusion:** Tumor incidence and mortality of high-fat diet TRAMP mice were significantly higher than those of normal diet TRAMP mice, indicating that high-fat diet can promote PCa development and progression. IGF-1 related pathway proteins, including IGF-1, IGF-1R $\alpha$ , IGF-1R $\beta$ , IGFBP3 and Akt, were found to have higher expression in high-fat diet TRAMP mice by ELISA and immunohistochemical staining study, suggesting that IGF-1 related pathway played an important role in high-fat diet promotion of TRAMP mouse PCa development and progression.

#### MP-20.03

##### Cytokines and High-Fat Diet Promotion of TRAMP Mouse Prostate Cancer Progression

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Huashan Hospital, Fudan University, Shanghai, China

**Introduction and Objectives:** We aimed to examine the effect of high-fat diet on PCa development and progression by comparing tumor incidence and mortality among TRAMP mice fed with high-fat diet or not. Then we investigate the role of cytokines in high-fat diet promotion of TRAMP mouse PCa progression. **Materials and Methods:** TRAMP (Transgenic Adenocarcinoma of the Mouse Prostate) mice were randomly divided into two groups: the experimental group - high-fat diet group ( $n=42$ ), and the control group - ordinary diet group ( $n=42$ ). High-fat diet group TRAMP mice were fed with high-fat forage to induce obesity and insulin resistance. TRAMP mice of both groups were sacrificed and sampled on the 20<sup>th</sup>, 24<sup>th</sup> and 28<sup>th</sup> week respectively. TRAMP mice were required to fast overnight before sacrifice. Prostate tissue of TRAMP mice was used for HE staining. Serum levels of different cytokines

including IL-1 $\alpha$ , IL-1 $\beta$ , IL-2, IL-4, IL-5, IL-6, IL-10, IL-13, IL-17, IL-21, IL-22, INF- $\gamma$ , and MCP-3 were examined by FlowCytomix.

**Results:** A total of 13 cases of death were observed in the study, including 3 TRAMP mice (7.14%) from the normal diet group, and 10 TRAMP mice (23.81%) from the high-fat diet group. The mortality of TRAMP mice from high-fat diet group was significantly higher than normal diet group ( $P=0.035$ ). When TRAMP mice were sacrificed at 20<sup>th</sup> week, there were 11 TRAMP mice (78.57%) from high-fat diet group had typical PCa, while there were only 5 TRAMP mice (35.71%) from ordinary diet group had typical PCa. The tumor incidence at 20<sup>th</sup> week of TRAMP mice from high-fat diet group was significantly higher than normal diet group ( $P=0.022$ ). Blood glucose, serum insulin and IGF-2 levels were not significantly different among normal diet group and high-fat diet group at 20<sup>th</sup>, 24<sup>th</sup> and 28<sup>th</sup> week. But serum IGF-1 level of high-fat diet TRAMP mice was significantly higher than that of normal diet TRAMP mice. The difference was even significant at 28<sup>th</sup> week with  $P$  value of 0.011. Several cytokines were significantly higher in high-fat diet TRAMP, including IL-5 (20<sup>th</sup> week 16.05 pg/mL vs. 3.61 pg/mL,  $P=0.028$ ; 28<sup>th</sup> week 35.07 pg/mL vs. 10.08 pg/mL,  $P=0.016$ ), IL-10 (20<sup>th</sup> week 9.08 pg/mL vs. 0.21 pg/mL,  $P<0.001$ ; 28<sup>th</sup> week 21.94 pg/mL vs. 2.63 pg/mL,  $P=0.019$ ), IL-17 (20<sup>th</sup> week 8.70 pg/mL vs. 5.27 pg/mL,  $P=0.040$ ), IL-21 (20<sup>th</sup> week 6.63 pg/mL vs. 0.98 pg/mL,  $P=0.002$ ; 28<sup>th</sup> week 13.39 pg/mL vs. 3.07 pg/mL,  $P=0.014$ ), IFN- $\gamma$  (20<sup>th</sup> week 5.92 pg/mL vs. 0.33 pg/mL,  $P<0.001$ ). MCP-3 was significantly lower in high-fat diet TRAMP (24<sup>th</sup> week 70.99 pg/mL vs. 295.69 pg/mL,  $P<0.001$ ; 28<sup>th</sup> week 65.16 pg/mL vs. 135.85 pg/mL,  $P=0.019$ ).

**Conclusion:** Tumor incidence and mortality of high-fat diet TRAMP mice were significantly higher than those of normal diet TRAMP mice, indicating that high-fat diet can promote PCa development and progression. Several cytokines were significantly higher in high-fat diet TRAMP, including IL-5, IL-10, IL-17, IL-21, and IFN- $\gamma$ , while MCP-3 was significantly lower in high-fat diet TRAMP, suggesting that cytokines played an important role in high-fat diet promotion of TRAMP mouse PCa development and progression.

#### MP-20.04 microRNA Dysregulation Supports Proliferative Inflammatory Atrophy as an Independent Precursor Lesion in Prostate Carcinogenesis

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Histopathology, St. James's Hospital, Dublin, Ireland; <sup>4</sup>Laboratory of Pathology, National Cancer Institute, National Institutes of Health, Bethesda, USA

**Introduction and Objectives:** High-grade prostatic intra-epithelial neoplasia (HGPIN) is currently the only accepted precursor lesion of prostate cancer (PCa). Proliferative Inflammatory Atrophy (PIA) describes focal atrophic lesions (containing proliferative epithelial cells that fail to differentiate into columnar epithelial cells), associated with chronic inflammation and has been suggested as a further PCa precursor lesion. Evidence supporting this hypothesis comes from both morphological and molecular data (*P53* mutations, chromosome 8 abnormalities and hypermethylation of *GSTP1*). Dysregulation of microRNAs is a common feature of PCa. However, microRNA data in PIA are lacking. The aim of this study was to examine expression of microRNAs in PIA and HGPIN in comparison to benign prostate tissue and invasive tumour.

**Materials and Methods:** Laser capture microdissection was performed on FFPE prostate tissue from 10 benign, 8 HGPIN, 4 PIA and 23 PCa samples (6 indolent, 8 aggressive and 9 metastatic tumours). Expression profiling of 752 human microRNAs annotated in miR-Base v18.0 was performed using the Exiqon miRCURY LNA<sup>TM</sup> qPCR panels I and II with ExiLent SYBR<sup>®</sup> Green. Validation of results was performed by qRT-PCR.

**Results:** Twenty one microRNAs showed significant differential expression in PIA vs. benign prostate tissue ( $P<0.05$  and fold change  $>1.5$ ). Of these 21 microRNAs, 3 showed similar differential expression patterns in PCa relative to benign tissue, indicating that these molecular events occur in early tumour initiation. Comparing PIA with tumour, 45 microRNAs were differentially expressed, 4 of which were also differentially expressed in PCa vs. benign, suggesting that these aberrations occur in the transition from PIA to invasive tumour. A similar stepwise progression was observed for HGPIN lesions. Notably, there was no other overlap in the expression profile of PIA and HGPIN, indicating divergent pre-invasive lesions. At both stages of the transition from benign prostate tissue to PIA to PCa, miR-205-5p was found to be down-regulated, supporting its role as driver in prostate carcinogenesis.

**Conclusion:** This is the first report of microRNA dysregulation in PIA, and lends support to the hypothesis that PIA is a precursor lesion to PCa. PIA shows a distinct miRNA expression profile, separate from that of HGPIN and benign prostate tissue.

#### MP-20.05

##### Predictive Values of Microvessel Densities Evaluated by CD31, CD34 and CD105 for Biochemical Recurrence after Prostatectomy with Neo-Adjuvant Therapy in Prostate Cancer Patients

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**Introduction and Objectives:** Neo-adjuvant hormonal therapy (NHT) is often performed to improve the outcomes in prostate cancer patients. However, change of angiogenic condition by NHT is not fully understood. In patients with NHT, we evaluated angiogenesis semiquantitatively by assessing microvessel density (MVD) using endothelial specific markers, CD31, CD34, or CD105. The aim was to clarify relationships between MVDs in NHT specimens and pathological change or biochemical recurrence.

**Materials and Methods:** Formalin-fixed specimens from 102 prostate cancer patients received prostatectomy were examined. NHT was performed in 54 patients (52.9%) and 48 patients that has similar Gleason score and pT stage were selected as non-NHT groups. MVD was measured by anti-CD31, anti-CD34, and anti-CD105 antibody. Relationships between each MVD in NHT specimens and anti-tumoral effects including histological effect and biochemical recurrence were investigated by multi-variate analyses. The histopathological effect was evaluated by the standardized rules for estimating the response of prostate cancer to NHT published by the Japanese Urological Association.

**Results:** CD34-MVD in NHT group (mean/SD=50.2/7.8) was similar to that in non-NHT group (51.6/7.0,  $P=0.338$ ). Although CD31-MVD in NHT group (36.2/9.9) was lower than in non-NHT group (39.0/8.2), this difference did not reach significant level ( $P=0.129$ ). CD105-MVD in NHT group (13.0/4.6) was remarkably lower ( $P<0.001$ ) than in non-NHT group (25.7/6.3). In non-NHT specimens, CD31-, CD34-, and CD105-MVD were significantly associated with Gleason score and pT stage. However, in NHT specimens, CD31- or CD34-MVD was not associated with these pathological features. CD105-MVD in NHT specimens was significantly associated with pT stage ( $P=0.003$ ). In addition, CD105-MVD in NHT specimens was correlated with histological effect ( $P=0.002$ ) and biochemical recurrence ( $P<0.001$ ). CD105-MVD was recognized as an independent predictor for biochemical recurrence (hazard ratio=3.2, 95% confidential interval=1.2-8.2,  $P=0.016$ ).

**Conclusion:** Our results showed CD105-MVD reflected angiogenic condition and malignant potential in patients with NHT. CD105-MVD was associated with anti-cancer effects and it was identified as a significant and

independent predictor for biochemical recurrence in patients received prostatectomy after hormonal therapy.

#### MP-20.06

##### Periprostatic Adipose Tissue, NGF and BDNF in Prostate Cancer

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**Introduction and Objectives:** Obesity is associated with larger size of prostate cancer and higher Gleason scores. The mechanism by which obesity promotes prostate cancer remains unknown. However, the prostate may be target at paracrine to some cancer-influencing adipokines. We focus on the potential role of NGF and related neurotrophins, secreted by the periprostatic and anterior perirectal adipose tissue in prostate cancer. We aimed to assess the immunohistochemical expression of NGF, BDNF, TrkA and p75 in the periprostatic and anterior perirectal adipose tissue in patients with prostate cancer.

**Materials and Methods:** Eight tissue samples from periprostatic (P-AT) and anterior perirectal (R-AT) adipose tissue were taken intraoperatively from patients with prostate cancer. Each sample contained both adipose and prostate tissue (P-PT and R-PT) from the tumor. Retropubic adipose tissue was taken from the same patients for a control (C). Tissue samples were routinely processed for H&E staining and peroxidase/antiperoxidase immunohistochemistry. For primary antibodies we used rabbit polyclonals (Santa Cruz Biotech) – NGF, BDNF, TrkA; mouse monoclonal (DAKO) – p75. The secondary antibody was EnVision Flex Vis Syst, DAKO. We measured the presence (+) and absence (-) of expression. Appropriate control stainings were used.

**Results:** All P-AT, R-AT and C samples showed expression of NGF. In the prostate all 174 patients but one P-PT sample expressed NGF. All R-PT samples were also positive. Only two P-AT samples showed positive expression. BDNF was not expressed in all R-AT samples. Only two C samples were positive. In contrast, all P-PT and R-PT samples showed positive expression of BDNF. All P-AT and R-AT samples were negative. Only one C sample showed positive expression. The results were more interesting in the prostate tissue. P-PT samples were all negative for TrkA, whereas all but two R-PT samples showed positive expression. All samples, both AT and PT were negative for p75.

**Conclusion:** NGF is expressed by both adipose tissue and prostate in periprostatic, perirectal and control samples. BDNF is expressed mostly in

the prostate and much less in adipose tissue. TrkA is expressed mostly in the posterior and not in both the anterior prostate and all adipose tissue samples. P75 has no expression in both adipose tissue and prostate. There is little or no significant difference between all adipose tissue samples.

#### MP-20.07

##### Low Expression of EphB6 Receptor Tyrosine Kinase in Prostate Cancer

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**Introduction and Objectives:** Loss of erythropoietin-producing hepatocyte (Eph) B6 gene expression has been correlated with poor prognosis in neuroblastoma, melanoma and other tumors. This study was evaluated expression of EphB6 Receptors tyrosine kinase (RTK) in normal and prostate cancer tissue using immunohistochemistry. The association among EphB6 expression, clinico-pathological findings, another prognostic marker of proliferating-cell nuclear antigen (PCNA) and progression of prostate cancer was analyzed.

**Materials and Methods:** Formalin fixed-paraffin embedded tissue blocks of normal prostatic tissue and prostate cancer tissue of 46 patients treated with radical prostatectomy for prostate cancer were included in this study. Polyclonal anti-EphB6 and monoclonal anti-PCNA antibodies were used to assess EphB6 and PCNA expression by immunohistochemistry.

**Results:** EphB6 expression was demonstrated in both normal and prostate cancer tissue. There is a significant reduction of EphB6 expression in prostate cancer tissue compared to normal prostatic tissue ( $p < 0.0001$ ), in high volume ( $\geq 4$  cc) of cancer compared to low volume ( $< 4$  cc) of cancer ( $p = 0.015$ ) and in pT3 stage compared to pT2 stage of the disease ( $p = 0.0007$ ). EphB6 expression has no correlation with expression of PCNA proliferation marker. Although there is no significant difference, there is a trend of short biochemical progression free survival in patients with low expression status of EphB6 protein ( $p = 0.157$ ).

**Conclusion:** This study revealed that early progression of prostatic cancer is significantly associated with low expression of EphB6 RTK. This could provide a new insight to use EphB6 receptor as a potential diagnostic/prognostic marker for prostate cancer.

#### MP-20.08

##### "Shared Care Model" for Patients with Stable Prostate Cancer

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**Introduction and Objectives:** The prevalence of PC is rising and patients PC are frequently checked and have their follow-up in hospitals after the diagnosis and the initial treatment is started. Guidelines recommend that patients with PC in stable phase can be monitored, controlled and treated in general practice. The main obstacles for transition of PC to the general practice include lack of national consensus on treatment and uncertainty among the patients whether the general practitioner can handle the follow-up. We want to describe a shared care model for patients with stable prostate cancer (PC) and intention to apply this model for control, care and treatment of PC in general practice in the County.

**Materials and Methods:** The strategy for shared care model was developed in collaboration with health professionals in the steering committee, which consisted of general practitioners, urologists and nurses from all urological departments in one county covering 1.2 mill people and people from the Department of General Medicine and patients'. An action plan included (1) the development of a shared care model for follow-up and treatment, (2) implementation of shared care model, by means of cooperation between the parties involved, (3) evaluation of the impact, change processes and contextual factors to transfer experience and model for other areas and (4) ensure specific referrals back to the department if necessary.

**Results:** The study started on 1 September 2013 and ending 30 March 2014. A total of 1079 patients with PC were enrolled in the study. There were 459 (43%) with disseminated disease, 388 (36%) who had curative therapy without relapse and 43 (4%) with recurrence, and 65 (6%) of watchful waiting and 102 (9%) with active surveillance. A total of 246 (23%) patients were completed to the GP, 833 (77%) could not be completed of which 22 patients wanted to continue in the outpatient clinic. The main reason that a patient could not be completed was unstable PSA 193 (27%), surgery 138 (19%) or radiation therapy 157 (22%). There was also socio-economic effect with this model e.g. cheaper for the society the patients were controlled by the GP, better for the patient to be followed by the same doctor.

**Conclusions:** The project has focused on factors that are essential for the successful transfer of responsibility for long-term follow-up in patients with prostate cancer with high professional quality and high patient satisfaction. The model had great socio-economic impact and can be applied to other urological disease areas.



Moderated Poster Session 21  
Stones: Various Topics  
Wednesday, October 15  
1300-1430

**MP-21.01**

**Genome-Wide Association Study Identifies Polymorphisms at Three Novel Loci as Urolithiasis Risk Factors by Using Single-Nucleotide Polymorphism Analysis**

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**Introduction and Objectives:** Identifying genetic predisposition to urolithiasis helps in the prophylaxis of the disease. To identify the genetic factors associated with urolithiasis, we conducted a 3-stage genome-wide association study (GWAS). To identify the risk factors for urolithiasis, a case-control study was conducted using single nucleotide polymorphism (SNP) analysis.

**Materials and Methods:** A 3-stage GWAS was conducted with 5,892 urolithiasis patients and 17,809 controls; all participants were of Japanese origin. We investigated the association of novel loci with urolithiasis through case-control association analysis of 601 cases and 201 control subjects from an independent Japanese sample set. Furthermore, the risk for urolithiasis was calculated from the total number of significant risk-allele associations by SNP analysis.

**Results:** Three novel loci for urolithiasis were identified in the GWAS: *RGS14-SLC34A1-PFN3-F12* on 5q35.3, *INMT-FAM188B-AQP1* on 7p14.3, and *DGKH* on 13q14.1. In the case-control association analysis, 7 SNPs were identified: rs12654812 and rs11746443 from 5q35.3 (*RGS14-SLC34A1-PFN3-F12*); rs12669187 and rs1000597 from 7p14.3 (*INMT-FAM188B-AQP1*); and rs7981733, rs1170155 and rs4142110 from 13q14.1 (*DGKH*). The SNPs rs12654812 at 5q35.3, rs12669187 at 7p14.3, and rs7981733 at 13p14.1 were significantly associated with urolithiasis, after applying Bonferroni correction ( $P = 3.12 \times 10^{-3}$ , odds ratio [OR] = 1.43;  $P = 6.40 \times 10^{-3}$ , OR = 1.57 and  $P = 5.00 \times 10^{-3}$ , OR = 1.41, respectively). In addition, meta-analysis of the current study and the GWAS exhibited significant associations, with  $p$ -values of  $1.42 \times 10^{-13}$ ,  $1.25 \times 10^{-15}$  and  $5.81 \times 10^{-10}$ . These three SNPs showed a cumulative effect: the risk of developing urolithiasis in individuals with  $\geq 3$  alleles associated with urolithiasis was 5.9-fold that in individuals with a single allele associated with the disease (35% control subjects). In addition, ORs were higher for patients with recurrent urolithiasis (OR = 6.4) than for those diagnosed with the disease for the first time (OR = 5.8).

**Conclusion:** The SLC34A1 gene on 5q35.1 encodes NPT2, a member of the type II

sodium-phosphate co-transporter family, and the AQP1 gene on 7p14.3 encodes aquaporin-1, a member of the major intrinsic protein family that is involved in regulation of water flow. Both of these proteins are highly expressed in kidneys. Three novel susceptibility loci at 5q35.5, 7p14.33, and 13q14 were identified to be associated with urolithiasis risk.

**MP-21.02**

**Vitamin D and Bone Mineral Density amongst Scottish Stones Formers**

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**Introduction and Objectives:** Along side other northern countries, Scotland faces long seasons of low levels of sunshine. Consequently, serum levels of vitamin D have been shown to be significantly lower than their more southerly European counterparts. Levels of vitamin D in patients who suffer from recurrent urolithiasis has not previously been looked at in the Scottish population. Inherently mineral losing, these patients may be at greater risk of developing bony complications. This would be exacerbated by low serum vitamin D levels. The aim of this study was assess if there was a difference between vitamin D levels in recurrent stone formers as compared to reference data and to comment on the incidence of low bone mineral density within this group.

**Materials and Methods:** From December 2011 to December 2013, all high risk patients in Ninewells Hospital, Dundee, Scotland identified with demonstrable biochemical, endocrine or genetic risk factors were referred to the joint Urology/Renal Medicine Metabolic Stone Team for extended screening and medical treatment. This included serum vitamin D levels for all and bone densitometry scanning where indicated. The results were collected and compared to regional reference data.

**Results:** Forty eight patient stone formers with demonstrable biochemical, endocrine or genetic risk factors were identified and referred. Geometric mean group Vitamin D levels in this group was 34.1 (29-40 95% CI) nmol/l. Population mean is 50 nmol/l in Scotland (60 nmol/l in England). Within this group, 26 patients had DEXA scanning (mean age 46) – 14 (54%) were osteopenic or osteoporotic and 4 (15%) were osteoporotic.

**Conclusion:** Metabolic stone screening continues to identify patients with important biochemical, genetic and endocrine conditions. Recurrent stone formers – particularly Scottish stone formers, appear to be at a high risk of osteopenia and osteoporosis. Future work regarding the efficacy and safety of Vitamin D replacement in this population needs to be evaluated.

**MP-21.03**

**Metabolic Evaluation of High Risk Stone Formers in a District General Hospital: What Difference Does It Make?**

Marri R, Housami F, Holliday M, McIlhenny C  
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**Introduction and Objectives:** The European Association of Urology Guidelines recommends metabolic evaluation (ME) for upper urinary tract calculi (UUTC) in all high risk stone formers. However ME is not regularly performed for these patients in many units. The aim of this study is to evaluate the prevalence of metabolic abnormalities in these patients and how this affects patient management.

**Materials and Methods:** All patients with high risk of recurrent UUTC (first stone at less than thirty years of age, two or more stones in less than five years duration or first stone with other risk factors like family history in patients more than thirty years old) seen in Forth Valley Royal hospital had ME. Stone analysis was done and ME was individually tailored as per stone composition. Data was collected regarding the patient risk factors, parameters tested during ME, its effect on patient management.

**Results:** Between September 2010 and August 2013, 39 patients at high risk of recurrent UUTC had ME. Stone composition in these patients was calcium oxalate (7), calcium phosphate (3), calcium oxalate and phosphate (5), urate (7) and unknown in rest. Three patients had low twenty four hour urine volumes. Metabolic abnormalities were identified in 19 patients (49%). Hypercalciuria, hyperuricosuria and hyperoxaluria were noted in 13, 5 and 1 patient respectively. Of the 13 patients with hypercalciuria, 8 were started on diuretics. Two patients failed to attend their subsequent appointments. One was referred to endocrinologists and another had parathyroid adenoma excised. Another patient with mild hypercalciuria and raised urine sodium was advised to reduce his salt intake. There was no significant change in treatment plan for the patient's with hyperuricosuria and hyperoxaluria. One patient started on diuretics for hypercalciuria had recurrent stones.

**Conclusion:** Metabolic abnormalities were identified in 49% of patients in this study. Identifying the underlying cause in these patients helps counsel them better regarding their risk of recurrent stones and discuss means for prevention of further recurrent UUTC. However longer duration follow-up is required to accurately assess stone recurrence in these patients.

**MP-21.04**

**HIV Patients and Stones: What Type Do They Form?**

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**Introduction and Objectives:** With the advent of newer antiretroviral medications, HIV is now viewed as a chronic disease rather than a life threatening illness. These patients have a lifetime risk of forming renal tract calculi secondary to antiretroviral treatment, classically due to Indinavir. We reviewed our experience with such patients over a 5 year period.

**Materials and Methods:** Over a five year period from 2008 to 2013 a total of 7 patients with a known diagnosis of HIV underwent endoscopic intervention in our unit for renal calculi. We reviewed the stone analysis in each case along with their antiretroviral medication.

**Results:** Median age was 46 years (range 39 – 68), with five male and two female patients in this group of stone formers. The results are outlined in Table 1. No stone analysis was available for one patient, and the other patient's radiolucent stone awaits treatment by PCNL. The pure atazanavir stones were not visible on either plain film; and one of these did not show on CT either despite a heavy stone burden confirmed by ureterorenoscopy.

**Conclusion:** HIV patients taking anti-retroviral medication such as atazanavir are at higher risk of forming pure radiolucent stones not visible on CT imaging, which can provide diagnostic challenges in such patients. The incidence of such stones are likely to increase due to the advances in management of HIV. It is important to establish whether there are differences between HIV treatments with regards to urolithiasis, and to develop guidelines for imaging and follow-up.

MP-21.04, Table 1.

| Type of stone                                          | Number |
|--------------------------------------------------------|--------|
| Pure Atazanavir                                        | 2      |
| Mixed Atazanavir / Calcium phosphate / calcium oxalate | 1      |
| Mixed Calcium Oxalate / Uric Acid                      | 1      |
| Calcium oxalate / calcium phosphate                    | 1      |

#### MP-21.05

##### Is Brand's Test Useful as a Screening in Adults with Renal Colic?

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**Introduction and Objectives:** The Cyanide-Nitroprusside test (Brand's test) is

recommended in pediatric recurrent renal stone formers or in adults, when the analysis of the stone reveals the presence of cystine. Patients affected by cystinuria suffer recurrence thus reducing their quality of life. We aimed to evaluate the incidence of cystinuria in the adult population referred by emergency department (ED) after an episode of renal colic.

**Materials and Methods:** After ERB approval, 300 consecutive adult patients presenting to clinic after ED admission were recruited. Urine was tested by Brand's test for cystinuria and repeated if equivocal. Exclusion criteria were: permanent catheters, aminoaciduria, Fanconi's syndrome or drugs (e.g. N-acetylcysteine or ampicillin).

**Results:** Of 300 patients, 193 were men and 107 were women. The mean age was 42.5 y (range 18-79 y). The percentage of patients with history of renal stones in each quartile is shown in Table 1. Three patients had a positive Brand test [2/193 men (1.03%) and 1/107 women (0.93%)] all of whom had a history of recurrent stone formation. When analysing the incidence of cystinuria in the 1<sup>st</sup> quartile (18-32 y, youngest group) with a history of calculi (2 cystinuria cases/36 pts) we found that this group had a statistically significant higher incidence when compared to the other subjects (5.5% vs. 0.4%; p=0.04 Fisher's exact test).

**Conclusion:** Given the significantly higher incidence of cystinuria (p=0.04) in the 18-32 year old group with a history of colic, and the clinical and social problems related to recurrent stone formation, we suggest the use of Brand's test as screening tool in this population.

#### MP-21.06

##### The Radiological Investigation of Suspected Ureteric Colic in a District General Hospital: Are the Right Patients Getting the Right Scans at the Right Time?

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<sup>2</sup>Dept. of Radiology, University Hospital, Ayr, UK

**Introduction and Objectives:** Non-contrast computed tomography (CT-KUB) is recommended by the British Association of Urological Surgeons (BAUS) and the Royal College of Radiologists (RCR) as the first line

investigation for suspected ureteric colic, allowing a rapid, definitive diagnosis. The primary aim was complete an audit cycle assessing the current practice for the investigation of suspected ureteric colic in a District General Hospital against the national BAUS/RCR standards. The secondary aim was to determine the diagnostic yield of imaging in younger female patients and assess whether a change in imaging protocol could maintain diagnostic sensitivity whilst reducing radiation exposure in this group.

**Materials and Methods:** An initial audit of all patients presenting with suspected ureteric colic within a 6-month period (n=130) was completed. After analysis and presentation of the results, interventions in the form of a new imaging algorithm for younger patients and new guidance for the trainee staff, were introduced. A re-audit was completed over a two-month period (n=46) and the data was analysed in an identical manner.

**Results:** CT-KUB was used to investigate suspected ureteric colic in 87.7% of patients (target = 100%). Among all, 64.9% underwent CT-KUB within 24 hours of admission (target = 100%). Mean time to CT-KUB was 24.3 hours. Calculi were detected in 54.4% (target 44-62%). A significant difference was noted in the proportion of young male and female patients with calculi confirmed on imaging (p=0.0028). On re-audit, CT-KUB was used as the initial investigation in 95.8% of cases. Significant reductions in the time to scan were noted, with a calculi detection rate within the target range. However, calculi detection rates were only 30% in young female patients.

**Conclusion:** This completed audit cycle illustrates some of the challenges in achieving national standards within a District General Hospital, as well as some of the strategies which can improve adherence. We have demonstrated an improvement in the use of the recommended first-line investigation via targeted interventions, with a diagnostic yield consistently within the target range. However, the use of CT-KUB as first line investigation in young female patients continues to be an area requiring further work to ensure the appropriate use of CT-KUB in these patients.

MP-21.05, Table 1.

| Quartile  | N° of patients | Patients with history of colic | Percentage |
|-----------|----------------|--------------------------------|------------|
| 1 (18-32) | 76             | 36                             | 47.4%      |
| 2 (33-47) | 122            | 64                             | 52.5%      |
| 3 (48-62) | 73             | 38                             | 52.1%      |
| 4 (63-79) | 29             | 16                             | 55.2%      |
| Total     | 300            | 154                            | 51.3%      |

**MP-21.07****Positive Rate of Ureteric Colic Referrals from the Emergency Department and Management**

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**Introduction and Objectives:** To determine the positive rate of ureteric colic referrals and to compare our diagnostic practice and management with the current guidelines.

**Materials and Methods:** A retrospective review was performed on all patients presenting to the Emergency Department (ED) and referred to Urology with suspected ureteric colic between February 2013 and April 2013. Information on imaging times, dates and reports were gathered from the Picture Archiving and Communication System (PACS) and patient management information was gathered from the Patient Pathway Manager (PPM) system.

**Results:** A total of 72 patients were referred to Urology from the ED with an average length of hospital stay of 44.4 hours; 45.8% were males and 54.2% were females. A total of 49 patients had a CTKUB and 32 patients had an AXR prior to a CTKUB which was requested by the ED. The average time taken to perform the CTKUB was 45 hours. Positive rate of urolithiasis was 38.9%, of which 60.1% of patients were male and 20.5% were female. Ureteric colic contributed 67.9% and the rest were renal calculi. Obstructive signs were found in 46.5% of the scans and 17.9% underwent either acute endoscopy or ESWL. Conservative treatment was successful in 28.6% of our patients, the remaining had intervention as an outpatient. Overall, 71.4% of our patients were successfully cleared of calculi. The remaining 18.1% patients who did not have urolithiasis had a different urological problem, 19.4% did not have a definitive diagnosis, 8.3% had musculoskeletal and general surgical diagnoses each and the remaining 6.9% were gynaecological.

**Conclusion:** Ureteric colic is a common presentation to the ED but the positive rate for ureteric calculi remains low. The female population is at higher risk of unnecessary radiation exposure and USS could be considered before CTKUB. As CTKUB is the gold standard for diagnosis of ureteric colic, AXRs should be avoided. The duration taken for CTKUB to be performed is longer than the BAUS guidelines of 24 hours, suggesting a pathway to perform CTKUB from the ED will be beneficial.

**MP-21.08****What Is the Role of Quantitative Measurement of Penicillamine-Cysteine Complexes in Cystinuria Patients on Penicillamine?**

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*Guy's and St Thomas' NHS Foundation Trust, London, UK*

**Introduction and Objectives:** With no clear guidance on Penicillamine dose administration and titration, utilisation of this drug to prevent stone formation risk in cystinurics remains a challenge. We assess the role of quantitative measurement of penicillamine-cysteine complexes (p-cc), correlations between p-cc measurements, Penicillamine dose, stone burden and urine cystine concentrations; and determine the potential role in guiding treatment.

**Materials and Methods:** Records were analysed using our clinic database. Multiple urine samples of patients on Penicillamine were measured using laboratory calculations, providing nominal p-cc concentrations. We correlated these with urinary cystine, ornithine, lysine, arginine concentrations and pH levels. Ratios of the penicillamine-cysteine complex to the total cystine level (p-cc:c) were calculated. Total daily doses of Penicillamine were noted and patients' stone status - "none"=no stones, "stable"=no new stones or "new" stones categorised.

**Results:** We identified 9 patients on Penicillamine; 5 males; 4 females. Age range was 29-65 years. Total daily doses ranged from 250-2000mg. Patients who had formed new stones had the lowest p-cc:c (median=0), compared to "none" (median=0.121) and "stable" groups (median=0.114). Relative p-cc levels for these respective groups are 85.5, 129 and 125.5umol/L. An inversely proportional correlation is seen between cystine levels and p-cc. Lowest cystine levels were amongst patients on a total daily dose of 1000mg, who had the highest p-cc:c. Median p-cc for patients on varying doses are 2000mg=183; 1500mg=121; 1000mg=125; 250mg=0umol/L. Relative p-cc:c ratios for these groups: 0.063, 0.093, 0.148 and 0. Patients who formed crystals had a lower median p-cc:c ratio of 0, compared to those who did not (median=0.148). Patients with SLC7A9 mutation show a better response to Penicillamine compared to SLC3A1 mutation (median p-cc:c ratios 0.145 and 0.061).

**Conclusion:** Patients on lowest dosage had the lowest p-cc levels and the highest urine cystine concentrations. Low p-cc:c ratios identifies a higher risk of new stone formation. Highest p-cc:c ratios are shown in patients with "none" or "stable" stones, no crystal formation, SLC7A9 mutation and those on 1000-1500mg daily dosage. Monitoring on Penicillamine remains challenging, however our results may provide a method for predicting response to treatment. We suggest p-cc:c ratios >0.10 and p-cc levels >125umol/L for favourable outcomes on Penicillamine.

**MP-21.09****The Predictability of Stone Type Based on Metabolic Stone Screen Results**

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**Introduction and Objectives:** Robertson (2003) described a risk factor model of stone formation, calculating a risk of future stone formation (PSF) in patients undergoing metabolic stone screening (MSS). Algorithms are available for use, calculating the PSF for certain stones types; uric acid, mixed calcium oxalate and uric acid, pure calcium oxalate, mixed calcium oxalate and calcium phosphate and pure calcium phosphate. There is no comparative study in the literature demonstrating the predictability of stone type based on PSF. The objective of this retrospective study was to investigate the ability of the risk factor model to predict the type of stone formed.

**Materials and Methods:** Two parallel datasets from one hospital trust representing patients undergoing MSS and stone samples sent for quantitative analysis were cross referenced according to patient number. A total of 122 patients were identified for whom at least one stone had been submitted, resulting in a total of 181 stones for correlation MSS. For those patients with more than one MSS recorded the most recent MSS was compared with stone analysis. Struvite and cysteine containing stones were excluded.

**Results:** Eighty six patients were male and 36 female (1: 2.4). The age range was 12-72 (mean 45). The time between MSS and stone analysis ranged from 0 – 140 months. MSS results, when applied according to the risk factor model, correlated with stone content in 43% of cases. The ability of the model to predict different types of stone varied substantially. None of 12 UA/CaOx stones were accurately predicted by the MSS result; however 92% of calcium phosphate stones correlated with a PSF (CaP) of  $\geq 0.5$  on MSS. There was no clear relationship between time between MSS and SA and correlation.

**Conclusion:** The risk factor model for stone formation predicts stone type in some patients. Calcium phosphate and mixed calcium containing stones were better predicted than calcium oxalate or uric acid containing stones. There was no clear effect of timing of MSS on correlation with stone analysis results. This has implications in clinical practice in terms of treatment choice and the approach to patient counselling on the basis of MSS results.

**MP-21.10****Stone Radiodensity Measured by Hounsfield Unit as a Predicting Factor for Success of Shock Wave Lithotripsy in Fragmentation of Renal Calculi**  
**Mahmoud M**, Elsayed M, Elgiouhy F,

Elsotohi I, Farid M, Ali M  
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**Introduction and Objectives:** To evaluate the usefulness of measuring urinary calculi attenuation values by non-contrast computed tomography (NCCT) for predicting the outcome of extracorporeal shockwave lithotripsy (ESWL) in the treatment of renal stones.

**Materials and Methods:** We included 60 patients with solitary renal stones up to 25 mm in this prospective study. Calculus density value was measured in Hounsfield units on NCCT. Patients were subsequently treated with ESWL using Dornier SII lithotripter with a maximum of three sessions. Patients were followed two weeks after each session of ESWL by plain X-ray for radio opaque stone or ultrasound for radiolucent stone to assess fragmentation. A "successful outcome" is defined as complete stone clearance or stone fragments < 4 mm. Absence of stone disintegration or presence of surgical fragment >4 mm after three sessions was considered failure of ESWL treatment.

**Results:** Patients were grouped according to calculus Hounsfield units (HU) as: Group (A) less than 500 HU, Group (B)—500 to 1,000 HU and Group (C)—greater than 1,000 HU. The rate of stone fragmentation and clearance was 100% (18 of 18 cases) in Group A, 95.6% (22 of 23) in Group B and 52.6% (10 of 19) in Group C. The success rate for stones with an attenuation value of greater than 1,000 HU was significantly lower than that for stones with a value of less than 1,000 HU (10 of 19 versus 40 of 41 cases,  $p < 0.001$ ). Patients in Group C required a greater median number of shock waves for stone fragmentation than those in groups A and B (9689, 3265, and 6331, respectively). The mean attenuation value (HU) and number of shock waves required for stone fragmentation correlated significantly ( $p < 0.001$ ).

**Conclusion:** H.U. of renal stones  $\leq 25$  mm detected by NCCT was the most predictive factor in the outcome of ESWL irrespective to location of stone or size. Stones < 1000 HU were treated effectively by ESWL while stones of 1000 H.U. or more did not respond effectively to ESWL. No correlation was found between radio density by plain X-ray and fragility and composition of the stones. Also there were no correlations between radio-opacity on plain X-ray and radiodensity on NCCT.

#### MP-21.11

##### **Does Thiazide Prophylaxis for Calcium-Based Stones Increase the Risk of New-Onset Diabetes Mellitus?**

**Knoedler J**, Krambeck A, Singh P, Lieske J, Li X, Bergstralh E, Rule A  
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**Introduction and Objectives:** Previous studies involving thiazides and thiazide-type diuretics (herein referred to as thiazides) used to treat

hypertension (HTN) have found an increased risk of new-onset diabetes mellitus. However, while thiazides used for the prevention of calcium based urolithiasis have been found to be associated with impaired fasting glucose, their association with diabetes mellitus (DM) remains to be defined. Herein we evaluate the association of thiazides used for calcium stone prophylaxis with new onset DM among incident stone formers.

**Materials and Methods:** We retrospectively identified 2,350 incident stone formers with calcium-based stones in Olmsted County, Minnesota, from 1984-2011 using the Rochester Epidemiology Project. Patients with a diagnosis of DM or thiazide usage predating their stone event were excluded. Kaplan-Meier analysis predicted survival free from DM, while multivariate models compared the association of clinical variables with the development of DM.

**Results:** A total of 329 patients (14%) were treated with thiazides following incident stone, while 2,021 (86%) did not receive thiazides. Median age for those receiving thiazides was 46.5 years (IQR 37.0-57.2) compared to 38.4 years (IQR 29.7-49.8) for those not receiving thiazides ( $p < 0.0001$ ). Median follow-up was 10 years (range 0.1-28 years). Incidence of new-onset DM 10 years after first stone event was 9.2% among the thiazide group, compared to 4.2% among the non-thiazide group ( $p < 0.001$ ). On multivariate analysis, use of thiazides (HR 1.73;  $p = 0.007$ ), increasing age (HR 1.03;  $p = 0.002$ ), and BMI at first stone event (HR 1.13;  $p < 0.001$ ) were associated with an increased risk of new-onset DM. The cohort of patients receiving thiazides after incident stone was then divided into those who received thiazides for HTN ( $n = 248$ ) and those who received thiazides for stone prophylaxis alone ( $n = 81$ ). On multivariate analysis thiazide administration for stone disease was not associated with new onset DM (HR 0.92;  $p = 0.87$ ).

**Conclusions:** While thiazide usage among stone formers with HTN is associated with an increased risk of new-onset DM, thiazides used for stone prophylaxis alone did not increase the risk of new-onset DM. To our knowledge, this study is the first to evaluate the role of thiazides in the development of DM among a population of stone formers.

#### MP-21.12

##### **Tamsulosin versus Nifedipine in Medical Expulsive Therapy for Distal Ureteral Stone and Predictive Value of Hounsfield Unit in Stone Expulsion**

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**Introduction and Objectives:** We evaluated the efficacy of tamsulosin and nifedipine in medical expulsive therapy (MET) in patients

with distal ureteral stone. Also, we tried to determine the predictive value of Hounsfield Unit (HU) of the stone in success of MET.

**Materials and Methods:** A total of 75 patients with 5 to 10 mm in diameter distal ureteral stone were randomly divided into 3 groups. Group 1 ( $n = 25$ ) received tamsulosin (Flomax®, Boehringer Ingelheim) orally 0.4 mg/d; Group 2 ( $n = 25$ ) received nifedipine (Adalat Crono®, Bayer) orally 10 mg/d and Group 3 ( $n = 25$ ) received only symptomatic therapy with diclofenac sodium (Voltaren®, Novartis) orally 50 mg when they needed. At the beginning of the treatments, HU of the stone was also measured with using non-contrast computerized tomography in all the patients. The treatments' results were evaluated at 4<sup>th</sup> week.

**Results:** The mean age of the patients was 36.8 (range; 16 to 68) years. Stone expulsion was observed in 19 (76%) patients in Group 1, (64%) 16 in Group 2 and 9 (36%) in Group 3. Stone expulsion rate significantly difference in groups 1 and 2 with respect to Group 3 ( $p_{\text{group 1-3}} = 0.004$ ,  $p_{\text{group 2-3}} = 0.048$ ,  $p_{\text{group 1-2}} = 0.355$ ). The mean stone expulsion time was 9, 9.1 and 10.3 days in Group 1, 2 and 3, respectively ( $p_{\text{group 1-3}} < 0.001$ ,  $p_{\text{group 2-3}} < 0.001$ ,  $p_{\text{group 1-2}} = 0.619$ ). The mean diclofenac sodium dosage per patient in groups 1 to 3 was 544, 602 and 1408 mg, respectively ( $p_{\text{group 1-3}} < 0.001$ ,  $p_{\text{group 2-3}} < 0.001$ ,  $p_{\text{group 1-2}} = 0.977$ ). Two patients in Group 1 had dizziness, 1 patient in Group 2 had dyspepsia while another one in the same group had diarrhea, and 1 patient in Group 3 had dyspepsia. None of the patients broke up from the study because of adverse effects. The mean HU of the stone in patients with and without successfully MET was 363 and 389, respectively ( $p = 0.462$ ).

**Conclusion:** Administration of tamsulosin and nifedipine in MET was determined to be equally safe and effective treatment options for distal ureteral stone. HU seems not to have a predictive value for stone expulsion rate in MET.

#### MP-21.13

##### **Dissolution Therapy for Radiolucent Kidney Stones: An Old Treatment Revisited**

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**Introduction and Objectives:** Uric acid calculi are amenable to medical therapy. Dissolution therapy for radiolucent kidney stones is based on urine alkalinisation and hydration. Different regimes to alkalinise the urine were proposed including sodium bicarbonate ( $\text{NaHCO}_3$ ) and potassium citrate. Dissolution therapy is a non invasive modality of treatment especially for patients with multiple comorbidities. We present our experience in medical dissolution therapy for radiolucent calculi.

**Materials and Methods:** Retrospective analysis of our experience. Complete or partial resolution results were noted as well as treatment



failures. Data on patient demographics in addition to stone size, location and serum uric acid levels were collected. NaHCO<sub>3</sub> was started at doses of 1 gm TDS oral then increased to a maximum of 3 gm TDS daily titrated to urinary pH. Patients were instructed to monitor their urinary pH daily to achieve at least pH of 7 (tending towards 8) and alter the dose of NaHCO<sub>3</sub> accordingly.

**Results:** Duration of therapy was two to three months. The majority of patients had renal ultrasound at the end of treatment period. We identified 27 patients who were diagnosed with radiolucent stones. Four received no sodium bicarbonate treatment and were treated with other modalities. Two are still awaiting results. Out of the 21 patients who completed the treatment regime, complete dissolution was noted in 33%, partial dissolution in 14% and treatment failed in 38%. Fifteen per cent were lost to follow-up or were non-compliant to therapy.

**Conclusion:** We recommend a trial of sodium bicarbonate dissolution therapy for treatment of radiolucent kidney stones which are not obvious on the initial KUB xray, in patients with hyperuricosuria or multiple comorbidities. Follow-up with renal u/s to check for dissolution is sufficient. Patients with partial response or failure will then be candidates for other treatment modalities like ESWL, flexible ureteroscopy or PCNL.

#### MP-21.14

##### **Incident and Recurrent Stone Formers Undergoing Percutaneous Nephrolithotomy: Clinical and Anatomic Differences**

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**Introduction and Objectives:** While interstitial Randall's plaque and duct of Bellini plugging have been implicated in urolithiasis formation, little is known how features change with recurrent stone events. Herein we compare the clinical characteristics and papillary architecture between incident and recurrent stone formers undergoing percutaneous nephrolithotomy (PCNL).

**Materials and Methods:** We prospectively enrolled a random cohort undergoing percutaneous nephrolithotomy. After stone removal, each calyx was endoscopically mapped with video imaging and the percent surface area involved by plaque and plug were digitally measured using image analysis software. Papillary biopsy of a representative papilla was performed. Stone composition was determined by micro-computed tomography and infrared analysis. All patients completed 24 hour urine studies and basic serum laboratory exams.

**Results:** We identified 178 patients who underwent PCNL including 52 (29.2%) incident stone formers and 126 (70.8%) recurrent stone

formers. There were no differences with regards to diabetes, previous bowel surgery, renal tubular acidosis, or primary hyperoxaluria. Compared to recurrent stone formers, incident stone formers were more likely to be female (78.8% vs. 51.6%;  $p=0.0008$ ), and have higher 24-hour urinary citrate (median 695 mg/day vs. 482 mg/day;  $p=0.05$ ). Mean papillary surface area involved by plaque (mean%involved $\pm$ SD) was  $1.8\pm 2.5$  for incident stone formers compared to  $2.7\pm 3.4$  for recurrent stone formers ( $p=0.04$ ). Meanwhile, the mean area involved by plug was  $0.2\pm 0.6$  for incident stone formers vs.  $0.9\pm 2.7$  for recurrent ( $p=0.004$ ). We then restricted the cohort to idiopathic calcium oxalate (CaOx) stones ( $n=112$ ), including 33 (29.5%) incident stone formers and 79 (70.5%) recurrent stone formers. Among incident CaOx, mean plaque was  $1.9\pm 2.7$  vs.  $3.2\pm 3.9$  for recurrent stone formers ( $p=0.04$ ). The difference in plug between incident and recurrent stone formers was no longer significant ( $0.3\pm 0.7$  vs.  $0.6\pm 1.3$ ;  $p=0.06$ ).

**Conclusions:** Among a prospective study of patients undergoing PCNL, we found significantly increased papillary plaque and plug among recurrent stone formers as compared to incident stone formers. However, when restricted to CaOx stones, only the percentage of plaque remained significantly higher among recurrent stone formers. Our findings support the theory that both plaque and plug contribute to recurrent stone formation, and that plaque specifically is associated with CaOx stone formation.

#### MP-21.16

##### **Assessment of Complications and Analysis of the Risk Factors for Percutaneous Nephrolithotomy Using the Clavien-Dindo Grading System**

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**Introduction and Objectives:** To evaluate the complications in percutaneous nephrolithotomy (PCNL) using Clavien-Dindo Grading System (CDGS) and to analyze the risk factors for complications in PCNL.

**Materials and Methods:** From January 2003 to April 2012, 2254 cases with calculus of upper urinary tract underwent PCNL in our center. The complications in PCNL were evaluated by the CDGS, and the risk factors for complications in PCNL were analyzed using the multiple logistic regression analysis.

**Results:** Among 591 (26.22%) cases with complications evaluated using the CDGS, grades I, II, IIIa, IIIb, IVa, IVb and V were 8.07% (182), 14.77% (333), 1.86% (42), 0.40% (9), 0.75% (17), 0.22% (5) and 0.13% (3), respectively. Hemorrhage was the most common complication, and followed by post-operative fever. The multiple logistic regression analysis showed that

the relevant risk factors of grade II and higher complications mainly included the body mass index (BMI), pre-operative hemoglobin levels, diabetes mellitus, degrees of renal insufficiency and hydronephrosis, staghorn calculi, calculi surface area, intra-operative amounts of channels, staging operation, operation time and styles of lithotripsy. There were 454 (20.14%) cases with staghorn calculi in a total of 2254 cases. The rank sum test showed that the complications incidence of PCNL for the staghorn calculi was obviously higher than that of the others ( $Z=-4.23$ ,  $P=0.000$ ).

**Conclusion:** The CDGS may objectively evaluate the complications of PCNL. BMI, Pre-operative hemoglobin levels, diabetes mellitus, degrees of renal insufficiency and hydronephrosis, staghorn calculi, calculi surface area, intra-operative amounts of channels, staging operation, operation time and styles of lithotripsy are the risk factors for grade II and higher complications of PCNL. The complications incidence of PCNL for the staghorn calculi is obviously higher than that of the others.

#### MP-21.17

##### **Fatal Venous Air Embolism as a Complication of Percutaneous Nephrolithotomy**

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**Introduction and Objectives:** Fatal air embolism during percutaneous nephrolithotomy (PCNL) was first reported in 1985 (Hobin, J Forensic Sciences 30(4): 1284-86). This complication was not mentioned in Campbell's Urology 8<sup>th</sup> or 9<sup>th</sup> editions (2002, 2007). Campbell-Walsh Urology 10<sup>th</sup> edition (2012) mentions 6 cases (one fatal). The trigger for the present study was a medico-legal inquest into a case of fatal air embolism during PCNL. The aim of this study was to assess the risk of this complication and awareness among urologists. **Materials and Methods:** A review of the English literature was performed using electronic search engines. An e-mail survey concerning air embolism as a complication of PCNL was sent to members our national urological association. Inspection of the PCNL equipment in various hospitals was performed to assess the safety aspects.

**Results:** The literature search revealed 6 cases of non-fatal venous air embolism during retrograde pneumopyelography to visualize a radiolucent stone, and 3 cases (2 fatal) not associated with pneumopyelography. One fatal case resulted from reversed connection of the rotary suction pump tubes, forcing air into the collecting system (Hobin 1985) but the other had no identifiable cause. (Turillazzi, Urology 2009; 73(3): 681). The survey was completed



by 70 urologists, 57% had been in practice >10 years, 64% performed 1-10 PCNLs per year, 28% were unaware of air embolism as a complication of PCNL, 40% knew of another urologist who had a patient with air embolism during PCNL, 51% thought it may result from reverse connection of the suction pump tubing.

Inspection of the PCNL equipment currently in use revealed that the inlet and outlet connections of the rotary suction pump are identical, so that inadvertent reverse connection can easily occur.

**Conclusion:** Air embolism is a rare but potentially fatal complication of PCNL. There

is limited awareness among urologists of this complication. The causes are pneumopyelography using air, or inadvertent reverse connection of the rotary suction pump. This can be avoided by using unidirectional wall suction during PCNL, but greater awareness among urologists is required.

Moderated Poster Session 22  
Kidney & Ureteral Cancer:  
Various Topics  
Wednesday, October 15  
1300-1430

**MP-22.01**

**Aberrant Promoter Methylation  
of the 16q23.1 Tumor Suppressor  
Gene ADAMTS18 in Clear Cell  
Renal Cell Carcinoma**

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**Introduction and Objectives:** To identify tumor suppressor genes (TSGs) silenced by aberrant promoter methylation and discover new epigenetic biomarkers for early cancer detection. *ADAMTS18*, located at 16q23.1 region, has been reported as a critical TSG for multiple primary tumors, but not in renal cell carcinoma (RCC) yet. We explore its epigenetic alteration in RCC and analyze the possible clinicopathological association.

**Materials and Methods:** We examined *ADAMTS18* gene expression and methylation by semiquantitative reverse transcriptase PCR (RT-PCR) and methylation specific polymerase chain reaction (MSP) in 5 RCC cell lines before and after treatment with 5-aza-2'-deoxycytidine (5-Aza) alone or and trichostatin A (TSA). MSP was further applied in 117 RCC primary tumors with 24 adjacent normal tissues. We also analyzed the relationship between *ADAMTS18* methylation and clinicopathological features in patients with RCC.

**Results:** *ADAMTS18* gene down-regulation and methylation was detected in RCC cell lines using RT-PCR and MSP. Treatment with 5-Aza alone or and TSA could reverse methylation and restore *ADAMTS18* expression. Aberrant methylation was further detected in 51 of 117 (43.6%) primary tumors and 4 of 24 (16.7%) adjacent normal tissues. However, a significant difference between both groups was observed ( $p=0.02$ ). Furthermore, *ADAMTS18* gene methylation status was not significantly associated with gender, age, location, tumor diameters, AJCC TNM stage or nuclear grade in patients with RCC ( $p>0.05$ ).

**Conclusions:** *ADAMTS18* gene is often down-regulated by aberrant promoter methylation in RCC cell lines and primary tumors, indicating its critical role as a TSG in RCC. We conclude that *ADAMTS18* gene methylation may be involved in the tumorigenesis of RCC and has a potential chance to be served as a novel biomarker for RCC.

**MP-22.02**

**FABP7 as a Potential Marker and Target  
in Clear Cell Renal Cell Carcinoma**

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**Introduction and Objectives:** To identify potential targets in clear cell renal cell carcinoma (ccRCC), we performed a transcriptome analysis

of ccRCC and normal kidney specimens. The transcriptome analysis revealed numerous highly overexpressed genes related to hypoxia, pH regulation and metabolisms in ccRCC. FABP7 (Fatty Acid Binding Protein 7) had the highest mean value of overexpressed mRNA in ccRCC specimens compared to normal kidney specimens. We confirmed the significance of FABP7 as a potential marker and target in ccRCC.

**Materials and Methods:** We performed a gene expression analysis against the total RNA extracted from 60 ccRCC and 20 normal kidney specimens on the human exon 1.0 ST array (Affimetrix). The expression profiles were analysed by unsupervised hierarchical average linkage clustering algorithm. Among the overexpressed genes, FABP7 had the highest mean difference between ccRCC and normal kidney in the profile. The expression levels were further studied by immunohistochemical analysis in 40 advanced ccRCC cases treated by cytokine and/or molecular targeted drugs. The effect of gene knockdown was studied in a cell line with overexpression of FABP7 (A498).

**Results:** Among the overexpressed genes, FABP7 had the highest mean expression level in mRNA, while the expression levels varied depending on the ccRCC cases. Our immunohistochemical analysis for the ccRCC specimens demonstrated significant correlation between FABP7 expression and corrected serum Ca level or distant metastasis (both  $p<0.05$ ). Based on proliferation assay (MTT assay) and invasion assay (Matrigel Invasion assay), functional suppression of FABP7 significantly reduced the proliferation and invasion of A498 cells (both  $p<0.05$ ).

**Conclusion:** FABP7 was highly overexpressed in ccRCC and was correlated with poor prognosis. Functional suppression of FABP7 downregulated the proliferation and invasion of a ccRCC cell line. FABP7 could be a prognostic marker and potential therapeutic target in ccRCC.

**MP-22.03**

**GOLPH3 Is a Novel Marker of Poor  
Prognosis and a Potential Therapeutic  
Target in Human Renal Cell Carcinoma**

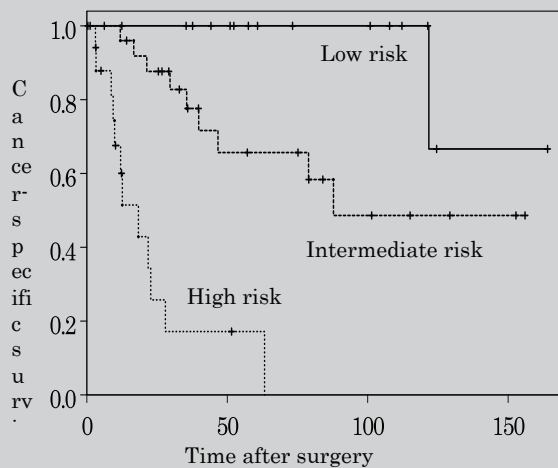
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**Introduction and Objectives:** Golgi phosphoprotein 3 (GOLPH3) has been reported to be involved in the development of several human cancers. The present study was conducted to investigate the expression of GOLPH3 and its prognostic significance in renal cell carcinoma (RCC). Meanwhile, the function of GOLPH3 in human RCC was further investigated in cell culture models.

**Materials and Methods:** Expression of GOLPH3 was examined in 43 fresh RCC

**MP-22.08, Figure 1.**



|                   |    |    |   |   |
|-------------------|----|----|---|---|
| No. at risk       |    |    |   |   |
| Low risk          | 17 | 13 | 8 | 2 |
| Intermediate risk | 26 | 12 | 6 | 3 |
| High risk         | 18 | 3  | 0 | 0 |

tissues and paired adjacent normal renal tissues by real-time quantitative PCR and western blotting. Immunohistochemistry for GOLPH3 was performed on additional 218 RCC tissues. The clinical significance of GOLPH3 expression was analysed. Downregulation of GOLPH3 was performed using small-interfering RNA (siRNA) in Caki-1 and 786-O cells with high abundance of GOLPH3, and the effects of GOLPH3 silencing on cell proliferation, migration, invasion *in vitro*, and tumour growth *in vivo* were evaluated.

**Results:** Expression of GOLPH3 was up-regulated in the majority of the RCC clinical tissue specimens at both mRNA and protein levels. Clinicopathological analysis showed that GOLPH3 expression was significantly correlated with T stage ( $P < 0.001$ ), lymph-node status ( $P = 0.003$ ), distant metastasis ( $P < 0.001$ ), tumour-node-metastasis (TNM) stage ( $P < 0.001$ ), and Fuhrman grade ( $P = 0.001$ ). Expression of GOLPH3 was inversely correlated with both overall and recurrence-free survival of RCC patients. Multivariate analysis showed that GOLPH3 expression was an independent prognostic indicator for patient's survival. Knockdown of the GOLPH3 expression reduced cell proliferation, anchorage-independent growth, migration, invasion, and tumour growth in xenograft model mice.

**Conclusion:** These results suggest that GOLPH3 expression is likely to have important roles in RCC development and progression, and that GOLPH3 is a prognostic biomarker and a promising therapeutic target for RCC.

#### MP-22.04

##### Anomalous Accumulation of Heavy Metals in Renal Cell Carcinoma: A Scanning Electron Microscopy and X-ray Microanalysis

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**Introduction and Objectives:** The association between heavy metals (HMs) contamination and cancer has been largely investigated. However, the role of HMs in renal carcinogenesis is far from being clearly understood. The aim of our study was to investigate the accumulation of particles of HMs in samples of renal cell carcinoma (RCC) and adjacent renal tissue (control samples) in patients submitted to radical or partial nephrectomy, using scanning electron microscopy and X-ray microanalysis (SEM-XRM).

**Materials and Methods:** Samples of RCC and adjacent renal parenchyma from 7 patients submitted to radical or partial nephrectomy were collected. They were fixed with glutaraldehyde and dehydrated in ethanol and critical point-dried in a Balzer's apparatus. The preparations

were mounted on metal stubs and coated by carbon under vacuum and examined in a JEOL JSM-6301F SEM, a high-resolution method, coupled with XRM. SEM-XRM was used to detect and semi-quantify HMs. Local Ethics Committee approved this study and all patients signed an informed consent.

**Results:** SEM-XRM revealed a marked accumulation of HMs (nickel, chromium, iron, tungsten, titanium, cadmium, copper, manganese, zinc and magnesium) particles in samples of RCC, but not in the surrounding parenchyma.

**Conclusion:** The chemical composition of RCC and the adjacent parenchyma is different. Inversely to the normal parenchyma, there was an anomalous accumulation of HMs in RCC samples. According to these findings, HMs might be implicated in renal carcinogenesis. In addition, they also may contribute to open a new venue for future studies on the kinetics of HMs in renal oncology.

#### MP-22.06

##### Challenging the Guidelines: Is Current Cumulative Radiation Exposure in Patients Treated with Partial Nephrectomy Safe?

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**Introduction and Objectives:** There is no consensus regarding the optimal surveillance of renal cell carcinoma (RCC) treated with partial nephrectomy (PN). Currently, the European Association of Urology (EAU) surveillance guidelines following PN involve patient exposure to ionizing radiation with six-month or annual computed tomography. There is however, a paucity of data examining cumulative radiation exposure (CRE) in this population. This is important as the additional risk of secondary malignancy is 0.04% for every 10 milli-Sieverts (mSv) of radiation exposure.

**Materials and Methods:** A prospective single centre review of 270 patients (mean age: 59, range 31-85) who underwent open or minimally invasive PN was performed between January 2002 and January 2014. Sixty four patients were excluded with benign histology or incomplete data. Follow-up CRE was calculated in mSv according to imaging modality and matched to tumour histology and stage. RCC recurrence was defined as radiologic evidence of local tumour recurrence or development of distant metastases.

**Results:** CRE was 15.8, 15.0, 50.3 and 12.9 mSv/patient/year for pT1a (n=154), pT1b (n=31), pT2 (n=3) and pT3 (n=18) tumour surveillance (mean follow-up: pT1a=42 months, pT1b=50 months, pT2=25 months, pT3=42 months). The mean time to radiological recurrence was 5.6 months (range 3-8) and this was observed in 3 patients (17%) all of

whom had pT3 (pT3a n=1, pT3b =2) disease. All recurrences demonstrated greater than Fuhrman grade-3 disease (n=1), tumour size over 3-cm (n=2), or a positive surgical margin (n=1). There was no evidence of disease recurrence or metastases in other tumour stages.

**Conclusion:** Current EAU follow-up protocols subject patients to un-necessary ionizing-radiation. Early recurrence rates are low for low-grade and stage disease and the diagnostic yield with current follow-up imaging protocols are poor. Surveillance regimens must reflect this to address radiation safety concerns. We propose the use of non-ionizing radiological follow-up for this cohort of patients.

#### MP-22.07

##### New Indication Approach for Partial Nephrectomy in Patients with RCC

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**Introduction and Objectives:** Nephrometry systems are good in predicting risk of complications but not indicating for nephron-sparing surgery (NSS) in RCC. Our objective was to develop a tool that will ease surgeons' choice of NSS treatment for RCC.

**Materials and Methods:** Retrospective analysis of 221 patients intended to undergo PN. Tumor size, location, R.E.N.A.L., remaining functional parenchyma volume (RFPV) of kidney with tumor and patients data were analyzed. RFPV calculated by formula:  $RFRV = (1 - [R_1 \cdot R_2 \cdot R_3 / 0.96 \cdot R_4 \cdot R_5 \cdot R_6]) \times 100\%$ , where  $R_1, R_2, R_3$  – kidney width, length and thickness;  $R_4, R_5, R_6$  – tumor sizes located within the kidney. RFPV was greater than 55% in all cases, but PN was performed in 171 (77.4%) and nephrectomy – in 50 (22.6%) patients.

**Results:** Comparing patients who underwent RN and PN showed no statistical difference in R.E.N.A.L. scores, but there was significant difference between lesions sizes and RFPV. Location analysis accounting for RFPV showed similar surgery type distribution between polar and lateral localizations (with RFPV more than 55% PN was performed in 133 [82.6%] of cases) and different results in medially located tumors, where RFPV did not show statistical dependency ( $p=0.13$ ). The main indication for PN in this situation was the size of the tumor. PN was performed in 92.3% with T1a lesions and only in 31.8% with tumors more than 4cm in size ( $p < 0.001$ ). Newly developed nephrometric system was named N.C.I.U.-nephrometry. Polar lines creating two segments U (upper) and I (inferior), third interpolar segment is divided by vertical line in interpolar segment forming N (nearness) - closer to pelvis and C (collateral). If tumor is located in one segment it is marked by capital letter of that segment and if there is a part of tumor growing into

other segment, it is marked with two letters, capital letter indicated majorly involved segment. Combining N.C.I.U. with RFPV gives information about location and percentage of remaining functional parenchyma.

**Conclusion:** Use of N.C.I.U. nephrometry with account of RFPV optimizes and eases surgeons' choice for PN in RCC. New indications for PN are lateral and polar tumors with RFPV more than 55%. In medially located lesions size less than 4 cm is the main indicator for PN.

#### MP-22.08

##### **Preoperative Risk Stratification for Cancer-Specific Survival in Patients with Renal Cell Carcinoma with Venous Involvement Who Underwent Nephrectomy**

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**Introduction and Objectives:** The aim of this study is to identify the preoperative prognostic factors and create a risk stratification model for patients with renal cell carcinoma (RCC) with extension into the renal vein (RV) or inferior vena cava (IVC).

**Materials and Methods:** The study cohort included 61 patients with RCC extending into the RV or IVC that underwent operations between 1993 and 2012. Cancer-specific survival (CSS) rates were estimated and univariate and multivariate analyses were carried out to determine the prognostic factors. A simple risk stratification model was developed for these patients.

**Results:** The median follow-up period of the current patient cohort was 33.7 months. Their 1, 3, and 5-year CSS was 89, 70, and 65%, respectively. On multivariate analysis, the level of tumor thrombus extension (extension into the supradiaphragm), presence of distant metastasis, and elevation of lactate dehydrogenase and C-reactive protein were independent negative prognostic factors for CSS. A simple scoring model to predict CSS after nephrectomy for RCC with venous involvement was developed using the regression coefficients from the multivariate analysis. The score was calculated as 2 (if the tumor extension into the supradiaphragm) + 1 (if distant metastasis) + 2 (if the level of LDH is abnormal) + 1 (if CRP is 5.0 mg/l or greater) and 0 (otherwise). The patients were divided into three groups according to the score: low- (0; n = 17), intermediate- (1-3; n = 26), and high-risk (4-6; n = 18) groups. CSS rates were clearly discriminated by the stratification according to the scoring model (p < 0.001). The concordance index of the new model was 0.80.

**Conclusions:** We demonstrated a simple risk stratification model with four preoperative independent prognostic factors for patients with RCC with venous involvement. This may be a useful decision-making model in the management of such patients.

#### MP-22.09

##### **Outcomes of Synchronous and Metachronous Bilateral Renal Cancers: A Population Based Study**

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**Introduction and Objectives:** Outcomes of bilateral renal cancers remains under reported especially for non-surgical management. The study reports a large population based cohort of patients with bilateral renal lesions.

**Materials and Methods:** Scottish Cancer Registry at Scottish National Services maintains datasets for all the renal cancers diagnosed since 1997. Altogether 10,200 patients with renal cancers were recorded during a period of 15 years. The incidence of bilateral cancer; both synchronous and metachronous (more than 6 months of unilateral lesion treatment) was calculated. Mean period for metachronous lesions were calculated. The large dataset was linked with departmental database to study in details the outcomes of patients with bilateral tumors including expectant treatment.

**Results:** A synchronous bilateral renal cell cancer was reported in 110 patients. A total of 40 metachronous bilateral cancers were recorded during mean follow-up of 8.7 y ± 7.3 years. Mean time for appearance of metachronous lesion was 62.32 ± 40.63 months (min.9.17; max 148.67). Majority of metachronous lesion had simultaneous appearance of metastatic disease. Radical nephrectomy remained the commonest procedure for synchronous lesions while nephron sparing surgical approaches were offered in smaller tumors. Younger age (<60), clear cell histology and size of primary tumor were significantly associated with development of metachronous lesions (p value 0.004). The increase in size of the synchronous lesion was 0.2cm/year compared to 1.46 cm/year for metachronous lesions.

**Conclusion:** Majority of bilateral renal masses are synchronous. Large tumor size and younger age had a higher risk for the development of metachronous lesions which significantly associated with metastatic disease elsewhere.

#### MP-22.10

##### **The Role of Image-Guided Renal Biopsy in Managing Small Renal Masses (SRM)**

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**Introduction and Objectives:** With the

expanding role of imaging in our aging population, incidental renal tumours are becoming increasingly prevalent. SRM's now represent 48-66% of renal cancers. We examine whether improved histopathological techniques change the potential role of diagnostic renal biopsies in determining appropriate management regimens. **Materials and Methods:** We prospectively collected data regarding all patients undergoing percutaneous image guided renal biopsy prior to nephron sparing surgery (NSS), to assess its use across our regional cancer network.

**Results:** Our cohort of SRM's ranged from 13-56 mm within a population aged 44-79 years. Of the 25 biopsied 68% were positive for malignancy, 20% benign and 12% failed (defined as an inadequate sample or technical fail). Overall 68% of patients were treated and 32% remain on radiological surveillance. Of those having NSS, the final histology matched the biopsy results in all but one case - where biopsy had shown "chromophobe" and histology confirmed "clear-cell" carcinoma. There were no recorded complications from the biopsies and no statistically significant difference in operative or ischaemic times between the biopsy and non-biopsy groups at the time of surgery. Subjectively, however, the dissection was felt to be more difficult than usual by the surgeon in 30% of the post-biopsy cases. The majority of patients also found that the biopsies themselves were more unpleasant than the subsequent surgery.

**Conclusions:** Image guided biopsy has a high success rate in terms of accurately targeting SRM's, and can be carried out safely to provide accurate histological results. There is no obvious negative impact upon clinical outcome, though its tolerability remains uncertain.

#### MP-22.11

##### **CCI-779 (mTOR Inhibitor) Alone Is More Effective than Combination with SB203580 (P38 MAPK Inhibitor) in RCC Cell Line**

Chauhan A, Singh S, Bhattacharyya S, Mandal A

Post Graduate Institute of Medical Education and Research, Chandigarh, India

**Introduction and Objectives:** Mammalian target of rapamycin (mTOR) induces Mitogen activated protein kinase (MAPK) which is involved in cell survival and proliferation. Combined inhibition of both the pathways caused increased apoptosis in prostate cancer. Present study was conducted to evaluate the effect of combination of mTOR inhibitor with MAPK inhibitor in renal cell cancer (RCC).

**Materials and Methods:** The effect of mTOR inhibitor (CCI-779) and p38 MAPK inhibitors (SB203580) alone and in combination was evaluated on RCC cell line, A-498. Subtoxic concentrations of inhibitors were selected by MTT assay. Cell viability was measured by



Annexin V/PI staining, sub-G1 population, ROS production and anchorage independent growth was measured by soft agar colony formation assay.

**Results:** Cells treated with CCI-779 increased the apoptosis from 10% to 38% which was found to be statistically significant ( $p < 0.05$ ). Treatment with SB203580 showed 16% apoptosis. Combined treatment with CCI-779 and SB203580 showed 26% apoptosis (Figure 1). Cells treated with CCI-779 showed higher ROS and lower anchorage independent growth as compare to SB203580 alone or in combination treated cells (Figure 2).

**Conclusion:** CCI-779 is more effective in RCC as compared to its combination with p38 MAPK inhibitor. This appears to be due to a complex crosstalk involved between mTOR

and MAPK pathways.

**MP-22.12**  
**Can Pre-Nephroureterectomy**  
**Ureteroscopy Reliably Diagnose**  
**Urothelial Carcinoma *in situ*?**

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**Introduction and Objectives:** Endoscopic management is expanding in upper tract urothelial carcinoma (UT-UCC). Reliable grading and staging still remains an essential element and presence of concomitant carcinoma *in situ* (CIS) on pre-nephroureterectomy ureteroscopic (PNU) will change management options. Whether white light PNU can

reliably diagnose concomitant CIS remains unproven. We review pre-operative endoscopic biopsy results with histology obtained at radical nephroureterectomy.

**Materials and Methods:** Patients who underwent radical nephroureterectomy at Ninewells Hospital, Dundee for UT-UCC between January 1998 and December 2012 were identified from our database Tayside Urological Cancers Network (TUCAN). PNU appearances, biopsy stage and grade were compared with final histology results of radical nephroureterectomy (RNU) for UT-UCC to assess the diagnostic accuracy of PNU for carcinoma *in situ* (CIS).

**Results:** A total of 138 patients underwent RNU for UT-UCC. Fifty six (40.6%) of these patients were evaluated via PNU using white light and 47 (83.9%) had a PNU biopsy (Group 1). Eighty two patients (59.4%) did not have PNU (Group 2). Post-nephroureterectomy histology showed CIS in 36 (26.1%) patients, 12 (33.3%) of these patients belonged to Group 1 and 24 belonged to Group 2. PNU, however failed to detect or suspect CIS in all 12 patients in Group 1. In fact, one patient had CIS on PNU biopsy which was not confirmed on the final RNU histology (false positive). Patients with a diagnosis of CIS showed shorter recurrence free survival, however this was not statistically significant ( $p=0.337$ ). Overall survival was also shorter in patients with CIS compared with those without; this showed strong statistical significance ( $p=0.004$ ). Although on multivariate cox proportional hazards model CIS was not found to be independently associated with survival ( $p=0.105$ ).

**Conclusions:** PNU using white light significantly under-diagnoses upper tract CIS and there is an urgent need for evaluating adjunct techniques (photodynamic diagnosis, narrow band imaging and autofluorescence) to detect this entity, especially in selecting patients for endoscopic management.

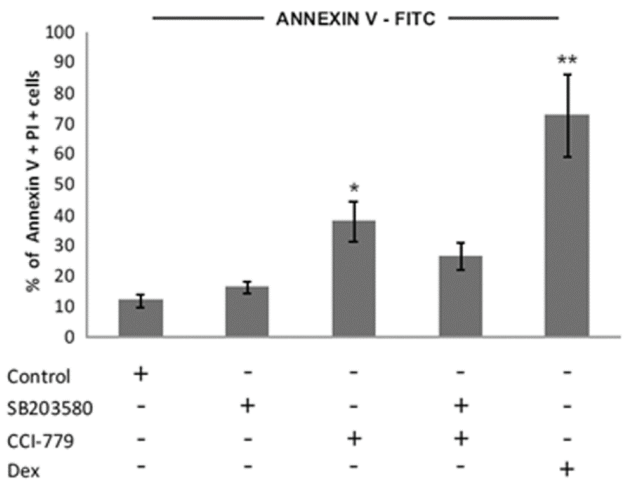
**MP-22.13**  
**Is It Time to Replace Ultrasound**  
**with Computed Tomography (CT)**  
**Urography as the Initial Investigation**  
**in Patients with Visible Haematuria?**

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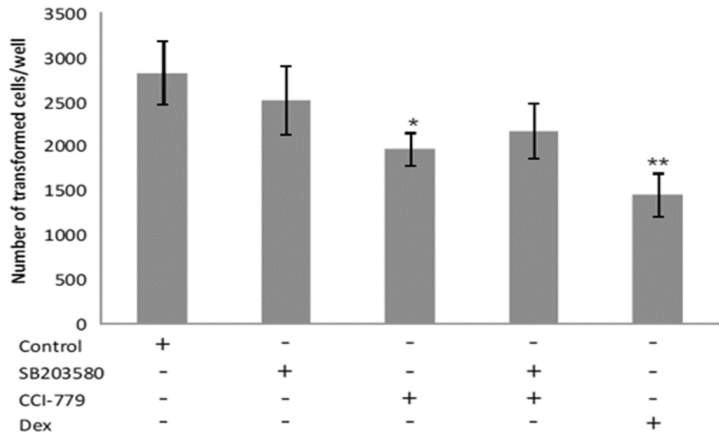
**Introduction and Objectives:** We investigated the role of CT urography (CTU) in patients with visible haematuria in a haematuria clinic. We focused on the detection of urological malignancies particularly upper tract transitional cell carcinomas (UT-TCC). The time delay from ultrasound kidneys, ureters and bladder (USS-KUB) to CTU was measured.

**Materials and Methods:** The data from 214 patients with visible haematuria between December 2012 and July 2013 was analysed. Patients underwent USS- KUB and cystoscopy;

**MP-22.11, Figures 1 and 2.**



**Fig. 1:** Effects of inhibitors in A-498 cells. (A) show cells stained with FITC-annexinV and PI (apoptotic cells) (B) Quantitative representation of apoptosis induced by inhibitors. Data is presented as mean  $\pm$  SD of three independent experiments. \*,  $p < 0.05$  compared to control. \*\*,  $p < 0.001$  compared to control.



**Fig. 2:** Anchorage independent growth of A-498 cells in A-498 cells. Dex= Dexamethasone as positive control. Data is presented as mean  $\pm$  SD. \*,  $p < 0.05$  compared to control. \*\*,  $p < 0.001$  compared to control.

CTU was offered to patients with negative USS-KUB or if otherwise clinically indicated. All patients diagnosed with high risk and invasive bladder cancers underwent upper tract evaluation.

**Results:** The mean age of patients was 63.6 yrs of which 76% were males. An USS-KUB was performed on 208 patients (97.2%) and 164 patients (76.6%) underwent a CT scan. Five UT-TCCs and three renal cancers were noted exclusively on CT. Additionally, CT provided upper tract evaluation for high grade bladder cancers (6), staging for muscle invasive bladder cancer (4) and in patients with tumours overlying the ureteric orifice (6). A malignant rectal cancer was noted incidentally on CT. CT also found 7 ureteric calculi necessitating urgent treatment, 15 significant renal stones of which 6 needed early treatment, and 4 complex renal cysts (Bosniak  $\geq 2f$ ) requiring follow-up. Other incidental lesions included an angiomyolipoma, a renal abscess, abdominal aortic and iliac aneurysms (3), a hiatus hernia, retroperitoneal lymphadenopathy (1) and an inferior vena caval thrombus. CT detected significant urological pathology and provided upper tract evaluation in 42 pathologies (22.0% of patients), and found 4.9% of patients to have other significant pathologies requiring follow-up or treatment. The median time delay from USS-KUB to CT was 50 days for cancers.

**Conclusion:** The majority of patients require a CT scan. CT detects additional cancers and provides superior delineation of the upper tract highlighting relevant information for high risk and invasive bladder cancers. Performing an initial CT instead of an USS-KUB shortens time delays by 7 weeks. This study provides strong evidence for replacing USS-KUB with CTU as the initial imaging of choice for patients with visible haematuria.

#### MP-22.14

##### **High Expression of GOLPH3 Defines Poor Prognosis in Non-Muscle-Invasive Bladder Cancer and in Patients with Invasive Bladder Cancer Undergoing Radical Cystectomy**

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**Introduction and Objectives:** Conventional clinicopathologic risk factors have failed to accurately predict the prognosis of patients with bladder cancer (BC). This study aimed to evaluate Golgi phosphoprotein 3 (GOLPH3) expression as a progression marker in patients with non-muscle-invasive BC (NMIBC) treated by conservative methods and as a prognostic marker in patients with invasive BC undergoing radical cystectomy (RC).

**Materials and Methods:** Two different tissue microarrays were constructed, one with

124 primary Ta/T1 tumours from patients treated by transurethral resection of the bladder and one with 92 tumours from RC patients. GOLPH3 expression was examined by immunohistochemistry.

**Results:** A high GOLPH3 expression in Ta/T1 patients was significantly correlated with a higher risk of progression that was independent of conventional risk factors in multivariate analysis. In patients undergoing RC, a high GOLPH3 expression was an independent predictor of poor prognosis. A high GOLPH3 expression was correlated with a higher risk of visceral metastasis rather than lymphatic spread. **Conclusion:** GOLPH3 expression is a marker for progression of NMIBC and a prognostic marker in patients undergoing RC.

#### MP-22.15

##### **Prognostic Factors for Poor Oncological Outcomes of Upper Tract Urothelial Carcinoma (UTUC) in the Chinese Population: 14-Year Experience from 2 Tertiary Centres**

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<sup>1</sup>Princess Margaret Hospital, Hong Kong, Hong Kong; <sup>2</sup>Queen Mary Hospital, Hong Kong, Hong Kong

**Introduction and Objectives:** To identify the prognostic factors for poor oncological outcomes, in terms of bladder recurrence, local recurrence, distant metastasis and 5-year cancer-specific survival (CSS), of UTUC in the Chinese population from 2 tertiary referral hospitals in Hong Kong.

**Materials and Methods:** Predictive factors were analyzed by retrospective review of all patients who were treated with nephroureterectomy (NU) or segmental ureterectomy (SU) for UTUC in two tertiary referral centres in Hong Kong from April 1999 to May 2013. Prognostic factors evaluated included three categories: pre-operative/ clinical factors, intra-operative/ surgical factors, and post-operative/ pathologic factors. Univariate and multivariate logistic regression analyses were used.

**Results:** A total of 124 patients who received NU (n=104, 83.9%) or SU (n=20, 16.1%) were reviewed. 72 (58%) patients had pathologic T2 or higher stage tumors. Overall survival was 57% at a median follow-up of 50 months and the actuarial CSS at 5 years was 77%. Both higher T staging (HR 1.841,  $p=0.014$ ) and lymphovascular invasion (HR 2.803,  $p=0.025$ ) were predictors for cancer related deaths. Fifty four (43.5%) patients developed bladder recurrence at a median interval of 8.5 months. Previous UTUC (HR 3.991,  $p=0.01$ ), multifocal tumors (HR 2.015,  $p=0.049$ ) and concurrent bladder tumor (HR 2.252,  $p=0.016$ ) were predictive factors for bladder recurrence in multivariate analysis. Local recurrence occurred in 8 (6.5%) patients

only in our series and analysis was therefore not performed. Thirty three (26.6%) patients had distant metastasis at a median interval of 13 months with T staging (HR 1.666,  $p=0.029$ ) being the only predictor for distant metastasis in multivariate analysis.

**Conclusion:** Higher T-staging and lymphovascular invasion were poor prognostic factors for CSS for UTUC in the Chinese population, while bladder recurrence tends to occur early. Patients with previous UTUC, multifocal tumors and concurrent bladder tumors should be closely followed-up for bladder recurrence after surgery.

#### MP-22.16

##### **Stratifying Cystoscopic Surveillance following Nephroureterectomy: Predictors of Recurrence according to Upper-Tract Tumour Characteristics**

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**Introduction and Objectives:** The current standard of care for resectable Upper Tract Urothelial Carcinomas (UTUC) is nephroureterectomy; however, post-operative bladder cancer (BCa) recurrence is common and quoted as high as 51%. The European Association of Urology have recommend follow-up cystoscopy at 3 months following surgery and then annually for 5 years. We present a study to assess whether primary UTUC pathological characteristics predict BCa recurrence and investigate whether current guidelines provide appropriate follow-up in this patient population.

**Materials and Methods:** BCa recurrence rates in patients undergoing nephroureterectomy over a period of 10 years in a single institution were reviewed. Pathological characteristics of the primary UTUC (including grade, stage, size, location, invasiveness) multiplicity and associated history of previous BCa were mapped against timing and stage of bladder recurrences. The minimum follow-up time was 12 months.

**Results:** Sixty-six patients had nephroureterectomy for UTUC. Six were excluded from analysis due to inadequate follow-up. The bladder recurrence rate was 26.7% (18 patients). There was no correlation between recurrence rate and upper tract tumour characteristics on single or multivariate regression analysis. Median time to bladder recurrence was 6.5 months (inter-quartile range 5.2-10.8) following nephroureterectomy for patients with high grade UTUC primary compared to 14.2 months (6.8-22.6) for grade one and two UTUC. This was not influenced by a prior history of BCa. In addition, there were 8 high-grade BCa recurrences (44%) all of which were in patients with high grade primary UTUC.

**Conclusion:** Histological grading of UTUC may help predict the timeframe and grade

of BCa recurrence. Patients with high-grade UTUC should be considered for more frequent cystoscopic surveillance than guidelines currently suggest; this is particularly so during the first year following nephroureterectomy.

#### MP-22.17

##### **The Curative Potential of Lymphadenectomy after Response to Chemotherapy in Patients with Urothelial Carcinoma Presenting with Regional or Distant Nodal Metastases: Analysis of a Series from a Tertiary Cancer Center**

Necchi A, Lo Vullo S, Giannatempo P, Farè E, Nicolai N, Piva L, Biasoni D, Torelli T, Catanzaro M, Stagni S, Crestani A, Colecchia M, Paolini B, Gianni A, Mariani L, Maffezzini M, Salvioni R

*Fondazione IRCCS Istituto Nazionale dei Tumori, Milano, Italy*

**Introduction and Objectives:** The available information would suggest a benefit from surgical removal of metastatic disease in selected patients with urothelial cancer, but the heterogeneity of data and the limited sample size hampered the level of evidence. We aimed to analyze the contribution of post-chemotherapy (CT) lymphadenectomy only on survival outcomes in responding patients from our center.

**Materials and Methods:** Between 1986 and 2012, 157 patients with locally advanced or metastatic urothelial cancer received first-line combination of methotrexate, vinblastine, doxorubicin, and cisplatin (MVAC). Of them, patients experiencing at least a stable disease of subdiaphragmatic nodal disease/local recurrence only were selected. For the sake of parsimony, the prognostic effect of singly taken covariates (surgery of tumor primary, site of nodal disease, extent of nodal sites [single vs. multiple]) upon survival was investigated using Cox proportional hazard regression models, with and without adjustment by treatment group (post-CT surgery vs. observation).

**Results:** Fifty nine patients were identified, 31 (52.5%) had regional nodes and 28 (47.5%) had metastatic disease. Forty two (71.2%) had multiple nodal sites, 15 patients (25.4%) had an upper tract tumor primary, 24 (40.7%) had received major surgery. Twenty-eight patients underwent post-chemotherapy pelvic (N=14) or retroperitoneal lymphadenectomy (N=14) after achieving a complete response (CR, N=7) or a partial response-stable disease (PR+SD, N=21). Eight out of 28 patients (28.6%) achieved a pathologic-CR. Median follow-up was 88 months (IQR: 24-211). Median progression-free (PFS) survival by treatment group (surgery vs. observation) was 18 (95% CI, 11-N.E.) and 11 (95% CI, 5-19) months, respectively (logrank test  $p=0.009$ ). Median overall survival (OS) was 37 (95% CI, 20-N.E.) and 19 (95% CI, 9-38) months, respectively

( $p=0.004$ ). Surgical consolidation was associated with better PFS (HR: 0.43, 95% CI, 0.22-0.84,  $p=0.013$ ) and OS (HR: 0.36, 95% CI, 0.17-0.76,  $p=0.007$ ) in univariable analysis (UVA). This was the only significance in UVA and it was retained in multivariable analysis when adjusting for each of the other covariates. No effect of pathologic status was found. Results are limited by small numbers.

**Conclusion:** In well selected patients with UC like those achieving a clinical benefit from chemotherapy and having exclusive nodal metastatic disease there was a clear survival advantage when removing disease residuals.

#### MP-22.18

##### **A Large Single Center Study Describing the Evolution of Renal Tumor Biopsy over a 13-Year Period**

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<sup>1</sup>Princess Margaret Cancer Centre, University Health Network, Toronto, Canada; <sup>2</sup>Toronto General Hospital, University Health Network, Toronto, Canada

**Introduction and Objectives:** The incidence of small renal masses (SRMs) has increased in recent years. SRMs are defined as enhancing solid lesions measuring  $\leq 4$ cm. We present our experience with percutaneous renal tumor biopsy (RTB) of SRMs, which to our knowledge, is the largest to date.

**Materials and Methods:** We reviewed database to ascertain the incidence of benign, malignant and non-diagnostic biopsies of SRMs performed in our institution from 2001 to December 2013. A total of 585 RTBs were performed (560 SRMs). The diagnostic rate was compared over time. The accuracy of the biopsies was correlated with the definitive pathology with regard to histology and grading (for conventional cell carcinoma only) in the cases where the biopsy led to surgery.

**Results:** The mean size of the 560 biopsied lesions was 2.5 cm (Interquartile range (IQR) 1.8-3.2). The diagnostic rate at the 1<sup>st</sup> biopsy was 89.8% (n=503). Of the diagnostic biopsies, 124 (24.7%) were benign and 379 (75.3%) were malignant. Of the 57 non-diagnostic biopsies, 25 (43.9%) had a repeat biopsy. A diagnosis was obtained in 21 cases [84.0%; 10 (47.6%) benign, 11 (52.4%) malignant]. Therefore RTB led to a diagnosis in 524 of the 560 SRMs (93.6%). Over time, there was a decrease in the rate of non-diagnostic biopsies from 17.8% (2001-2005) to 9.1% (2011-2012;  $p=0.09$ ). Of the available surgical pathology (n=171), there was agreement between histology in 93.6% of the cases (kappa 0.89, 95% CI 0.83-0.96). For the conventional cell carcinoma (n=102), there was concordance between Fuhrman grades in 94.3% of the cases (kappa 0.70, 95% CI 0.48-0.92).

**Conclusions:** RTB of SRMs provided a

diagnosis in 93.6% of the cases. Surgery was avoided in almost 25% of patients following RTB because of benign histology. The rate of non-diagnostic biopsies diminished over time. Finally, there was a high correlation between the biopsy and the final pathology with regards to both histology and grade. In an era where overdiagnosis and overtreatment of favorable cancers is gaining attention, routine RTB for SRM leads to diminished intervention and going forward personalization of care.

#### MP-22.19

##### **Evolution of Operative Time and Intraoperative Bleeding in the First 100 Laparoscopic Partial Nephrectomies at an Argentinean Community Hospital**

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**Introduction and Objectives:** Minimally invasive partial nephrectomy is in continuous development. This technique is currently the best option for the treatment of localized renal tumors. However, it requires hard training and a long learning curve. Surgeon's skill directly affects surgical time, ischemia time and intraoperative bleeding. Also surgeon's ability may determine case selection.

**Materials and Methods:** We present our first 100 laparoscopic partial nephrectomies (LPN) performed between November 2007 and March 2014 in a single center by the same surgeon or by residents at his care. Surgical time and intraoperative bleeding is compared between the first 50 (Group A) and second 50 (Group B) laparoscopic partial nephrectomies. We point out the cases that didn't require vascular clamping and the T1b tumors operated in each group.

**Results:** Mean surgical time decreased from 128 minutes in Group A to 101 minutes in Group B (21%). A total of 5 and 8 T1b tumors were operated in Groups A and B respectively. Mean warm ischemia time decreased from 17.7 minutes in Group A to 13.7 minutes in Group B (22%). A total of 12 and 9 surgeries were performed without vascular clamping in Groups A and B respectively. Mean operative bleeding decreased from 141 ml in Group A to 75 ml in Group B (47%). Three patients required blood transfusion on Group A. In 2 cases of Group A, laparoscopic radical nephrectomy had to be performed due to secondary bleeding during immediate postoperative.

**Conclusions:** Surgical time, warm ischemia time and intraoperative bleeding have been significantly reduced by 21, 22 and 47% respectively between the first 50 and second 50 laparoscopic partial nephrectomies. There was no need for transfusions or laparoscopic radical nephrectomies in the second 50 surgeries.

**MP-22.20****Complete Ureterectomy:  
Necessary or Optional?**

Nemade H<sup>1</sup>, Cuckow P<sup>2</sup>, Mushtaq I<sup>2</sup>, Smeulders N<sup>2</sup>, Cherian A<sup>2</sup>

<sup>1</sup>Basildon and Thurrock University Hospital, Basildon, UK; <sup>2</sup>Great Ormond Street Hospital, London, UK

**Introduction and Objectives:** The aim of this study was to assess the incidence and risk factors for residual ureteric stump excision following hemi and total nephrectomy in children.

**Materials and Methods:** Retrospective review of children who underwent hemi-nephrectomy/nephrectomy over 10 years was carried out. Indications for stump excision were noted. Follow-up included clinical review, ultrasound, MAG-3 renogram and bladder function assessment as required.

**Results:** Hemi-nephrectomies (total-76, Laparoscopic-16) and nephrectomies (total-116, Laparoscopic-30) were reviewed. Mean age at primary procedure: 30 months (range=3-90) and mean-interval from primary procedure for residual stump excision: 36 months (range=12-72). Residual stump excised: hemi-nephrectomies – 7 (9%) and total nephrectomies – 3 (2.5%) with resolution of symptoms in all. Male:female ratio (2:8); Seven duplex and 3 simplex renal units; Eight (2 simplex) were refluxing and 2 ectopic ureters. Indications for stump excision were recurrent urinary tract infection (UTI) (n=6), persistent vaginal discharge (n=2), voiding dysfunction (VD) with incomplete emptying (n=1) and UTI with VD (n=1). All had persistently dilated ureteric stump on ultrasound before re-intervention. Ureterocele excision (n=2) and ureteric reimplantation (n=1) of contralateral side was done along with stump excision. Interestingly in the re-intervention group all had open primary procedure. One of 10 stumps was excised laparoscopically.

**Conclusion:** Subtotal ureterectomy remains the policy in our institution, however children with risk factors such as reflux and duplex systems will require follow-up for at least 5 years. In simplex units every attempt must be made to excise the refluxing ureter completely and this could be facilitated laparoscopically.

**MP-22.21****Complete Off-Clamp Zero Ischemia  
Partial Nephrectomy without the  
Exposure of Renal Artery under Normal  
Blood Pressure: Initial Experiences  
Have Revealed Short Operation Time**

Yanaihara H, Ogihara K, Kaguyama H, Hanashima F, Sakamoto H, Aonuma K, Matsuda K, Nakahira Y, Asakura H  
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**Introduction and Objectives:** Complete off-clamp Zero Ischemia partial nephrectomy (ZIPN) is one of the most challenging procedures among several variation of partial nephrectomy. The exposure of renal artery is usually obligated to prepare shutting the blood stream in emergency situation. But it has been already reported that partial nephrectomy can be done without massive bleeding for small tumors. We tried Zero Ischemia partial nephrectomies without the exposure of renal hilum.

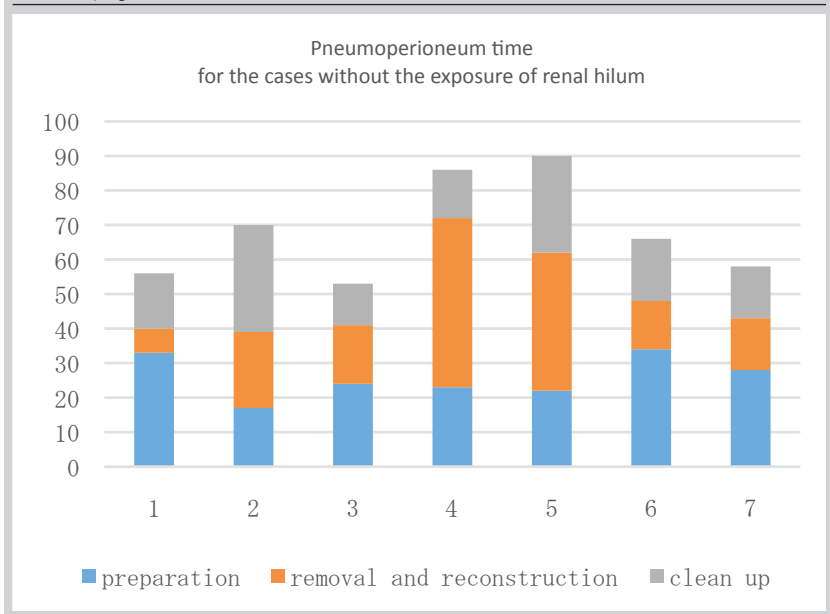
**Materials and Methods:** Subjects are ten T1a tumors and one T1b tumor. Tumor size was 12-44 mm (median 20). RENAL score was from

4 to 8 (median 6). All tumors were removed without clamping renal artery. The exposure of renal artery was not done in seven cases. Renal artery was exposed to prepare clamping for the first case, T1b tumor case and two retroperitoneal approach cases.

**Result:** Blood loss was 0-280 ml (median 40 ml). Pneumoperitoneal time was 53-127 min (median 70 min). In the cases without the exposure of renal artery, Pneumoperitoneal time was 53-90 min (median 70 min). 7-50 min (median 22 min) was taken from the removal of tumor to the end of renorrhaphy. All procedure was completed without any complication.

**Conclusion:** Complete off-clamp ZIPN could be done under normal blood pressure. The exposure of renal hilum can be omitted safely. It made the procedure simple, which can contribute to shorten the operation time of partial nephrectomy. Besides, laparoscopic ZIPN has an advantage of pneumoperitoneum pressure to reduce blood loss from incised parenchyma. From our experience, this procedure could be a valuable option.

MP-22.21, Figure 1.





## Moderated Poster Session 23

### Training and Education

Wednesday, October 15  
1300-1430

#### MP-23.01

##### The Importance of Patient and Physician Education in Public Health Campaigns: Haematuria, a Case in Point

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<sup>2</sup>Imperial College London, UK

**Introduction and Objectives:** Public health campaigns seek to raise public awareness of a given condition, prompting presentation usually at a primary healthcare provider. A recent public health campaign in the United Kingdom focused on haematuria, instructing members of the public to seek medical advice if they had experienced an episode of 'blood in the pee'. This campaign ran over October and November 2012. In this study we examine the relationship between haematuria referrals to a large hospital trust, with a catchment of over 2 million people, and the number of new diagnoses of renal and urothelial malignancy.

**Materials and Methods:** Data regarding new referrals to the trust's dedicated haematuria clinics and new renal and urothelial cancer diagnoses were obtained for the period February 2012 to January 2014. Paired t-tests were undertaken comparing referral and diagnoses for the February 2012 to January 2013 and February 2013 to January 2014 time periods. The time periods

were split into pre-campaign (Feb-Sept) and post-campaign (Oct-Jan) cohorts for analysis. **Results:** When data analysis was performed a trend towards a significant increase in the number of referrals, month-on-month, post the start of the public health campaign was seen ( $p = 0.056$ ) compared to a non-significant difference in the number of pre campaign referrals ( $p = 0.46$ ). This increase in referrals was not associated with a concurrent increase in cancer diagnoses ( $p = 0.84$ ) (see Figure 1). **Conclusions:** Although public health campaigns are invaluable in increasing public awareness of pathology they must be accompanied by appropriate education of healthcare professionals to ensure that initial investigation are carried out in the most appropriate healthcare setting.

#### MP-23.02

##### Impact of the Lack of Community Catheter Care Services on Emergency Department

Tay L, Lyons H, Raison N, Khan A, Taylor C, Thompson P

King's College Hospital, London, UK

**Introduction and Objectives:** Although catheter care is provided by the community nurses, long-term catheter (LTC) problems are a common presentation to the emergency departments (ED) and are often a significant burden to the services. We carried out an audit to assess the patients presenting with LTC associated problems to our ED and to identify the availability of community nursing support for their urinary catheter.

**Materials and Methods:** A prospective audit was carried out over two separate periods of all

patients presenting to ED with a urinary catheter problem.

**Results:** A total of 78 patients presented to ED over a cumulative period of 69 days with 42% of them ( $n=33$ ) presenting between 9-5pm. The mean age was 74 years (range 42-93). Median duration of urinary catheter in situ was 11 months (range 1-120). The most common reasons for attendance were blocked catheter (40%,  $n=31$ ) and catheter-bypass (23%,  $n=18$ ). Only 28% ( $n=22$ ) had community nurse involvement. The majority of the remaining patients self-referred to ED. Fourteen (18%) needed urological intervention during acute presentation as their catheters could not be replaced by ED nurse or doctors.

**Conclusion:** The high morbidity of long-term catheter causes a considerable demand on ED services. The majority of patients had minimal community nurse input, although their catheter problems were easily dealt with by ED. Hence, more community nurses need to be trained in the management of the catheterized patient. We also need to improve the access to community support and ensure patients are informed of their available options, instead of defaulting to ED.

#### MP-23.03

##### Quality of Testosterone Replacement Prescribing and Monitoring in the Dayton, Ohio, USA Veterans Affairs Medical Center

Moore D<sup>1</sup>, Ullah S<sup>2</sup>, Bansal A<sup>2</sup>, Gupta A<sup>1,2</sup>

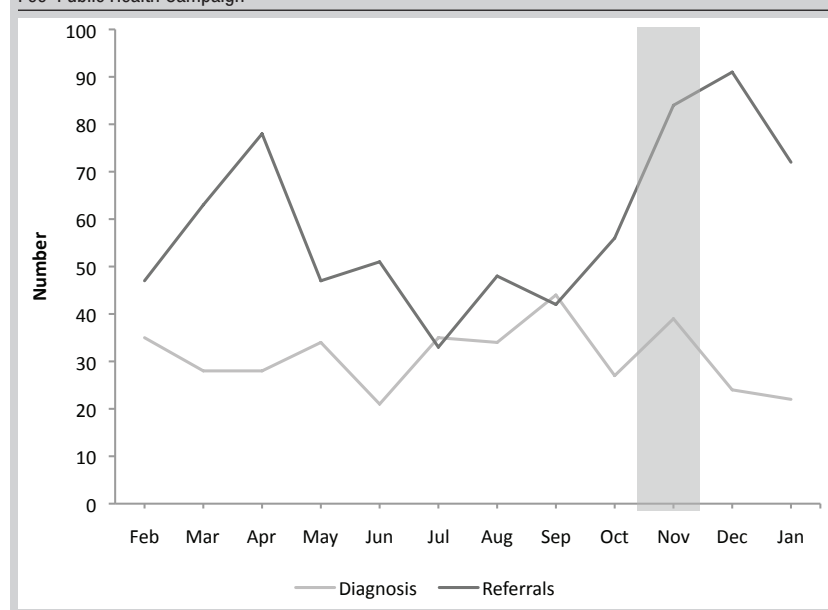
<sup>1</sup>VAMC, Dayton, USA; <sup>2</sup>Wright State Medical School, Dayton, USA

**Introduction and Objectives:** Providers are replacing Testosterone (T) in men 50+ years of age due to ill-defined conditions. We noted men in the private sector and at our facility were started on T replacement without recommended labs and screening. We also found dosing and laboratory follow up was not appropriate. We sought to enable providers to appropriately identify patients for T replacement and ensure it was done safely under current guidelines.

**Materials and Methods:** We developed a template to educate and assist in prescribing and monitoring of T replacement using the Endocrine Society's guidelines. Retrospectively, 100 patients receiving injectable T replacement were reviewed comparing pre template vs post template for quality of prescribing and monitoring.

**Results:** Men age 33 to 77 underwent T replacement. Body mass index ranged from 21.70 to 52.60. Among all the patients, 91 were white, 9 were of African descent. Pre-template data is as follows: 31 patients had 2 separate AM testosterone levels drawn prior to treatment, 69 did not. Three patients had LH, FSH, and prolactin levels drawn, 97 did not. Seventy five had a prostate specific antigen lab within one year, 25 did not. Twenty had a digital rectal exam (DRE) before treatment, 80 did not. Forty three

**MP-23.01, Figure 1.** Total Number of Patients Referred with Haematuria and the Number of New Diagnoses of Renal or Urothelial Carcinoma: Shaded Area Represents the Duration of the 'Blood in the Pee' Public Health Campaign



patients had the correct dosage of T prescribed, 50 did not. Forty eight patients had T levels checked post initiation of T replacement while 52 did not. Overall, pre-template evaluation showed 100% lack of compliance to guidelines before initiating T replacement. Post template prescribing was complete on all parameters but the DRE. Data to verify appropriate monitoring post T replacement is in progress.

**Conclusion:** Testosterone replacement promises the “fountain of youth”. T replacement has become ubiquitous worldwide. Risks remain unknown in the over 50 year old population. We believe that we need to closely adhere to guidelines to not only correctly identify men that will benefit from T replacement but also prevent untoward events. We have found that the quality of prescribing is enhanced by a prescribing template. Evaluation of quality of monitoring post template is underway.

#### MP-23.04

##### Variability in the Performance of Urology Multidisciplinary Teams: What Can We Learn?

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**Introduction and Objectives:** Multidisciplinary team meetings are a mandatory component of cancer care in the UK. Their effectiveness is dependent upon quality of information exchanged and communication amongst team members. We assessed how these variables differed across time within one Urology Multidisciplinary Team (MDT) and their impact on decision-making.

**Materials and Methods:** This was a cross-sectional, observation study with prospective data collection over 10 weeks on 382 consecutive patients discussed in a Urology MDT. Two trained observers used a validated assessment tool (MDT *MODE*) to capture data in real-time on the quality of information presented and quality of discussion by MDT members (min score 1, max 5). Variability of outcomes across the 10-week study period was assessed using Kruskal-Wallis tests (all  $p < 0.05$ ).

**Results:** Mean number of cases discussed per meeting was 38. Mean time per case was 2.5 minutes. A mean of 11 core MDT members were present per meeting. Kruskal-Wallis showed significant variability between 10 consecutive meetings ( $N=382$ ) on psychosocial aspects ( $H(9)=34.54$ ,  $p < 0.001$ ) and comorbidity ( $H(9)=34.54$ ,  $p < 0.001$ ), as well as on chair's ( $H(9)=87.52$ ,  $p < 0.001$ ), oncologist's ( $H(9)=18.17$ ,  $p < 0.05$ ), nurse's ( $H(9)=50.95$ ,  $p < 0.001$ ) and radiologist's ( $H(9)=24.7$ ,  $p < 0.01$ ) contributions. In addition, significant variability was also evident for the outcome variable, i.e., decision made ( $H(9)=26.78$ ,  $p < 0.01$ ). In

contrast, variability appeared stable for patient history, x-ray information, pathological information, patient views, as well as surgeon and coordinator's contributions (all  $p > 0.05$ ).

**Conclusion:** This study showed that quality of information exchange, communication and decision-making is variable within urology MDTs across a period of time. Understanding these differences may help standardise best practices and identify areas to target improvement efforts to optimise cancer care for patients in urology.

#### MP-23.05

##### Integrated Skills Development for Ureteroscopy: Development and Validation of a Curriculum

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MRC Centre for Transplantation, King's College, London, UK and Dept. of Urology, Guy's Hospital, London, UK

**Introduction and Objectives:** Ureteroscopy training modalities available contain sufficient validation evidence for utilisation within training. However, a structured and proficiency based curriculum is required. We therefore aimed to:

1. Develop a simulation based ureteroscopy curriculum integrating technical and non-technical skills teaching.
2. Validate the curriculum in terms of feasibility, educational impact and content validity.

**Materials and Methods:** The developed proficiency based curriculum was divided into four modules: 1. Knowledge. 2. Technical Skills. 3. Integrated skills development. 4. Non-technical skills. Three modalities of training were incorporated into this including a virtual

reality simulator, a bench-top model and the Distributed simulator, an inflatable high-fidelity training environment. Content validation of the developed curriculum was sought from 7 experts from 3 separate institutions. Thirty two ureteroscopy naïve medical students from 7 medical institutions (average age 20.8) were recruited. The study protocol is highlighted in Figure 1. Time to completion and subjective questionnaires were utilised as outcome measures.

**Results:** A total of 86% of experts agreed the content of the curriculum was appropriate and 100% of respondents agreed that integration of non-technical skills via the distributed simulator was useful. Time to completion of the task was significantly lower within the group that had undertaken the simulation-based curriculum (932 vs. 1616 seconds,  $p < 0.001$ ). On a 10-point likert scale, feasibility of delivery was rated at an average of 9.3 and content of curriculum was rated at 9.1. One hundred percent of participants believed the addition of the full immersive environment was useful for the development of key non-technical skills.

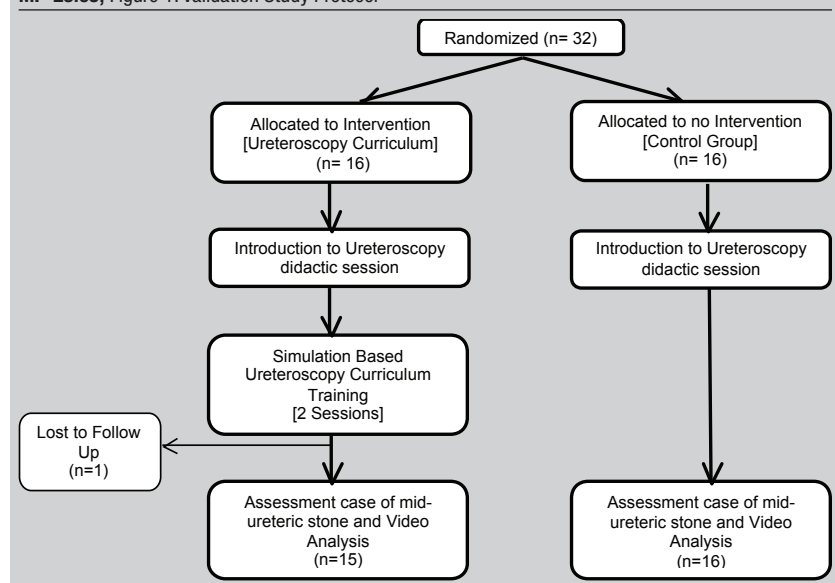
**Conclusion:** This preliminary data suggests the developed curriculum is a feasible and acceptable modality for both technical and non-technical skills training with a proven educational impact. This curriculum could be adapted and used as a framework for the teaching of other procedures for a standardized and feasible simulation based teaching program.

#### MP-23.06

##### Face Validity Study of Cadavers Using Thiel Method of Embalming for Endoscopic Surgery in Urology

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MP-23.05, Figure 1. Validation Study Protocol



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**Introduction and Objectives:** Human Cadavers forms the cornerstone of surgical training. Previous embalming techniques have however failed to display endoscopic anatomy adequately. Thiel Embalming method is a novel technique, which maintains normal tissue consistency with lifelike display of endoscopic anatomy. The study describes face validity of ureterorenoscopy (URS) and transurethral resection of prostate (TURP) in human cadaveric models embalmed using Thiel's method.

**Materials and Methods:** The University of Dundee has developed a facility for embalming cadavers using Thiels method for embalming specifically to address challenge of surgical skills training and research. These cadavers were used as models for assessing suitability for endoscopic (upper and lower urinary tract) skills training. Eleven practicing urologists (4 experts and seven trainees) were recruited to evaluate this model. Respondents were asked to perform part of whole procedure and complete a rating survey (Linkert scale 1 to five) upon completion specifically assessing positioning, tissue quality, resemblance to normal and preservation anatomical landmarks.

**Results:** In using a Likert scale between 1 and 5, results for anatomic preservation and resemblance to "real-life" were favourable, with the best scores for overall anatomical landmarks for ureteroscopy and stent insertion (4.7) and for patients positioning and endoscopic placement of instruments (4.5). The poorest scores were for the use of Thiel specific fluid for endoscopic resection (3.3), however this was improved using alternate fluid for irrigation. Scores for haptic realism were good as well. The average score for the question "overall, how valuable is the model as a surgical simulator" was 4.1. The experts felt that trainees (specialist trainees year 1-3) would benefit most from this surgical education model.

**Conclusions:** The cadavers using Thiel's method for embalming can be considered to have face validity for lower and upper tract endoscopy. Improvements will continue to be made to address some of the deficiencies in the type of fluid to be used for endoscopic resection using this model.

#### MP-23.07

##### Development and Content Validation of a Safety Checklist for Robot Assisted Radical Prostatectomy

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Institute, Buffalo, USA; <sup>4</sup>OLV Robotic Surgery Institute, Aalst, Belgium; <sup>5</sup>Depts of Urology and Nephrology & Renal Transplantation, Guy's & St Thomas' Hospital, London, UK; <sup>6</sup>Dept. of Urology, Netherlands Cancer Institute, Amsterdam, The Netherlands; <sup>7</sup>Henry Ford Hospital, Detroit, USA

**Introduction and Objectives:** Novel technologies require up-to-date technical and non-technical skills from surgeons to prevent adverse effects. Education and assessment of trainees must be adapted to equip surgeons with these skills, ensuring safe practice. This study aims to develop and validate a checklist via Healthcare Failure Mode and Effect Analysis (HFMEA) for training and assessing surgeons performing robot assisted radical prostatectomy (RARP).

**Materials and Methods:** This is a multi-institutional, observational, prospective study. HFMEA methodology identified the high-risk steps of RARP. HFMEA is a specific branch of risk assessment that aims to reduce the occurrence of adverse errors causing effects by adopting a pre-emptive approach. After assembling a multi-disciplinary team and observing surgeries, a procedural map was created which was further analyzed by an expert focus group using HFMEA scoring system. Steps with Hazard scores above the defined threshold were included in the checklist that was then further distributed to experts in the field for content validation.

**Results:** Five surgeons were observed for 42 hours to create a diagrammatic representation of RARP. In collaboration with an international focus group of 3 expert surgeons, consensus on the definition of each step was reached. HFMEA was undertaken at a further focus-group meeting with 4 expert surgeons. Eighty four failure modes were identified and associated with 46 potential causes with a "Hazard score" ≥8. After content validation with 4 experts, a 3-stage checklist was produced including 17 processes and 40 sub-processes. This encompassed "Preparation of the Operative Field" (4 processes, 6 sub-processes), "Dissection of the Bladder and Prostate" (9 processes, 24 sub-processes) and "Anastomosis and Closure" (4 processes, 10 sub-processes).

**Conclusion:** HFMEA identified hazardous steps specific to RARP and content validation produced a checklist. This checklist can be used in training and assessment of robotic surgeons in urology.

#### MP-23.08

##### IVUmed: A Nonprofit Model for Surgical Training in Low Resource Countries

DeLong J<sup>1</sup>, Wood J<sup>2</sup>, Steele A<sup>2</sup>, Fredley M<sup>2</sup>, McCammon K<sup>1</sup>, deVries C<sup>3</sup>

<sup>1</sup>Eastern Virginia Medical School, Norfolk, USA; <sup>2</sup>IVUmed, Salt Lake City, USA; <sup>3</sup>The University of Utah Center for Global Surgery, Salt Lake City, USA

**Introduction and Objectives:** Low Resource Countries (LRCs) face both training and infrastructural challenges for surgical care. This is particularly true for specialty care such as urology. Local practitioners charged with caring for these patients have few options for basic or advanced study: travel abroad for hands-on training is virtually impossible due to certification regulations in the U.S. and Europe. IVUmed is a nonprofit organization that has supported urological educational programs in over 30 countries worldwide over the last 20 years. IVUmed's motto "Teach One, Reach Many" has emphasized a teach-the-teacher approach to sustainability.

**Materials and Methods:** The most limited resources for collaborative surgical training are time and administrative support. For most academic training departments even in wealthy countries, small numbers of specialized faculty mean that each has a very limited time available to train surgeons in LRCs. In order to maximize the short time available for teaching, IVUmed developed multi-faceted approach:

1. Long-term memoranda of understanding with partner programs in LRCs.
2. Expert training teams including surgeons, anesthesiologists, and ancillary staff.
3. Specific objectives for training based on procedures, problems or type of patients.
4. Scholarships for American residents to build sustainable leadership through experience working in LRCs during training.
5. Close relationships with regional, national and international urological associations such as the American Urological Association, the Société Internationale d'Urologie, the Pan African Urological Surgeons Association and the Société Haitienne d'Urologie.
6. Telemedicine support for interim case conferences and training between workshops.

**Results:** IVUmed partner sites have built sustainable training programs to address common and complex urological problems. IVUmed started with two workshops in 1992 with a single team leader. In 2012, the organization lead 21 workshops treating more than 800 patients under 16 different team leaders.

**Conclusions:** Successful collaborations for surgical training benefit from dedicated nonprofit involvement to coordinate volunteers, maintain standards of education and research and to leverage support from consortia of institutions, industry and individuals. The IVUmed model has shown sustainability and applicability to a variety of sites around the world.

#### MP-23.09

##### Modular Pelvic Surgery in Urology: Improving the Learning Curve

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<sup>1</sup>Dept. of Urology, Main-Kinzig-Kliniken, Gelnhausen, Germany; <sup>2</sup>Dept. of Urology, Johanniter Krankenhaus, Stendal, Germany



**Introduction and Objectives:** The “anatomical approach” to radical retropubic prostatectomy (RRP), designed to spare the nerve bundles and better control the bleeding during surgery, was developed by Patrick Walsh in the early 1980s. It rapidly became a standard surgical procedure and one of the most often performed operations in the oncological field of urology. A great part of the knowledge and efforts of the urological societies is aimed at standardizing the radical prostatectomy and its oncological results. Working in the same field we describe our results of performing RRP as part of our ‘modular pelvic surgery’ (MPS), an approach dividing the surgery into simple steps according to the anatomical topography. We have designed it to standardize the pelvic surgical procedures and achieve constant outcomes and a steep learning curve.

**Materials and Methods:** We performed a retrospective analysis of the last 130 Patients who underwent RRP at our clinic. The outcome meet the criteria of the German Association of Urology for prostate cancer treatment. The procedures were executed by 6 different surgeons, to 5 of whom the modular surgery approach had been unknown. The tutor surgeon operated 55 men (group S1) and the MPS-trainees 77 men (group S2). From the second group only one urologist have had performed a RRP before. **Results:** We compared group S1 with S2. Using the independent samples *t*-test or Z-test we found no statistical difference in the following parameters: blood loss (as Hemoglobin-loss in g/dl 3.77 vs. 3.70;  $p=0.785$ ), catheterisation period (7.66 vs. 7.93 days;  $p=0.472$ ), postoperative lymphocele (6% vs. 6.5%;  $p=0.456$ ), number of removed lymph nodes (10.54 vs. 9.53;  $p=0.250$ ), PSA-progression free survival (at 36 months  $p=4.19$ ) and positive surgical margins (12% vs. 11.7%;  $p=0.480$ ). The operative time was the only parameter with statistically significant difference (132.6 min vs. 173 min;  $p<0.001$ ). In terms of BMI and prostate size there was no significant difference between the two groups.

**Conclusions:** The results showed almost no statistical difference in the compared parameters which evaluate the surgical technique, and most important no difference in the very valuable oncologic parameter - the surgical margins. We are convinced that the modular approach in the pelvic surgery simplifies the radical prostatectomy and leads to a steeper learning curve.

#### MP-23.10

**Incorporation of the GreenLight-SIM™ Simulator at the Annual Quebec Urology Objective Structured Clinical Examinations**  
Nourelidin Y<sup>1,2</sup>, Elkoushy M<sup>1</sup>, Fahmy N<sup>1</sup>, Carrier S<sup>1</sup>, Elhilali M<sup>1</sup>, Andonian S<sup>1</sup>

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**Introduction and Objectives:** To assess laser prostatectomy skills of postgraduate trainees (PGTs) during the annual Quebec Urology Objective Structured Clinical Examinations (OSCEs).

**Materials and Methods:** After obtaining Institutional Review Board (IRB) approval and written informed consent, urology PGTs in Post-Graduate Years (PGY-3 to PGY-5) from all five urology training programs in Quebec were recruited to participate in assessment of their laser Photoselective Vaporization of the Prostate (PVP) skills using the GreenLight-SIM™ (GL-SIM) during two annual OSCEs on Dec 1<sup>st</sup> 2012 and Dec 7<sup>th</sup> 2013. PGTs were asked to perform two exercises: anatomical identification and PVP of a 30 g normal prostate within a 20-minute station. Grams vaporised, global scores and number of correct anatomical landmarks were recorded and correlated with PGY level, training on the GL-SIM and previous PVP experience.

**Results:** Twenty five PGTs were recruited at each OSCE with 13 PGTs participating in both OSCEs. PGTs had performed on average 2.8 and 4.5 PVP cases ( $p>0.05$ ) prior to the 1<sup>st</sup> and 2<sup>nd</sup> OSCEs, respectively. When comparing scores from the 1<sup>st</sup> to the 2<sup>nd</sup> OSCE, there was a significant increase in the number of grams vaporised ( $2.9\pm0.2$  vs.  $4.3\pm0.4$ ;  $p=0.003$ ) and global score ( $100\pm15$  vs.  $165\pm26$ ;  $p=0.03$ ). There was good correlation between the number of previously performed PVP cases and the global score ( $r=0.4$ ,  $p=0.04$ ). PGTs with previous practice on the GL-SIM (27/50) showed significantly increased global score ( $100.6\pm19.6$  vs.  $162.6\pm22.4$ ;  $p=0.04$ ) and grams vaporised ( $3.1\pm0.27$  vs.  $4.1\pm0.36$ ;  $p=0.04$ ) compared with those who did not practice on GL-SIM before. PGY level did not significantly affect grams vaporised and global score ( $p>0.05$ ). For the 13 PGTs who participated at both OSCEs, there was significant improvement in the global score ( $107\pm21$  vs.  $219\pm31$ ;  $p=0.003$ ) and grams vaporised ( $2.9\pm0.3$  vs.  $5.2\pm0.5$ ;  $p=0.001$ ) from the 1<sup>st</sup> to the 2<sup>nd</sup> OSCE.

**Conclusion:** Performance on the GreenLight-SIM at OSCEs significantly correlated with previous practice on the GL-SIM simulator and number of previous PVP cases performed rather than PGY level.

#### MP-23.11

**How Are Multidisciplinary Team Decisions Implemented in Prostate Cancer Patients?**

Thakare N, Njoku V, Turk S, Chingewundoh F  
Barts Health NHS Trust, London, UK

**Introduction and Objectives:** The multidisciplinary team (MDT) approach is of paramount importance in improving quality of care in cancer patients. Observational tools are being developed to assess performance of MDT meetings. To truly improve the efficacy of MDT pathways, it is necessary to study how decisions are implemented. We aim to review MDT decision-making and implementation in prostate cancer patients and to evaluate reasons for non-implementation.

**Materials and Methods:** Data were retrospectively collected for patients diagnosed with prostate cancer within a 12-month period in a single urology unit. MDT meeting documentation and patient records were assessed for details including PSA, Stage and Gleason score, MDT meeting recommendations, subsequent discussion with patients in MDT clinic and final management.

**Results:** MDT decisions were reviewed for 104 men with median age 74 and median PSA 15. Out of these, 65 (62.5%) had localized prostate cancer, 28 (27%) locally advanced and 11 advanced. MDT recommendations were made for all radical treatment options including prostatectomy and radiotherapy in 35 (33.6%) patients. Radiotherapy (+/- hormones) recommendation was made for patients including those not suitable for surgical options. Active surveillance was recommended as an option for low risk disease. No decision was made in 2 patients due to lack of information. Overall 4 (3.9%) decisions were not implemented; 1 was patient choice and 3 were physician decisions. Results are summarized in Table 1.

**Conclusion:** Implementation of MDT decisions in prostate cancer patients is good and the majority of decisions are concordant due to patient compliance. Physician change of decisions should be re-evaluated. It will be interesting to explore patient beliefs and degree of involvement in decision-making. We recommend a national governance framework to audit the MDT process in urological cancer management.

MP-23.11, Table 1. Summary of MDT Recommendations and Implementation

| MDT Recommendation          | No. (%age) | Implementation |
|-----------------------------|------------|----------------|
| All Radical Options         | 35 (33.6%) | 34/35          |
| Radiotherapy (+/- hormones) | 20 (19.2%) | 17/20          |
| Hormone therapy only        | 26 (25%)   | 26/26          |
| Active Surveillance         | 21 (20.2%) | 21/21          |
| No Decision                 | 2          | Not applicable |



## Moderated Poster Session 24

### Assessment of LUT/LUTD

Wednesday, October 15  
1300-1430

#### MP-24.01

##### Clinical Significance of NIH-Chronic Prostatitis Symptom Index Pain Score in Patients with Lower Urinary Tract Symptoms

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<sup>1</sup>Dept. of Urology, Yonsei University Wonju College of Medicine, Wonju, South Korea;

<sup>2</sup>Dept. of Urology, Cheongju St. Mary's Hospital, Cheongju, South Korea

**Introduction and Objectives:** Recently, 16% of healthy men reported a history of prostatitis and there was a large overlap with men reporting benign prostatic hyperplasia. Therefore, we evaluated the change in International Prostate Symptom Score (IPSS) according to NIH-Chronic Prostatitis Symptom Index (NIH-CPSI) pain score in patients with lower urinary tract symptoms (LUTS).

**Materials and Methods:** From March 2011 to February 2013, 134 patients with Lower Urinary Tract Symptoms (LUTS) were analyzed retrospectively. The presence of metabolic syndrome, prostate specific antigen, prostate volume, IPSS, NIH-CPSI, uroflowmetry, was collected.

**Results:** Forty one (30.5%) patients reported pain/discomfort on NIH-CPSI. The patients were classified into Group 1 (pain score  $\leq 3$ ) and Group 2 (pain score  $> 3$ ). Mean IPSS were  $16.79 \pm 8.27$  and  $19.91 \pm 6.93$  between the 2 groups, respectively ( $p=0.020$ ). Mean storage symptoms on IPSS were  $6.53 \pm 4.16$  and  $8.46 \pm 3.56$  in each group, respectively ( $p=0.005$ ) (Table 1). Logistic regression analysis showed that NIH-CPSI pain score was a significant predictive factor, Moderate IPSS (R, 8.989: 95%CI, 1.056 – 76.539) and Severe IPSS (R, 11.359: 95%CI, 1.306-98.751), whereas age, metabolic SD did not prove to be significant in this study (Table 2). In addition, 13 (9.7%) patients have pain/discomfort on ejaculation that was correlated with IPSS ( $p=0.021$ ).

**Conclusions:** The presence and severity of pain scores are independent risk factors for men with LUTS. Particularly, specific prostatitis-like symptoms of pain/discomfort on ejaculation was correlated with severity of LUTS.

#### MP-24.02

##### Effect of Diabetes Mellitus on the Development or Presence of LUTS and Bladder Outlet Obstruction

Nale D<sup>1,2</sup>, Bojanic N<sup>1,2</sup>, Nikic P<sup>1</sup>

<sup>1</sup>Clinic of Urology, Belgrade, Serbia; <sup>2</sup>Faculty of Medicine, University of Belgrade, Belgrade, Serbia

**Introduction and Objectives:** Treatment of

diabetic cystopathy may be complicated by frequently occurring coexisting urologic conditions. The aim of our study was to establish the influence of diabetes mellitus (DM) on the development or presence of lower urinary tract symptoms (LUTS) and bladder outlet obstruction (BOO).

**Materials and Methods:** Ninety four diabetic patients (cross-sectional study), mean age  $53 \pm 13$  years, were investigated by: Frequency-Volume Chart, neurological examination, electromyography of lower extremities (EMG), digito-rectal examination, urinoculture, ultrasound examination including prostatic weight, detrusor wall thickness, and PVR, and urodynamic examination. The patients were classified based on presence or absence of LUTS. Of the 94 diabetic patients, 43 patients had LUTS (symptomatic group) and 51 patients were without LUTS (asymptomatic group). ICS nomogram was used as a criterion for diagnosis of the bladder outlet obstruction. BOO was defined as a sustained detrusor contraction of

greater than 40 cm water, a catheterized urine flow rate of less than 12ml /sec.

**Results:** BOO was present in 6 men (11.76%) without LUTS and in 20 men (46.51%) with LUTS using pressure-flow-EMG study ( $p<0.01$ ). One-way analysis of variance revealed that duration of DM had no influence on onset of BOO ( $p=0.23$ ) and on onset of voiding symptoms in men ( $p=0.68$ ), but it significantly influenced the onset of increased daytime frequency and nocturia in men. BOO had no influence on onset of storage symptoms ( $p=0.12$ ), but it significantly influenced the onset of voiding symptoms ( $p<0.01$ ).

**Conclusion:** Diabetes may affect storage more than voiding function in men. Alterations of proprioceptive sensations of the bladder and increased PVR by diabetes are a probable hypothesis to explain these findings. Duration of DM does not have influence on onset of BOO. Perception of voiding difficulties by diabetics indicate onset of BOO.

MP-24.01, Table 1. Baseline Characteristics (n = 134) of Patients

|                      | Pain score $\leq 3$ | Pain score $> 3$  | p-value |
|----------------------|---------------------|-------------------|---------|
| Age (years)          | $64.82 \pm 7.97$    | $65.29 \pm 9.09$  | 0.752   |
| Metabolic SD         | 31                  | 34                | 0.879   |
| Prostate volume (gm) | $33.11 \pm 15.55$   | $30.15 \pm 12.10$ | 0.290   |
| PSA (ng/dL)          | $1.29 \pm 1.00$     | $1.15 \pm 0.85$   | 0.398   |
| IPSS total           | $16.79 \pm 8.27$    | $19.91 \pm 6.93$  | 0.020   |
| voiding              | $10.25 \pm 5.51$    | $11.43 \pm 4.48$  | 0.178   |
| storage              | $6.53 \pm 4.16$     | $8.46 \pm 3.56$   | 0.005   |
| QoL                  | $4.03 \pm 1.26$     | $4.33 \pm 1.37$   | 0.195   |
| IIEF 5               | $9.14 \pm 6.84$     | $10.58 \pm 6.58$  | 0.331   |
| Qmax (ml/s)          | $11.43 \pm 6.56$    | $11.14 \pm 6.04$  | 0.806   |

Metabolic SD = Metabolic syndrome; PSA = Prostate-specific antigen; IPSS = International Prostate Symptom Score; QoL = Quality of life; IIEF-5 total, International Index of Erectile Function total score; Qmax, peak flow rate.

MP-24.01, Table 2. Potential Risk Factor for NIH-CPSI Pain Score

|                                          | OR (95%CI)              |
|------------------------------------------|-------------------------|
| Age                                      |                         |
| $\leq 60$                                | —                       |
| $61 \leq 70$                             | 0.559 (0.233 – 1.340)   |
| $\geq 71$                                | 1.148 (0.530 – 3.850)   |
| IPSS                                     |                         |
| Mild                                     | —                       |
| Moderate                                 | 8.989 (1.056 – 76.539)  |
| Severe                                   | 11.359 (1.306 – 98.751) |
| Metabolic SD                             | 1.193 (0.557 – 2.556)   |
| OR, odds ratio; CI, confidence interval. |                         |

#### MP-24.03

##### Visual Prostate Score (VPSS) and Maximum Flow Rate (Qmax) in Men with Lower Urinary Tract Symptoms

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**Introduction and Objectives:** To determine the relationship between a visual prostate score (VPSS) and Maximum flow rate (Qmax) in men with lower urinary tract symptoms.

**Material and Methods:** Sixty-seven adult male patients  $> 40$  years of age were included in the study following an informed consent. Following induction a complete clinical history and physical examination was performed. The data was collected on a pre-approved proforma

filled duly by the patient in the clinic or as an in-patient. Qmax and voided volume recorded from the uroflowmetry graph and at the same time VPSS was assessed.

**Results:** The education level was assessed in various defined groups. Mean age was  $66.1 \pm 10.1$  years (median 68). There were 47.7% uneducated, 2.9% had primary education only, 8.9% had education up to middle level, 10.4% had education up to higher level, 7.4% up to intermediate level, and 22.3% had graduated from university program. The mean voided volume on uroflowmetry was  $268 \pm 160$  ml (median 208), the mean Qmax was  $9.6 \pm 4.96$  ml/sec (median 9.0). The mean VPSS score was  $11.4 \pm 2.72$  (11.0). In the univariate linear regression analysis there was strong negative Pearson correlation between VPSS and Qmax ( $r = -0.848$ ,  $p < 0.001$ ). In the multiple linear regression analysis there was a significant correlation between VPSS and Qmax ( $\beta -0.466$  with ( $p < 0.001$ ) after adjusting the effect of age, voided volume and level of education. Multiple linear regressions analysis done for independent variable and results showed that there was no significant correlation between the VPSS and independent factors included age ( $p = 0.27$ ), LOE ( $p = 0.941$ ) and V.V ( $p = 0.082$ ).

**Conclusion:** Our study showed a significant negative correlation between VPSS and Qmax. The VPSS can be used in lieu of IPSS score due to its other advantages. Our study proved that VPSS can be completed without assistance by a greater proportion of men even with limited educations.

#### MP-24.04

##### Development of Computer-Aided Diagnosis System to Detect Lesions for Voiding Dysfunction Using Cystourethroscopy

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**Introduction and Objectives:** Lower urinary tract consists of hollow and tubular organs, which is closing in the continent condition and deformed under internal pressure during micturition. Among various methods, cystourethroscopy constantly provides shape and color information of the urethra together with a role as a potent therapeutic tool. We developed software to generate a panoramic image from cystourethroscopic video image and construct a three-dimensional (3D) model of the intra-urethral information for a Computer-Aided Diagnosis (CAD) system to determine candidate lesions for voiding dysfunction, and tested its validity.

**Materials and Methods:** Cystourethroscopy was performed in nineteen patients with benign prostate hyperplasia around administration of

alpha-1 adrenoceptor blockade under approval of local ethic committee. Cystourethroscopic video image was recorded from bladder outlet to the distal part of the verumontanum by slowly withdrawing the scope in about 10 seconds. An opened panoramic image was processed by retrieving and connecting color information on each video frame. At the same time, the reflection intensity of each pixel was measured to estimate the shape of the urethral wall. Using the information, a virtual 3D model of urethra was generated under an assumption that the urethra is a cylindrical organ. The software was designed to call back the video frame by putting a pointer on the panoramic image to identify the region of interest. In addition, fluid dynamics through the 3D model was computed to simulate the urine flow in the prostatic urethra.

**Results:** 3D image processing was successfully achieved in 17 of 19 patients. In the urine stream simulation, turbulence and vortex formation was observed adjacent to obstacles in 3D model in the 17 patients before therapy, and was diminished after therapy in 9 patients who showed improvement of urine flow. The obstacles can be identified in the panoramic image, and then indicate the "candidate lesion" for voiding dysfunction in the original cystourethroscopic video image.

**Conclusion:** Processing 3D endoscopic image of the male urethra, linked to the original cystourethroscopic video image would contribute to establish novel therapeutic modality to minimize its drawbacks in patients with voiding dysfunction.

#### MP-24.05

##### Urethral Catheter in Urodynamic: Does It Affect the Urine Flow?

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**Introduction and Objectives:** The importance of pressure flow studies in the evaluation of patients with LUTS has long been recognized, however the use of urethral catheter during the study can influence the flow rate leading to misinterpretation of the results. Previous

research have demonstrated various effects depending on the catheter size, in our work we have evaluated the effect of 8F catheter on the Maximum flow ( $Q_{max}$ ).

**Materials and Methods:** Eighty patients were included in the study, of those only 22 passed >150 ml and so were included. The patients underwent urodynamic study and had their  $Q_{max}$  measured before and during the study, the results were compared using paired student *t* test.

**Results:** The means of the  $Q_{max}$  in the free-flow study (FFS) and pressure-flow study (PFS) were 19.7 ml/sec (range 4-55) and 13 ml/sec (range 4-33) respectively. The difference was statistically significant ( $p = 0.004$ ); on the other hand the difference between the voided volume of the FFS and PFS was not statistically significant, (226 and 290 ml respectively;  $p = 0.14$ ).  $Q_{max}$  decreased in 73%, increased in 23% and was stable in 4% of cases at the time of pressure flow study.

**Conclusion:** Our results are in line with previous studies, however we demonstrated that the difference can be statistically significant even with 8F catheter, further work is need to further establish the relation between catheter size and the urine flow.

#### MP-24.06

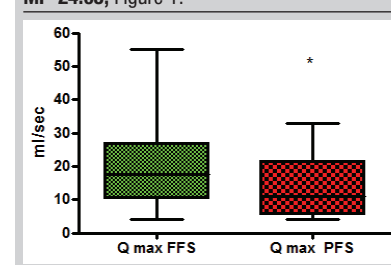
##### Community Management of Male Patients with Lower Urinary Tract Symptoms (LUTS) in the UK

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**Introduction and Objectives:** Providing a cost-effective and patient-centered service has

MP-24.05, Figure 1.



MP-24.05, Table 1.

| Patients's Number | Qmax FFS Ml/Sec | Qmax PFS Ml/Sec | P value | VV FFS     | VV PFS     | P value |
|-------------------|-----------------|-----------------|---------|------------|------------|---------|
| 22                | 19.7+/-2.6      | 13+/-1.7        | 0.004   | 226+/-20.7 | 290+/-35.6 | NS      |

been a highlighting feature in the UK National Health System. Simple interventions can significantly improve management and reduce unnecessary costs. The cost of treatment for benign prostatic hyperplasia (BPH), the most common cause of LUTS, is more than £180 million each year of which 60% is incurred in secondary care. Eighty percent of male LUTS patients attending primary care would need no further referral on to secondary care with effective management within the primary care setting. We looked into inappropriate referrals and management of LUTS in the community.

**Materials and Methods:** Information was gathered regarding inappropriate referrals to Prostate assessment clinic at BTUH during the month of October 2013. An audit was then conducted to see whether general practitioners were adhering to NICE guidelines for management of male LUTS and therefore referring appropriately to secondary care. A questionnaire based on the NICE guidelines was developed and distributed amongst local general practitioners.

**Results:** This audit aimed to have a 100% correct response rate in the following areas: performing a PSA where appropriate (92%), calculating International Prostate Symptom score (50%), offering urinary frequency volume charts (23%) and initiating an alpha blocker prior to referral (79%). The data was analyzed showing that in this small random population, overall, they did not adhere to the recommended pathway.

**Conclusion:** As the pathway was not being adhered to, an action was agreed upon to produce a local urology pathway highlighting the recommended management. The launch of the local LUTS pathway has been used as an educational tool for fellow GPs, with discussions around practical implementation of the recommendations being discussed at a GP Forum, led by a local urologist. This discussion is aligned with the guideline, which encourages communication and alignment between primary and secondary care in the management of LUTS patients. A re-audit has been commenced to see if the intervention has been effective in ensuring better management of patients and reducing costs in the outpatient setting.

#### MP-24.07

##### Alpha-1 Blocker Treatment for Lower Urinary Tract Symptoms Improve the Index of Depression and Anxiety: A Prospective Multicenter Study

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South Korea; <sup>6</sup>Danhook University College of Medicine, Cheonan, South Korea

**Introduction and Objectives:** We investigated that depression and anxiety could be improved by alpha-1 blocker treatment to patients with lower urinary tract symptoms (LUTS).

**Materials and Methods:** A total of 201 patients were enrolled. The 96 patients with LUTS (40 ~ 80 years old) were evaluated using the Beck Anxiety Inventory (BAI) questionnaire of the Beck Depression Inventory (BDI) Scale and anxiety and depression degree, international prostate symptom index (IPSS) questionnaire before and after a 12-week administration period. The other 105 patients with no LUTS were evaluated by same methods. They visited our hospital for the health screening and had no administration period. We analyzed the correlation of IPSS, BAI, and BDI score.

**Results:** The mean IPSS score was 16.8 in the LUTS group while was 4.0 in the control group, was 4.0. The mean BDI and BAI score in the control group was 8.5 and 5.2, respectively, compared to 10.7 and 8.8 in the LUTS group. After a 12-week administration period, the LUTS group's mean IPSS score dropped to 10.3. The mean BDI and BAI score were reduced to 9.0 and 7.1, respectively. IPSS score was significantly correlated with BDI and BAI score in the total 201 patients with the correlation analysis. After treatment, the mean score of IPSS, BDI and BAI shifted to more mild degree (Table 1).

**Conclusion:** The Patients with LUTS have higher degree of depression and anxiety. Alpha-1 blocker treatment significantly reduces BDI and BAI scores as well as IPSS and life scores in LUTS patients.

#### MP-24.08

##### A Randomized Control Trial to Assess Efficacy of Diclofenac Suppository during Flexible Cystoscopy on Pain Control: DUF Trail

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**Introduction and Objectives:** To compare mean VAS for pain during flexible cystoscopy between patients undergoing the procedure with plain gel only and with diclofenac suppository besides plain gel.

**Materials and Methods:** It is a prospective, randomized, controlled study. A total of 60 male patients with an indication of flexible cystoscopy were enrolled and randomized in two equal groups. In group "A", patients received diclofenac suppository one hour prior to the procedure while group "B" did not receive diclofenac suppository. Both groups received 10 mls of intra-urethral plain gel for lubrication during flexible cystoscopy. Operating surgeon and pain score assessor were blinded to randomization. Pain score was recorded immediately after the procedure using visual analogue scale. Statistical analyses were performed using chi-Square test and student t-test. Regression analysis was performed to address the confounding variables.

**Results:** Mean age of the patients was 46.75 years with SD of +/- 16.12 years. Most common indication for flexible cystoscopy was removal of double J stent. Variables including age, duration of procedure, indication of procedure, level of operating surgeon were comparable. Mean Pain score in group A and group B was 3.16+/-1.53 and 4.10+/-1.24 respectively. This difference was found to be statistically significant ( $p=0.012$ ). Regression analysis showed

MP-24.07, Table 1. The Score Shifting of BDI and BAI after alpha-1 Blocker Treatment

|      | Degree                    | Control<br>(n=105) | Pre-<br>Treatment<br>(n=96) | Post-<br>Treatment<br>(n=96) | P<br>value* |
|------|---------------------------|--------------------|-----------------------------|------------------------------|-------------|
| IPSS | 0-7 mild                  | 105 (100%)         | 0                           | 35 (36.5%)                   | 0.000       |
|      | 8-19 moderate             | 0                  | 71 (74.0%)                  | 55 (57.3%)                   |             |
|      | 20-35 severe              | 0                  | 25 (26.0%)                  | 6 (6.3%)                     |             |
| BDI  | 1-10 normal               | 77 (73.3%)         | 53 (55.2%)                  | 66 (68.8%)                   | 0.000       |
|      | 11-16 mood confusion      | 7 (6.7%)           | 24 (25.0%)                  | 15 (15.6%)                   |             |
|      | 17-20 borderline          | 2 (1.9%)           | 8 (8.3%)                    | 4 (4.2%)                     |             |
|      | 21-30 moderate depression | 19 (18.1%)         | 9 (9.4%)                    | 10 (10.4%)                   |             |
|      | 31-40 severe depression   | 0                  | 2 (2.1%)                    | 1 (1.0%)                     |             |
|      | 41-63 extreme depression  | 0                  | 0                           | 0                            |             |
| BAI  | 0-4 normal                | 67 (63.8%)         | 35 (36.5%)                  | 39 (40.6%)                   | 0.000       |
|      | 5-10 borderline           | 21 (20.0%)         | 34 (35.4%)                  | 40 (41.7%)                   |             |
|      | 11-20 mild anxiety        | 14 (13.3%)         | 20 (20.8%)                  | 12 (12.5%)                   |             |
|      | 21-30 moderate anxiety    | 3 (2.9%)           | 5 (5.2%)                    | 4 (4.2%)                     |             |
|      | 31-50 severe anxiety      | 0                  | 2 (2.1%)                    | 1 (1.0%)                     |             |
|      | 51-99 extreme anxiety     | 0                  | 0                           | 0                            |             |

that none of the confounding variable was significantly affecting the pain perception.

**Conclusions:** Intra Rectal diclofenac Suppository is simple and effective pre emptive analgesia. We recommend its routine use during flexible cystoscopy for better pain control.

#### MP-24.09

##### Unexpected Problems in Men

##### Post TUR Prostate

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**Introduction and Objectives:** Patients can find the immediate post-operative period following transurethral resection of prostate (TURP) surgery troublesome. There are few studies in

the literature that focus on patients' experiences immediately post-TURP. In this study we determine what complications patients experience in the 6 weeks post-surgery, and establish the impact of TURP on patients' quality of life, symptomatology and sexual function.

**Materials and Methods:** A validated patient questionnaire was used to collect data. Questionnaires were mailed to 55 patients at six weeks post-TURP surgery.

**Results:** Forty five out of 55 (81.8%) patients returned the questionnaire. Thirty five patients (77.8%) had no requirement for a urethral catheter after discharge. Three patients (6.7%) were sent home with a catheter that was still in situ 6 weeks post-operatively. Five patients (11.1%) had a catheter in place for less than a month. Fourteen patients (31.1%) visited a doctor at least once because of lower urinary

tract symptoms. Overall, 10 patients (22.2%) were prescribed antibiotics. Only 2 of these patients had bacterial growth on urine culture. Three patients (6.7%) needed to be re-catheterised for retention of urine following discharge. Three patients (6.7%) were re-admitted to hospital due to a problem relating to the operation. Thirteen patients (30.2%) leaked troublesome amounts of urine or had to wear pads to keep dry. Ten patients (23.3%) were completely dry following their operation. There was no statistically significant difference between the numbers of patients who were sexually active in the month preceding the operation and those attempting intercourse in the immediate 6 weeks following TURP, 19.5% and 21.4% respectively (p<0.05). There was also no significant difference in ability to achieve and maintain erections pre- and post-TURP. Most

MP-24.10, Table 1. Results for Efficacy Variables (FAS)

|                                                                    | Solifenacin 2.5 mg           |                              | Solifenacin 5 mg             |                              |
|--------------------------------------------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
|                                                                    | Mirabegron 25 mg<br>(n = 33) | Mirabegron 50 mg<br>(n = 37) | Mirabegron 25 mg<br>(n = 55) | Mirabegron 50 mg<br>(n = 93) |
| <b>Number of micturitions/24 h, mean ± SD (n)</b>                  |                              |                              |                              |                              |
| Baseline†                                                          | 10.38 ± 2.128                | 11.12 ± 2.722                | 9.81 ± 1.918                 | 10.15 ± 2.331                |
| Change from baseline to EOT                                        | -1.89 ± 1.783***<br>(33)     | -2.36 ± 2.106***<br>(37)     | -1.94 ± 1.792***<br>(55)     | -2.12 ± 2.094***<br>(93)     |
| <b>Number of urgency episodes/24 h, mean ± SD (n)</b>              |                              |                              |                              |                              |
| Baseline                                                           | 2.86 ± 2.433<br>(30)         | 4.12 ± 2.586<br>(36)         | 2.86 ± 1.997<br>(49)         | 3.63 ± 2.846<br>(90)         |
| Change from baseline to EOT                                        | -1.57 ± 1.382***<br>(30)     | -2.59 ± 2.201***<br>(36)     | -1.93 ± 1.877***<br>(49)     | -2.02 ± 2.392***<br>(90)     |
| <b>Number of incontinence episodes/24 h, mean ± SD (n)</b>         |                              |                              |                              |                              |
| Baseline                                                           | 1.45 ± 1.454<br>(14)         | 1.54 ± 1.248<br>(28)         | 2.09 ± 2.121<br>(29)         | 1.51 ± 1.520<br>(60)         |
| Change from baseline to EOT                                        | -1.07 ± 1.064**<br>(14)      | -1.08 ± 1.168***<br>(28)     | -1.32 ± 1.505***<br>(29)     | -1.06 ± 1.055***<br>(60)     |
| <b>Number of urgency incontinence episodes/24 h, mean ± SD (n)</b> |                              |                              |                              |                              |
| Baseline                                                           | 1.42 ± 1.506<br>(11)         | 1.42 ± 1.243<br>(28)         | 1.69 ± 1.366<br>(26)         | 1.30 ± 1.138<br>(52)         |
| Change from baseline to EOT                                        | -0.97 ± 0.948**<br>(11)      | -1.00 ± 1.089***<br>(28)     | -1.04 ± 1.412***<br>(26)     | -0.94 ± 1.027***<br>(52)     |
| <b>Number of nocturia episodes/24 h, mean ± SD (n)</b>             |                              |                              |                              |                              |
| Baseline                                                           | 1.79 ± 1.075<br>(28)         | 1.55 ± 0.783<br>(29)         | 1.85 ± 1.006<br>(50)         | 1.72 ± 1.084<br>(81)         |
| Change from baseline to EOT                                        | -0.43 ± 0.847*<br>(28)       | -0.31 ± 0.674*<br>(29)       | -0.44 ± 0.812***<br>(50)     | -0.60 ± 0.982***<br>(81)     |
| <b>Volume voided/micturition (mL), mean ± SD (n)</b>               |                              |                              |                              |                              |
| Baseline†                                                          | 158.627 ± 47.0511            | 158.633 ± 43.6678            | 182.362 ± 56.3405            | 173.658 ± 47.5973            |
| Change from baseline to EOT                                        | 29.865 ± 32.5400***<br>(31)  | 33.031 ± 38.2492***<br>(37)  | 34.118 ± 32.6790***<br>(50)  | 36.957 ± 43.6344***<br>(93)  |

\*P<0.05; \*\*P<0.01; \*\*\*P<0.001; †n = as for FAS; EOT, End of Treatment



men (33 patients, 73.3%) were not concerned by any change in sexual function.

**Conclusion:** A significant proportion of patients are concerned enough post-TURP surgery to seek medical attention. Appropriately preparing patients for the more minor, but frequently occurring, complications of TURP may help reduce the unnecessary utilisation of antibiotics. This study also presents data

relating to sexual activity in the immediate 6 weeks following TURP. This is something that is under-represented in the literature.

#### MP-24.10

**Safety and Efficacy of Mirabegron as Add-on Therapy in Patients with Overactive Bladder Treated with Solifenacin (MILAI Study)**  
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Igawa Y<sup>4</sup>, Takeda M<sup>5</sup>, Nishizawa O<sup>6</sup>, Gotoh M<sup>7</sup>, Yoshida M<sup>8</sup>, Yokoyama O<sup>9</sup>, Seki N<sup>10</sup>, Okitsu A<sup>11</sup>, Hamada T<sup>11</sup>, Kobayashi A<sup>11</sup>, Kuroishi K<sup>11</sup>  
<sup>1</sup>Nihon University, School of Engineering, Koriyama, Japan; <sup>2</sup>Dept. of Renal and Urologic Surgery, Asahikawa Medical University, Asahikawa, Japan; <sup>3</sup>Dept. of Urology, The University of Tokyo Graduate School of Medicine, Tokyo, Japan; <sup>4</sup>Dept. of Continence Medicine, The

**MP-24.11**, Table 1. Responder Analyses for Micturition Frequency and PROs at End of Treatment

|                                                                                                                                                                                                                                                                                                                                                          | PBO                             | MIRA 25 mg                      | MIRA 50 mg                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------|-------------------------------|
| <b>Micturition frequency normalization</b>                                                                                                                                                                                                                                                                                                               | <b>N=79</b>                     | <b>N=75</b>                     | <b>N=76</b>                   |
| Responders, n (%)                                                                                                                                                                                                                                                                                                                                        | 42 (53.2)                       | 32 (42.7)                       | 36 (47.4)                     |
| Odds ratio vs. solifenacin 5 mg (95% CI) [p value]                                                                                                                                                                                                                                                                                                       | --                              | --                              | --                            |
| Odds ratio vs. placebo (95% CI) [p value]                                                                                                                                                                                                                                                                                                                | --                              | 0.89 (0.45, 1.76) [0.73]        | 0.88 (0.45, 1.72) [0.70]      |
| <b>≥ 1 PPBC</b>                                                                                                                                                                                                                                                                                                                                          | <b>N=78</b>                     | <b>N=73</b>                     | <b>N=76</b>                   |
| Responders, n (%)                                                                                                                                                                                                                                                                                                                                        | 54 (69.2)                       | 56 (76.7)                       | 60 (78.9)                     |
| Odds ratio vs. solifenacin 5 mg (95% CI) [p value]                                                                                                                                                                                                                                                                                                       | --                              | --                              | --                            |
| Odds ratio vs. placebo (95% CI) [p value]                                                                                                                                                                                                                                                                                                                | --                              | 1.25 (0.58, 2.71) [0.57]        | 1.67 (0.77, 3.64) [0.19]      |
| <b>≥ 10 in Symptom Bothers</b>                                                                                                                                                                                                                                                                                                                           | <b>N=79</b>                     | <b>N=75</b>                     | <b>N=76</b>                   |
| Responders, n (%)                                                                                                                                                                                                                                                                                                                                        | 58 (73.4)                       | 63 (84.0)                       | 60 (78.9)                     |
| Odds ratio vs. solifenacin 5 mg (95% CI) [p value]                                                                                                                                                                                                                                                                                                       | --                              | --                              | --                            |
| Odds ratio vs. placebo (95% CI) [p value]                                                                                                                                                                                                                                                                                                                | --                              | 1.75 (0.75, 4.08) [0.19]        | 1.25 (0.56, 2.77) [0.59]      |
| <b>≥ 10 in HRQoL</b>                                                                                                                                                                                                                                                                                                                                     | <b>N=79</b>                     | <b>N=75</b>                     | <b>N=76</b>                   |
| Responders, n (%)                                                                                                                                                                                                                                                                                                                                        | 53 (67.1)                       | 43 (57.3)                       | 54 (71.1)                     |
| Odds ratio vs. solifenacin 5 mg (95% CI) [p value]                                                                                                                                                                                                                                                                                                       | --                              | --                              | --                            |
| Odds ratio vs. placebo (95% CI) [p value]                                                                                                                                                                                                                                                                                                                | --                              | 0.50 (0.24, 1.04) [0.064]       | 1.24 (0.59, 2.61) [0.57]      |
|                                                                                                                                                                                                                                                                                                                                                          | <b>SOLI 2.5 mg + MIRA 25 mg</b> | <b>SOLI 2.5 mg + MIRA 50 mg</b> | <b>SOLI 5 mg + MIRA 25 mg</b> |
| <b>Micturition frequency normalization</b>                                                                                                                                                                                                                                                                                                               | <b>N=144</b>                    | <b>N=145</b>                    | <b>N=140</b>                  |
| Responders, n (%)                                                                                                                                                                                                                                                                                                                                        | 78 (54.2)                       | 77 (53.1)                       | 73 (52.1)                     |
| Odds ratio vs. solifenacin 5 mg (95% CI) [p value]                                                                                                                                                                                                                                                                                                       | 1.20 (0.72, 2.00) [0.48]        | 1.11 (0.67, 1.83) [0.69]        | 1.02 (0.61, 1.69) [0.95]      |
| Odds ratio vs. placebo (95% CI) [p value]                                                                                                                                                                                                                                                                                                                | 1.36 (0.75, 2.45) [0.31]        | 1.25 (0.70, 2.25) [0.45]        | 1.15 (0.64, 2.07) [0.64]      |
| <b>≥ 1 PPBC</b>                                                                                                                                                                                                                                                                                                                                          | <b>N=141</b>                    | <b>N=143</b>                    | <b>N=136</b>                  |
| Responders, n (%)                                                                                                                                                                                                                                                                                                                                        | 105 (74.5)                      | 119 (83.2)                      | 106 (77.9)                    |
| Odds ratio vs. solifenacin 5 mg (95% CI) [p value]                                                                                                                                                                                                                                                                                                       | 1.10 (0.63, 1.92) [0.73]        | 1.82 (1.00, 3.30) [0.050]       | 1.45 (0.81, 2.59) [0.21]      |
| Odds ratio vs. placebo (95% CI) [p value]                                                                                                                                                                                                                                                                                                                | 1.15 (0.60, 2.22) [0.67]        | 1.90 (0.95, 3.78) [0.070]       | 1.51 (0.77, 2.98) [0.23]      |
| <b>≥ 10 in Symptom Bothers</b>                                                                                                                                                                                                                                                                                                                           | <b>N=144</b>                    | <b>N=145</b>                    | <b>N=139</b>                  |
| Responders, n (%)                                                                                                                                                                                                                                                                                                                                        | 119 (82.6)                      | 124 (85.5)                      | 119 (85.6)                    |
| Odds ratio vs. solifenacin 5 mg (95% CI) [p value]                                                                                                                                                                                                                                                                                                       | 1.04 (0.55, 1.97) [0.89]        | 1.21 (0.63, 2.33) [0.57]        | 1.36 (0.70, 2.66) [0.36]      |
| Odds ratio vs. placebo (95% CI) [p value]                                                                                                                                                                                                                                                                                                                | 1.64 (0.81, 3.33) [0.17]        | 1.90 (0.92, 3.92) [0.084]       | 2.14 (1.02, 4.48) [0.043*]    |
| <b>≥ 10 in HRQoL</b>                                                                                                                                                                                                                                                                                                                                     | <b>N=144</b>                    | <b>N=145</b>                    | <b>N=139</b>                  |
| Responders, n (%)                                                                                                                                                                                                                                                                                                                                        | 105 (72.9)                      | 114 (78.6)                      | 107 (77.0)                    |
| Odds ratio vs. solifenacin 5 mg (95% CI) [p value]                                                                                                                                                                                                                                                                                                       | 1.01 (0.58, 1.77) [0.98]        | 1.25 (0.70, 2.23) [0.45]        | 1.41 (0.79, 2.53) [0.24]      |
| Odds ratio vs. placebo (95% CI) [p value]                                                                                                                                                                                                                                                                                                                | 1.12 (0.58, 2.15) [0.74]        | 1.39 (0.71, 2.71) [0.34]        | 1.57 (0.80, 3.07) [0.19]      |
| P values were from logistic regression model with 2 main factors mirabegron dose (0, 25 and 50 mg) and solifenacin dose (0, 2.5, 5 and 10 mg) and their corresponding dose combination interaction term to reflect the factorial design. The model included the main factors of sex, age group, geographic region and baseline measurement as covariate. |                                 |                                 |                               |
| *Statistically significantly superior compared to solifenacin 5 mg or placebo (p<0.05).                                                                                                                                                                                                                                                                  |                                 |                                 |                               |

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Obu, Japan; <sup>9</sup>Dept. of Urology, University of Fukui Faculty of Medical Sciences, Fukui, Japan; <sup>10</sup>Dept. of Urology, Kyushu Central Hospital of the Mutual Aid Association of Public School Teachers, Fukuoka, Japan; <sup>11</sup>Astellas Pharma Inc, Tokyo, Japan

**Introduction and Objectives:** To evaluate safety and efficacy of mirabegron as add-on

therapy in patients with OAB treated with solifenacin.

**Materials and Methods:** This 16-week, multicenter, open-label Phase IV study, performed in accordance with ICH-GCP, enrolled patients who had been treated with solifenacin at a stable dose of 2.5 mg or 5 mg for at least 4 weeks, and who had an overactive bladder symptom score (OABSS) total score of  $\geq 3$  points and an OABSS Question 3 score of  $\geq 2$  points. In addition to solifenacin 2.5 mg or 5 mg, patients initially received mirabegron at a dose of 25 mg but this dose could be increased to 50 mg at week 8 if patients met dose escalation criteria. Efficacy was evaluated based on patient diary and changes in overactive bladder questionnaire short form (OAB-q SF) score. Safety assessments were based on AEs, laboratory values, vital signs, 12-lead ECG, corrected QT (QTc) interval, and postvoid residual volume.

**Results:** Of 223 patients entering the treatment period, mean age was 64.6 years, and approximately 55% were aged  $\geq 65$  years. At end of treatment, significant improvements from baseline were seen in all efficacy endpoints in all groups (Table 1). Overall incidence of drug-related TEAEs was 23.3%. Almost all TEAEs were mild or moderate. The most common TEAE was constipation; incidence in groups where mirabegron dose was increased was similar to that in mirabegron dose-maintained groups. Changes in post-void residual volume, QTcF interval, pulse rate, and blood pressure were clinically insignificant and there were no cases of urinary retention.

**Conclusion:** Add-on therapy with mirabegron once daily for 16 weeks, with an optional dose increase from 25 to 50 mg at week 8, was well tolerated and significantly improved OAB symptoms from baseline in patients with OAB treated with solifenacin 2.5 or 5 mg once daily.

#### MP-24.11

#### Combination Treatment with Mirabegron and Solifenacin in Patients with Overactive Bladder: Exploratory Responder Analyses of Efficacy and Evaluation of Patient Reported Outcomes from a Randomised, Double-Blind, Dose-Ranging, Phase 2 Study (Symphony)

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**Introduction and Objectives:** To assess responders for incontinence episodes and micturition frequency and patient reported outcomes (PROs) with mirabegron and solifenacin monotherapy and combination therapy in patients

| SOLI 2.5 mg                | SOLI 5 mg                  | SOLI 10 mg                 |
|----------------------------|----------------------------|----------------------------|
| N=73                       | N=146                      | N=75                       |
| 40 (54.8)                  | 70 (47.9)                  | 41 (54.7)                  |
| --                         | --                         | 1.29 (0.69, 2.39) [0.42]   |
| 1.49 (0.74, 2.99) [0.26]   | 1.13 (0.63, 2.04) [0.69]   | 1.45 (0.73, 2.89) [0.29]   |
| N=75                       | N=147                      | N=73                       |
| 51 (68.0)                  | 107 (72.8)                 | 62 (84.9)                  |
| --                         | --                         | 2.26 (1.04, 4.95) [0.040*] |
| 0.86 (0.41, 1.79) [0.69]   | 1.04 (0.55, 1.99) [0.90]   | 2.36 (1.01, 5.55) [0.048*] |
| N=77                       | N=150                      | N=75                       |
| 66 (85.7)                  | 122 (81.3)                 | 64 (85.3)                  |
| --                         | --                         | 1.42 (0.63, 3.20) [0.40]   |
| 1.97 (0.83, 4.71) [0.13]   | 1.57 (0.78, 3.16) [0.20]   | 2.24 (0.94, 5.34) [0.070]  |
| N=77                       | N=150                      | N=75                       |
| 57 (74.0)                  | 109 (72.7)                 | 56 (74.7)                  |
| --                         | --                         | 1.04 (0.52, 2.08) [0.92]   |
| 1.26 (0.59, 2.68) [0.55]   | 1.11 (0.58, 2.13) [0.75]   | 1.15 (0.53, 2.49) [0.72]   |
| SOLI 5 mg + MIRA 50 mg     | SOLI 10 mg + MIRA 25 mg    | SOLI 10 mg + MIRA 50 mg    |
| N=146                      | N=78                       | N=79                       |
| 90 (61.6)                  | 51 (65.4)                  | 46 (58.2)                  |
| 1.91 (1.14, 3.21) [0.015*] | 2.06 (1.11, 3.84) [0.023*] | 1.54 (0.84, 2.84) [0.16]   |
| 2.16 (1.18, 3.94) [0.013*] | 2.33 (1.16, 4.66) [0.017*] | 1.74 (0.88, 3.45) [0.11]   |
| N=144                      | N=76                       | N=78                       |
| 119 (82.6)                 | 63 (82.9)                  | 59 (75.6)                  |
| 2.02 (1.11, 3.66) [0.021*] | 1.64 (0.79, 3.40) [0.19]   | 1.12 (0.57, 2.19) [0.74]   |
| 2.11 (1.06, 4.19) [0.034*] | 1.71 (0.76, 3.84) [0.20]   | 1.17 (0.55, 2.49) [0.68]   |
| N=146                      | N=78                       | N=78                       |
| 129 (88.4)                 | 69 (88.5)                  | 65 (83.3)                  |
| 1.66 (0.83, 3.32) [0.15]   | 1.55 (0.67, 3.59) [0.31]   | 0.91 (0.42, 1.95) [0.81]   |
| 2.61 (1.22, 5.58) [0.013*] | 2.43 (0.99, 5.97) [0.053]  | 1.43 (0.63, 3.26) [0.39]   |
| N=146                      | N=78                       | N=78                       |
| 123 (84.2)                 | 62 (79.5)                  | 64 (82.1)                  |
| 2.21 (1.19, 4.09) [0.012*] | 1.43 (0.70, 2.90) [0.32]   | 1.51 (0.72, 3.14) [0.27]   |
| 2.45 (1.22, 4.94) [0.012*] | 1.59 (0.72, 3.47) [0.25]   | 1.67 (0.74, 3.75) [0.21]   |

with overactive bladder (OAB).

**Materials and Methods:** Adult patients with symptoms of OAB for  $\geq 3$  months were randomized to one of 12 treatment groups for 12 weeks (placebo; solifenacin 2.5/5/10 mg or mirabegron 25/50 mg alone or in combination). A responder for micturition frequency was defined as a patient with  $\geq 8$  micturitions/24 h at baseline and  $\leq 8$  micturitions/24 h post-baseline and a reduction from baseline. There were two responder analyses for patients who were incontinent at baseline: (a) proportion who became continent during treatment and (b) proportion who had a  $\geq 50\%$  reduction in incontinence episodes/24 h during treatment.

Responder analyses were also performed for PROs (a) Patient Perception of Bladder Condition (PPBC), based on a minimally important difference (MID) of 1 point; (b) the symptom bother score and (c) health related quality of life (HRQoL) subscales of the OAB-q, based on a MID of 10 points.

**Results:** Micturition frequency normalization increased with increasing dose of mirabegron monotherapy, and was highest among the 5 + 50 mg and 10 + 25 mg combinations. Due to the low number of incontinent patients (13-35 patients/group), and low baseline incontinence episodes, no meaningful conclusions can be drawn from the incontinence responder

analyses. Combination therapy, mirabegron 25 and 50 mg, and solifenacin 10 mg monotherapies were associated with higher responder rates vs. solifenacin 5 mg and vs. placebo in terms of PPBC, symptom bother score, and HRQoL (Table 1).

**Conclusions:** Combination therapy with mirabegron and solifenacin is associated with higher responder rates in terms of micturition frequency normalization and clinically relevant PRO measurements of total HRQoL, PPBC and symptom bother compared with solifenacin 5 mg monotherapy.